PhD in Children and Adolescent Counseling

CAC.901

Advanced Developmental Psychology

A doctoral lecture coursebook

Developmental plasticity Cognition Emotion Trajectories Longitudinal research

Twenty lecture chapters with real-life applications, original diagrams, seminar tasks, a working glossary, and attributed foundational readings.

PDF-only course edition 2026

Prepared for the supplied curriculum. This independent educational volume is not an official university publication, accreditation document, or clinical practice guideline.

How to Use This Book

Purpose and scope

This volume covers CAC.901 Advanced Developmental Psychology within the supplied PhD curriculum. It connects developmental science to counseling, family relationships, school participation, research interpretation, and ethical reflection. It is a complete course volume, not a claim to contain the entire discipline.

Two clearly identified layers

Each chapter begins with a newly written counseling lecture, a real-life application, and a doctoral seminar task. An original vector diagram makes a central relationship visible. The chapter then provides attributed foundational readings adapted from OpenStax Psychology 2e. The readings supply broader psychological background; the new lectures provide the counseling interpretation and critical cautions.

Simple English with doctoral reasoning

Technical terms are introduced in context and collected in the glossary. Simple language is not treated as a reason to avoid uncertainty, competing explanations, measurement limitations, or structural influences. The seminar tasks ask the reader to evaluate evidence and revise decisions rather than memorize labels.

How to study a chapter

Read the learning aims, study the lecture, explain the diagram in your own words, and complete the application task. Then read the foundational material critically. Record which claims are supported by the source, which require newer evidence, and which depend on local law or professional guidance. Finish with a short reflective note about implications for practice.

Cases and diagrams

All named cases in the new lectures are fictional teaching examples. They are not research participants or evidence of treatment effectiveness. The twenty numbered course diagrams are original conceptual illustrations. Graphs without data are explicitly labeled as schematics. Source photographs and illustrations are not reproduced; their descriptions are retained only as reading support.

Clinical and legal caution

This book is for education and professional development. It does not provide individualized diagnosis, treatment, legal advice, or emergency care. The foundational readings come from a 2020 introductory textbook. Diagnostic terminology, prevalence estimates, and examples may be dated. Use current professional standards, applicable law, qualified supervision, and appropriate referral for real cases.

Safeguarding and participation

Explain confidentiality and its limits in understandable language. Seek appropriate permission and the child s informed assent. Follow local safeguarding requirements and urgent safety procedures. Do not use a teaching exercise to investigate an allegation or pressure a child into disclosure. Share only information needed for a legitimate protective or care purpose.

Source transparency

The OpenStax readings are used under Creative Commons Attribution 4.0. They have been selected, reorganized, and typeset for this course, with new lectures and diagrams added. They are not presented as wholly original text. Module-level attribution and a pinned source snapshot appear in the attribution register. Inherited references are identified separately from the supplemental sources checked for this edition.

Length and verification

The publication check counts lexical words in the actual PDF s instructional pages and excludes the reference section, front matter, running headers, and page numbers. The minimum is 150,000 words for this individual book. The verification report is saved with the production files; the PDF is the reader-facing deliverable.

Course coverage at a glance

Developmental plasticity: Chapters 1, 4, 6, 7, 16, and 20.

Cognitive development: Chapters 4, 5, 8, 9, 10, and 11.

Emotional and social development: Chapters 5, 12, 13, 14, and 15.

Developmental trajectories: Chapters 1, 3, 16, 17, and 20.

Longitudinal research and ethics: Chapters 2, 3, 10, 19, and 20.

Integrated clinical assessment and doctoral reflection: Chapters 17 20.

Contents

Development through the lifespan 6

Observation, evidence, and research ethics 15

Designing and interpreting longitudinal research 28

Developmental theories in counseling perspective 51

Prenatal development, childhood, and adolescence 63

Genes, neurons, and developing systems 85

Brain development and biological context 100

Learning, conditioning, and modeling 115

Language, thought, and problem solving 143

Intelligence: measurement and interpretation 161

Memory development and children s accounts 178

Emotion, motivation, and regulation 202

Self, personality, and culture 219

Social cognition, self-presentation, and peer influence 238

Belonging, prejudice, aggression, and helping 260

Stress, coping, and adaptation 280

Developmental pathways to distress and disorder 306

Neurodevelopmental differences and support 330

Developmentally informed intervention 339

Integration, reflection, and doctoral practice 359

Applied Developmental Workbook 362

Working Glossary 368

References and Source Notes 372

Attribution and Production Record 436

CHAPTER 01 / CAC.901

Development through the lifespan

Lecture: Development is a moving relationship

Advanced developmental psychology begins with a deceptively simple question: how does a person change over time? A counselor cannot answer that question by looking only at a child s present behavior. The behavior has a history, a current setting, and an expected future. A seven-year-old who cries when her mother leaves may be showing a developmentally familiar response, a response to a recent separation, a reaction to an unfamiliar room, or several of these at once. The counselor s task is not to attach the first available label. It is to understand the pattern, the conditions under which it occurs, and its meaning for this child.

Development includes growth, reorganization, stability, and loss. Some changes are easy to see, such as an increase in height. Others are changes in how an existing ability is used. An adolescent may not acquire an entirely new capacity for thought overnight. Instead, she may become better at considering several possibilities, questioning an assumption, or connecting a school task to a future goal. Emotional development also involves reorganization. A young child may need an adult nearby to calm down; an older child may use words, memories, and plans to help regulate the same distress. The earlier need for support does not simply disappear. It becomes part of a changing system.

A lifespan perspective treats development as continuing beyond childhood. This matters in counseling for two reasons. First, children live within developing adults and institutions. A caregiver s health, employment, identity, and relationships can change the child s environment. Second, early experience influences later possibilities without making every later outcome inevitable. A difficult beginning deserves attention, but it should not become a prophecy. Equally, an apparently successful childhood does not guarantee later well-being. Counselors need hopeful language that remains accurate: change is possible, but it is constrained by biology, history, resources, and the support available.

Three domains, one child

Developmental descriptions often distinguish physical, cognitive, and psychosocial domains. These categories help organize observation. Physical development includes the brain, body, health, movement, and sensory capacities. Cognitive development includes attention, memory, language, reasoning, and learning. Psychosocial development includes emotion, identity, relationships, and participation in social life. The categories are analytic tools, not separate compartments inside the child.

Consider a child with intermittent hearing loss. A hearing challenge belongs partly to physical development, but it may affect access to spoken instructions, participation in group play, and a teacher s judgment about attention. A cognitive interpretation such as does not listen may miss the sensory and environmental pathway. A psychosocial interpretation such as oppositional may miss repeated experiences of confusion. The domains interact. Good assessment follows those interactions rather than forcing the child into one explanatory box.

The same principle applies to sleep. Insufficient sleep can affect emotional control, learning, and family conflict. A counselor who treats only the argument at breakfast may miss a modifiable contributor. This does not mean that every emotional problem is a sleep problem. It means that a developmental formulation considers several plausible pathways and seeks evidence that can distinguish them.

Plasticity is not unlimited change

Plasticity means that development can be modified. Learning, supportive relationships, rehabilitation, and changes in context can alter functioning. Plasticity is a reason to provide intervention rather than assume that a difficulty is fixed. It is not a promise that any outcome can be produced through effort. Different capacities have different constraints, and the same intervention may help one person more than another.

Imagine two children who experience a disruption in schooling. One returns to a stable home, an accessible classroom, and familiar friends. The other faces continuing displacement, financial strain, and inconsistent adult availability. It would be misleading to describe both children as equally plastic and then explain their different outcomes by motivation. The capacity for change exists within conditions. A responsible counselor asks what would make change more possible, not merely whether the child is trying hard enough.

Plasticity also has a temporal dimension. An experience may matter differently depending on when it occurs and how long it lasts. However, counselors should be cautious with dramatic claims that a particular opportunity closes forever at a single age. Developmental timing can matter without supporting a simple deadline for every skill. When making a claim about a sensitive period, specify the domain, the evidence, and the limits of generalization.

A trajectory is not a destiny

A developmental trajectory describes a pattern over time. It might concern reading, anxious symptoms, peer connection, or sleep. A single measurement is a point, not a trajectory. Even two measurements can be misleading if they capture an unusual week. Repeated observation helps distinguish a temporary state from a more persistent pattern, although persistence alone does not establish the cause.

Figure 1.1 shows several possible paths. Its lines are teaching illustrations, not results from a study. One child may improve after a change in support; another may show a delayed difficulty when school demands increase; a third may fluctuate with circumstances. The practical lesson is to ask what changed around the change in functioning. Did teaching change? Did a friendship end? Did a health problem improve? Did the measurement method change? A graph invites questions rather than ending them.

A developmental counselor also distinguishes age, cohort, and historical period. Age refers to time in an individual life. A cohort is a group that moves through time together under partly shared conditions. A historical period includes events that can affect several age groups. Adolescents who experienced disrupted schooling are not simply the usual fifteen-year-olds. Their age-related development occurred within a particular history. Nor should every member of that cohort be assumed to have had the same experience.

From description to explanation

Suppose a parent says, He has always been difficult. This statement communicates strain and may contain useful information, but it is not yet a developmental account. Ask about always. When did the difficulty first become noticeable? In which settings? With which adults? What was happening in the family at that time? What does the child do well? What helps? The purpose is not to challenge the parent s credibility. It is to turn a broad narrative into observations that can guide support.

A doctoral-level formulation distinguishes description, hypothesis, and evidence. The child leaves his seat frequently is a description. The behavior helps him escape a difficult task is a hypothesis. Repeated observation showing that seat-leaving increases during difficult work and results in removal from that work would support the hypothesis, though further investigation may still be needed. Confusing these levels leads to confident but weak reasoning.

The child s own account matters. Children are not merely collections of adult ratings. Their explanations may be incomplete, but so may adults explanations. A child may describe a fear of reading aloud that no behavior chart records. Use developmentally appropriate communication, allow time, and avoid treating silence as agreement. Participation is both an ethical commitment and a source of information.

Application: the new school term

A fictional nine-year-old, Amara, begins complaining of stomachaches on school mornings. Her parent worries that she is becoming defiant. A teacher reports that she is quiet and sometimes asks to visit the sick bay. Before choosing an explanation, map the timing. Did the symptoms begin with a new class, a health change, a bullying incident, or a family event? Are there mornings without symptoms? What happens after she reaches school? What does Amara herself notice?

The map may reveal several contributors. Perhaps reading tasks have become harder, a friendship has changed, and worried adults unintentionally provide relief through avoidance. No single observation proves the whole account. The counselor can coordinate proportionate health assessment, talk with Amara, examine school demands, and agree on a plan that supports attendance without shame. The developmental question is how the pattern formed and what might change it, not which adult has correctly identified a bad attitude.

Doctoral seminar task

Select a behavior that adults often describe as a fixed trait. Write one purely descriptive account, two competing developmental hypotheses, and the observations that would help distinguish them. Include the child s likely perspective and one structural constraint. Then identify what your preferred explanation would imply for intervention. Finally, state what evidence would make you revise that explanation. This exercise trains flexibility rather than diagnostic certainty alone.

The chapter s central habit is disciplined curiosity. Describe before explaining; locate the person in time and context; treat plasticity as conditional rather than magical; and use repeated evidence without turning a trajectory into a verdict. These habits prepare the counselor to read research critically and to intervene without ignoring the child s lived world.

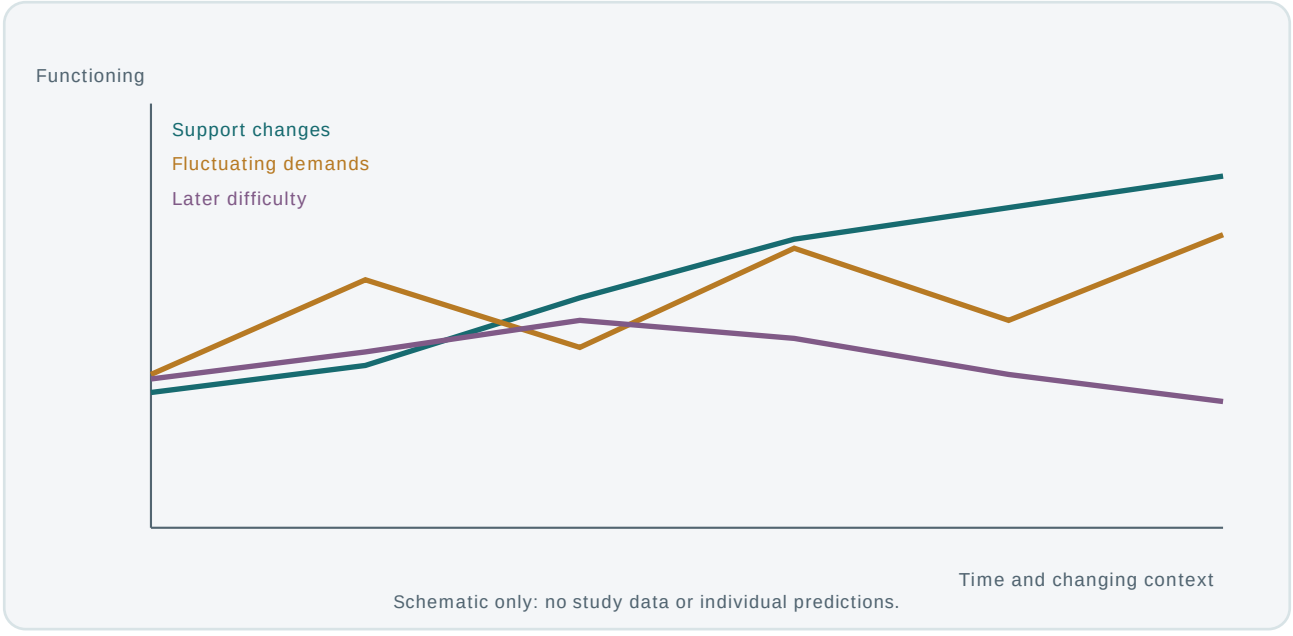


Figure 1.1. Different paths are possible. Illustrative developmental trajectories, not observed study data. The same starting point can be followed by improvement, fluctuation, or later difficulty. A trajectory describes a pattern; it does not determine an individual future. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

What Is Lifespan Development?

OpenStax source module m82218 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82218/index.cnxml

My heart leaps up when I behold A rainbow in the sky: So was it when my life began; So is it now I am a man; So be it when I shall grow old, Or let me die! The Child is father of the Man; I could wish my days to be Bound each to each by natural piety. (Wordsworth, 1802)

In this poem, William Wordsworth writes, the child is father of the man. What does this seemingly incongruous statement mean, and what does it have to do with lifespan development? Wordsworth might be suggesting that the person he is as an adult depends largely on the experiences he had in childhood. Consider the following questions: To what extent is the adult you are today influenced by the child you once were? To what extent is a child fundamentally different from the adult he grows up to be?

These are the types of questions developmental psychologists try to answer, by studying how humans change and grow from conception through childhood, adolescence, adulthood, and death. They view development as a lifelong process that can be studied scientifically across three developmental domains physical, cognitive, and psychosocial development. Physical development involves growth and changes in the body and brain, the senses, motor skills, and health and wellness. Cognitive development involves learning, attention, memory, language, thinking, reasoning, and creativity. Psychosocial development involves emotions, personality, and social relationships. We refer to these domains throughout the chapter.

Connect the Concepts

You ve learned about a variety of research methods used by psychologists. Developmental psychologists use many of these approaches in order to better understand how individuals change mentally and physically over time. These methods include naturalistic observations, case studies, surveys, and experiments, among others.

Naturalistic observations involve observing behavior in its natural context. A developmental psychologist might observe how children behave on a playground, at a daycare center, or in the child s own home. While this research approach provides a glimpse into how children behave in their natural settings, researchers have very little control over the types and/or frequencies of displayed behavior.

In a case study, developmental psychologists collect a great deal of information from one individual in order to better understand physical and psychological changes over the lifespan. This particular approach is an excellent way to better understand individuals, who are exceptional in some way, but it is especially prone to researcher bias in interpretation, and it is difficult to generalize conclusions to the larger population.

In one classic example of this research method being applied to a study of lifespan development Sigmund Freud analyzed the development of a child known as Little Hans (Freud, 1909/1949). Freud s findings helped inform his theories of psychosexual development in children, which you will learn about later in this chapter. Little Genie, the subject of a case study discussed in the chapter on thinking and intelligence, provides another example of how psychologists examine developmental milestones through detailed research on a single individual. In Genie s case, her neglectful and abusive upbringing led to her being unable to speak until, at age 13, she was removed from that

harmful environment. As she learned to use language, psychologists were able to compare how her language acquisition abilities differed when occurring in her late-stage development compared to the typical acquisition of those skills during the ages of infancy through early childhood (Fromkin, Krashen, Curtiss, Rigler, & Rigler, 1974; Curtiss, 1981).

The survey method asks individuals to self-report important information about their thoughts, experiences, and beliefs. This particular method can provide large amounts of information in relatively short amounts of time; however, validity of data collected in this way relies on honest self-reporting, and the data is relatively shallow when compared to the depth of information collected in a case study. An example of comprehensive survey was the research done by Ruth W. Howard. In 1934, she obtained her doctorate by surveying 229 sets of triplets, the most comprehensive research of triplets completed at the time. This pioneering woman was also the first African-American woman to earn a PhD in psychology (American Psychological Association, 2019).

Experiments involve significant control over extraneous variables and manipulation of the independent variable. As such, experimental research allows developmental psychologists to make causal statements about certain variables that are important for the developmental process. Because experimental research must occur in a controlled environment, researchers must be cautious about whether behaviors observed in the laboratory translate to an individual s natural environment.

Later in this chapter, you will learn about several experiments in which toddlers and young children observe scenes or actions so that researchers can determine at what age specific cognitive abilities develop. For example, children may observe a quantity of liquid poured from a short, fat glass into a tall, skinny glass. As the experimenters question the children about what occurred, the subjects answers help psychologists understand at what age a child begins to comprehend that the volume of liquid remained the same although the shapes of the containers differs.

Across these three domains physical, cognitive, and psychosocial the normative approach to development is also discussed. This approach asks, What is normal development? In the early decades of the 20th century, normative psychologists studied large numbers of children at various ages to determine norms (i.e., average ages) of when most children reach specific developmental milestones in each of the three domains (Gesell, 1933, 1939, 1940; Gesell & Ilg, 1946; Hall, 1904). Although children develop at slightly different rates, we can use these age-related averages as general guidelines to compare children with same-age peers to determine the approximate ages they should reach specific normative events called developmental milestones (e.g., crawling, walking, writing, dressing, naming colors, speaking in sentences, and starting puberty).

Not all normative events are universal, meaning they are not experienced by all individuals across all cultures. Biological milestones, such as puberty, tend to be universal, but social milestones, such as the age when children begin formal schooling, are not necessarily universal; instead, they affect most individuals in a particular culture (Gesell & Ilg, 1946). For example, in developed countries children begin school around 5 or 6 years old, but in developing countries, like Nigeria, children often enter school at an advanced age, if at all (Huebler, 2005; United Nations Educational, Scientific, and Cultural Organization [UNESCO], 2013).

To better understand the normative approach, imagine two new mothers, Louisa and Kimberly, who are close friends and have children around the same age. Louisa s daughter is 14 months old, and Kimberly s son is 12 months old. According to the normative approach, the average age a child starts to walk is 12 months. However, at 14 months Louisa s daughter still isn t walking. She tells Kimberly she is worried that something might be wrong with her baby. Kimberly is surprised because her son started walking when he was only 10 months old. Should Louisa be worried? Should she be concerned if her daughter is not walking by 15 months or 18 months?

Issues in Developmental Psychology

There are many different theoretical approaches regarding human development. As we evaluate them in this chapter, recall that developmental psychology focuses on how people change, and keep in mind that all the approaches that we present in this chapter address questions of change: Is the change smooth or uneven (continuous versus discontinuous)? Is this pattern of change the same for everyone, or are there many different patterns of change (one course of development versus many courses)? How do genetics and environment interact to influence development (nature versus nurture)?

Is Development Continuous or Discontinuous?

Continuous development views development as a cumulative process, gradually improving on existing skills (the accompanying source-figure description). With this type of development, there is gradual change. Consider, for example, a child s physical growth: adding inches to height year by year. In contrast, theorists who view development as discontinuous believe that development takes place in unique stages: It occurs at specific times or ages. With this type of development, the change is more sudden, such as an infant s ability to conceive object permanence.

Source illustration description (image not reproduced): Continuous and Discontinuous development are shown side by side using two separate pictures. The first picture is a triangle labeled Continuous Development which slopes upward from Infancy to Adulthood in a straight line. The second picture is 4 bars side by side labeled Discontinuous Development which get higher from Infancy to Adulthood. These bars resemble a staircase. The concept of continuous development can be visualized as a smooth slope of progression, whereas discontinuous development sees growth in more discrete stages.

Is There One Course of Development or Many?

Is development essentially the same, or universal, for all children (i.e., there is one course of development) or does development follow a different course for each child, depending on the child s specific genetics and environment (i.e., there are many courses of development)? Do people across the world share more similarities or more differences in their development? How much do culture and genetics influence a child s behavior?

Stage theories hold that the sequence of development is universal. For example, in cross-cultural studies of language development, children from around the world reach language milestones in a similar sequence (Gleitman & Newport, 1995). Infants in all cultures coo before they babble. They begin babbling at about the same age and utter their first word around 12 months old. Yet we live in diverse contexts that have a unique effect on each of us. For example, researchers once believed that motor development follows one course for all children regardless of culture. However, child care practices vary by culture, and different practices have been found to accelerate or inhibit achievement of developmental milestones such as sitting, crawling, and walking (Karasik, Adolph, Tamis-LeMonda, & Bornstein, 2010).

For instance, let s look at the Ach society in Paraguay. They spend a significant amount of time foraging in forests. While foraging, Ach mothers carry their young children, rarely putting them down in order to protect them from getting hurt in the forest. Consequently, their children walk much later: They walk around 23 25 months old, in comparison to infants in Western cultures who begin to walk around 12 months old. However, as Ach children become older, they are allowed more freedom to move about, and by about age 9, their motor skills surpass those of U.S. children of the same age: Ach children are able to climb trees up to 25 feet tall and use machetes to chop their way through the forest (Kaplan & Dove, 1987). As you can see, our development is influenced by multiple contexts, so the timing of basic motor functions may vary across cultures. However, the functions themselves are present in all societies (the accompanying source-figure description).

Source illustration description (image not reproduced): Photograph A shows two children wearing inner tubes playing in the shallow water at the beach. Photograph B shows two children playing in the sand at a beach. All children across the world love to play. Whether in (a) Florida or (b) South Africa, children enjoy exploring sand, sunshine, and the sea. (credit a: modification of work by Visit St. Pete/Clearwater /Flickr; credit b: modification of work by

"stringer_bel"/Flickr)

How Do Nature and Nurture Influence Development?

Are we who we are because of nature (biology and genetics), or are we who we are because of nurture (our environment and culture)? This longstanding question is known in psychology as the nature versus nurture debate. It seeks to understand how our personalities and traits are the product of our genetic makeup and biological factors, and how they are shaped by our environment, including our guardians, peers, and culture. For instance, why do biological children sometimes act like their parents is it because of genetics or because of early childhood environment and what the child has learned from the parents? What about children who are adopted are they more like their biological families or more like their adoptive families? And how can siblings from the same family be so different?

We are all born with specific genetic traits inherited from our parents, such as eye color, height, and certain personality traits. Beyond our basic genotype, however, there is a deep interaction between our genes and our environment: Our unique experiences in our environment influence whether and how particular traits are expressed, and at the same time, our genes influence how we interact with our environment (Diamond, 2009; Lobo, 2008). This chapter will show that there is a reciprocal interaction between nature and nurture as they both shape who we become, but the debate continues as to the relative contributions of each.

Dig Deeper

The achievement gap refers to the persistent difference in grades, test scores, and graduation rates that exist among students of different ethnicities, races, and in certain subjects genders (Winerman, 2011). Research suggests that these achievement gaps are strongly influenced by differences in socioeconomic factors that exist among the families of these children. While the researchers acknowledge that programs aimed at reducing such socioeconomic discrepancies would likely aid in equalizing the aptitude and performance of children from different backgrounds, they recognize that such large-scale interventions would be difficult to achieve. Therefore, it is recommended that programs aimed at fostering aptitude and achievement among disadvantaged children may be the best option for dealing with issues related to academic achievement gaps (Duncan & Magnuson, 2005).

Children from low-income households perform significantly more poorly than their middle- and high-income peers on a number of educational variables: They have significantly lower standardized test scores, graduation rates, and college entrance rates, and they have much higher school dropout rates. There have been attempts to correct the achievement gap through state and federal legislation, but what if the problems start before the children even enter school?

Psychologists Betty Hart and Todd Risley (2006) spent their careers looking at early language ability and progression of children in various income levels. In one longitudinal study, they found that although all the parents in the study engaged and interacted with their children, middle- and high-income parents interacted with their children differently than low-income parents. After analyzing 1,300 hours of parent-child interactions, the researchers found that middle- and high-income parents talk to their children significantly more, starting when the children are infants. By 3 years old, high-income children knew almost double the number of words known by their low-income counterparts, and they had heard an estimated total of 30 million more words than the low-income counterparts (Hart & Risley, 2003). And the gaps only become more pronounced. Before entering kindergarten, high-income children score 60% higher on achievement tests than their low-income peers (Lee & Burkam, 2002).

There are solutions to this problem. At the University of Chicago, experts are working with low-income families, visiting them at their homes, and encouraging them to speak more to their children on a daily and hourly basis. Other experts are designing preschools in which students from diverse economic backgrounds are placed in the same classroom. In this research, low-income children made significant gains in their language development, likely as a result of attending the specialized preschool (Schechter & Byeb, 2007). What other methods or interventions could be used to decrease the achievement gap? What types of activities could be implemented to help the children of your community or a neighboring community?

Summary

Lifespan development explores how we change and grow from conception to death. This field of psychology is studied by developmental psychologists. They view development as a lifelong process that can be studied scientifically across three developmental domains: physical, cognitive development, and psychosocial. There are several theories of development that focus on the following issues: whether development is continuous or discontinuous, whether development follows one course or many, and the relative influence of nature versus nurture on development.

Review Questions

OpenStax learning exercise

Q. The view that development is a cumulative process, gradually adding to the same type of skills is known as _____.

A. C

OpenStax learning exercise

Q. Developmental psychologists study human growth and development across three domains. Which of the following is not one of these domains?

A. B

OpenStax learning exercise

Q. How is lifespan development defined?

A. A

Critical Thinking Questions

OpenStax learning exercise

Q. Describe the nature versus nurture controversy, and give an example of a trait and how it might be influenced by each?

A. The nature versus nurture controversy seeks to understand whether our personalities and traits are the product of our genetic makeup and biological factors, or whether they are shaped by our environment, which includes such things as our parents, peers, and culture. Today, psychologists agree that both nature and nurture interact to shape who we become, but the debate over the relative contributions of each continues. An example would be a child learning to walk: Nature influences when the physical ability occurs, but culture can influence when a child masters this skill, as in Ach culture.

OpenStax learning exercise

Q. Compare and contrast continuous and discontinuous development.

A. Continuous development sees our development as a cumulative process: Changes are gradual. On the other hand, discontinuous development sees our development as taking place in specific steps or stages: Changes are sudden.

OpenStax learning exercise

Q. Why should developmental milestones only be used as a general guideline for normal child development?

A. Children develop at different rates. For example, some children may walk and talk as early as 8 months old, while others may not do so until well after their first birthday. Each child s unique contexts will influence when he reaches these milestones.

Personal Application Questions

OpenStax learning exercise

Q. How are you different today from the person you were at 6 years old? What about at 16 years old? How are you the same as the person you were at those ages?

OpenStax learning exercise

Q. Your 3-year-old daughter is not yet potty trained. Based on what you know about the normative approach, should you be concerned? Why or why not?

CHAPTER 02 / CAC.901

Observation, evidence, and research ethics

Lecture: Evidence, permission, and the child s voice

A caring intention does not make a claim true. A counselor may sincerely believe that a new school program reduces anxiety, yet the belief requires investigation. Research helps by making claims answerable to evidence. It does not remove values from decision-making. Researchers still choose which questions matter, which outcomes count as improvement, whose voices are heard, and which risks are acceptable. Good developmental science combines systematic methods with explicit ethical reasoning.

Begin by separating a practical problem from a research question. This child is struggling identifies a concern. What happens to this child s attendance when the classroom entry routine is made more predictable? is a more specific question. It identifies a person or population, an intervention or condition, an outcome, and a time frame. Specificity makes evidence easier to collect and interpret. It also prevents a broad promise, such as improving well-being, from hiding the absence of a measurable outcome.

What an observation can and cannot tell you

Observation is essential when working with children, but it is never a view from nowhere. A child who speaks little in a clinic may speak freely at home. A child who appears calm may have learned not to show distress around unfamiliar adults. An observer s presence can change the setting. The observer also chooses what to record. Two people may watch the same playground and notice different events because they hold different expectations.

Improve observation by defining behavior clearly. Aggressive is broad. Hits or kicks another child during unstructured play is easier to count. Record the setting, activity, people present, and time. When appropriate, examine what happens before and after the behavior. Use more than one observation when feasible, and check whether independent observers agree. Agreement does not guarantee that the category is meaningful, but disagreement warns that the measurement needs attention.

Self-report and adult report answer different questions. A caregiver can describe sleep routines and developmental history; a teacher can describe classroom demands; a child can describe private worry, pain, or belonging. Disagreement is not automatically evidence that someone is wrong. It may reflect real variation across settings or different access to internal experience. A multimethod assessment asks what each source can reasonably know.

Research permission is a process

In research involving minors, permission from a parent or legally authorized representative and the child s assent are distinct considerations. Exact legal requirements vary by jurisdiction and by the nature of the study. Ethics review is required where applicable; counselor enthusiasm or school approval does not replace it. The research team must explain the purpose, procedures, foreseeable risks, possible benefits, alternatives, privacy protections, and the voluntary nature of participation in language the participants can understand.

Assent should not be treated as a signature obtained quickly before data collection. A child needs a genuine opportunity to ask questions and decline. Understanding can be checked by asking the child to explain the study in their own words. The explanation should be revisited when the study changes or when a young participant becomes distressed. Signs of unwillingness matter even if a caregiver has given permission. A child s participation should not be secured through pressure disguised as encouragement.

Confidentiality in research also needs precise explanation. What information will be stored? Who can access it? Will quotations be used? Could a school or family recognize a participant from a description? What would happen if the team learned of a serious safety concern? Avoid promising absolute secrecy. Explain limits before participation, and design reporting so that small groups are not made identifiable through unnecessary detail.

Evidence and power

Research relationships contain power. A teacher may ask a class to participate, and children may assume that refusal will affect grades. A service provider may invite clients into a study, and families may fear losing support. Recruitment should separate participation from access to ordinary services wherever possible. An independent person can sometimes help communicate that declining will not bring a penalty. Payment or other incentives should encourage participation without becoming so large that they distort voluntary choice.

The history of research makes these protections more than administrative formalities. Communities have experienced extraction, misrepresentation, and interventions imposed without meaningful participation. A culturally responsive research plan asks whether the question is important to the community, whether local concepts have been understood, and how findings will be returned. Community consultation does not transfer every ethical responsibility away from the researcher, but it can reveal harms that a distant protocol missed.

A practical claim under scrutiny

A school advertises that a resilience workshop works because participants report feeling more confident afterward. This is a beginning, not a complete evaluation. Confidence may improve because of the workshop, because participants expected improvement, because the term became less stressful, or because the most distressed students were absent at follow-up. Without an appropriate comparison, the workshop s specific contribution is uncertain.

Figure 2.1 separates the claim, the evidence offered, alternative explanations, and the next evidence needed. The point is not to dismiss every promising program. It is to match the strength of the conclusion to the design. A pre-post survey can show change and acceptability. It cannot, by itself, isolate the effect of the program from other changes occurring at the same time.

A counselor can communicate this distinction without using discouraging language. Students reported improvement, and we need a comparison to estimate how much the program contributed is both honest and useful. It preserves the value of the initial information while identifying the next step. Families and school leaders are better served by calibrated confidence than by a promise that later proves unsupported.

Application: evaluating a homework club

A fictional community center wants to know whether its homework club improves school engagement. The team first defines engagement. Is it attendance, completed assignments, student-reported belonging, or teacher-rated participation? Choosing one outcome changes the meaning of success. The team then considers selection: students who attend voluntarily may already have more support. Comparing them with nonattendees without accounting for that difference could exaggerate the club s effect.

A stronger evaluation might use an appropriate comparison group, collect baseline information, track attendance at the club, and ask students about barriers. If random assignment is feasible and ethical, it can reduce selection bias. If it is not, the team should describe the limits of causal interpretation. Implementation matters too. A club that operates twice a week differs from one that rarely opens, even if both have the same name.

Children can help interpret the findings. They may explain that transportation, hunger, or caregiving duties affected attendance more than motivation did. Their insight can prevent the team from turning a structural barrier into an individual deficit. Research ethics and research quality meet at this point: listening carefully can improve both the fairness and the validity of the account.

Doctoral seminar task

Draft a one-page participant explanation for a study of a school-based coping program. Use short sentences without hiding important uncertainty. Then write a separate assent conversation for a ten-year-old and a permission explanation for a caregiver. Identify three ways the recruitment setting could create pressure and propose a protection for each. Do not assume that the same legal rules apply everywhere.

Next, design two measures of the program s intended outcome, one from the child and one from another source. Explain what each measure can capture and what it may miss. State how disagreement will be handled. The exercise should make clear that ethical research is not a form attached to good science; it is part of the conditions under which trustworthy knowledge is produced.



Figure 2.1. From a claim to a proportionate conclusion. A research claim becomes more trustworthy when the question, measurement, design, ethics, and alternative explanations are made explicit. The final conclusion must remain within the limits of the evidence. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

Why Is Research Important?

OpenStax source module m82167 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82167/index.cnxml

Scientific research is a critical tool for successfully navigating our complex world. Without it, we would be forced to rely solely on intuition, other people s authority, and blind luck. While many of us feel confident in our abilities to decipher and interact with the world around us, history is filled with examples of how very wrong we can be when we fail to recognize the need for evidence in supporting claims. At various times in history, we would have been certain that the sun revolved around a flat earth, that the earth s continents did not move, and that mental illness was caused by possession (the accompanying source-figure description). It is through systematic scientific research that we divest ourselves of our preconceived notions and superstitions and gain an objective understanding of ourselves and our world.

Source illustration description (image not reproduced): A skull has a large hole bored through the forehead. Some of our ancestors, across the world and over the centuries, believed that trephination the practice of making a hole in the skull, as shown here allowed evil spirits to leave the body, thus curing mental illness and other disorders. (credit: taiproject /Flickr)

The goal of all scientists is to better understand the world around them. Psychologists focus their attention on understanding behavior, as well as the cognitive (mental) and physiological (body) processes that underlie behavior. In contrast to other methods that people use to understand the behavior of others, such as intuition and personal experience, the hallmark of scientific research is that there is evidence to support a claim. Scientific knowledge is empirical: It is grounded in objective, tangible evidence that can be observed time and time again, regardless of who is observing.

While behavior is observable, the mind is not. If someone is crying, we can see behavior. However, the reason for the behavior is more difficult to determine. Is the person crying due to being sad, in pain, or happy? Sometimes we can learn the reason for someone s behavior by simply asking a question, like Why are you crying? However, there are situations in which an individual is either uncomfortable or unwilling to answer the question honestly, or is incapable of answering. For example, infants would not be able to explain why they are crying. In such circumstances, the psychologist must be creative in finding ways to better understand behavior. This chapter explores how scientific knowledge is generated, and how important that knowledge is in forming decisions in our personal lives and in the public domain.

Use of Research Information

Trying to determine which theories are and are not accepted by the scientific community can be difficult, especially in an area of research as broad as psychology. More than ever before, we have an incredible amount of information at our fingertips, and a simple internet search on any given research topic might result in a number of contradictory studies. In these cases, we are witnessing the scientific community going through the process of reaching a consensus, and it could be quite some time before a consensus emerges. For example, the explosion in our use of technology has led researchers to question whether this ultimately helps or hinders us. The use and implementation of technology in educational settings has become widespread over the last few decades. Researchers are coming to different conclusions regarding the use of technology. To illustrate this point, a study investigating a smartphone app targeting surgery residents (graduate students in surgery training) found that the use of this app can increase student engagement and raise test scores (Shaw & Tan, 2015). Conversely, another study found that the use of technology in undergraduate student populations had negative impacts on sleep, communication, and time management skills (Massimini & Peterson, 2009). Until sufficient amounts of research have been conducted, there will be no clear consensus on the effects that technology has on a student's acquisition of knowledge, study skills, and mental health.

In the meantime, we should strive to think critically about the information we encounter by exercising a degree of healthy skepticism. When someone makes a claim, we should examine the claim from a number of different perspectives: what is the expertise of the person making the claim, what might they gain if the claim is valid, does the claim seem justified given the evidence, and what do other researchers think of the claim? This is especially important when we consider how much information in advertising campaigns and on the internet claims to be based on scientific evidence when in actuality it is a belief or perspective of just a few individuals trying to sell a product or draw attention to their perspectives.

We should be informed consumers of the information made available to us because decisions based on this information have significant consequences. One such consequence can be seen in politics and public policy. Imagine that you have been elected as the governor of your state. One of your responsibilities is to manage the state budget and determine how to best spend your constituents tax dollars. As the new governor, you need to decide whether to continue funding early intervention programs. These programs are designed to help children who come from low-income backgrounds, have special needs, or face other disadvantages. These programs may involve providing a wide variety of services to maximize the children's development and position them for optimal levels of success in school and later in life (Blann, 2005). While such programs sound appealing, you would want to be sure that they also proved effective before investing additional money in these programs. Fortunately, psychologists and other scientists have conducted vast amounts of research on such programs and, in general, the programs are found to be effective (Neil & Christensen, 2009; Peters-Scheffer, Didden, Korzilius, & Sturmey, 2011). While not all programs are equally effective, and the short-term effects of many such programs are more pronounced, there is reason to believe that many of these programs produce long-term benefits for participants (Barnett, 2011). If you are committed to being a good steward of taxpayer money, you would want to look at research. Which programs are most effective? What characteristics of these programs make them effective? Which programs promote the best outcomes? After examining the research, you would be best equipped to make decisions about which programs to fund.

Ultimately, it is not just politicians who can benefit from using research in guiding their decisions. We all might look to research from time to time when making decisions in our lives. Imagine you just found out that a close friend has breast cancer or that one of your young relatives has recently been diagnosed with autism. In either case, you want to know which treatment or support options are most successful with the fewest negative impacts. How would you find that out? You would probably talk with your doctor regarding the research that has been done on various treatment options always with a critical eye to ensure that you are as informed as possible.

In the end, research is what makes the difference between facts and opinions. Facts are observable realities, and opinions are personal judgments, conclusions, or attitudes that may or may not be accurate. In the scientific community, facts can be established only using evidence collected through empirical research.

NOTABLE RESEARCHERS

Psychological research has a long history involving important figures from diverse backgrounds. While the introductory chapter discussed several researchers who made significant contributions to the discipline, there are many more individuals who deserve attention in considering how psychology has advanced as a science through their work (the accompanying source-figure description). For instance, Margaret Floy Washburn (1871 1939) was the first woman to earn a PhD in psychology. Her research focused on animal behavior and cognition (Margaret Floy Washburn, PhD, n.d.). Mary Whiton Calkins (1863 1930) was a preeminent first-generation American psychologist who opposed the behaviorist movement, conducted significant research into memory, and established one of the earliest experimental psychology labs in the United States (Mary Whiton Calkins, n.d.).

Francis Sumner (1895 1954) was the first African American to receive a PhD in psychology in 1920. His dissertation focused on issues related to psychoanalysis. Sumner also had research interests in racial bias and educational justice. Sumner was one of the founders of Howard University s department of psychology, and because of his accomplishments, he is sometimes referred to as the Father of Black Psychology. Thirteen years later, Inez Beverly Prosser (1895 1934) became the first African American woman to receive a PhD in psychology. Prosser s research highlighted issues related to education in segregated versus integrated schools, and ultimately, her work was very influential in the hallmark Brown v. Board of Education Supreme Court ruling that segregation of public schools was unconstitutional (Ethnicity and Health in America Series: Featured Psychologists, n.d.).

Source illustration description (image not reproduced): Figure a is a portrait of Margaret Floy Washburn. Figure b is the front page of the Implementation Decree from the Supreme Court for the Brown vs. Board of Education case. (a) Margaret Floy Washburn was the first woman to earn a doctorate degree in psychology. (b) The outcome of Brown v. Board of Education was influenced by the research of psychologist Inez Beverly Prosser, who was the first African American woman to earn a PhD in psychology.

Although the establishment of psychology s scientific roots occurred first in Europe and the United States, it did not take much time until researchers from around the world began to establish their own laboratories and research programs. For example, some of the first experimental psychology laboratories in South America were founded by Horatio Pi ero (1869 1919) at two institutions in Buenos Aires, Argentina (Godoy & Brussino, 2010). In India, Gunamudian David Boaz (1908 1965) and Narendra Nath Sen Gupta (1889 1944) established the first independent departments of psychology at the University of Madras and the University of Calcutta, respectively. These developments provided an opportunity for Indian researchers to make important contributions to the field (Gunamudian David Boaz, n.d.; Narendra Nath Sen Gupta, n.d.).

When the American Psychological Association (APA) was first founded in 1892, all of the members were White males (Women and Minorities in Psychology, n.d.). However, by 1905, Mary Whiton Calkins was elected as the first female president of the APA, and by 1946, nearly one-quarter of American psychologists were female. Psychology became a popular degree option for students enrolled in the nation s historically Black higher education institutions, increasing the number of Black Americans who went on to become psychologists. Given demographic shifts occurring in the United States and increased access to higher educational opportunities among historically underrepresented populations, there is reason to hope that the diversity of the field will increasingly match the larger population, and that the research contributions made by the psychologists of the future will better serve people of all backgrounds (Women and Minorities in Psychology, n.d.).

The Process of Scientific Research

Scientific knowledge is advanced through a process known as the scientific method. Basically, ideas (in the form of theories and hypotheses) are tested against the real world (in the form of empirical observations), and those empirical observations lead to more ideas that are tested against the real world, and so on. In this sense, the scientific process is circular. The types of reasoning within the circle are called deductive and inductive. In deductive reasoning, ideas are tested in the real world; in inductive reasoning, real-world observations lead to new ideas (the accompanying source-figure description). These processes are inseparable, like inhaling and exhaling, but different research approaches place different emphasis on the deductive and inductive aspects.

Source illustration description (image not reproduced): A diagram has a box at the top labeled hypothesis or general premise and a box at the bottom labeled empirical observations. On the left, an arrow labeled inductive reasoning goes from the bottom to top box. On the right, an arrow labeled deductive reasoning goes from the top to the bottom box. Psychological research relies on both inductive and deductive reasoning.

In the scientific context, deductive reasoning begins with a generalization one hypothesis that is then used to reach logical conclusions about the real world. If the hypothesis is correct, then the logical conclusions reached through deductive reasoning should also be correct. A deductive reasoning argument might go something like this: All living things require energy to survive (this would be your hypothesis). Ducks are living things. Therefore, ducks require energy to survive (logical conclusion). In this example, the hypothesis is correct; therefore, the conclusion is correct as well. Sometimes, however, an incorrect hypothesis may lead to a logical but incorrect conclusion. Consider this argument: all ducks are born with the ability to see. Quackers is a duck. Therefore, Quackers was born with the ability to see. Scientists use deductive reasoning to empirically test their hypotheses. Returning to the example of the ducks, researchers might design a study to test the hypothesis that if all living things require energy to survive, then ducks will be found to require energy to survive.

Deductive reasoning starts with a generalization that is tested against real-world observations; however, inductive reasoning moves in the opposite direction. Inductive reasoning uses empirical observations to construct broad generalizations. Unlike deductive reasoning, conclusions drawn from inductive reasoning may or may not be correct, regardless of the observations on which they are based. For instance, you may notice that your favorite fruits apples, bananas, and oranges all grow on trees; therefore, you assume that all fruit must grow on trees. This would be an example of inductive reasoning, and, clearly, the existence of strawberries, blueberries, and kiwi demonstrate that this generalization is not correct despite it being based on a number of direct observations. Scientists use inductive reasoning to formulate theories, which in turn generate hypotheses that are tested with deductive reasoning. In the end, science involves both deductive and inductive processes.

For example, case studies, which you will read about in the next section, are heavily weighted on the side of empirical observations. Thus, case studies are closely associated with inductive processes as researchers gather massive amounts of observations and seek interesting patterns (new ideas) in the data. Experimental research, on the other hand, puts great emphasis on deductive reasoning.

We ve stated that theories and hypotheses are ideas, but what sort of ideas are they, exactly? A theory is a well-developed set of ideas that propose an explanation for observed phenomena. Theories are repeatedly checked against the world, but they tend to be too complex to be tested all at once; instead, researchers create hypotheses to test specific aspects of a theory.

A hypothesis is a testable prediction about how the world will behave if our idea is correct, and it is often worded as an if-then statement (e.g., if I study all night, I will get a passing grade on the test). The hypothesis is extremely important because it bridges the gap between the realm of ideas and the real world. As specific hypotheses are tested, theories are modified and refined to reflect and incorporate the result of these tests the accompanying source-figure description.

Source illustration description (image not reproduced): A diagram has seven labeled boxes with arrows to show the progression in the flow chart. The chart starts at Theory and moves to Generate hypothesis, Collect data,

Analyze data, and Summarize data and report findings. There are two arrows coming from Summarize data and report findings to show two options. The first arrow points to Confirm theory. The second arrow points to Modify theory, which has an arrow that points back to Generate hypothesis. The scientific method involves deriving hypotheses from theories and then testing those hypotheses. If the results are consistent with the theory, then the theory is supported. If the results are not consistent, then the theory should be modified and new hypotheses will be generated.

To see how this process works, let s consider a specific theory and a hypothesis that might be generated from that theory. As you ll learn in a later chapter, the James-Lange theory of emotion asserts that emotional experience relies on the physiological arousal associated with the emotional state. If you walked out of your home and discovered a very aggressive snake waiting on your doorstep, your heart would begin to race and your stomach churn. According to the James-Lange theory, these physiological changes would result in your feeling of fear. A hypothesis that could be derived from this theory might be that a person who is unaware of the physiological arousal that the sight of the snake elicits will not feel fear.

A scientific hypothesis is also falsifiable, or capable of being shown to be incorrect. Recall from the introductory chapter that Sigmund Freud had lots of interesting ideas to explain various human behaviors (the accompanying source-figure description). However, a major criticism of Freud s theories is that many of his ideas are not falsifiable; for example, it is impossible to imagine empirical observations that would disprove the existence of the id, the ego, and the superego the three elements of personality described in Freud s theories. Despite this, Freud s theories are widely taught in introductory psychology texts because of their historical significance for personality psychology and psychotherapy, and these remain the root of all modern forms of therapy.

Source illustration description (image not reproduced): (a)A photograph shows Freud holding a cigar. (b) The mind s conscious and unconscious states are illustrated as an iceberg floating in water. Beneath the water s surface in the unconscious area are the id, ego, and superego. The area just below the water s surface is labeled preconscious. The area above the water s surface is labeled conscious. Many of the specifics of (a) Freud's theories, such as (b) his division of the mind into id, ego, and superego, have fallen out of favor in recent decades because they are not falsifiable. In broader strokes, his views set the stage for much of psychological thinking today, such as the unconscious nature of the majority of psychological processes.

In contrast, the James-Lange theory does generate falsifiable hypotheses, such as the one described above. Some individuals who suffer significant injuries to their spinal columns are unable to feel the bodily changes that often accompany emotional experiences. Therefore, we could test the hypothesis by determining how emotional experiences differ between individuals who have the ability to detect these changes in their physiological arousal and those who do not. In fact, this research has been conducted and while the emotional experiences of people deprived of an awareness of their physiological arousal may be less intense, they still experience emotion (Chwalisz, Diener, & Gallagher, 1988).

Scientific research s dependence on falsifiability allows for great confidence in the information that it produces. Typically, by the time information is accepted by the scientific community, it has been tested repeatedly.

Summary

Scientists are engaged in explaining and understanding how the world around them works, and they are able to do so by coming up with theories that generate hypotheses that are testable and falsifiable. Theories that stand up to their tests are retained and refined, while those that do not are discarded or modified. In this way, research enables scientists to separate fact from simple opinion. Having good information generated from research aids in making wise decisions both in public policy and in our personal lives.

Review Questions

OpenStax learning exercise

Q. Scientific hypotheses are _____ and falsifiable.

A. D

OpenStax learning exercise

Q. _____ are defined as observable realities.

A. B

OpenStax learning exercise

Q. Scientific knowledge is _____.

A. B

OpenStax learning exercise

Q. A major criticism of Freud s early theories involves the fact that his theories _____.

A. D

Critical Thinking Questions

OpenStax learning exercise

Q. In this section, there was a discussion about researchers arriving at different conclusions about the use of technology in certain student populations, with one study determining that a smartphone app for surgery students seemed effective, but another study finding negative impacts on sleep, communication, and time management resulting from the use of technology by undergraduates. How might an educational leader best work through these discrepancies?

OpenStax learning exercise

Q. The scientific method is often described as self-correcting and cyclical. Briefly describe your understanding of the scientific method with regard to these concepts.

A. This cyclical, self-correcting process is primarily a function of the empirical nature of science. Theories are generated as explanations of real-world phenomena. From theories, specific hypotheses are developed and tested. As a function of this testing, theories will be revisited and modified or refined to generate new hypotheses that are again tested. This cyclical process ultimately allows for more and more precise (and presumably accurate) information to be collected.

Personal Application Questions

OpenStax learning exercise

Q. Healthcare professionals cite an enormous number of health problems related to obesity, and many people have an understandable desire to attain a healthy weight. There are many diet programs, services, and products on the market to aid those who wish to lose weight. If a close friend was considering purchasing or participating in one of these products, programs, or services, how would you make sure your friend was fully aware of the potential consequences of this decision? What sort of information would you want to review before making such an investment or lifestyle change yourself?

Ethics

OpenStax source module m82170 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82170/index.cnxml

Today, scientists agree that good research is ethical in nature and is guided by a basic respect for human dignity and safety. However, as you will read in the feature box, this has not always been the case. Modern researchers must demonstrate that the research they perform is ethically sound. This section presents how ethical considerations affect the design and implementation of research conducted today.

Research Involving Human Participants

Any experiment involving the participation of human subjects is governed by extensive, strict guidelines designed to ensure that the experiment does not result in harm. Any research institution that receives federal support for research involving human participants must have access to an institutional review board (IRB). The IRB is a committee of individuals often made up of members of the institution s administration, scientists, and community members (the accompanying source-figure description). The purpose of the IRB is to review proposals for research that involves human participants. The IRB reviews these proposals with the principles mentioned above in mind, and generally, approval from the IRB is required in order for the experiment to proceed.

Source illustration description (image not reproduced): A photograph shows a group of people seated around tables in a meeting room. An institution s IRB meets regularly to review experimental proposals that involve human participants. (credit: International Hydropower Association/Flickr)

An institution s IRB requires several components in any experiment it approves. For one, each participant must sign an informed consent form before they can participate in the experiment. An informed consent form provides a written description of what participants can expect during the experiment, including potential risks and implications of the research. It also lets participants know that their involvement is completely voluntary and can be discontinued without penalty at any time. Furthermore, the informed consent guarantees that any data collected in the experiment will remain completely confidential. In cases where research participants are under the age of 18, the parents or legal guardians are required to sign the informed consent form.

While the informed consent form should be as honest as possible in describing exactly what participants will be doing, sometimes deception is necessary to prevent participants knowledge of the exact research question from affecting the results of the study. Deception involves purposely misleading experiment participants in order to maintain the integrity of the experiment, but not to the point where the deception could be considered harmful. For example, if we are interested in how our opinion of someone is affected by their attire, we might use deception in describing the experiment to prevent that knowledge from affecting participants responses. In cases where deception is involved, participants must receive a full debriefing upon conclusion of the study complete, honest information about the purpose of the experiment, how the data collected will be used, the reasons why deception was necessary, and information about how to obtain additional information about the study.

Dig Deeper

Unfortunately, the ethical guidelines that exist for research today were not always applied in the past. In 1932, rural, Black men from Tuskegee, Alabama, were recruited to participate in an experiment conducted by the U.S. Public Health Service, with the aim of studying syphilis in Black men (the accompanying source-figure description). In exchange for free medical care, meals, and burial insurance, 600 men agreed to participate in the study. A little more than half of the men tested positive for syphilis, and they served as the experimental group (given that the researchers could not randomly assign participants to groups, this represents a quasi-experiment). The remaining syphilis-free individuals served as the control group. However, those individuals that tested positive for syphilis were never informed that they had the disease.

While there was no treatment for syphilis when the study began, by 1947 penicillin was recognized as an effective treatment for the disease. Despite this, no penicillin was administered to the participants in this study, and the participants were not allowed to seek treatment at any other facilities if they continued in the study. Over the course of 40 years, many of the participants unknowingly spread syphilis to their wives (and subsequently their children born from their wives) and eventually died because they never received treatment for the disease. This study was discontinued in 1972 when the experiment was discovered by the national press (Tuskegee University, n.d.). The resulting outrage over the experiment led directly to the National Research Act of 1974 and the strict ethical guidelines for research on humans described in this chapter. Why is this study unethical? How were the men who participated and their families harmed as a function of this research?

Source illustration description (image not reproduced): A photograph shows a person administering an injection. A participant in the Tuskegee Syphilis Study receives an injection.

Research Involving Animal Subjects

Many psychologists conduct research involving animal subjects. Often, these researchers use rodents (the accompanying source-figure description) or birds as the subjects of their experiments the APA estimates that 90% of all animal research in psychology uses these species (American Psychological Association, n.d.). Because many basic processes in animals are sufficiently similar to those in humans, these animals are acceptable substitutes for research that would be considered unethical in human participants.

Source illustration description (image not reproduced): A photograph shows a rat. Rats, like the one shown here, often serve as the subjects of animal research.

This does not mean that animal researchers are immune to ethical concerns. Indeed, the humane and ethical treatment of animal research subjects is a critical aspect of this type of research. Researchers must design their experiments to minimize any pain or distress experienced by animals serving as research subjects.

Whereas IRBs review research proposals that involve human participants, animal experimental proposals are reviewed by an Institutional Animal Care and Use Committee (IACUC). An IACUC consists of institutional administrators, scientists, veterinarians, and community members. This committee is charged with ensuring that all experimental proposals require the humane treatment of animal research subjects. It also conducts semi-annual inspections of all animal facilities to ensure that the research protocols are being followed. No animal research project can proceed without the committee s approval.

Summary

Ethics in research is an evolving field, and some practices that were accepted or tolerated in the past would be considered unethical today. Researchers are expected to adhere to basic ethical guidelines when conducting experiments that involve human participants. Any experiment involving human participants must be approved by an IRB. Participation in experiments is voluntary and requires informed consent of the participants. If any deception is involved in the experiment, each participant must be fully debriefed upon the conclusion of the study.

Animal research is also held to a high ethical standard. Researchers who use animals as experimental subjects must design their projects so that pain and distress are minimized. Animal research requires the approval of an IACUC, and all animal facilities are subject to regular inspections to ensure that animals are being treated humanely.

Review Questions

OpenStax learning exercise

Q. _____ is to animal research as _____ is to human research.

A. B

OpenStax learning exercise

Q. Researchers might use _____ when providing participants with the full details of the experiment could skew their responses.

A. B

OpenStax learning exercise

Q. A person s participation in a research project must be _____.

A. C

OpenStax learning exercise

Q. Before participating in an experiment, individuals should read and sign the _____ form.

A. A

Critical Thinking Questions

OpenStax learning exercise

Q. Some argue that animal research is inherently flawed in terms of being ethical because unlike human participants, animals do not consent to be involved in research. Do you agree with this perspective? Given that animals do not consent to be involved in research projects, what sorts of extra precautions should be taken to ensure that they receive the most humane treatment possible?

A. In general, the fact that consent cannot be obtained from animal research subjects places extra responsibility on the researcher to ensure that the animal is treated as humanely as possible and to respect the sacrifice that the animal is making for the advancement of science. Like human research, the animals themselves should also receive some of the benefits of the research, and they do in the form of advanced veterinary medicine, and so on.

OpenStax learning exercise

Q. At the end of the last section, you were asked to design a basic experiment to answer some question of interest. What ethical considerations should be made with the study you proposed to ensure that your experiment would conform to the scientific community s expectations of ethical research?

A. The research should be designed in such a way to adhere to the principles described in this section depending on the type of study that was proposed.

Personal Application Questions

OpenStax learning exercise

Q. Take a few minutes to think about all of the advancements that our society has achieved as a function of research involving animal subjects. How have you, a friend, or a family member benefited directly from this kind of research?

CHAPTER 03 / CAC.901

Designing and interpreting longitudinal research

Lecture: Time must be represented, not merely mentioned

Longitudinal research follows the same people over time. That simple feature gives it special value for developmental psychology. It allows researchers to describe individual change, examine the order of events, and investigate whether earlier experiences predict later outcomes. Yet collecting repeated measurements does not automatically answer a causal question. Time helps organize evidence; it does not remove confounding, measurement error, or selective participation.

The first design decision concerns timing. Measurements should match the process being studied. A fear response may change within minutes, a classroom adjustment within weeks, and an identity narrative over years. Asking about a slowly changing outcome every day can burden participants without adding useful information. Asking once a year about a rapidly changing process can miss the most important variation. A theoretical account should explain why the chosen intervals are informative.

Different questions hide inside the word change

Consider adolescent sleep and distress. A between-person question asks whether adolescents who generally sleep less also tend to report more distress than adolescents who generally sleep more. A within-person question asks whether an adolescent reports more distress than usual during periods when that adolescent sleeps less than usual. These are not the same question. The first concerns differences among people; the second concerns fluctuations within a person.

Figure 3.1 illustrates why the distinction matters. Suppose adolescents with heavier academic demands both sleep less and report more distress. A between-person association could arise partly from those demands. Within a particular adolescent, a short-term change in sleep might or might not be followed by a change in distress. A model that mixes the two sources of variation can produce an estimate that is difficult to interpret. Curran and Bauer s discussion of disaggregation and the methodological critique by Hamaker and colleagues are important entry points to this issue.

A counselor should therefore ask what level a research claim addresses. Young people with more support tend to have better outcomes does not necessarily mean that increasing support by a particular amount next week will produce a particular improvement in one child. Group-level patterns inform practice, but individual intervention still needs a plausible mechanism, local assessment, and monitoring.

Growth curves and individual paths

A growth model represents repeated outcomes as a function of time. In a simple linear model, an intercept describes a starting or reference level and a slope describes a constant rate of change. Real development may not be linear. A skill may improve rapidly and then level off; distress may rise during a transition and later decline. The model should reflect the expected pattern and the available measurement occasions.

The coding of time changes the meaning of the intercept. If time is zero at the first assessment, the intercept refers to the beginning of the study. If time is zero at school transition, it refers to that

transition. Neither coding is universally correct. The choice should make the parameter answer a meaningful question. Researchers should report it clearly so readers do not mistake a reference score for a general trait.

Individual paths can vary around an average path. An average improvement may conceal children who worsen, children who remain stable, and children who improve greatly. Examine variability and clinically meaningful outcomes as well as the mean. However, do not treat every estimated individual slope as equally precise. Sparse or noisy measurements can produce unstable estimates. A sophisticated graph does not eliminate uncertainty.

Missing data are part of the story

Children miss assessments for many reasons. Families move, appointments clash, devices fail, and participants withdraw. Missingness matters because the remaining sample may no longer represent the original sample. If the most distressed families are more likely to leave a study, an apparent improvement may partly reflect who stayed. Report participation at each wave and compare available baseline characteristics where appropriate.

Modern methods can handle some missing-data problems under stated assumptions, but no method can recover information that was never observed with certainty. Complete-case analysis can be wasteful and biased. Multiple imputation and likelihood-based approaches may be useful, depending on the design and assumptions. Sensitivity analyses ask whether conclusions change under plausible alternative assumptions. The key is transparency, not the appearance of a technical solution.

Measurement invariance raises another question: does a score mean the same thing across ages, languages, or groups? A behavior that indicates distress at one age may be developmentally ordinary at another. A translated item may carry a different implication. Longitudinal comparison becomes weaker when the construct or measurement changes without recognition. Researchers should examine invariance where appropriate and explain what comparisons remain defensible when full invariance is not supported.

Temporal order is not a causal guarantee

Imagine that peer difficulties at age ten predict anxiety at age twelve. The order is consistent with a pathway in which exclusion contributes to anxiety. It is also possible that earlier subtle anxiety affected peer relationships, that family stress influenced both, or that an unmeasured factor shaped the association. A cross-lagged model can represent competing temporal paths, but its interpretation depends on design, timing, measurement, and assumptions.

A randomized intervention can strengthen causal inference when it is feasible and well conducted. Even then, nonadherence, attrition, and differences in implementation require attention. Observational longitudinal studies remain essential for questions that cannot be randomized. Their value lies in careful design and bounded interpretation, not in pretending to be experiments.

Application: a three-wave school study

A fictional study measures belonging, attendance, and distress before a school transition, three months later, and one year later. The team expects belonging to change quickly and distress to respond more slowly. It chooses those waves for practical reasons and acknowledges that important short-term changes may be missed. At each wave, it records school changes and major disruptions rather than assuming that time alone explains development.

Before analysis, the team writes a question for each level. Between students, is higher belonging associated with lower distress? Within students, do periods of higher-than-usual belonging precede lower-than-usual distress? Between schools, do staffing and program implementation help explain

differences in average outcomes? These questions require different parts of the model. Combining them without a clear structure would make the results difficult to translate into practice.

The team also plans how to report uncertainty. It will present estimates with confidence intervals, describe missingness, and avoid declaring that a nonsignificant result proves no effect. A small estimated difference with wide intervals may be uninformative rather than evidence of equivalence. The report will distinguish statistical evidence from practical importance.

Doctoral seminar task

Draw a repeated-measures design for a developmental process you want to understand. Mark each measurement occasion and explain why it occurs then. Identify one within-person question, one between-person question, and one contextual question. For each, state the minimum evidence needed and a plausible threat to interpretation.

Then write two conclusions using the same imaginary pattern: one that overstates causality and one that accurately reflects the design. Explain the difference in plain language. This is a useful skill when preparing reports for caregivers, schools, policy teams, and research audiences. The aim is not to make every sentence cautious beyond usefulness. It is to make every claim proportionate to what was actually observed.

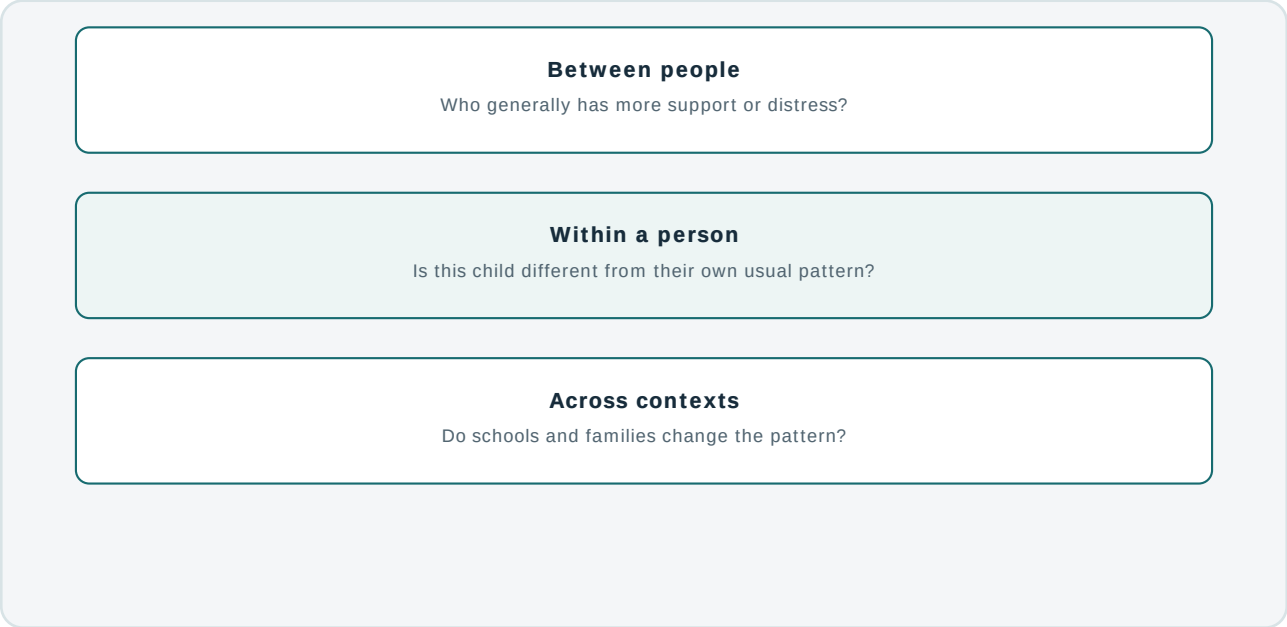


Figure 3.1. Two different questions about change. Between-person differences and within-person fluctuations are distinct sources of variation. A longitudinal analysis should state which question its parameters answer. This schematic contains no empirical estimates. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

Approaches to Research

OpenStax source module m82168 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82168/index.cnxml

There are many research methods available to psychologists in their efforts to understand, describe, and explain behavior and the cognitive and biological processes that underlie it. Some methods rely on observational techniques. Other approaches involve interactions between the researcher and the individuals who are being studied ranging from a series of simple questions to extensive, in-depth interviews to well-controlled experiments.

Each of these research methods has unique strengths and weaknesses, and each method may only be appropriate for certain types of research questions. For example, studies that rely primarily on observation produce incredible amounts of information, but the ability to apply this information to the larger population is somewhat limited because of small sample sizes. Survey research, on the other hand, allows researchers to easily collect data from relatively large samples. While this allows for results to be generalized to the larger population more easily, the information that can be collected on any given survey is somewhat limited and subject to problems associated with any type of self-reported data. Some researchers conduct archival research by using existing records. While this can be a fairly inexpensive way to collect data that can provide insight into a number of research questions, researchers using this approach have no control on how or what kind of data was collected. All of the methods described thus far are correlational in nature. This means that researchers can speak to important relationships that might exist between two or more variables of interest. However, correlational data cannot be used to make claims about cause-and-effect relationships.

Correlational research can find a relationship between two variables, but the only way a researcher can claim that the relationship between the variables is cause and effect is to perform an experiment. In experimental research, which will be discussed later in this chapter, there is a tremendous amount of control over variables of interest. While this is a powerful approach, experiments are often conducted in artificial settings. This calls into question the validity of experimental findings with regard to how they would apply in real-world settings. In addition, many of the questions that psychologists would like to answer cannot be pursued through experimental research because of ethical concerns.

Clinical or Case Studies

In 2011, the New York Times published a feature story on Krista and Tatiana Hogan, Canadian twin girls. These particular twins are unique because Krista and Tatiana are conjoined twins, connected at the head. There is evidence that the two girls are connected in a part of the brain called the thalamus, which is a major sensory relay center. Most incoming sensory information is sent through the thalamus before reaching higher regions of the cerebral cortex for processing.

The implications of this potential connection mean that it might be possible for one twin to experience the sensations of the other twin. For instance, if Krista is watching a particularly funny television program, Tatiana might smile or laugh even if she is not watching the program. This particular

possibility has piqued the interest of many neuroscientists who seek to understand how the brain uses sensory information.

These twins represent an enormous resource in the study of the brain, and since their condition is very rare, it is likely that as long as their family agrees, scientists will follow these girls very closely throughout their lives to gain as much information as possible (Dominus, 2011).

Over time, it has become clear that while Krista and Tatiana share some sensory experiences and motor control, they remain two distinct individuals, which provides invaluable insight for researchers interested in the mind and the brain (Egnor, 2017).

In observational research, scientists are conducting a clinical or case study when they focus on one person or just a few individuals. Indeed, some scientists spend their entire careers studying just 10 20 individuals. Why would they do this? Obviously, when they focus their attention on a very small number of people, they can gain a precious amount of insight into those cases. The richness of information that is collected in clinical or case studies is unmatched by any other single research method. This allows the researcher to have a very deep understanding of the individuals and the particular phenomenon being studied.

If clinical or case studies provide so much information, why are they not more frequent among researchers? As it turns out, the major benefit of this particular approach is also a weakness. As mentioned earlier, this approach is often used when studying individuals who are interesting to researchers because they have a rare characteristic. Therefore, the individuals who serve as the focus of case studies are not like most other people. If scientists ultimately want to explain all behavior, focusing attention on such a special group of people can make it difficult to generalize any observations to the larger population as a whole. Generalizing refers to the ability to apply the findings of a particular research project to larger segments of society. Again, case studies provide enormous amounts of information, but since the cases are so specific, the potential to apply what s learned to the average person may be very limited.

Naturalistic Observation

If you want to understand how behavior occurs, one of the best ways to gain information is to simply observe the behavior in its natural context. However, people might change their behavior in unexpected ways if they know they are being observed. How do researchers obtain accurate information when people tend to hide their natural behavior? As an example, imagine that your professor asks everyone in your class to raise their hand if they always wash their hands after using the restroom. Chances are that almost everyone in the classroom will raise their hand, but do you think hand washing after every trip to the restroom is really that universal?

This is very similar to the phenomenon mentioned earlier in this chapter: many individuals do not feel comfortable answering a question honestly. But if we are committed to finding out the facts about hand washing, we have other options available to us.

Suppose we send a classmate into the restroom to actually watch whether everyone washes their hands after using the restroom. Will our observer blend into the restroom environment by wearing a white lab coat, sitting with a clipboard, and staring at the sinks? We want our researcher to be inconspicuous perhaps standing at one of the sinks pretending to put in contact lenses while secretly recording the relevant information. This type of observational study is called naturalistic observation: observing behavior in its natural setting. To better understand peer exclusion, Suzanne Fanger collaborated with colleagues at the University of Texas to observe the behavior of preschool children on a playground. How did the observers remain inconspicuous over the duration of the study? They equipped a few of the children with wireless microphones (which the children quickly forgot about) and observed while taking notes from a distance. Also, the children in that particular preschool (a laboratory preschool) were accustomed to having observers on the playground (Fanger, Frankel, & Hazen, 2012).

It is critical that the observer be as unobtrusive and as inconspicuous as possible: when people know they are being watched, they are less likely to behave naturally. If you have any doubt about this, ask yourself how your driving behavior might differ in two situations: In the first situation, you are driving down a deserted highway during the middle of the day; in the second situation, you are being followed by a police car down the same deserted highway (the accompanying source-figure description).

Source illustration description (image not reproduced): A photograph shows two police cars driving, one with its lights flashing. Seeing a police car behind you would probably affect your driving behavior. (credit: Michael Gil)

It should be pointed out that naturalistic observation is not limited to research involving humans. Indeed, some of the best-known examples of naturalistic observation involve researchers going into the field to observe various kinds of animals in their own environments. As with human studies, the researchers maintain their distance and avoid interfering with the animal subjects so as not to influence their natural behaviors. Scientists have used this technique to study social hierarchies and interactions among animals ranging from ground squirrels to gorillas. The information provided by these studies is invaluable in understanding how those animals organize socially and communicate with one another. The anthropologist Jane Goodall, for example, spent nearly five decades observing the behavior of chimpanzees in Africa (the accompanying source-figure description). As an illustration of the types of concerns that a researcher might encounter in naturalistic observation, some scientists criticized Goodall for giving the chimps names instead of referring to them by numbers using names was thought to undermine the emotional detachment required for the objectivity of the study (McKie, 2010).

Source illustration description (image not reproduced): (a) A photograph shows Jane Goodall speaking from a lectern. (b) A photograph shows a chimpanzee s face. (a) Jane Goodall made a career of conducting naturalistic observations of (b) chimpanzee behavior. (credit Jane Goodall : modification of work by Erik Hersman; chimpanzee : modification of work by Afrika Force /Flickr.com)

The greatest benefit of naturalistic observation is the validity, or accuracy, of information collected unobtrusively in a natural setting. Having individuals behave as they normally would in a given situation means that we have a higher degree of ecological validity, or realism, than we might achieve with other research approaches. Therefore, our ability to generalize the findings of the research to real-world situations is enhanced. If done correctly, we need not worry about people or animals modifying their behavior simply because they are being observed. Sometimes, people may assume that reality programs give us a glimpse into authentic human behavior. However, the principle of inconspicuous observation is violated as reality stars are followed by camera crews and are interviewed on camera for personal confessionals. Given that environment, we must doubt how natural and realistic their behaviors are.

The major downside of naturalistic observation is that they are often difficult to set up and control. In our restroom study, what if you stood in the restroom all day prepared to record people s hand washing behavior and no one came in? Or, what if you have been closely observing a troop of gorillas for weeks only to find that they migrated to a new place while you were sleeping in your tent? The benefit of realistic data comes at a cost. As a researcher you have no control of when (or if) you have behavior to observe. In addition, this type of observational research often requires significant investments of time, money, and a good dose of luck.

Sometimes studies involve structured observation. In these cases, people are observed while engaging in set, specific tasks. An excellent example of structured observation comes from Strange Situation by Mary Ainsworth (you will read more about this in the chapter on lifespan development). The Strange Situation is a procedure used to evaluate attachment styles that exist between an infant and caregiver. In this scenario, caregivers bring their infants into a room filled with toys. The Strange Situation involves a number of phases, including a stranger coming into the room, the caregiver leaving the room, and the caregiver s return to the room. The infant s behavior is closely monitored at each phase, but it is the behavior of the infant upon being reunited with the caregiver that is most telling in terms of characterizing the infant s attachment style with the caregiver.

Another potential problem in observational research is observer bias. Generally, people who act as observers are closely involved in the research project and may unconsciously skew their observations to

fit their research goals or expectations. To protect against this type of bias, researchers should have clear criteria established for the types of behaviors recorded and how those behaviors should be classified. In addition, researchers often compare observations of the same event by multiple observers, in order to test inter-rater reliability: a measure of reliability that assesses the consistency of observations by different observers.

Surveys

Often, psychologists develop surveys as a means of gathering data. Surveys are lists of questions to be answered by research participants, and can be delivered as paper-and-pencil questionnaires, administered electronically, or conducted verbally (the accompanying source-figure description). Generally, the survey itself can be completed in a short time, and the ease of administering a survey makes it easy to collect data from a large number of people.

Surveys allow researchers to gather data from larger samples than may be afforded by other research methods. A sample is a subset of individuals selected from a population, which is the overall group of individuals that the researchers are interested in. Researchers study the sample and seek to generalize their findings to the population. Generally, researchers will begin this process by calculating various measures of central tendency from the data they have collected. These measures provide an overall summary of what a typical response looks like. There are three measures of central tendency: mode, median, and mean. The mode is the most frequently occurring response, the median lies at the middle of a given data set, and the mean is the arithmetic average of all data points. Means tend to be most useful in conducting additional analyses like those described below; however, means are very sensitive to the effects of outliers, and so one must be aware of those effects when making assessments of what measures of central tendency tell us about a data set in question.

Source illustration description (image not reproduced): A sample online survey reads, Dear visitor, your opinion is important to us. We would like to invite you to participate in a short survey to gather your opinions and feedback on your news consumption habits. The survey will take approximately 10-15 minutes. Simply click the Yes button below to launch the survey. Would you like to participate? Two buttons are labeled yes and no. Surveys can be administered in a number of ways, including electronically administered research, like the survey shown here. (credit: Robert Nyman)

There is both strength and weakness of the survey in comparison to case studies. By using surveys, we can collect information from a larger sample of people. A larger sample is better able to reflect the actual diversity of the population, thus allowing better generalizability. Therefore, if our sample is sufficiently large and diverse, we can assume that the data we collect from the survey can be generalized to the larger population with more certainty than the information collected through a case study. However, given the greater number of people involved, we are not able to collect the same depth of information on each person that would be collected in a case study.

Another potential weakness of surveys is something we touched on earlier in this chapter: People don't always give accurate responses. They may lie, misremember, or answer questions in a way that they think makes them look good. For example, people may report drinking less alcohol than is actually the case.

Any number of research questions can be answered through the use of surveys. One real-world example is the research conducted by Jenkins, Ruppel, Kizer, Yehl, and Griffin (2012) about the backlash against the US Arab-American community following the terrorist attacks of September 11, 2001. Jenkins and colleagues wanted to determine to what extent these negative attitudes toward Arab-Americans still existed nearly a decade after the attacks occurred. In one study, 140 research participants filled out a survey with 10 questions, including questions asking directly about the participant s overt prejudicial attitudes toward people of various ethnicities. The survey also asked indirect questions about how likely the participant would be to interact with a person of a given ethnicity in a variety of settings (such as, How likely do you think it is that you would introduce yourself to a person of Arab-American descent?). The results of the research suggested that participants were unwilling to report prejudicial

attitudes toward any ethnic group. However, there were significant differences between their pattern of responses to questions about social interaction with Arab-Americans compared to other ethnic groups: they indicated less willingness for social interaction with Arab-Americans compared to the other ethnic groups. This suggested that the participants harbored subtle forms of prejudice against Arab-Americans, despite their assertions that this was not the case (Jenkins et al., 2012).

Archival Research

Some researchers gain access to large amounts of data without interacting with a single research participant. Instead, they use existing records to answer various research questions. This type of research approach is known as archival research. Archival research relies on looking at past records or data sets to look for interesting patterns or relationships.

For example, a researcher might access the academic records of all individuals who enrolled in college within the past ten years and calculate how long it took them to complete their degrees, as well as course loads, grades, and extracurricular involvement. Archival research could provide important information about who is most likely to complete their education, and it could help identify important risk factors for struggling students (the accompanying source-figure description).

Source illustration description (image not reproduced): (a) A photograph shows stacks of paper files on shelves. (b) A photograph shows a computer. A researcher doing archival research examines records, whether archived as a (a) hardcopy or (b) electronically. (credit paper files : modification of work by Newtown graffiti /Flickr; computer : modification of work by INPIVIC Family/Flickr)

In comparing archival research to other research methods, there are several important distinctions. For one, the researcher employing archival research never directly interacts with research participants. Therefore, the investment of time and money to collect data is considerably less with archival research. Additionally, researchers have no control over what information was originally collected. Therefore, research questions have to be tailored so they can be answered within the structure of the existing data sets. There is also no guarantee of consistency between the records from one source to another, which might make comparing and contrasting different data sets problematic.

Longitudinal and Cross-Sectional Research

Sometimes we want to see how people change over time, as in studies of human development and lifespan. When we test the same group of individuals repeatedly over an extended period of time, we are conducting longitudinal research. Longitudinal research is a research design in which data-gathering is administered repeatedly over an extended period of time. For example, we may survey a group of individuals about their dietary habits at age 20, retest them a decade later at age 30, and then again at age 40.

Another approach is cross-sectional research. In cross-sectional research, a researcher compares multiple segments of the population at the same time. Using the dietary habits example above, the researcher might directly compare different groups of people by age. Instead of studying a group of people for 20 years to see how their dietary habits changed from decade to decade, the researcher would study a group of 20-year-old individuals and compare them to a group of 30-year-old individuals and a group of 40-year-old individuals. While cross-sectional research requires a shorter-term investment, it is also limited by differences that exist between the different generations (or cohorts) that have nothing to do with age per se, but rather reflect the social and cultural experiences of different generations of individuals make them different from one another.

To illustrate this concept, consider the following survey findings. In recent years there has been significant growth in the popular support of same-sex marriage. Many studies on this topic break down survey participants into different age groups. In general, younger people are more supportive of same-sex marriage than are those who are older (Jones, 2013). Does this mean that as we age we become less open to the idea of same-sex marriage, or does this mean that older individuals have

different perspectives because of the social climates in which they grew up? Longitudinal research is a powerful approach because the same individuals are involved in the research project over time, which means that the researchers need to be less concerned with differences among cohorts affecting the results of their study.

Often longitudinal studies are employed when researching various diseases in an effort to understand particular risk factors. Such studies often involve tens of thousands of individuals who are followed for several decades. Given the enormous number of people involved in these studies, researchers can feel confident that their findings can be generalized to the larger population. The Cancer Prevention Study-3 (CPS-3) is one of a series of longitudinal studies sponsored by the American Cancer Society aimed at determining predictive risk factors associated with cancer. When participants enter the study, they complete a survey about their lives and family histories, providing information on factors that might cause or prevent the development of cancer. Then every few years the participants receive additional surveys to complete. In the end, hundreds of thousands of participants will be tracked over 20 years to determine which of them develop cancer and which do not.

Clearly, this type of research is important and potentially very informative. For instance, earlier longitudinal studies sponsored by the American Cancer Society provided some of the first scientific demonstrations of the now well-established links between increased rates of cancer and smoking (American Cancer Society, n.d.) (the accompanying source-figure description).

Source illustration description (image not reproduced): A photograph shows pack of cigarettes and cigarettes in an ashtray. The pack of cigarettes reads, Surgeon general s warning: smoking causes lung cancer, heart disease, emphysema, and may complicate pregnancy. Longitudinal research like the CPS-3 help us to better understand how smoking is associated with cancer and other diseases. (credit: CDC/Debora Cartagena)

As with any research strategy, longitudinal research is not without limitations. For one, these studies require an incredible time investment by the researcher and research participants. Given that some longitudinal studies take years, if not decades, to complete, the results will not be known for a considerable period of time. In addition to the time demands, these studies also require a substantial financial investment. Many researchers are unable to commit the resources necessary to see a longitudinal project through to the end.

Research participants must also be willing to continue their participation for an extended period of time, and this can be problematic. People move, get married and take new names, get ill, and eventually die. Even without significant life changes, some people may simply choose to discontinue their participation in the project. As a result, the attrition rates, or reduction in the number of research participants due to dropouts, in longitudinal studies are quite high and increases over the course of a project. For this reason, researchers using this approach typically recruit many participants fully expecting that a substantial number will drop out before the end. As the study progresses, they continually check whether the sample still represents the larger population, and make adjustments as necessary.

Summary

The clinical or case study involves studying just a few individuals for an extended period of time. While this approach provides an incredible depth of information, the ability to generalize these observations to the larger population is problematic. Naturalistic observation involves observing behavior in a natural setting and allows for the collection of valid, true-to-life information from realistic situations. However, naturalistic observation does not allow for much control and often requires quite a bit of time and money to perform. Researchers strive to ensure that their tools for collecting data are both reliable (consistent and replicable) and valid (accurate).

Surveys can be administered in a number of ways and make it possible to collect large amounts of data quickly. However, the depth of information that can be collected through surveys is somewhat limited compared to a clinical or case study.

Archival research involves studying existing data sets to answer research questions.

Longitudinal research has been incredibly helpful to researchers who need to collect data on how people change over time. Cross-sectional research compares multiple segments of a population at a single time.

Review Questions

OpenStax learning exercise

Q. Sigmund Freud developed his theory of human personality by conducting in-depth interviews over an extended period of time with a few clients. This type of research approach is known as a(n): _____.

A. B

OpenStax learning exercise

Q. _____ involves observing behavior in individuals in their natural environments.

A. C

OpenStax learning exercise

Q. The major limitation of case studies is _____.

A. C

OpenStax learning exercise

Q. The benefit of naturalistic observation studies is _____.

A. A

OpenStax learning exercise

Q. Using existing records to try to answer a research question is known as _____.

A. D

OpenStax learning exercise

Q. _____ involves following a group of research participants for an extended period of time.

A. B

OpenStax learning exercise

Q. A(n) _____ is a list of questions developed by a researcher that can be administered in paper form.

A. D

OpenStax learning exercise

Q. Longitudinal research is complicated by high rates of _____.

A. C

Critical Thinking Questions

OpenStax learning exercise

Q. In this section, conjoined twins, Krista and Tatiana, were described as being potential participants in a case study. In what other circumstances would you think that this particular research approach would be especially helpful and why?

A. Case studies might prove especially helpful using individuals who have rare conditions. For instance, if one wanted to study multiple personality disorder then the case study approach with individuals diagnosed with multiple personality disorder would be helpful.

OpenStax learning exercise

Q. Presumably, reality television programs aim to provide a realistic portrayal of the behavior displayed by the characters featured in such programs. This section pointed out why this is not really the case. What changes could be made in the way that these programs are produced that would result in more honest portrayals of realistic behavior?

A. The behavior displayed on these programs would be more realistic if the cameras were mounted in hidden locations, or if the people who appear on these programs did not know when they were being recorded.

OpenStax learning exercise

Q. Which of the research methods discussed in this section would be best suited to study the impact of diet and exercise on the prevalence of a disease such as diabetes? Why?

OpenStax learning exercise

Q. Aside from biomedical research, what other areas of research could greatly benefit by both longitudinal and archival research?

A. Answers will vary. Possibilities include research on hiring practices based on human resource records, and research that follows former prisoners to determine if the time that they were incarcerated provided any sort of positive influence on their likelihood of engaging in criminal behavior in the future.

Personal Application Questions

OpenStax learning exercise

Q. A friend of yours is working part-time in a local pet store. Your friend has become increasingly interested in how dogs normally communicate and interact with each other, and is thinking of visiting a local veterinary clinic to see how dogs interact in the waiting room. After reading this section, do you think this is the best way to better understand such interactions? Do you have any suggestions that might result in more valid data?

OpenStax learning exercise

Q. As a college student, you are no doubt concerned about the grades that you earn while completing your coursework. If you wanted to know how overall GPA is related to success in life after college, how would you choose to approach this question and what kind of resources would you need to conduct this research?

Analyzing Findings

OpenStax source module m82169 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82169/index.cnxml

Did you know that as sales in ice cream increase, so does the overall rate of crime? Is it possible that indulging in your favorite flavor of ice cream could send you on a crime spree? Or, after committing crime do you think you might decide to treat yourself to a cone? There is no question that a relationship exists between ice cream and crime (e.g., Harper, 2013), but it would be pretty foolish to decide that one thing actually caused the other to occur.

It is much more likely that both ice cream sales and crime rates are related to the temperature outside. When the temperature is warm, there are lots of people out of their houses, interacting with each other, getting annoyed with one another, and sometimes committing crimes. Also, when it is warm outside, we are more likely to seek a cool treat like ice cream. How do we determine if there is indeed a relationship between two things? And when there is a relationship, how can we discern whether it is attributable to coincidence or causation?

Correlational Research

Correlation means that there is a relationship between two or more variables (such as ice cream consumption and crime), but this relationship does not necessarily imply cause and effect. When two variables are correlated, it simply means that as one variable changes, so does the other. We can measure correlation by calculating a statistic known as a correlation coefficient. A correlation coefficient is a number from -1 to +1 that indicates the strength and direction of the relationship between variables. The correlation coefficient is usually represented by the letter r.

The number portion of the correlation coefficient indicates the strength of the relationship. The closer the number is to 1 (be it negative or positive), the more strongly related the variables are, and the more predictable changes in one variable will be as the other variable changes. The closer the number is to zero, the weaker the relationship, and the less predictable the relationships between the variables becomes. For instance, a correlation coefficient of 0.9 indicates a far stronger relationship than a correlation coefficient of 0.3. If the variables are not related to one another at all, the correlation coefficient is 0. The example above about ice cream and crime is an example of two variables that we might expect to have no relationship to each other.

The sign positive or negative of the correlation coefficient indicates the direction of the relationship (the accompanying source-figure description). A positive correlation means that the variables move in the same direction. Put another way, it means that as one variable increases so does the other, and conversely, when one variable decreases so does the other. A negative correlation means that the variables move in opposite directions. If two variables are negatively correlated, a decrease in one variable is associated with an increase in the other and vice versa.

The example of ice cream and crime rates is a positive correlation because both variables increase when temperatures are warmer. Other examples of positive correlations are the relationship between an individual s height and weight or the relationship between a person s age and number of wrinkles. One might expect a negative correlation to exist between someone s tiredness during the day and the number of hours they slept the previous night: the amount of sleep decreases as the feelings of tiredness increase. In a real-world example of negative correlation, student researchers at the University of Minnesota found a weak negative correlation (r = -0.29) between the average number of days per week that students got fewer than 5 hours of sleep and their GPA (Lowry, Dean, & Manders, 2010). Keep in mind that a negative correlation is not the same as no correlation. For example, we would probably find no correlation between hours of sleep and shoe size.

As mentioned earlier, correlations have predictive value. Imagine that you are on the admissions committee of a major university. You are faced with a huge number of applications, but you are able to accommodate only a small percentage of the applicant pool. How might you decide who should be admitted? You might try to correlate your current students college GPA with their scores on standardized tests like the SAT or ACT. By observing which correlations were strongest for your current students, you could use this information to predict relative success of those students who have applied for admission into the university.

Source illustration description (image not reproduced): Three scatterplots are shown. Scatterplot (a) is labeled positive correlation and shows scattered dots forming a rough line from the bottom left to the top right; the x-axis is labeled weight and the y-axis is labeled height. Scatterplot (b) is labeled negative correlation and shows scattered dots forming a rough line from the top left to the bottom right; the x-axis is labeled tiredness and the y-axis is labeled hours of sleep. Scatterplot (c) is labeled no correlation and shows scattered dots having no pattern; the x-axis is labeled shoe size and the y-axis is labeled hours of sleep. Scatterplots are a graphical view of the strength and direction of correlations. The stronger the correlation, the closer the data points are to a straight line. In these examples, we see that there is (a) a positive correlation between weight and height, (b) a negative correlation between tiredness and hours of sleep, and (c) no correlation between shoe size and hours of sleep.

Correlation Does Not Indicate Causation

Correlational research is useful because it allows us to discover the strength and direction of relationships that exist between two variables. However, correlation is limited because establishing the existence of a relationship tells us little about cause and effect. While variables are sometimes correlated because one does cause the other, it could also be that some other factor, a confounding variable, is actually causing the systematic movement in our variables of interest. In the ice cream/crime rate example mentioned earlier, temperature is a confounding variable that could account for the relationship between the two variables.

Even when we cannot point to clear confounding variables, we should not assume that a correlation between two variables implies that one variable causes changes in another. This can be frustrating when a cause-and-effect relationship seems clear and intuitive. Think back to our discussion of the research done by the American Cancer Society and how their research projects were some of the first demonstrations of the link between smoking and cancer. It seems reasonable to assume that smoking causes cancer, but if we were limited to correlational research, we would be overstepping our bounds by making this assumption.

Unfortunately, people mistakenly make claims of causation as a function of correlations all the time. Such claims are especially common in advertisements and news stories. For example, recent research found that people who eat cereal on a regular basis achieve healthier weights than those who rarely eat cereal (Frantzen, Trevi o, Echon, Garcia-Dominic, & DiMarco, 2013; Barton et al., 2005). Guess how the cereal companies report this finding. Does eating cereal really cause an individual to maintain a healthy weight, or are there other possible explanations, such as, someone at a healthy weight is more likely to regularly eat a healthy breakfast than someone who is obese or someone who avoids meals in an attempt to diet (the accompanying source-figure description)? While correlational research is invaluable in identifying relationships among variables, a major limitation is the inability to establish causality. Psychologists want to make statements about cause and effect, but the only way to do that is to conduct an experiment to answer a research question. The next section describes how scientific experiments incorporate methods that eliminate, or control for, alternative explanations, which allow researchers to explore how changes in one variable cause changes in another variable.

Source illustration description (image not reproduced): A photograph shows a bowl of cereal. Does eating cereal really cause someone to be a healthy weight? (credit: Tim Skillern)

Illusory Correlations

The temptation to make erroneous cause-and-effect statements based on correlational research is not the only way we tend to misinterpret data. We also tend to make the mistake of illusory correlations, especially with unsystematic observations. Illusory correlations, or false correlations, occur when people believe that relationships exist between two things when no such relationship exists. One well-known illusory correlation is the supposed effect that the moon s phases have on human behavior. Many people passionately assert that human behavior is affected by the phase of the moon, and specifically, that people act strangely when the moon is full (the accompanying source-figure description).

Source illustration description (image not reproduced): A photograph shows the moon. Many people believe that a full moon makes people behave oddly. (credit: Cory Zanker)

There is no denying that the moon exerts a powerful influence on our planet. The ebb and flow of the ocean s tides are tightly tied to the gravitational forces of the moon. Many people believe, therefore, that it is logical that we are affected by the moon as well. After all, our bodies are largely made up of water. A meta-analysis of nearly 40 studies consistently demonstrated, however, that the relationship between the moon and our behavior does not exist (Rotton & Kelly, 1985). While we may pay more attention to odd behavior during the full phase of the moon, the rates of odd behavior remain constant throughout the lunar cycle.

Why are we so apt to believe in illusory correlations like this? Often we read or hear about them and simply accept the information as valid. Or, we have a hunch about how something works and then look for evidence to support that hunch, ignoring evidence that would tell us our hunch is false; this is known as confirmation bias. Other times, we find illusory correlations based on the information that comes most easily to mind, even if that information is severely limited. And while we may feel confident that we can use these relationships to better understand and predict the world around us, illusory correlations can have significant drawbacks. For example, research suggests that illusory correlations in which certain behaviors are inaccurately attributed to certain groups are involved in the formation of prejudicial attitudes that can ultimately lead to discriminatory behavior (Fiedler, 2004).

Causality: Conducting Experiments and Using the Data

As you ve learned, the only way to establish that there is a cause-and-effect relationship between two variables is to conduct a scientific experiment. Experiment has a different meaning in the scientific context than in everyday life. In everyday conversation, we often use it to describe trying something for the first time, such as experimenting with a new hair style or a new food. However, in the scientific context, an experiment has precise requirements for design and implementation.

The Experimental Hypothesis

In order to conduct an experiment, a researcher must have a specific hypothesis to be tested. As you ve learned, hypotheses can be formulated either through direct observation of the real world or after careful review of previous research. For example, if you think that the use of technology in the classroom has negative impacts on learning, then you have basically formulated a hypothesis namely, that the use of technology in the classroom should be limited because it decreases learning. How might you have arrived at this particular hypothesis? You may have noticed that your classmates who take notes on their laptops perform at lower levels on class exams than those who take notes by hand, or those who receive a lesson via a computer program versus via an in-person teacher have different levels of performance when tested (the accompanying source-figure description).

Source illustration description (image not reproduced): Many rows of students are in a classroom. One student has an open laptop on his desk. How might the use of technology in the classroom impact learning? (credit: modification of work by Nikolay Georgiev/Pixabay)

These sorts of personal observations are what often lead us to formulate a specific hypothesis, but we cannot use limited personal observations and anecdotal evidence to rigorously test our hypothesis. Instead, to find out if real-world data supports our hypothesis, we have to conduct an experiment.

Designing an Experiment

The most basic experimental design involves two groups: the experimental group and the control group. The two groups are designed to be the same except for one difference experimental manipulation. The experimental group gets the experimental manipulation that is, the treatment or variable being tested (in this case, the use of technology) and the control group does not. Since experimental manipulation is the only difference between the experimental and control groups, we can be sure that any differences between the two are due to experimental manipulation rather than chance.

In our example of how the use of technology should be limited in the classroom, we have the experimental group learn algebra using a computer program and then test their learning. We measure the learning in our control group after they are taught algebra by a teacher in a traditional classroom. It is important for the control group to be treated similarly to the experimental group, with the exception that the control group does not receive the experimental manipulation.

We also need to precisely define, or operationalize, how we measure learning of algebra. An operational definition is a precise description of our variables, and it is important in allowing others to understand exactly how and what a researcher measures in a particular experiment. In operationalizing learning, we might choose to look at performance on a test covering the material on which the individuals were taught by the teacher or the computer program. We might also ask our participants to summarize the information that was just presented in some way. Whatever we determine, it is important that we operationalize learning in such a way that anyone who hears about our study for the first time knows exactly what we mean by learning. This aids peoples ability to interpret our data as well as their capacity to repeat our experiment should they choose to do so.

Once we have operationalized what is considered use of technology and what is considered learning in our experiment participants, we need to establish how we will run our experiment. In this case, we might have participants spend 45 minutes learning algebra (either through a computer program or with an in-person math teacher) and then give them a test on the material covered during the 45 minutes.

Ideally, the people who score the tests are unaware of who was assigned to the experimental or control group, in order to control for experimenter bias. Experimenter bias refers to the possibility that a researcher s expectations might skew the results of the study. Remember, conducting an experiment requires a lot of planning, and the people involved in the research project have a vested interest in supporting their hypotheses. If the observers knew which child was in which group, it might influence how they interpret ambiguous responses, such as sloppy handwriting or minor computational mistakes. By being blind to which child is in which group, we protect against those biases. This situation is a single-blind study, meaning that one of the groups (participants) are unaware as to which group they are in (experiment or control group) while the researcher who developed the experiment knows which participants are in each group.

In a double-blind study, both the researchers and the participants are blind to group assignments. Why would a researcher want to run a study where no one knows who is in which group? Because by doing so, we can control for both experimenter and participant expectations. If you are familiar with the phrase placebo effect, you already have some idea as to why this is an important consideration. The placebo effect occurs when people's expectations or beliefs influence or determine their experience in a given situation. In other words, simply expecting something to happen can actually make it happen.

The placebo effect is commonly described in terms of testing the effectiveness of a new medication. Imagine that you work in a pharmaceutical company, and you think you have a new drug that is effective in treating depression. To demonstrate that your medication is effective, you run an experiment with two groups: The experimental group receives the medication, and the control group does not. But you don t want participants to know whether they received the drug or not.

Why is that? Imagine that you are a participant in this study, and you have just taken a pill that you think will improve your mood. Because you expect the pill to have an effect, you might feel better simply because you took the pill and not because of any drug actually contained in the pill this is the placebo effect.

To make sure that any effects on mood are due to the drug and not due to expectations, the control group receives a placebo (in this case a sugar pill). Now everyone gets a pill, and once again neither the researcher nor the experimental participants know who got the drug and who got the sugar pill. Any differences in mood between the experimental and control groups can now be attributed to the drug itself rather than to experimenter bias or participant expectations (the accompanying source-figure description).

Source illustration description (image not reproduced): A photograph shows three glass bottles of pills labeled as placebos. Providing the control group with a placebo treatment protects against bias caused by expectancy. (credit: Elaine and Arthur Shapiro)

Independent and Dependent Variables

In a research experiment, we strive to study whether changes in one thing cause changes in another. To achieve this, we must pay attention to two important variables, or things that can be changed, in any experimental study: the independent variable and the dependent variable. An independent variable is manipulated or controlled by the experimenter. In a well-designed experimental study, the independent variable is the only important difference between the experimental and control groups. In our example of how technology use in the classroom affects learning, the independent variable is the type of learning by participants in the study (the accompanying source-figure description). A dependent variable is what the researcher measures to see how much effect the independent variable had. In our example, the dependent variable is the learning exhibited by our participants.

Source illustration description (image not reproduced): A box labeled independent variable: taking notes on a laptop or by hand contains a photograph of a classroom of students with an open laptop on one student's desk. An arrow labeled influences change in the leads to a second box. The second box is labeled dependent variable: performance on measure of learning and has a photograph of a student at a desk, taking a test. In an experiment, manipulations of the independent variable are expected to result in changes in the dependent variable. (credit: classroom modification of work by Nikolay Georgiev/Pixabay; credit note taking : modification of work by KF/Wikimedia)

We expect that the dependent variable will change as a function of the independent variable. In other words, the dependent variable depends on the independent variable. A good way to think about the relationship between the independent and dependent variables is with this question: What effect does the independent variable have on the dependent variable? Returning to our example, what is the effect of being taught a lesson through a computer program versus through an in-person instructor?

Selecting and Assigning Experimental Participants

Now that our study is designed, we need to obtain a sample of individuals to include in our experiment. Our study involves human participants so we need to determine who to include. Participants are the subjects of psychological research, and as the name implies, individuals who are involved in psychological research actively participate in the process. Often, psychological research projects rely on college students to serve as participants. In fact, the vast majority of research in psychology subfields has historically involved students as research participants (Sears, 1986; Arnett, 2008). But are college students truly representative of the general population? College students tend to be younger, more educated, more liberal, and less diverse than the general population. Although using students as test subjects is an accepted practice, relying on such a limited pool of research participants can be problematic because it is difficult to generalize findings to the larger population.

Our hypothetical experiment involves high school students, and we must first generate a sample of students. Samples are used because populations are usually too large to reasonably involve every member in our particular experiment (the accompanying source-figure description). If possible, we should use a random sample (there are other types of samples, but for the purposes of this chapter, we

will focus on random samples). A random sample is a subset of a larger population in which every member of the population has an equal chance of being selected. Random samples are preferred because if the sample is large enough we can be reasonably sure that the participating individuals are representative of the larger population. This means that the percentages of characteristics in the sample sex, ethnicity, socioeconomic level, and any other characteristics that might affect the results are close to those percentages in the larger population.

In our example, let s say we decide our population of interest is algebra students. But all algebra students is a very large population, so we need to be more specific; instead we might say our population of interest is all algebra students in a particular city. We should include students from various income brackets, family situations, races, ethnicities, religions, and geographic areas of town. With this more manageable population, we can work with the local schools in selecting a random sample of around 200 algebra students who we want to participate in our experiment.

In summary, because we cannot test all of the algebra students in a city, we want to find a group of about 200 that reflects the composition of that city. With a representative group, we can generalize our findings to the larger population without fear of our sample being biased in some way.

Source illustration description (image not reproduced): (a) A photograph shows an aerial view of crowds on a street. (b) A photograph shows s small group of children. Researchers may work with (a) a large population or (b) a sample group that is a subset of the larger population. (credit crowd : modification of work by James Cridland; credit students : modification of work by Laurie Sullivan)

Now that we have a sample, the next step of the experimental process is to split the participants into experimental and control groups through random assignment. With random assignment, all participants have an equal chance of being assigned to either group. There is statistical software that will randomly assign each of the algebra students in the sample to either the experimental or the control group.

Random assignment is critical for sound experimental design. With sufficiently large samples, random assignment makes it unlikely that there are systematic differences between the groups. So, for instance, it would be very unlikely that we would get one group composed entirely of males, a given ethnic identity, or a given religious ideology. This is important because if the groups were systematically different before the experiment began, we would not know the origin of any differences we find between the groups: Were the differences preexisting, or were they caused by manipulation of the independent variable? Random assignment allows us to assume that any differences observed between experimental and control groups result from the manipulation of the independent variable.

Issues to Consider

While experiments allow scientists to make cause-and-effect claims, they are not without problems. True experiments require the experimenter to manipulate an independent variable, and that can complicate many questions that psychologists might want to address. For instance, imagine that you want to know what effect sex (the independent variable) has on spatial memory (the dependent variable). Although you can certainly look for differences between males and females on a task that taps into spatial memory, you cannot directly control a person s sex. We categorize this type of research approach as quasi-experimental and recognize that we cannot make cause-and-effect claims in these circumstances.

Experimenters are also limited by ethical constraints. For instance, you would not be able to conduct an experiment designed to determine if experiencing abuse as a child leads to lower levels of self-esteem among adults. To conduct such an experiment, you would need to randomly assign some experimental participants to a group that receives abuse, and that experiment would be unethical.

Interpreting Experimental Findings

Once data is collected from both the experimental and the control groups, a statistical analysis is conducted to find out if there are meaningful differences between the two groups. A statistical analysis determines how likely any difference found is due to chance (and thus not meaningful). For example, if an experiment is done on the effectiveness of a nutritional supplement, and those taking a placebo pill (and not the supplement) have the same result as those taking the supplement, then the experiment has shown that the nutritional supplement is not effective. Generally, psychologists consider differences to be statistically significant if there is less than a five percent chance of observing them if the groups did not actually differ from one another. Stated another way, psychologists want to limit the chances of making false positive claims to five percent or less.

The greatest strength of experiments is the ability to assert that any significant differences in the findings are caused by the independent variable. This occurs because random selection, random assignment, and a design that limits the effects of both experimenter bias and participant expectancy should create groups that are similar in composition and treatment. Therefore, any difference between the groups is attributable to the independent variable, and now we can finally make a causal statement. If we find that watching a violent television program results in more violent behavior than watching a nonviolent program, we can safely say that watching violent television programs causes an increase in the display of violent behavior.

Reporting Research

When psychologists complete a research project, they generally want to share their findings with other scientists. The American Psychological Association (APA) publishes a manual detailing how to write a paper for submission to scientific journals. Unlike an article that might be published in a magazine like Psychology Today, which targets a general audience with an interest in psychology, scientific journals generally publish peer-reviewed journal articles aimed at an audience of professionals and scholars who are actively involved in research themselves.

A peer-reviewed journal article is read by several other scientists (generally anonymously) with expertise in the subject matter. These peer reviewers provide feedback to both the author and the journal editor regarding the quality of the draft. Peer reviewers look for a strong rationale for the research being described, a clear description of how the research was conducted, and evidence that the research was conducted in an ethical manner. They also look for flaws in the study's design, methods, and statistical analyses. They check that the conclusions drawn by the authors seem reasonable given the observations made during the research. Peer reviewers also comment on how valuable the research is in advancing the discipline s knowledge. This helps prevent unnecessary duplication of research findings in the scientific literature and, to some extent, ensures that each research article provides new information. Ultimately, the journal editor will compile all of the peer reviewer feedback and determine whether the article will be published in its current state (a rare occurrence), published with revisions, or not accepted for publication.

Peer review provides some degree of quality control for psychological research. Poorly conceived or executed studies can be weeded out, and even well-designed research can be improved by the revisions suggested. Peer review also ensures that the research is described clearly enough to allow other scientists to replicate it, meaning they can repeat the experiment using different samples to determine reliability. Sometimes replications involve additional measures that expand on the original finding. In any case, each replication serves to provide more evidence to support the original research findings. Successful replications of published research make scientists more apt to adopt those findings, while repeated failures tend to cast doubt on the legitimacy of the original article and lead scientists to look elsewhere. For example, it would be a major advancement in the medical field if a published study indicated that taking a new drug helped individuals achieve a healthy weight without changing their diet. But if other scientists could not replicate the results, the original study s claims would be questioned.

In recent years, there has been increasing concern about a replication crisis that has affected a number of scientific fields, including psychology. Some of the most well-known studies and scientists have produced research that has failed to be replicated by others (as discussed in Shrout & Rodgers, 2018). In fact, even a famous Nobel Prize-winning scientist has recently retracted a published paper because she had difficulty replicating her results (Nobel Prize-winning scientist Frances Arnold retracts paper, 2020 January 3). These kinds of outcomes have prompted some scientists to begin to work together and more openly, and some would argue that the current crisis is actually improving the ways in which science is conducted and in how its results are shared with others (Aschwanden, 2018).

Dig Deeper

Some scientists have claimed that routine childhood vaccines cause some children to develop autism, and, in fact, several peer-reviewed publications published research making these claims. Since the initial reports, large-scale epidemiological research has indicated that vaccinations are not responsible for causing autism and that it is much safer to have your child vaccinated than not. Furthermore, several of the original studies making this claim have since been retracted.

A published piece of work can be rescinded when data is called into question because of falsification, fabrication, or serious research design problems. Once rescinded, the scientific community is informed that there are serious problems with the original publication. Retractions can be initiated by the researcher who led the study, by research collaborators, by the institution that employed the researcher, or by the editorial board of the journal in which the article was originally published. In the vaccine-autism case, the retraction was made because of a significant conflict of interest in which the leading researcher had a financial interest in establishing a link between childhood vaccines and autism (Offit, 2008). Unfortunately, the initial studies received so much media attention that many parents around the world became hesitant to have their children vaccinated (the accompanying source-figure description). Continued reliance on such debunked studies has significant consequences. For instance, between January and October of 2019, there were 22 measles outbreaks across the United States and more than a thousand cases of individuals contracting measles (Patel et al., 2019). This is likely due to the anti-vaccination movements that have risen from the debunked research. For more information about how the vaccine/autism story unfolded, as well as the repercussions of this story, take a look at Paul Offit s book, Autism s False Prophets: Bad Science, Risky Medicine, and the Search for a Cure.

Source illustration description (image not reproduced): A photograph shows a child being given an oral vaccine. Some people still think vaccinations cause autism. (credit: modification of work by UNICEF Sverige)

Reliability and Validity

Reliability and validity are two important considerations that must be made with any type of data collection. Reliability refers to the ability to consistently produce a given result. In the context of psychological research, this would mean that any instruments or tools used to collect data do so in consistent, reproducible ways. There are a number of different types of reliability. Some of these include inter-rater reliability (the degree to which two or more different observers agree on what has been observed), internal consistency (the degree to which different items on a survey that measure the same thing correlate with one another), and test-retest reliability (the degree to which the outcomes of a particular measure remain consistent over multiple administrations).

Unfortunately, being consistent in measurement does not necessarily mean that you have measured something correctly. To illustrate this concept, consider a kitchen scale that would be used to measure the weight of cereal that you eat in the morning. If the scale is not properly calibrated, it may consistently under- or overestimate the amount of cereal that s being measured. While the scale is highly reliable in producing consistent results (e.g., the same amount of cereal poured onto the scale produces the same reading each time), those results are incorrect. This is where validity comes into play. Validity refers to the extent to which a given instrument or tool accurately measures what it s supposed to measure, and once again, there are a number of ways in which validity can be expressed. Ecological validity (the degree to which research results generalize to real-world applications), construct validity (the degree to which a given variable actually captures or measures what it is

intended to measure), and face validity (the degree to which a given variable seems valid on the surface) are just a few types that researchers consider. While any valid measure is by necessity reliable, the reverse is not necessarily true. Researchers strive to use instruments that are both highly reliable and valid.

Everyday Connection

Standardized tests like the SAT and ACT are supposed to measure an individual s aptitude for a college education, but how reliable and valid are such tests? Research conducted by the College Board suggests that scores on the SAT have high predictive validity for first-year college students GPA (Kobrin, Patterson, Shaw, Mattern, & Barbuti, 2008). In this context, predictive validity refers to the test s ability to effectively predict the GPA of college freshmen. Given that many institutions of higher education require the SAT or ACT for admission, this high degree of predictive validity might be comforting.

However, the emphasis placed on SAT or ACT scores in college admissions is changing based on a number of factors. For one, some researchers assert that these tests are biased, and students from historically marginalized populations are at a disadvantage that unfairly reduces the likelihood of being admitted into a college (Santelices & Wilson, 2010). Additionally, some research has suggested that the predictive validity of these tests is grossly exaggerated in how well they are able to predict the GPA of first-year college students. In fact, it has been suggested that the SAT s predictive validity may be overestimated by as much as 150% (Rothstein, 2004). Many institutions of higher education are beginning to consider de-emphasizing the significance of SAT scores in making admission decisions (Rimer, 2008).

Recent examples of high profile cheating scandals both domestically and abroad have only increased the scrutiny being placed on these types of tests, and as of March 2019, more than 1000 institutions of higher education have either relaxed or eliminated the requirements for SAT or ACT testing for admissions (Strauss, 2019, March 19).

Summary

A correlation is described with a correlation coefficient, r, which ranges from -1 to 1. The correlation coefficient tells us about the nature (positive or negative) and the strength of the relationship between two or more variables. Correlations do not tell us anything about causation regardless of how strong the relationship is between variables. In fact, the only way to demonstrate causation is by conducting an experiment. People often make the mistake of claiming that correlations exist when they really do not.

Researchers can test cause-and-effect hypotheses by conducting experiments. Ideally, experimental participants are randomly selected from the population of interest. Then, the participants are randomly assigned to their respective groups. Sometimes, the researcher and the participants are blind to group membership to prevent their expectations from influencing the results.

In ideal experimental design, the only difference between the experimental and control groups is whether participants are exposed to the experimental manipulation. Each group goes through all phases of the experiment, but each group will experience a different level of the independent variable: the experimental group is exposed to the experimental manipulation, and the control group is not exposed to the experimental manipulation. The researcher then measures the changes that are produced in the dependent variable in each group. Once data is collected from both groups, it is analyzed statistically to determine if there are meaningful differences between the groups.

Psychologists report their research findings in peer-reviewed journal articles. Research published in this format is checked by several other psychologists who serve as a filter separating ideas that are supported by evidence from ideas that are not. Replication has an important role in ensuring the legitimacy of published research. In the long run, only those findings that are capable of being replicated consistently will achieve consensus in the scientific community.

Review Questions

OpenStax learning exercise

Q. Height and weight are positively correlated. This means that:

A. D

OpenStax learning exercise

Q. Which of the following correlation coefficients indicates the strongest relationship between two variables?

A. A

OpenStax learning exercise

Q. Which statement best illustrates a negative correlation between the number of hours spent watching TV the week before an exam and the grade on that exam?

A. D

OpenStax learning exercise

Q. The correlation coefficient indicates the weakest relationship when _____.

A. A

OpenStax learning exercise

Q. _____ means that everyone in the population has the same likelihood of being asked to participate in the study.

A. D

OpenStax learning exercise

Q. The _____ is controlled by the experimenter, while the _____ represents the information collected and statistically analyzed by the experimenter.

A. B

OpenStax learning exercise

Q. Researchers must _____ important concepts in their studies so others would have a clear understanding of exactly how those concepts were defined.

A. C

OpenStax learning exercise

Q. Sometimes, researchers will administer a(n) _____ to participants in the control group to control for the effects that participant expectation might have on the experiment.

A. D

Critical Thinking Questions

OpenStax learning exercise

Q. Earlier in this section, we read about research suggesting that there is a correlation between eating cereal and weight. Cereal companies that present this information in their advertisements could lead someone to believe that eating more cereal causes healthy weight. Why would they make such a claim and what arguments could you make to counter this cause-and-effect claim?

A. The cereal companies are trying to make a profit, so framing the research findings in this way would improve their bottom line. However, it could be that people who forgo more fatty options for breakfast are health conscious and engage in a variety of other behaviors that help them maintain a healthy weight.

OpenStax learning exercise

Q. Recently a study was published in the journal, Nutrition and Cancer, which established a negative correlation between coffee consumption and breast cancer. Specifically, it was found that women consuming more than 5 cups of coffee a day were less likely to develop breast cancer than women who never consumed coffee (Lowcock, Cotterchio, Anderson, Boucher, & El-Sohemy, 2013). Imagine you see a newspaper story about this research that says, Coffee Protects Against Cancer. Why is this headline misleading and why would a more accurate headline draw less interest?

A. Using the word protects seems to suggest causation as a function of correlation. If the headline were more accurate, it would be less interesting because indicating that two things are associated is less powerful than indicating that doing one thing causes a change in the other.

OpenStax learning exercise

Q. Sometimes, true random sampling can be very difficult to obtain. Many researchers make use of convenience samples as an alternative. For example, one popular convenience sample would involve students enrolled in Introduction to Psychology courses. What are the implications of using this sampling technique?

A. If research is limited to students enrolled in Introduction to Psychology courses, then our ability to generalize to the larger population would be dramatically reduced. One could also argue that students enrolled in Introduction to Psychology courses may not be representative of the larger population of college students at their school, much less the larger general population.

OpenStax learning exercise

Q. Peer review is an important part of publishing research findings in many scientific disciplines. This process is normally conducted anonymously; in other words, the author of the article being reviewed does not know who is reviewing the article, and the reviewers are unaware of the author s identity. Why would this be an important part of this process?

A. Anonymity protects against personal biases interfering with the reviewer s opinion of the research. Allowing the reviewer to remain anonymous would mean that they can be honest in their appraisal of the manuscript without fear of reprisal.

Personal Application Questions

OpenStax learning exercise

Q. We all have a tendency to make illusory correlations from time to time. Try to think of an illusory correlation that is held by you, a family member, or a close friend. How do you think this illusory correlation came about and what can be done in the future to combat them?

OpenStax learning exercise

Q. Are there any questions about human or animal behavior that you would really like to answer? Generate a hypothesis and briefly describe how you would conduct an experiment to answer your question.

CHAPTER 04 / CAC.901

Developmental theories in counseling perspective

Lecture: Theories are tools with boundaries

A theory helps organize observations and generate explanations. It directs attention toward some processes and away from others. A cognitive theory may highlight how a child interprets an event. A behavioral account may focus on learning conditions. A family-systems account may examine repeated interactions. An ecological account may locate those interactions within institutions and social conditions. None of these perspectives should be treated as a complete description of a child simply because it is familiar to the counselor.

Theories are useful when they improve understanding, suggest testable expectations, and guide action. They become harmful when they turn into labels that explain everything after the fact. If a child s protest is called resistance, compliance is called dependence, and silence is called denial, the theory has become impossible to challenge. A doctoral practitioner should ask what evidence would count against an interpretation and whether another explanation fits the observations better.

Stage descriptions and their limits

Stage theories describe qualitatively different forms of functioning. They can help adults avoid expecting an abstract explanation from a child who is still building the relevant skills. Piaget s account of cognitive development, for example, draws attention to how children actively construct understanding. It encourages counselors to examine the child s reasoning rather than merely score an answer as right or wrong.

However, a stage is not a rigid container. Performance varies with task, familiarity, language, support, and context. A child may reason successfully in a meaningful everyday situation and struggle with an unfamiliar formal task. Cultural practices also shape which abilities are exercised and valued. Use stage-related ideas as prompts for investigation, not as reasons to dismiss a child s capacity. Ask what the child can do with support and how the task could be made more accessible.

Erikson s psychosocial framework highlights recurring life tasks, such as trust, autonomy, initiative, industry, and identity. These themes can help a counselor connect an individual concern to a broader developmental challenge. An adolescent exploring identity may need room for questions rather than immediate correction. A younger child learning initiative may benefit from safe opportunities to choose and contribute. Yet the framework should not be used to assume that every family follows the same route toward independence or that a person who misses a presumed task is permanently damaged.

Learning and meaning

Behavioral theories explain how behavior changes in relation to antecedents and consequences. They are especially useful when a problem has become tied to particular settings. A child s avoidance may reduce distress immediately, making avoidance more likely. An adult s repeated reassurance may also reduce distress briefly while leaving the underlying fear untouched. These are process hypotheses that can be examined through observation and collaborative planning.

Cognitive perspectives add attention to interpretation. Two children may experience the same classroom correction differently. One hears information about a task; the other hears proof of being

stupid. The counselor does not need to decide that one interpretation is merely irrational. The child may have experienced humiliation, discrimination, or repeated failure. The work is to understand how the meaning developed, evaluate its current fit, and create experiences that support a broader account.

A theory of learning without meaning can miss the child s experience. A theory of meaning without context can overlook hunger, inaccessible teaching, or unsafe relationships. Integration should therefore be purposeful. It should explain why several processes matter in this case and how the proposed interventions relate to them. Combining popular techniques without a coherent formulation is not the same as theoretical integration.

Ecological and transactional thinking

Ecological perspectives locate development within nested and connected settings. Family, school, community, policy, and culture can all influence daily experience. The familiar nested-circle diagram is a starting point, but real systems are not simply concentric containers. A caregiver s workplace may affect school pickup. A national examination policy may change classroom pressure. A community organization may connect a family to reliable support. The arrows and boundaries matter.

Transactional thinking adds change over time. A child s behavior influences adults, and adult responses influence the child s next behavior. A slow-to-warm child may evoke extra protection; extra protection may reduce opportunities to practice; reduced practice may increase hesitation. This account does not blame the child or caregiver. It identifies a pattern that can be interrupted by changing opportunities and responses.

Figure 4.1 compares several theoretical lenses applied to one concern. Each lens suggests different questions and possible actions. The figure is not a menu from which the counselor chooses a favorite at random. It is a reminder to match the lens to the problem, evidence, and level of intervention.

Cultural assumptions inside theory

Theories emerge from particular histories. Some were developed using narrow samples and then presented as universal. Concepts such as maturity, independence, healthy attachment, and appropriate emotional expression can carry cultural assumptions. This does not make every theory unusable. It makes explicit examination necessary.

Ask whose development was treated as the standard, which family forms were visible, and which outcomes were valued. Consult local knowledge without romanticizing every practice. A practice can be culturally meaningful and still require examination if it causes harm. Equally, a difference from a counselor s expectation is not automatically a deficit. Cultural humility combines respect, curiosity, and accountability to evidence and rights.

Application: three explanations of school avoidance

A fictional twelve-year-old, Tolu, sometimes refuses to enter school. A cognitive lens asks what Tolu expects will happen in the classroom. A learning lens examines whether leaving school reduces distress. A family-systems lens explores morning interactions and caregiver responses. An ecological lens asks about bullying, transport, teaching demands, and available support. Each question can reveal something important.

A weak integration would declare that all theories are equally true and recommend everything at once. A stronger integration develops a sequence. First assess immediate safety and the reasons for absence. Then identify maintaining processes and the child s priorities. If bullying is present, do not treat avoidance only as an anxiety-maintenance habit. If a learning difficulty contributes, coordinate educational support. Use the theory to organize a plan that can be revised as information emerges.

The counselor should also specify outcomes. Attendance alone may be too narrow if the child enters school but remains unsafe or unable to learn. Include distress, participation, learning access, and the child s experience. A theory-driven plan should make improvement visible without reducing the child to a single number.

Doctoral seminar task

Choose one presenting concern and formulate it through three theories. For each formulation, identify the central mechanism, supporting observations, missing information, and a possible intervention. Then identify one cultural assumption that could distort the formulation. Finally, write an integrated account that explains why the selected elements belong together.

A strong answer will not merely list techniques. It will show how evidence changes the weight given to each explanation and how the child s goals influence the plan. It will also state a stopping or revision rule: what would show that the current formulation is not helping? Theoretical sophistication is demonstrated by disciplined selection and revision, not by the number of theorists named.

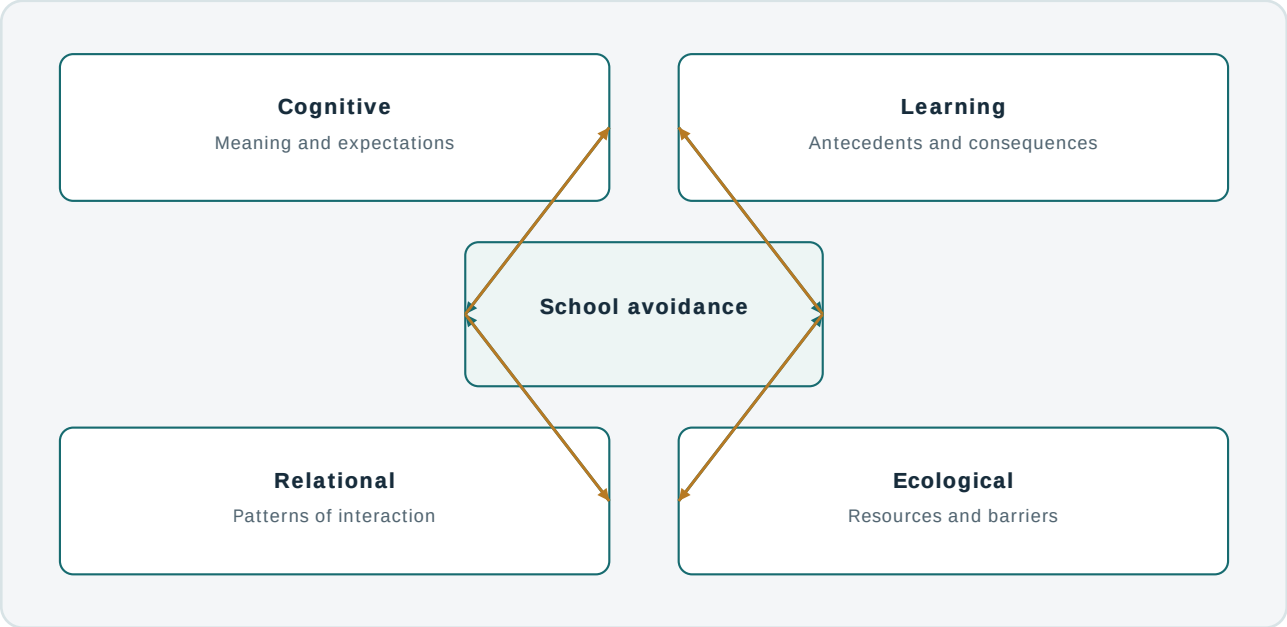


Figure 4.1. One concern, several theoretical lenses. Theories direct attention to different mechanisms. An integrated formulation explains why particular lenses are useful for this child and setting, rather than treating every explanation as equally supported. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

Lifespan Theories

OpenStax source module m82219 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82219/index.cnxml

There are many theories regarding how babies and children grow and develop into happy, healthy adults. We explore several of these theories in this section.

Psychosexual Theory of Development

Sigmund Freud (1856 1939) believed that personality develops during early childhood. For Freud, childhood experiences shape our personalities and behavior as adults. Freud viewed development as discontinuous; he believed that each of us must pass through a series of stages during childhood, and that if we lack proper nurturance and parenting during a stage, we may become stuck, or fixated, in that stage. Freud s stages are called the stages of psychosexual development. According to Freud, children s pleasure-seeking urges are focused on a different area of the body, called an erogenous zone, at each of the five stages of development: oral, anal, phallic, latency, and genital.

While most of Freud s ideas have not found support in modern research, we cannot discount the contributions that Freud has made to the field of psychology. Psychologists today dispute Freud's psychosexual stages as a legitimate explanation for how one's personality develops, but what we can take away from Freud s theory is that personality is shaped, in some part, by experiences we have in childhood. These stages are discussed in detail in the chapter on personality.

Psychosocial Theory of Development

Erik Erikson (1902 1994) (the accompanying source-figure description), another stage theorist, took Freud s theory and modified it as psychosocial theory. Erikson s psychosocial development theory emphasizes the social nature of our development rather than its sexual nature. While Freud believed that personality is shaped only in childhood, Erikson proposed that personality development takes place all through the lifespan. Erikson suggested that how we interact with others is what affects our sense of self, or what he called the ego identity.

Source illustration description (image not reproduced): A photograph depicts Erik Erikson in his later years. Erik Erikson proposed the psychosocial theory of development. In each stage of Erikson s theory, there is a psychosocial task that we must master in order to feel a sense of competence.

Erikson proposed that we are motivated by a need to achieve competence in certain areas of our lives. According to psychosocial theory, we experience eight stages of development over our lifespan, from infancy through late adulthood. At each stage there is a conflict, or task, that we need to resolve. Successful completion of each developmental task results in a sense of competence and a healthy personality. Failure to master these tasks leads to feelings of inadequacy.

According to Erikson (1963), trust is the basis of our development during infancy (birth to 12 months). Therefore, the primary task of this stage is trust versus mistrust. Infants are dependent upon their caregivers, so caregivers who are responsive and sensitive to their infant s needs help their baby to develop a sense of trust; their baby will see the world as a safe, predictable place. Unresponsive caregivers who do not meet their baby s needs can engender feelings of anxiety, fear, and mistrust; their baby may see the world as unpredictable.

As toddlers (ages 1 3 years) begin to explore their world, they learn that they can control their actions and act on the environment to get results. They begin to show clear preferences for certain elements of the environment, such as food, toys, and clothing. A toddler s main task is to resolve the issue of autonomy versus shame and doubt, by working to establish independence. This is the me do it stage. For example, we might observe a budding sense of autonomy in a 2-year-old child who wants to choose her clothes and dress herself. Although her outfits might not be appropriate for the situation, her input in such basic decisions has an effect on her sense of independence. If denied the opportunity to act on her environment, she may begin to doubt her abilities, which could lead to low self-esteem and feelings of shame.

Once children reach the preschool stage (ages 3 6 years), they are capable of initiating activities and asserting control over their world through social interactions and play. According to Erikson, preschool children must resolve the task of initiative versus guilt. By learning to plan and achieve goals while interacting with others, preschool children can master this task. Those who do will develop self-confidence and feel a sense of purpose. Those who are unsuccessful at this stage with their initiative misfiring or stifled may develop feelings of guilt. How might over-controlling parents stifle a child s initiative?

During the elementary school stage (ages 7 11), children face the task of industry versus inferiority. Children begin to compare themselves to their peers to see how they measure up. They either develop a sense of pride and accomplishment in their schoolwork, sports, social activities, and family life, or they feel inferior and inadequate when they don t measure up. What are some things parents and teachers can do to help children develop a sense of competence and a belief in themselves and their abilities?

In adolescence (ages 12 18), children face the task of identity versus role confusion. According to Erikson, an adolescent s main task is developing a sense of self. Adolescents struggle with questions such as Who am I? and What do I want to do with my life? Along the way, most adolescents try on many different selves to see which ones fit. Adolescents who are successful at this stage have a strong sense of identity and are able to remain true to their beliefs and values in the face of problems and other people s perspectives. What happens to apathetic adolescents, who do not make a conscious search for identity, or those who are pressured to conform to their parents ideas for the future? These teens will have a weak sense of self and experience role confusion. They are unsure of their identity and confused about the future.

People in early adulthood (i.e., 20s through early 40s) are concerned with intimacy versus isolation. After we have developed a sense of self in adolescence, we are ready to share our life with others. Erikson said that we must have a strong sense of self before developing intimate relationships with others. Adults who do not develop a positive self-concept in adolescence may experience feelings of loneliness and emotional isolation.

When people reach their 40s, they enter the time known as middle adulthood, which extends to the mid-60s. The social task of middle adulthood is generativity versus stagnation. Generativity involves finding your life s work and contributing to the development of others, through activities such as volunteering, mentoring, and raising children. Those who do not master this task may experience stagnation, having little connection with others and little interest in productivity and self-improvement.

From the mid-60s to the end of life, we are in the period of development known as late adulthood. Erikson s task at this stage is called integrity versus despair. He said that people in late adulthood reflect on their lives and feel either a sense of satisfaction or a sense of failure. People who feel proud of

their accomplishments feel a sense of integrity, and they can look back on their lives with few regrets. However, people who are not successful at this stage may feel as if their life has been wasted. They focus on what would have, should have, and could have been. They face the end of their lives with feelings of bitterness, depression, and despair. the accompanying source-figure description summarizes the stages of Erikson s theory.

Source comparison table

Erikson s Psychosocial Stages of Development			
Stage	Age (years)	Developmental Task	Description
1	0 1	Trust vs. mistrust	Trust (or mistrust) that basic needs, such as nourishment and affection, will be met
2	1 3	Autonomy vs. shame/doubt	Develop a sense of independence in many tasks
3	3 6	Initiative vs. guilt	Take initiative on some activities may develop guilt when unsuccessful or boundaries overstepped
4	7 11	Industry vs. inferiority	Develop self-confidence in abilities when competent or sense of inferiority when not
5	12 18	Identity vs. confusion	Experiment with and develop identity and roles
6	19 29	Intimacy vs. isolation	Establish intimacy and relationships with others
7	30 64	Generativity vs. stagnation	Contribute to society and be part of a family
8	65	Integrity vs. despair	Assess and make sense of life and meaning of contributions

Cognitive Theory of Development

Jean Piaget (1896 1980) is another stage theorist who studied childhood development (the accompanying source-figure description). Instead of approaching development from a psychoanalytical or psychosocial perspective, Piaget focused on children s cognitive growth. He believed that thinking is a central aspect of development and that children are naturally inquisitive. However, he said that children do not think and reason like adults (Piaget, 1930, 1932). His theory of cognitive development holds that our cognitive abilities develop through specific stages, which exemplifies the discontinuity approach to development. As we progress to a new stage, there is a distinct shift in how we think and reason.

Source illustration description (image not reproduced): A photograph depicts Jean Piaget in his later years. Jean Piaget spent over 50 years studying children and how their minds develop.

Piaget said that children develop schemata to help them understand the world. Schemata are concepts (mental models) that are used to help us categorize and interpret information. By the time children have reached adulthood, they have created schemata for almost everything. When children learn new information, they adjust their schemata through two processes: assimilation and accommodation. First, they assimilate new information or experiences in terms of their current schemata: assimilation is when they take in information that is comparable to what they already know. Accommodation describes when they change their schemata based on new information. This process continues as children interact with their environment.

For example, 2-year-old Abdul learned the schema for dogs because his family has a Labrador retriever. When Abdul sees other dogs in his picture books, he says, Look mommy, dog! Thus, he has assimilated them into his schema for dogs. One day, Abdul sees a sheep for the first time and says, Look mommy, dog! Having a basic schema that a dog is an animal with four legs and fur, Abdul thinks all furry, four-legged creatures are dogs. When Abdul s mom tells him that the animal he sees is a sheep, not a dog, Abdul must accommodate his schema for dogs to include more information based on his new experiences. Abdul s schema for dog was too broad, since not all furry, four-legged creatures are dogs. He now modifies his schema for dogs and forms a new one for sheep.

Like Freud and Erikson, Piaget thought development unfolds in a series of stages approximately associated with age ranges. He proposed a theory of cognitive development that unfolds in four stages: sensorimotor, preoperational, concrete operational, and formal operational (the accompanying source-figure description).

Source comparison table

Piaget s Stages of Cognitive Development			
Age (years)	Stage	Description	Developmental issues
0 2	Sensorimotor	World experienced through senses and actions	Object permanence Stranger anxiety
2 6	Preoperational	Use words and images to represent things, but lack logical reasoning	Pretend play Egocentrism Language development
7 11	Concrete operational	Understand concrete events and analogies logically; perform arithmetical operations	Conservation Mathematical transformations
12	Formal operational	Formal operations Utilize abstract reasoning	Abstract logic Moral reasoning

The first stage is the sensorimotor stage, which lasts from birth to about 2 years old. During this stage, children learn about the world through their senses and motor behavior. Young children put objects in their mouths to see if the items are edible, and once they can grasp objects, they may shake or bang them to see if they make sounds. Between 5 and 8 months old, the child develops object permanence, which is the understanding that even if something is out of sight, it still exists (Bogartz, Shinskey, & Schilling, 2000). According to Piaget, young infants do not remember an object after it has been removed from sight. Piaget studied infants reactions when a toy was first shown to an infant and then hidden under a blanket. Infants who had already developed object permanence would reach for the hidden toy, indicating that they knew it still existed, whereas infants who had not developed object permanence would appear confused.

In Piaget s view, around the same time children develop object permanence, they also begin to exhibit stranger anxiety, which is a fear of unfamiliar people. Babies may demonstrate this by crying and turning away from a stranger, by clinging to a caregiver, or by attempting to reach their arms toward familiar faces such as parents. Stranger anxiety results when a child is unable to assimilate the stranger into an existing schema; therefore, she can t predict what her experience with that stranger will be like, which results in a fear response.

Piaget s second stage is the preoperational stage, which is from approximately 2 to 7 years old. In this stage, children can use symbols to represent words, images, and ideas, which is why children in this stage engage in pretend play. A child s arms might become airplane wings as they zoom around the room, or a child with a stick might become a brave knight with a sword. Children also begin to use language in the preoperational stage, but they cannot understand adult logic or mentally manipulate information (the term operational refers to logical manipulation of information, so children at this stage are considered to be pre-operational). Children s logic is based on their own personal knowledge of the world so far, rather than on conventional knowledge. For example, dad gave a slice of pizza to 10-year-old Keiko and another slice to her 3-year-old brother, Kenny. Kenny s pizza slice was cut into five pieces, so Kenny told his sister that he got more pizza than she did. Children in this stage cannot perform mental operations because they have not developed an understanding of conservation, which is the idea that even if you change the appearance of something, it is still equal in size as long as nothing has been removed or added.

During this stage, we also expect children to display egocentrism, which means that the child is not able to take the perspective of others. A child at this stage thinks that everyone sees, thinks, and feels just as they do. Let s look at Kenny and Keiko again. Keiko s birthday is coming up, so their mom takes Kenny to the toy store to choose a present for his sister. He selects an Iron Man action figure for her, thinking that if he likes the toy, his sister will too. An egocentric child is not able to infer the perspective of other people and instead attributes his own perspective.

Piaget s third stage is the concrete operational stage, which occurs from about 7 to 11 years old. In this stage, children can think logically about real (concrete) events; they have a firm grasp on the use of numbers and start to employ memory strategies. They can perform mathematical operations and understand transformations, such as addition is the opposite of subtraction, and multiplication is the opposite of division. In this stage, children also master the concept of conservation: Even if something changes shape, its mass, volume, and number stay the same. For example, if you pour water from a tall, thin glass to a short, fat glass, you still have the same amount of water. Remember Keiko and Kenny and the pizza? How did Keiko know that Kenny was wrong when he said that he had more pizza?

Children in the concrete operational stage also understand the principle of reversibility, which means that objects can be changed and then returned back to their original form or condition. Take, for example, water that you poured into the short, fat glass: You can pour water from the fat glass back to the thin glass and still have the same amount (minus a couple of drops).

The fourth, and last, stage in Piaget s theory is the formal operational stage, which is from about age 11 to adulthood. Whereas children in the concrete operational stage are able to think logically only about concrete events, children in the formal operational stage can also deal with abstract ideas and hypothetical situations. Children in this stage can use abstract thinking to problem solve, look at alternative solutions, and test these solutions. In adolescence, a renewed egocentrism occurs. For example, a 15-year-old with a very small pimple on her face might think it is huge and incredibly visible, under the mistaken impression that others must share her perceptions.

Beyond Formal Operational Thought

As with other major contributors of theories of development, several of Piaget s ideas have come under criticism based on the results of further research. For example, several contemporary studies support a model of development that is more continuous than Piaget s discrete stages (Courage & Howe, 2002; Siegler, 2005, 2006). Many others suggest that children reach cognitive milestones earlier than Piaget describes (Baillargeon, 2004; de Hevia & Spelke, 2010).

According to Piaget, the highest level of cognitive development is formal operational thought, which develops between 11 and 20 years old. However, many developmental psychologists disagree with Piaget, suggesting a fifth stage of cognitive development, known as the postformal stage (Basseches, 1984; Commons & Bresette, 2006; Sinnott, 1998). In postformal thinking, decisions are made based on situations and circumstances, and logic is integrated with emotion as adults develop principles that depend on contexts. One way that we can see the difference between an adult in postformal thought and an adolescent in formal operations is in terms of how they handle emotionally charged issues.

It seems that once we reach adulthood our problem solving abilities change: As we attempt to solve problems, we tend to think more deeply about many areas of our lives, such as relationships, work, and politics (Labouvie-Vief & Diehl, 1999). Because of this, postformal thinkers are able to draw on past experiences to help them solve new problems. Problem-solving strategies using postformal thought vary, depending on the situation. What does this mean? Adults can recognize, for example, that what seems to be an ideal solution to a problem at work involving a disagreement with a colleague may not be the best solution to a disagreement with a significant other.

Connect the Concepts

The genetic environmental correlation you ve learned about concerning the bidirectional influence of genes and the environment has been explored in more recent theories (Newcombe, 2011). One such theory, neuroconstructivism, suggests that neural brain development influences cognitive development. Experiences that a child encounters can impact or change the way that neural pathways develop in response to the environment. An individual s behavior is based on how one understands the world. There is interaction between neural and cognitive networks at and between each level, consisting of these:

- genes
- neurons
- brain
- body
- social environment

These interactions shape mental representations in the brain and are dependent on context that individuals actively explore throughout their lifetimes (Westermann, Mareschal, Johnson, Sirois, Spratling, & Thomas, 2007).

An example of this would be a child who may be genetically predisposed to a difficult temperament. They may have parents who provide a social environment in which they are encouraged to express themselves in an optimal manner. The child's brain would form neural connections enhanced by that environment, thus influencing the brain. The brain gives information to the body about how it will experience the environment. Thus, neural and cognitive networks work together to influence genes (i.e., attenuating temperament), body (i.e., may be less prone to high blood pressure), and social environment (i.e., may seek people who are similar to them).

SOCIOCULTURAL THEORY OF DEVELOPMENT

Lev Vygotsky was a Russian psychologist who proposed a sociocultural theory of development. He suggested that human development is rooted in one s culture. A child s social world, for example, forms the basis for the formation of language and thought. The language one speaks and the ways a person thinks about things is dependent on one s cultural background. Vygotsky also considered historical influences as key to one s development. He was interested in the process of development and the individual s interactions with their environment (John-Steiner & Mahn, 1996).

Moral Theory Of Development

A major task beginning in childhood and continuing into adolescence is discerning right from wrong. Psychologist Lawrence Kohlberg (1927 1987) extended upon the foundation that Piaget built regarding cognitive development. Kohlberg believed that moral development, like cognitive development, follows a series of stages. To develop this theory, Kohlberg posed moral dilemmas to people of all ages, and then he analyzed their answers to find evidence of their particular stage of moral development. Before reading about the stages, take a minute to consider how you would answer one of Kohlberg's best-known moral dilemmas, commonly known as the Heinz dilemma:

In Europe, a woman was near death from a special kind of cancer. There was one drug that the doctors thought might save her. It was a form of radium that a druggist in the same town had recently discovered. The drug was expensive to make, but the druggist was charging ten times what the drug cost him to make. He paid $200 for the radium and charged $2,000 for a small dose of the drug. The sick woman's husband, Heinz, went to everyone he knew to borrow the money, but he could only get together about $1,000, which is half of what it cost. He told the druggist that his wife was dying and asked him to sell it cheaper or let him pay later. But the druggist said: No, I discovered the drug and I'm going to make money from it. So Heinz got desperate and broke into the man's store to steal the drug for his wife. Should the husband have done that? (Kohlberg, 1969, p. 379)

How would you answer this dilemma? Kohlberg was not interested in whether you answer yes or no to the dilemma: Instead, he was interested in the reasoning behind your answer.

After presenting people with this and various other moral dilemmas, Kohlberg reviewed people s responses and placed them in different stages of moral reasoning (the accompanying source-figure description). According to Kohlberg, an individual progresses from the capacity for pre-conventional morality (before age 9) to the capacity for conventional morality (early adolescence), and toward attaining post-conventional morality (once formal operational thought is attained), which only a few fully achieve. Kohlberg placed in the highest stage responses that reflected the reasoning that Heinz should steal the drug because his wife s life is more important than the pharmacist making money. The value of a human life overrides the pharmacist s greed.

Source illustration description (image not reproduced): Nine boxes are arranged in rows and columns of three. The top left box contains Level 1, Pre-conventional Morality. A line connects this box with another box to the right containing Stage 1, Obedience and punishment: behavior driven by avoiding punishment. To the right is another box connected by a line containing Stage 2, Individual interest: behavior driven by self-interest and rewards. The middle left box contains Level 2, Conventional Morality. A line connects this box with another box to the right containing Stage 3, Interpersonal: behavior driven by social approval. To the right is another box connected by a line containing Stage 4, Authority: behavior driven by obeying authority and conforming to social order. The lower left box contains Level 3, Post-conventional Morality. A line connects this box with another box to the right containing Stage 5, Social contract: behavior driven by balance of social order and individual rights. To the right is another box connected by a line containing Stage 6, Universal ethics: behavior driven by internal moral principles. Kohlberg identified three levels of moral reasoning: pre-conventional, conventional, and post-conventional: Each level is associated with increasingly complex stages of moral development.

It is important to realize that even those people who have the most sophisticated, post-conventional reasons for some choices may make other choices for the simplest of pre-conventional reasons. Many psychologists agree with Kohlberg's theory of moral development but point out that moral reasoning is very different from moral behavior. Sometimes what we say we would do in a situation is not what we actually do in that situation. In other words, we might talk the talk, but not walk the walk.

How does this theory apply to people of different genders? Kohlberg (1969) felt that more men than women move past stage four in their moral development. He went on to note that women seem to be deficient in their moral reasoning abilities. These ideas were not well received by Carol Gilligan, a research assistant of Kohlberg, who consequently developed her own ideas of moral development. In her groundbreaking book, In a Different Voice: Psychological Theory and Women s Development, Gilligan (1982) criticized her former mentor s theory because it was based only on upper class White men and boys. She argued that women are not deficient in their moral reasoning she proposed that

different genders reason differently. Girls and women focus more on staying connected and the importance of interpersonal relationships. Therefore, in the Heinz dilemma, many girls and women respond that Heinz should not steal the medicine. Their reasoning is that if he steals the medicine, is arrested, and is put in jail, then he and his wife will be separated, and she could die while he is still in prison.

Summary

There are many theories regarding how babies and children grow and develop into happy, healthy adults. Sigmund Freud suggested that we pass through a series of psychosexual stages in which our energy is focused on certain erogenous zones on the body. Eric Erikson modified Freud s ideas and suggested a theory of psychosocial development. Erikson said that our social interactions and successful completion of social tasks shape our sense of self. Jean Piaget proposed a theory of cognitive development that explains how children think and reason as they move through various stages. Finally, Lawrence Kohlberg turned his attention to moral development. He said that we pass through three levels of moral thinking that build on our cognitive development.

Review Questions

OpenStax learning exercise

Q. The idea that even if something is out of sight, it still exists is called _____.

A. B

OpenStax learning exercise

Q. Which theorist proposed that moral thinking proceeds through a series of stages?

A. D

OpenStax learning exercise

Q. According to Erikson s theory of psychosocial development, what is the main task of the adolescent?

A. C

Critical Thinking Questions

OpenStax learning exercise

Q. What is the difference between assimilation and accommodation? Provide examples of each.

A. Assimilation is when we take in information that is comparable to what we already know. Accommodation is when we change our schemata based on new information. An example of assimilation is a child s schema of dog based on the family s golden retriever being expanded to include two newly adopted golden retrievers. An example of accommodation is that same child s schema of dog being adjusted to exclude other four-legged furry animals such as sheep and foxes.

OpenStax learning exercise

Q. Why was Carol Gilligan critical of Kohlberg s theory of moral development?

A. Gilligan criticized Kohlberg because his theory was based on the responses of upper class White men and boys, arguing that it was biased against women. While Kohlberg concluded that women must be deficient in their moral reasoning abilities, Gilligan disagreed, suggesting that female moral reasoning is not deficient, just different.

OpenStax learning exercise

Q. What is egocentrism? Provide an original example.

A. Egocentrism is the inability to take the perspective of another person. This type of thinking is common in young children in the preoperational stage of cognitive development. An example might be that upon seeing his mother crying, a young child gives her his favorite stuffed animal to make her feel better.

Personal Application Questions

OpenStax learning exercise

Q. Explain how you would use your understanding of one of the major developmental theories to deal with each of the difficulties listed below:

CHAPTER 05 / CAC.901

Prenatal development, childhood, and adolescence

Lecture: Childhood is not one developmental setting

Children of different ages encounter different bodies, tasks, relationships, and social expectations. Counselors need developmental knowledge because the same concern can require different forms of communication and support. A preschooler s separation distress, a nine-year-old s worry about competence, and a fifteen-year-old s concern about identity may share emotional processes, but they do not call for identical conversations.

Developmental periods are useful approximations. Chronological age is easy to record, but it does not fully describe a child s communication, health, learning, responsibilities, or cultural context. Two ten-year-olds may differ greatly in language access, autonomy, and experience of adversity. Age should guide questions, not replace them. The counselor asks what this child understands, what supports participation, and which expectations are creating difficulty.

Early development and the caregiving environment

Early development occurs through repeated interactions. Feeding, sleep, play, comfort, and communication are not isolated tasks. They form routines in which a child learns what to expect and caregivers learn how to respond. Health conditions, caregiver stress, housing, and work patterns can shape those routines. A developmental assessment should examine the caregiving environment without turning ordinary difficulty into a moral judgment about parents.

Milestones can help identify when further assessment may be useful, but variation is expected. A responsible account distinguishes a delay in one skill from a broad pattern, considers opportunities to learn, and examines hearing, vision, health, and communication access. Avoid using a single milestone as a complete measure of a child s potential. Avoid the opposite error as well: dismissing a persistent concern by saying that every child develops differently. Variation is real, and so is the value of timely support.

Play provides a context for practice. Children explore objects, rehearse roles, negotiate meaning, and express concerns that may be difficult to state directly. A counselor can observe how a child begins play, handles uncertainty, seeks help, and recovers after frustration. Observation should not be turned into unsupported symbolic certainty. A drawing of a storm is not proof of a particular experience. Meaning is explored with the child and considered alongside other information.

Middle childhood and expanding comparison

During middle childhood, school and peer settings become increasingly important. Children compare their performance, friendships, and social standing. They also develop more complex ways to describe themselves. A difficulty that was previously vague may become I am bad at school or Nobody wants me. These statements deserve attention because they organize future participation.

Cognitive growth can support planning and perspective-taking, but stress may reduce access to those capacities. A child who can explain a coping plan in a calm session may not retrieve it during a public confrontation. Practice should therefore occur across realistic conditions, with adult support where needed. Teaching a skill once is not the same as making it usable under pressure.

Family relationships remain central even as peers become more influential. Avoid describing development as a simple transfer of loyalty from parents to friends. Many children move among several meaningful relationships. Cultural expectations about responsibility, respect, and interdependence vary. The counselor should ask how the child and family understand these relationships rather than imposing a single model of healthy separation.

Adolescence and the reorganization of identity

Adolescence involves physical maturation, changing social expectations, and increasing exploration of identity and autonomy. It is not accurately described as an inevitable period of reckless behavior. Risk and vulnerability vary with context, and young people also show creativity, commitment, moral concern, and capacity for care. A deficit-only account can damage the working relationship and miss resources.

Puberty is biological, but its meaning is social. Young people compare their timing and appearance with peers and with cultural ideals. A counselor should use inclusive language and avoid assuming that every adolescent s experience fits a binary or heterosexual pattern. Private questions can create room for concerns that a young person may not raise in front of caregivers. Explain confidentiality and its limits clearly, and follow applicable law and professional standards.

Decision-making also depends on conditions. An adolescent may reason carefully in a quiet conversation yet act differently under peer pressure, strong emotion, or immediate reward. Rather than concluding that adolescents cannot make sound decisions, examine the conditions that support their capacities. Counseling can help create time, reduce pressure, identify trusted support, and connect choices to personally meaningful goals.

Transitions as developmental events

Starting school, changing schools, moving home, entering adolescence, and preparing for adulthood can reorganize daily life. A transition is not automatically a crisis. It can create opportunity as well as strain. The effect depends partly on continuity: familiar relationships, predictable routines, accessible information, and a sense of influence over what happens next.

Figure 5.1 maps developmental periods alongside changing tasks and contexts. The boundaries are deliberately flexible. The diagram helps identify where expectations may have changed faster than support. For example, a student may move to a school that assumes independent organization before anyone has taught the relevant routines. The resulting difficulty should not be interpreted only as low motivation.

Application: the suddenly immature adolescent

A fictional fourteen-year-old, Sade, manages homework independently in one subject but becomes overwhelmed in another. Her caregivers call this immaturity. A developmental assessment examines the tasks. The successful subject may provide a clear schedule, shorter assignments, and a trusted teacher. The difficult subject may require planning across several weeks and public performance. Sade s capacity is interacting with task structure.

The intervention can preserve expectations while changing support. Break a long project into visible stages, rehearse how to ask for clarification, and agree on a review point. Ask Sade which support feels useful and which feels intrusive. Monitor whether independence increases rather than assuming that assistance creates dependence. The developmental aim is not permanent control by adults; it is growing capability under realistic conditions.

Caregivers may need help distinguishing capacity from consistency. Growing adolescents do not perform at their best in every setting. Fatigue, conflict, and social worry matter. At the same time,

persistent difficulties across settings deserve assessment rather than repeated moral labeling. A balanced account makes room for both support and expectation.

Doctoral seminar task

Create a developmental communication plan for the same concern at three ages: early childhood, middle childhood, and adolescence. Specify how you would explain the counseling purpose, invite participation, gather the young person s view, and involve caregivers. Include one visual or activity-based method and explain what it can and cannot establish.

Then identify a transition that could change the meaning of the concern. Describe the supports that would make the transition more manageable and the signs that would prompt reassessment. The exercise should show that developmental appropriateness is an active fit between person, task, and environment, not a fixed script attached to an age.

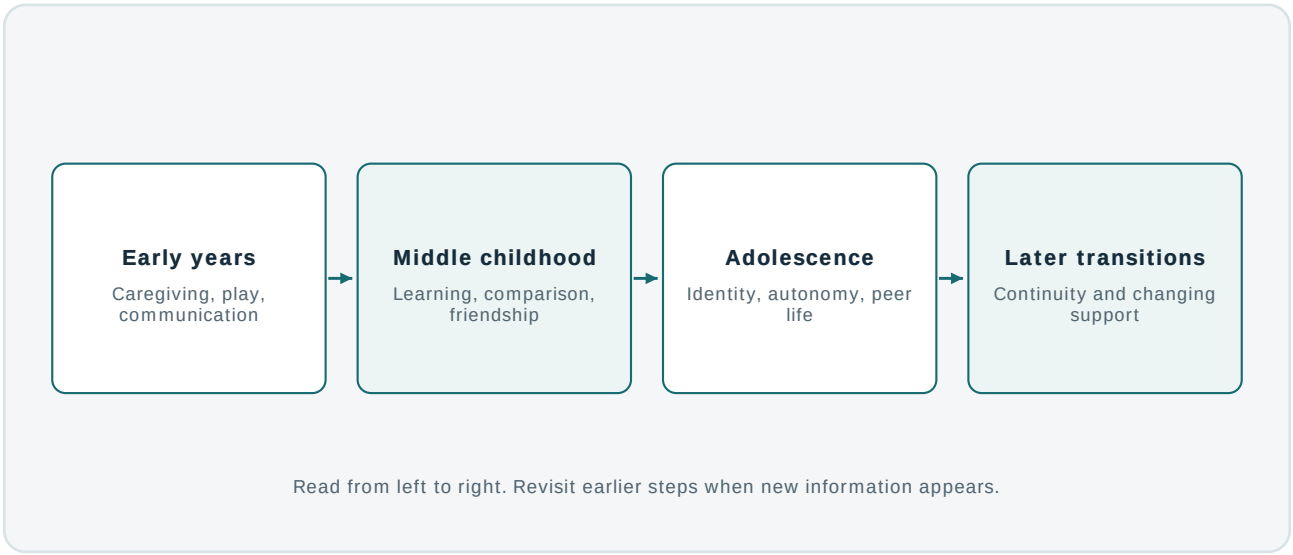


Figure 5.1. Developmental tasks and changing contexts. Age bands are approximate and overlap. Communication, culture, health, and responsibility affect how a child experiences each transition. The diagram is a guide to questions, not a checklist of universal milestones. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

Stages of Development

OpenStax source module m82220 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82220/index.cnxml

From the moment we are born until the moment we die, we continue to develop.

As discussed at the beginning of this chapter, developmental psychologists often divide our development into three areas: physical development, cognitive development, and psychosocial development. Mirroring Erikson s stages, lifespan development is divided into different stages that are based on age. We will discuss prenatal, infant, child, adolescent, and adult development.

Prenatal Development

How did you come to be who you are? From beginning as a one-cell structure to your birth, your prenatal development occurred in an orderly and delicate sequence.

There are three stages of prenatal development: germinal, embryonic, and fetal. Let s take a look at what happens to the developing baby in each of these stages.

Germinal Stage (Weeks 1 2)

In the discussion of biopsychology earlier in the book, you learned about genetics and DNA. DNA is passed on to the child at the moment of conception. Conception occurs when sperm fertilizes an egg and forms a zygote (the accompanying source-figure description). A zygote begins as a one-cell structure that is created when a sperm and egg merge. The genetic makeup and sex of the zygote are set at this point. During the first week after conception, the zygote divides and multiplies, going from a one-cell structure to two cells, then four cells, then eight cells, and so on. This process of cell division is called mitosis. Mitosis is a fragile process, and fewer than one-half of all zygotes survive beyond the first two weeks (Hall, 2004). After 5 days of mitosis there are 100 cells, and after 9 months there are billions of cells. As the cells divide, they become more specialized, forming different organs and body parts. In the germinal stage, the mass of cells has yet to attach itself to the lining of the mother s uterus. Once it does, the next stage begins.

Source illustration description (image not reproduced): A microscopic picture shows a single sperm fusing with the ovum. Sperm and ovum fuse at the point of conception.

Embryonic Stage (Weeks 3 8)

After the zygote divides for about 7 10 days and has 150 cells, it travels down the fallopian tubes and implants itself in the lining of the uterus. Upon implantation, this multi-cellular organism is called an embryo. Now blood vessels grow, forming the placenta. The placenta is a structure connected to the uterus that provides nourishment and oxygen from the mother to the developing embryo via the umbilical cord. Basic structures of the embryo start to develop into areas that will become the head, chest, and abdomen. During the embryonic stage, the heart begins to beat and organs form and begin to function. The neural tube forms along the back of the embryo, developing into the spinal cord and brain.

Fetal Stage (Weeks 9 40)

When the organism is about nine weeks old, the embryo is called a fetus. At this stage, the fetus is about the size of a kidney bean and begins to take on the recognizable form of a human being as the tail begins to disappear.

From 9 12 weeks, the sex organs begin to differentiate. At about 16 weeks, the fetus is approximately 4.5 inches long. Fingers and toes are fully developed, and fingerprints are visible. By the time the fetus reaches the sixth month of development (24 weeks), it weighs up to 1.4 pounds. Hearing has developed, so the fetus can respond to sounds. The internal organs, such as the lungs, heart, stomach, and intestines, have formed enough that a fetus born prematurely at this point has a chance to survive outside of the mother s womb. Throughout the fetal stage the brain continues to grow and develop, nearly doubling in size from weeks 16 to 28. Around 36 weeks, the fetus is almost ready for birth. It weighs about 6 pounds and is about 18.5 inches long, and by week 37 all of the fetus s organ systems are developed enough that it could survive outside the uterus without many of the risks associated with premature birth. The fetus continues to gain weight and grow in length until approximately 40 weeks. By then, the fetus has very little room to move around and birth becomes imminent. The progression through the stages is shown in the accompanying source-figure description.

Source illustration description (image not reproduced): The growth of a fetus is shown using nine pictures in different stages of development. For each stage, there is a picture of a fetus which gets progressively larger and more mature. The first stage is labeled 9 weeks; fetal stage begins. The second stage is labeled 12 weeks; sex organs differentiate. The third stage is labeled 16 weeks; fingers and toes develop. The fourth stage is labeled 20 weeks; hearing begins. The fifth stage is labeled 24 weeks; lungs begin to develop. The sixth stage is labeled 28 weeks; brain grows rapidly. The seventh stage is labeled 32 weeks; bones fully develop. The eighth stage is labeled 36 weeks; muscles fully develop. The ninth stage is labeled 40 weeks; full-term development. During the fetal stage, the baby's brain develops and the body adds size and weight, until the fetus reaches full-term development.

Prenatal Influences

During each prenatal stage, genetic and environmental factors can affect development. The developing fetus is completely dependent on the mother for life. It is important that the mother takes good care of herself and receives prenatal care, which is medical care during pregnancy that monitors the health of both the mother and the fetus (the accompanying source-figure description). According to the National Institutes of Health ([NIH], 2013), routine prenatal care is important because it can reduce the risk of complications to the mother and fetus during pregnancy. In fact, women who are trying to become pregnant or who may become pregnant should discuss pregnancy planning with their doctor. They may be advised, for example, to take a vitamin containing folic acid, which helps prevent certain birth defects, or to monitor aspects of their diet or exercise routines.

Source illustration description (image not reproduced): A technician uses ultrasound equipment to examine a pregnant woman's belly. A pregnant person receives an ultrasound as part of their prenatal care. (credit: "MIKI Yoshihito_Flickr"/Flickr)

Recall that when the zygote attaches to the wall of the mother s uterus, the placenta is formed. The placenta provides nourishment and oxygen to the fetus. Most everything the mother ingests, including food, liquid, and even medication, travels through the placenta to the fetus, hence the common phrase

eating for two. Anything the mother is exposed to in the environment affects the fetus; if the mother is exposed to something harmful, the child can show life-long effects.

A teratogen is any environmental agent biological, chemical, or physical that causes damage to the developing embryo or fetus. There are different types of teratogens. Alcohol and most drugs cross the placenta and affect the fetus. Alcohol is not safe to drink in any amount during pregnancy. Alcohol use during pregnancy has been found to be the leading preventable cause of intellectual disabilities in children in the United States (Maier & West, 2001). Excessive maternal drinking while pregnant can cause fetal alcohol spectrum disorders with life-long consequences for the child ranging in severity from minor to major (the accompanying source-figure description). Fetal alcohol spectrum disorders (FASD) are a collection of birth defects associated with heavy consumption of alcohol during pregnancy. Physically, children with FASD may have a small head size and abnormal facial features. Cognitively, these children may have poor judgment, poor impulse control, higher rates of ADHD, learning issues, and lower IQ scores. These developmental problems and delays persist into adulthood (Streissguth et al., 2004). Based on studies conducted on animals, it also has been suggested that a parent's alcohol consumption during pregnancy may predispose their child to like alcohol (Youngentob et al., 2007).

Source comparison table

Fetal Alcohol Syndrome Facial Features	
Facial Feature	Potential Effect of Fetal Alcohol Syndrome
Head size	Below-average head circumference
Eyes	Smaller than average eye opening, skin folds at corners of eyes
Nose	Low nasal bridge, short nose
Midface	Smaller than average midface size
Lip and philtrum	Thin upper lip, indistinct philtrum

Smoking is also considered a teratogen because nicotine travels through the placenta to the fetus. When the mother smokes, the developing baby experiences a reduction in blood oxygen levels. According to the Centers for Disease Control and Prevention (2013), smoking while pregnant can result in premature birth, low-birth-weight infants, stillbirth, and sudden infant death syndrome (SIDS).

Heroin, cocaine, methamphetamine, almost all prescription medicines, and most over-the counter medications are also considered teratogens. Babies born with a heroin addiction need heroin just like an adult addict. The child will need to be gradually weaned from the heroin under medical supervision; otherwise, the child could have seizures and die. Other teratogens include radiation, viruses such as HIV and herpes, and rubella (German measles). Women in the United States are much less likely to be afflicted with rubella because most women received childhood immunizations or vaccinations that protect the body from disease.

Each organ of the fetus develops during a specific period in the pregnancy, called the critical or sensitive period (the accompanying source-figure description). For example, research with primate models of FASD has demonstrated that the time during which a developing fetus is exposed to alcohol can dramatically affect the appearance of facial characteristics associated with fetal alcohol syndrome. Specifically, this research suggests that alcohol exposure that is limited to day 19 or 20 of gestation can lead to significant facial abnormalities in the offspring (Ashley, Magnuson, Omnell, & Clarren, 1999). Given regions of the brain also show sensitive periods during which they are most susceptible to the teratogenic effects of alcohol (Tran & Kelly, 2003).

What Do You Think?

As you now know, people who use drugs or alcohol during pregnancy can cause serious lifelong harm to their child. Some have advocated mandatory screenings for pregnant people and have a history of problematic drug use, and if they continue using, to arrest, prosecute, and incarcerate them (Figdor & Kaeser, 1998). This policy was tried in Charleston, South Carolina, as recently as 20 years ago. The policy was called the Interagency Policy on Management of Substance Abuse During Pregnancy, and had disastrous results.

The Interagency Policy applied to patients attending the obstetrics clinic at MUSC, which primarily serves patients who are indigent or on Medicaid. It did not apply to private obstetrical patients. The policy required patient education about the harmful effects of substance abuse during pregnancy. . . . [A] statement also warned patients that protection of unborn and newborn children from the harms of illegal drug abuse could involve the Charleston police, the Solicitor of the Ninth Judicial Court, and the Protective Services Division of the Department of Social Services (DSS). (Jos, Marshall, & Perlmutter, 1995, pp. 120 121)

This policy seemed to deter pregnant people from seeking prenatal care, deterred them from seeking other social services, and was applied solely to people with low incomes, resulting in lawsuits. The program was canceled after 5 years, during which 42 women were arrested. A federal agency later determined that the program involved human experimentation without the approval and oversight of an institutional review board (IRB). What were the flaws in the program and how would you correct them? What are the ethical implications of charging pregnant people with child abuse?

Infancy Through Childhood

The average newborn weighs approximately 7.5 pounds. Although small, newborns are not completely helpless because their reflexes and sensory capacities help them interact with the environment from the moment of birth. All healthy babies are born with newborn reflexes: inborn automatic responses to particular forms of stimulation. Reflexes help the newborn survive until it is capable of more complex behaviors these reflexes are crucial to survival. They are present in babies whose brains are developing normally and usually disappear around 4 5 months old. Let s take a look at some of these newborn reflexes. The rooting reflex is the newborn s response to anything that touches their cheek: When you stroke a baby s cheek, the baby naturally turns the head in that direction and begins to suck. The sucking reflex is the automatic, unlearned, sucking motions that infants do with their mouths. Several other interesting newborn reflexes can be observed. For instance, if you put your finger into a newborn s hand, you will witness the grasping reflex, in which a baby automatically grasps anything that touches the palms. The Moro reflex is the newborn s response to the sensation of falling. The baby spreads the arms, pulls them back in, and then (usually) cries. How do you think these reflexes promote survival in the first months of life?

What can young infants see, hear, and smell? Newborn infants sensory abilities are significant, but their senses are not yet fully developed. Many of a newborn s innate preferences facilitate interaction with caregivers and other humans. Although vision is their least developed sense, newborns already show a preference for faces. Babies who are just a few days old also prefer human voices, they will listen to voices longer than sounds that do not involve speech (Vouloumanos & Werker, 2004), and they seem to prefer their mother s voice over a stranger s voice (Mills & Melhuish, 1974). In an interesting experiment, 3-week-old babies were given pacifiers that played a recording of the infant s mother s voice and of a stranger s voice. When the infants heard their mother s voice, they sucked more strongly at the pacifier (Mills & Melhuish, 1974). Newborns also have a strong sense of smell. For instance, newborn babies can distinguish the smell of their own mother from that of others. In a study by MacFarlane (1978), 1-week-old babies who were being breastfed were placed between two gauze pads. One gauze pad was from the bra of a nursing mother who was a stranger, and the other gauze pad was from the bra of the infant s own mother. More than two-thirds of the week-old babies turned toward the gauze pad with their mother s scent.

Physical Development

In infancy, toddlerhood, and early childhood, the body s physical development is rapid (the accompanying source-figure description). On average, newborns weigh between 5 and 10 pounds, and a newborn s weight typically doubles in six months and triples in one year. By 2 years old the weight will have quadrupled, so we can expect that a 2 year old should weigh between 20 and 40 pounds. The average length of a newborn is 19.5 inches, increasing to 29.5 inches by 12 months and 34.4 inches by 2 years old (WHO Multicentre Growth Reference Study Group, 2006).

Source illustration description (image not reproduced): A collage of four photographs depicting babies is shown. From left to right they get progressively older. The far left photograph is a bundled up sleeping newborn. To the right is a picture of a toddler next to a toy giraffe. To the right is a baby blowing out a single candle. To the far right is a child on a swing set. Children experience rapid physical changes through infancy and early childhood. (credit "left": modification of work by Kerry Ceszyk; credit "middle-left": modification of work by Kristi Fausel; credit "middle-right": modification of work by "devinf"/Flickr; credit "right": modification of work by Rose Spielman)

During infancy and childhood, growth does not occur at a steady rate (Carel, Lahlou, Roger, & Chaussain, 2004). Growth slows between 4 and 6 years old: During this time children gain 5 7 pounds and grow about 2 3 inches per year. Once females reach 8 9 years old, their growth rate outpaces that of males due to a pubertal growth spurt. This growth spurt continues until around 12 years old, coinciding with the start of the menstrual cycle. By 10 years old, the average female weighs 88 pounds, and the average male weighs 85 pounds.

It was previously believed that we are born with all of the brain cells we will ever have. More recent research suggests that neurogenesis (the formation of neurons) can continue through adulthood. However, the vast majority of neural connections and pathways occur during the first few years of a child s life (National Institute of Neurological Disorders and Stroke, 2019). This period of rapid neural growth is called blooming. Neural pathways continue to develop through puberty. The blooming period of neural growth is then followed by a period of pruning, where neural connections are reduced. It is thought that pruning causes the brain to function more efficiently, allowing for mastery of more complex skills (Hutchinson, 2011). Blooming occurs during the first few years of life, and pruning continues through childhood and into adolescence in various areas of the brain.

The size of our brains increases rapidly. For example, the brain of a 2-year-old is 55% of its adult size, and by 6 years old the brain is about 90% of its adult size (Tanner, 1978). During early childhood (ages 3 6), the frontal lobes grow rapidly. Recalling our discussion of the 4 lobes of the brain earlier in this book, the frontal lobes are associated with planning, reasoning, memory, and impulse control. Therefore, by the time children reach school age, they are developmentally capable of controlling their attention and behavior. Through the elementary school years, the frontal, temporal, occipital, and parietal lobes all grow in size. The brain growth spurts experienced in childhood tend to follow Piaget s sequence of cognitive development, so that significant changes in neural functioning account for cognitive advances (Kolb & Whishaw, 2009; Overman, Bachevalier, Turner, & Peuster, 1992).

Motor development occurs in an orderly sequence as infants move from reflexive reactions (e.g., sucking and rooting) to more advanced motor functioning. For instance, babies first learn to hold their heads up, then to sit with assistance, and then to sit unassisted, followed later by crawling and then walking.

Motor skills refer to our ability to move our bodies and manipulate objects. Fine motor skills focus on the muscles in our fingers, toes, and eyes, and enable coordination of small actions (e.g., grasping a toy, writing with a pencil, and using a spoon). Gross motor skills focus on large muscle groups that control our arms and legs and involve larger movements (e.g., balancing, running, and jumping).

As motor skills develop, there are certain developmental milestones that young children should achieve (the accompanying source-figure description). For each milestone there is an average age, as well as a range of ages in which the milestone should be reached. An example of a developmental milestone is sitting. On average, most babies sit alone at 7 months old. Sitting involves both coordination and muscle strength, and 90% of babies achieve this milestone between 5 and 9 months old. In another example,

babies on average are able to hold up their head at 6 weeks old, and 90% of babies achieve this between 3 weeks and 4 months old. If a baby is not holding up their head by 4 months old, they are showing a delay. If the child is displaying delays on several milestones, that is reason for concern, and the parent or caregiver should discuss this with the child s pediatrician. Some developmental delays can be identified and addressed through early intervention.

Source comparison table

Developmental Milestones, Ages 2 5 Years				
Age (years)	Physical	Personal/Social	Language	Cognitive
2	Kicks a ball; walks up and down stairs	Plays alongside other children; copies adults	Points to objects when named; puts 2 4 words together in a sentence	Sorts shapes and colors; follows 2-step instructions
3	Climbs and runs; pedals tricycle	Takes turns; expresses many emotions; dresses self	Names familiar things; uses pronouns	Plays make believe; works toys with parts (levers, handles)
4	Catches balls; uses scissors	Prefers social play to solo play; knows likes and interests	Knows songs and rhymes by memory	Names colors and numbers; begins writing letters
5	Hops and swings; uses fork and spoon	Distinguishes real from pretend; likes to please friends	Speaks clearly; uses full sentences	Counts to 10 or higher; prints some letters and copies basic shapes

Cognitive Development

In addition to rapid physical growth, young children also exhibit significant development of their cognitive abilities. Piaget thought that children s ability to understand objects such as learning that a rattle makes a noise when shaken was a cognitive skill that develops slowly as a child matures and interacts with the environment. Today, developmental psychologists think Piaget was incorrect. Researchers have found that even very young children understand objects and how they work long before they have experience with those objects (Baillargeon, 1987; Baillargeon, Li, Gertner, & Wu, 2011). For example, children as young as 3 months old demonstrated knowledge of the properties of objects that they had only viewed and did not have prior experience with them. In one study, 3-month-old infants were shown a truck rolling down a track and behind a screen. The box, which appeared solid but was actually hollow, was placed next to the track. The truck rolled past the box as would be expected. Then the box was placed on the track to block the path of the truck. When the truck was rolled down the track this time, it continued unimpeded. The infants spent significantly more time looking at this impossible event (the accompanying source-figure description). Baillargeon (1987) concluded that they knew solid objects cannot pass through each other. Baillargeon s findings suggest that very young children have an understanding of objects and how they work, which Piaget (1954) would have said is beyond their cognitive abilities due to their limited experiences in the world.

Source illustration description (image not reproduced): Image A shows a toy truck coasting along a track unobstructed. Image B shows a toy truck coasting along a track with a box in the background. Image C shows a truck coasting along a track and going through what appears to be an obstruction. In Baillargeon s study, infants observed a truck (a) roll down an unobstructed track, (b) roll down an unobstructed track with an obstruction (box) beside it, and (c) roll down and pass through what appeared to be an obstruction.

Just as there are physical milestones that we expect children to reach, there are also cognitive milestones. It is helpful to be aware of these milestones as children gain new abilities to think, problem solve, and communicate. For example, infants shake their head no around 6 9 months, and they respond to verbal requests to do things like wave bye-bye or blow a kiss around 9 12 months. Remember Piaget s ideas about object permanence? We can expect children to grasp the concept that objects continue to exist even when they are not in sight by around 8 months old. Because toddlers (i.e., 12 24 months old) have mastered object permanence, they enjoy games like hide and seek, and they realize that when someone leaves the room they will come back (Loop, 2013). Toddlers also point to pictures in books and look in appropriate places when you ask them to find objects.

Preschool-age children (i.e., 3 5 years old) also make steady progress in cognitive development. Not only can they count, name colors, and tell you their name and age, but they can also make some decisions on their own, such as choosing an outfit to wear. Preschool-age children understand basic time concepts and sequencing (e.g., before and after), and they can predict what will happen next in a story. They also begin to enjoy the use of humor in stories. Because they can think symbolically, they enjoy pretend play and inventing elaborate characters and scenarios. One of the most common examples of their cognitive growth is their blossoming curiosity. Preschool-age children love to ask Why?

An important cognitive change occurs in children this age. Recall that Piaget described 2 3 year olds as egocentric, meaning that they do not have an awareness of others points of view. Between 3 and 5 years old, children come to understand that people have thoughts, feelings, and beliefs that are different from their own. This is known as theory-of-mind (TOM). Children can use this skill to tease others, persuade their parents to purchase a candy bar, or understand why a sibling might be angry. When children develop TOM, they can recognize that others have false beliefs (Dennett, 1987; Callaghan et al., 2005).

Cognitive skills continue to expand in middle and late childhood (6 11 years old). Thought processes become more logical and organized when dealing with concrete information (the accompanying source-figure description). Children at this age understand concepts such as the past, present, and future, giving them the ability to plan and work toward goals. Additionally, they can process complex ideas such as addition and subtraction and cause-and-effect relationships. However, children s attention spans tend to be very limited until they are around 11 years old. After that point, it begins to improve through adulthood.

Source illustration description (image not reproduced): A photograph of children playing baseball is shown. Five children are in the picture, two on one team, and three on the other. Because they understand luck and fairness, children in middle and late childhood (6 11 years old) are able to follow rules for games. (credit: Edwin Martinez)

One well-researched aspect of cognitive development is language acquisition. As mentioned earlier, the order in which children learn language structures is consistent across children and cultures (Hatch, 1983). You ve also learned that some psychological researchers have proposed that children possess a biological predisposition for language acquisition.

Starting before birth, babies begin to develop language and communication skills. At birth, babies apparently recognize their mother s voice and can discriminate between the language(s) spoken by their mothers and foreign languages, and they show preferences for faces that are moving in synchrony with audible language (Blossom & Morgan, 2006; Pickens, 1994; Spelke & Cortelyou, 1981).

Children communicate information through gesturing long before they speak, and there is some evidence that gesture usage predicts subsequent language development (Iverson & Goldin-Meadow, 2005). In terms of producing spoken language, babies begin to coo almost immediately. Cooing is a one-syllable combination of a consonant and a vowel sound (e.g., coo or ba). Interestingly, babies replicate sounds from their own languages. A baby whose parents speak French will coo in a different tone than a baby whose parents speak Spanish or Urdu. After cooing, the baby starts to babble. Babbling begins with repeating a syllable, such as ma-ma, da-da, or ba-ba. When babies are about 12

months old, we expect them to say their first word for meaning, and to start combining words for meaning at about 18 months.

At about 2 years old, a toddler uses between 50 and 200 words; by 3 years old they have a vocabulary of up to 1,000 words and can speak in sentences. During the early childhood years, children's vocabulary increases at a rapid pace. This is sometimes referred to as the vocabulary spurt and has been claimed to involve an expansion in vocabulary at a rate of 10 20 new words per week. Recent research may indicate that while some children experience these spurts, it is far from universal (as discussed in Ganger & Brent, 2004). It has been estimated that, 5 year olds understand about 6,000 words, speak 2,000 words, and can define words and question their meanings. They can rhyme and name the days of the week. Seven year olds speak fluently and use slang and clich s (Stork & Widdowson, 1974).

What accounts for such dramatic language learning by children? Behaviorist B. F. Skinner thought that we learn language in response to reinforcement or feedback, such as through parental approval or through being understood. For example, when a two-year-old child asks for juice, he might say, me juice, to which his mother might respond by giving him a cup of apple juice. Noam Chomsky (1957) criticized Skinner s theory and proposed that we are all born with an innate capacity to learn language. Chomsky called this mechanism a language acquisition device (LAD). Who is correct? Both Chomsky and Skinner are right. Remember that we are a product of both nature and nurture. Researchers now believe that language acquisition is partially inborn and partially learned through our interactions with our linguistic environment (Gleitman & Newport, 1995; Stork & Widdowson, 1974).

Attachment

Psychosocial development occurs as children form relationships, interact with others, and understand and manage their feelings. In social and emotional development, forming healthy attachments is very important and is the major social milestone of infancy. Attachment is a long-standing connection or bond with others. Developmental psychologists are interested in how infants reach this milestone. They ask such questions as: How do parent and infant attachment bonds form? How does neglect affect these bonds? What accounts for children s attachment differences?

Researchers Harry Harlow, John Bowlby, and Mary Ainsworth conducted studies designed to answer these questions. In the 1950s, Harlow conducted a series of experiments on monkeys. He separated newborn monkeys from their mothers. Each monkey was presented with two surrogate mothers. One surrogate monkey was made out of wire mesh, and she could dispense milk. The other monkey was softer and made from cloth: This monkey did not dispense milk. Research shows that the monkeys preferred the soft, cuddly cloth monkey, even though she did not provide any nourishment. The baby monkeys spent their time clinging to the cloth monkey and only went to the wire monkey when they needed to be fed. Prior to this study, the medical and scientific communities generally thought that babies become attached to the people who provide their nourishment. However, Harlow (1958) concluded that there was more to the mother-child bond than nourishment. Feelings of comfort and security are the critical components to maternal-infant bonding, which leads to healthy psychosocial development.

Building on the work of Harlow and others, John Bowlby developed the concept of attachment theory. He defined attachment as the affectional bond or tie that an infant forms with the mother (Bowlby, 1969). An infant must form this bond with a primary caregiver in order to have normal social and emotional development. In addition, Bowlby proposed that this attachment bond is very powerful and continues throughout life. He used the concept of secure base to define a healthy attachment between parent and child (1988). A secure base is a parental presence that gives the child a sense of safety as he explores his surroundings. Bowlby said that two things are needed for a healthy attachment: The caregiver must be responsive to the child s physical, social, and emotional needs; and the caregiver and child must engage in mutually enjoyable interactions (Bowlby, 1969) (the accompanying source-figure description).

Source illustration description (image not reproduced): A man holds a baby in the air. Mutually enjoyable interactions promote the parent-infant bond. (credit: "balouriarajesh_Pixabay"/Pixabay)

While Bowlby thought attachment was an all-or-nothing process, Mary Ainsworth s (1970) research showed otherwise. Ainsworth wanted to know if children differ in the ways they bond, and if so, why. To find the answers, she used the Strange Situation procedure to study attachment between mothers and their infants (1970). In the Strange Situation, the mother (or primary caregiver) and the infant (age 12-18 months) are placed in a room together. There are toys in the room, and the caregiver and child spend some time alone in the room. After the child has had time to explore her surroundings, a stranger enters the room. The mother then leaves her baby with the stranger. After a few minutes, she returns to comfort her child.

Based on how the infants/toddlers responded to the separation and reunion, Ainsworth identified three types of parent-child attachments: secure, avoidant, and resistant (Ainsworth & Bell, 1970). A fourth style, known as disorganized attachment, was later described (Main & Solomon, 1990). The most common type of attachment also considered the healthiest is called secure attachment (the accompanying source-figure description). In this type of attachment, the toddler prefers his parent over a stranger. The attachment figure is used as a secure base to explore the environment and is sought out in times of stress. Securely attached children were distressed when their caregivers left the room in the Strange Situation experiment, but when their caregivers returned, the securely attached children were happy to see them. Securely attached children have caregivers who are sensitive and responsive to their needs.

Source illustration description (image not reproduced): A photograph shows a person squatting down next to a small child who is standing up. In secure attachment, the parent provides a secure base for the toddler, allowing him to securely explore his environment. (credit: Kerry Ceszyk)

With avoidant attachment, the child is unresponsive to the parent, does not use the parent as a secure base, and does not care if the parent leaves. The toddler reacts to the parent the same way she reacts to a stranger. When the parent does return, the child is slow to show a positive reaction. Ainsworth theorized that these children were most likely to have a caregiver who was insensitive and inattentive to their needs (Ainsworth, Blehar, Waters, & Wall, 1978).

In cases of resistant attachment, children tend to show clingy behavior, but then they reject the attachment figure s attempts to interact with them (Ainsworth & Bell, 1970). These children do not explore the toys in the room, as they are too fearful. During separation in the Strange Situation, they became extremely disturbed and angry with the parent. When the parent returns, the children are difficult to comfort. Resistant attachment is the result of the caregivers inconsistent level of response to their child.

Finally, children with disorganized attachment behaved oddly in the Strange Situation. They freeze, run around the room in an erratic manner, or try to run away when the caregiver returns (Main & Solomon, 1990). This type of attachment is seen most often in kids who have been abused. Research has shown that abuse disrupts a child s ability to regulate their emotions.

While Ainsworth s research has found support in subsequent studies, it has also met criticism. Some researchers have pointed out that a child s temperament may have a strong influence on attachment (Gervai, 2009; Harris, 2009), and others have noted that attachment varies from culture to culture, a factor not accounted for in Ainsworth s research (Rothbaum, Weisz, Pott, Miyake, & Morelli, 2000; van Ijzendoorn & Sagi-Schwartz, 2008).

Self-Concept

Just as attachment is the main psychosocial milestone of infancy, the primary psychosocial milestone of childhood is the development of a positive sense of self. How does self-awareness develop? Infants don t have a self-concept, which is an understanding of who they are. If you place a baby in front of a mirror, they will reach out to touch their image, thinking it is another baby. However, by about 18 months a toddler will recognize that the person in the mirror is themself. How do we know this? In a well-known experiment, a researcher placed a red dot of paint on children s noses before putting them in front of a mirror (Amsterdam, 1972). Commonly known as the mirror test, this behavior is demonstrated by humans and a few other species and is considered evidence of self-recognition (Archer, 1992). At 18 months old they would touch their own noses when they saw the paint, surprised to see a spot on their faces. By 24 36 months old children can name and/or point to themselves in pictures, clearly indicating self-recognition.

Children from 2 4 years old display a great increase in social behavior once they have established a self-concept. They enjoy playing with other children, but they have difficulty sharing their possessions. Also, through play children explore and come to understand their gender roles and can label themselves as a girl or boy (Chick, Heilman-Houser, & Hunter, 2002). By 4 years old, children can cooperate with other children, share when asked, and separate from parents with little anxiety. Children at this age also exhibit autonomy, initiate tasks, and carry out plans. Success in these areas contributes to a positive sense of self. Once children reach 6 years old, they can identify themselves in terms of group memberships: I m a first grader! School-age children compare themselves to their peers and discover that they are competent in some areas and less so in others (recall Erikson s task of industry versus inferiority). At this age, children recognize their own personality traits as well as some other traits they would like to have. For example, 10-year-old Layla says, I m kind of shy. I wish I could be more talkative like my friend Alexa.

Development of a positive self-concept is important to healthy development. Children with a positive self-concept tend to be more confident, do better in school, act more independently, and are more willing to try new activities (Maccoby, 1980; Ferrer & Fugate, 2003). Formation of a positive self-concept begins in Erikson s toddlerhood stage, when children establish autonomy and become confident in their abilities. Development of self-concept continues in elementary school, when children compare themselves to others. When the comparison is favorable, children feel a sense of competence and are motivated to work harder and accomplish more. Self-concept is re-evaluated in Erikson s adolescence stage, as teens form an identity. They internalize the messages they have received regarding their strengths and weaknesses, keeping some messages and rejecting others. Adolescents who have achieved identity formation are capable of contributing positively to society (Erikson, 1968).

Dig Deeper

Kenneth and Mamie Clark were pioneering psychologists responsible for the first psychological study used in a Supreme Court case. Their research with African American children and doll choices was used to highlight the harmful effects of segregation and provided support for the Browns and the NAACP in their lawsuit against the Board of Education. The finding that African American children were more likely to choose a White doll over a Black doll, in both northern and southern states, led them to theorize that the children did not have a healthy concept of themselves (Clark & Clark, 1950).

The Clarks research differed from that of Inez Beverly Prosser, who also studied African American children in segregated and integrated schools in Cincinnati. Parents could choose either environment for their children during the 1930s. She found, among other factors, that the self-concept of children at segregated schools was more positive versus those in integrated schools, partly due to teachers low expectations. Prosser also noted that the child s personality should be considered when choosing a segregated school or an integrated school (Benjamin, Henry, & McMahon, 2005).

Later researchers suggested that African American children choosing a doll that did not look like them was not an indication of their self-esteem or their self-image. For instance, Rogers and Meltzoff (2017) found that gender identity was more important than race in their study of diverse children whose average age was about 10 years old. Thus, for children that young, the meaning of race is an evolving process, as opposed to adolescents search for

identity. The ethnic minority children in the study did view racial identity as important, compared to their White counterparts.

For teenagers who are members of ethnic minority groups, racial/ethnic/cultural identity can be paramount, depending on the family s processes. Racial socialization involves teaching them the positive aspects of their in-group, usually by caregivers. Most of the students in a study by Neblett, Smalls, Ford, Nguyen, and Sellers (2009) reported having received such messages but a few received no racial socialization messages. They found that these messages played a role in how they felt about their in-group.

Some theories have been developed to explain the behaviors of ethnic minority youth. One such theory is the Phenomenological Variant of Ecological Systems Theory (PVEST), put forth by Margaret Beale Spencer. It is a merging of phenomenology and Bronfenbrenner s ecological systems theory. A phenomenological approach is based on how a person makes meaning of their experiences. For example, young African American boys have different experiences in educational settings compared to African American girls. Consequently, the meaning they assign to those experiences differs. Bronfenbrenner s ecological systems theory suggests that development occurs based on interactions among environments such as school, family, and community (Bronfenbrenner, 1977).

The research that Spencer, Dupree, and Hartmann (1997) conducted with African American adolescent boys and girls was explained by PVEST. They found that negative learning attitudes were predicted by unpopularity with peers for girls and boys. Additionally, for boys, more stress predicted a less negative attitude toward learning, possibly due to focus on the school environment instead of on personal issues. This occurred along with perceiving that teachers had positive expectations of African American boys. The researchers surmised that PVEST accounted for how others perceptions and their subsequent attitudes were related and worked both ways.

What can parents do to nurture a healthy self-concept? Diana Baumrind (1971, 1991) thinks parenting style may be a factor. The way we parent is an important factor in a child s socioemotional growth. Baumrind developed and refined a theory describing four parenting styles: authoritative, authoritarian, permissive, and uninvolved. With the authoritative style, the parent gives reasonable demands and consistent limits, expresses warmth and affection, and listens to the child s point of view. Parents set rules and explain the reasons behind them. They are also flexible and willing to make exceptions to the rules in certain cases for example, temporarily relaxing bedtime rules to allow for a nighttime swim during a family vacation. Of the four parenting styles, the authoritative style is the one that is most encouraged in modern American society. American children raised by authoritative parents tend to have high self-esteem and social skills. However, effective parenting styles vary as a function of culture and, as Small (1999) points out, the authoritative style is not necessarily preferred or appropriate in all cultures.

In authoritarian style, the parent places high value on conformity and obedience. The parents are often strict, tightly monitor their children, and express little warmth. In contrast to the authoritative style, authoritarian parents probably would not relax bedtime rules during a vacation because they consider the rules to be set, and they expect obedience. This style can create anxious, withdrawn, and unhappy kids. However, it is important to point out that authoritarian parenting is as beneficial as the authoritative style in some ethnic groups (Russell, Crockett, & Chao, 2010). For instance, first-generation Chinese American children raised by authoritarian parents did just as well in school as their peers who were raised by authoritative parents (Russell et al., 2010).

For parents who employ the permissive style of parenting, the kids run the show and anything goes. Permissive parents make few demands and rarely use punishment. They tend to be very nurturing and loving, and may play the role of friend rather than parent. In terms of our example of vacation bedtimes, permissive parents might not have bedtime rules at all instead they allow the child to choose their bedtime whether on vacation or not. Not surprisingly, children raised by permissive parents tend to lack self-discipline, and the permissive parenting style is negatively associated with grades (Dornbusch, Ritter, Leiderman, Roberts, & Fraleigh, 1987). The permissive style may also contribute to other risky behaviors such as alcohol abuse (Bahr & Hoffman, 2010), risky sexual behavior especially among female children (Donenberg, Wilson, Emerson, & Bryant, 2002), and increased display of disruptive behaviors by male children (Parent et al., 2011). However, there are some positive outcomes associated with children raised by permissive parents. They tend to have higher self-esteem, better social skills, and report lower levels of depression (Darling, 1999).

With the uninvolved style of parenting, the parents are indifferent, uninvolved, and sometimes referred to as neglectful. They don t respond to the child s needs and make relatively few demands. This could be because of severe depression or substance abuse, or other factors such as the parents extreme focus on work. These parents may provide for the child s basic needs, but little else. The children raised in this parenting style are usually emotionally withdrawn, fearful, anxious, perform poorly in school, and are at an increased risk of substance abuse (Darling, 1999).

As you can see, parenting styles influence childhood adjustment, but could a child s temperament likewise influence parenting? Temperament refers to innate traits that influence how one thinks, behaves, and reacts with the environment. Children with easy temperaments demonstrate positive emotions, adapt well to change, and are capable of regulating their emotions. Conversely, children with difficult temperaments demonstrate negative emotions and have difficulty adapting to change and regulating their emotions. Difficult children are much more likely to challenge parents, teachers, and other caregivers (Thomas, 1984). Therefore, it s possible that easy children (i.e., social, adaptable, and easy to soothe) tend to elicit warm and responsive parenting, while demanding, irritable, withdrawn children evoke irritation in their parents or cause their parents to withdraw (Sanson & Rothbart, 1995).

Everyday Connection

According to the American Academy of Pediatrics (2007), unstructured play is an integral part of a child s development. It builds creativity, problem solving skills, and social relationships. Play also allows children to develop a theory-of-mind as they imaginatively take on the perspective of others.

Outdoor play allows children the opportunity to directly experience and sense the world around them. While doing so, they may collect objects that they come across and develop lifelong interests and hobbies. They also benefit from increased exercise, and engaging in outdoor play can actually increase how much they enjoy physical activity. This helps support the development of a healthy heart and brain. Unfortunately, research suggests that today s children are engaging in less and less outdoor play (Clements, 2004). Perhaps, it is no surprise to learn that lowered levels of physical activity in conjunction with easy access to calorie-dense foods with little nutritional value are contributing to alarming levels of childhood obesity (Karnik & Kanekar, 2012).

Despite the adverse consequences associated with reduced play, some children are over scheduled and have little free time to engage in unstructured play. In addition, some schools have taken away recess time for children in a push for students to do better on standardized tests, and many schools commonly use loss of recess as a form of punishment. Do you agree with these practices? Why or why not?

Adolescence

Adolescence is a socially constructed concept. In pre-industrial society, children were considered adults when they reached physical maturity, but today we have an extended time between childhood and adulthood called adolescence. Adolescence is the period of development that begins at puberty and ends at emerging adulthood, which is discussed later. In the United States, adolescence is seen as a time to develop independence from parents while remaining connected to them (the accompanying source-figure description). The typical age range of adolescence is from 12 to 18 years, and this stage of development also has some predictable physical, cognitive, and psychosocial milestones.

Source illustration description (image not reproduced): A photograph captures four people midair as they hold hands and jump. Peers are a primary influence on our development in adolescence. (credit: "manseok_Pixabay"/ Pixabay)

Physical Development

As noted above, adolescence begins with puberty. While the sequence of physical changes in puberty is predictable, the onset and pace of puberty vary widely. Several physical changes occur during puberty, such as adrenarche and gonadarche, the maturing of the adrenal glands and sex glands, respectively. Also during this time, primary and secondary sexual characteristics develop and mature. Primary sexual characteristics are organs specifically needed for reproduction, like the uterus and ovaries in females and testes in males. Secondary sexual characteristics are physical signs of sexual maturation that do not directly involve sex organs, such as development of breasts and hips in females, and development of facial hair and a deepened voice in males. Females experience menarche, the beginning of menstrual periods, usually around 12 13 years old, and males experience spermarche, the first ejaculation, around 13 14 years old.

During puberty, people experience a rapid increase in height (i.e., growth spurt). For females this begins between 8 and 13 years old, with adult height reached between 10 and 16 years old. Males begin their growth spurt slightly later, usually between 10 and 16 years old, and reach their adult height between 13 and 17 years old. Both nature (i.e., genes) and nurture (e.g., nutrition, medications, and medical conditions) can influence height.

Because rates of physical development vary so widely among teenagers, puberty can be a source of pride or embarrassment. Early maturing boys tend to be stronger, taller, and more athletic than their later maturing peers. They are usually more popular, confident, and independent, but they are also at a greater risk for substance abuse and early sexual activity (Flannery, Rowe, & Gulley, 1993; Kaltiala-Heino, Rimpela, Rissanen, & Rantanen, 2001). Early maturing girls may be teased or overtly admired, which can cause them to feel self-conscious about their developing bodies. These girls are at a higher risk for depression, substance abuse, and eating disorders (Ge, Conger, & Elder, 2001; Graber, Lewinsohn, Seeley, & Brooks-Gunn, 1997; Striegel-Moore & Cachelin, 1999). Late blooming boys and girls (i.e., they develop more slowly than their peers) may feel self-conscious about their lack of physical development. Negative feelings are particularly a problem for late maturing boys, who are at a higher risk for depression and conflict with parents (Graber et al., 1997) and more likely to be bullied (Pollack & Shuster, 2000).

The adolescent brain also remains under development. Up until puberty, brain cells continue to bloom in the frontal region. Adolescents engage in increased risk-taking behaviors and emotional outbursts possibly because the frontal lobes of their brains are still developing (the accompanying source-figure description). Recall that this area is responsible for judgment, impulse control, and planning, and it is still maturing into early adulthood (Casey, Tottenham, Liston, & Durston, 2005).

Source illustration description (image not reproduced): An illustration of a brain is shown with the frontal lobe labeled. Brain growth continues into the early 20s. The development of the frontal lobe, in particular, is important during this stage.

Cognitive Development

More complex thinking abilities emerge during adolescence. Some researchers suggest this is due to increases in processing speed and efficiency rather than as the result of an increase in mental capacity in other words, due to improvements in existing skills rather than development of new ones (Bjorkland, 1987; Case, 1985). During adolescence, teenagers move beyond concrete thinking and become capable of abstract thought. Recall that Piaget refers to this stage as formal operational thought. Teen thinking is also characterized by the ability to consider multiple points of view, imagine hypothetical situations, debate ideas and opinions (e.g., politics, religion, and justice), and form new ideas (the accompanying source-figure description). In addition, it s not uncommon for adolescents to question authority or challenge established societal norms.

Cognitive empathy, also known as theory-of-mind (which we discussed earlier with regard to egocentrism), relates to the ability to take the perspective of others and feel concern for others (Shamay-Tsoory, Tomer, & Aharon-Peretz, 2005). Cognitive empathy begins to increase in adolescence

and is an important component of social problem solving and conflict avoidance. According to one longitudinal study, levels of cognitive empathy begin rising in girls around 13 years old, and around 15 years old in boys (Van der Graaff et al., 2013). Teens who reported having supportive fathers with whom they could discuss their worries were found to be better able to take the perspective of others (Miklikowska, Duriez, & Soenens, 2011).

Source illustration description (image not reproduced): A picture shows four people gathered around a table attempting to figure out a problem together. Teenage thinking is characterized by the ability to reason logically and solve hypothetical problems such as how to design, plan, and build a structure. (credit: U.S. Army RDECOM)

Psychosocial Development

Adolescents continue to refine their sense of self as they relate to others. Erikson referred to the task of the adolescent as one of identity versus role confusion. Thus, in Erikson s view, an adolescent s main questions are Who am I? and Who do I want to be? Some adolescents adopt the values and roles that their parents expect for them. Other teens develop identities that are in opposition to their parents but align with a peer group. This is common as peer relationships become a central focus in adolescents lives.

As adolescents work to form their identities, they pull away from their parents, and the peer group becomes very important (Shanahan, McHale, Osgood, & Crouter, 2007). Despite spending less time with their parents, most teens report positive feelings toward them (Moore, Guzman, Hair, Lippman, & Garrett, 2004). Warm and healthy parent-child relationships have been associated with positive child outcomes, such as better grades and fewer school behavior problems, in the United States as well as in other countries (Hair et al., 2005).

It appears that most teens don t experience adolescent storm and stress to the degree once famously suggested by G. Stanley Hall, a pioneer in the study of adolescent development. Only small numbers of teens have major conflicts with their parents (Steinberg & Morris, 2001), and most disagreements are minor. For example, in a study of over 1,800 parents of adolescents from various cultural and ethnic groups, Barber (1994) found that conflicts occurred over day-to-day issues such as homework, money, curfews, clothing, chores, and friends. These types of arguments tend to decrease as teens develop (Galambos & Almeida, 1992). There is emerging research on the adolescent brain. Galvan, Hare, Voss, Glover and Casey (2007) examined its role in risk-taking behavior. They used fMRI to assess the readings relationship to risk-taking, risk perception, and impulsivity. The researchers found that there was no correlation between brain activity in the neural reward center and impulsivity and risk perception. However, activity in that part of the brain was correlated to risk taking. In other words, risk-taking adolescents experienced brain activity in the reward center. The idea that adolescents, however, are more impulsive than other demographics was challenged in their research, which included children and adults.

Emerging Adulthood

The next stage of development is emerging adulthood. This is a relatively newly defined period of lifespan development spanning from 18 years old to the mid-20s, characterized as an in-between time where identity exploration is focused on work and love.

When does a person become an adult? There are many ways to answer this question. In the United States, you are legally considered an adult at 18 years old. But other definitions of adulthood vary widely; in sociology, for example, a person may be considered an adult when they become self-supporting, choose a career, get married, or start a family. The ages at which we achieve these milestones vary from person to person as well as from culture to culture. For example, in the African country of Malawi, 15-year-old Njemile was married at 14 years old and had her first child at 15 years old. In her culture she is considered an adult. Children in Malawi take on adult responsibilities such as marriage and work (e.g., carrying water, tending babies, and working fields) as early as 10 years old. In stark contrast, independence in Western cultures is taking longer and longer, effectively delaying the onset of adult life.

What factors are leading to these changes regarding financial and familial independence? It seems that emerging adulthood is a product of both Western culture and our current times (Arnett, 2000). People in higher income and more industrialized countries are living longer, allowing the freedom to take an extra decade to start a career and family. Changes in the workforce also play a role. For example, 50 years ago, a young adult with a high school diploma could immediately enter the work force and climb the economic ladder. That is no longer the case. Bachelor s and even graduate degrees are required more and more often even for entry-level jobs (Arnett, 2000). In addition, many students are taking longer (five or six years) to complete a college degree as a result of working and going to school at the same time. After graduation, many young adults return to the family home because they have difficulty finding a job. The term Boomerang Generation describes recent college graduates for whom lack of adequate employment upon college graduation often leads to a return to the parental home (Davidson, 2014). Changing cultural expectations may be the most important reason for the delay in entering adult roles. Young people are spending more time exploring their options, so they are delaying marriage and work as they change majors and jobs multiple times, putting them on a much later timetable than their parents (Arnett, 2000).

Adulthood

Adulthood begins around 20 years old and has three distinct stages: early, middle, and late. Each stage brings its own set of rewards and challenges.

Physical Development

By the time we reach early adulthood (20 to early 40s), our physical maturation is complete, although our height and weight may increase slightly. In young adulthood, our physical abilities are at their peak, including muscle strength, reaction time, sensory abilities, and cardiac functioning. Most professional athletes are at the top of their game during this stage. Many women have children in the young adulthood years, so they may see additional weight gain and breast changes.

Middle adulthood extends from the 40s to the 60s (the accompanying source-figure description). Physical decline is gradual. The skin loses some elasticity, and wrinkles are among the first signs of aging. Visual acuity decreases during this time. Women experience a gradual decline in fertility as they approach the onset of menopause, the end of the menstrual cycle, around 50 years old. Both men and women tend to gain weight: in the abdominal area for men and in the hips and thighs for women. Hair begins to thin and turn gray.

Source illustration description (image not reproduced): A picture shows a person in a harness ascending a climbing wall. Physical declines of middle and late adulthood can be minimized with proper exercise, nutrition, and an active lifestyle. (credit: modification of work by Peter Stevens)

Late adulthood is considered to extend from the 60s on. This is the last stage of physical change. The skin continues to lose elasticity, reaction time slows further, and muscle strength diminishes. Smell, taste, hearing, and vision, so sharp in our twenties, decline significantly. The brain may also no longer function at optimal levels, leading to problems like memory loss, dementia, and Alzheimer s disease in later years.

Cognitive Development

Because we spend so many years in adulthood (more than any other stage), cognitive changes are numerous. In fact, research suggests that adult cognitive development is a complex, ever changing process that may be even more active than cognitive development in infancy and early childhood (Fischer, Yan, & Stewart, 2003).

Unlike our physical abilities, which peak in our mid-20s and then begin a slow decline, our cognitive abilities remain steady throughout early and middle adulthood. Our crystallized intelligence (information, skills, and strategies we have gathered through a lifetime of experience) tends to hold steady as we age it may even improve. For example, adults show relatively stable to increasing scores

on intelligence tests until their mid-30s to mid-50s (Bayley & Oden, 1955). However, in late adulthood we begin to experience a decline in another area of our cognitive abilities fluid intelligence (information processing abilities, reasoning, and memory). These processes become slower. How can we delay the onset of cognitive decline? Mental and physical activity seems to play a part (the accompanying source-figure description). Research has found adults who engage in mentally and physically stimulating activities experience less cognitive decline and have a reduced incidence of mild cognitive impairment and dementia (Hertzog, Kramer, Wilson, & Lindenberger, 2009; Larson et al., 2006; Podewils et al., 2005).

Source illustration description (image not reproduced): A picture shows three people at a table leaning over a board game. Cognitive activities such as playing mahjong, chess, or other games, can keep you mentally fit. The same is true for solo pastimes like reading and completing crossword puzzles. (credit: Philippe Put)

Researchers have examined the aging brain by comparing it to brain functioning in younger people. Forstmann and colleagues (2011) compared elderly participants to younger participants, who in the study were asked to report the direction of movement of a set of dots. They were given feedback regarding speed and accuracy. The researchers found that older participants made more errors and were slower due to degeneration of corticostriatal connections. In other words, the decreased ability typically assigned to elderly people may be due to circumstances in the brain beyond their control. Interestingly, other researchers have found similarities in spatial representations when comparing children aged 6 7 to those over the age of 80. Ruggiero, D Errico, and Iachini (2016) reported that this is due to neurodegeneration in older adults and immature neurology in young children.

Many elderly people experience dementia, changes in the brain that negatively affect cognition. Alzheimer s disease is one type of dementia, initially studied by medical researcher Solomon Carter Fuller. Alzheimer s disease has a genetic basis. Plaques in the brain are due to cell death, which then causes those affected with the disease severe forgetfulness. A person can forget how to walk, talk, and eventually eat. The disease can be mitigated by assessing environmental factors (exposure to lead, iron, and zinc increase risk) and nutritional factors (the Mediterranean diet lowers risk) (Arora, Mittal, & Kakkar, 2015). Although there is no cure, there is hope. Cognitive rehabilitation can offset mild cognitive impairment, as it can evolve into dementia. Garcia-Betances, Jimenez-Mixco, Arredondo, and Cabrera-Umpierrez (2015) examined the use of virtual reality as a possible cognitive rehabilitative method. They suggested that virtual reality technology should involve daily living activities, memory, and language, among other considerations.

Psychosocial Development

There are many theories about the social and emotional aspects of aging. Some aspects of healthy aging include activities, social connectedness, and the role of a person s culture. According to many theorists, including George Vaillant (2002), who studied and analyzed over 50 years of data, we need to have and continue to find meaning throughout our lives. For those in early and middle adulthood, meaning is found through work (Sterns & Huyck, 2001) and family life (Markus, Ryff, Curan, & Palmersheim, 2004). These areas relate to the tasks that Erikson referred to as generativity and intimacy. As mentioned previously, adults tend to define themselves by what they do their careers. Earnings peak during this time, yet job satisfaction is more closely tied to work that involves contact with other people, is interesting, provides opportunities for advancement, and allows some independence (Mohr & Zoghi, 2006) than it is to salary (Iyengar, Wells, & Schwartz, 2006). How might being unemployed or being in a dead-end job challenge adult well-being?

Positive relationships with significant others in our adult years have been found to contribute to a state of well-being (Ryff & Singer, 2009). Most adults in the United States identify themselves through their relationships with family particularly with spouses, children, and parents (Markus et al., 2004). While raising children can be stressful, especially when they are young, research suggests that parents reap the rewards down the road, as adult children tend to have a positive effect on parental well-being (Umberson, Pudrovska, & Reczek, 2010). Having a stable marriage has also been found to contribute to well-being throughout adulthood (Vaillant, 2002).

Another aspect of positive aging is believed to be social connectedness and social support. As we get older, socioemotional selectivity theory suggests that our social support and friendships dwindle in number, but remain as close, if not more close than in our earlier years (Carstensen, 1992) (the accompanying source-figure description).

Source illustration description (image not reproduced): Four people are sitting on a bench looking off in the same direction. Social support is important as we age. (credit: Gabriel Rocha)

Summary

At conception the egg and sperm cell are united to form a zygote, which will begin to divide rapidly. This marks the beginning of the first stage of prenatal development (germinal stage), which lasts about two weeks. Then the zygote implants itself into the lining of the uterus, marking the beginning of the second stage of prenatal development (embryonic stage), which lasts about six weeks. The embryo begins to develop body and organ structures, and the neural tube forms, which will later become the brain and spinal cord. The third phase of prenatal development (fetal stage) begins at 9 weeks and lasts until birth. The body, brain, and organs grow rapidly during this stage. During all stages of pregnancy it is important that the parent receive prenatal care to reduce health risks to themself and to the developing baby.

Newborn infants weigh about 7.5 pounds. Doctors assess a newborn s reflexes, such as the sucking, rooting, and Moro reflexes. Our physical, cognitive, and psychosocial skills grow and change as we move through developmental stages from infancy through late adulthood. Attachment in infancy is a critical component of healthy development. Parenting styles have been found to have an effect on childhood outcomes of well-being. The transition from adolescence to adulthood can be challenging due to the timing of puberty, and due to the extended amount of time spent in emerging adulthood. Although physical decline begins in middle adulthood, cognitive decline does not begin until later. Activities that keep the body and mind active can help maintain good physical and cognitive health as we age. Social supports through family and friends remain important as we age.

Review Questions

OpenStax learning exercise

Q. Which of the following is the correct order of prenatal development?

A. D

OpenStax learning exercise

Q. The time during fetal growth when specific parts or organs develop is known as _____.

A. A

OpenStax learning exercise

Q. What begins as a single-cell structure that is created when a sperm and egg merge at conception?

A. C

OpenStax learning exercise

Q. Using scissors to cut out paper shapes is an example of _____.

A. B

OpenStax learning exercise

Q. The child uses the parent as a base from which to explore her world in which attachment style?

A. A

OpenStax learning exercise

Q. The frontal lobes become fully developed _____.

A. D

Critical Thinking Questions

OpenStax learning exercise

Q. What are some known teratogens, and what kind of damage can they do to the developing fetus?

A. Alcohol is a teratogen. Excessive drinking can cause mental retardation in children. The child can also have a small head and abnormal facial features, which are characteristic of fetal alcohol syndrome (FAS). Another teratogen is nicotine. Smoking while pregnant can lead to low-birth weight, premature birth, stillbirth, and SIDS.

OpenStax learning exercise

Q. What is prenatal care and why is it important?

A. Prenatal care is medical care during pregnancy that monitors the health of both the mother and fetus. It s important to receive prenatal care because it can reduce complications to the mother and fetus during pregnancy.

OpenStax learning exercise

Q. Describe what happens in the embryonic stage of development. Describe what happens in the fetal stage of development.

A. In the embryonic stage, basic structures of the embryo start to develop into areas that will become the head, chest, and abdomen. The heart begins to beat and organs form and begin to function. The neural tube forms along the back of the embryo, developing into the spinal cord and brain. In the fetal stage, the brain and body continue to develop. Fingers and toes develop along with hearing, and internal organs form.

OpenStax learning exercise

Q. What makes a personal quality part of someone s personality?

A. The particular quality or trait must be part of an enduring behavior pattern, so that it is a consistent or predictable quality.

OpenStax learning exercise

Q. Describe some of the newborn reflexes. How might they promote survival?

A. The sucking reflex is the automatic, unlearned sucking motions that infants do with their mouths. It may help promote survival because this action helps the baby take in nourishment. The rooting reflex is the newborn s response to anything that touches her cheek. When you stroke a baby s cheek she will naturally turn her head that way and begin to suck. This may aid survival because it helps the newborn locate a source of food.

OpenStax learning exercise

Q. Compare and contrast the four parenting styles and describe the kinds of childhood outcomes we can expect with each.

A. With the authoritative style, children are given reasonable demands and consistent limits, warmth and affection are expressed, the parent listens to the child s point of view, and the child initiates positive standards. Children raised by authoritative parents tend to have high self-esteem and social skills. Another parenting style is authoritarian: The parent places a high value on conformity and obedience. The parents are often strict, tightly monitor their children, and express little warmth. This style can create anxious, withdrawn, and unhappy kids. The third parenting style is permissive: Parents make few demands, rarely use punishment, and give their children free rein. Children raised by permissive parents tend to lack self-discipline, which contributes to poor grades and alcohol abuse. However, they have higher self-esteem, better social skills, and lower levels of depression. The fourth style is the uninvolved parent: They are indifferent, uninvolved, and sometimes called neglectful. The children raised in this parenting style are usually emotionally withdrawn, fearful, anxious, perform poorly in school, and are at an increased risk of substance abuse.

OpenStax learning exercise

Q. What is emerging adulthood and what are some factors that have contributed to this new stage of development?

A. Emerging adulthood is a relatively new period of lifespan development from 18 years old to the mid-20s, characterized as a transitional time in which identity exploration focuses on work and love. According to Arnett, changing cultural expectations facilitate the delay to full adulthood. People are spending more time exploring their options, so they are delaying marriage and work as they change majors and jobs multiple times, putting them on a much later timetable than their parents.

Personal Application Questions

OpenStax learning exercise

Q. Which parenting style describes how you were raised? Provide an example or two to support your answer.

OpenStax learning exercise

Q. Would you describe your experience of puberty as one of pride or embarrassment? Why?

OpenStax learning exercise

Q. Your best friend is a smoker who just found out she is pregnant. What would you tell her about smoking and pregnancy?

OpenStax learning exercise

Q. Imagine you are a nurse working at a clinic that provides prenatal care for pregnant women. Your patient, Anna, has heard that it s a good idea to play music for her unborn baby, and she wants to know when her baby s hearing will develop. What will you tell her?

CHAPTER 06 / CAC.901

Genes, neurons, and developing systems

Lecture: Biology acts through relationships and time

Biological explanations can reduce blame, but they can also create a new kind of fatalism. A caregiver who hears that a difficulty is in the brain may conclude that nothing can be done. A counselor who hears that a behavior is genetic may stop examining the environment. Both reactions misunderstand developmental biology. Genes, neural systems, bodies, and environments act together over time. Biological does not mean unchangeable, and environmental does not mean easy to change.

A gene is not a miniature instruction for a complex behavior. Genes participate in biological processes through regulation and interaction with other genes and conditions. A developmental outcome usually reflects many influences, each contributing in ways that depend on context. It is therefore unsafe to infer a child s future from a single genetic association. It is equally unsafe to assume that a trait appearing in several family members must be genetic. Families also share environments, opportunities, stressors, and ways of interpreting experience.

Heritability is often misunderstood

Heritability is a population-level statistic concerning variation in a particular population and environment. It does not tell us what proportion of one child s behavior was caused by genes. It does not establish that a group difference is genetic. It does not show that an intervention cannot work. A highly heritable characteristic may still respond strongly to environmental change, and a low heritability estimate does not identify the best intervention.

Consider height as a familiar comparison. Genetic differences contribute to variation in height, yet nutrition and health matter. The example does not make psychological development identical to height; it shows why a statistic about variation is not a statement about an individual s fixed destiny. In counseling, the useful question is usually not whether biology or environment wins. It is which processes are relevant, which can be assessed, and which supports can improve functioning.

Gene environment correlation adds complexity. A child s interests may influence the activities adults provide. A child who enjoys reading may seek books, and a book-rich environment may further support reading. Evocative processes occur when a child s characteristics influence others responses. These patterns are not necessarily conscious or intentional. They help explain why separating person and context into independent boxes can be misleading.

Neurons and networks

Neurons communicate through electrical and chemical processes. Synapses allow signals to pass between cells, and networks support perception, movement, learning, and regulation. Counselors do not need to practice neurology, but they should understand enough to avoid exaggerated brain claims. A colorful brain image does not prove that a teaching method is effective. A reference to dopamine does not by itself explain a complex social problem.

Development involves changes in neural connections and coordination, not simply the addition of more brain material. Experience matters, but not every experience produces a dramatic or permanent neural effect. The claim that ordinary parenting must provide constant enrichment or a

window will close forever is not a responsible translation of developmental neuroscience. Children need safe relationships, adequate nutrition, opportunities to learn, rest, and protection from harm. These are stronger practical foundations than a promise to optimize every neural connection.

Figure 6.1 represents influences as a changing loop rather than a one-way arrow. A child s characteristics affect participation; participation changes experience; experience contributes to learning and biological adaptation. The diagram is a conceptual teaching model, not a complete map of causal mechanisms. Its value is that it discourages the question Is it biology or the environment? when the more useful question is How are they acting together here?

Stress biology and restraint

Stress-response systems help people respond to challenge. Activation can be useful in the short term. Problems may arise when demands are severe, prolonged, or unmanageable without adequate support. Counselors should avoid using stress biology as a universal explanation for every difficulty. Two children exposed to a similar event may differ in appraisal, prior experience, health, relationships, and current resources.

Supportive relationships can buffer stress, but they should not be described as a guarantee against harm. Nor should a child s distress be treated as evidence that caregivers failed to provide enough buffering. Structural conditions matter. A family may offer warmth while facing unsafe housing or inaccessible healthcare. An accurate formulation recognizes care and constraint at the same time.

Biological information can support referral and collaboration. Persistent changes in sleep, appetite, energy, attention, or functioning may warrant health assessment. A counselor should describe observations without diagnosing outside their competence. Medication decisions belong with appropriately qualified prescribers and the family s informed participation. The counselor can contribute developmental history, functional observations, and monitoring information within an agreed role.

Application: It runs in the family

A fictional caregiver says that several relatives were late talkers, so the child s communication difficulty must be inherited. The statement is relevant history, but it should not end assessment. The counselor can explore what late talking meant, whether hearing was checked, how the child communicates now, and what participation is difficult. A developmental history should include strengths as well as concerns.

The plan may include appropriate hearing assessment, observation across settings, and collaboration with professionals qualified to evaluate communication. Family history can inform the account without becoming a reason to wait indefinitely. At the same time, the counselor should avoid frightening the family with predictions that the assessment cannot support. A clear next step is more useful than a debate about whether genes are to blame.

The child s experience remains important. Communication difficulty may affect play, frustration, and belonging. Support should not depend on proving a single cause. Provide accessible communication, reduce avoidable pressure, and help adults respond to the child s intended meaning. These actions can be useful while assessment continues.

Doctoral seminar task

Find a popular statement that explains a child s behavior using a gene, a brain region, or a neurotransmitter. Rewrite it as a testable developmental question. Identify the level of evidence required, the difference between association and mechanism, and at least two alternative explanations. Then write a caregiver-friendly version that preserves uncertainty without sounding dismissive.

Finally, propose an intervention that could be helpful even while the biological account remains incomplete. Explain how you would monitor benefit and unintended effects. This task reinforces a central principle: biological knowledge should expand thoughtful care, not replace assessment or narrow a child s future.

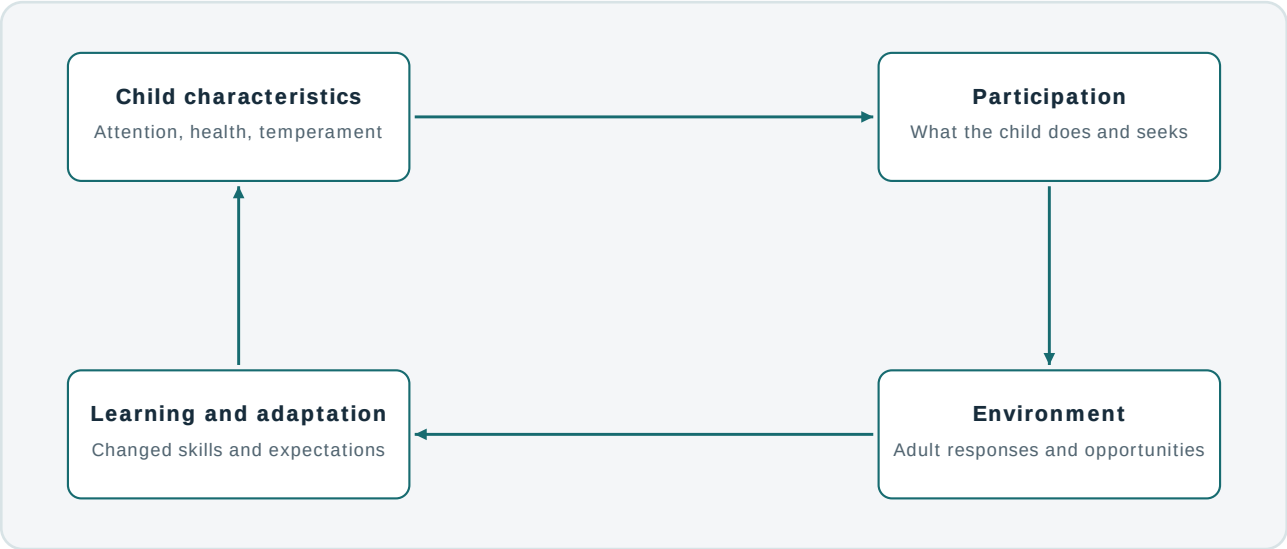


Figure 6.1. Development is a transaction over time. A child s characteristics and participation influence the environment, and the environment influences later learning and adaptation. Biological and social processes are connected rather than competing explanations. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

Human Genetics

OpenStax source module m82172 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82172/index.cnxml

Psychological researchers study genetics in order to better understand the biological factors that contribute to certain behaviors. While all humans share certain biological mechanisms, we are each unique. And while our bodies have many of the same parts brains and hormones and cells with genetic codes these are expressed in a wide variety of behaviors, thoughts, and reactions.

Why do two people infected by the same disease have different outcomes: one surviving and one succumbing to the ailment? How are genetic diseases passed through family lines? Are there genetic components to psychological disorders, such as depression or schizophrenia? To what extent might there be a psychological basis to health conditions such as childhood obesity?

To explore these questions, let s start by focusing on a specific genetic disorder, sickle cell anemia, and how it might manifest in two affected sisters. Sickle-cell anemia is a genetic condition in which red blood cells, which are normally round, take on a crescent-like shape (the accompanying source-figure description). The changed shape of these cells affects how they function: sickle-shaped cells can clog blood vessels and block blood flow, leading to high fever, severe pain, swelling, and tissue damage.

Source illustration description (image not reproduced): An illustration shows round and sickle-shaped blood cells. Normal blood cells travel freely through the blood vessels, while sickle-shaped cells form blockages preventing blood flow.

Many people with sickle-cell anemia and the particular genetic mutation that causes it die at an early age. While the notion of survival of the fittest may suggest that people with this disorder have a low survival rate and therefore the disorder will become less common, this is not the case. Despite the negative evolutionary effects associated with this genetic mutation, the sickle-cell gene remains relatively common among people of African descent. Why is this? The explanation is illustrated with the following scenario.

Imagine two young women Luwi and Sena sisters in rural Zambia, Africa. Luwi carries the gene for sickle-cell anemia; Sena does not carry the gene. Sickle-cell carriers have one copy of the sickle-cell gene but do not have full-blown sickle-cell anemia. They experience symptoms only if they are severely dehydrated or are deprived of oxygen (as in mountain climbing). Carriers are thought to be immune from malaria (an often deadly disease that is widespread in tropical climates) because changes in their blood chemistry and immune functioning prevent the malaria parasite from having its effects (Gong, Parikh, Rosenthal, & Greenhouse, 2013). However, full-blown sickle-cell anemia, with two copies of the sickle-cell gene, does not provide immunity to malaria.

While walking home from school, both sisters are bitten by mosquitoes carrying the malaria parasite. Luwi is protected against malaria because she carries the sickle-cell mutation. Sena, on the other hand, develops malaria and dies just two weeks later. Luwi survives and eventually has children, to whom she may pass on the sickle-cell mutation.

Malaria is rare in the United States, so the sickle-cell gene benefits nobody: the gene manifests primarily in minor health problems for carriers with one copy, or a severe full-blown disease with no health benefits for carriers with two copies. However, the situation is quite different in other parts of the world. In parts of Africa where malaria is prevalent, having the sickle-cell mutation does provide health benefits for carriers (protection from malaria).

The story of malaria fits with Charles Darwin's theory of evolution by natural selection (the accompanying source-figure description). In simple terms, the theory states that organisms that are better suited for their environment will survive and reproduce, while those that are poorly suited for their environment will die off. In our example, we can see that, as a carrier, Luwi s mutation is highly adaptive in her African homeland; however, if she resided in the United States (where malaria is rare), her mutation could prove costly with a high probability of the disease in her descendants and minor health problems of her own.

Source illustration description (image not reproduced): Image (a) is a painted portrait of Darwin. Image (b) is a sketch of lines that split apart into branched structures. (a) In 1859, Charles Darwin proposed his theory of evolution by natural selection in his book, On the Origin of Species. (b) The book contains just one illustration: this diagram that shows how species evolve over time through natural selection.

Dig Deeper

It s easy to get confused about two fields that study the interaction of genes and the environment, such as the fields of evolutionary psychology and behavioral genetics. How can we tell them apart?

In both fields, it is understood that genes not only code for particular traits, but also contribute to certain patterns of cognition and behavior. Evolutionary psychology focuses on how universal patterns of behavior and cognitive processes have evolved over time. Therefore, variations in cognition and behavior would make individuals more or less successful in reproducing and passing those genes on to their offspring. Evolutionary psychologists study a variety of psychological phenomena that may have evolved as adaptations, including fear response, food preferences, mate selection, and cooperative behaviors (Confer et al., 2010).

Whereas evolutionary psychologists focus on universal patterns that evolved over millions of years, behavioral geneticists study how individual differences arise, in the present, through the interaction of genes and the environment. When studying human behavior, behavioral geneticists often employ twin and adoption studies to research questions of interest. Twin studies compare the likelihood that a given behavioral trait is shared among identical and fraternal twins; adoption studies compare those rates among biologically related relatives and adopted relatives. Both approaches provide some insight into the relative importance of genes and environment for the expression of a given trait.

Genetic Variation

Genetic variation, the genetic difference between individuals, is what contributes to a species adaptation to its environment. In humans, genetic variation begins with an egg, about 100 million sperm, and fertilization. Roughly once per month, active ovaries release an egg from follicles. During the egg's journey from the ovary through the fallopian tubes, to the uterus, a sperm may fertilize the egg.

The egg and the sperm each contain 23 chromosomes. Chromosomes are long strings of genetic material known as deoxyribonucleic acid (DNA). DNA is a helix-shaped molecule made up of nucleotide base pairs. In each chromosome, sequences of DNA make up genes that control or partially control a number of visible characteristics, known as traits, such as eye color, hair color, and so on. A single gene may have multiple possible variations, or alleles. An allele is a specific version of a gene. So, a given gene may code for the trait of hair color, and the different alleles of that gene affect which hair color an individual has.

When a sperm and egg fuse, their 23 chromosomes combine to create a zygote with 46 chromosomes (23 pairs). Therefore, each parent contributes half the genetic information carried by the offspring; the resulting physical characteristics of the offspring (called the phenotype) are determined by the

interaction of genetic material supplied by the sperm and egg (called the genotype). A person s genotype is the genetic makeup of that individual. Phenotype, on the other hand, refers to the individual s inherited physical characteristics, which are a combination of genetic and environmental influences (the accompanying source-figure description).

Source illustration description (image not reproduced): Image (a) shows the helical structure of DNA. Image (b) shows a person s face. (a) Genotype refers to the genetic makeup of an individual based on the genetic material (DNA) inherited from one s genetic contributors. (b) Phenotype describes an individual s observable characteristics, such as hair color, skin color, height, and build. (credit a: modification of work by Caroline Davis; credit b: modification of work by Cory Zanker)

Note that, in genetics and reproduction, "parent" is often used to describe the individual organisms that contribute genetic material to offspring, usually in the form of gamete cells (sperm and egg). The concept of a genetic parent is distinct from social and legal concepts of parenthood, and may differ from those whom people consider their parents.

Most traits are controlled by multiple genes, but some traits are controlled by one gene. A characteristic like cleft chin, for example, is influenced by a single gene from each parent. In this example, we will call the gene for cleft chin B, and the gene for smooth chin b. Cleft chin is a dominant trait, which means that having the dominant allele either from one parent (Bb) or both parents (BB) will always result in the phenotype associated with the dominant allele. When someone has two copies of the same allele, they are said to be homozygous for that allele. When someone has a combination of alleles for a given gene, they are said to be heterozygous. For example, smooth chin is a recessive trait, which means that an individual will only display the smooth chin phenotype if they are homozygous for that recessive allele (bb).

Imagine that a person with a cleft chin mates with a person with a smooth chin. What type of chin will their offspring have? The answer to that depends on which alleles each parent carries. If the person with a cleft is homozygous for cleft chin (BB), their offspring will always have cleft chin. It gets a little more complicated, however, if the person is heterozygous for this gene (Bb). Since the other person has a smooth chin therefore homozygous for the recessive allele (bb) we can expect the offspring to have a 50% chance of having a cleft chin and a 50% chance of having a smooth chin (the accompanying source-figure description).

Source illustration description (image not reproduced): Image (a) is a Punnett square showing the four possible combinations (Bb, bb, Bb, bb) resulting from the pairing of a bb parent and a Bb parent. Image (b) is a close-up photograph showing a cleft chin. (a) A Punnett square is a tool used to predict how genes will interact in the production of offspring. The capital B represents the dominant allele, and the lowercase b represents the recessive allele. In the example of the cleft chin, where B is cleft chin (dominant allele), wherever a pair contains the dominant allele, B, you can expect a cleft chin phenotype. You can expect a smooth chin phenotype only when there are two copies of the recessive allele, bb. (b) A cleft chin, shown here, is an inherited trait.

In sickle cell anemia, heterozygous carriers (like Luwi from the example) can develop blood resistance to malaria infection while those who are homozygous (like Sena) have a potentially lethal blood disorder. Sickle-cell anemia is just one of many genetic disorders caused by the pairing of two recessive genes. For example, phenylketonuria (PKU) is a condition in which individuals lack an enzyme that normally converts harmful amino acids into harmless byproducts. If someone with this condition goes untreated, they will experience significant deficits in cognitive function, seizures, and an increased risk of various psychiatric disorders. Because PKU is a recessive trait, each parent must have at least one copy of the recessive allele in order to produce a child with the condition (the accompanying source-figure description).

So far, we have discussed traits that involve just one gene, but few human characteristics are controlled by a single gene. Most traits are polygenic: controlled by more than one gene. Height is one example of a polygenic trait, as are skin color and weight.

Source illustration description (image not reproduced): A Punnett square shows the four possible combinations (NN, Np, Np, pp) resulting from the pairing of two Np parents. In this Punnett square, N represents the normal allele, and p

represents the recessive allele that is associated with PKU. If two individuals mate who are both heterozygous for the allele associated with PKU, their offspring have a 25% chance of expressing the PKU phenotype.

Where do harmful genes that contribute to diseases like PKU come from? Gene mutations provide one source of harmful genes. A mutation is a sudden, permanent change in a gene. While many mutations can be harmful or lethal, once in a while, a mutation benefits an individual by giving that person an advantage over those who do not have the mutation. Recall that the theory of evolution asserts that individuals best adapted to their particular environments are more likely to reproduce and pass on their genes to future generations. In order for this process to occur, there must be competition more technically, there must be variability in genes (and resultant traits) that allow for variation in adaptability to the environment. If a population consisted of identical individuals, then any dramatic changes in the environment would affect everyone in the same way, and there would be no variation in selection. In contrast, diversity in genes and associated traits allows some individuals to perform slightly better than others when faced with environmental change. This creates a distinct advantage for individuals best suited for their environments in terms of successful reproduction and genetic transmission.

Dig Deeper

This chapter focuses on biology. Later in this course you will learn about social psychology and issues of race, prejudice, and discrimination. When we focus strictly on biology, race becomes a weak construct. After the sequencing of the human genome at the turn of the millennium, many scientists began to argue that race was not a useful variable in genetic research and that its continued use represents a potential source of confusion and harm. The racial categories that some believed to be helpful in studying genetic diversity in humans are largely irrelevant. A person's skin tone, eye color, and hair texture are functions of their genetic makeups, but there is actually more genetic variation within a given racial category than there is between racial categories. In some cases, focus on race has led to difficulties with misdiagnoses and/or under-diagnoses of diseases ranging from sickle cell anemia to cystic fibrosis. Some argue that we need to distinguish between ancestry and race and then focus on ancestry. This approach would facilitate greater understanding of human genetic diversity (Yudell, Roberts, DeSalle, & Tishkoff, 2016).

Gene-Environment Interactions

Genes do not exist in a vacuum. Although we are all biological organisms, we also exist in an environment that is incredibly important in determining not only when and how our genes express themselves, but also in what combination. Each of us represents a unique interaction between our genetic makeup and our environment; range of reaction is one way to describe this interaction. Range of reaction asserts that our genes set the boundaries within which we can operate, and our environment interacts with the genes to determine where in that range we will fall. For example, if an individual s genetic makeup predisposes them to high levels of intellectual potential and they are reared in a rich, stimulating environment, then they will be more likely to achieve full potential than if they were raised under conditions of significant deprivation. According to the concept of range of reaction, genes set definite limits on potential, and environment determines how much of that potential is achieved. Some disagree with this theory and argue that genes do not set a limit on a person s potential with reaction norms being determined by the environment. For example, when individuals experience neglect or abuse early in life, they are more likely to exhibit adverse psychological and/or physical conditions that can last throughout their lives. These conditions may develop as a function of the negative environmental experiences in individuals from dissimilar genetic backgrounds (Miguel, Pereira, Silveira, & Meaney, 2019; Short & Baram, 2019).

Another perspective on the interaction between genes and the environment is the concept of genetic environmental correlation. Stated simply, our genes influence our environment, and our environment influences the expression of our genes (the accompanying source-figure description). Not only do our genes and environment interact, as in range of reaction, but they also influence one another bidirectionally. For example, the child of an NBA player would probably be exposed to basketball from

an early age. Such exposure might allow the child to realize their full genetic, athletic potential. Thus, the parents genes, which the child shares, influence the child s environment, and that environment, in turn, is well suited to support the child s genetic potential.

Source illustration description (image not reproduced): Two jigsaw puzzle pieces are shown; one depicts images of houses, and the other depicts a helical DNA strand. Nature and nurture work together like complex pieces of a human puzzle. The interaction of our environment and genes makes us the individuals we are. (credit "puzzle": modification of work by Cory Zanker)

In another approach to gene-environment interactions, the field of epigenetics looks beyond the genotype itself and studies how the same genotype can be expressed in different ways. In other words, researchers study how the same genotype can lead to very different phenotypes. As mentioned earlier, gene expression is often influenced by environmental context in ways that are not entirely obvious. For instance, identical twins share the same genetic information (identical twins develop from a single fertilized egg that split, so the genetic material is exactly the same in each; in contrast, fraternal twins usually result from two different eggs fertilized by different sperm, so the genetic material varies as with non-twin siblings). But even with identical genes, there remains an incredible amount of variability in how gene expression can unfold over the course of each twin s life. Sometimes, one twin will develop a disease and the other will not. In one example, Aliya, an identical twin, died from cancer at age 7, but her twin, now 19 years old, has never had cancer. Although these individuals share an identical genotype, their phenotypes differ as a result of how that genetic information is expressed over time and through their unique environmental interactions. The epigenetic perspective is very different from range of reaction, because here the genotype is not fixed and limited.

Genes affect more than our physical characteristics. Indeed, scientists have found genetic linkages to a number of behavioral characteristics, ranging from basic personality traits to sexual orientation to spirituality (for examples, see Mustanski et al., 2005; Comings, Gonzales, Saucier, Johnson, & MacMurray, 2000). Genes are also associated with temperament and a number of psychological disorders, such as depression and schizophrenia. So while it is true that genes provide the biological blueprints for our cells, tissues, organs, and body, they also have a significant impact on our experiences and our behaviors.

Let s look at the following findings regarding schizophrenia in light of our three views of gene-environment interactions. Which view do you think best explains this evidence?

In a 2004 study by Tienari and colleagues, adoptees whose biological mothers had schizophrenia and who had been raised in a disturbed family environment were much more likely to develop schizophrenia or another psychotic disorder than were any of the other groups in the study:

- Of adoptees whose biological mothers had schizophrenia (high genetic risk) and who were raised in disturbed family environments, 36.8% were likely to develop schizophrenia.
- Of adoptees whose biological mothers had schizophrenia (high genetic risk) and who were raised in healthy family environments, 5.8% were likely to develop schizophrenia.
- Of adoptees with a low genetic risk (whose mothers did not have schizophrenia) and who were raised in disturbed family environments, 5.3% were likely to develop schizophrenia.
- Of adoptees with a low genetic risk (whose mothers did not have schizophrenia) and who were raised in healthy family environments, 4.8% were likely to develop schizophrenia.

The study shows that adoptees with high genetic risk were most likely to develop schizophrenia if they were raised in disturbed home environments. This research lends credibility to the notion that both genetic vulnerability and environmental stress are necessary for schizophrenia to develop, and that genes alone do not tell the full tale.

Summary

Genes are sequences of DNA that code for a particular trait. Different versions of a gene are called alleles sometimes alleles can be classified as dominant or recessive. A dominant allele always results in the dominant phenotype. In order to exhibit a recessive phenotype, an individual must be homozygous for the recessive allele. Genes affect both physical and psychological characteristics. Ultimately, how and when a gene is expressed, and what the outcome will be in terms of both physical and psychological characteristics is a function of the interaction between our genes and our environments.

Review Questions

OpenStax learning exercise

Q. A(n) _____ is a sudden, permanent change in a sequence of DNA.

A. D

OpenStax learning exercise

Q. _____ refers to a person s genetic makeup, while _____ refers to a person s physical characteristics.

A. B

OpenStax learning exercise

Q. _____ is the field of study that focuses on genes and their expression.

A. C

OpenStax learning exercise

Q. Humans have _____ pairs of chromosomes.

A. B

Critical Thinking Questions

OpenStax learning exercise

Q. The theory of evolution by natural selection requires variability of a given trait. Why is variability necessary and where does it come from?

A. Variability is essential for natural selection to work. If all individuals are the same on a given trait, there will be no relative difference in their reproductive success because everyone will be equally adapted to their environments on that trait. Mutations are one source of variability, but sexual reproduction is another important source of variation given that individuals inherit half of their genetic makeup from each of their parents.

Personal Application Questions

OpenStax learning exercise

Q. You share half of your genetic makeup with each of your parents, but you are no doubt very different from both of them. Spend a few minutes jotting down the similarities and differences between you and your parents. How do you think your unique environment and experiences have contributed to some of the differences you see?

Cells of the Nervous System

OpenStax source module m82173 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82173/index.cnxml

Psychologists striving to understand the human mind may study the nervous system. Learning how the body's cells and organs function can help us understand the biological basis of human psychology. The nervous system is composed of two basic cell types: glial cells (also known as glia) and neurons. Glial cells are traditionally thought to play a supportive role to neurons, both physically and metabolically. Glial cells provide scaffolding on which the nervous system is built, help neurons line up closely with each other to allow neuronal communication, provide insulation to neurons, transport nutrients and waste products, and mediate immune responses. For years, researchers believed that there were many more glial cells than neurons; however, more recent work from Suzanna Herculano-Houzel's laboratory has called this long-standing assumption into question and has provided important evidence that there may be a nearly 1:1 ratio of glia cells to neurons. This is important because it suggests that human brains are more similar to other primate brains than previously thought (Azevedo et al, 2009; Herculano-Houzel, 2012; Herculano-Houzel, 2009). Neurons, on the other hand, serve as interconnected information processors that are essential for all of the tasks of the nervous system. This section briefly describes the structure and function of neurons.

Neuron Structure

Neurons are the central building blocks of the nervous system, 100 billion strong at birth. Like all cells, neurons consist of several different parts, each serving a specialized function (the accompanying source-figure description). A neuron s outer surface is made up of a semipermeable membrane. This membrane allows smaller molecules and molecules without an electrical charge to pass through it, while stopping larger or highly charged molecules.

Source illustration description (image not reproduced): An illustration shows a neuron with labeled parts for the cell membrane, dendrite, cell body, axon, and terminal buttons. A myelin sheath covers part of the neuron. This illustration shows a prototypical neuron, which is being myelinated by a glial cell.

The nucleus of the neuron is located in the soma, or cell body. The soma has branching extensions known as dendrites. The neuron is a small information processor, and dendrites serve as input sites where signals are received from other neurons. These signals are transmitted electrically across the soma and down a major extension from the soma known as the axon, which ends at multiple terminal buttons. The terminal buttons contain synaptic vesicles that house neurotransmitters, the chemical messengers of the nervous system.

Axons range in length from a fraction of an inch to several feet. In some axons, glial cells form a fatty substance known as the myelin sheath, which coats the axon and acts as an insulator, increasing the speed at which the signal travels. The myelin sheath is not continuous and there are small gaps that occur down the length of the axon. These gaps in the myelin sheath are known as the Nodes of Ranvier. The myelin sheath is crucial for the normal operation of the neurons within the nervous system: the loss of the insulation it provides can be detrimental to normal function. To understand how this works, let s consider an example. PKU, a genetic disorder discussed earlier, causes a reduction in myelin and abnormalities in white matter cortical and subcortical structures. The disorder is associated with a variety of issues including severe cognitive deficits, exaggerated reflexes, and seizures (Anderson &

Leuzzi, 2010; Huttenlocher, 2000). Another disorder, multiple sclerosis (MS), an autoimmune disorder, involves a large-scale loss of the myelin sheath on axons throughout the nervous system. The resulting interference in the electrical signal prevents the quick transmittal of information by neurons and can lead to a number of symptoms, such as dizziness, fatigue, loss of motor control, and sexual dysfunction. While some treatments may help to modify the course of the disease and manage certain symptoms, there is currently no known cure for multiple sclerosis.

In healthy individuals, the neuronal signal moves rapidly down the axon to the terminal buttons, where synaptic vesicles release neurotransmitters into the synaptic cleft (the accompanying source-figure description). The synaptic cleft is a very small space between two neurons and is an important site where communication between neurons occurs. Once neurotransmitters are released into the synaptic cleft, they travel across it and bind with corresponding receptors on the dendrite of an adjacent neuron. Receptors, proteins on the cell surface where neurotransmitters attach, vary in shape, with different shapes matching different neurotransmitters.

How does a neurotransmitter know which receptor to bind to? The neurotransmitter and the receptor have what is referred to as a lock-and-key relationship specific neurotransmitters fit specific receptors similar to how a key fits a lock. The neurotransmitter binds to any receptor that it fits.

Source illustration description (image not reproduced): Image (a) shows the synaptic space between two neurons, with neurotransmitters being released into the synapse and attaching to receptors. Image (b) is a micrograph showing a spherical terminal button with part of the exterior removed, revealing a solid interior of small round parts. (a) The synaptic cleft is the space between the terminal button of one neuron and the dendrite of another neuron. (b) In this pseudo-colored image from a scanning electron microscope, a terminal button (green) has been opened to reveal the synaptic vesicles (orange and blue) inside. Each vesicle contains about 10,000 neurotransmitter molecules. (credit b: modification of work by Tina Carvalho, NIH-NIGMS; scale-bar data from Matt Russell)

Neuronal Communication

Now that we have learned about the basic structures of the neuron and the role that these structures play in neuronal communication, let s take a closer look at the signal itself how it moves through the neuron and then jumps to the next neuron, where the process is repeated.

We begin at the neuronal membrane. The neuron exists in a fluid environment it is surrounded by extracellular fluid and contains intracellular fluid (i.e., cytoplasm). The neuronal membrane keeps these two fluids separate a critical role because the electrical signal that passes through the neuron depends on the intra- and extracellular fluids being electrically different. This difference in charge across the membrane, called the membrane potential, provides energy for the signal.

The electrical charge of the fluids is caused by charged molecules (ions) dissolved in the fluid. The semipermeable nature of the neuronal membrane somewhat restricts the movement of these charged molecules, and, as a result, some of the charged particles tend to become more concentrated either inside or outside the cell.

Between signals, the neuron membrane s potential is held in a state of readiness, called the resting potential. Like a rubber band stretched out and waiting to spring into action, ions line up on either side of the cell membrane, ready to rush across the membrane when the neuron goes active and the membrane opens its gates. Ions in high-concentration areas are ready to move to low-concentration areas, and positive ions are ready to move to areas with a negative charge.

In the resting state, sodium (Na+) is at higher concentrations outside the cell, so it will tend to move into the cell. Potassium (K+), on the other hand, is more concentrated inside the cell, and will tend to move out of the cell (the accompanying source-figure description). In addition, the inside of the cell is slightly negatively charged compared to the outside, due to the activity of the sodium-potassium pump. This pump actively transports three sodium ions out of the cell for every two potassium ions in, creating a net negative charge inside the cell. This provides an additional force on sodium, causing it to move into the cell.

Source illustration description (image not reproduced): A close-up illustration depicts the difference in charges across the cell membrane, and shows how Na+ and K+ cells concentrate more closely near the membrane. At resting potential, Na+ (blue pentagons) is more highly concentrated outside the cell in the extracellular fluid (shown in blue), whereas K+ (purple squares) is more highly concentrated near the membrane in the cytoplasm or intracellular fluid. Other molecules, such as chloride ions (yellow circles) and negatively charged proteins (brown squares), help contribute to a positive net charge in the extracellular fluid and a negative net charge in the intracellular fluid.

From this resting potential state, the neuron receives a signal and its state changes abruptly (the accompanying source-figure description). When a neuron receives signals at the dendrites due to neurotransmitters from an adjacent neuron binding to its receptors small pores, or gates, open on the neuronal membrane, allowing Na+ ions, propelled by both charge and concentration differences, to move into the cell. With this influx of positive ions, the internal charge of the cell becomes more positive. If that charge reaches a certain level, called the threshold of excitation, the neuron becomes active and the action potential begins.

Many additional pores open, causing a massive influx of Na+ ions and a huge positive spike in the membrane potential, the peak action potential. At the peak of the spike, the sodium gates close and the potassium gates open. As positively charged potassium ions leave, the cell quickly begins repolarization. At first, it hyperpolarizes, becoming slightly more negative than the resting potential, and then it levels off, returning to the resting potential.

Source illustration description (image not reproduced): A graph shows the increase, peak, and decrease in membrane potential. The millivolts through the phases are approximately -70mV at resting potential, -55mV at threshold of excitation, 30mV at peak action potential, 5mV at repolarization, and -80mV at hyperpolarization. During the action potential, the electrical charge across the membrane changes dramatically.

This positive spike constitutes the action potential: the electrical signal that typically moves from the cell body down the axon to the axon terminals. The electrical signal moves down the axon with the impulses jumping in a leapfrog fashion between the Nodes of Ranvier. The Nodes of Ranvier are natural gaps in the myelin sheath. At each point, some of the sodium ions that enter the cell diffuse to the next section of the axon, raising the charge past the threshold of excitation and triggering a new influx of sodium ions. The action potential moves all the way down the axon in this fashion until reaching the terminal buttons.

The action potential is an all-or-none phenomenon. In simple terms, this means that an incoming signal from another neuron is either sufficient or insufficient to reach the threshold of excitation. There is no in-between, and there is no turning off an action potential once it starts. Think of it like sending an email or a text message. You can think about sending it all you want, but the message is not sent until you hit the send button. Furthermore, once you send the message, there is no stopping it.

Because it is all or none, the action potential is recreated, or propagated, at its full strength at every point along the axon. Much like the lit fuse of a firecracker, it does not fade away as it travels down the axon. It is this all-or-none property that explains the fact that your brain perceives an injury to a distant body part like your toe as equally painful as one to your nose.

As noted earlier, when the action potential arrives at the terminal button, the synaptic vesicles release their neurotransmitters into the synaptic cleft. The neurotransmitters travel across the synapse and bind to receptors on the dendrites of the adjacent neuron, and the process repeats itself in the new neuron (assuming the signal is sufficiently strong to trigger an action potential). Once the signal is delivered, excess neurotransmitters in the synaptic cleft drift away, are broken down into inactive fragments, or are reabsorbed in a process known as reuptake. Reuptake involves the neurotransmitter being pumped back into the neuron that released it, in order to clear the synapse (the accompanying source-figure description). Clearing the synapse serves both to provide a clear on and off state between signals and to regulate the production of neurotransmitter (full synaptic vesicles provide signals that no additional neurotransmitters need to be produced).

Source illustration description (image not reproduced): The synaptic space between two neurons is shown. Some neurotransmitters that have been released into the synapse are attaching to receptors while others undergo reuptake

into the axon terminal. Reuptake involves moving a neurotransmitter from the synapse back into the axon terminal from which it was released.

Neuronal communication is often referred to as an electrochemical event. The movement of the action potential down the length of the axon is an electrical event, and movement of the neurotransmitter across the synaptic space represents the chemical portion of the process. However, there are some specialized connections between neurons that are entirely electrical. In such cases, the neurons are said to communicate via an electrical synapse. In these cases, two neurons physically connect to one another via gap junctions, which allows the current from one cell to pass into the next. There are far fewer electrical synapses in the brain, but those that do exist are much faster than the chemical synapses that have been described above (Connors & Long, 2004).

Neurotransmitters and Drugs

There are several different types of neurotransmitters released by different neurons, and we can speak in broad terms about the kinds of functions associated with different neurotransmitters (the accompanying source-figure description). Much of what psychologists know about the functions of neurotransmitters comes from research on the effects of drugs in psychological disorders. Psychologists who take a biological perspective and focus on the physiological causes of behavior assert that psychological disorders like depression and schizophrenia are associated with imbalances in one or more neurotransmitter systems. In this perspective, psychotropic medications can help improve the symptoms associated with these disorders. Psychotropic medications are drugs that treat psychiatric symptoms by restoring neurotransmitter balance.

Source comparison table

Major Neurotransmitters and How They Affect Behavior		
Neurotransmitter	Involved in	Potential Effect on Behavior
Acetylcholine	Muscle action, memory	Increased arousal, enhanced cognition
Beta-endorphin	Pain, pleasure	Decreased anxiety, decreased tension
Dopamine	Mood, sleep, learning	Increased pleasure, suppressed appetite
Gamma-aminobutyric acid (GABA)	Brain function, sleep	Decreased anxiety, decreased tension
Glutamate	Memory, learning	Increased learning, enhanced memory
Norepinephrine	Heart, intestines, alertness	Increased arousal, suppressed appetite
Serotonin	Mood, sleep	Modulated mood, suppressed appetite

Psychoactive drugs can act as agonists or antagonists for a given neurotransmitter system. Agonists are chemicals that mimic a neurotransmitter at the receptor site. An antagonist, on the other hand, blocks or impedes the normal activity of a neurotransmitter at the receptor. Agonists and antagonists represent drugs that are prescribed to correct the specific neurotransmitter imbalances underlying a person s condition. For example, Parkinson's disease, a progressive nervous system disorder, is associated with low levels of dopamine. Therefore, a common treatment strategy for Parkinson's

disease involves using dopamine agonists, which mimic the effects of dopamine by binding to dopamine receptors.

Certain symptoms of schizophrenia are associated with overactive dopamine neurotransmission. The antipsychotics used to treat these symptoms are antagonists for dopamine they block dopamine s effects by binding its receptors without activating them. Thus, they prevent dopamine released by one neuron from signaling information to adjacent neurons.

In contrast to agonists and antagonists, which both operate by binding to receptor sites, reuptake inhibitors prevent unused neurotransmitters from being transported back to the neuron. This allows neurotransmitters to remain active in the synaptic cleft for longer durations, increasing their effectiveness. Depression, which has been consistently linked with reduced serotonin levels, is commonly treated with selective serotonin reuptake inhibitors (SSRIs). By preventing reuptake, SSRIs strengthen the effect of serotonin, giving it more time to interact with serotonin receptors on dendrites. Common SSRIs on the market today include Prozac, Paxil, and Zoloft. The drug LSD is structurally very similar to serotonin, and it affects the same neurons and receptors as serotonin. Psychotropic drugs are not instant solutions for people suffering from psychological disorders. Often, an individual must take a drug for several weeks before seeing improvement, and many psychoactive drugs have significant negative side effects. Furthermore, individuals vary dramatically in how they respond to the drugs. To improve chances for success, it is not uncommon for people receiving pharmacotherapy to undergo psychological and/or behavioral therapies as well. Some research suggests that combining drug therapy with other forms of therapy tends to be more effective than any one treatment alone (for one such example, see March et al., 2007).

Summary

Glia and neurons are the two cell types that make up the nervous system. While glia generally play supporting roles, the communication between neurons is fundamental to all of the functions associated with the nervous system. Neuronal communication is made possible by the neuron s specialized structures. The soma contains the cell nucleus, and the dendrites extend from the soma in tree-like branches. The axon is another major extension of the cell body; axons are often covered by a myelin sheath, which increases the speed of transmission of neural impulses. At the end of the axon are terminal buttons that contain synaptic vesicles filled with neurotransmitters.

Neuronal communication is an electrochemical event. The dendrites contain receptors for neurotransmitters released by nearby neurons. If the signals received from other neurons are sufficiently strong, an action potential will travel down the length of the axon to the terminal buttons, resulting in the release of neurotransmitters into the synaptic cleft. Action potentials operate on the all-or-none principle and involve the movement of Na+ and K+ across the neuronal membrane.

Different neurotransmitters are associated with different functions. Often, psychological disorders involve imbalances in a given neurotransmitter system. Therefore, psychotropic drugs are prescribed in an attempt to bring the neurotransmitters back into balance. Drugs can act either as agonists or as antagonists for a given neurotransmitter system.

Review Questions

OpenStax learning exercise

Q. The _____ receive(s) incoming signals from other neurons.

A. D

OpenStax learning exercise

Q. A(n) _____ facilitates or mimics the activity of a given neurotransmitter system.

A. C

OpenStax learning exercise

Q. Multiple sclerosis involves a breakdown of the _____.

A. B

OpenStax learning exercise

Q. An action potential involves Na+ moving _____ the cell and K+ moving _____ the cell.

A. A

Critical Thinking Questions

OpenStax learning exercise

Q. Cocaine has two effects on synaptic transmission: it impairs reuptake of dopamine and it causes more dopamine to be released into the synaptic cleft. Would cocaine be classified as an agonist or antagonist? Why?

A. As a reuptake inhibitor, cocaine blocks the normal activity of dopamine at the receptor. The function causing more dopamine to be released into the synapse is agonist because it mimics and strengthens the effect of the neurotransmitter. Cocaine would be considered an agonist because by preventing the enzymatic degradation of the neurotransmitters, it increases the potential time that these neurotransmitters might be active in the synapse.

OpenStax learning exercise

Q. Drugs such as lidocaine and novocaine act as Na+ channel blockers. In other words, they prevent sodium from moving across the neuronal membrane. Why would this particular effect make these drugs such effective local anesthetics?

A. The action potential is initiated by an influx of Na+ into the neuron. If this process is prevented, then no action potentials in neurons in a given area will occur. Therefore, any painful stimuli would not result in action potentials carrying that information to the brain.

Personal Application Queation

OpenStax learning exercise

Q. Have you or someone you know ever been prescribed a psychotropic medication? If so, what side effects were associated with the treatment?

CHAPTER 07 / CAC.901

Brain development and biological context

Lecture: The developing brain is not a courtroom verdict

Adolescent brain development is often used to explain behavior in public discussion. Such explanations can encourage patience and support, but they can also become oversimplified stories. The teenage brain is unfinished may sound compassionate, yet it can imply that adolescents are incapable of meaningful judgment. The frontal lobe explains risk may sound scientific, yet it can hide the importance of context, learning, relationships, and individual differences.

The brain develops through coordinated changes across systems. Different regions and networks contribute to attention, planning, emotional salience, memory, and control. No single region is the sole home of responsibility or maturity. Development is not a switch that turns on at a legal birthday. It is also not identical for every capacity. A young person may show strong reasoning in a familiar domain while needing support in a new or emotionally charged situation.

Avoiding the maturity shortcut

Counselors sometimes use neuroscience to argue that adolescents should receive understanding rather than punishment. The humane intention is valuable, but the reasoning needs care. Rights and dignity should not depend on proving that a young person s brain is immature. Adults also act under emotion, social pressure, and incomplete information. Developmental knowledge should inform proportionate responses without creating a hierarchy in which only adults deserve agency.

In practice, ask about the conditions surrounding a decision. Was there time to think? Were alternatives visible? Was the young person trying to gain peer approval, avoid humiliation, or respond to a threat? What did they understand at the time? What supports helped them reflect afterward? These questions produce a more useful account than a general statement about brain age.

Sensitive periods and practical expectations

Some learning processes are influenced by timing, but sensitive periods are domain-specific and vary in strength. Counselors should distinguish evidence about sensory or language development from broad claims about personality, counseling, or academic achievement. The existence of developmental timing does not justify abandoning older children or adolescents. It supports providing appropriate opportunities as early as feasible while recognizing continuing capacity for change.

Sleep is a practical example of biology interacting with social conditions. Adolescents may experience changes in sleep timing while school schedules, homework, work responsibilities, and digital activity shape actual sleep opportunity. A counselor should not reduce the issue to poor choices or assume that removing a phone will solve every sleep problem. Assess the full routine, including safety, overcrowding, caregiving, anxiety, and health.

Figure 7.1 maps several influences on a young person s capacity to use a skill in the moment. It separates having a capacity from accessing it under particular conditions. This distinction matters in school and family conflicts. A young person who can plan during a calm session may still need

environmental prompts during a stressful morning. Support can scaffold performance while competence develops.

Biological claims in assessment

A brain scan is not a routine explanation for every counseling concern. Clinical imaging has specific medical purposes, and research differences between groups do not automatically provide diagnostic information about an individual. Counselors should be cautious about services that promise to identify a child s emotional future from a scan or simple biological test. Ask about validity, intended use, independent evidence, and potential harms.

Neurodevelopmental assessment relies on developmental history, observations, appropriate measures, and functioning across settings, with medical collaboration when indicated. A diagnosis may help explain a pattern and guide support, but it does not describe the whole child. Two children with the same diagnosis may have different strengths, communication needs, learning environments, and goals. Intervention should be individualized rather than derived mechanically from a label.

Application: a risky decision after school

A fictional fifteen-year-old, Kunle, joins a dangerous activity after classmates challenge him. Later, he explains the risks clearly and regrets his decision. An adult concludes that the later explanation must be insincere because he understood the danger. A developmental account recognizes that knowledge can be available yet difficult to use under social pressure. This does not erase accountability; it changes what an effective response should include.

The counselor can examine the moment before the decision, the role of status and belonging, and possible exit strategies. Role-play can make refusal more practicable, but it should not assume that a young person can safely resist every group. Environmental changes and trusted adult support may also be needed. The plan should address both skill and situation.

When discussing the event with caregivers, avoid a lecture about an immature brain. Describe the specific conditions, ask about patterns, and involve Kunle in designing a safer response. A developmentally informed conversation preserves responsibility while creating the conditions for better decisions.

Doctoral seminar task

Write two explanations of an adolescent decision: one using a broad brain-maturity claim and one using a contextual developmental account. Compare their predictions. Which suggests a broader range of interventions? Which better accounts for variation across situations? Which risks stereotyping?

Then design a brief monitoring plan for a support intended to improve decision-making in daily life. Include the young person s goals, observable indicators, and a review point. Do not use a neuroscience claim as a substitute for evidence that the support is helping. The standard is practical usefulness with accurate explanation.

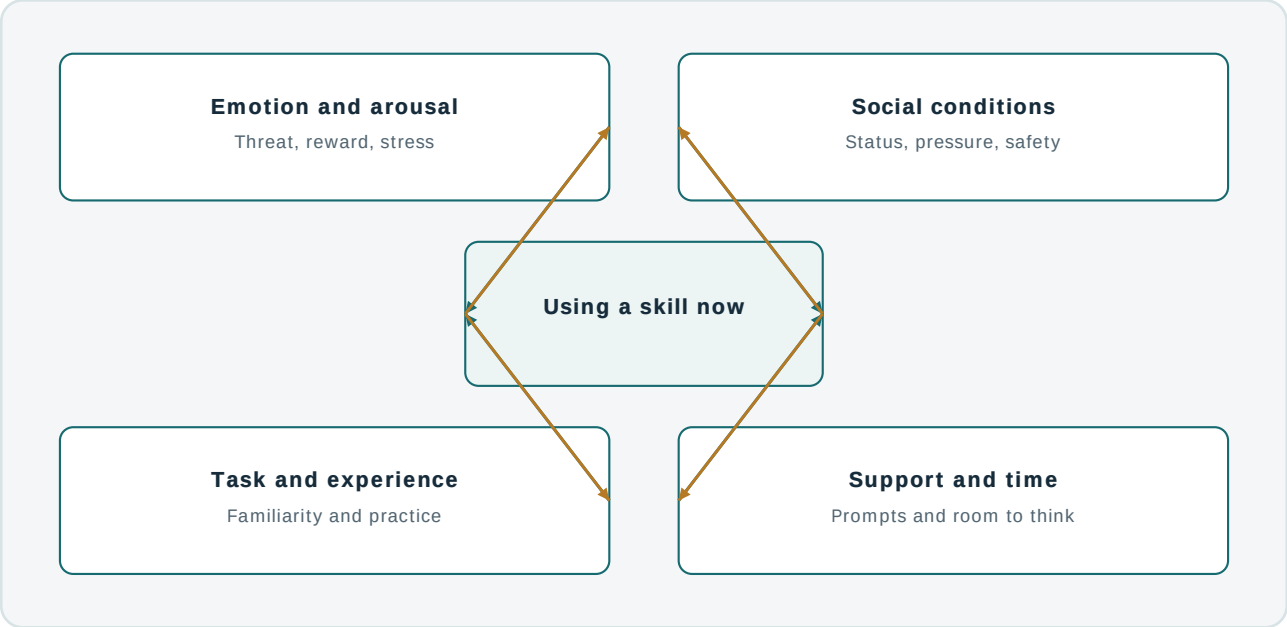


Figure 7.1. Capacity is used under conditions. A young person s ability to use a skill depends on more than whether the skill exists. Emotion, peer pressure, fatigue, familiarity, and support can affect performance in the moment. This is a conceptual map, not an anatomical brain model. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

The Brain and Spinal Cord

OpenStax source module m82175 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82175/index.cnxml

The brain is a remarkably complex organ comprised of billions of interconnected neurons and glia. It is a bilateral, or two-sided, structure that can be separated into distinct lobes. Each lobe is associated with certain types of functions, but, ultimately, all of the areas of the brain interact with one another to provide the foundation for our thoughts and behaviors. In this section, we discuss the overall organization of the brain and the functions associated with different brain areas, beginning with what can be seen as an extension of the brain, the spinal cord.

The Spinal Cord

It can be said that the spinal cord is what connects the brain to the outside world. Because of it, the brain can act. The spinal cord is like a relay station, but a very smart one. It not only routes messages to and from the brain, but it also has its own system of automatic processes, called reflexes.

The top of the spinal cord is a bundle of nerves that merges with the brain stem, where the basic processes of life are controlled, such as breathing and digestion. In the opposite direction, the spinal cord ends just below the ribs contrary to what we might expect, it does not extend all the way to the base of the spine.

The spinal cord is functionally organized in 30 segments, corresponding with the vertebrae. Each segment is connected to a specific part of the body through the peripheral nervous system. Nerves branch out from the spine at each vertebra. Sensory nerves bring messages in; motor nerves send messages out to the muscles and organs. Messages travel to and from the brain through every segment.

Some sensory messages are immediately acted on by the spinal cord, without any input from the brain. Withdrawal from a hot object and the knee jerk are two examples. When a sensory message meets certain parameters, the spinal cord initiates an automatic reflex. The signal passes from the sensory nerve to a simple processing center, which initiates a motor command. Seconds are saved, because messages don t have to go the brain, be processed, and get sent back. In matters of survival, the spinal reflexes allow the body to react extraordinarily fast.

The spinal cord is protected by bony vertebrae and cushioned in cerebrospinal fluid, but injuries still occur. When the spinal cord is damaged in a particular segment, all lower segments are cut off from the brain, causing paralysis. Therefore, the lower on the spine damage occurs, the fewer functions an injured individual will lose.

Neuroplasticity

Bob Woodruff, a reporter for ABC, suffered a traumatic brain injury after a bomb exploded next to the vehicle he was in while covering a news story in Iraq. As a consequence of these injuries, Woodruff experienced many cognitive deficits including difficulties with memory and language. However, over time and with the aid of intensive amounts of cognitive and speech therapy, Woodruff has shown an incredible recovery of function (Fernandez, 2008, October 16).

One of the factors that made this recovery possible was neuroplasticity. Neuroplasticity refers to how the nervous system can change and adapt. Neuroplasticity can occur in a variety of ways including personal experiences, developmental processes, or, as in Woodruff's case, in response to some sort of damage or injury that has occurred. Neuroplasticity can involve creation of new synapses, pruning of synapses that are no longer used, changes in glial cells, and even the birth of new neurons. Because of neuroplasticity, our brains are constantly changing and adapting, and while our nervous system is most plastic when we are very young, as Woodruff's case suggests, it is still capable of remarkable changes later in life.

The Two Hemispheres

The surface of the brain, known as the cerebral cortex, is very uneven, characterized by a distinctive pattern of folds or bumps, known as gyri (singular: gyrus), and grooves, known as sulci (singular: sulcus), shown in the accompanying source-figure description. These gyri and sulci form important landmarks that allow us to separate the brain into functional centers. The most prominent sulcus, known as the longitudinal fissure, is the deep groove that separates the brain into two halves or hemispheres: the left hemisphere and the right hemisphere.

Source illustration description (image not reproduced): An illustration of the brain s exterior surface shows the ridges and depressions, and the deep fissure that runs through the center. The surface of the brain is covered with gyri and sulci. A deep sulcus is called a fissure, such as the longitudinal fissure that divides the brain into left and right hemispheres. (credit: modification of work by Bruce Blaus)

There is evidence of specialization of function referred to as lateralization in each hemisphere, mainly regarding differences in language functions. The left hemisphere controls the right half of the body, and the right hemisphere controls the left half of the body. Decades of research on lateralization of function by Michael Gazzaniga and his colleagues suggest that a variety of functions ranging from cause-and-effect reasoning to self-recognition may follow patterns that suggest some degree of hemispheric dominance (Gazzaniga, 2005). For example, the left hemisphere has been shown to be superior for forming associations in memory, selective attention, and positive emotions. The right hemisphere, on the other hand, has been shown to be superior in pitch perception, arousal, and negative emotions (Ehret, 2006). However, it should be pointed out that research on which hemisphere is dominant in a variety of different behaviors has produced inconsistent results, and therefore, it is probably better to think of how the two hemispheres interact to produce a given behavior rather than attributing certain behaviors to one hemisphere versus the other (Banich & Heller, 1998).

The two hemispheres are connected by a thick band of neural fibers known as the corpus callosum, consisting of about 200 million axons. The corpus callosum allows the two hemispheres to communicate with each other and allows for information being processed on one side of the brain to be shared with the other side.

Normally, we are not aware of the different roles that our two hemispheres play in day-to-day functions, but there are people who come to know the capabilities and functions of their two hemispheres quite well. In some cases of severe epilepsy, doctors elect to sever the corpus callosum as a means of controlling the spread of seizures (the accompanying source-figure description). While this is an effective treatment option, it results in individuals who have "split brains." After surgery, these split-brain patients show a variety of interesting behaviors. For instance, a split-brain patient is unable to name a picture that is shown in the patient s left visual field because the information is only available in the largely nonverbal right hemisphere. However, they are able to recreate the picture with their left

hand, which is also controlled by the right hemisphere. When the more verbal left hemisphere sees the picture that the hand drew, the patient is able to name it (assuming the left hemisphere can interpret what was drawn by the left hand).

Source illustration description (image not reproduced): Illustrations (a) and (b) show the corpus callosum s location in the brain in front and side views. Photograph (c) shows the corpus callosum in a dissected brain. (a, b) The corpus callosum connects the left and right hemispheres of the brain. (c) A scientist spreads this dissected sheep brain apart to show the corpus callosum between the hemispheres. (credit c: modification of work by Aaron Bornstein)

Much of what we know about the functions of different areas of the brain comes from studying changes in the behavior and ability of individuals who have suffered damage to the brain. For example, researchers study the behavioral changes caused by strokes to learn about the functions of specific brain areas. A stroke, caused by an interruption of blood flow to a region in the brain, causes a loss of brain function in the affected region. The damage can be in a small area, and, if it is, this gives researchers the opportunity to link any resulting behavioral changes to a specific area. The types of deficits displayed after a stroke will be largely dependent on where in the brain the damage occurred.

Consider Theona, an intelligent, self-sufficient woman, who is 62 years old. Recently, she suffered a stroke in the front portion of her right hemisphere. As a result, she has great difficulty moving her left leg. (As you learned earlier, the right hemisphere controls the left side of the body; also, the brain s main motor centers are located at the front of the head, in the frontal lobe.) Theona has also experienced behavioral changes. For example, while in the produce section of the grocery store, she sometimes eats grapes, strawberries, and apples directly from their bins before paying for them. This behavior which would have been very embarrassing to her before the stroke is consistent with damage in another region in the frontal lobe the prefrontal cortex, which is associated with judgment, reasoning, and impulse control.

Forebrain Structures

The two hemispheres of the cerebral cortex are part of the forebrain (the accompanying source-figure description), which is the largest part of the brain. The forebrain contains the cerebral cortex and a number of other structures that lie beneath the cortex (called subcortical structures): thalamus, hypothalamus, pituitary gland, and the limbic system (a collection of structures). The cerebral cortex, which is the outer surface of the brain, is associated with higher level processes such as consciousness, thought, emotion, reasoning, language, and memory. Each cerebral hemisphere can be subdivided into four lobes, each associated with different functions.

Source illustration description (image not reproduced): An illustration shows the position and size of the forebrain (the largest portion), midbrain (a small central portion), and hindbrain (a portion in the lower back part of the brain). The brain and its parts can be divided into three main categories: the forebrain, midbrain, and hindbrain.

Lobes of the Brain

The four lobes of the brain are the frontal, parietal, temporal, and occipital lobes (the accompanying source-figure description). The frontal lobe is located in the forward part of the brain, extending back to a fissure known as the central sulcus. The frontal lobe is involved in reasoning, motor control, emotion, and language. It contains the motor cortex, which is involved in planning and coordinating movement; the prefrontal cortex, which is responsible for higher-level cognitive functioning; and Broca s area, which is essential for language production.

Source illustration description (image not reproduced): An illustration shows the four lobes of the brain. The lobes of the brain are shown.

People who suffer damage to Broca s area have great difficulty producing language of any form (the accompanying source-figure description). For example, Padma was an electrical engineer who was socially active and a caring, involved parent. About twenty years ago, she was in a car accident and suffered damage to her Broca s area. She completely lost the ability to speak and form any kind of meaningful language. There is nothing wrong with her mouth or her vocal cords, but she is unable to

produce words. She can follow directions but can t respond verbally, and she can read but no longer write. She can do routine tasks like running to the market to buy milk, but she could not communicate verbally if a situation called for it.

Probably the most famous case of frontal lobe damage is that of a man by the name of Phineas Gage. On September 13, 1848, Gage (age 25) was working as a railroad foreman in Vermont. He and his crew were using an iron rod to tamp explosives down into a blasting hole to remove rock along the railway s path. Unfortunately, the iron rod created a spark and caused the rod to explode out of the blasting hole, into Gage s face, and through his skull (the accompanying source-figure description). Although lying in a pool of his own blood with brain matter emerging from his head, Gage was conscious and able to get up, walk, and speak. But in the months following his accident, people noticed that his personality had changed. Many of his friends described him as no longer being himself. Before the accident, it was said that Gage was a well-mannered, soft-spoken man, but he began to behave in odd and inappropriate ways after the accident. Such changes in personality would be consistent with loss of impulse control a frontal lobe function.

Beyond the damage to the frontal lobe itself, subsequent investigations into the rod's path also identified probable damage to pathways between the frontal lobe and other brain structures, including the limbic system. With connections between the planning functions of the frontal lobe and the emotional processes of the limbic system severed, Gage had difficulty controlling his emotional impulses.

However, there is some evidence suggesting that the dramatic changes in Gage s personality were exaggerated and embellished. Gage's case occurred in the midst of a 19th century debate over localization regarding whether certain areas of the brain are associated with particular functions. On the basis of extremely limited information about Gage, the extent of his injury, and his life before and after the accident, scientists tended to find support for their own views, on whichever side of the debate they fell (Macmillan, 1999).

Source illustration description (image not reproduced): Image (a) is a photograph of Phineas Gage holding a metal rod. Image (b) is an illustration of a skull with a metal rod passing through it from the cheek area to the top of the skull. (a) Phineas Gage holds the iron rod that penetrated his skull in an 1848 railroad construction accident. (b) Gage s prefrontal cortex was severely damaged in the left hemisphere. The rod entered Gage s face on the left side, passed behind his eye, and exited through the top of his skull, before landing about 80 feet away. (credit a: modification of work by Jack and Beverly Wilgus)

The brain s parietal lobe is located immediately behind the frontal lobe, and is involved in processing information from the body s senses. It contains the somatosensory cortex, which is essential for processing sensory information from across the body, such as touch, temperature, and pain. The somatosensory cortex is an area of the brain which processes touch and sensation. The somatosensory cortex is fascinating because each different area of the cortex processes sensations from a different part of your body. Furthermore, the larger the surface area of the specific body part and the greater amount of nerves in that body part, the larger the area dedicated to processing sensation from that body part in the somatosensory cortex. For example, fingers take up a lot more space than toes. As you can notice from (the accompanying source-figure description), the amount of space to process sensation from fingers is much greater than that of toes.

Source illustration description (image not reproduced): A diagram shows the organization in the somatosensory cortex, with functions for these parts in this proximal sequential order: genitals, toes, ankles, knees, hips, trunk, shoulders, elbows, wrists, hands, fingers, thumbs, neck, eyebrows and eyelids, eyeballs, face, lips, jaw, tongue, salivation, chewing, and swallowing. Spatial relationships in the body are mirrored in the organization of the somatosensory cortex.

The temporal lobe is located on the side of the head (temporal means near the temples), and is associated with hearing, memory, emotion, and some aspects of language. The auditory cortex, the main area responsible for processing auditory information, is located within the temporal lobe. Wernicke s area, important for speech comprehension, is also located here. Whereas individuals with

damage to Broca s area have difficulty producing language, those with damage to Wernicke s area can produce sensible language, but they are unable to understand it (the accompanying source-figure description).

Source illustration description (image not reproduced): An illustration shows the locations of Broca s and Wernicke s areas. Damage to either Broca s area or Wernicke s area can result in language deficits. The types of deficits are very different, however, depending on which area is affected.

The occipital lobe is located at the very back of the brain, and contains the primary visual cortex, which is responsible for interpreting incoming visual information. The occipital cortex is organized retinotopically, which means there is a close relationship between the position of an object in a person s visual field and the position of that object s representation on the cortex. You will learn much more about how visual information is processed in the occipital lobe when you study sensation and perception.

Other Areas of the Forebrain

Other areas of the forebrain, located beneath the cerebral cortex, include the thalamus and the limbic system. The thalamus is a sensory relay for the brain. All of our senses, with the exception of smell, are routed through the thalamus before being directed to other areas of the brain for processing (the accompanying source-figure description).

Source illustration description (image not reproduced): An illustration shows the location of the thalamus in the brain. The thalamus serves as the relay center of the brain where most senses are routed for processing.

The limbic system is involved in processing both emotion and memory. Interestingly, the sense of smell projects directly to the limbic system; therefore, not surprisingly, smell can evoke emotional responses in ways that other sensory modalities cannot. The limbic system is made up of a number of different structures, but three of the most important are the hippocampus, the amygdala, and the hypothalamus (the accompanying source-figure description). The hippocampus is an essential structure for learning and memory. The amygdala is involved in our experience of emotion and in tying emotional meaning to our memories. The hypothalamus regulates a number of homeostatic processes, including the regulation of body temperature, appetite, and blood pressure. The hypothalamus also serves as an interface between the nervous system and the endocrine system and in the regulation of sexual motivation and behavior.

Source illustration description (image not reproduced): An illustration shows the locations of parts of the brain involved in the limbic system: the hypothalamus, amygdala, and hippocampus. The limbic system is involved in mediating emotional response and memory.

The Case of Henry Molaison (H.M.)

In 1953, Henry Gustav Molaison (H. M.) was a 27-year-old man who experienced severe seizures. In an attempt to control his seizures, H. M. underwent brain surgery to remove his hippocampus and amygdala. Following the surgery, H.M s seizures became much less severe, but he also suffered some unexpected and devastating consequences of the surgery: he lost his ability to form many types of new memories. For example, he was unable to learn new facts, such as who was president of the United States. He was able to learn new skills, but afterward he had no recollection of learning them. For example, while he might learn to use a computer, he would have no conscious memory of ever having used one. He could not remember new faces, and he was unable to remember events, even immediately after they occurred. Researchers were fascinated by his experience, and he is considered one of the most studied cases in medical and psychological history (Hardt, Einarsson, & Nader, 2010; Squire, 2009). Indeed, his case has provided tremendous insight into the role that the hippocampus plays in the consolidation of new learning into explicit memory.

Midbrain and Hindbrain Structures

The midbrain is comprised of structures located deep within the brain, between the forebrain and the hindbrain. The reticular formation is centered in the midbrain, but it actually extends up into the forebrain and down into the hindbrain. The reticular formation is important in regulating the sleep/wake cycle, arousal, alertness, and motor activity.

The substantia nigra (Latin for black substance) and the ventral tegmental area (VTA) are also located in the midbrain (the accompanying source-figure description). Both regions contain cell bodies that produce the neurotransmitter dopamine, and both are critical for movement. Degeneration of the substantia nigra and VTA is involved in Parkinson s disease. In addition, these structures are involved in mood, reward, and addiction (Berridge & Robinson, 1998; Gardner, 2011; George, Le Moal, & Koob, 2012).

Source illustration description (image not reproduced): An illustration shows the location of the substantia nigra and VTA in the brain. The substantia nigra and ventral tegmental area (VTA) are located in the midbrain.

The hindbrain is located at the back of the head and looks like an extension of the spinal cord. It contains the medulla, pons, and cerebellum (the accompanying source-figure description). The medulla controls the automatic processes of the autonomic nervous system, such as breathing, blood pressure, and heart rate. The word pons literally means bridge, and as the name suggests, the pons serves to connect the hindbrain to the rest of the brain. It also is involved in regulating brain activity during sleep. The medulla, pons, and various structures are known as the brainstem, and aspects of the brainstem span both the midbrain and the hindbrain.

Source illustration description (image not reproduced): An illustration shows the location of the pons, medulla, and cerebellum. The pons, medulla, and cerebellum make up the hindbrain.

The cerebellum (Latin for little brain) receives messages from muscles, tendons, joints, and structures in our ear to control balance, coordination, movement, and motor skills. The cerebellum is also thought to be an important area for processing some types of memories. In particular, procedural memory, or memory involved in learning and remembering how to perform tasks, is thought to be associated with the cerebellum. Recall that H. M. was unable to form new explicit memories, but he could learn new tasks. This is likely due to the fact that H. M. s cerebellum remained intact.

What Do You Think?

What would you do if your spouse or loved one was declared brain dead but their body was being kept alive by medical equipment? Whose decision should it be to remove a feeding tube? Should medical care costs be a factor?

On February 25, 1990, a Florida woman named Terri Schiavo went into cardiac arrest, apparently triggered by a bulimic episode. She was eventually revived, but her brain had been deprived of oxygen for a long time. Brain scans indicated that there was no activity in her cerebral cortex, and she suffered from severe and permanent cerebral atrophy. Basically, Schiavo was in a vegetative state. Medical professionals determined that she would never again be able to move, talk, or respond in any way. To remain alive, she required a feeding tube, and there was no chance that her situation would ever improve.

On occasion, Schiavo s eyes would move, and sometimes she would groan. Despite the doctors insistence to the contrary, her parents believed that these were signs that she was trying to communicate with them.

After 12 years, Schiavo s husband argued that his wife would not have wanted to be kept alive with no feelings, sensations, or brain activity. Her parents, however, were very much against removing her feeding tube. Eventually, the case made its way to the courts, both in the state of Florida and at the federal level. By 2005, the courts found in favor of Schiavo s husband, and the feeding tube was removed on March 18, 2005. Schiavo died 13 days later.

Why did Schiavo s eyes sometimes move, and why did she groan? Although the parts of her brain that control thought, voluntary movement, and feeling were completely damaged, her brainstem was still intact. Her medulla and pons maintained her breathing and caused involuntary movements of her eyes and the occasional groans. Over the 15-year period that she was on a feeding tube, Schiavo s medical costs may have topped $7 million (Arnst, 2003).

These questions were brought to popular conscience decades ago in the case of Terri Schiavo, and they have persisted. In 2013, a 13-year-old girl who suffered complications after tonsil surgery was declared brain dead. There was a battle between her family, who wanted her to remain on life support, and the hospital s policies regarding persons declared brain dead. In another complicated 2013 14 case in Texas, a pregnant EMT professional declared brain dead was kept alive for weeks, despite her spouse s directives, which were based on her wishes should this situation arise. In this case, state laws designed to protect an unborn fetus came into consideration until doctors determined the fetus unviable.

Decisions surrounding the medical response to patients declared brain dead are complex. What do you think about these issues?

Brain Imaging

You have learned how brain injury can provide information about the functions of different parts of the brain. Increasingly, however, we are able to obtain that information using brain imaging techniques on individuals who have not suffered brain injury. In this section, we take a more in-depth look at some of the techniques that are available for imaging the brain, including techniques that rely on radiation, magnetic fields, or electrical activity within the brain.

Techniques Involving Radiation

A computerized tomography (CT) scan involves taking a number of x-rays of a particular section of a person s body or brain (the accompanying source-figure description). The x-rays pass through tissues of different densities at different rates, allowing a computer to construct an overall image of the area of the body being scanned. A CT scan is often used to determine whether someone has a tumor or significant brain atrophy.

Source illustration description (image not reproduced): Image (a) shows a brain scan where the brain matter s appearance is fairly uniform. Image (b) shows a section of the brain that looks different from the surrounding tissue and is labeled tumor. A CT scan can be used to show brain tumors. (a) The image on the left shows a healthy brain, whereas (b) the image on the right indicates a brain tumor in the left frontal lobe. (credit a: modification of work by "Aceofhearts1968"/Wikimedia Commons; credit b: modification of work by Roland Schmitt et al)

Positron emission tomography (PET) scans create pictures of the living, active brain (the accompanying source-figure description). An individual receiving a PET scan drinks or is injected with a mildly radioactive substance, called a tracer. Once in the bloodstream, the amount of tracer in any given region of the brain can be monitored. As a brain area becomes more active, more blood flows to that area. A computer monitors the movement of the tracer and creates a rough map of active and inactive areas of the brain during a given behavior. PET scans show little detail, are unable to pinpoint events precisely in time, and require that the brain be exposed to radiation; therefore, this technique has been replaced by the fMRI as an alternative diagnostic tool. However, combined with CT, PET technology is still being used in certain contexts. For example, CT/PET scans allow better imaging of the activity of neurotransmitter receptors and open new avenues in schizophrenia research. In this hybrid CT/PET technology, CT contributes clear images of brain structures, while PET shows the brain s activity.

Source illustration description (image not reproduced): A brain scan shows different parts of the brain in different colors. A PET scan is helpful for showing activity in different parts of the brain. (credit: Health and Human Services Department, National Institutes of Health)

Techniques Involving Magnetic Fields

In magnetic resonance imaging (MRI), a person is placed inside a machine that generates a strong magnetic field. The magnetic field causes the hydrogen atoms in the body s cells to move. When the magnetic field is turned off, the hydrogen atoms emit electromagnetic signals as they return to their original positions. Tissues of different densities give off different signals, which a computer interprets and displays on a monitor. Functional magnetic resonance imaging (fMRI) operates on the same principles, but it shows changes in brain activity over time by tracking blood flow and oxygen levels. The fMRI provides more detailed images of the brain s structure, as well as better accuracy in time, than is possible in PET scans (the accompanying source-figure description). With their high level of detail, MRI and fMRI are often used to compare the brains of healthy individuals to the brains of individuals diagnosed with psychological disorders. This comparison helps determine what structural and functional differences exist between these populations.

Source illustration description (image not reproduced): A brain scan shows brain tissue in gray with some small areas highlighted red. An fMRI shows activity in the brain over time. This image represents a single frame from an fMRI. (credit: modification of work by Kim J, Matthews NL, Park S.)

Techniques Involving Electrical Activity

In some situations, it is helpful to gain an understanding of the overall activity of a person s brain, without needing information on the actual location of the activity. Electroencephalography (EEG) serves this purpose by providing a measure of a brain s electrical activity. An array of electrodes is placed around a person s head (the accompanying source-figure description). The signals received by the electrodes result in a printout of the electrical activity of their brain, or brainwaves, showing both the frequency (number of waves per second) and amplitude (height) of the recorded brainwaves, with an accuracy within milliseconds. Such information is especially helpful to researchers studying sleep patterns among individuals with sleep disorders.

Source illustration description (image not reproduced): A photograph depicts a person looking at a computer screen and using the keyboard and mouse. The person wears a white cap covered in electrodes and wires. Using caps with electrodes, modern EEG research can study the precise timing of overall brain activities. (credit: SMI Eye Tracking)

Summary

The brain consists of two hemispheres, each controlling the opposite side of the body. Each hemisphere can be subdivided into different lobes: frontal, parietal, temporal, and occipital. In addition to the lobes of the cerebral cortex, the forebrain includes the thalamus (sensory relay) and limbic system (emotion and memory circuit). The midbrain contains the reticular formation, which is important for sleep and arousal, as well as the substantia nigra and ventral tegmental area. These structures are important for movement, reward, and addictive processes. The hindbrain contains the structures of the brainstem (medulla, pons, and midbrain), which control automatic functions like breathing and blood pressure. The hindbrain also contains the cerebellum, which helps coordinate movement and certain types of memories.

Individuals with brain damage have been studied extensively to provide information about the role of different areas of the brain, and recent advances in technology allow us to glean similar information by imaging brain structure and function. These techniques include CT, PET, MRI, fMRI, and EEG.

Review Questions

OpenStax learning exercise

Q. The _____ is a sensory relay station where all sensory information, except for smell, goes before being sent to other areas of the brain for further processing.

A. D

OpenStax learning exercise

Q. Damage to the _____ disrupts one s ability to comprehend language, but it leaves one s ability to produce words intact.

A. C

OpenStax learning exercise

Q. A(n) _____ uses magnetic fields to create pictures of a given tissue.

A. B

OpenStax learning exercise

Q. Which of the following is not a structure of the forebrain?

A. D

Critical Thinking Questions

OpenStax learning exercise

Q. Before the advent of modern imaging techniques, scientists and clinicians relied on autopsies of people who suffered brain injury with resultant change in behavior to determine how different areas of the brain were affected. What are some of the limitations associated with this kind of approach?

A. The same limitations associated with any case study would apply here. In addition, it is possible that the damage caused changes in other areas of the brain, which might contribute to the behavioral deficits. Such changes would not necessarily be obvious to someone performing an autopsy, as they may be functional in nature, rather than structural.

OpenStax learning exercise

Q. Which of the techniques discussed would be viable options for you to determine how activity in the reticular formation is related to sleep and wakefulness? Why?

A. The most viable techniques are fMRI and PET because of their ability to provide information about brain activity and structure simultaneously.

Personal Application Questions

OpenStax learning exercise

Q. You read about H. M. s memory deficits following the bilateral removal of his hippocampus and amygdala. Have you encountered a character in a book, television program, or movie that suffered memory deficits? How was that character similar to and different from H. M.?

The Endocrine System

OpenStax source module m82176 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82176/index.cnxml

The endocrine system consists of a series of glands that produce chemical substances known as hormones (the accompanying source-figure description). Like neurotransmitters, hormones are chemical messengers that must bind to a receptor in order to send their signal. However, unlike

neurotransmitters, which are released in close proximity to cells with their receptors, hormones are secreted into the bloodstream and travel throughout the body, affecting any cells that contain receptors for them. Thus, whereas neurotransmitters effects are localized, the effects of hormones are widespread. Also, hormones are slower to take effect, and tend to be longer lasting.

Source illustration description (image not reproduced): A diagram of the human body illustrates the locations of the thymus, several parts within the brain (pineal gland, thalamus, hypothalamus, pituitary gland), several parts within the thyroid (cartilage of the larynx, thyroid gland, parathyroid glands, trachea), the adrenal glands, pancreas, uterus (female), ovaries (female), and testes (male). The major glands of the endocrine system are shown.

Hormones are involved in regulating all sorts of bodily functions, and they are ultimately controlled through interactions between the hypothalamus (in the central nervous system) and the pituitary gland (in the endocrine system). Imbalances in hormones are related to a number of disorders. This section explores some of the major glands that make up the endocrine system and the hormones secreted by these glands (the accompanying source-figure description).

Major Glands

The pituitary gland descends from the hypothalamus at the base of the brain, and acts in close association with it. The pituitary is often referred to as the master gland because its messenger hormones control all the other glands in the endocrine system, although it mostly carries out instructions from the hypothalamus. In addition to messenger hormones, the pituitary also secretes growth hormone, endorphins for pain relief, and a number of key hormones that regulate fluid levels in the body.

Located in the neck, the thyroid gland releases hormones that regulate growth, metabolism, and appetite. In hyperthyroidism, or Grave s disease, the thyroid secretes too much of the hormone thyroxine, causing agitation, bulging eyes, and weight loss. In hypothyroidism, reduced hormone levels cause sufferers to experience tiredness, and they often complain of feeling cold. Fortunately, thyroid disorders are often treatable with medications that help reestablish a balance in the hormones secreted by the thyroid.

The adrenal glands sit atop our kidneys and secrete hormones involved in the stress response, such as epinephrine (adrenaline) and norepinephrine (noradrenaline). The pancreas is an internal organ that secretes hormones that regulate blood sugar levels: insulin and glucagon. These pancreatic hormones are essential for maintaining stable levels of blood sugar throughout the day by lowering blood glucose levels (insulin) or raising them (glucagon). People who suffer from diabetes do not produce enough insulin; therefore, they must take medications that stimulate or replace insulin production, and they must closely control the amount of sugars and carbohydrates they consume.

The gonads secrete sexual hormones, which are important in reproduction, and mediate both sexual motivation and behavior. The female gonads are the ovaries; the male gonads are the testes. Ovaries secrete estrogens and progesterone, and the testes secrete androgens, such as testosterone.

Source comparison table

Major Endocrine Glands and Associated Hormone Functions		
Endocrine Gland	Associated Hormones	Function
Pituitary	Growth hormone, releasing and inhibiting hormones (such as thyroid stimulating hormone)	Regulate growth, regulate hormone release
Thyroid	Thyroxine, triiodothyronine	Regulate metabolism and appetite

Major Endocrine Glands and Associated Hormone Functions		
Pineal	Melatonin	Regulate some biological rhythms such as sleep cycles
Adrenal	Epinephrine, norepinephrine	Stress response, increase metabolic activities
Pancreas	Insulin, glucagon	Regulate blood sugar levels
Ovaries	Estrogen, progesterone	Mediate sexual motivation and behavior, reproduction
Testes	Androgens, such as testosterone	Mediate sexual motivation and behavior, reproduction

Dig Deeper

Although it is against Federal laws and many professional athletic associations (The National Football League, for example) have banned their use, anabolic steroid drugs continue to be used by amateur and professional athletes. The drugs are believed to enhance athletic performance. Anabolic steroid drugs mimic the effects of the body s own steroid hormones, like testosterone and its derivatives. These drugs have the potential to provide a competitive edge by increasing muscle mass, strength, and endurance, although not all users may experience these results. Moreover, use of performance-enhancing drugs (PEDs) does not come without risks. Anabolic steroid use has been linked with a wide variety of potentially negative outcomes, ranging in severity from largely cosmetic (acne) to life threatening (heart attack). Furthermore, use of these substances can result in profound changes in mood and can increase aggressive behavior (National Institute on Drug Abuse, 2001).

Baseball player Alex Rodriguez (A-Rod) spent the latter part of his playing career at the center of a media storm regarding his use of illegal PEDs. Rodriguez excelled while using the drugs; his success played a large role in negotiating a contract that made him the highest paid player in professional baseball. A subsequent scandal and suspension tarnished his reputation and, according to a statement he made once retired, cost him over $40 million. Even lower-profile athletes, particularly in cycling and Olympic sports, have been revealed as steroid users. What are your thoughts on athletes and doping? Why or why not should the use of PEDs be banned? What advice would you give an athlete who was considering using PEDs?

Summary

The glands of the endocrine system secrete hormones to regulate normal body functions. The hypothalamus serves as the interface between the nervous system and the endocrine system, and it controls the secretions of the pituitary. The pituitary serves as the master gland, controlling the secretions of all other glands. The thyroid secretes thyroxine, which is important for basic metabolic processes and growth; the adrenal glands secrete hormones involved in the stress response; the pancreas secretes hormones that regulate blood sugar levels; and the ovaries and testes produce sex hormones that regulate sexual motivation and behavior.

Review Questions

OpenStax learning exercise

Q. The two major hormones secreted from the pancreas are:

A. D

OpenStax learning exercise

Q. The _____ secretes messenger hormones that direct the function of the rest of the endocrine glands.

A. C

OpenStax learning exercise

Q. The _____ gland secretes epinephrine.

A. A

OpenStax learning exercise

Q. The _____ secretes hormones that regulate the body s fluid levels.

A. B

Critical Thinking Questions

OpenStax learning exercise

Q. Hormone secretion is often regulated through a negative feedback mechanism, which means that once a hormone is secreted it will cause the hypothalamus and pituitary to shut down the production of signals necessary to secrete the hormone in the first place. Most oral contraceptives are made of small doses of estrogen and/or progesterone. Why would this be an effective means of contraception?

A. The introduction of relatively low, yet constant, levels of gonadal hormones places the hypothalamus and pituitary under inhibition via negative feedback mechanisms. This prevents the alterations in both estrogen and progesterone concentrations that are necessary for successful ovulation and implantation.

OpenStax learning exercise

Q. Chemical messengers are used in both the nervous system and the endocrine system. What properties do these two systems share? What properties are different? Which one would be faster? Which one would result in long-lasting changes?

A. Both systems involve chemical messengers that must interact with receptors in order to have an effect. The relative proximity of the release site and target tissue varies dramatically between the two systems. In neurotransmission, reuptake and enzymatic breakdown immediately clear the synapse. Metabolism of hormones must occur in the liver. Therefore, while neurotransmission is much more rapid in signaling information, hormonal signaling can persist for quite some time as the concentrations of the hormone in the bloodstream vary gradually over time.

Personal Application Questions

OpenStax learning exercise

Q. Given the negative health consequences associated with the use of anabolic steroids, what kinds of considerations might be involved in a person s decision to use them?

CHAPTER 08 / CAC.901

Learning, conditioning, and modeling

Lecture: Learning is visible in the pattern

When a child repeats a behavior, adults often search for an internal trait. Learning perspectives invite a different first question: what happens around the behavior, and how does that pattern change over time? This is not a denial of thought or emotion. It is a method for examining how experience and consequences shape what a child does.

Classical conditioning concerns associations between events. A place, sound, or routine may become linked with comfort or fear. Operant conditioning concerns how consequences influence the future likelihood of behavior. Observational learning concerns what people learn by watching others. These processes can operate together. A child may learn that a classroom predicts embarrassment, discover that leaving reduces distress, and observe other students avoiding the same task.

Reinforcement is not a reward sticker

Reinforcement means that a consequence is associated with an increase in behavior. It is defined by the pattern, not by the adult s intention. Praise does not reinforce every child. Public attention may be welcome for one child and humiliating for another. A token may support participation in one setting and feel meaningless in another. The child s experience and the observed change matter.

Negative reinforcement is often confused with punishment. Negative reinforcement involves the removal or avoidance of something aversive, which can increase a behavior. If a child leaves a difficult task and distress falls, leaving may become more likely. The child is not necessarily manipulating adults. The pattern can develop naturally because relief is a powerful consequence.

Punishment refers to a consequence that reduces behavior. It does not necessarily teach an alternative, repair harm, or address the reason for the behavior. Harsh or humiliating punishment can create additional fear and damage trust. Ethical intervention prioritizes understanding, prevention, teaching useful alternatives, and reinforcement that respects the child. Restrictions should be proportionate, necessary, and reviewed rather than automatically intensified.

Define the behavior and the function

A useful behavioral description is specific enough to observe. Disrespectful all day is difficult to measure. Interrupts adult instructions with comments during whole-class teaching identifies a pattern that can be examined. Record what happens before, what the behavior looks like, and what follows. Consider whether the behavior helps the child gain attention, escape a demand, access an activity, or manage sensory experience. These are hypotheses, not labels to impose.

Figure 8.1 shows an antecedent behavior consequence sequence with feedback into the next occasion. It also includes setting events such as fatigue or conflict. A complete account does not treat the immediate trigger as the whole cause. A task may be the occasion for a behavior, while sleep loss and repeated failure influence its likelihood.

Functional thinking should include the child s report. A child may say, I go outside because everyone can see I cannot do it. That account suggests escape from public failure, not simply escape from work. The intervention may need private practice, accessible instruction, or a different

way to request help. Changing the consequence without changing the underlying demand may leave the child s problem untouched.

Learning through models

Children learn from caregivers, teachers, peers, media, and institutional behavior. They notice not only what adults say but what adults do under stress. A teacher who apologizes after an unfair correction models repair. A caregiver who pauses before responding models regulation. A school that publicly humiliates students while teaching respect provides conflicting lessons.

Observational learning is not simple imitation. Children attend selectively, interpret what they see, and consider likely consequences. They may imitate a behavior when it appears to bring status or safety. Counselors can use modeling thoughtfully, demonstrating a skill in manageable steps and allowing the child to practice without shame. A video or story can help, but real change usually needs opportunities in the child s own environment.

Application: the reading refusal

A fictional eight-year-old, Binta, pushes her reading book away and is sent to the corridor. The behavior appears to increase. A learning hypothesis is that removal from reading provides relief. Before intervening, examine whether the text is too difficult, whether reading aloud has led to teasing, and whether Binta has a way to request help.

A plan could include an appropriate reading assessment, a predictable private practice time, a respectful help signal, and reinforcement for using the signal or attempting a manageable step. The aim is not to reward bad behavior. It is to make communication and participation more effective than refusal. Track both the behavior and reading access. If the behavior becomes quieter but learning does not improve, the plan needs revision.

Caregivers and teachers should agree on the target and avoid changing several conditions without recording them. This makes the pattern easier to understand. Review whether the support is acceptable to Binta and whether it preserves dignity. A technically effective plan that humiliates a child is not an ethically successful intervention.

Doctoral seminar task

Observe or imagine a recurring interaction. Write an operational definition, identify setting events, and record several possible antecedents and consequences. Generate two functional hypotheses. For each, propose a different intervention and a measurable indicator that would support or weaken the hypothesis.

Then audit the plan for ethics. Does it increase the child s agency? Does it teach a useful alternative? Are consequences proportionate? Could the target behavior be a reasonable response to an inaccessible or unsafe environment? Behavioral thinking is strongest when it changes conditions thoughtfully rather than demanding compliance for its own sake.

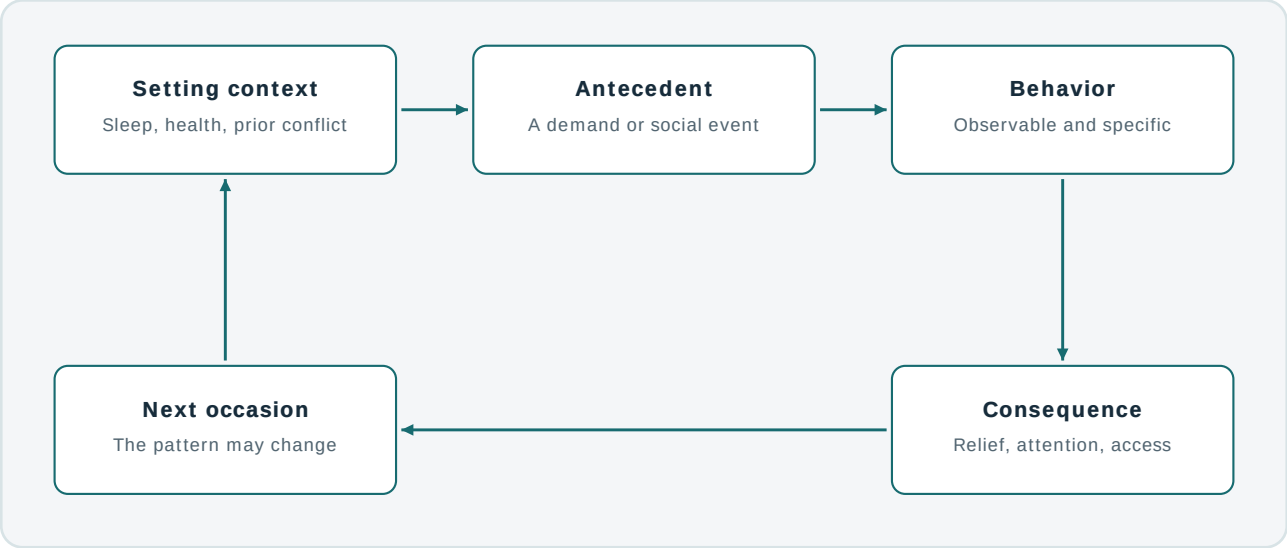


Figure 8.1. The learning sequence has feedback. Setting events influence how an immediate antecedent is experienced. The consequence can change the likelihood of the behavior on a later occasion. The child s account helps distinguish escape, access, communication, and other possible functions. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

What Is Learning?

OpenStax source module m82199 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82199/index.cnxml

Birds build nests and migrate as winter approaches. Infants suckle for nurishment. Dogs shake water off wet fur. Salmon swim upstream to spawn, and spiders spin intricate webs. What do these seemingly unrelated behaviors have in common? They all are unlearned behaviors. Both instincts and reflexes are innate (unlearned) behaviors that organisms are born with. Reflexes are a motor or neural reaction to a specific stimulus in the environment. They tend to be simpler than instincts, involve the activity of specific body parts and systems (e.g., the knee-jerk reflex and the contraction of the pupil in bright light), and involve more primitive centers of the central nervous system (e.g., the spinal cord and the medulla). In contrast, instincts are innate behaviors that are triggered by a broader range of events, such as maturation and the change of seasons. They are more complex patterns of behavior, involve movement of the organism as a whole (e.g., sexual activity and migration), and involve higher brain centers.

Both reflexes and instincts help an organism adapt to its environment and do not have to be learned. For example, every healthy human baby has a sucking reflex, present at birth. Babies are born knowing how to suck on a nipple, whether artificial (from a bottle) or human. Nobody teaches the baby to suck, just as no one teaches a sea turtle hatchling to move toward the ocean. Learning, like reflexes and instincts, allows an organism to adapt to its environment. But unlike instincts and reflexes, learned behaviors involve change and experience: learning is a relatively permanent change in behavior or knowledge that results from experience. In contrast to the innate behaviors discussed above, learning involves acquiring knowledge and skills through experience. Looking back at our surfing scenario, Julian will have to spend much more time training with his surfboard before he learns how to ride the waves like his father.

Learning to surf, as well as any complex learning process (e.g., learning about the discipline of psychology), involves a complex interaction of conscious and unconscious processes. Learning has traditionally been studied in terms of its simplest components the associations our minds automatically make between events. Our minds have a natural tendency to connect events that occur closely together or in sequence. Associative learning occurs when an organism makes connections between stimuli or events that occur together in the environment. You will see that associative learning is central to all three basic learning processes discussed in this chapter; classical conditioning tends to involve unconscious processes, operant conditioning tends to involve conscious processes, and observational learning adds social and cognitive layers to all the basic associative processes, both conscious and unconscious. These learning processes will be discussed in detail later in the chapter, but it is helpful to have a brief overview of each as you begin to explore how learning is understood from a psychological perspective.

In classical conditioning, also known as Pavlovian conditioning, organisms learn to associate events or stimuli that repeatedly happen together. We experience this process throughout our daily lives. For

example, you might see a flash of lightning in the sky during a storm and then hear a loud boom of thunder. The sound of the thunder naturally makes you jump (loud noises have that effect by reflex). Because lightning reliably predicts the impending boom of thunder, you may associate the two and jump when you see lightning. Psychological researchers study this associative process by focusing on what can be seen and measured behaviors. Researchers ask if one stimulus triggers a reflex, can we train a different stimulus to trigger that same reflex? In operant conditioning, organisms learn, again, to associate events a behavior and its consequence (reinforcement or punishment). A pleasant consequence encourages more of that behavior in the future, whereas a punishment deters the behavior. Imagine you are teaching your dog, Hodor, to sit. You tell Hodor to sit, and give him a treat when he does. After repeated experiences, Hodor begins to associate the act of sitting with receiving a treat. He learns that the consequence of sitting is that he gets a doggie biscuit (the accompanying source-figure description). Conversely, if the dog is punished when exhibiting a behavior, it becomes conditioned to avoid that behavior (e.g., receiving a small shock when crossing the boundary of an invisible electric fence).

Source illustration description (image not reproduced): A photograph shows a dog standing at attention and smelling a treat in a person s hand. In operant conditioning, a response is associated with a consequence. This dog has learned that certain behaviors result in receiving a treat. (credit: Crystal Rolfe)

Observational learning extends the effective range of both classical and operant conditioning. In contrast to classical and operant conditioning, in which learning occurs only through direct experience, observational learning is the process of watching others and then imitating what they do. A lot of learning among humans and other animals comes from observational learning. To get an idea of the extra effective range that observational learning brings, consider Ben and his son Julian from the introduction. How might observation help Julian learn to surf, as opposed to learning by trial and error alone? By watching his father, he can imitate the moves that bring success and avoid the moves that lead to failure. Can you think of something you have learned how to do after watching someone else?

All of the approaches covered in this chapter are part of a particular tradition in psychology, called behaviorism, which we discuss in the next section. However, these approaches do not represent the entire study of learning. Separate traditions of learning have taken shape within different fields of psychology, such as memory and cognition, so you will find that other chapters will round out your understanding of the topic. Over time these traditions tend to converge. For example, in this chapter you will see how cognition has come to play a larger role in behaviorism, whose more extreme adherents once insisted that behaviors are triggered by the environment with no intervening thought.

Summary

Instincts and reflexes are innate behaviors they occur naturally and do not involve learning. In contrast, learning is a change in behavior or knowledge that results from experience. There are three main types of learning: classical conditioning, operant conditioning, and observational learning. Both classical and operant conditioning are forms of associative learning where associations are made between events that occur together. Observational learning is just as it sounds: learning by observing others.

Review Questions

OpenStax learning exercise

Q. Which of the following is an example of a reflex that occurs at some point in the development of a human being?

A. C

OpenStax learning exercise

Q. Learning is best defined as a relatively permanent change in behavior that _____.

A. B

OpenStax learning exercise

Q. Two forms of associative learning are _____ and _____.

A. A

OpenStax learning exercise

Q. In _____ the stimulus or experience occurs before the behavior and then gets paired with the behavior.

A. D

Critical Thinking Questions

OpenStax learning exercise

Q. Compare and contrast classical and operant conditioning. How are they alike? How do they differ?

A. Both classical and operant conditioning involve learning by association. In classical conditioning, responses are involuntary and automatic; however, responses are voluntary and learned in operant conditioning. In classical conditioning, the event that drives the behavior (the stimulus) comes before the behavior; in operant conditioning, the event that drives the behavior (the consequence) comes after the behavior. Also, whereas classical conditioning involves an organism forming an association between an involuntary (reflexive) response and a stimulus, operant conditioning involves an organism forming an association between a voluntary behavior and a consequence.

OpenStax learning exercise

Q. What is the difference between a reflex and a learned behavior?

A. A reflex is a behavior that humans are born knowing how to do, such as sucking or blushing; these behaviors happen automatically in response to stimuli in the environment. Learned behaviors are things that humans are not born knowing how to do, such as swimming and surfing. Learned behaviors are not automatic; they occur as a result of practice or repeated experience in a situation.

Personal Application Questions

OpenStax learning exercise

Q. What is your personal definition of learning? How do your ideas about learning compare with the definition of learning presented in this text?

OpenStax learning exercise

Q. What kinds of things have you learned through the process of classical conditioning? Operant conditioning? Observational learning? How did you learn them?

Classical Conditioning

OpenStax source module m82200 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82200/index.cnxml

Does the name Ivan Pavlov ring a bell? Even if you are new to the study of psychology, chances are that you have heard of Pavlov and his famous dogs.

Pavlov (1849 1936), a Russian scientist, performed extensive research on dogs and is best known for his experiments in classical conditioning (the accompanying source-figure description). As we discussed briefly in the previous section, classical conditioning is a process by which we learn to associate stimuli and, consequently, to anticipate events.

Source illustration description (image not reproduced): A portrait shows Ivan Pavlov. Ivan Pavlov s research on the digestive system of dogs unexpectedly led to his discovery of the learning process now known as classical conditioning.

Pavlov came to his conclusions about how learning occurs completely by accident. Pavlov was a physiologist, not a psychologist. Physiologists study the life processes of organisms, from the molecular level to the level of cells, organ systems, and entire organisms. Pavlov s area of interest was the digestive system (Hunt, 2007). In his studies with dogs, Pavlov measured the amount of saliva produced in response to various foods. Over time, Pavlov (1927) observed that the dogs began to salivate not only at the taste of food, but also at the sight of food, at the sight of an empty food bowl, and even at the sound of the laboratory assistants' footsteps. Salivating to food in the mouth is reflexive, so no learning is involved. However, dogs don t naturally salivate at the sight of an empty bowl or the sound of footsteps.

These unusual responses intrigued Pavlov, and he wondered what accounted for what he called the dogs' psychic secretions (Pavlov, 1927). To explore this phenomenon in an objective manner, Pavlov designed a series of carefully controlled experiments to see which stimuli would cause the dogs to salivate. He was able to train the dogs to salivate in response to stimuli that clearly had nothing to do with food, such as the sound of a bell, a light, and a touch on the leg. Through his experiments, Pavlov realized that an organism has two types of responses to its environment: (1) unconditioned (unlearned) responses, or reflexes, and (2) conditioned (learned) responses.

In Pavlov s experiments, the dogs salivated each time meat powder was presented to them. The meat powder in this situation was an unconditioned stimulus (UCS): a stimulus that elicits a reflexive response in an organism. The dogs salivation was an unconditioned response (UCR): a natural (unlearned) reaction to a given stimulus. Before conditioning, think of the dogs stimulus and response like this:

Meat powder (UCS) Salivation (UCR)

In classical conditioning, a neutral stimulus is presented immediately before an unconditioned stimulus. Pavlov would sound a tone (like ringing a bell) and then give the dogs the meat powder (the accompanying source-figure description). The tone was the neutral stimulus (NS), which is a stimulus that does not naturally elicit a response. Prior to conditioning, the dogs did not salivate when they just heard the tone because the tone had no association for the dogs.

Tone (NS) + Meat Powder (UCS) Salivation (UCR)

When Pavlov paired the tone with the meat powder over and over again, the previously neutral stimulus (the tone) also began to elicit salivation from the dogs. Thus, the neutral stimulus became the conditioned stimulus (CS), which is a stimulus that elicits a response after repeatedly being paired with an unconditioned stimulus. Eventually, the dogs began to salivate to the tone alone, just as they previously had salivated at the sound of the assistants footsteps. The behavior caused by the conditioned stimulus is called the conditioned response (CR). In the case of Pavlov s dogs, they had learned to associate the tone (CS) with being fed, and they began to salivate (CR) in anticipation of food.

Tone (CS) Salivation (CR)

Source illustration description (image not reproduced): Two illustrations are labeled before conditioning and show a dog salivating over a dish of food, and a dog not salivating while a bell is rung. An illustration labeled during conditioning shows a dog salivating over a bowl of food while a bell is rung. An illustration labeled after conditioning shows a dog salivating while a bell is rung. Before conditioning, an unconditioned stimulus (food) produces an unconditioned response (salivation), and a neutral stimulus (bell) does not produce a response. During conditioning, the unconditioned stimulus (food) is presented repeatedly just after the presentation of the neutral stimulus (bell). After conditioning, the neutral stimulus alone produces a conditioned response (salivation), thus becoming a conditioned stimulus.

Real World Application of Classical Conditioning

How does classical conditioning work in the real world? Consider the case of Moisha, who was diagnosed with cancer. When she received her first chemotherapy treatment, she vomited shortly after the chemicals were injected. In fact, every trip to the doctor for chemotherapy treatment shortly after the drugs were injected, she vomited. Moisha s treatment was a success and her cancer went into remission. Now, when she visits her oncologist's office every 6 months for a check-up, she becomes nauseous. In this case, the chemotherapy drugs are the unconditioned stimulus (UCS), vomiting is the unconditioned response (UCR), the doctor s office is the conditioned stimulus (CS) after being paired with the UCS, and nausea is the conditioned response (CR). Let's assume that the chemotherapy drugs that Moisha takes are given through a syringe injection. After entering the doctor's office, Moisha sees a syringe, and then gets her medication. In addition to the doctor's office, Moisha will learn to associate the syringe with the medication and will respond to syringes with nausea. This is an example of higher-order (or second-order) conditioning, when the conditioned stimulus (the doctor's office) serves to condition another stimulus (the syringe). It is hard to achieve anything above second-order conditioning. For example, if someone rang a bell every time Moisha received a syringe injection of chemotherapy drugs in the doctor's office, Moisha likely will never get sick in response to the bell.

Consider another example of classical conditioning. Let s say you have a cat named Tiger, who is quite spoiled. You keep her food in a separate cabinet, and you also have a special electric can opener that you use only to open cans of cat food. For every meal, Tiger hears the distinctive sound of the electric can opener (zzhzhz) and then gets her food. Tiger quickly learns that when she hears zzhzhz she is about to get fed. What do you think Tiger does when she hears the electric can opener? She will likely get excited and run to where you are preparing her food. This is an example of classical conditioning. In this case, what are the UCS, CS, UCR, and CR?

What if the cabinet holding Tiger s food becomes squeaky? In that case, Tiger hears squeak (the cabinet), zzhzhz (the electric can opener), and then she gets her food. Tiger will learn to get excited when she hears the squeak of the cabinet. Pairing a new neutral stimulus (squeak) with the conditioned stimulus (zzhzhz) is called higher-order conditioning, or second-order conditioning. This means you are using the conditioned stimulus of the can opener to condition another stimulus: the squeaky cabinet (the accompanying source-figure description). It is hard to achieve anything above second-order conditioning. For example, if you ring a bell, open the cabinet (squeak), use the can opener (zzhzhz), and then feed Tiger, Tiger will likely never get excited when hearing the bell alone.

Source illustration description (image not reproduced): A diagram is labeled Higher-Order / Second-Order Conditioning and has three rows. The first row shows an electric can opener labeled conditioned stimulus followed by a plus sign and then a dish of food labeled unconditioned stimulus, followed by an equal sign and a picture of a salivating cat labeled unconditioned response. The second row shows a squeaky cabinet door labeled second-order stimulus followed by a plus sign and then an electric can opener labeled conditioned stimulus, followed by an equal sign and a picture of a salivating cat labeled conditioned response. The third row shows a squeaky cabinet door labeled second-order stimulus followed by an equal sign and a picture of a salivating cat labeled conditioned response. In higher-order conditioning, an established conditioned stimulus is paired with a new neutral stimulus (the second-order stimulus), so that eventually the new stimulus also elicits the conditioned response, without the initial conditioned stimulus being presented.

Everyday Connection

Kate and her spouse recently vacationed in the Cayman Islands, and booked a boat tour to Stingray City, where they could feed and swim with the southern stingrays. The boat captain explained how the normally solitary stingrays have become accustomed to interacting with humans. About 40 years ago, people began to clean fish and conch (unconditioned stimulus) at a particular sandbar near a barrier reef, and large numbers of stingrays would swim in to eat (unconditioned response) what the people threw into the water; this continued for years. By the late 1980s, word of the large group of stingrays spread among scuba divers, who then started feeding them by hand. Over time, the southern stingrays in the area were classically conditioned much like Pavlov s dogs. When they hear the sound of a boat engine (neutral stimulus that becomes a conditioned stimulus), they know that they will get to eat (conditioned response).

As soon as they reached Stingray City, over two dozen stingrays surrounded their tour boat. The couple slipped into the water with bags of squid, the stingrays favorite treat. The swarm of stingrays bumped and rubbed up against their legs like hungry cats (the accompanying source-figure description). Kate was able to feed, pet, and even kiss (for luck) these amazing creatures. Then all the squid was gone, and so were the stingrays.

Source illustration description (image not reproduced): A photograph shows a woman standing in the ocean holding a stingray. Kate holds a southern stingray at Stingray City in the Cayman Islands. These stingrays have been classically conditioned to associate the sound of a boat motor with food provided by tourists. (credit: Kathryn Dumper)

Classical conditioning also applies to humans, even babies. For example, Elan buys formula in blue canisters for their six-month-old daughter, Angelina. Whenever Elan takes out a formula container, Angelina gets excited, tries to reach toward the food, and most likely salivates. Why does Angelina get excited when she sees the formula canister? What are the UCS, CS, UCR, and CR here?

So far, all of the examples have involved food, but classical conditioning extends beyond the basic need to be fed. Consider our earlier example of a dog whose owners install an invisible electric dog fence. A small electrical shock (unconditioned stimulus) elicits discomfort (unconditioned response). When the unconditioned stimulus (shock) is paired with a neutral stimulus (the edge of a yard), the dog associates the discomfort (unconditioned response) with the edge of the yard (conditioned stimulus) and stays within the set boundaries. In this example, the edge of the yard elicits fear and anxiety in the dog. Fear and anxiety are the conditioned response.

General Processes in Classical Conditioning

Now that you know how classical conditioning works and have seen several examples, let s take a look at some of the general processes involved. In classical conditioning, the initial period of learning is known as acquisition, when an organism learns to connect a neutral stimulus and an unconditioned stimulus. During acquisition, the neutral stimulus begins to elicit the conditioned response, and eventually the neutral stimulus becomes a conditioned stimulus capable of eliciting the conditioned response by itself. Timing is important for conditioning to occur. Typically, there should only be a brief interval between presentation of the conditioned stimulus and the unconditioned stimulus. Depending on what is being conditioned, sometimes this interval is as little as five seconds (Chance, 2009). However, with other types of conditioning, the interval can be up to several hours.

Taste aversion is a type of conditioning in which an interval of several hours may pass between the conditioned stimulus (something ingested) and the unconditioned stimulus (nausea or illness). Here s how it works. Between classes, you and a friend grab a quick lunch from a food cart on campus. You share a dish of chicken curry and head off to your next class. A few hours later, you feel nauseous and become ill. Although your friend is fine and you determine that you have intestinal flu (the food is not the culprit), you ve developed a taste aversion; the next time you are at a restaurant and someone orders curry, you immediately feel ill. While the chicken dish is not what made you sick, you are experiencing taste aversion: you ve been conditioned to be averse to a food after a single, bad experience.

How does this occur conditioning based on a single instance and involving an extended time lapse between the event and the negative stimulus? Research into taste aversion suggests that this response

may be an evolutionary adaptation designed to help organisms quickly learn to avoid harmful foods (Garcia & Rusiniak, 1980; Garcia & Koelling, 1966). Not only may this contribute to species survival via natural selection, but it may also help us develop strategies for challenges such as helping cancer patients through the nausea induced by certain treatments (Holmes, 1993; Jacobsen et al., 1993; Hutton, Baracos, & Wismer, 2007; Skolin et al., 2006). Garcia and Koelling (1966) showed not only that taste aversions could be conditioned, but also that there were biological constraints to learning. In their study, separate groups of rats were conditioned to associate either a flavor with illness, or lights and sounds with illness. Results showed that all rats exposed to flavor-illness pairings learned to avoid the flavor, but none of the rats exposed to lights and sounds with illness learned to avoid lights or sounds. This added evidence to the idea that classical conditioning could contribute to species survival by helping organisms learn to avoid stimuli that posed real dangers to health and welfare.

Robert Rescorla demonstrated how powerfully an organism can learn to predict the UCS from the CS. Take, for example, the following two situations. Ari s dad always has dinner on the table every day at 6:00. Soraya s mom switches it up so that some days they eat dinner at 6:00, some days they eat at 5:00, and other days they eat at 7:00. For Ari, 6:00 reliably and consistently predicts dinner, so Ari will likely start feeling hungry every day right before 6:00, even if he's had a late snack. Soraya, on the other hand, will be less likely to associate 6:00 with dinner, since 6:00 does not always predict that dinner is coming. Rescorla, along with his colleague at Yale University, Alan Wagner, developed a mathematical formula that could be used to calculate the probability that an association would be learned given the ability of a conditioned stimulus to predict the occurrence of an unconditioned stimulus and other factors; today this is known as the Rescorla-Wagner model (Rescorla & Wagner, 1972)

Once we have established the connection between the unconditioned stimulus and the conditioned stimulus, how do we break that connection and get the dog, cat, or child to stop responding? In Tiger s case, imagine what would happen if you stopped using the electric can opener for her food and began to use it only for human food. Now, Tiger would hear the can opener, but she would not get food. In classical conditioning terms, you would be giving the conditioned stimulus, but not the unconditioned stimulus. Pavlov explored this scenario in his experiments with dogs: sounding the tone without giving the dogs the meat powder. Soon the dogs stopped responding to the tone. Extinction is the decrease in the conditioned response when the unconditioned stimulus is no longer presented with the conditioned stimulus. When presented with the conditioned stimulus alone, the dog, cat, or other organism would show a weaker and weaker response, and finally no response. In classical conditioning terms, there is a gradual weakening and disappearance of the conditioned response.

What happens when learning is not used for a while when what was learned lies dormant? As we just discussed, Pavlov found that when he repeatedly presented the bell (conditioned stimulus) without the meat powder (unconditioned stimulus), extinction occurred; the dogs stopped salivating to the bell. However, after a couple of hours of resting from this extinction training, the dogs again began to salivate when Pavlov rang the bell. What do you think would happen with Tiger s behavior if your electric can opener broke, and you did not use it for several months? When you finally got it fixed and started using it to open Tiger s food again, Tiger would remember the association between the can opener and her food she would get excited and run to the kitchen when she heard the sound. The behavior of Pavlov s dogs and Tiger illustrates a concept Pavlov called spontaneous recovery: the return of a previously extinguished conditioned response following a rest period (the accompanying source-figure description).

Source illustration description (image not reproduced): A chart has an x-axis labeled time and a y-axis labeled strength of CR; there are four columns of graphed data. The first column is labeled acquisition (CS + UCS) and the line rises steeply from the bottom to the top. The second column is labeled Extinction (CS alone) and the line drops rapidly from the top to the bottom. The third column is labeled Pause and has no line. The fourth column has a line that begins midway and drops sharply to the bottom. At the point where the line begins, it is labeled Spontaneous recovery of CR ; the halfway point on the line is labeled Extinction (CS alone). This is the curve of acquisition, extinction, and spontaneous recovery. The rising curve shows the conditioned response quickly getting stronger through the repeated pairing of the conditioned stimulus and the unconditioned stimulus (acquisition). Then the curve

decreases, which shows how the conditioned response weakens when only the conditioned stimulus is presented (extinction). After a break or pause from conditioning, the conditioned response reappears (spontaneous recovery).

Of course, these processes also apply in humans. For example, let s say that every day when you walk to campus, an ice cream truck passes your route. Day after day, you hear the truck s music (neutral stimulus), so you finally stop and purchase a chocolate ice cream bar. You take a bite (unconditioned stimulus) and then your mouth waters (unconditioned response). This initial period of learning is known as acquisition, when you begin to connect the neutral stimulus (the sound of the truck) and the unconditioned stimulus (the taste of the chocolate ice cream in your mouth). During acquisition, the conditioned response gets stronger and stronger through repeated pairings of the conditioned stimulus and unconditioned stimulus. Several days (and ice cream bars) later, you notice that your mouth begins to water (conditioned response) as soon as you hear the truck s musical jingle even before you bite into the ice cream bar. Then one day you head down the street. You hear the truck s music (conditioned stimulus), and your mouth waters (conditioned response). However, when you get to the truck, you discover that they are all out of ice cream. You leave disappointed. The next few days you pass by the truck and hear the music, but don t stop to get an ice cream bar because you re running late for class. You begin to salivate less and less when you hear the music, until by the end of the week, your mouth no longer waters when you hear the tune. This illustrates extinction. The conditioned response weakens when only the conditioned stimulus (the sound of the truck) is presented, without being followed by the unconditioned stimulus (chocolate ice cream in the mouth). Then the weekend comes. You don t have to go to class, so you don t pass the truck. Monday morning arrives and you take your usual route to campus. You round the corner and hear the truck again. What do you think happens? Your mouth begins to water again. Why? After a break from conditioning, the conditioned response reappears, which indicates spontaneous recovery.

Acquisition and extinction involve the strengthening and weakening, respectively, of a learned association. Two other learning processes stimulus discrimination and stimulus generalization are involved in determining which stimuli will trigger learned responses. Animals (including humans) need to distinguish between stimuli for example, between sounds that predict a threatening event and sounds that do not so that they can respond appropriately (such as running away if the sound is threatening). When an organism learns to respond differently to various stimuli that are similar, it is called stimulus discrimination. In classical conditioning terms, the organism demonstrates the conditioned response only to the conditioned stimulus. Pavlov s dogs discriminated between the basic tone that sounded before they were fed and other tones (e.g., the doorbell), because the other sounds did not predict the arrival of food. Similarly, Tiger, the cat, discriminated between the sound of the can opener and the sound of the electric mixer. When the electric mixer is going, Tiger is not about to be fed, so she does not come running to the kitchen looking for food. In our other example, Moisha, the cancer patient, discriminated between oncologists and other types of doctors. She learned not to feel ill when visiting doctors for other types of appointments, such as her annual physical.

On the other hand, when an organism demonstrates the conditioned response to stimuli that are similar to the condition stimulus, it is called stimulus generalization, the opposite of stimulus discrimination. The more similar a stimulus is to the condition stimulus, the more likely the organism is to give the conditioned response. For instance, if the electric mixer sounds very similar to the electric can opener, Tiger may come running after hearing its sound. But if you do not feed her following the electric mixer sound, and you continue to feed her consistently after the electric can opener sound, she will quickly learn to discriminate between the two sounds (provided they are sufficiently dissimilar that she can tell them apart). In our other example, Moisha continued to feel ill whenever visiting other oncologists or other doctors in the same building as her oncologist.

Behaviorism

John B. Watson, shown in the accompanying source-figure description, is considered the founder of behaviorism. Behaviorism is a school of thought that arose during the first part of the 20th century, which incorporates elements of Pavlov s classical conditioning (Hunt, 2007). In stark contrast with Freud, who considered the reasons for behavior to be hidden in the unconscious, Watson championed the idea that all behavior can be studied as a simple stimulus-response reaction, without regard for internal processes. Watson argued that in order for psychology to become a legitimate science, it must shift its concern away from internal mental processes because mental processes cannot be seen or measured. Instead, he asserted that psychology must focus on outward observable behavior that can be measured.

Source illustration description (image not reproduced): A photograph shows John B. Watson. John B. Watson used the principles of classical conditioning in the study of human emotion.

Watson s ideas were influenced by Pavlov s work. According to Watson, human behavior, just like animal behavior, is primarily the result of conditioned responses. Whereas Pavlov s work with dogs involved the conditioning of reflexes, Watson believed the same principles could be extended to the conditioning of human emotions (Watson, 1919).

In 1920, while chair of the psychology department at Johns Hopkins University, Watson and his graduate student, Rosalie Rayner, conducted research on a baby nicknamed Little Albert. Rayner and Watson s experiments with Little Albert demonstrated how fears can be conditioned using classical conditioning. Through these experiments, Little Albert was exposed to and conditioned to fear certain things. Initially he was presented with various neutral stimuli, including a rabbit, a dog, a monkey, masks, cotton wool, and a white rat. He was not afraid of any of these things. Then Watson, with the help of Rayner, conditioned Little Albert to associate these stimuli with an emotion fear. For example, Watson handed Little Albert the white rat, and Little Albert enjoyed playing with it. Then Watson made a loud sound, by striking a hammer against a metal bar hanging behind Little Albert s head, each time Little Albert touched the rat. Little Albert was frightened by the sound demonstrating a reflexive fear of sudden loud noises and began to cry. Watson repeatedly paired the loud sound with the white rat. Soon Little Albert became frightened by the white rat alone. In this case, what are the UCS, CS, UCR, and CR? Days later, Little Albert demonstrated stimulus generalization he became afraid of other furry things: a rabbit, a furry coat, and even a Santa Claus mask (the accompanying source-figure description). Watson had succeeded in conditioning a fear response in Little Albert, thus demonstrating that emotions could become conditioned responses. It had been Watson s intention to produce a phobia a persistent, excessive fear of a specific object or situation through conditioning alone, thus countering Freud s view that phobias are caused by deep, hidden conflicts in the mind. However, there is no evidence that Little Albert experienced phobias in later years. Little Albert s mother moved away, ending the experiment. While Watson s research provided new insight into conditioning, it would be considered unethical by today s standards.

Source illustration description (image not reproduced): A photograph shows a man wearing a mask with a white beard; his face is close to a baby who is crawling away. A caption reads, Now he fears even Santa Claus. Through stimulus generalization, Little Albert came to fear furry things, including Watson in a Santa Claus mask.

Everyday Connection

Advertising executives are pros at applying the principles of associative learning. Think about the car commercials you have seen on television. Many of them feature an attractive model. By associating the model with the car being advertised, you come to see the car as being desirable (Cialdini, 2008). You may be asking yourself, does this advertising technique actually work? According to Cialdini (2008), men who viewed a car commercial that included an attractive model later rated the car as being faster, more appealing, and better designed than did men who viewed an advertisement for the same car minus the model.

Have you ever noticed how quickly advertisers cancel contracts with a famous athlete following a scandal? As far as the advertiser is concerned, that athlete is no longer associated with positive feelings; therefore, the athlete

cannot be used as an unconditioned stimulus to condition the public to associate positive feelings (the unconditioned response) with their product (the conditioned stimulus).

Now that you are aware of how associative learning works, see if you can find examples of these types of advertisements on television, in magazines, or on the Internet.

Summary

Pavlov s pioneering work with dogs contributed greatly to what we know about learning. His experiments explored the type of associative learning we now call classical conditioning. In classical conditioning, organisms learn to associate events that repeatedly happen together, and researchers study how a reflexive response to a stimulus can be mapped to a different stimulus by training an association between the two stimuli. Pavlov s experiments show how stimulus-response bonds are formed. Watson, the founder of behaviorism, was greatly influenced by Pavlov s work. He tested humans by conditioning fear in an infant known as Little Albert. His findings suggest that classical conditioning can explain how some fears develop.

Review Questions

OpenStax learning exercise

Q. A stimulus that does not initially elicit a response in an organism is a(n) _____.

A. B

OpenStax learning exercise

Q. In Watson and Rayner s experiments, Little Albert was conditioned to fear a white rat, and then he began to be afraid of other furry white objects. This demonstrates _____.

A. D

OpenStax learning exercise

Q. Extinction occurs when _____.

A. A

OpenStax learning exercise

Q. In Pavlov s work with dogs, the psychic secretions were _____.

A. B

Critical Thinking Questions

OpenStax learning exercise

Q. If the sound of your toaster popping up toast causes your mouth to water, what are the UCS, CS, and CR?

A. The food being toasted is the UCS; the sound of the toaster popping up is the CS; salivating to the sound of the toaster is the CR.

OpenStax learning exercise

Q. Explain how the processes of stimulus generalization and stimulus discrimination are considered opposites.

A. In stimulus generalization, an organism responds to new stimuli that are similar to the original conditioned stimulus. For example, a dog barks when the doorbell rings. He then barks when the oven timer dings because it sounds very similar to the doorbell. On the other hand, stimulus discrimination occurs when an organism learns a response to a specific stimulus, but does not respond the same way to new stimuli that are similar. In this case, the dog would bark when he hears the doorbell, but he would not bark when he hears the oven timer ding because they sound different; the dog is able to distinguish between the two sounds.

OpenStax learning exercise

Q. How does a neutral stimulus become a conditioned stimulus?

A. This occurs through the process of acquisition. A human or an animal learns to connect a neutral stimulus and an unconditioned stimulus. During the acquisition phase, the neutral stimulus begins to elicit the conditioned response. The neutral stimulus is becoming the conditioned stimulus. At the end of the acquisition phase, learning has occurred and the neutral stimulus becomes a conditioned stimulus capable of eliciting the conditioned response by itself.

Personal Application Question

OpenStax learning exercise

Q. Can you think of an example in your life of how classical conditioning has produced a positive emotional response, such as happiness or excitement? How about a negative emotional response, such as fear, anxiety, or anger?

Operant Conditioning

OpenStax source module m82201 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82201/index.cnxml

The previous section of this chapter focused on the type of associative learning known as classical conditioning. Remember that in classical conditioning, something in the environment triggers a reflex automatically, and researchers train the organism to react to a different stimulus. Now we turn to the second type of associative learning, operant conditioning. In operant conditioning, organisms learn to associate a behavior and its consequence (the accompanying source-figure description). A pleasant consequence makes that behavior more likely to be repeated in the future. For example, Spirit, a dolphin at the National Aquarium in Baltimore, does a flip in the air when her trainer blows a whistle. The consequence is that she gets a fish.

Source comparison table

Classical and Operant Conditioning Compared		
	Classical Conditioning	Operant Conditioning

Classical and Operant Conditioning Compared		
Conditioning approach	An unconditioned stimulus (such as food) is paired with a neutral stimulus (such as a bell). The neutral stimulus eventually becomes the conditioned stimulus, which brings about the conditioned response (salivation).	The target behavior is followed by reinforcement or punishment to either strengthen or weaken it, so that the learner is more likely to exhibit the desired behavior in the future.
Stimulus timing	The stimulus occurs immediately before the response.	The stimulus (either reinforcement or punishment) occurs soon after the response.

Psychologist B. F. Skinner saw that classical conditioning is limited to existing behaviors that are reflexively elicited, and it doesn t account for new behaviors such as riding a bike. He proposed a theory about how such behaviors come about. Skinner believed that behavior is motivated by the consequences we receive for the behavior: the reinforcements and punishments. His idea that learning is the result of consequences is based on the law of effect, which was first proposed by psychologist Edward Thorndike. According to the law of effect, behaviors that are followed by consequences that are satisfying to the organism are more likely to be repeated, and behaviors that are followed by unpleasant consequences are less likely to be repeated (Thorndike, 1911). Essentially, if an organism does something that brings about a desired result, the organism is more likely to do it again. If an organism does something that does not bring about a desired result, the organism is less likely to do it again. An example of the law of effect is in employment. One of the reasons (and often the main reason) we show up for work is because we get paid to do so. If we stop getting paid, we will likely stop showing up even if we love our job.

Working with Thorndike s law of effect as his foundation, Skinner began conducting scientific experiments on animals (mainly rats and pigeons) to determine how organisms learn through operant conditioning (Skinner, 1938). He placed these animals inside an operant conditioning chamber, which has come to be known as a Skinner box (the accompanying source-figure description). A Skinner box contains a lever (for rats) or disk (for pigeons) that the animal can press or peck for a food reward via the dispenser. Speakers and lights can be associated with certain behaviors. A recorder counts the number of responses made by the animal.

Source illustration description (image not reproduced): A photograph shows B.F. Skinner. An illustration shows a rat in a Skinner box: a chamber with a speaker, lights, a lever, and a food dispenser. (a) B. F. Skinner developed operant conditioning for systematic study of how behaviors are strengthened or weakened according to their consequences. (b) In a Skinner box, a rat presses a lever in an operant conditioning chamber to receive a food reward. (credit a: modification of work by "Silly rabbit"/Wikimedia Commons)

In discussing operant conditioning, we use several everyday words positive, negative, reinforcement, and punishment in a specialized manner. In operant conditioning, positive and negative do not mean good and bad. Instead, positive means you are adding something, and negative means you are taking something away. Reinforcement means you are increasing a behavior, and punishment means you are decreasing a behavior. Reinforcement can be positive or negative, and punishment can also be positive or negative. All reinforcers (positive or negative) increase the likelihood of a behavioral response. All punishers (positive or negative) decrease the likelihood of a behavioral response. Now let s combine these four terms: positive reinforcement, negative reinforcement, positive punishment, and negative punishment (the accompanying source-figure description).

Source comparison table

Positive and Negative Reinforcement and Punishment		
	Reinforcement	Punishment
Positive	Something is added to increase the likelihood of a behavior.	Something is added to decrease the likelihood of a behavior.
Negative	Something is removed to increase the likelihood of a behavior.	Something is removed to decrease the likelihood of a behavior.

Reinforcement

The most effective way to teach a person or animal a new behavior is with positive reinforcement. In positive reinforcement, a desirable stimulus is added to increase a behavior.

For example, you tell your five-year-old son, Jerome, that if he cleans his room, he will get a toy. Jerome quickly cleans his room because he wants a new art set. Let s pause for a moment. Some people might say, Why should I reward my child for doing what is expected? But in fact we are constantly and consistently rewarded in our lives. Our paychecks are rewards, as are high grades and acceptance into our preferred school. Being praised for doing a good job and for passing a driver s test is also a reward. Positive reinforcement as a learning tool is extremely effective. It has been found that one of the most effective ways to increase achievement in school districts with below-average reading scores was to pay the children to read. Specifically, second-grade students in Dallas were paid $2 each time they read a book and passed a short quiz about the book. The result was a significant increase in reading comprehension (Fryer, 2010). What do you think about this program? If Skinner were alive today, he would probably think this was a great idea. He was a strong proponent of using operant conditioning principles to influence students behavior at school. In fact, in addition to the Skinner box, he also invented what he called a teaching machine that was designed to reward small steps in learning (Skinner, 1961) an early forerunner of computer-assisted learning. His teaching machine tested students knowledge as they worked through various school subjects. If students answered questions correctly, they received immediate positive reinforcement and could continue; if they answered incorrectly, they did not receive any reinforcement. The idea was that students would spend additional time studying the material to increase their chance of being reinforced the next time (Skinner, 1961).

In negative reinforcement, an undesirable stimulus is removed to increase a behavior. For example, car manufacturers use the principles of negative reinforcement in their seatbelt systems, which go beep, beep, beep until you fasten your seatbelt. The annoying sound stops when you exhibit the desired behavior, increasing the likelihood that you will buckle up in the future. Negative reinforcement is also used frequently in horse training. Riders apply pressure by pulling the reins or squeezing their legs and then remove the pressure when the horse performs the desired behavior, such as turning or speeding up. The pressure is the negative stimulus that the horse wants to remove.

Punishment

Many people confuse negative reinforcement with punishment in operant conditioning, but they are two very different mechanisms. Remember that reinforcement, even when it is negative, always increases a behavior. In contrast, punishment always decreases a behavior. In positive punishment, you add an undesirable stimulus to decrease a behavior. An example of positive punishment is scolding a student to get the student to stop texting in class. In this case, a stimulus (the reprimand) is added in order to decrease the behavior (texting in class). In negative punishment, you remove a pleasant stimulus to decrease behavior. For example, when a child misbehaves, a parent can take away a favorite toy. In this case, a stimulus (the toy) is removed in order to decrease the behavior.

Punishment, especially when it is immediate, is one way to decrease undesirable behavior. For example, imagine your four-year-old son, Brandon, hit his younger brother. You have Brandon write 100 times I will not hit my brother" (positive punishment). Chances are he won t repeat this behavior. While strategies like this are common today, in the past children were often subject to physical punishment, such as spanking. It s important to be aware of some of the drawbacks in using physical punishment on children. First, punishment may teach fear. Brandon may become fearful of the street, but he also may become fearful of the person who delivered the punishment you, his parent. Similarly, children who are punished by teachers may come to fear the teacher and try to avoid school (Gershoff et al., 2010). Consequently, most schools in the United States have banned corporal punishment. Second, punishment may cause children to become more aggressive and prone to antisocial behavior and delinquency (Gershoff, 2002). They see their parents resort to spanking when they become angry and frustrated, so, in turn, they may act out this same behavior when they become angry and frustrated. For example, because you spank Brenda when you are angry with her for her misbehavior, she might start hitting her friends when they won t share their toys.

While positive punishment can be effective in some cases, Skinner suggested that the use of punishment should be weighed against the possible negative effects. Today s psychologists and parenting experts favor reinforcement over punishment they recommend that you catch your child doing something good and reward them for it.

Shaping

In his operant conditioning experiments, Skinner often used an approach called shaping. Instead of rewarding only the target behavior, in shaping, we reward successive approximations of a target behavior. Why is shaping needed? Remember that in order for reinforcement to work, the organism must first display the behavior. Shaping is needed because it is extremely unlikely that an organism will display anything but the simplest of behaviors spontaneously. In shaping, behaviors are broken down into many small, achievable steps. The specific steps used in the process are the following: Reinforce any response that resembles the desired behavior. Then reinforce the response that more closely resembles the desired behavior. You will no longer reinforce the previously reinforced response. Next, begin to reinforce the response that even more closely resembles the desired behavior. Continue to reinforce closer and closer approximations of the desired behavior. Finally, only reinforce the desired behavior.

Shaping is often used in teaching a complex behavior or chain of behaviors. Skinner used shaping to teach pigeons not only such relatively simple behaviors as pecking a disk in a Skinner box, but also many unusual and entertaining behaviors, such as turning in circles, walking in figure eights, and even playing ping pong; the technique is commonly used by animal trainers today. An important part of shaping is stimulus discrimination. Recall Pavlov s dogs he trained them to respond to the tone of a bell, and not to similar tones or sounds. This discrimination is also important in operant conditioning and in shaping behavior.

It s easy to see how shaping is effective in teaching behaviors to animals, but how does shaping work with humans? Let s consider parents whose goal is to have their child learn to clean his room. They use shaping to help him master steps toward the goal. Instead of performing the entire task, they set up these steps and reinforce each step. First, he cleans up one toy. Second, he cleans up five toys. Third, he chooses whether to pick up ten toys or put his books and clothes away. Fourth, he cleans up everything except two toys. Finally, he cleans his entire room.

Primary and Secondary Reinforcers

Rewards such as stickers, praise, money, toys, and more can be used to reinforce learning. Let s go back to Skinner s rats again. How did the rats learn to press the lever in the Skinner box? They were rewarded with food each time they pressed the lever. For animals, food would be an obvious reinforcer.

What would be a good reinforcer for humans? For your child Chris, it was the promise of a toy when they cleaned their room. How about Sydney, the soccer player? If you gave Sydney a piece of candy every time Sydney scored a goal, you would be using a primary reinforcer. Primary reinforcers are reinforcers that have innate reinforcing qualities. These kinds of reinforcers are not learned. Water, food, sleep, shelter, sex, and touch, among others, are primary reinforcers. Pleasure is also a primary reinforcer. Organisms do not lose their drive for these things. For most people, jumping in a cool lake on a very hot day would be reinforcing and the cool lake would be innately reinforcing the water would cool the person off (a physical need), as well as provide pleasure.

A secondary reinforcer has no inherent value and only has reinforcing qualities when linked with a primary reinforcer. Praise, linked to affection, is one example of a secondary reinforcer, as when you called out Great shot! every time Sydney made a goal. Another example, money, is only worth something when you can use it to buy other things either things that satisfy basic needs (food, water, shelter all primary reinforcers) or other secondary reinforcers. If you were on a remote island in the middle of the Pacific Ocean and you had stacks of money, the money would not be useful if you could not spend it. What about the stickers on the behavior chart? They also are secondary reinforcers.

Sometimes, instead of stickers on a sticker chart, a token is used. Tokens, which are also secondary reinforcers, can then be traded in for rewards and prizes. Entire behavior management systems, known as token economies, are built around the use of these kinds of token reinforcers. Token economies have been found to be very effective at modifying behavior in a variety of settings such as schools, prisons, and mental hospitals. For example, a study by Adibsereshki and Abkenar (2014) found that use of a token economy increased appropriate social behaviors and reduced inappropriate behaviors in a group of eight grade students. Similar studies show demonstrable gains on behavior and academic achievement for groups ranging from first grade to high school, and representing a wide array of abilities and disabilities. For example, during studies involving younger students, when children in the study exhibited appropriate behavior (not hitting or pinching), they received a quiet hands token. When they hit or pinched, they lost a token. The children could then exchange specified amounts of tokens for minutes of playtime.

Everyday Connection

Parents and teachers often use behavior modification to change a child s behavior. Behavior modification uses the principles of operant conditioning to accomplish behavior change so that undesirable behaviors are switched for more socially acceptable ones. Some teachers and parents create a sticker chart, in which several behaviors are listed (the accompanying source-figure description). Sticker charts are a form of token economies, as described in the text. Each time children perform the behavior, they get a sticker, and after a certain number of stickers, they get a prize, or reinforcer. The goal is to increase acceptable behaviors and decrease misbehavior. Remember, it is best to reinforce desired behaviors, rather than to use punishment. In the classroom, the teacher can reinforce a wide range of behaviors, from students raising their hands, to walking quietly in the hall, to turning in their homework. At home, parents might create a behavior chart that rewards children for things such as putting away toys, brushing their teeth, and helping with dinner. In order for behavior modification to be effective, the reinforcement needs to be connected with the behavior; the reinforcement must matter to the child and be done consistently.

Source illustration description (image not reproduced): A photograph shows a child placing stickers on a chart hanging on the wall. Sticker charts are a form of positive reinforcement and a tool for behavior modification. Once this child earns a certain number of stickers for demonstrating a desired behavior, she will be rewarded with a trip to the ice cream parlor. (credit: Abigail Batchelder)

Time-out is another popular technique used in behavior modification with children. It operates on the principle of negative punishment. When a child demonstrates an undesirable behavior, they are removed from the desirable activity at hand (the accompanying source-figure description). For example, say that Sophia and her brother Mario are playing with building blocks. Sophia throws some blocks at her brother, so you give her a warning that she will go to time-out if she does it again. A few minutes later, she throws more blocks at Mario. You remove Sophia from the room for a few minutes. When she comes back, she doesn t throw blocks.

There are several important points that you should know if you plan to implement time-out as a behavior modification technique. First, make sure the child is being removed from a desirable activity and placed in a less desirable location. If the activity is something undesirable for the child, this technique will backfire because it is more enjoyable for the child to be removed from the activity. Second, the length of the time-out is important. The general rule of thumb is one minute for each year of the child s age. Sophia is five; therefore, she sits in a time-out for five minutes. Setting a timer helps children know how long they have to sit in time-out. Finally, as a caregiver, keep several guidelines in mind over the course of a time-out: remain calm when directing your child to time-out; ignore your child during time-out (because caregiver attention may reinforce misbehavior); and give the child a hug or a kind word when time-out is over.

Source illustration description (image not reproduced): Photograph A shows several children climbing on playground equipment. Photograph B shows a child sitting alone on a bench. Time-out is a popular form of negative punishment used by caregivers. When a child misbehaves, they are removed from a desirable activity in an effort to decrease the unwanted behavior. For example, (a) a child might be playing on the playground with friends and push another child; (b) the child who misbehaved would then be removed from the activity for a short period of time. (credit a: modification of work by Simone Ramella; credit b: modification of work by Spring Dew /Flickr)

Reinforcement Schedules

Remember, the best way to teach a person or animal a behavior is to use positive reinforcement. For example, Skinner used positive reinforcement to teach rats to press a lever in a Skinner box. At first, the rat might randomly hit the lever while exploring the box, and out would come a pellet of food. After eating the pellet, what do you think the hungry rat did next? It hit the lever again, and received another pellet of food. Each time the rat hit the lever, a pellet of food came out. When an organism receives a reinforcer each time it displays a behavior, it is called continuous reinforcement. This reinforcement schedule is the quickest way to teach someone a behavior, and it is especially effective in training a new behavior. Let s look back at the dog that was learning to sit earlier in the chapter. Now, each time he sits, you give him a treat. Timing is important here: you will be most successful if you present the reinforcer immediately after he sits, so that he can make an association between the target behavior (sitting) and the consequence (getting a treat).

Once a behavior is trained, researchers and trainers often turn to another type of reinforcement schedule partial reinforcement. In partial reinforcement, also referred to as intermittent reinforcement, the person or animal does not get reinforced every time they perform the desired behavior. There are several different types of partial reinforcement schedules (the accompanying source-figure description). These schedules are described as either fixed or variable, and as either interval or ratio. Fixed refers to the number of responses between reinforcements, or the amount of time between reinforcements, which is set and unchanging. Variable refers to the number of responses or amount of time between reinforcements, which varies or changes. Interval means the schedule is based on the time between reinforcements, and ratio means the schedule is based on the number of responses between reinforcements.

Source comparison table

Reinforcement Schedules			
Reinforcement Schedule	Description	Result	Example
Fixed interval	Reinforcement is delivered at predictable time intervals (e.g., after 5, 10, 15, and 20 minutes).	Moderate response rate with significant pauses after reinforcement	Hospital patient uses patient-controlled, doctor-timed pain relief

Reinforcement Schedules			
Variable interval	Reinforcement is delivered at unpredictable time intervals (e.g., after 5, 7, 10, and 20 minutes).	Moderate yet steady response rate	Checking social media
Fixed ratio	Reinforcement is delivered after a predictable number of responses (e.g., after 2, 4, 6, and 8 responses).	High response rate with pauses after reinforcement	Piecework factory worker getting paid for every x number of items manufactured
Variable ratio	Reinforcement is delivered after an unpredictable number of responses (e.g., after 1, 4, 5, and 9 responses).	High and steady response rate	Gambling

Now let s combine these four terms. A fixed interval reinforcement schedule is when behavior is rewarded after a set amount of time. For example, June undergoes major surgery in a hospital. During recovery, they are expected to experience pain and will require prescription medications for pain relief. June is given an IV drip with a patient-controlled painkiller. Their doctor sets a limit: one dose per hour. June pushes a button when pain becomes difficult, and they receive a dose of medication. Since the reward (pain relief) only occurs on a fixed interval, there is no point in exhibiting the behavior when it will not be rewarded.

With a variable interval reinforcement schedule, the person or animal gets the reinforcement based on varying amounts of time, which are unpredictable. Say that Manuel is the manager at a fast-food restaurant. Every once in a while someone from the quality control division comes to Manuel s restaurant. If the restaurant is clean and the service is fast, everyone on that shift earns a $20 bonus. Manuel never knows when the quality control person will show up, so he always tries to keep the restaurant clean and ensures that his employees provide prompt and courteous service. His productivity regarding prompt service and keeping a clean restaurant are steady because he wants his crew to earn the bonus.

With a fixed ratio reinforcement schedule, there are a set number of responses that must occur before the behavior is rewarded. Carla sells glasses at an eyeglass store, and she earns a commission every time she sells a pair of glasses. She always tries to sell people more pairs of glasses, including prescription sunglasses or a backup pair, so she can increase her commission. She does not care if the person really needs the prescription sunglasses, Carla just wants her bonus. The quality of what Carla sells does not matter because her commission is not based on quality; it s only based on the number of pairs sold. This distinction in the quality of performance can help determine which reinforcement method is most appropriate for a particular situation. Fixed ratios are better suited to optimize the quantity of output, whereas a fixed interval, in which the reward is not quantity based, can lead to a higher quality of output.

In a variable ratio reinforcement schedule, the number of responses needed for a reward varies. This is the most powerful partial reinforcement schedule. An example of the variable ratio reinforcement schedule is gambling. Imagine that Sarah generally a smart, thrifty woman visits Las Vegas for the first time. She is not a gambler, but out of curiosity she puts a quarter into the slot machine, and then another, and another. Nothing happens. Two dollars in quarters later, her curiosity is fading, and she is just about to quit. But then, the machine lights up, bells go off, and Sarah gets 50 quarters back. That s more like it! Sarah gets back to inserting quarters with renewed interest, and a few minutes later she has used up all her gains and is $10 in the hole. Now might be a sensible time to quit. And yet, she keeps putting money into the slot machine because she never knows when the next reinforcement is coming. She keeps thinking that with the next quarter she could win $50, or $100, or even more.

Because the reinforcement schedule in most types of gambling has a variable ratio schedule, people keep trying and hoping that the next time they will win big. This is one of the reasons that gambling is so addictive and so resistant to extinction.

In operant conditioning, extinction of a reinforced behavior occurs at some point after reinforcement stops, and the speed at which this happens depends on the reinforcement schedule. In a variable ratio schedule, the point of extinction comes very slowly, as described above. But in the other reinforcement schedules, extinction may come quickly. For example, if June presses the button for the pain relief medication before the allotted time the doctor has approved, no medication is administered. They are on a fixed interval reinforcement schedule (dosed hourly), so extinction occurs quickly when reinforcement doesn t come at the expected time. Among the reinforcement schedules, variable ratio is the most productive and the most resistant to extinction. Fixed interval is the least productive and the easiest to extinguish (the accompanying source-figure description).

Source illustration description (image not reproduced): A graph has an x-axis labeled Time and a y-axis labeled Cumulative number of responses. Two lines labeled Variable Ratio and Fixed Ratio have similar, steep slopes. The variable ratio line remains straight and is marked in random points where reinforcement occurs. The fixed ratio line has consistently spaced marks indicating where reinforcement has occurred, but after each reinforcement, there is a small drop in the line before it resumes its overall slope. Two lines labeled Variable Interval and Fixed Interval have similar slopes at roughly a 45-degree angle. The variable interval line remains straight and is marked in random points where reinforcement occurs. The fixed interval line has consistently spaced marks indicating where reinforcement has occurred, but after each reinforcement, there is a drop in the line. The four reinforcement schedules yield different response patterns. The variable ratio schedule is unpredictable and yields high and steady response rates, with little if any pause after reinforcement (e.g., gambler). A fixed ratio schedule is predictable and produces a high response rate, with a short pause after reinforcement (e.g., eyeglass saleswoman). The variable interval schedule is unpredictable and produces a moderate, steady response rate (e.g., restaurant manager). The fixed interval schedule yields a scallop-shaped response pattern, reflecting a significant pause after reinforcement (e.g., surgery patient).

Connect the Concepts

Skinner (1953) stated, If the gambling establishment cannot persuade a patron to turn over money with no return, it may achieve the same effect by returning part of the patron's money on a variable-ratio schedule (p. 397).

Skinner uses gambling as an example of the power of the variable-ratio reinforcement schedule for maintaining behavior even during long periods without any reinforcement. In fact, Skinner was so confident in his knowledge of gambling addiction that he even claimed he could turn a pigeon into a pathological gambler (Skinner s Utopia, 1971). It is indeed true that variable-ratio schedules keep behavior quite persistent just imagine the frequency of a child s tantrums if a parent gives in even once to the behavior. The occasional reward makes it almost impossible to stop the behavior.

Recent research in rats has failed to support Skinner s idea that training on variable-ratio schedules alone causes pathological gambling (Laskowski et al., 2019). However, other research suggests that gambling does seem to work on the brain in the same way as most addictive drugs, and so there may be some combination of brain chemistry and reinforcement schedule that could lead to problem gambling (the accompanying source-figure description). Specifically, modern research shows the connection between gambling and the activation of the reward centers of the brain that use the neurotransmitter (brain chemical) dopamine (Murch & Clark, 2016). Interestingly, gamblers don t even have to win to experience the rush of dopamine in the brain. Near misses, or almost winning but not actually winning, also have been shown to increase activity in the ventral striatum and other brain reward centers that use dopamine (Chase & Clark, 2010). These brain effects are almost identical to those produced by addictive drugs like cocaine and heroin (Murch & Clark, 2016). Based on the neuroscientific evidence showing these similarities, the DSM-5 now considers gambling an addiction, while earlier versions of the DSM classified gambling as an impulse control disorder.

Source illustration description (image not reproduced): A photograph shows four digital gaming machines. Some research suggests that pathological gamblers use gambling to compensate for abnormally low levels of the hormone norepinephrine, which is associated with stress and is secreted in moments of arousal and thrill. (credit: Ted Murphy)

In addition to dopamine, gambling also appears to involve other neurotransmitters, including norepinephrine and serotonin (Potenza, 2013). Norepinephrine is secreted when a person feels stress, arousal, or thrill. It may be that pathological gamblers use gambling to increase their levels of this neurotransmitter. Deficiencies in serotonin might also contribute to compulsive behavior, including a gambling addiction (Potenza, 2013).

It may be that pathological gamblers brains are different than those of other people, and perhaps this difference may somehow have led to their gambling addiction, as these studies seem to suggest. However, it is very difficult to ascertain the cause because it is impossible to conduct a true experiment (it would be unethical to try to turn randomly assigned participants into problem gamblers). Therefore, it may be that causation actually moves in the opposite direction perhaps the act of gambling somehow changes neurotransmitter levels in some gamblers brains. It also is possible that some overlooked factor, or confounding variable, played a role in both the gambling addiction and the differences in brain chemistry.

Cognition and Latent Learning

Strict behaviorists like Watson and Skinner focused exclusively on studying behavior rather than cognition (such as thoughts and expectations). In fact, Skinner was such a staunch believer that cognition didn't matter that his ideas were considered radical behaviorism. Skinner considered the mind a "black box" something completely unknowable and, therefore, something not to be studied. However, another behaviorist, Edward C. Tolman, had a different opinion. Tolman s experiments with rats demonstrated that organisms can learn even if they do not receive immediate reinforcement (Tolman & Honzik, 1930; Tolman, Ritchie, & Kalish, 1946). This finding was in conflict with the prevailing idea at the time that reinforcement must be immediate in order for learning to occur, thus suggesting a cognitive aspect to learning.

In the experiments, Tolman placed hungry rats in a maze with no reward for finding their way through it. He also studied a comparison group that was rewarded with food at the end of the maze. As the unreinforced rats explored the maze, they developed a cognitive map: a mental picture of the layout of the maze (the accompanying source-figure description). After 10 sessions in the maze without reinforcement, food was placed in a goal box at the end of the maze. As soon as the rats became aware of the food, they were able to find their way through the maze quickly, just as quickly as the comparison group, which had been rewarded with food all along. This is known as latent learning: learning that occurs but is not observable in behavior until there is a reason to demonstrate it.

Source illustration description (image not reproduced): An illustration shows three rats in a maze, with a starting point and food at the end. Psychologist Edward Tolman found that rats use cognitive maps to navigate through a maze. Have you ever worked your way through various levels on a video game? You learned when to turn left or right, move up or down. In that case you were relying on a cognitive map, just like the rats in a maze. (credit: modification of work by "FutUndBeidl"/Flickr)

Latent learning also occurs in humans. Children may learn by watching the actions of their parents but only demonstrate it at a later date, when the learned material is needed. For example, suppose that Ravi s dad drives him to school every day. In this way, Ravi learns the route from his house to his school, but he s never driven there himself, so he has not had a chance to demonstrate that he s learned the way. One morning Ravi s dad has to leave early for a meeting, so he can t drive Ravi to school. Instead, Ravi follows the same route on his bike that his dad would have taken in the car. This demonstrates latent learning. Ravi had learned the route to school, but had no need to demonstrate this knowledge earlier.

Everyday Connection

Have you ever gotten lost in a building and couldn t find your way back out? While that can be frustrating, you re not alone. At one time or another we ve all gotten lost in places like a museum, hospital, or university library. Whenever we go someplace new, we build a mental representation or cognitive map of the location, as Tolman s rats built a cognitive map of their maze. However, some buildings are confusing because they include many areas that look alike or have short lines of sight. Because of this, it s often difficult to predict what s around a corner or decide whether to turn left or right to get out of a building. Psychologist Laura Carlson (2010) suggests that what we place in our cognitive map can impact our success in navigating through the environment. She suggests that paying attention to specific features upon entering a building, such as a picture on the wall, a fountain, a statue, or an escalator, adds information to our cognitive map that can be used later to help find our way out of the building.

Summary

Operant conditioning is based on the work of B. F. Skinner. Operant conditioning is a form of learning in which the motivation for a behavior happens after the behavior is demonstrated. An animal or a human receives a consequence after performing a specific behavior. The consequence is either a reinforcer or a punisher. All reinforcement (positive or negative) increases the likelihood of a behavioral response. All punishment (positive or negative) decreases the likelihood of a behavioral response. Several types of reinforcement schedules are used to reward behavior depending on either a set or variable period of time.

Review Questions

OpenStax learning exercise

Q. _____ is when you take away a pleasant stimulus to stop a behavior.

A. D

OpenStax learning exercise

Q. Which of the following is not an example of a primary reinforcer?

A. B

OpenStax learning exercise

Q. Rewarding successive approximations toward a target behavior is _____.

A. A

OpenStax learning exercise

Q. Slot machines reward gamblers with money according to which reinforcement schedule?

A. B

Critical Thinking Questions

OpenStax learning exercise

Q. What is a Skinner box and what is its purpose?

A. A Skinner box is an operant conditioning chamber used to train animals such as rats and pigeons to perform certain behaviors, like pressing a lever. When the animals perform the desired behavior, they receive a reward: food or water.

OpenStax learning exercise

Q. What is the difference between negative reinforcement and punishment?

A. In negative reinforcement you are taking away an undesirable stimulus in order to increase the frequency of a certain behavior (e.g., buckling your seat belt stops the annoying beeping sound in your car and increases the likelihood that you will wear your seatbelt). Punishment is designed to reduce a behavior (e.g., you scold your child for running into the street in order to decrease the unsafe behavior.)

OpenStax learning exercise

Q. What is shaping and how would you use shaping to teach a dog to roll over?

A. Shaping is an operant conditioning method in which you reward closer and closer approximations of the desired behavior. If you want to teach your dog to roll over, you might reward him first when he sits, then when he lies down, and then when he lies down and rolls onto his back. Finally, you would reward him only when he completes the entire sequence: lying down, rolling onto his back, and then continuing to roll over to his other side.

Personal Application Questions

OpenStax learning exercise

Q. Explain the difference between negative reinforcement and punishment, and provide several examples of each based on your own experiences.

OpenStax learning exercise

Q. Think of a behavior that you have that you would like to change. How could you use behavior modification, specifically positive reinforcement, to change your behavior? What is your positive reinforcer?

Observational Learning (Modeling)

OpenStax source module m82202 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82202/index.cnxml

Previous sections of this chapter focused on classical and operant conditioning, which are forms of associative learning. In observational learning, we learn by watching others and then imitating, or modeling, what they do or say. For instance, have you ever gone to YouTube to find a video showing you how to do something? The individuals performing the imitated behavior are called models. Research suggests that this imitative learning involves a specific type of neuron, called a mirror neuron (Hickock, 2010; Rizzolatti, Fadiga, Fogassi, & Gallese, 2002; Rizzolatti, Fogassi, & Gallese, 2006).

Humans and other animals are capable of observational learning. As you will see, the phrase monkey see, monkey do really is accurate (the accompanying source-figure description). The same could be said about other animals. For example, in a study of social learning in chimpanzees, researchers gave juice boxes with straws to two groups of captive chimpanzees. The first group dipped the straw into the juice box, and then sucked on the small amount of juice at the end of the straw. The second group sucked through the straw directly, getting much more juice. When the first group, the dippers, observed the second group, the suckers, what do you think happened? All of the dippers in the first group switched to sucking through the straws directly. By simply observing the other chimps and modeling their behavior, they learned that this was a more efficient method of getting juice (Yamamoto, Humle, and Tanaka, 2013).

Source illustration description (image not reproduced): A photograph shows a person drinking from a water bottle, and a monkey next to the person drinking water from a bottle in the same manner. This spider monkey learned to drink water from a plastic bottle by seeing the behavior modeled by a human. (credit: U.S. Air Force, Senior Airman Kasey Close)

Imitation is much more obvious in humans, but is imitation really the sincerest form of flattery? Consider Claire s experience with observational learning. Claire s nine-year-old son, Jay, was getting into trouble at school and was defiant at home. Claire feared that Jay would end up like her brothers, two of whom were in prison. One day, after yet another bad day at school and another negative note

from the teacher, Claire, at her wit s end, beat her son with a belt to get him to behave. Later that night, as she put her children to bed, Claire witnessed her four-year-old daughter, Anna, take a belt to her teddy bear and whip it. Claire was horrified, realizing that Anna was imitating her mother. It was then that Claire knew she wanted to discipline her children in a different manner.

Like Tolman, whose experiments with rats suggested a cognitive component to learning, psychologist Albert Bandura s ideas about learning were different from those of strict behaviorists. Bandura and other researchers proposed a brand of behaviorism called social learning theory, which took cognitive processes into account. According to Bandura, pure behaviorism could not explain why learning can take place in the absence of external reinforcement. He felt that internal mental states must also have a role in learning and that observational learning involves much more than imitation. In imitation, a person simply copies what the model does. Observational learning is much more complex. According to Lefran ois (2012) there are several ways that observational learning can occur: You learn a new response. After watching your coworker get chewed out by your boss for coming in late, you start leaving home 10 minutes earlier so that you won t be late. You choose whether or not to imitate the model depending on what you saw happen to the model. Remember Julian and his father? When learning to surf, Julian might watch how his father pops up successfully on his surfboard and then attempt to do the same thing. On the other hand, Julian might learn not to touch a hot stove after watching his father get burned on a stove. You learn a general rule that you can apply to other situations.

Bandura identified three kinds of models: live, verbal, and symbolic. A live model demonstrates a behavior in person, as when Ben stood up on his surfboard so that Julian could see how he did it. A verbal instructional model does not perform the behavior, but instead explains or describes the behavior, as when a soccer coach tells his young players to kick the ball with the side of the foot, not with the toe. A symbolic model can be fictional characters or real people who demonstrate behaviors in books, movies, television shows, video games, or Internet sources (the accompanying source-figure description).

Source illustration description (image not reproduced): Photograph A shows a yoga instructor demonstrating a yoga pose while a group of students observes her and copies the pose. Photo B shows a child watching television. (a) Yoga students learn by observation as their yoga instructor demonstrates the correct stance and movement for her students (live model). (b) Models don t have to be present for learning to occur: through symbolic modeling, this child can learn a behavior by watching someone demonstrate it on television. (credit a: modification of work by Tony Cecala; credit b: modification of work by Andrew Hyde)

Steps in the Modeling Process

Of course, we don t learn a behavior simply by observing a model. Bandura described specific steps in the process of modeling that must be followed if learning is to be successful: attention, retention, reproduction, and motivation. First, you must be focused on what the model is doing you have to pay attention. Next, you must be able to retain, or remember, what you observed; this is retention. Then, you must be able to perform the behavior that you observed and committed to memory; this is reproduction. Finally, you must have motivation. You need to want to copy the behavior, and whether or not you are motivated depends on what happened to the model. If you saw that the model was reinforced for their behavior, you will be more motivated to copy them. This is known as vicarious reinforcement. On the other hand, if you observed the model being punished, you would be less motivated to copy them. This is called vicarious punishment. For example, imagine that four-year-old Allison watched her older sister Kaitlyn playing in their mother s makeup, and then saw Kaitlyn get a time out when their mother came in. After their mother left the room, Allison was tempted to play in the make-up, but she did not want to get a time-out from her mother. What do you think she did? Once you actually demonstrate the new behavior, the reinforcement you receive plays a part in whether or not you will repeat the behavior.

Bandura researched modeling behavior, particularly children s modeling of adults aggressive and violent behaviors (Bandura, Ross, & Ross, 1961). He conducted an experiment with a five-foot inflatable

doll that he called a Bobo doll. In the experiment, children s aggressive behavior was influenced by whether the teacher was punished for her behavior. In one scenario, a teacher acted aggressively with the doll, hitting, throwing, and even punching the doll, while a child watched. There were two types of responses by the children to the teacher s behavior. When the teacher was punished for her bad behavior, the children decreased their tendency to act as she had. When the teacher was praised or ignored (and not punished for her behavior), the children imitated what she did, and even what she said. They punched, kicked, and yelled at the doll.

What are the implications of this study? Bandura concluded that we watch and learn, and that this learning can have both prosocial and antisocial effects. Prosocial (positive) models can be used to encourage socially acceptable behavior. Parents in particular should take note of this finding. If you want your children to read, then read to them. Let them see you reading. Keep books in your home. Talk about your favorite books. If you want your children to be healthy, then let them see you eat right and exercise, and spend time engaging in physical fitness activities together. The same holds true for qualities like kindness, courtesy, and honesty. The main idea is that children observe and learn from their parents, even their parents morals, so be consistent and toss out the old adage Do as I say, not as I do, because children tend to copy what you do instead of what you say. Besides parents, many public figures, such as Martin Luther King, Jr. and Mahatma Gandhi, are viewed as prosocial models who are able to inspire global social change. Can you think of someone who has been a prosocial model in your life?

The antisocial effects of observational learning are also worth mentioning. As you saw from the example of Claire at the beginning of this section, her daughter viewed Claire s aggressive behavior and copied it. Research suggests that this may help to explain why victims of abuse often grow up to be abusers themselves (Murrell, Christoff, & Henning, 2007). In fact, about 30% of child abuse victims become abusive parents (U.S. Department of Health & Human Services, 2013). We tend to do what we know. Children who grow up witnessing their parents deal with anger and frustration through violent and aggressive acts often learn to behave in that manner themselves.

Some studies suggest that violent television shows, movies, and video games may also have antisocial effects (the accompanying source-figure description) although further research needs to be done to understand the correlational and causational aspects of media violence and behavior. Some studies have found a link between viewing violence and aggression seen in children (Anderson & Gentile, 2008; Kirsch, 2010; Miller, Grabell, Thomas, Bermann, & Graham-Bermann, 2012). These findings may not be surprising, given that a child graduating from high school has been exposed to around 200,000 violent acts including murder, robbery, torture, bombings, beatings, and rape through various forms of media (Huston et al., 1992). Not only might viewing media violence affect aggressive behavior by teaching people to act that way in real life situations, but it has also been suggested that repeated exposure to violent acts also desensitizes people to it. Psychologists are working to understand this dynamic.

Source illustration description (image not reproduced): A photograph shows two children playing a video game and pointing a gun-like object toward a screen. Can viewing violent media make us violent? Psychological researchers study this topic. (credit: "woodleywonderworks"/Flickr)

What Do You Think?

Does watching violent media or playing violent video games cause aggression? Albert Bandura's early studies suggested television violence increased aggression in children, and more recent studies support these findings. For example, research by Craig Anderson and colleagues (Anderson, Bushman, Donnerstein, Hummer, & Warburton, 2015; Anderson et al., 2010; Bushman et al., 2016) found extensive evidence to suggest a causal link between hours of exposure to violent media and aggressive thoughts and behaviors. However, studies by Christopher Ferguson and others suggests that while there may be a link between violent media exposure and aggression, research to date has not accounted for other risk factors for aggression including mental health and family life (Ferguson, 2011; Gentile, 2016). What do think?

Summary

According to Bandura, learning can occur by watching others and then modeling what they do or say. This is known as observational learning. There are specific steps in the process of modeling that must be followed if learning is to be successful. These steps include attention, retention, reproduction, and motivation. Through modeling, Bandura has shown that children learn many things both good and bad simply by watching their parents, siblings, and others.

Review Questions

OpenStax learning exercise

Q. The person who performs a behavior that serves as an example is called a _____.

A. B

OpenStax learning exercise

Q. In Bandura s Bobo doll study, when the children who watched the aggressive model were placed in a room with the doll and other toys, they _____.

A. D

OpenStax learning exercise

Q. Which is the correct order of steps in the modeling process?

A. A

OpenStax learning exercise

Q. Who proposed observational learning?

A. C

Critical Thinking Questions

OpenStax learning exercise

Q. What is the effect of prosocial modeling and antisocial modeling?

A. Prosocial modeling can prompt others to engage in helpful and healthy behaviors, while antisocial modeling can prompt others to engage in violent, aggressive, and unhealthy behaviors.

OpenStax learning exercise

Q. Cara is 17 years old. Cara s mother and father both drink alcohol every night. They tell Cara that drinking is bad and she shouldn t do it. Cara goes to a party where beer is being served. What do you think Cara will do? Why?

A. Cara is more likely to drink at the party because she has observed her parents drinking regularly. Children tend to follow what a parent does rather than what they say.

Personal Application Question

OpenStax learning exercise

Q. What is something you have learned how to do after watching someone else?

CHAPTER 09 / CAC.901

Language, thought, and problem solving

Lecture: Understanding the task before judging the child

A child s answer can reveal more about the task than a score reveals. Suppose an adult asks, What would you do if you felt angry? A child may understand the emotion but struggle with the open-ended question, the expected social answer, or the language in which it is asked. Another child may provide a fluent answer without being able to use the strategy in daily life. Counselors should examine the demands of communication before drawing conclusions about capacity.

Cognitive development includes how children represent information, direct attention, solve problems, and use language. These processes are connected but not identical. A child may have strong vocabulary and weak planning, or strong practical reasoning and limited access to the language of testing. A developmental account looks for a pattern rather than converting one performance into a general judgment.

Language is participation

Language allows children to describe experience, ask for help, remember events, and negotiate relationships. Multilingual children may distribute their vocabulary and experience across languages. A word known at home may not be known in the school language, and an emotional experience may be expressed more naturally in one language than another. Assessment through only one language can underestimate what a child knows.

Use interpreters or appropriately qualified bilingual professionals when needed, rather than relying on a sibling to translate sensitive material. Ask about the child s language history and current opportunities. Distinguish difference from disorder through appropriate assessment, not by assuming that an accent or grammatical pattern reflects limited thought. Respect the family s language while ensuring that the child can communicate privately and safely.

In counseling, concrete aids can reduce language demands. A timeline, drawing, emotion card, or sequence of pictures can support discussion. These tools do not guarantee accuracy. A child may select a card because it is nearest or because they think the adult expects it. Offer choices, allow I don t know, and check meaning in more than one way.

Working memory and instructions

Working memory supports holding and using information over a short period. A long sequence of instructions can overload a child even when the child understands each part. Reduce unnecessary load by giving one or two steps, using visual prompts, and asking the child to explain the plan. This is not lowering expectations. It is making the route to the task accessible.

Executive demands include planning, inhibition, shifting, and monitoring. These skills develop over time and vary with context. A child may organize a favorite activity well but struggle with a task that feels meaningless or threatening. Counselors should not infer a global executive deficit from one setting. Examine demands, motivation, teaching, and opportunities to practice.

Problem solving as a shared process

Adults often move too quickly from a child s problem to an adult solution. A collaborative process begins by defining the problem in terms the child recognizes. You need to stop being lazy is not a workable definition. You want to finish the project, but starting feels confusing after school creates a problem that can be solved together.

Generate several possibilities before choosing one. Consider what might make the plan difficult, and identify a small first step. Review what happened without blame. This process teaches a transferable skill and provides information about barriers. It also models that plans can be revised rather than treated as tests of character.

Figure 9.1 shows the path from task demands to performance, with language, memory, emotion, and support influencing the route. The figure helps counselors ask where a breakdown occurs. Is the child unable to understand, unable to remember the sequence, afraid to attempt, or unsupported in organizing the task? Different answers suggest different supports.

Application: the child who knows but doesn t do

A fictional eleven-year-old, Musa, can explain how to complete a science assignment during counseling but rarely completes it at home. His caregiver sees deliberate avoidance. A task analysis reveals that the assignment requires choosing a topic, locating materials, reading several pages, and recording observations over a week. Musa struggles at the first planning step, then avoids the rest.

The intervention can use a visible planning sheet, a brief check-in, and a prepared set of materials. Musa can choose among topics that meet the learning objective. The counselor and caregiver monitor whether the support helps him begin and whether he gradually takes more responsibility. The goal is not permanent adult management; it is access to a process he can learn.

This example also shows why competence and performance should be distinguished. A child may possess part of a skill but lack the supports needed to coordinate it. Conversely, repeated adult completion of a task may hide a child s need for guided practice. Review the level of assistance and fade it thoughtfully when the child is ready.

Doctoral seminar task

Choose a common counseling or school task and list its hidden cognitive and language demands. Include understanding instructions, remembering steps, interpreting social expectations, and communicating an answer. Redesign the task in two ways: one that reduces unnecessary demands without changing the intended skill, and one that teaches the skill through supported practice.

Then explain how you would evaluate whether the redesign improves access rather than simply making the task easier. Include the child s perspective and an indicator of independent use. The exercise trains counselors to see performance as an interaction between person and task.

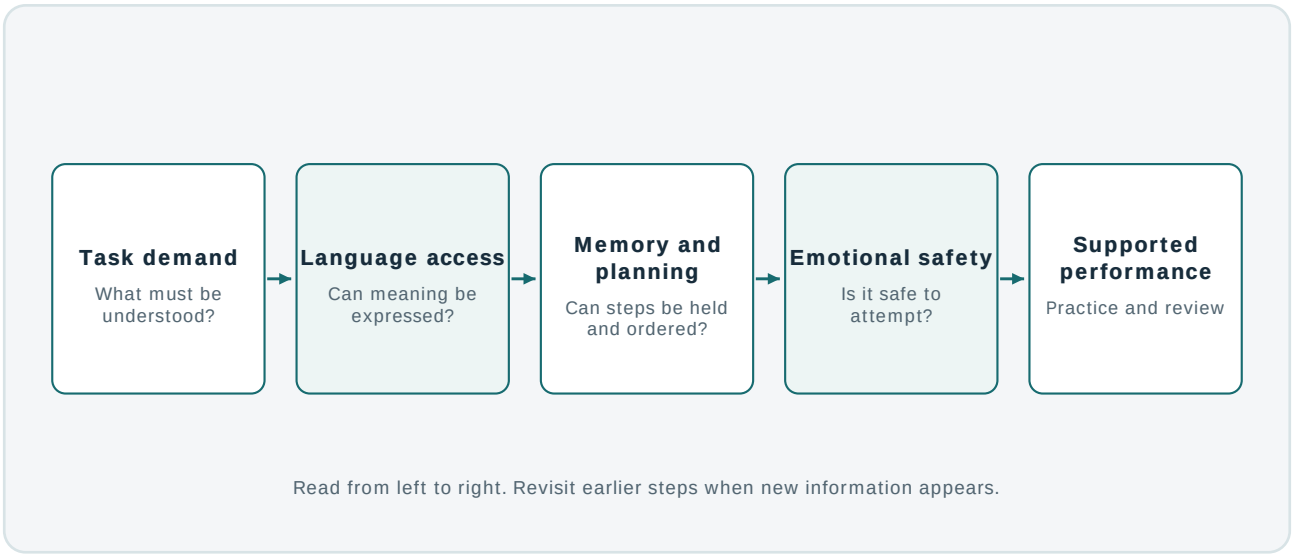


Figure 9.1. Performance depends on the whole task. Understanding, language access, working memory, emotional safety, and organization all contribute to observed performance. Assess where the task breaks down before attributing the difficulty to motivation or general ability. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

What Is Cognition?

OpenStax source module m82205 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82205/index.cnxml

Imagine all of your thoughts as if they were physical entities, swirling rapidly inside your mind. How is it possible that the brain is able to move from one thought to the next in an organized, orderly fashion? The brain is endlessly perceiving, processing, planning, organizing, and remembering it is always active. Yet, you don t notice most of your brain s activity as you move throughout your daily routine. This is only one facet of the complex processes involved in cognition. Simply put, cognition is thinking, and it encompasses the processes associated with perception, knowledge, problem solving, judgment, language, and memory. Scientists who study cognition are searching for ways to understand how we integrate, organize, and utilize our conscious cognitive experiences without being aware of all of the unconscious work that our brains are doing (for example, Kahneman, 2011).

Cognition

Upon waking each morning, you begin thinking contemplating the tasks that you must complete that day. In what order should you run your errands? Should you go to the bank, the cleaners, or the grocery store first? Can you get these things done before you head to class or will they need to wait until school is done? These thoughts are one example of cognition at work. Exceptionally complex, cognition is an essential feature of human consciousness, yet not all aspects of cognition are consciously experienced.

Cognitive psychology is the field of psychology dedicated to examining how people think. It attempts to explain how and why we think the way we do by studying the interactions among human thinking, emotion, creativity, language, and problem solving, in addition to other cognitive processes. Cognitive psychologists strive to determine and measure different types of intelligence, why some people are better at problem solving than others, and how emotional intelligence affects success in the workplace, among countless other topics. They also sometimes focus on how we organize thoughts and information gathered from our environments into meaningful categories of thought, which will be discussed later.

Concepts and Prototypes

The human nervous system is capable of handling endless streams of information. The senses serve as the interface between the mind and the external environment, receiving stimuli and translating it into nervous impulses that are transmitted to the brain. The brain then processes this information and uses the relevant pieces to create thoughts, which can then be expressed through language or stored in memory for future use. To make this process more complex, the brain does not gather information from external environments only. When thoughts are formed, the mind synthesizes information from emotions and memories (the accompanying source-figure description). Emotion and memory are powerful influences on both our thoughts and behaviors.

Source illustration description (image not reproduced): The outline of a human head is shown. There is a box containing Information, sensations in front of the head. An arrow from this box points to another box containing Emotions,

memories located where the front of the person's brain would be. An arrow from this second box points to a third box containing Thoughts located where the back of the person's brain would be. There are two arrows coming from Thoughts. One arrow points back to the second box, Emotions, memories, and the other arrow points to a fourth box, Behavior. Sensations and information are received by our brains, filtered through emotions and memories, and processed to become thoughts.

In order to organize this staggering amount of information, the mind has developed a "file cabinet" of sorts in the mind. The different files stored in the file cabinet are called concepts. Concepts are categories or groupings of linguistic information, images, ideas, or memories, such as life experiences. Concepts are, in many ways, big ideas that are generated by observing details, and categorizing and combining these details into cognitive structures. You use concepts to see the relationships among the different elements of your experiences and to keep the information in your mind organized and accessible.

Concepts are informed by our semantic memory (you will learn more about semantic memory in a later chapter) and are present in every aspect of our lives; however, one of the easiest places to notice concepts is inside a classroom, where they are discussed explicitly. When you study United States history, for example, you learn about more than just individual events that have happened in America s past. You absorb a large quantity of information by listening to and participating in discussions, examining maps, and reading first-hand accounts of people s lives. Your brain analyzes these details and develops an overall understanding of American history. In the process, your brain gathers details that inform and refine your understanding of related concepts like democracy, power, and freedom.

Concepts can be complex and abstract, like justice, or more concrete, like types of birds. In psychology, for example, Piaget s stages of development are abstract concepts. Some concepts, like tolerance, are agreed upon by many people, because they have been used in various ways over many years. Other concepts, like the characteristics of your ideal friend or your family s birthday traditions, are personal and individualized. In this way, concepts touch every aspect of our lives, from our many daily routines to the guiding principles behind the way governments function.

Another technique used by your brain to organize information is the identification of prototypes for the concepts you have developed. A prototype is the best example or representation of a concept. For example, what comes to your mind when you think of a dog? Most likely your early experiences with dogs will shape what you imagine. If your first pet was a Golden Retriever, there is a good chance that this would be your prototype for the category of dogs.

Natural and Artificial Concepts

In psychology, concepts can be divided into two categories, natural and artificial. Natural concepts are created naturally through your experiences and can be developed from either direct or indirect experiences. For example, if you live in Essex Junction, Vermont, you have probably had a lot of direct experience with snow. You ve watched it fall from the sky, you ve seen lightly falling snow that barely covers the windshield of your car, and you ve shoveled out 18 inches of fluffy white snow as you ve thought, This is perfect for skiing. You ve thrown snowballs at your best friend and gone sledding down the steepest hill in town. In short, you know snow. You know what it looks like, smells like, tastes like, and feels like. If, however, you ve lived your whole life on the island of Saint Vincent in the Caribbean, you may never have actually seen snow, much less tasted, smelled, or touched it. You know snow from the indirect experience of seeing pictures of falling snow or from watching films that feature snow as part of the setting. Either way, snow is a natural concept because you can construct an understanding of it through direct observations, experiences with snow, or indirect knowledge (such as from films or books) (the accompanying source-figure description).

Source illustration description (image not reproduced): Photograph A shows a snow covered landscape with the sun shining over it. Photograph B shows a sphere shaped object perched atop the corner of a cube shaped object. There is also a triangular object shown. (a) Our concept of snow is an example of a natural concept one that we understand through direct observation and experience. (b) In contrast, artificial concepts are ones that we know by a specific set of characteristics that they always exhibit, such as what defines different basic shapes. (credit a: modification of work by

Maarten Takens; credit b: modification of work by Shayan (USA) /Flickr)

An artificial concept, on the other hand, is a concept that is defined by a specific set of characteristics. Various properties of geometric shapes, like squares and triangles, serve as useful examples of artificial concepts. A triangle always has three angles and three sides. A square always has four equal sides and four right angles. Mathematical formulas, like the equation for area (length width) are artificial concepts defined by specific sets of characteristics that are always the same. Artificial concepts can enhance the understanding of a topic by building on one another. For example, before learning the concept of area of a square (and the formula to find it), you must understand what a square is. Once the concept of area of a square is understood, an understanding of area for other geometric shapes can be built upon the original understanding of area. The use of artificial concepts to define an idea is crucial to communicating with others and engaging in complex thought. According to Goldstone and Kersten (2003), concepts act as building blocks and can be connected in countless combinations to create complex thoughts.

Schemata

A schema is a mental construct consisting of a cluster or collection of related concepts (Bartlett, 1932). There are many different types of schemata, and they all have one thing in common: schemata are a method of organizing information that allows the brain to work more efficiently. When a schema is activated, the brain makes immediate assumptions about the person or object being observed.

There are several types of schemata. A role schema makes assumptions about how individuals in certain roles will behave (Callero, 1994). For example, imagine you meet someone who introduces himself as a firefighter. When this happens, your brain automatically activates the firefighter schema and begins making assumptions that this person is brave, selfless, and community-oriented. Despite not knowing this person, already you have unknowingly made judgments about them. Schemata also help you fill in gaps in the information you receive from the world around you. While schemata allow for more efficient information processing, there can be problems with schemata, regardless of whether they are accurate: Perhaps this particular firefighter is not brave, they just work as a firefighter to pay the bills while studying to become a children s librarian.

An event schema, also known as a cognitive script, is a set of behaviors that can feel like a routine. Think about what you do when you walk into an elevator (the accompanying source-figure description). First, the doors open and you wait to let exiting passengers leave the elevator car. Then, you step into the elevator and turn around to face the doors, looking for the correct button to push. You never face the back of the elevator, do you? And when you re riding in a crowded elevator and you can t face the front, it feels uncomfortable, doesn t it? Interestingly, event schemata can vary widely among different cultures and countries. For example, while it is quite common for people to greet one another with a handshake in the United States, in Tibet, you greet someone by sticking your tongue out at them, and in Belize, you bump fists (Cairns Regional Council, n.d.)

Source illustration description (image not reproduced): A crowded elevator is shown. There are many people standing close to one another. What event schema do you perform when riding in an elevator? (credit: Gideon /Flickr)

Because event schemata are automatic, they can be difficult to change. Imagine that you are driving home from work or school. This event schema involves getting in the car, shutting the door, and buckling your seatbelt before putting the key in the ignition. You might perform this script two or three times each day. As you drive home, you hear your phone s ring tone. Typically, the event schema that occurs when you hear your phone ringing involves locating the phone and answering it or responding to your latest text message. So without thinking, you reach for your phone, which could be in your pocket, in your bag, or on the passenger seat of the car. This powerful event schema is informed by your pattern of behavior and the pleasurable stimulation that a phone call or text message gives your brain. Because it is a schema, it is extremely challenging for us to stop reaching for the phone, even though

we know that we endanger our own lives and the lives of others while we do it (Neyfakh, 2013) (the accompanying source-figure description).

Source illustration description (image not reproduced): A person s right hand is holding a cellular phone. The person is in the driver s seat of an automobile while on the road. Texting while driving is dangerous, but it is a difficult event schema for some people to resist.

Remember the elevator? It feels almost impossible to walk in and not face the door. Our powerful event schema dictates our behavior in the elevator, and it is no different with our phones. Current research suggests that it is the habit, or event schema, of checking our phones in many different situations that makes refraining from checking them while driving especially difficult (Bayer & Campbell, 2012). Because texting and driving has become a dangerous epidemic in recent years, psychologists are looking at ways to help people interrupt the phone schema while driving. Event schemata like these are the reason why many habits are difficult to break once they have been acquired. As we continue to examine thinking, keep in mind how powerful the forces of concepts and schemata are to our understanding of the world.

Summary

In this section, you were introduced to cognitive psychology, which is the study of cognition, or the brain s ability to think, perceive, plan, analyze, and remember. Concepts and their corresponding prototypes help us quickly organize our thinking by creating categories into which we can sort new information. We also develop schemata, which are clusters of related concepts. Some schemata involve routines of thought and behavior, and these help us function properly in various situations without having to think twice about them. Schemata show up in social situations and routines of daily behavior.

Review Questions

OpenStax learning exercise

Q. Cognitive psychology is the branch of psychology that focuses on the study of _____.

A. B

OpenStax learning exercise

Q. Which of the following is an example of a prototype for the concept of leadership on an athletic team?

A. C

OpenStax learning exercise

Q. Which of the following is an example of an artificial concept?

A. B

OpenStax learning exercise

Q. An event schema is also known as a cognitive _____.

A. C

Critical Thinking Questions

OpenStax learning exercise

Q. Describe an event schema that you would notice at a sporting event.

A. Answers will vary. When attending a basketball game, it is typical to support your team by wearing the team colors and sitting behind their bench.

OpenStax learning exercise

Q. Explain why event schemata have so much power over human behavior.

A. Event schemata are rooted in the social fabric of our communities. We expect people to behave in certain ways in certain types of situations, and we hold ourselves to the same social standards. It is uncomfortable to go against an event schema it feels almost like we are breaking the rules.

Personal Application Question

OpenStax learning exercise

Q. Describe a natural concept that you know fully but that would be difficult for someone else to understand and explain why it would be difficult.

Language

OpenStax source module m82206 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82206/index.cnxml

Language is a communication system that involves using words and systematic rules to organize those words to transmit information from one individual to another. While language is a form of communication, not all communication is language. Many species communicate with one another through their postures, movements, odors, or vocalizations. This communication is crucial for species that need to interact and develop social relationships with their conspecifics. However, many people have asserted that it is language that makes humans unique among all of the animal species (Corballis & Suddendorf, 2007; Tomasello & Rakoczy, 2003). This section will focus on what distinguishes language as a special form of communication, how the use of language develops, and how language affects the way we think.

Components of Language

Language, be it spoken, signed, or written, has specific components: a lexicon and grammar. Lexicon refers to the words of a given language. Thus, lexicon is a language s vocabulary. Grammar refers to the set of rules that are used to convey meaning through the use of the lexicon (Fern ndez & Cairns, 2011). For instance, English grammar dictates that most verbs receive an -ed at the end to indicate past tense.

Words are formed by combining the various phonemes that make up the language. A phoneme (e.g., the sounds ah vs. eh) is a basic sound unit of a given language, and different languages have different sets of phonemes. Phonemes are combined to form morphemes, which are the smallest units of language that convey some type of meaning (e.g., I is both a phoneme and a morpheme). We use semantics and syntax to construct language. Semantics and syntax are part of a language s grammar. Semantics refers to the process by which we derive meaning from morphemes and words. Syntax refers to the way words are organized into sentences (Chomsky, 1965; Fern ndez & Cairns, 2011).

We apply the rules of grammar to organize the lexicon in novel and creative ways, which allow us to communicate information about both concrete and abstract concepts. We can talk about our immediate and observable surroundings as well as the surface of unseen planets. We can share our innermost

thoughts, our plans for the future, and debate the value of a college education. We can provide detailed instructions for cooking a meal, fixing a car, or building a fire. Through our use of words and language, we are able to form, organize, and express ideas, schema, and artificial concepts.

Language Development

Given the remarkable complexity of a language, one might expect that mastering a language would be an especially arduous task; indeed, for those of us trying to learn a second language as adults, this might seem to be true. However, young children master language very quickly with relative ease. B. F. Skinner (1957) proposed that language is learned through reinforcement. Noam Chomsky (1965) criticized this behaviorist approach, asserting instead that the mechanisms underlying language acquisition are biologically determined. The use of language develops in the absence of formal instruction and appears to follow a very similar pattern in children from vastly different cultures and backgrounds. It would seem, therefore, that we are born with a biological predisposition to acquire a language (Chomsky, 1965; Fern ndez & Cairns, 2011). Moreover, it appears that there is a critical period for language acquisition, such that this proficiency at acquiring language is maximal early in life; generally, as people age, the ease with which they acquire and master new languages diminishes (Johnson & Newport, 1989; Lenneberg, 1967; Singleton, 1995).

Children begin to learn about language from a very early age (the accompanying source-figure description). In fact, it appears that this is occurring even before we are born. Newborns show preference for their mother s voice and appear to be able to discriminate between the language spoken by their mother and other languages. Babies are also attuned to the languages being used around them and show preferences for videos of faces that are moving in synchrony with the audio of spoken language versus videos that do not synchronize with the audio (Blossom & Morgan, 2006; Pickens, 1994; Spelke & Cortelyou, 1981).

Source comparison table

Stages of Language and Communication Development		
Stage	Age	Developmental Language and Communication
1	0 3 months	Reflexive communication
2	3 8 months	Reflexive communication; interest in others
3	8 13 months	Intentional communication; sociability
4	12 18 months	First words
5	18 24 months	Simple sentences of two words
6	2 3 years	Sentences of three or more words
7	3 5 years	Complex sentences; has conversations

Dig Deeper

In the fall of 1970, a social worker in the Los Angeles area found a 13-year-old girl who was being raised in extremely neglectful and abusive conditions. The girl, who came to be known as Genie, had lived most of her life tied to a potty chair or confined to a crib in a small room that was kept closed with the curtains drawn. For a little over a decade, Genie had virtually no social interaction and no access to the outside world. As a result of these conditions, Genie was unable to stand up, chew solid food, or speak (Fromkin, Krashen, Curtiss, Rigler, & Rigler, 1974; Rymer, 1993). The police took Genie into protective custody.

Genie s abilities improved dramatically following her removal from her abusive environment, and early on, it appeared she was acquiring language much later than would be predicted by critical period hypotheses that had been posited at the time (Fromkin et al., 1974). Genie managed to amass an impressive vocabulary in a relatively short amount of time. However, she never developed a mastery of the grammatical aspects of language (Curtiss, 1981). Perhaps being deprived of the opportunity to learn language during a critical period impeded Genie s ability to fully acquire and use language.

You may recall that each language has its own set of phonemes that are used to generate morphemes, words, and so on. Babies can discriminate among the sounds that make up a language (for example, they can tell the difference between the s in vision and the ss in fission); early on, they can differentiate between the sounds of all human languages, even those that do not occur in the languages that are used in their environments. However, by the time that they are about 1 year old, they can only discriminate among those phonemes that are used in the language or languages in their environments (Jensen, 2011; Werker & Lalonde, 1988; Werker & Tees, 1984).

After the first few months of life, babies enter what is known as the babbling stage, during which time they tend to produce single syllables that are repeated over and over. As time passes, more variations appear in the syllables that they produce. During this time, it is unlikely that the babies are trying to communicate; they are just as likely to babble when they are alone as when they are with their caregivers (Fern ndez & Cairns, 2011). Interestingly, babies who are raised in environments in which sign language is used will also begin to show babbling in the gestures of their hands during this stage (Petitto, Holowka, Sergio, Levy, & Ostry, 2004).

Generally, a child s first word is uttered sometime between the ages of 1 year to 18 months, and for the next few months, the child will remain in the one word stage of language development. During this time, children know a number of words, but they only produce one-word utterances. The child s early vocabulary is limited to familiar objects or events, often nouns. Although children in this stage only make one-word utterances, these words often carry larger meaning (Fern ndez & Cairns, 2011). So, for example, a child saying cookie could be identifying a cookie or asking for a cookie.

As a child s lexicon grows, they begin to utter simple sentences and to acquire new vocabulary at a very rapid pace. In addition, children begin to demonstrate a clear understanding of the specific rules that apply to their language(s). Even the mistakes that children sometimes make provide evidence of just how much they understand about those rules. This is sometimes seen in the form of overgeneralization. In this context, overgeneralization refers to an extension of a language rule to an exception to the rule. For example, in English, it is usually the case that an s is added to the end of a word to indicate plurality. For example, we speak of one dog versus two dogs. Young children will overgeneralize this rule to cases that are exceptions to the add an s to the end of the word rule and say things like those two gooses or three mouses. Clearly, the rules of the language are understood, even if the exceptions to the rules are still being learned (Moskowitz, 1978).

Language and Thought

When we speak one language, we agree that words are representations of ideas, people, places, and events. The given language that children learn is connected to their culture and surroundings. But can words themselves shape the way we think about things? Psychologists have long investigated the question of whether language shapes thoughts and actions, or whether our thoughts and beliefs shape our language. Two researchers, Edward Sapir and Benjamin Lee Whorf, began this investigation in the 1940s. They wanted to understand how the language habits of a community encourage members of that community to interpret language in a particular manner (Sapir, 1941/1964). Sapir and Whorf proposed that language determines thought. For example, in some languages there are many different words for love. However, in English we use the word love for all types of love. Does this affect how we think about love depending on the language that we speak (Whorf, 1956)? Researchers have since identified this view as too absolute, pointing out a lack of empiricism behind what Sapir and Whorf proposed (Abler, 2013; Boroditsky, 2011; van Troyer, 1994). Today, psychologists continue to study and debate the relationship between language and thought.

What Do You Think?

Think about what you know of other languages; perhaps you even speak multiple languages. Imagine for a moment that your closest friend fluently speaks more than one language. Do you think that friend thinks differently, depending on which language is being spoken? You may know a few words that are not translatable from their original language into English. For example, the Portuguese word saudade originated during the 15th century, when Portuguese sailors left home to explore the seas and travel to Africa or Asia. Those left behind described the emptiness and fondness they felt as saudade (the accompanying source-figure description). The word came to express many meanings, including loss, nostalgia, yearning, warm memories, and hope. There is no single word in English that includes all of those emotions in a single description. Do words such as saudade indicate that different languages produce different patterns of thought in people? What do you think??

Source illustration description (image not reproduced): Photograph A shows a painting of a person leaning against a ledge, slumped sideways over a box. Photograph B shows a painting of a person reading by a window. These two works of art depict saudade. (a) Saudade de N poles, which is translated into missing Naples, was painted by Bertha Worms in 1895. (b) Almeida J nior painted Saudade in 1899.

Language may indeed influence the way that we think, an idea known as linguistic determinism. One recent demonstration of this phenomenon involved differences in the way that English and Mandarin Chinese speakers talk and think about time. English speakers tend to talk about time using terms that describe changes along a horizontal dimension, for example, saying something like I m running behind schedule or Don t get ahead of yourself. While Mandarin Chinese speakers also describe time in horizontal terms, it is not uncommon to also use terms associated with a vertical arrangement. For example, the past might be described as being up and the future as being down. It turns out that these differences in language translate into differences in performance on cognitive tests designed to measure how quickly an individual can recognize temporal relationships. Specifically, when given a series of tasks with vertical priming, Mandarin Chinese speakers were faster at recognizing temporal relationships between months. Indeed, Boroditsky (2001) sees these results as suggesting that habits in language encourage habits in thought (p. 12).

One group of researchers who wanted to investigate how language influences thought compared how English speakers and the Dani people of Papua New Guinea think and speak about color. The Dani have two words for color: one word for light and one word for dark. In contrast, the English language has 11 color words. Researchers hypothesized that the number of color terms could limit the ways that the Dani people conceptualized color. However, the Dani were able to distinguish colors with the same ability as English speakers, despite having fewer words at their disposal (Berlin & Kay, 1969). A recent review of research aimed at determining how language might affect something like color perception suggests that language can influence perceptual phenomena, especially in the left hemisphere of the brain. You may recall from earlier chapters that the left hemisphere is associated with language for

most people. However, the right (less linguistic hemisphere) of the brain is less affected by linguistic influences on perception (Regier & Kay, 2009)

Summary

Language is a communication system that has both a lexicon and a system of grammar. Language acquisition occurs naturally and effortlessly during the early stages of life, and this acquisition occurs in a predictable sequence for individuals around the world. Language has a strong influence on thought, and the concept of how language may influence cognition remains an area of study and debate in psychology.

Review Questions

OpenStax learning exercise

Q. _____ provides general principles for organizing words into meaningful sentences.

A. D

OpenStax learning exercise

Q. _____ are the smallest unit of language that carry meaning.

A. C

OpenStax learning exercise

Q. The meaning of words and phrases is determined by applying the rules of _____.

A. D

OpenStax learning exercise

Q. _____ is (are) the basic sound units of a spoken language.

A. B

Critical Thinking Questions

OpenStax learning exercise

Q. How do words not only represent our thoughts but also represent our values?

A. People tend to talk about the things that are important to them or the things they think about the most. What we talk about, therefore, is a reflection of our values.

OpenStax learning exercise

Q. How could grammatical errors actually be indicative of language acquisition in children?

A. People tend to talk about the things that are important to them or the things they think about the most. What we talk about, therefore, is a reflection of our values.

OpenStax learning exercise

Q. How could a specific learning disability, such as dysgraphia or dyslexia, impact a child s educational and school experience?

Personal Application Question

OpenStax learning exercise

Q. Can you think of examples of how language affects cognition?

Problem Solving

OpenStax source module m82207 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82207/index.cnxml

People face problems every day usually, multiple problems throughout the day. Sometimes these problems are straightforward: To double a recipe for pizza dough, for example, all that is required is that each ingredient in the recipe be doubled. Sometimes, however, the problems we encounter are more complex. For example, say you have a work deadline, and you must mail a printed copy of a report to your supervisor by the end of the business day. The report is time-sensitive and must be sent overnight. You finished the report last night, but your printer will not work today. What should you do? First, you need to identify the problem and then apply a strategy for solving the problem.

Problem-Solving Strategies

When you are presented with a problem whether it is a complex mathematical problem or a broken printer, how do you solve it? Before finding a solution to the problem, the problem must first be clearly identified. After that, one of many problem solving strategies can be applied, hopefully resulting in a solution.

A problem-solving strategy is a plan of action used to find a solution. Different strategies have different action plans associated with them (the accompanying source-figure description). For example, a well-known strategy is trial and error. The old adage, If at first you don t succeed, try, try again describes trial and error. In terms of your broken printer, you could try checking the ink levels, and if that doesn t work, you could check to make sure the paper tray isn t jammed. Or maybe the printer isn t actually connected to your laptop. When using trial and error, you would continue to try different solutions until you solved your problem. Although trial and error is not typically one of the most time-efficient strategies, it is a commonly used one.

Source comparison table

Problem-Solving Strategies		
Method	Description	Example
Trial and error	Continue trying different solutions until problem is solved	Restarting phone, turning off WiFi, turning off bluetooth in order to determine why your phone is malfunctioning
Algorithm	Step-by-step problem-solving formula	Instructional video for installing new software on your computer
Heuristic	General problem-solving framework	Working backwards; breaking a task into steps

Another type of strategy is an algorithm. An algorithm is a problem-solving formula that provides you with step-by-step instructions used to achieve a desired outcome (Kahneman, 2011). You can think of an algorithm as a recipe with highly detailed instructions that produce the same result every time they are performed. Algorithms are used frequently in our everyday lives, especially in computer science. When

you run a search on the Internet, search engines like Google use algorithms to decide which entries will appear first in your list of results. Facebook also uses algorithms to decide which posts to display on your newsfeed. Can you identify other situations in which algorithms are used?

A heuristic is another type of problem solving strategy. While an algorithm must be followed exactly to produce a correct result, a heuristic is a general problem-solving framework (Tversky & Kahneman, 1974). You can think of these as mental shortcuts that are used to solve problems. A rule of thumb is an example of a heuristic. Such a rule saves the person time and energy when making a decision, but despite its time-saving characteristics, it is not always the best method for making a rational decision. Different types of heuristics are used in different types of situations, but the impulse to use a heuristic occurs when one of five conditions is met (Pratkanis, 1989):

- When one is faced with too much information
- When the time to make a decision is limited
- When the decision to be made is unimportant
- When there is access to very little information to use in making the decision
- When an appropriate heuristic happens to come to mind in the same moment

Working backwards is a useful heuristic in which you begin solving the problem by focusing on the end result. Consider this example: You live in Washington, D.C. and have been invited to a wedding at 4 PM on Saturday in Philadelphia. Knowing that Interstate 95 tends to back up any day of the week, you need to plan your route and time your departure accordingly. If you want to be at the wedding service by 3:30 PM, and it takes 2.5 hours to get to Philadelphia without traffic, what time should you leave your house? You use the working backwards heuristic to plan the events of your day on a regular basis, probably without even thinking about it.

Another useful heuristic is the practice of accomplishing a large goal or task by breaking it into a series of smaller steps. Students often use this common method to complete a large research project or long essay for school. For example, students typically brainstorm, develop a thesis or main topic, research the chosen topic, organize their information into an outline, write a rough draft, revise and edit the rough draft, develop a final draft, organize the references list, and proofread their work before turning in the project. The large task becomes less overwhelming when it is broken down into a series of small steps.

Everyday Connection

Problem-solving abilities can improve with practice. Many people challenge themselves every day with puzzles and other mental exercises to sharpen their problem-solving skills. Sudoku puzzles appear daily in most newspapers. Typically, a sudoku puzzle is a 9 9 grid. The simple sudoku below (the accompanying source-figure description) is a 4 4 grid. To solve the puzzle, fill in the empty boxes with a single digit: 1, 2, 3, or 4. Here are the rules: The numbers must total 10 in each bolded box, each row, and each column; however, each digit can only appear once in a bolded box, row, and column. Time yourself as you solve this puzzle and compare your time with a classmate.

Source illustration description (image not reproduced): A four column by four row Sudoku puzzle is shown. The top left cell contains the number 3. The top right cell contains the number 2. The bottom right cell contains the number 1. The bottom left cell contains the number 4. The cell at the intersection of the second row and the second column contains the number 4. The cell to the right of that contains the number 1. The cell below the cell containing the number 1 contains the number 2. The cell to the left of the cell containing the number 2 contains the number 3. How long did it take you to solve this sudoku puzzle? (You can see the answer at the end of this section.)

Here is another popular type of puzzle (the accompanying source-figure description) that challenges your spatial reasoning skills. Connect all nine dots with four connecting straight lines without lifting your pencil from the paper:

Source illustration description (image not reproduced): A square shaped outline contains three rows and three columns of dots with equal space between them. Did you figure it out? (The answer is at the end of this section.) Once you understand how to crack this puzzle, you won t forget.

Take a look at the Puzzling Scales logic puzzle below (the accompanying source-figure description). Sam Loyd, a well-known puzzle master, created and refined countless puzzles throughout his lifetime (Cyclopedia of Puzzles, n.d.).

Source illustration description (image not reproduced): A puzzle involving a scale is shown. At the top of the figure it reads: Sam Loyds Puzzling Scales. The first row of the puzzle shows a balanced scale with 3 blocks and a top on the left and 12 marbles on the right. Below this row it reads: Since the scales now balance. The next row of the puzzle shows a balanced scale with just the top on the left, and 1 block and 8 marbles on the right. Below this row it reads: And balance when arranged this way. The third row shows an unbalanced scale with the top on the left side, which is much lower than the right side. The right side is empty. Below this row it reads: Then how many marbles will it require to balance with that top? What steps did you take to solve this puzzle? You can read the solution at the end of this section.

Pitfalls to Problem Solving

Not all problems are successfully solved, however. What challenges stop us from successfully solving a problem? Imagine a person in a room that has four doorways. One doorway that has always been open in the past is now locked. The person, accustomed to exiting the room by that particular doorway, keeps trying to get out through the same doorway even though the other three doorways are open. The person is stuck but they just need to go to another doorway, instead of trying to get out through the locked doorway. A mental set is where you persist in approaching a problem in a way that has worked in the past but is clearly not working now.

Functional fixedness is a type of mental set where you cannot perceive an object being used for something other than what it was designed for. Duncker (1945) conducted foundational research on functional fixedness. He created an experiment in which participants were given a candle, a book of matches, and a box of thumbtacks. They were instructed to use those items to attach the candle to the wall so that it did not drip wax onto the table below. Participants had to use functional fixedness to solve the problem (the accompanying source-figure description). During the Apollo 13 mission to the moon, NASA engineers at Mission Control had to overcome functional fixedness to save the lives of the astronauts aboard the spacecraft. An explosion in a module of the spacecraft damaged multiple systems. The astronauts were in danger of being poisoned by rising levels of carbon dioxide because of problems with the carbon dioxide filters. The engineers found a way for the astronauts to use spare plastic bags, tape, and air hoses to create a makeshift air filter, which saved the lives of the astronauts.

Source illustration description (image not reproduced): Figure a shows a book of matches, a box of thumbtacks, and a candle. Figure b shows the candle standing in the box that held the thumbtacks. A thumbtack attaches the box holding the candle to the wall. In Duncker's classic study, participants were provided the three objects in the top panel and asked to solve the problem. The solution is shown in the bottom portion.

Researchers have investigated whether functional fixedness is affected by culture. In one experiment, individuals from the Shuar group in Ecuador were asked to use an object for a purpose other than that for which the object was originally intended. For example, the participants were told a story about a bear and a rabbit that were separated by a river and asked to select among various objects, including a spoon, a cup, erasers, and so on, to help the animals. The spoon was the only object long enough to span the imaginary river, but if the spoon was presented in a way that reflected its normal usage, it took participants longer to choose the spoon to solve the problem. (German & Barrett, 2005). The researchers wanted to know if exposure to highly specialized tools, as occurs with individuals in industrialized nations, affects their ability to transcend functional fixedness. It was determined that functional fixedness is experienced in both industrialized and nonindustrialized cultures (German & Barrett, 2005).

In order to make good decisions, we use our knowledge and our reasoning. Often, this knowledge and reasoning is sound and solid. Sometimes, however, we are swayed by biases or by others manipulating a situation. For example, let s say you and three friends wanted to rent a house and had a combined target budget of $1,600. The realtor shows you only very run-down houses for $1,600 and then shows

you a very nice house for $2,000. Might you ask each person to pay more in rent to get the $2,000 home? Why would the realtor show you the run-down houses and the nice house? The realtor may be challenging your anchoring bias. An anchoring bias occurs when you focus on one piece of information when making a decision or solving a problem. In this case, you re so focused on the amount of money you are willing to spend that you may not recognize what kinds of houses are available at that price point.

The confirmation bias is the tendency to focus on information that confirms your existing beliefs. For example, if you think that your professor is not very nice, you notice all of the instances of rude behavior exhibited by the professor while ignoring the countless pleasant interactions he is involved in on a daily basis. Hindsight bias leads you to believe that the event you just experienced was predictable, even though it really wasn t. In other words, you knew all along that things would turn out the way they did. Representative bias describes a faulty way of thinking, in which you unintentionally stereotype someone or something; for example, you may assume that your professors spend their free time reading books and engaging in intellectual conversation, because the idea of them spending their time playing volleyball or visiting an amusement park does not fit in with your stereotypes of professors.

Finally, the availability heuristic is a heuristic in which you make a decision based on an example, information, or recent experience that is that readily available to you, even though it may not be the best example to inform your decision. Biases tend to preserve that which is already established to maintain our preexisting knowledge, beliefs, attitudes, and hypotheses (Aronson, 1995; Kahneman, 2011). These biases are summarized in the accompanying source-figure description.

Source comparison table

Summary of Decision Biases	
Bias	Description
Anchoring	Tendency to focus on one particular piece of information when making decisions or problem-solving
Confirmation	Focuses on information that confirms existing beliefs
Hindsight	Belief that the event just experienced was predictable
Representative	Unintentional stereotyping of someone or something
Availability	Decision is based upon either an available precedent or an example that may be faulty

Were you able to determine how many marbles are needed to balance the scales in the accompanying source-figure description? You need nine. Were you able to solve the problems in the accompanying source-figure description and the accompanying source-figure description? Here are the answers (the accompanying source-figure description).

Source illustration description (image not reproduced): The first puzzle is a Sudoku grid of 16 squares (4 rows of 4 squares) is shown. Half of the numbers were supplied to start the puzzle and are colored blue, and half have been filled in as the puzzle s solution and are colored red. The numbers in each row of the grid, left to right, are as follows. Row 1: blue 3, red 1, red 4, blue 2. Row 2: red 2, blue 4, blue 1, red 3. Row 3: red 1, blue 3, blue 2, red 4. Row 4: blue 4, red 2, red 3, blue 1.The second puzzle consists of 9 dots arranged in 3 rows of 3 inside of a square. The solution, four straight lines made without lifting the pencil, is shown in a red line with arrows indicating the direction of movement. In order to solve the puzzle, the lines must extend beyond the borders of the box. The four connecting lines are drawn as follows. Line 1 begins at the top left dot, proceeds through the middle and right dots of the top row, and extends to the right beyond the border of the square. Line 2 extends from the end of line 1, through the right dot of the horizontally centered row, through the middle dot of the bottom row, and beyond the square s border ending in the space beneath the left dot of the bottom row. Line 3 extends from the end of line 2 upwards through the left dots of the bottom, middle, and top rows. Line 4 extends from the end of line 3 through the middle dot in the middle row and ends at the right dot of the bottom row.

Summary

Many different strategies exist for solving problems. Typical strategies include trial and error, applying algorithms, and using heuristics. To solve a large, complicated problem, it often helps to break the problem into smaller steps that can be accomplished individually, leading to an overall solution. Roadblocks to problem solving include a mental set, functional fixedness, and various biases that can cloud decision making skills.

Review Questions

OpenStax learning exercise

Q. A specific formula for solving a problem is called _____.

A. A

OpenStax learning exercise

Q. A mental shortcut in the form of a general problem-solving framework is called _____.

A. B

OpenStax learning exercise

Q. Which type of bias involves becoming fixated on a single trait of a problem?

A. A

OpenStax learning exercise

Q. Which type of bias involves relying on a false stereotype to make a decision?

A. C

Critical Thinking Questions

OpenStax learning exercise

Q. What is functional fixedness and how can overcoming it help you solve problems?

A. Functional fixedness occurs when you cannot see a use for an object other than the use for which it was intended. For example, if you need something to hold up a tarp in the rain, but only have a pitchfork, you must overcome your expectation that a pitchfork can only be used for garden chores before you realize that you could stick it in the ground and drape the tarp on top of it to hold it up.

OpenStax learning exercise

Q. How does an algorithm save you time and energy when solving a problem?

A. An algorithm is a proven formula for achieving a desired outcome. It saves time because if you follow it exactly, you will solve the problem without having to figure out how to solve the problem. It is a bit like not reinventing the wheel.

Personal Application Question

OpenStax learning exercise

Q. Which type of bias do you recognize in your own decision making processes? How has this bias affected how you ve made decisions in the past and how can you use your awareness of it to improve your decisions making skills in the future?

CHAPTER 10 / CAC.901

Intelligence: measurement and interpretation

Lecture: A score is evidence, not an identity

Tests can organize information, compare performance with an appropriate reference, and identify areas for further investigation. They can also create false certainty. A number may appear precise even when the underlying interpretation is uncertain. Developmental psychology requires counselors to understand what a score represents, how it was obtained, and which decisions it can reasonably support.

Intelligence is a contested construct, not a complete measure of human worth. Different theories emphasize different abilities. Tests sample particular kinds of performance under particular conditions. They may be useful for identifying learning needs, but they do not capture every form of creativity, practical knowledge, social understanding, or cultural competence. A responsible report explains both the usefulness and the limits of the instrument.

Norms and reference groups

A norm-referenced score compares a child s performance with a reference sample. Ask who was included, when the norms were collected, and whether the child s language, age, educational experience, and context are represented. A score does not become valid simply because the test is standardized. Standardization improves consistency; validity concerns the meaning and use of the interpretation.

Reliability and validity are related but distinct. Reliability concerns consistency or measurement precision. Validity concerns whether evidence supports the intended interpretation and decision. A measure can be reliably wrong for a particular purpose. Counselors should examine the manual, independent evidence, and the fit between the referral question and the test.

Avoid interpreting small score differences as meaningful without considering measurement error. Confidence intervals and the standard error of measurement help communicate uncertainty. When two scores differ, ask whether the difference is reliable and whether it is common or clinically important. A profile should not be turned into a story about every peak and valley without evidence.

Conditions of performance

Anxiety, fatigue, pain, unfamiliarity with testing, language demands, and the relationship with the examiner can affect performance. So can motivation and the child s understanding of why the assessment matters. Record relevant conditions without assuming that they explain the whole score. A child should not be denied support because an adult believes the test underestimates ability, but the uncertainty should be acknowledged.

Dynamic assessment or guided observation can sometimes reveal how a child responds to support. This may inform teaching, but it should not be presented as a substitute for validated assessment when a standardized decision is required. Different methods answer different questions. The assessment plan should begin with the decision to be made, not with a favorite test.

Figure 10.1 shows the route from referral question to test selection, administration, interpretation, and decision. Feedback loops remind the clinician to reconsider earlier steps when results conflict with daily functioning. A test score should meet other evidence, not replace it.

Fairness and cultural meaning

Fairness involves more than removing obviously offensive items. It includes access to learning, language, familiarity with testing practices, and the consequences of interpretation. A lower score may reflect restricted opportunity rather than lower capacity. At the same time, dismissing all group differences as bias can prevent a child from receiving needed support. The task is careful investigation rather than a predetermined conclusion.

Use culturally and linguistically appropriate assessment where available, and explain when an instrument s validity is uncertain. Do not invent local norms or casually adjust scores. Consult qualified professionals and relevant guidance. When evidence is limited, say so and use a broader assessment strategy. Uncertainty should lead to caution, not to confident labels.

Application: conflicting accounts of ability

A fictional ten-year-old, Zainab, scores lower on a group-administered test than her classroom work suggests. The school proposes a fixed low-expectation placement. A developmental review examines the testing conditions, language, hearing, attendance, and the nature of classroom tasks. It also asks whether Zainab receives substantial help that the group test does not provide.

The appropriate response is not automatically to reject the score or to accept it. Clarify the referral question and gather additional evidence. If a learning difficulty is present, support should address it. If the score reflects access or measurement problems, the placement may be inappropriate. The decision should be revisited as the child responds to teaching.

Communicate findings in plain language. This score estimates performance on these tasks under these conditions is more accurate than This is her intelligence. Describe strengths, needs, uncertainty, and practical recommendations. A report should help adults act differently, not merely attach a number to the child.

Doctoral seminar task

Create a checklist for reviewing a test report. Include the referral question, normative sample, language, reliability, validity, administration conditions, measurement uncertainty, functional evidence, and potential consequences. Apply it to a fictional or properly de-identified report.

Write two summaries of the same finding: one for a caregiver and one for an interdisciplinary team. Preserve the uncertainty in both. Then identify a decision that should not be made from the score alone. The exercise develops the professional discipline of using measurement without surrendering judgment to it.

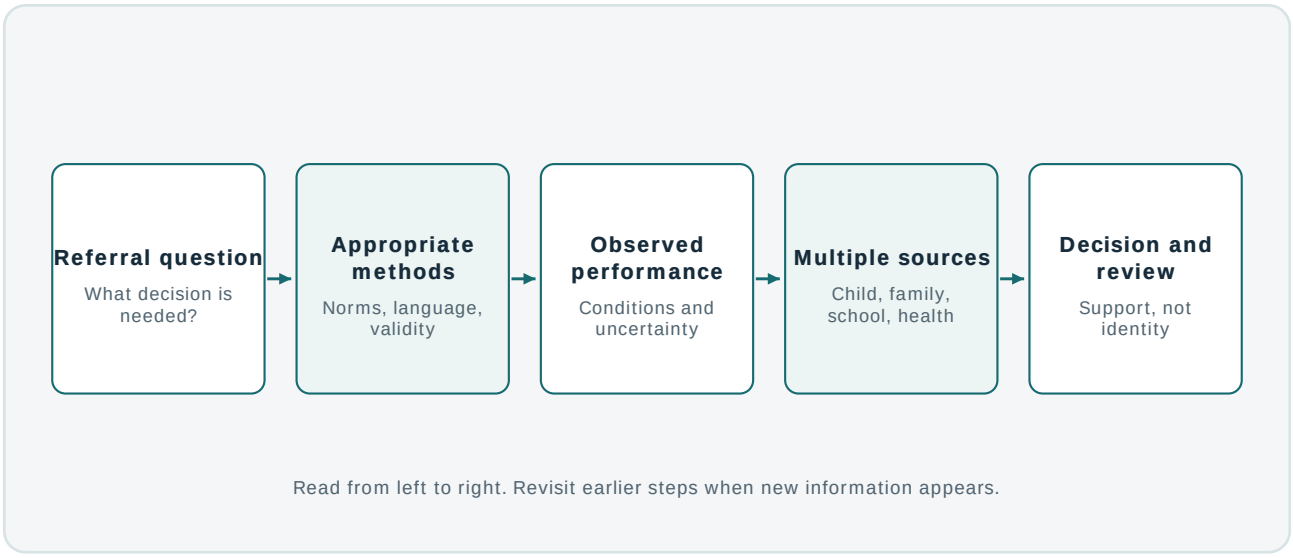


Figure 10.1. A score is one part of a decision. Responsible assessment begins with a referral question and checks the fit, quality, and uncertainty of measurement. Test results meet developmental and functional evidence before a decision is made, and the decision is reviewed. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

What Are Intelligence and Creativity?

OpenStax source module m82208 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82208/index.cnxml

A four-and-a-half-year-old boy sits at the kitchen table with his father, who is reading a new story aloud to him. He turns the page to continue reading, but before he can begin, the boy says, Wait, Daddy! He points to the words on the new page and reads aloud, Go, Pig! Go! The father stops and looks at his son. Can you read that? he asks. Yes, Daddy! And he points to the words and reads again, Go, Pig! Go!

This father was not actively teaching his son to read, even though the child constantly asked questions about letters, words, and symbols that they saw everywhere: in the car, in the store, on the television. The dad wondered about what else his son might understand and decided to try an experiment. Grabbing a sheet of blank paper, he wrote several simple words in a list: mom, dad, dog, bird, bed, truck, car, tree. He put the list down in front of the boy and asked him to read the words. Dad, dog, bird, bed, truck, car, tree, he read, slowing down to carefully pronounce bird and truck. Then, Did I do it, Daddy? You sure did! That is very good. The father gave his little boy a warm hug and continued reading the story about the pig, all the while wondering if his son s abilities were an indication of exceptional intelligence or simply a normal pattern of linguistic development. Like the father in this example, psychologists have wondered what constitutes intelligence and how it can be measured.

Classifying Intelligence

What exactly is intelligence? The way that researchers have defined the concept of intelligence has been modified many times since the birth of psychology. British psychologist Charles Spearman believed intelligence consisted of one general factor, called g, which could be measured and compared among individuals. Spearman focused on the commonalities among various intellectual abilities and de-emphasized what made each unique. Long before modern psychology developed, however, ancient philosophers, such as Aristotle, held a similar view (Cianciolo & Sternberg, 2004).

Others psychologists believe that instead of a single factor, intelligence is a collection of distinct abilities. In the 1940s, Raymond Cattell proposed a theory of intelligence that divided general intelligence into two components: crystallized intelligence and fluid intelligence (Cattell, 1963). Crystallized intelligence is characterized as acquired knowledge and the ability to retrieve it. When you learn, remember, and recall information, you are using crystallized intelligence. You use crystallized intelligence all the time in your coursework by demonstrating that you have mastered the information covered in the course. Fluid intelligence encompasses the ability to see complex relationships and solve problems. Navigating your way home after being detoured onto an unfamiliar route because of road construction would draw upon your fluid intelligence. Fluid intelligence helps you tackle complex, abstract challenges in your daily life, whereas crystallized intelligence helps you overcome concrete, straightforward problems (Cattell, 1963).

Other theorists and psychologists believe that intelligence should be defined in more practical terms. For example, what types of behaviors help you get ahead in life? Which skills promote success? Think about this for a moment. Being able to recite all 45 presidents of the United States in order is an excellent party trick, but will knowing this make you a better person?

Robert Sternberg developed another theory of intelligence, which he titled the triarchic theory of intelligence because it sees intelligence as comprised of three parts (Sternberg, 1988): practical, creative, and analytical intelligence (the accompanying source-figure description).

Source illustration description (image not reproduced): Three boxes are arranged in a triangle. The top box contains Analytical intelligence; academic problem solving and computation. There is a line with arrows on both ends connecting this box to another box containing Practical intelligence; street smarts and common sense. Another line with arrows on both ends connects this box to another box containing Creative intelligence; imaginative and innovative problem solving. Another line with arrows on both ends connects this box to the first box described, completing the triangle. Sternberg s theory identifies three types of intelligence: practical, creative, and analytical.

Practical intelligence, as proposed by Sternberg, is sometimes compared to street smarts. Being practical means you find solutions that work in your everyday life by applying knowledge based on your experiences. This type of intelligence appears to be separate from traditional understanding of IQ; individuals who score high in practical intelligence may or may not have comparable scores in creative and analytical intelligence (Sternberg, 1988).

Analytical intelligence is closely aligned with academic problem solving and computations. Sternberg says that analytical intelligence is demonstrated by an ability to analyze, evaluate, judge, compare, and contrast. When reading a classic novel for literature class, for example, it is usually necessary to compare the motives of the main characters of the book or analyze the historical context of the story. In a science course such as anatomy, you must study the processes by which the body uses various minerals in different human systems. In developing an understanding of this topic, you are using analytical intelligence. When solving a challenging math problem, you would apply analytical intelligence to analyze different aspects of the problem and then solve it section by section.

Creative intelligence is marked by inventing or imagining a solution to a problem or situation. Creativity in this realm can include finding a novel solution to an unexpected problem or producing a beautiful work of art or a well-developed short story. Imagine for a moment that you are camping in the woods with some friends and realize that you ve forgotten your camp coffee pot. The person in your group who figures out a way to successfully brew coffee for everyone would be credited as having higher creative intelligence.

Multiple Intelligences Theory was developed by Howard Gardner, a Harvard psychologist and former student of Erik Erikson. In Gardner s theory, each person possesses at least eight intelligences. The eight intelligences are linguistic intelligence, logical-mathematical intelligence, musical intelligence, bodily kinesthetic intelligence, spatial intelligence, interpersonal intelligence, intrapersonal intelligence, and naturalistic intelligence. Among cognitive psychologists, Gardner s theory has been heavily criticized for lacking empirical evidence. However, educators continue to study and use Gardner s theory, with some colleges even discussing how they integrate Gardner s theory into their classrooms. Gottfredson describes one possible reason for the continued use of Gardner s theory: . . . that there are multiple independent intelligences, suggesting that everyone can be smart in some way. This is, understandably, a very attractive idea in democratic societies (2004).

Gardner s inter- and intrapersonal intelligences are often combined into a single type: emotional intelligence. Emotional intelligence encompasses the ability to understand the emotions of yourself and others, show empathy, understand social relationships and cues, and regulate your own emotions and respond in culturally appropriate ways (Parker, Saklofske, & Stough, 2009). People with high emotional intelligence typically have well-developed social skills. Some researchers, including Daniel Goleman, the author of Emotional Intelligence: Why It Can Matter More than IQ, argue that emotional intelligence is a better predictor of success than traditional intelligence (Goleman, 1995). However, emotional intelligence has been widely debated, with researchers pointing out inconsistencies in how it

is defined and described, as well as questioning results of studies on a subject that is difficult to measure and study empirically (Locke, 2005; Mayer, Salovey, & Caruso, 2004)

The most comprehensive theory of intelligence to date is the Cattell-Horn-Carroll (CHC) theory of cognitive abilities (Schneider & McGrew, 2018). In this theory, abilities are related and arranged in a hierarchy with general abilities at the top, broad abilities in the middle, and narrow (specific) abilities at the bottom. The narrow abilities are the only ones that can be directly measured; however, they are integrated within the other abilities. At the general level is general intelligence. Next, the broad level consists of general abilities such as fluid reasoning, short-term memory, and processing speed. Finally, as the hierarchy continues, the narrow level includes specific forms of cognitive abilities. For example, short-term memory would further break down into memory span and working memory capacity.

Intelligence can also have different meanings and values in different cultures. If you live on a small island, where most people get their food by fishing from boats, it would be important to know how to fish and how to repair a boat. If you were an exceptional angler, your peers would probably consider you intelligent. If you were also skilled at repairing boats, your intelligence might be known across the whole island. Think about your own family s culture. What values are important for Latinx families? Italian families? In Irish families, hospitality and telling an entertaining story are marks of the culture. If you are a skilled storyteller, other members of Irish culture are likely to consider you intelligent.

Some cultures place a high value on working together as a collective. In these cultures, the importance of the group supersedes the importance of individual achievement. When you visit such a culture, how well you relate to the values of that culture exemplifies your cultural intelligence, sometimes referred to as cultural competence.

Creativity

Creativity is the ability to generate, create, or discover new ideas, solutions, and possibilities. Very creative people often have intense knowledge about something, work on it for years, look at novel solutions, seek out the advice and help of other experts, and take risks. Although creativity is often associated with the arts, it is actually a vital form of intelligence that drives people in many disciplines to discover something new. Creativity can be found in every area of life, from the way you decorate your residence to a new way of understanding how a cell works.

Creativity is often connected to a person s ability to engage in divergent thinking. Divergent thinking can be described as thinking outside the box; it allows an individual to arrive at unique, multiple solutions to a given problem. In contrast, convergent thinking describes the ability to provide a correct or well-established answer or solution to a problem (Cropley, 2006; Gilford, 1967)

Everyday Connection

Dr. Tom Steitz, former Sterling Professor of Biochemistry and Biophysics at Yale University, spent his career looking at the structure and specific aspects of RNA molecules and how their interactions could help produce antibiotics and ward off diseases. As a result of his lifetime of work, he won the Nobel Prize in Chemistry in 2009. He wrote, Looking back over the development and progress of my career in science, I am reminded how vitally important good mentorship is in the early stages of one's career development and constant face-to-face conversations, debate and discussions with colleagues at all stages of research. Outstanding discoveries, insights and developments do not happen in a vacuum (Steitz, 2010, para. 39). Based on Steitz s comment, it becomes clear that someone s creativity, although an individual strength, benefits from interactions with others. Think of a time when your creativity was sparked by a conversation with a friend or classmate. How did that person influence you and what problem did you solve using creativity?

Summary

Intelligence is a complex characteristic of cognition. Many theories have been developed to explain what intelligence is and how it works. Sternberg generated his triarchic theory of intelligence, whereas Gardner posits that intelligence is comprised of many factors. Still others focus on the importance of emotional intelligence. Finally, creativity seems to be a facet of intelligence, but it is extremely difficult to measure objectively.

Review Questions

OpenStax learning exercise

Q. Fluid intelligence is characterized by _____.

A. D

OpenStax learning exercise

Q. Which of the following is not one of Gardner s Multiple Intelligences?

A. A

OpenStax learning exercise

Q. Which theorist put forth the triarchic theory of intelligence?

A. C

OpenStax learning exercise

Q. When you are examining data to look for trends, which type of intelligence are you using most?

A. B

Critical Thinking Questions

OpenStax learning exercise

Q. Describe a situation in which you would need to use practical intelligence.

A. You are out with friends and it is getting late. You need to make it home before your curfew, but you don t have a ride home. You need to get in touch with your parents, but your cell phone is dead. So, you enter a nearby convenience store and explain your situation to the clerk. He allows you to use the store s phone to call your parents, and they come and pick you and your friends up, and take all of you home.

OpenStax learning exercise

Q. Describe a situation in which cultural intelligence would help you communicate better.

A. You are visiting Madrid, Spain, on a language immersion trip. Your Spanish is okay, but you still not sure about some of the facial expressions and body language of the native speakers. When faced with a sticky social situation, you do not engage immediately as you might back home. Instead, you hold back and observe what others are doing before reacting.

Personal Application Question

OpenStax learning exercise

Q. What influence do you think emotional intelligence plays in your personal life?

Measures of Intelligence

OpenStax source module m82209 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82209/index.cnxml

While you re likely familiar with the term IQ and associate it with the idea of intelligence, what does IQ really mean? IQ stands for intelligence quotient and describes a score earned on a test designed to measure intelligence. You ve already learned that there are many ways psychologists describe intelligence (or more aptly, intelligences). Similarly, IQ tests the tools designed to measure intelligence have been the subject of debate throughout their development and use.

When might an IQ test be used? What do we learn from the results, and how might people use this information? While there are certainly many benefits to intelligence testing, it is important to also note the limitations and controversies surrounding these tests. For example, IQ tests have sometimes been used as arguments in support of insidious purposes, such as the eugenics movement (Severson, 2011). The infamous Supreme Court Case, Buck v. Bell, legalized the forced sterilization of some people deemed feeble-minded through this type of testing, resulting in about 65,000 sterilizations (Buck v. Bell, 274 U.S. 200; Ko, 2016). Today, only professionals trained in psychology can administer IQ tests, and the purchase of most tests requires an advanced degree in psychology. Other professionals in the field, such as social workers and psychiatrists, cannot administer IQ tests. In this section, we will explore what intelligence tests measure, how they are scored, and how they were developed.

Measuring Intelligence

It seems that the human understanding of intelligence is somewhat limited when we focus on traditional or academic-type intelligence. How then, can intelligence be measured? And when we measure intelligence, how do we ensure that we capture what we re really trying to measure (in other words, that IQ tests function as valid measures of intelligence)? In the following paragraphs, we will explore how intelligence tests were developed and the history of their use.

The IQ test has been synonymous with intelligence for over a century. In the late 1800s, Sir Francis Galton developed the first broad test of intelligence (Flanagan & Kaufman, 2004). Although he was not a psychologist, his contributions to the concepts of intelligence testing are still felt today (Gordon, 1995). Reliable intelligence testing (you may recall from earlier chapters that reliability refers to a test s ability to produce consistent results) began in earnest during the early 1900s with a researcher named Alfred Binet (the accompanying source-figure description). Binet was asked by the French government to develop an intelligence test to use on children to determine which ones might have difficulty in school; it included many verbally based tasks. American researchers soon realized the value of such testing. Louis Terman, a Stanford professor, modified Binet s work by standardizing the administration of the test and tested thousands of different-aged children to establish an average score for each age. As a result, the test was normed and standardized, which means that the test was administered consistently to a large enough representative sample of the population that the range of scores resulted in a bell curve (bell curves will be discussed later). Standardization means that the manner of administration, scoring, and interpretation of results is consistent. Norming involves giving a test to a large population so data can be collected comparing groups, such as age groups. The resulting data provide norms, or referential scores, by which to interpret future scores. Norms are not expectations of what a given group should know but a demonstration of what that group does know. Norming and standardizing the test ensures that new scores are reliable. This new version of the test was called the Stanford-Binet Intelligence Scale (Terman, 1916). Remarkably, an updated version of this test is still widely used today.

Source illustration description (image not reproduced): Photograph A shows a portrait of Alfred Binet. Photograph B shows six sketches of human faces. Above these faces is the label Guide for Binet-Simon Scale. 223 The faces are arranged in three rows of two, and these rows are labeled 1, 2, and 3. At the bottom it reads: The psychological clinic is indebted for the loan of these cuts and those on p. 225 to the courtesy of Dr. Oliver P. Cornman, Associate Superintendent of Schools of Philadelphia, and Chairman of Committee on Backward Children Investigation. See Report of Committee, Dec. 31, 1910, appendix. French psychologist Alfred Binet helped to develop intelligence testing. (b) This page is from a 1908 version of the Binet-Simon Intelligence Scale. Children being tested were asked which face, of each pair, was prettier.

In 1939, David Wechsler, a psychologist who spent part of his career working with World War I veterans, developed a new IQ test in the United States. Wechsler combined several subtests from other intelligence tests used between 1880 and World War I. These subtests tapped into a variety of verbal and nonverbal skills, because Wechsler believed that intelligence encompassed the global capacity of a person to act purposefully, to think rationally, and to deal effectively with his environment (Wechsler, 1958, p. 7). He named the test the Wechsler-Bellevue Intelligence Scale (Wechsler, 1981). This combination of subtests became one of the most extensively used intelligence tests in the history of psychology. Although its name was later changed to the Wechsler Adult Intelligence Scale (WAIS) and has been revised several times, the aims of the test remain virtually unchanged since its inception (Boake, 2002). Today, there are three intelligence tests credited to Wechsler, the Wechsler Adult Intelligence Scale-fourth edition (WAIS-IV), the Wechsler Intelligence Scale for Children (WISC-V), and the Wechsler Preschool and Primary Scale of Intelligence IV (WPPSI-IV) (Wechsler, 2012). These tests are used widely in schools and communities throughout the United States, and they are periodically normed and standardized as a means of recalibration. As a part of the recalibration process, the WISC-V was given to thousands of children across the country, and children taking the test today are compared with their same-age peers (the accompanying source-figure description).

The WISC-V is composed of 14 subtests, which comprise five indices, which then render an IQ score. The five indices are Verbal Comprehension, Visual Spatial, Fluid Reasoning, Working Memory, and Processing Speed. When the test is complete, individuals receive a score for each of the five indices and a Full Scale IQ score. The method of scoring reflects the understanding that intelligence is comprised of multiple abilities in several cognitive realms and focuses on the mental processes that the child used to arrive at their answers to each test item.

Interestingly, the periodic recalibrations have led to an interesting observation known as the Flynn effect. Named after James Flynn, who was among the first to describe this trend, the Flynn effect refers to the observation that each generation has a significantly higher IQ than the last. Flynn himself argues, however, that increased IQ scores do not necessarily mean that younger generations are more intelligent per se (Flynn, Shaughnessy, & Fulgham, 2012).

Ultimately, we are still left with the question of how valid intelligence tests are. Certainly, the most modern versions of these tests tap into more than verbal competencies, yet the specific skills that should be assessed in IQ testing, the degree to which any test can truly measure an individual s intelligence, and the use of the results of IQ tests are still issues of debate (Gresham & Witt, 1997; Flynn, Shaughnessy, & Fulgham, 2012; Richardson, 2002; Schlinger, 2003).

What Do You Think?

The case of Atkins v. Virginia was a landmark case in the United States Supreme Court. On August 16, 1996, two men, Daryl Atkins and William Jones, robbed, kidnapped, and then shot and killed Eric Nesbitt, a local airman from the U.S. Air Force. A clinical psychologist evaluated Atkins and testified at the trial that Atkins had an IQ of 59. The mean IQ score is 100. The psychologist concluded that Atkins had an intellectual disability.

The jury found Atkins guilty, and he was sentenced to death. Atkins and his attorneys appealed to the Supreme Court. In June 2002, the Supreme Court reversed a previous decision and ruled that executions of people with intellectual disabilities are cruel and unusual punishments prohibited by the Eighth Amendment. Using the accepted language of the time, the court wrote in their decision:

Clinical definitions of mental retardation require not only subaverage intellectual functioning, but also significant limitations in adaptive skills. Mentally retarded persons frequently know the difference between right and wrong and are competent to stand trial. Because of their impairments, however, by definition they have diminished capacities to understand and process information, to communicate, to abstract from mistakes and learn from experience, to engage in logical reasoning, to control impulses, and to understand others reactions. Their deficiencies do not warrant an exemption from criminal sanctions, but diminish their personal culpability (Atkins v. Virginia, 2002, par. 5).

The court also decided that there was a state legislature consensus against the execution of people with intellectual disabilities and that this consensus should stand for all of the states. The Supreme Court ruling left it up to the states to determine their own definitions of intellectual disability. The definitions vary among states as to who can be executed. In the Atkins case, a jury decided that because he had many contacts with his lawyers and thus was provided with intellectual stimulation, his IQ had reportedly increased to a level where the state could execute him. He was given an execution date and then received a stay of execution after it was revealed that lawyers for co-defendant, William Jones, coached Jones to produce a testimony against Mr. Atkins that did match the evidence (Liptak, 2008). After the revelation of this misconduct, Atkins was re-sentenced to life imprisonment.

Atkins v. Virginia (2002) highlights several issues regarding society s beliefs around intelligence. In the Atkins case, the Supreme Court decided that intellectual disability does affect decision making and therefore should affect the nature of the punishment such criminals receive. Where, however, should the lines of intellectual disability be drawn? In May 2014, the Supreme Court ruled in a related case (Hall v. Florida) that IQ scores cannot be used as a final determination of a prisoner s eligibility for the death penalty (Roberts, 2014).

The Bell Curve

The results of intelligence tests follow the bell curve, a graph in the general shape of a bell. When the bell curve is used in psychological testing, the graph demonstrates a normal distribution of a trait, in this case, intelligence, in the human population. Many human traits naturally follow the bell curve. For example, if you lined up all your female schoolmates according to height, it is likely that a large cluster of them would be the average height for an American woman: 5 4 5 6 . This cluster would fall in the center of the bell curve, representing the average height for American women (the accompanying source-figure description). There would be fewer women who stand closer to 4 11 . The same would be true for women of above-average height: those who stand closer to 5 11 . The trick to finding a bell curve in nature is to use a large sample size. Without a large sample size, it is less likely that the bell curve will represent the wider population. A representative sample is a subset of the population that accurately represents the general population. If, for example, you measured the height of the women in your classroom only, you might not actually have a representative sample. Perhaps the women s basketball team wanted to take this course together, and they are all in your class. Because basketball players tend to be taller than average, the women in your class may not be a good representative sample of the population of American women. But if your sample included all the women at your school, it is likely that their heights would form a natural bell curve.

Source illustration description (image not reproduced): A graph of a bell curve is labeled Height of U.S. Women. The x axis is labeled Height and the y axis is labeled Frequency. Between the heights of five feet tall and five feet and five inches tall, the frequency rises to a curved peak, then begins dropping off at the same rate until it hits five feet ten inches tall. Are you of below-average, average, or above-average height?

The same principles apply to intelligence tests scores. Individuals earn a score called an intelligence quotient (IQ). Over the years, different types of IQ tests have evolved, but the way scores are interpreted remains the same. On most IQ tests, the average (or mean) IQ score is 100. Standard deviations describe how data are dispersed in a population and give context to large data sets. The bell curve uses the standard deviation to show how all scores are dispersed from the average score (the accompanying source-figure description). In modern IQ testing, one standard deviation is 15 points. So a score of 85 would be described as one standard deviation below the mean. How would you describe a score of 115 and a score of 70? Any IQ score that falls within one standard deviation above and below the mean (between 85 and 115) is considered average, and 68% of the population has IQ scores in this range. An IQ score of 130 or above is considered a superior level.

Source illustration description (image not reproduced): A graph of a bell curve is labeled Intelligence Quotient Score. The x axis is labeled IQ, and the y axis is labeled Population. Beginning at an IQ of 60, the population rises to a curved peak at an IQ of 100 and then drops off at the same rate ending near zero at an IQ of 140. The majority of people have an IQ score between 85 and 115.

Only 2.2% of the population has an IQ score below 70 (American Psychiatric Association [APA], 2013). If a person earns a score approximately two standard deviations below the mean on an intelligence test, (about 70 on a test with a mean of 100), has major deficits in adaptive functioning, and these cognitive and adaptive deficits were present before the age of 18, they could be diagnosed as having an intellectual disability (ID). Formerly known as mental retardation, the accepted term now is intellectual disability, and it has four subtypes: mild, moderate, severe, and profound (the accompanying source-figure description). The Diagnostic and Statistical Manual of Psychological Disorders lists criteria for each subgroup (APA, 2013).

Source comparison table

Characteristics of Cognitive Disorders		
Intellectual Disability Subtype	Percentage of Population with Intellectual Disabilities	Description
Mild	85%	3rd- to 6th-grade skill level in reading, writing, and math; may be employed and live independently
Moderate	10%	Basic reading and writing skills; functional self-care skills; requires some oversight
Severe	5%	Functional self-care skills; requires oversight of daily environment and activities
Profound	<1%	May be able to communicate verbally or nonverbally; requires intensive oversight

On the other end of the intelligence spectrum are those individuals whose IQs fall into the highest ranges. Consistent with the bell curve, about 2% of the population falls into this category. People are considered to have a higher aptitude for learning (and may be classified as "gifted" within educational systems) if they have an IQ score of 130 or higher, or superior intelligence in a particular area. Long ago, popular belief suggested that people of high intelligence were maladjusted. This idea was disproven through a groundbreaking study of these children. In 1921, Lewis Terman began a longitudinal study of over 1500 children with IQs over 135 (Terman, 1925). His findings showed that these children became well-educated, successful adults who were, in fact, well-adjusted (Terman & Oden, 1947). Additionally, Terman s study showed that the subjects were above average in physical build and attractiveness, dispelling an earlier popular notion that highly intelligent people were weaklings. Some people with very high IQs elect to join Mensa, an organization dedicated to identifying, researching, and fostering intelligence. Members must have an IQ score in the top 2% of the population, and they may be required to pass other exams in their application to join the group.

Dig Deeper

In the past, individuals with IQ scores below 70 and significant adaptive and social functioning delays were diagnosed with mental retardation. When this diagnosis was first named, it was replacing more negative and insensitive terms, and the title held no social stigma; several prominent research and support organizations even used the word in their names and mission statements. However, members of those populations as well as their families and supporting professionals found that the term was not only inaccurate, but demeaning and insulting. As such, the DSM-5 now labels this diagnosis as intellectual disability. Many states once had a Department of Mental Retardation to serve those diagnosed with such cognitive delays, but most have changed their name to Department of Developmental Disabilities or something similar in language. Due to the passage of "Rosa's Law" in 2010 and to the growing support for changing the terminology, most U.S. federal agencies formally adopted the words "intellectual disability." While the change was widely supported, you can view in the Federal Register several counterpoints from parents of people with intellectual disabilities, who felt that the new term was imprecise and less applicable to their children. Earlier in the chapter, we discussed how language affects how we think. Do you think changing the title of this department has any impact on how people regard those with developmental disabilities? Does a different name give people more dignity, and if so, how? Do you think the terminology is likely to change again? Why or why not?

Why Measure Intelligence?

The value of IQ testing is most evident in educational or clinical settings. Children who seem to be experiencing learning difficulties or severe behavioral problems can be tested to ascertain whether the child s difficulties can be partly attributed to an IQ score that is significantly different from the mean for their age group. Without IQ testing or another measure of intelligence children and adults needing extra support might not be identified effectively. In addition, IQ testing is used in courts to determine whether a defendant has special or extenuating circumstances that preclude them from participating in some way in a trial. People also use IQ testing results to seek disability benefits from the Social Security Administration.

The following case study demonstrates the usefulness and benefits of IQ testing. Candace, a 14-year-old girl experiencing problems at school in Connecticut, was referred for a court-ordered psychological evaluation. She was in regular education classes in ninth grade and was failing every subject. Candace had never been a stellar student but had always been passed to the next grade. Frequently, she would curse at any of her teachers who called on her in class. She also got into fights with other students and occasionally shoplifted. When she arrived for the evaluation, Candace immediately said that she hated everything about school, including the teachers, the rest of the staff, the building, and the homework. Her parents stated that they felt their daughter was picked on, because she was of a different ethnicity than the teachers and most of the other students. When asked why she cursed at her teachers, Candace replied, They only call on me when I don t know the answer. I don t want to say, I don t know all of the time and look like an idiot in front of my friends. The teachers embarrass me. She was given a battery of tests, including an IQ test. Her score on the IQ test was 68. What does Candace s score say about her ability to excel or even succeed in regular education classes without assistance? Why were her difficulties never noticed or addressed?

Summary

In this section, we learned about the history of intelligence testing and some of the challenges regarding intelligence testing. Intelligence tests began in earnest with Binet; Wechsler later developed intelligence tests that are still in use today: the WAIS-IV and WISC-V. The Bell curve shows the range of scores that encompass average intelligence as well as standard deviations.

Review Questions

OpenStax learning exercise

Q. In order for a test to be normed and standardized it must be tested on _____.

A. B

OpenStax learning exercise

Q. The mean score for a person with an average IQ is _____.

A. D

OpenStax learning exercise

Q. Who developed the IQ test most widely used today?

A. D

OpenStax learning exercise

Q. The DSM-5 now uses _____ as a diagnostic label for what was once referred to as mental retardation.

A. C

Critical Thinking Questions

OpenStax learning exercise

Q. Why do you think different theorists have defined intelligence in different ways?

A. Since cognitive processes are complex, ascertaining them in a measurable way is challenging. Researchers have taken different approaches to define intelligence in an attempt to comprehensively describe and measure it.

OpenStax learning exercise

Q. Compare and contrast the benefits of the Stanford-Binet IQ test and Wechsler s IQ tests.

A. The Wechsler-Bellevue IQ test combined a series of subtests that tested verbal and nonverbal skills into a single IQ test in order to get a reliable, descriptive score of intelligence. While the Stanford-Binet test was normed and standardized, it focused more on verbal skills than variations in other cognitive processes.

Personal Application Question

OpenStax learning exercise

Q. In thinking about the case of Candace described earlier, do you think that Candace benefitted or suffered as a result of consistently being passed on to the next grade?

The Source of Intelligence

OpenStax source module m82211 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82211/index.cnxml

A young girl, born of teenage parents, lives with her grandmother in rural Mississippi. They have a very low income, but they do their best to get by with what they have. She learns to read when she is just 3 years old. As she grows older, she longs to live with her mother, who now resides in Wisconsin. She moves there at the age of 6 years. At 9 years of age, she is raped. During the next several years, several

different male relatives repeatedly molest her. Her life unravels. She turns to drugs and sex to fill the deep, lonely void inside her. Her mother then sends her to Nashville to live with her father, who imposes strict behavioral expectations upon her, and over time, her wild life settles once again. She begins to experience success in school, and at 19 years old, becomes the youngest and first African-American female news anchor (Dates and Events, n.d.). The woman Oprah Winfrey goes on to become a media giant known for both her intelligence and her empathy.

High Intelligence: Nature or Nurture?

Where does high intelligence come from? Some researchers believe that intelligence is a trait inherited from a person s parents. Scientists who research this topic typically use twin studies to determine the heritability of intelligence. The Minnesota Study of Twins Reared Apart is one of the most well-known twin studies. In this investigation, researchers found that identical twins raised together and identical twins raised apart exhibit a higher correlation between their IQ scores than siblings or fraternal twins raised together (Bouchard, Lykken, McGue, Segal, & Tellegen, 1990). The findings from this study reveal a genetic component to intelligence (the accompanying source-figure description). At the same time, other psychologists believe that intelligence is shaped by a child s developmental environment. If parents were to provide their children with intellectual stimuli from before they are born, it is likely that they would absorb the benefits of that stimulation, and it would be reflected in intelligence levels.

Source illustration description (image not reproduced): A chart shows correlations of IQs for people of varying relationships. The bottom is labeled Percent IQ Correlation and the left side is labeled Relationship. The percent IQ Correlation for relationships where no genes are shared, including adoptive parent-child pairs, similarly aged unrelated children raised together, and adoptive siblings are around 21 percent, 30 percent, and 32 percent, respectively. The percent IQ Correlation for relationships where 25 percent of genes are shared, as in half-siblings, is around 33 percent. The percent IQ Correlation for relationships where 50 percent of genes are shared, including parent-children pairs, and fraternal twins raised together, are roughly 44 percent and 62 percent, respectively. A relationship where 100 percent of genes are shared, as in identical twins raised apart, results in a nearly 80 percent IQ correlation. The correlations of IQs of unrelated versus related persons reared apart or together suggest a genetic component to intelligence.

The reality is that aspects of each idea are probably correct. In fact, one study suggests that although genetics seem to be in control of the level of intelligence, the environmental influences provide both stability and change to trigger manifestation of cognitive abilities (Bartels, Rietveld, Van Baal, & Boomsma, 2002). Certainly, there are behaviors that support the development of intelligence, but the genetic component of high intelligence should not be ignored. As with all heritable traits, however, it is not always possible to isolate how and when high intelligence is passed on to the next generation.

Range of Reaction is the theory that each person responds to the environment in a unique way based on their genetic makeup. According to this idea, your genetic potential is a fixed quantity, but whether you reach your full intellectual potential is dependent upon the environmental stimulation you experience, especially in childhood. Think about this scenario: A couple adopts a child who has average genetic intellectual potential. They raise her in an extremely stimulating environment. What will happen to the couple s new daughter? It is likely that the stimulating environment will improve her intellectual outcomes over the course of her life. But what happens if this experiment is reversed? If a child with an extremely strong genetic background is placed in an environment that does not stimulate him: What happens? Interestingly, according to a longitudinal study of highly gifted individuals, it was found that the two extremes of optimal and pathological experience are both represented disproportionately in the backgrounds of creative individuals ; however, those who experienced supportive family environments were more likely to report being happy (Csikszentmihalyi & Csikszentmihalyi, 1993, p. 187).

Another challenge to determining origins of high intelligence is the confounding nature of our human social structures. It is troubling to note that some ethnic groups perform better on IQ tests than others and it is likely that the results do not have much to do with the quality of each ethnic group s intellect. The same is true for socioeconomic status. Children who live in poverty experience more pervasive, daily stress than children who do not worry about the basic needs of safety, shelter, and food.

These worries can negatively affect how the brain functions and develops, causing a dip in IQ scores. Mark Kishiyama and his colleagues determined that children living in poverty demonstrated reduced prefrontal brain functioning comparable to children with damage to the lateral prefrontal cortex (Kishiyama, Boyce, Jimenez, Perry, & Knight, 2009).

The debate around the foundations and influences on intelligence exploded in 1969, when an educational psychologist named Arthur Jensen published the article How Much Can We Boost I.Q. and Achievement in the Harvard Educational Review. Jensen had administered IQ tests to diverse groups of students, and his results led him to the conclusion that IQ is determined by genetics. He also posited that intelligence was made up of two types of abilities: Level I and Level II. In his theory, Level I is responsible for rote memorization, whereas Level II is responsible for conceptual and analytical abilities. According to his findings, Level I remained consistent among the human race. Level II, however, exhibited differences among ethnic groups (Modgil & Routledge, 1987). Jensen s most controversial conclusion was that Level II intelligence is prevalent among Asian people, then White people, then Black people. Robert Williams was among those who called out racial bias in Jensen s results (Williams, 1970).

Obviously, Jensen s interpretation of his own data caused an intense response in a nation that continued to grapple with the effects of racism (Fox, 2012). While even some who took issue with Jensen's findings indicated that they did not detect overt racism in his work, Jensen himself made a number of racist statements during an interview with a White nationalist publication, American Renaissance. He indicated his belief that many Black people were not "educable up to a level for which there s any economic demand," while also having a higher birth rate than other groups, which would lead to a shift in population and a deterioration of the nation (Taylor, 1992). However, Jensen s ideas were not solitary or unique; rather, they represented one of many examples of psychologists asserting racial differences in IQ and cognitive ability. In fact, Rushton and Jensen (2005) reviewed three decades worth of research on the relationship between race and cognitive ability. Jensen s belief in the inherited nature of intelligence and the validity of the IQ test to be the truest measure of intelligence are at the core of his conclusions.

In a related story, parents of African American students filed a case against the State of California in 1979, because they believed that the testing method used to identify students with learning disabilities was culturally unfair as the tests were normed and standardized using White children (Larry P. v. Riles). The testing method used by the state disproportionately identified African American children as mentally retarded," which resulted in many students being incorrectly classified. According to a summary of the case, Larry P. v. Riles:

In violation of Title VI of the Civil Rights Act of 1964, the Rehabilitation Act of 1973, and the Education for All Handicapped Children Act of 1975, defendants have utilized standardized intelligence tests that are racially and culturally biased, have a discriminatory impact against Black children, and have not been validated for the purpose of essentially permanent placements of Black children into educationally dead-end, isolated, and stigmatizing classes for the so-called educable mentally retarded. Further, these federal laws have been violated by defendants' general use of placement mechanisms that, taken together, have not been validated and result in a large over-representation of Black children in the special E.M.R. classes. (Larry P. v. Riles, par. 6)

Once again, the limitations of intelligence testing were revealed.

What are Learning Disabilities?

Learning disabilities are cognitive disorders that affect different areas of cognition, particularly language or reading. It should be pointed out that learning disabilities are not the same thing as intellectual disabilities. Learning disabilities are considered specific neurological impairments rather than global intellectual or developmental disabilities. A person with a language disability has difficulty understanding or using spoken language, whereas someone with a reading disability, such as dyslexia, has difficulty processing what they are reading.

Often, learning disabilities are not recognized until a child reaches school age. One confounding aspect of learning disabilities is that they most often affect children with average to above-average intelligence. In other words, the disability is specific to a particular area and not a measure of overall intellectual ability. At the same time, learning disabilities tend to exhibit comorbidity with other disorders, like attention-deficit hyperactivity disorder (ADHD). Anywhere between 30 70% of individuals with diagnosed cases of ADHD also have some sort of learning disability (Riccio, Gonzales, & Hynd, 1994). Let s take a look at three examples of common learning disabilities: dysgraphia, dyslexia, and dyscalculia.

Dysgraphia

Children with dysgraphia have a learning disability that results in a struggle to write legibly. The physical task of writing with a pen and paper is extremely challenging for the person. These children often have extreme difficulty putting their thoughts down on paper (Smits-Engelsman & Van Galen, 1997). This difficulty is inconsistent with a person s IQ. That is, based on the child s IQ and/or abilities in other areas, a child with dysgraphia should be able to write, but can t. Children with dysgraphia may also have problems with spatial abilities.

Students with dysgraphia need academic accommodations to help them succeed in school. These accommodations can provide students with alternative assessment opportunities to demonstrate what they know (Barton, 2003). For example, a student with dysgraphia might be permitted to take an oral exam rather than a traditional paper-and-pencil test. Treatment is usually provided by an occupational therapist, although there is some question as to how effective such treatment is (Zwicker, 2005).

Dyslexia

Dyslexia is the most common learning disability in children. An individual with dyslexia exhibits an inability to correctly process letters. The neurological mechanism for sound processing does not work properly in someone with dyslexia. As a result, dyslexic children may not understand sound-letter correspondence. A child with dyslexia may mix up letters within words and sentences letter reversals, such as those shown in the accompanying source-figure description, are a hallmark of this learning disability or skip whole words while reading. A dyslexic child may have difficulty spelling words correctly while writing. Because of the disordered way that the brain processes letters and sound, learning to read is a frustrating experience. Some dyslexic individuals cope by memorizing the shapes of most words, but they never actually learn to read (Berninger, 2008).

Source illustration description (image not reproduced): Two columns and five rows all containing the word teapot are shown. Teapot is written ten times with the letters jumbled, sometimes appearing backwards and upside down. These written words show variations of the word teapot as written by individuals with dyslexia.

Dyscalculia

Dyscalculia is difficulty in learning or comprehending arithmetic. This learning disability is often first evident when children exhibit difficulty discerning how many objects are in a small group without counting them. Other symptoms may include struggling to memorize math facts, organize numbers, or fully differentiate between numerals, math symbols, and written numbers (such as 3 and three).

Summary

Genetics and environment affect intelligence and the challenges of certain learning disabilities. The intelligence levels of all individuals seem to benefit from rich stimulation in their early environments. Highly intelligent individuals, however, may have a built-in resiliency that allows them to overcome difficult obstacles in their upbringing. Learning disabilities can cause major challenges for children who are learning to read and write. Unlike developmental disabilities, learning disabilities are strictly neurological in nature and are not related to intelligence levels. Students with dyslexia, for example, may have extreme difficulty learning to read, but their intelligence levels are typically average or above average.

Review Questions

OpenStax learning exercise

Q. Where does high intelligence come from?

A. C

OpenStax learning exercise

Q. Arthur Jensen believed that _____.

A. A

OpenStax learning exercise

Q. What is a learning disability?

A. B

OpenStax learning exercise

Q. Which of the following statements is true?

A. D

Critical Thinking Questions

OpenStax learning exercise

Q. What evidence exists for a genetic component to an individual s IQ?

A. Twin studies are one strong indication that IQ has a genetic component. Another indication is anecdotal evidence in the form of stories about highly intelligent individuals who come from difficult backgrounds yet still become highly successful adults.

OpenStax learning exercise

Q. Describe the relationship between learning disabilities and intellectual disabilities to intelligence.

A. Learning disabilities are specific neurological problems within the brain and are separate from intelligence. Intellectual disabilities are pervasive and related to intelligence.

Personal Application Question

OpenStax learning exercise

Q. Do you believe your level of intelligence was improved because of the stimuli in your childhood environment? Why or why not?

CHAPTER 11 / CAC.901

Memory development and children s accounts

Lecture: Memory is constructed, communicated, and checked

Memory is not a recording that can be replayed without change. Children encode selected information, retain it through biological and cognitive processes, and retrieve it in a context. What they remember can be influenced by attention, knowledge, emotion, language, and the questions asked afterward. This does not mean that children s accounts are generally unreliable. It means that the conditions of remembering and reporting deserve careful attention.

Counselors often need information about events that cannot be observed directly. The ethical and investigative stakes can be high. A developmental understanding of memory helps professionals avoid two opposite errors: treating every statement as a literal record, or dismissing a child because memory is imperfect. Neither approach is responsible. The quality of the account, the interviewing process, and corroborating information all matter.

Remembering is not one ability

Recognition and recall make different demands. A child may recognize a familiar person without being able to provide a detailed description. A prompt may help retrieval, but it may also introduce information. Open invitations generally allow the child to organize an account, while highly specific questions can narrow attention or imply an expected answer. The appropriate method depends on the purpose and the professional role.

Source memory concerns where information came from. A child may remember an event, something heard afterward, or a mixture of both. Repeated suggestion can affect confidence and detail. Counselors should avoid repeated interviews that rehearse a presumed story. When forensic investigation is possible, follow local protocols and coordinate with professionals trained to conduct investigative interviews. The counseling role is not to prove an allegation through repeated questioning.

Developmental differences affect narrative length, temporal language, and the amount of detail a child can provide. A sparse account is not necessarily false, and a detailed account is not necessarily complete. Confidence alone should not be treated as a guarantee. Consider consistency at the level of central meaning while recognizing that peripheral details may vary.

Emotion and memory

Strong emotion can influence what is attended to and later remembered. Stress may narrow attention or make retrieval more difficult, but effects vary. Avoid simple claims that trauma always produces perfect memories or always prevents coherent recall. A child s developmental level, relationship to the event, repeated exposure, and current safety can all matter.

The counseling setting should prioritize safety, regulation, and the child s control over disclosure. Do not demand a detailed trauma narrative simply because an adult believes it will be therapeutic. Evidence-based trauma treatments have structured approaches and require appropriate training and assessment. Unnecessary pressure for detail can distress a child without improving understanding.

Supporting communication without shaping an account

Use neutral invitations, allow silence, and avoid supplying names or events the child has not mentioned. Record the child s exact words where appropriate, and distinguish them from the counselor s interpretation. Document the context, questions asked, and people present. If an interpreter is involved, record that role and avoid treating translation as word-for-word certainty.

Figure 11.1 separates experience, encoding, later information, retrieval, and reporting. It shows several points at which the final account can be influenced. The diagram is not a reason to distrust children automatically. It is a reason to protect the conditions under which they can communicate as freely and accurately as possible.

Corroboration should be proportionate and lawful. Not every counseling concern requires an investigation, but serious safety concerns require appropriate action under local requirements. Explain confidentiality limits in advance. Share only what is necessary with those responsible for protection, and avoid turning a child s disclosure into broad community discussion.

Application: a concerning school statement

A fictional seven-year-old tells a counselor that someone hurt me and then stops speaking. The counselor s first tasks are to respond calmly, ensure immediate safety, and allow the child to communicate without pressure. The counselor should not begin a sequence of leading questions to establish a complete story. Follow safeguarding procedures and document the spontaneous statement accurately.

Later disagreements about details should not be handled by arguing with the child. Different professionals may need to clarify the account through appropriate roles. The counselor can continue providing support while avoiding contamination of an investigation. The child should not feel that care depends on producing a satisfactory narrative.

This example illustrates the distinction between therapeutic listening and evidence gathering. Both require respect, but they have different purposes and methods. A counselor who confuses them may unintentionally compromise the child s welfare or the interpretation of the account.

Doctoral seminar task

Review a fictional interview transcript containing open invitations, specific questions, and leading assumptions. Identify how each question could influence the response. Rewrite the sequence using neutral language while preserving the counselor s duty to respond to safety concerns. Explain which decisions belong to another professional role.

Then write a documentation note that separates exact quotation, observed behavior, and interpretation. Avoid adding a causal conclusion that the interview cannot support. The exercise demonstrates that careful language is a clinical and ethical skill, not merely a reporting style.

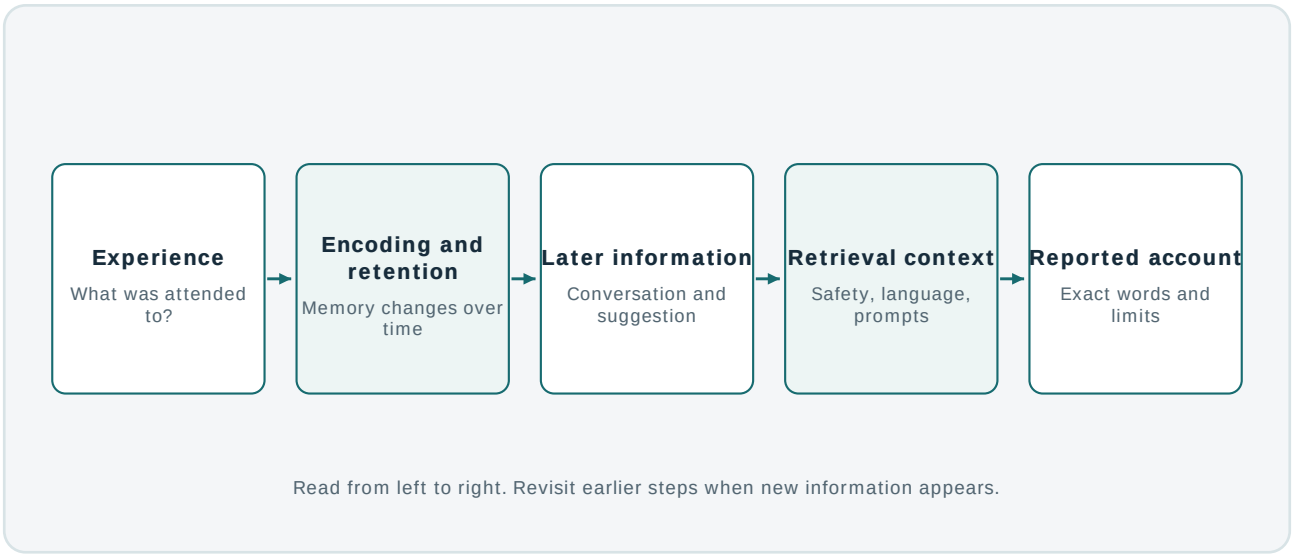


Figure 11.1. An account has a communication history. Experience, attention, later information, retrieval conditions, and questioning can influence what is reported. Protect neutral communication and distinguish therapeutic support from investigative evidence gathering. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

How Memory Functions

OpenStax source module m82213 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82213/index.cnxml

Memory is an information processing system; therefore, we often compare it to a computer. Memory is the set of processes used to encode, store, and retrieve information over different periods of time (the accompanying source-figure description).

Source illustration description (image not reproduced): A diagram shows three boxes, placed in a row from left to right, respectively titled Encoding, Storage, and Retrieval. One right-facing arrow connects Encoding to Storage and another connects Storage to Retrieval. Encoding involves the input of information into the memory system. Storage is the retention of the encoded information. Retrieval, or getting the information out of memory and back into awareness, is the third function.

Encoding

We get information into our brains through a process called encoding, which is the input of information into the memory system. Once we receive sensory information from the environment, our brains label or code it. We organize the information with other similar information and connect new concepts to existing concepts. Encoding information occurs through automatic processing and effortful processing.

If someone asks you what you ate for lunch today, more than likely you could recall this information quite easily. This is known as automatic processing, or the encoding of details like time, space, frequency, and the meaning of words. Automatic processing is usually done without any conscious awareness. Recalling the last time you studied for a test is another example of automatic processing. But what about the actual test material you studied? It probably required a lot of work and attention on your part in order to encode that information. This is known as effortful processing (the accompanying source-figure description).

Source illustration description (image not reproduced): A photograph shows a person driving a car. When you first learn new skills such as driving a car, you have to put forth effort and attention to encode information about how to start a car, how to brake, how to handle a turn, and so on. Once you know how to drive, you can encode additional information about this skill automatically. (credit: Robert Couse-Baker)

What are the most effective ways to ensure that important memories are well encoded? Even a simple sentence is easier to recall when it is meaningful (Anderson, 1984). Read the following sentences (Bransford & McCarrell, 1974), then look away and count backwards from 30 by threes to zero, and then try to write down the sentences (no peeking back at this page!).

- The notes were sour because the seams split.
- The voyage wasn't delayed because the bottle shattered.
- The haystack was important because the cloth ripped.

How well did you do? By themselves, the statements that you wrote down were most likely confusing and difficult for you to recall. Now, try writing them again, using the following prompts: bagpipe, ship

christening, and parachutist. Next count backwards from 40 by fours, then check yourself to see how well you recalled the sentences this time. You can see that the sentences are now much more memorable because each of the sentences was placed in context. Material is far better encoded when you make it meaningful.

There are three types of encoding. The encoding of words and their meaning is known as semantic encoding. It was first demonstrated by William Bousfield (1935) in an experiment in which he asked people to memorize words. The 60 words were actually divided into 4 categories of meaning, although the participants did not know this because the words were randomly presented. When they were asked to remember the words, they tended to recall them in categories, showing that they paid attention to the meanings of the words as they learned them.

Visual encoding is the encoding of images, and acoustic encoding is the encoding of sounds, words in particular. To see how visual encoding works, read over this list of words: car, level, dog, truth, book, value. If you were asked later to recall the words from this list, which ones do you think you d most likely remember? You would probably have an easier time recalling the words car, dog, and book, and a more difficult time recalling the words level, truth, and value. Why is this? Because you can recall images (mental pictures) more easily than words alone. When you read the words car, dog, and book you created images of these things in your mind. These are concrete, high-imagery words. On the other hand, abstract words like level, truth, and value are low-imagery words. High-imagery words are encoded both visually and semantically (Paivio, 1986), thus building a stronger memory.

Now let s turn our attention to acoustic encoding. You are driving in your car and a song comes on the radio that you haven t heard in at least 10 years, but you sing along, recalling every word. In the United States, children often learn the alphabet through song, and they learn the number of days in each month through rhyme: Thirty days hath September, / April, June, and November; / All the rest have thirty-one, / Save February, with twenty-eight days clear, / And twenty-nine each leap year. These lessons are easy to remember because of acoustic encoding. We encode the sounds the words make. This is one of the reasons why much of what we teach young children is done through song, rhyme, and rhythm.

Which of the three types of encoding do you think would give you the best memory of verbal information? Some years ago, psychologists Fergus Craik and Endel Tulving (1975) conducted a series of experiments to find out. Participants were given words along with questions about them. The questions required the participants to process the words at one of the three levels. The visual processing questions included such things as asking the participants about the font of the letters. The acoustic processing questions asked the participants about the sound or rhyming of the words, and the semantic processing questions asked the participants about the meaning of the words. After participants were presented with the words and questions, they were given an unexpected recall or recognition task.

Words that had been encoded semantically were better remembered than those encoded visually or acoustically. Semantic encoding involves a deeper level of processing than the shallower visual or acoustic encoding. Craik and Tulving concluded that we process verbal information best through semantic encoding, especially if we apply what is called the self-reference effect. The self-reference effect is the tendency for an individual to have better memory for information that relates to oneself in comparison to material that has less personal relevance (Rogers, Kuiper, & Kirker, 1977). Could semantic encoding be beneficial to you as you attempt to memorize the concepts in this chapter?

Storage

Once the information has been encoded, we have to somehow retain it. Our brains take the encoded information and place it in storage. Storage is the creation of a permanent record of information.

In order for a memory to go into storage (i.e., long-term memory), it has to pass through three distinct stages: Sensory Memory, Short-Term Memory, and finally Long-Term Memory. These stages were first

proposed by Richard Atkinson and Richard Shiffrin (1968). Their model of human memory (the accompanying source-figure description), called Atkinson and Shiffrin's model, is based on the belief that we process memories in the same way that a computer processes information.

Source illustration description (image not reproduced): A flow diagram consists of four boxes with connecting arrows. The first box is labeled sensory input. An arrow leads to the second box, which is labeled sensory memory. An arrow leads to the third box which is labeled short-term memory (STM). An arrow points to the fourth box, labeled long-term memory (LTM), and an arrow points in the reverse direction from the fourth to the third box. Above the short-term memory box, an arrow leaves the top-right of the box and curves around to point back to the top-left of the box; this arrow is labeled rehearsal. Both the sensory memory and short-term memory boxes have an arrow beneath them pointing to the text information not transferred is lost. According to the Atkinson-Shiffrin model of memory, information passes through three distinct stages in order for it to be stored in long-term memory.

Atkinson and Shiffrin's model is not the only model of memory. Baddeley and Hitch (1974) proposed a working memory model in which short-term memory has different forms. In their model, storing memories in short-term memory is like opening different files on a computer and adding information. The working memory files hold a limited amount of information. The type of short-term memory (or computer file) depends on the type of information received. There are memories in visual-spatial form, as well as memories of spoken or written material, and they are stored in three short-term systems: a visuospatial sketchpad, an episodic buffer (Baddeley, 2000), and a phonological loop. According to Baddeley and Hitch, a central executive part of memory supervises or controls the flow of information to and from the three short-term systems, and the central executive is responsible for moving information into long-term memory.

Sensory Memory

In the Atkinson-Shiffrin model, stimuli from the environment are processed first in sensory memory: storage of brief sensory events, such as sights, sounds, and tastes. It is very brief storage up to a couple of seconds. We are constantly bombarded with sensory information. We cannot absorb all of it, or even most of it. And most of it has no impact on our lives. For example, what was your professor wearing the last class period? As long as the professor was dressed appropriately, it does not really matter what she was wearing. Sensory information about sights, sounds, smells, and even textures, which we do not view as valuable information, we discard. If we view something as valuable, the information will move into our short-term memory system.

Short-Term Memory

Short-term memory (STM) is a temporary storage system that processes incoming sensory memory. The terms short-term and working memory are sometimes used interchangeably, but they are not exactly the same. Short-term memory is more accurately described as a component of working memory. Short-term memory takes information from sensory memory and sometimes connects that memory to something already in long-term memory. Short-term memory storage lasts 15 to 30 seconds. Think of it as the information you have displayed on your computer screen, such as a document, spreadsheet, or website. Then, information in STM goes to long-term memory (you save it to your hard drive), or it is discarded (you delete a document or close a web browser).

Rehearsal moves information from short-term memory to long-term memory. Active rehearsal is a way of attending to information to move it from short-term to long-term memory. During active rehearsal, you repeat (practice) the information to be remembered. If you repeat it enough, it may be moved into long-term memory. For example, this type of active rehearsal is the way many children learn their ABCs by singing the alphabet song. Alternatively, elaborative rehearsal is the act of linking new information you are trying to learn to existing information that you already know. For example, if you meet someone at a party and your phone is dead but you want to remember his phone number, which starts with area code 203, you might remember that your uncle Abdul lives in Connecticut and has a 203 area code. This way, when you try to remember the phone number of your new prospective friend, you will easily remember the area code. Craik and Lockhart (1972) proposed the levels of processing hypothesis that states the deeper you think about something, the better you remember it.

You may find yourself asking, How much information can our memory handle at once? To explore the capacity and duration of your short-term memory, have a partner read the strings of random numbers (the accompanying source-figure description) out loud to you, beginning each string by saying, Ready? and ending each by saying, Recall, at which point you should try to write down the string of numbers from memory.

Source illustration description (image not reproduced): A series of numbers includes two rows, with six numbers in each row. From left to right, the numbers increase from four digits to five, six, seven, eight, and nine digits. The first row includes 9754, 68259, 913825, 5316842, 86951372, and 719384273, and the second row includes 6419, 67148, 648327, 5963827, 51739826, and 163875942. Work through this series of numbers using the recall exercise explained above to determine the longest string of digits that you can store.

Note the longest string at which you got the series correct. For most people, the capacity will probably be close to 7 plus or minus 2. In 1956, George Miller reviewed most of the research on the capacity of short-term memory and found that people can retain between 5 and 9 items, so he reported the capacity of short-term memory was the "magic number" 7 plus or minus 2. However, more contemporary research has found working memory capacity is 4 plus or minus 1 (Cowan, 2010). Generally, recall is somewhat better for random numbers than for random letters (Jacobs, 1887) and also often slightly better for information we hear (acoustic encoding) rather than information we see (visual encoding) (Anderson, 1969).

Memory trace decay and interference are two factors that affect short-term memory retention. Peterson and Peterson (1959) investigated short-term memory using the three letter sequences called trigrams (e.g., CLS) that had to be recalled after various time intervals between 3 and 18 seconds. Participants remembered about 80% of the trigrams after a 3-second delay, but only 10% after a delay of 18 seconds, which caused them to conclude that short-term memory decayed in 18 seconds. During decay, the memory trace becomes less activated over time, and the information is forgotten. However, Keppel and Underwood (1962) examined only the first trials of the trigram task and found that proactive interference also affected short-term memory retention. During proactive interference, previously learned information interferes with the ability to learn new information. Both memory trace decay and proactive interference affect short-term memory. Once the information reaches long-term memory, it has to be consolidated at both the synaptic level, which takes a few hours, and into the memory system, which can take weeks or longer.

Long-term Memory

Long-term memory (LTM) is the continuous storage of information. Unlike short-term memory, long-term memory storage capacity is believed to be unlimited. It encompasses all the things you can remember that happened more than just a few minutes ago. One cannot really consider long-term memory without thinking about the way it is organized. Really quickly, what is the first word that comes to mind when you hear peanut butter ? Did you think of jelly? If you did, you probably have associated peanut butter and jelly in your mind. It is generally accepted that memories are organized in semantic (or associative) networks (Collins & Loftus, 1975). A semantic network consists of concepts, and as you may recall from what you ve learned about memory, concepts are categories or groupings of linguistic information, images, ideas, or memories, such as life experiences. Although individual experiences and expertise can affect concept arrangement, concepts are believed to be arranged hierarchically in the mind (Anderson & Reder, 1999; Johnson & Mervis, 1997, 1998; Palmer, Jones, Hennessy, Unze, & Pick, 1989; Rosch, Mervis, Gray, Johnson, & Boyes-Braem, 1976; Tanaka & Taylor, 1991). Related concepts are linked, and the strength of the link depends on how often two concepts have been associated.

Semantic networks differ depending on personal experiences. Importantly for memory, activating any part of a semantic network also activates the concepts linked to that part to a lesser degree. The process is known as spreading activation (Collins & Loftus, 1975). If one part of a network is activated, it is easier to access the associated concepts because they are already partially activated. When you remember or recall something, you activate a concept, and the related concepts are more easily remembered because they are partially activated. However, the activations do not spread in just one

direction. When you remember something, you usually have several routes to get the information you are trying to access, and the more links you have to a concept, the better your chances of remembering.

There are two types of long-term memory: explicit and implicit (the accompanying source-figure description). Understanding the difference between explicit memory and implicit memory is important because aging, particular types of brain trauma, and certain disorders can impact explicit and implicit memory in different ways. Explicit memories are those we consciously try to remember, recall, and report. For example, if you are studying for your chemistry exam, the material you are learning will be part of your explicit memory. In keeping with the computer analogy, some information in your long-term memory would be like the information you have saved on the hard drive. It is not there on your desktop (your short-term memory), but most of the time you can pull up this information when you want it. Not all long-term memories are strong memories, and some memories can only be recalled using prompts. For example, you might easily recall a fact, such as the capital of the United States, but you might struggle to recall the name of the restaurant at which you had dinner when you visited a nearby city last summer. A prompt, such as that the restaurant was named after its owner, might help you recall the name of the restaurant. Explicit memory is sometimes referred to as declarative memory, because it can be put into words. Explicit memory is divided into episodic memory and semantic memory.

Episodic memory is information about events we have personally experienced (i.e., an episode). For instance, the memory of your last birthday is an episodic memory. Usually, episodic memory is reported as a story. The concept of episodic memory was first proposed about in the 1970s (Tulving, 1972). Since then, Tulving and others have reformulated the theory, and currently scientists believe that episodic memory is memory about happenings in particular places at particular times the what, where, and when of an event (Tulving, 2002). It involves recollection of visual imagery as well as the feeling of familiarity (Hassabis & Maguire, 2007). Semantic memory is knowledge about words, concepts, and language-based knowledge and facts. Semantic memory is typically reported as facts. Semantic means having to do with language and knowledge about language. For example, answers to the following questions like what is the definition of psychology and who was the first African American president of the United States are stored in your semantic memory.

Implicit memories are long-term memories that are not part of our consciousness. Although implicit memories are learned outside of our awareness and cannot be consciously recalled, implicit memory is demonstrated in the performance of some task (Roediger, 1990; Schacter, 1987). Implicit memory has been studied with cognitive demand tasks, such as performance on artificial grammars (Reber, 1976), word memory (Jacoby, 1983; Jacoby & Witherspoon, 1982), and learning unspoken and unwritten contingencies and rules (Greenspoon, 1955; Giddan & Eriksen, 1959; Krieckhaus & Eriksen, 1960). Returning to the computer metaphor, implicit memories are like a program running in the background, and you are not aware of their influence. Implicit memories can influence observable behaviors as well as cognitive tasks. In either case, you usually cannot put the memory into words that adequately describe the task. There are several types of implicit memories, including procedural, priming, and emotional conditioning.

Source illustration description (image not reproduced): A diagram consists of three rows of boxes. The box in the top row is labeled long-term memory; a line from the box separates into two lines leading to two boxes on the second row, labeled explicit memory and implicit memory. From each of the second row boxes, lines split and lead to additional boxes. From the explicit memory box are two boxes labeled episodic (events and experiences) and semantic (concepts and facts). From the implicit memory box are three boxes labeled procedural (How to do things), Priming (stimulus exposure affects responses to a later stimulus), and emotional conditioning (Classically conditioned emotional responses). There are two components of long-term memory: explicit and implicit. Explicit memory includes episodic and semantic memory. Implicit memory includes procedural memory and things learned through conditioning.

Implicit procedural memory is often studied using observable behaviors (Adams, 1957; Lacey & Smith, 1954; Lazarus & McCleary, 1951). Implicit procedural memory stores information about the way to do something, and it is the memory for skilled actions, such as brushing your teeth, riding a bicycle, or driving a car. You were probably not that good at riding a bicycle or driving a car the first time you

tried, but you were much better after doing those things for a year. Your improved bicycle riding was due to learning balancing abilities. You likely thought about staying upright in the beginning, but now you just do it. Moreover, you probably are good at staying balanced, but cannot tell someone the exact way you do it. Similarly, when you first learned to drive, you probably thought about a lot of things that you just do now without much thought. When you first learned to do these tasks, someone may have told you how to do them, but everything you learned since those instructions that you cannot readily explain to someone else as the way to do it is implicit memory.

Implicit priming is another type of implicit memory (Schacter, 1992). During priming exposure to a stimulus affects the response to a later stimulus. Stimuli can vary and may include words, pictures, and other stimuli to elicit a response or increase recognition. For instance, some people really enjoy picnics. They love going into nature, spreading a blanket on the ground, and eating a delicious meal. Now, unscramble the following letters to make a word.

AETPL

What word did you come up with? Chances are good that it was "plate."

Had you read, Some people really enjoy growing flowers. They love going outside to their garden, fertilizing their plants, and watering their flowers, you probably would have come up with the word "petal" instead of plate.

Do you recall the earlier discussion of semantic networks? The reason people are more likely to come up with plate after reading about a picnic is that plate is associated (linked) with picnic. Plate was primed by activating the semantic network. Similarly, petal is linked to flower and is primed by flower. Priming is also the reason you probably said jelly in response to peanut butter.

Implicit emotional conditioning is the type of memory involved in classically conditioned emotion responses (Olson & Fazio, 2001). These emotional relationships cannot be reported or recalled but can be associated with different stimuli. For example, specific smells can cause specific emotional responses for some people. If there is a smell that makes you feel positive and nostalgic, and you don't know where that response comes from, it is an implicit emotional response. Similarly, most people have a song that causes a specific emotional response. That song's effect could be an implicit emotional memory (Yang, Xu, Du, Shi, & Fang, 2011).

Everyday Connection

Episodic memories are also called autobiographical memories. Let s quickly test your autobiographical memory. What were you wearing exactly five years ago today? What did you eat for lunch on April 10, 2009? You probably find it difficult, if not impossible, to answer these questions. Can you remember every event you have experienced over the course of your life meals, conversations, clothing choices, weather conditions, and so on? Most likely none of us could even come close to answering these questions; however, American actress Marilu Henner, best known for the television show Taxi, can remember. She has an amazing and highly superior autobiographical memory (the accompanying source-figure description).

Source illustration description (image not reproduced): A photograph shows Marilu Henner. Marilu Henner s super autobiographical memory is known as hyperthymesia. (credit: Mark Richardson)

Very few people can recall events in this way; right now, fewer than 20 have been identified as having this ability, and only a few have been studied (Parker, Cahill & McGaugh 2006). And although hyperthymesia normally appears in adolescence, two children in the United States appear to have memories from well before their tenth birthdays.

Retrieval

So you have worked hard to encode (via effortful processing) and store some important information for your upcoming final exam. How do you get that information back out of storage when you need it? The act of getting information out of memory storage and back into conscious awareness is known as retrieval. This would be similar to finding and opening a paper you had previously saved on your computer s hard drive. Now it s back on your desktop, and you can work with it again. Our ability to retrieve information from long-term memory is vital to our everyday functioning. You must be able to retrieve information from memory in order to do everything from knowing how to brush your hair and teeth, to driving to work, to knowing how to perform your job once you get there.

There are three ways you can retrieve information out of your long-term memory storage system: recall, recognition, and relearning. Recall is what we most often think about when we talk about memory retrieval: it means you can access information without cues. For example, you would use recall for an essay test. Recognition happens when you identify information that you have previously learned after encountering it again. It involves a process of comparison. When you take a multiple-choice test, you are relying on recognition to help you choose the correct answer. Here is another example. Let s say you graduated from high school 10 years ago, and you have returned to your hometown for your 10-year reunion. You may not be able to recall all of your classmates, but you recognize many of them based on their yearbook photos.

The third form of retrieval is relearning, and it s just what it sounds like. It involves learning information that you previously learned. Whitney took Spanish in high school, but after high school she did not have the opportunity to speak Spanish. Whitney is now 31, and her company has offered her an opportunity to work in their Mexico City office. In order to prepare herself, she enrolls in a Spanish course at the local community center. She s surprised at how quickly she s able to pick up the language after not speaking it for 13 years; this is an example of relearning.

Summary

Memory is a system or process that stores what we learn for future use.

Our memory has three basic functions: encoding, storing, and retrieving information. Encoding is the act of getting information into our memory system through automatic or effortful processing. Storage is retention of the information, and retrieval is the act of getting information out of storage and into conscious awareness through recall, recognition, and relearning. The idea that information is processed through three memory systems is called the Atkinson-Shiffrin model of memory. First, environmental stimuli enter our sensory memory for a period of less than a second to a few seconds. Those stimuli that we notice and pay attention to then move into short-term memory. According to the Atkinson-Shiffrin model, if we rehearse this information, then it moves into long-term memory for permanent storage. Other models like that of Baddeley and Hitch suggest there is more of a feedback loop between short-term memory and long-term memory. Long-term memory has a practically limitless storage capacity and is divided into implicit and explicit memory.

Review Questions

OpenStax learning exercise

Q. _____ is a memory store with a phonological loop, visuospatial sketchpad, episodic buffer, and a central executive.

A. C

OpenStax learning exercise

Q. The storage capacity of long-term memory is _____.

A. D

OpenStax learning exercise

Q. The three functions of memory are _____.

A. D

Critical Thinking Questions

OpenStax learning exercise

Q. Compare and contrast implicit and explicit memory.

A. Both are types of long-term memory. Explicit memories are memories we consciously try to remember and recall. Explicit memory is also called declarative memory and is subdivided into episodic memory (life events) and semantic memory (words, ideas, and concepts). Implicit memories are memories that are not part of our consciousness; they are memories formed from behaviors. Implicit memory is also called non-declarative memory and includes procedural memory as well as things learned through classical conditioning.

OpenStax learning exercise

Q. According to the Atkinson-Shiffrin model, name and describe the three stages of memory.

A. According to the Atkinson-Shiffrin model, memory is processed in three stages. The first is sensory memory; this is very brief: 1 2 seconds. Anything not attended to is ignored. The stimuli we pay attention to then move into our short-term memory. Short-term memory can hold approximately 7 bits of information for around 20 seconds. Information here is either forgotten, or it is encoded into long-term memory through the process of rehearsal. Long-term memory is the permanent storage of information its capacity is basically unlimited.

OpenStax learning exercise

Q. Compare and contrast the two ways in which we encode information.

A. Information is encoded through automatic or effortful processing. Automatic processing refers to all information that enters long-term memory without conscious effort. This includes things such as time, space, and frequency for example, your ability to remember what you ate for breakfast today or the fact that you remember that you ran into your best friend in the supermarket twice this week. Effortful processing refers to encoding information through conscious attention and effort. Material that you study for a test requires effortful processing.

Personal Application Questions

OpenStax learning exercise

Q. Describe something you have learned that is now in your procedural memory. Discuss how you learned this information.

OpenStax learning exercise

Q. Describe something you learned in high school that is now in your semantic memory.

Problems with Memory

OpenStax source module m82215 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82215/index.cnxml

You may pride yourself on your amazing ability to remember the birthdates and ages of all of your friends and family members, or you may be able recall vivid details of your 5th birthday party at Chuck E. Cheese s. However, all of us have at times felt frustrated, and even embarrassed, when our memories have failed us. There are several reasons why this happens.

Amnesia

Amnesia is the loss of long-term memory that occurs as the result of disease, physical trauma, or psychological trauma. Endel Tulving (2002) and his colleagues at the University of Toronto studied K. C. for years. K. C. suffered a traumatic head injury in a motorcycle accident and then had severe amnesia. Tulving writes,

the outstanding fact about K.C.'s mental make-up is his utter inability to remember any events, circumstances, or situations from his own life. His episodic amnesia covers his whole life, from birth to the present. The only exception is the experiences that, at any time, he has had in the last minute or two. (Tulving, 2002, p. 14)

Anterograde Amnesia

There are two common types of amnesia: anterograde amnesia and retrograde amnesia (the accompanying source-figure description). Anterograde amnesia is commonly caused by brain trauma, such as a blow to the head. With anterograde amnesia, you cannot remember new information, although you can remember information and events that happened prior to your injury. The hippocampus is usually affected (McLeod, 2011). This suggests that damage to the brain has resulted in the inability to transfer information from short-term to long-term memory; that is, the inability to consolidate memories.

Many people with this form of amnesia are unable to form new episodic or semantic memories, but are still able to form new procedural memories (Bayley & Squire, 2002). This was true of H. M., which was discussed earlier. The brain damage caused by his surgery resulted in anterograde amnesia. H. M. would read the same magazine over and over, having no memory of ever reading it it was always new to him. He also could not remember people he had met after his surgery. If you were introduced to H. M. and then you left the room for a few minutes, he would not know you upon your return and would introduce himself to you again. However, when presented the same puzzle several days in a row, although he did not remember having seen the puzzle before, his speed at solving it became faster each day (because of relearning) (Corkin, 1965, 1968).

Source illustration description (image not reproduced): A single-line flow diagram compares two types of amnesia. In the center is a box labeled event with arrows extending from both sides. Extending to the left is an arrow pointing left to the word past ; the arrow is labeled retrograde amnesia. Extending to the right is an arrow pointing right to the word present ; the arrow is labeled anterograde amnesia. This diagram illustrates the timeline of retrograde and anterograde amnesia. Memory problems that extend back in time before the injury and prevent retrieval of information previously stored in long-term memory are known as retrograde amnesia. Conversely, memory problems that extend forward in time from the point of injury and prevent the formation of new memories are called anterograde amnesia.

Retrograde Amnesia

Retrograde amnesia is loss of memory for events that occurred prior to the trauma. People with retrograde amnesia cannot remember some or even all of their past. They have difficulty remembering episodic memories. What if you woke up in the hospital one day and there were people surrounding your bed claiming to be your spouse, your children, and your parents? The trouble is you don t recognize any of them. You were in a car accident, suffered a head injury, and now have retrograde amnesia. You don t remember anything about your life prior to waking up in the hospital. This may sound like the stuff of Hollywood movies, and Hollywood has been fascinated with the amnesia plot for nearly a century, going all the way back to the film Garden of Lies from 1915 to more recent movies such as the Jason Bourne spy thrillers. However, for real-life sufferers of retrograde amnesia, like former NFL football player Scott Bolzan, the story is not a Hollywood movie. Bolzan fell, hit his head, and deleted 46 years of his life in an instant. He is now living with one of the most extreme cases of retrograde amnesia on record.

Memory Construction and Reconstruction

The formulation of new memories is sometimes called construction, and the process of bringing up old memories is called reconstruction. Yet as we retrieve our memories, we also tend to alter and modify them. A memory pulled from long-term storage into short-term memory is flexible. New events can be added and we can change what we think we remember about past events, resulting in inaccuracies and distortions. People may not intend to distort facts, but it can happen in the process of retrieving old memories and combining them with new memories (Roediger & DeSoto, 2015).

Suggestibility

When someone witnesses a crime, that person s memory of the details of the crime is very important in catching the suspect. Because memory is so fragile, witnesses can be easily (and often accidentally) misled due to the problem of suggestibility. Suggestibility describes the effects of misinformation from external sources that leads to the creation of false memories. In the fall of 2002, a sniper in the DC area shot people at a gas station, leaving Home Depot, and walking down the street. These attacks went on in a variety of places for over three weeks and resulted in the deaths of ten people. During this time, as you can imagine, people were terrified to leave their homes, go shopping, or even walk through their neighborhoods. Police officers and the FBI worked frantically to solve the crimes, and a tip hotline was set up. Law enforcement received over 140,000 tips, which resulted in approximately 35,000 possible suspects (Newseum, n.d.).

Most of the tips were dead ends, until a white van was spotted at the site of one of the shootings. The police chief went on national television with a picture of the white van. After the news conference, several other eyewitnesses called to say that they too had seen a white van fleeing from the scene of the shooting. At the time, there were more than 70,000 white vans in the area. Police officers, as well as the general public, focused almost exclusively on white vans because they believed the eyewitnesses. Other tips were ignored. When the suspects were finally caught, they were driving a blue sedan.

As illustrated by this example, we are vulnerable to the power of suggestion, simply based on something we see on the news. Or we can claim to remember something that in fact is only a suggestion someone made. It is the suggestion that is the cause of the false memory.

Eyewitness Misidentification

Even though memory and the process of reconstruction can be fragile, police officers, prosecutors, and the courts often rely on eyewitness identification and testimony in the prosecution of criminals. However, faulty eyewitness identification and testimony can lead to wrongful convictions (the accompanying source-figure description).

Source illustration description (image not reproduced): A bar graph is titled Leading cause of wrongful conviction in DNA exoneration cases (source: Innocence Project). The x-axis is labeled leading cause, and the y-axis is labeled percentage of wrongful convictions (first 239 DNA exonerations). Four bars show data: eyewitness misidentification

is the leading cause in about 75% of cases, forensic science in about 49% of cases, false confession in about 23% of cases, and informant in about 18% of cases. In studying cases where DNA evidence has exonerated people from crimes, the Innocence Project discovered that eyewitness misidentification is the leading cause of wrongful convictions (Benjamin N. Cardozo School of Law, Yeshiva University, 2009).

How does this happen? In 1984, Jennifer Thompson, then a 22-year-old college student in North Carolina, was brutally raped at knifepoint. As she was being raped, she tried to memorize every detail of her rapist s face and physical characteristics, vowing that if she survived, she would help get him convicted. After the police were contacted, a composite sketch was made of the suspect, and Jennifer was shown six photos. She chose two, one of which was of Ronald Cotton. After looking at the photos for 4 5 minutes, she said, Yeah. This is the one, and then she added, I think this is the guy. When questioned about this by the detective who asked, You re sure? Positive? She said that it was him. Then she asked the detective if she did OK, and he reinforced her choice by telling her she did great. These kinds of unintended cues and suggestions by police officers can lead witnesses to identify the wrong suspect. The district attorney was concerned about her lack of certainty the first time, so she viewed a lineup of seven men. She said she was trying to decide between numbers 4 and 5, finally deciding that Cotton, number 5, Looks most like him. He was 22 years old.

By the time the trial began, Jennifer Thompson had absolutely no doubt that she was raped by Ronald Cotton. She testified at the court hearing, and her testimony was compelling enough that it helped convict him. How did she go from, I think it s the guy and it Looks most like him, to such certainty? Gary Wells and Deah Quinlivan (2009) assert it s suggestive police identification procedures, such as stacking lineups to make the defendant stand out, telling the witness which person to identify, and confirming witnesses choices by telling them Good choice, or You picked the guy.

After Cotton was convicted of the rape, he was sent to prison for life plus 50 years. After 4 years in prison, he was able to get a new trial. Jennifer Thompson once again testified against him. This time Ronald Cotton was given two life sentences. After serving 11 years in prison, DNA evidence finally demonstrated that Ronald Cotton did not commit the rape, was innocent, and had served over a decade in prison for a crime he did not commit.

Ronald Cotton s story, unfortunately, is not unique. There are also people who were convicted and placed on death row, who were later exonerated. The Innocence Project is a non-profit group that works to exonerate falsely convicted people, including those convicted by eyewitness testimony. To learn more, you can visit http://www.innocenceproject.org.

Dig Deeper

Contrast the Cotton case with what happened in the Elizabeth Smart case. When Elizabeth was 14 years old and fast asleep in her bed at home, she was abducted at knifepoint. Her nine-year-old sister, Mary Katherine, was sleeping in the same bed and watched, terrified, as her beloved older sister was abducted. Mary Katherine was the sole eyewitness to this crime and was very fearful. In the following weeks, the Salt Lake City police and the FBI proceeded with caution with Mary Katherine. They did not want to implant any false memories or mislead her in any way. They did not show her police line-ups or push her to do a composite sketch of the abductor. They knew if they corrupted her memory, Elizabeth might never be found. For several months, there was little or no progress on the case. Then, about 4 months after the kidnapping, Mary Katherine first recalled that she had heard the abductor s voice prior to that night (he had worked exactly one day as a handyman at the family s home) and then she was able to name the person whose voice it was. The family contacted the press and others recognized him after a total of nine months, the suspect was caught and Elizabeth Smart was returned to her family.

The Misinformation Effect

Cognitive psychologist Elizabeth Loftus has conducted extensive research on memory. She has studied false memories as well as recovered memories of childhood sexual abuse. Loftus also developed the misinformation effect paradigm, which holds that after exposure to additional and possibly inaccurate information, a person may misremember the original event.

According to Loftus, an eyewitness s memory of an event is very flexible due to the misinformation effect. To test this theory, Loftus and John Palmer (1974) asked 45 U.S. college students to estimate the speed of cars using different forms of questions (the accompanying source-figure description). The participants were shown films of car accidents and were asked to play the role of the eyewitness and describe what happened. They were asked, About how fast were the cars going when they (smashed, collided, bumped, hit, contacted) each other? The participants estimated the speed of the cars based on the verb used.

Participants who heard the word smashed estimated that the cars were traveling at a much higher speed than participants who heard the word contacted. The implied information about speed, based on the verb they heard, had an effect on the participants memory of the accident. In a follow-up one week later, participants were asked if they saw any broken glass (none was shown in the accident pictures). Participants who had been in the smashed group were more than twice as likely to indicate that they did remember seeing glass. Loftus and Palmer demonstrated that a leading question encouraged them to not only remember the cars were going faster, but to also falsely remember that they saw broken glass.

Source illustration description (image not reproduced): Photograph A shows two cars that have crashed into each other. Part B is a bar graph titled perceived speed based on questioner s verb (source: Loftus and Palmer, 1974). The x-axis is labeled questioner s verb, and the y-axis is labeled perceived speed (mph). Five bars share data: smashed was perceived at about 41 mph, collided at about 39 mph, bumped at about 37 mph, hit at about 34 mph, and contacted at about 32 mph. When people are asked leading questions about an event, their memory of the event may be altered. (credit a: modification of work by Rob Young)

Controversies over Repressed and Recovered Memories

Other researchers have described how whole events, not just words, can be falsely recalled, even when they did not happen. The idea that memories of traumatic events could be repressed has been a theme in the field of psychology, beginning with Sigmund Freud, and the controversy surrounding the idea continues today.

Recall of false autobiographical memories is called false memory syndrome. This syndrome has received a lot of publicity, particularly as it relates to memories of events that do not have independent witnesses often the only witnesses to the abuse are the perpetrator and the victim (e.g., sexual abuse).

On one side of the debate are those who have recovered memories of childhood abuse years after it occurred. These researchers argue that some children s experiences have been so traumatizing and distressing that they must lock those memories away in order to lead some semblance of a normal life. They believe that repressed memories can be locked away for decades and later recalled intact through hypnosis and guided imagery techniques (Devilly, 2007).

Research suggests that having no memory of childhood sexual abuse is quite common in adults. For instance, one large-scale study conducted by John Briere and Jon Conte (1993) revealed that 59% of 450 men and women who were receiving treatment for sexual abuse that had occurred before age 18 had forgotten their experiences. Ross Cheit (2007) suggested that repressing these memories created psychological distress in adulthood. The Recovered Memory Project was created so that victims of childhood sexual abuse can recall these memories and allow the healing process to begin (Cheit, 2007; Devilly, 2007).

On the other side, Loftus has challenged the idea that individuals can repress memories of traumatic events from childhood, including sexual abuse, and then recover those memories years later through therapeutic techniques such as hypnosis, guided visualization, and age regression.

Loftus is not saying that childhood sexual abuse doesn t happen, but she does question whether or not those memories are accurate, and she is skeptical of the questioning process used to access these memories, given that even the slightest suggestion from the therapist can lead to misinformation effects. For example, researchers Stephen Ceci and Maggie Brucks (1993, 1995) asked three-year-old children to use an anatomically correct doll to show where their pediatricians had touched them during

an exam. Fifty-five percent of the children pointed to the genital/anal area on the dolls, even when they had not received any form of genital exam.

Ever since Loftus published her first studies on the suggestibility of eyewitness testimony in the 1970s, social scientists, police officers, therapists, and legal practitioners have been aware of the flaws in interview practices. Consequently, steps have been taken to decrease suggestibility of witnesses. One way is to modify how witnesses are questioned. When interviewers use neutral and less leading language, children more accurately recall what happened and who was involved (Goodman, 2006; Pipe, 1996; Pipe, Lamb, Orbach, & Esplin, 2004). Another change is in how police lineups are conducted. It s recommended that a blind photo lineup be used. This way the person administering the lineup doesn t know which photo belongs to the suspect, minimizing the possibility of giving leading cues. Additionally, judges in some states now inform jurors about the possibility of misidentification. Judges can also suppress eyewitness testimony if they deem it unreliable.

Forgetting

I ve a grand memory for forgetting, quipped Robert Louis Stevenson. Forgetting refers to loss of information from long-term memory. We all forget things, like a loved one s birthday, someone s name, or where we put our car keys. As you ve come to see, memory is fragile, and forgetting can be frustrating and even embarrassing. But why do we forget? To answer this question, we will look at several perspectives on forgetting.

Encoding Failure

Sometimes memory loss happens before the actual memory process begins, which is encoding failure. We can t remember something if we never stored it in our memory in the first place. This would be like trying to find a book on your e-reader that you never actually purchased and downloaded. Often, in order to remember something, we must pay attention to the details and actively work to process the information (effortful encoding). Lots of times we don t do this. For instance, think of how many times in your life you ve seen a penny. Can you accurately recall what the front of a U.S. penny looks like? When researchers Raymond Nickerson and Marilyn Adams (1979) asked this question, they found that most Americans don t know which one it is. The reason is most likely encoding failure. Most of us never encode the details of the penny. We only encode enough information to be able to distinguish it from other coins. If we don t encode the information, then it s not in our long-term memory, so we will not be able to remember it.

Source illustration description (image not reproduced): Four illustrations of nickels have minor differences in the placement and orientation of text. Can you tell which coin, (a), (b), (c), or (d) is the accurate depiction of a US nickel? The correct answer is (c).

Memory Errors

Psychologist Daniel Schacter (2001), a well-known memory researcher, offers seven ways our memories fail us. He calls them the seven sins of memory and categorizes them into three groups: forgetting, distortion, and intrusion (the accompanying source-figure description).

Source comparison table

Schacter s Seven Sins of Memory			
Sin	Type	Description	Example
Transience	Forgetting	Accessibility of memory decreases over time	Forget events that occurred long ago
absentmindedness	Forgetting	Forgetting caused by lapses in attention	Forget where your phone is

Schacter s Seven Sins of Memory			
Blocking	Forgetting	Accessibility of information is temporarily blocked	Tip of the tongue
Misattribution	Distortion	Source of memory is confused	Recalling a dream memory as a waking memory
Suggestibility	Distortion	False memories	Result from leading questions
Bias	Distortion	Memories distorted by current belief system	Align memories to current beliefs
Persistence	Intrusion	Inability to forget undesirable memories	Traumatic events

Let s look at the first sin of the forgetting errors: transience, which means that memories can fade over time. Here s an example of how this happens. Nathan s English teacher has assigned his students to read the novel To Kill a Mockingbird. Nathan comes home from school and tells his mom he has to read this book for class. Oh, I loved that book! she says. Nathan asks her what the book is about, and after some hesitation she says, Well . . . I know I read the book in high school, and I remember that one of the main characters is named Scout, and her father is an attorney, but I honestly don t remember anything else. Nathan wonders if his mother actually read the book, and his mother is surprised she can t recall the plot. What is going on here is storage decay: unused information tends to fade with the passage of time.

In 1885, German psychologist Hermann Ebbinghaus analyzed the process of memorization. First, he memorized lists of nonsense syllables. Then he measured how much he learned (retained) when he attempted to relearn each list. He tested himself over different periods of time from 20 minutes later to 30 days later. The result is his famous forgetting curve (the accompanying source-figure description). Due to storage decay, an average person will lose 50% of the memorized information after 20 minutes and 70% of the information after 24 hours (Ebbinghaus, 1885/1964). Your memory for new information decays quickly and then eventually levels out.

Source illustration description (image not reproduced): A line graph has an x-axis labeled elapsed time since learning with a scale listing these intervals: 0, 20, and 60 minutes; 9, 24, and 48 hours; and 6 and 31 days. The y-axis is labeled retention (%) with a scale of zero to 100. The line reflects these approximate data points: 0 minutes is 100%, 20 minutes is 55%, 60 minutes is 40%, 9 hours is 37%, 24 hours is 30%, 48 hours is 25%, 6 days is 20%, and 31 days is 10%. The Ebbinghaus forgetting curve shows how quickly memory for new information decays.

Are you constantly losing your cell phone? Have you ever driven back home to make sure you turned off the stove? Have you ever walked into a room for something, but forgotten what it was? You probably answered yes to at least one, if not all, of these examples but don t worry, you are not alone. We are all prone to committing the memory error known as absentmindedness, which describes lapses in memory caused by breaks in attention or our focus being somewhere else.

Cynthia, a psychologist, recalls a time when she recently committed the memory error of absentmindedness.

When I was completing court-ordered psychological evaluations, each time I went to the court, I was issued a temporary identification card with a magnetic strip which would open otherwise locked doors. As you can imagine, in a courtroom, this identification is valuable and important and no one wanted it to be lost or be picked up by a criminal. At the end of the day, I would hand in my temporary identification. One day, when I was almost done with an evaluation, my daughter s day care called and said she was

sick and needed to be picked up. It was flu season, I didn t know how sick she was, and I was concerned. I finished up the evaluation in the next ten minutes, packed up my briefcase, and rushed to drive to my daughter s day care. After I picked up my daughter, I could not remember if I had handed back my identification or if I had left it sitting out on a table. I immediately called the court to check. It turned out that I had handed back my identification. Why could I not remember that? (personal communication, September 5, 2013)

When have you experienced absentmindedness?

I just streamed this movie called Oblivion, and it had that famous actor in it. Oh, what s his name? He s been in all of those movies, like The Shawshank Redemption and The Dark Knight trilogy. I think he s even won an Oscar. Oh gosh, I can picture his face in my mind, and hear his distinctive voice, but I just can t think of his name! This is going to bug me until I can remember it! This particular error can be so frustrating because you have the information right on the tip of your tongue. Have you ever experienced this? If so, you ve committed the error known as blocking: you can t access stored information (the accompanying source-figure description).

Source illustration description (image not reproduced): A photograph shows Morgan Freeman. Blocking is also known as tip-of-the-tongue (TOT) phenomenon. The memory is right there, but you can t seem to recall it, just like not being able to remember the name of that very famous actor, Morgan Freeman. (credit: modification of work by D. Miller)

Now let s take a look at the three errors of distortion: misattribution, suggestibility, and bias. Misattribution happens when you confuse the source of your information. Let s say Alejandra was dating Lucia and they saw the first Hobbit movie together. Then they broke up and Alejandra saw the second Hobbit movie with someone else. Later that year, Alejandra and Lucia get back together. One day, they are discussing how the Hobbit books and movies are different and Alejandra says to Lucia, I loved watching the second movie with you and seeing you jump out of your seat during that super scary part. When Lucia responded with a puzzled and then angry look, Alejandra realized she d committed the error of misattribution.

What if someone is a victim of rape shortly after watching a television program? Is it possible that the victim could actually blame the rape on the person she saw on television because of misattribution? This is exactly what happened to Donald Thomson.

Australian eyewitness expert Donald Thomson appeared on a live TV discussion about the unreliability of eyewitness memory. He was later arrested, placed in a lineup and identified by a victim as the man who had raped her. The police charged Thomson although the rape had occurred at the time he was on TV. They dismissed his alibi that he was in plain view of a TV audience and in the company of the other discussants, including an assistant commissioner of police. . . . Eventually, the investigators discovered that the rapist had attacked the woman as she was watching TV the very program on which Thomson had appeared. Authorities eventually cleared Thomson. The woman had confused the rapist's face with the face that she had seen on TV. (Baddeley, 2004, p. 133)

The second distortion error is suggestibility. Suggestibility is similar to misattribution, since it also involves false memories, but it s different. With misattribution you create the false memory entirely on your own, which is what the victim did in the Donald Thomson case above. With suggestibility, it comes from someone else, such as a therapist or police interviewer asking leading questions of a witness during an interview.

Memories can also be affected by bias, which is the final distortion error. Schacter (2001) says that your feelings and view of the world can actually distort your memory of past events. There are several types of bias: Stereotypical bias involves racial and gender biases. For example, when Asian American and European American research participants were presented with a list of names, they more frequently incorrectly remembered typical African American names such as Jamal and Tyrone to be associated with the occupation basketball player, and they more frequently incorrectly remembered typical White names such as Greg and Howard to be associated with the occupation of politician (Payne, Jacoby, & Lambert, 2004). Egocentric bias involves enhancing our memories of the past (Payne et al.,

2004). Did you really score the winning goal in that big soccer match, or did you just assist? Hindsight bias happens when we think an outcome was inevitable after the fact. This is the I knew it all along phenomenon. The reconstructive nature of memory contributes to hindsight bias (Carli, 1999). We remember untrue events that seem to confirm that we knew the outcome all along.

Have you ever had a song play over and over in your head? How about a memory of a traumatic event, something you really do not want to think about? When you keep remembering something, to the point where you can t get it out of your head and it interferes with your ability to concentrate on other things, it is called persistence. It s Schacter s seventh and last memory error. It s actually a failure of our memory system because we involuntarily recall unwanted memories, particularly unpleasant ones (the accompanying source-figure description). For instance, you witness a horrific car accident on the way to work one morning, and you can t concentrate on work because you keep remembering the scene.

Source illustration description (image not reproduced): A photograph shows two soldiers physically fighting. Many veterans of military conflicts involuntarily recall unwanted, unpleasant memories. (credit: Department of Defense photo by U.S. Air Force Tech. Sgt. Michael R. Holzworth)

Interference

Sometimes information is stored in our memory, but for some reason it is inaccessible. This is known as interference, and there are two types: proactive interference and retroactive interference (the accompanying source-figure description). Have you ever gotten a new phone number or moved to a new address, but right after you tell people the old (and wrong) phone number or address? When the new year starts, do you find you accidentally write the previous year? These are examples of proactive interference: when old information hinders the recall of newly learned information. Retroactive interference happens when information learned more recently hinders the recall of older information. For example, this week you are studying about memory and learn about the Ebbinghaus forgetting curve. Next week you study lifespan development and learn about Erikson's theory of psychosocial development, but thereafter have trouble remembering Ebbinghaus's work because you can only remember Erickson's theory.

Source illustration description (image not reproduced): A diagram shows two types of interference. A box with the text learn combination to high school locker, 17 04 32 is followed by an arrow pointing right toward a box labeled memory of old locker combination interferes with recall of new gym locker combination, ?? ?? ?? ; the arrow connecting the two boxes contains the text proactive interference (old information hinders recall of new information. Beneath that is a second part of the diagram. A box with the text knowledge of new email address interferes with recall of old email address, nvayala@??? is followed by an arrow pointing left toward the early event box and away from another box labeled learn sibling s new college email address, npatel@siblingcollege.edu ; the arrow connecting the two boxes contains the text retroactive interference (new information hinders recall of old information. Sometimes forgetting is caused by a failure to retrieve information. This can be due to interference, either retroactive or proactive.

Summary

All of us at times have felt dismayed, frustrated, and even embarrassed when our memories have failed us. Our memory is flexible and prone to many errors, which is why eyewitness testimony has been found to be largely unreliable. There are several reasons why forgetting occurs. In cases of brain trauma or disease, forgetting may be due to amnesia. Another reason we forget is due to encoding failure. We can t remember something if we never stored it in our memory in the first place. Schacter presents seven memory errors that also contribute to forgetting. Sometimes, information is actually stored in our memory, but we cannot access it due to interference. Proactive interference happens when old information hinders the recall of newly learned information. Retroactive interference happens when information learned more recently hinders the recall of older information.

Review Questions

OpenStax learning exercise

Q. _____ is when our recollections of the past are done in a self-enhancing manner.

A. B

OpenStax learning exercise

Q. Tip-of-the-tongue phenomenon is also known as _____.

A. D

OpenStax learning exercise

Q. The formulation of new memories is sometimes called _____, and the process of bringing up old memories is called _____.

A. A

Critical Thinking Questions

OpenStax learning exercise

Q. Compare and contrast the two types of interference.

A. There are two types of interference: retroactive and proactive. Both are types of forgetting caused by a failure to retrieve information. With retroactive interference, new information hinders the ability to recall older information. With proactive interference, it s the opposite: old information hinders the recall of newly learned information.

OpenStax learning exercise

Q. Compare and contrast the two types of amnesia.

A. There are two types of amnesia: retrograde and anterograde. Both involve the loss of long-term memory that occurs as the result of disease, physical trauma, or psychological trauma. With anterograde amnesia, you cannot remember new information; however, you can remember information and events that happened prior to your injury. Retrograde amnesia is the exact opposite: you experience loss of memory for events that occurred before the trauma.

Personal Application Questions

OpenStax learning exercise

Q. Which of the seven memory errors presented by Schacter have you committed? Provide an example of each one.

OpenStax learning exercise

Q. Jurors place a lot of weight on eyewitness testimony. Imagine you are an attorney representing a defendant who is accused of robbing a convenience store. Several eyewitnesses have been called to testify against your client. What would you tell the jurors about the reliability of eyewitness testimony?

Ways to Enhance Memory

OpenStax source module m82216 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82216/index.cnxml

Most of us suffer from memory failures of one kind or another, and most of us would like to improve our memories so that we don t forget where we put the car keys or, more importantly, the material we need to know for an exam. In this section, we ll look at some ways to help you remember better, and at some strategies for more effective studying.

Memory-Enhancing Strategies

What are some everyday ways we can improve our memory, including recall? To help make sure information goes from short-term memory to long-term memory, you can use memory-enhancing strategies. One strategy is rehearsal, or the conscious repetition of information to be remembered (Craik & Watkins, 1973). Think about how you learned your multiplication tables as a child. You may recall that 6 x 6 = 36, 6 x 7 = 42, and 6 x 8 = 48. Memorizing these facts is rehearsal.

Another strategy is chunking: you organize information into manageable bits or chunks (Bodie, Powers, & Fitch-Hauser, 2006). Chunking is useful when trying to remember information like dates and phone numbers. Instead of trying to remember 5205550467, you remember the number as 520-555-0467. So, if you met an interesting person at a party and you wanted to remember his phone number, you would naturally chunk it, and you could repeat the number over and over, which is the rehearsal strategy.

You could also enhance memory by using elaborative rehearsal: a technique in which you think about the meaning of new information and its relation to knowledge already stored in your memory (Tigner, 1999). Elaborative rehearsal involves both linking the information to knowledge already stored and repeating the information. For example, in this case, you could remember that 520 is an area code for Arizona and the person you met is from Arizona. This would help you better remember the 520 prefix. If the information is retained, it goes into long-term memory.

Mnemonic devices are memory aids that help us organize information for encoding (the accompanying source-figure description). They are especially useful when we want to recall larger bits of information such as steps, stages, phases, and parts of a system (Bellezza, 1981). Brian needs to learn the order of the planets in the solar system, but he s having a hard time remembering the correct order. His friend Kelly suggests a mnemonic device that can help him remember. Kelly tells Brian to simply remember the name Mr. VEM J. SUN, and he can easily recall the correct order of the planets: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune. You might use a mnemonic device to help you remember someone s name, a mathematical formula, or the order of mathematical operations.

Source illustration description (image not reproduced): A photograph shows a person s two hands clenched into fists so the knuckles show. The knuckles are labeled with the months and the number of days in each month, with the knuckle protrusions corresponding to the months with 31 days, and the indentations between knuckles corresponding to February and the months with 30 days. This is a knuckle mnemonic to help you remember the number of days in each month. Months with 31 days are represented by the protruding knuckles and shorter months fall in the spots between knuckles. (credit: modification of work by Cory Zanker)

If you have ever watched the television show Modern Family, you might have seen Phil Dunphy explain how he remembers names:

The other day I met this guy named Carl. Now, I might forget that name, but he was wearing a Grateful Dead t-shirt. What s a band like the Grateful Dead? Phish. Where do fish live? The ocean. What else lives in the ocean? Coral. Hello, Co-arl. (Wrubel & Spiller, 2010)

It seems the more vivid or unusual the mnemonic, the easier it is to remember. The key to using any mnemonic successfully is to find a strategy that works for you.

Some other strategies that are used to improve memory include expressive writing and saying words aloud. Expressive writing helps boost your short-term memory, particularly if you write about a

traumatic experience in your life. Masao Yogo and Shuji Fujihara (2008) had participants write for 20-minute intervals several times per month. The participants were instructed to write about a traumatic experience, their best possible future selves, or a trivial topic. The researchers found that this simple writing task increased short-term memory capacity after five weeks, but only for the participants who wrote about traumatic experiences. Psychologists can t explain why this writing task works, but it does.

What if you want to remember items you need to pick up at the store? Simply say them out loud to yourself. A series of studies (MacLeod, Gopie, Hourihan, Neary, & Ozubko, 2010) found that saying a word out loud improves your memory for the word because it increases the word s distinctiveness. Feel silly, saying random grocery items aloud? This technique works equally well if you just mouth the words. Using these techniques increased participants memory for the words by more than 10%. These techniques can also be used to help you study.

How to Study Effectively

Based on the information presented in this chapter, here are some strategies and suggestions to help you hone your study techniques (the accompanying source-figure description). The key with any of these strategies is to figure out what works best for you.

Source illustration description (image not reproduced): A photograph shows students studying. Memory techniques can be useful when studying for class. (credit: Barry Pousman)

- Use elaborative rehearsal: In a famous article, Fergus Craik and Robert Lockhart (1972) discussed their belief that information we process more deeply goes into long-term memory. Their theory is called levels of processing. If we want to remember a piece of information, we should think about it more deeply and link it to other information and memories to make it more meaningful. For example, if we are trying to remember that the hippocampus is involved with memory processing, we might envision a hippopotamus with excellent memory and then we could better remember the hippocampus.
- Apply the self-reference effect: As you go through the process of elaborative rehearsal, it would be even more beneficial to make the material you are trying to memorize personally meaningful to you. In other words, make use of the self-reference effect. Write notes in your own words. Write definitions from the text, and then rewrite them in your own words. Relate the material to something you have already learned for another class, or think how you can apply the concepts to your own life. When you do this, you are building a web of retrieval cues that will help you access the material when you want to remember it.
- Use distributed practice: Study across time in short durations rather than trying to cram it all in at once. Memory consolidation takes time, and studying across time allows time for memories to consolidate. In addition, cramming can cause the links between concepts to become so active that you get stuck in a link, and it prevents you from accessing the rest of the information that you learned.
- Rehearse, rehearse, rehearse: Review the material over time, in spaced and organized study sessions. Organize and study your notes, and take practice quizzes/exams. Link the new information to other information you already know well.
- Study efficiently: Students are great highlighters, but highlighting is not very efficient because students spend too much time studying the things they already learned. Instead of highlighting, use index cards. Write the question on one side and the answer on the other side. When you study, separate your cards into those you got right and those you got wrong. Study the ones you got wrong and keep sorting. Eventually, all your cards will be in the pile you answered correctly.
- Be aware of interference: To reduce the likelihood of interference, study during a quiet time without interruptions or distractions (like television or music).

- Keep moving: Of course you already know that exercise is good for your body, but did you also know it s also good for your mind? Research suggests that regular aerobic exercise (anything that gets your heart rate elevated) is beneficial for memory (van Praag, 2008). Aerobic exercise promotes neurogenesis: the growth of new brain cells in the hippocampus, an area of the brain known to play a role in memory and learning.
- Get enough sleep: While you are sleeping, your brain is still at work. During sleep the brain organizes and consolidates information to be stored in long-term memory (Abel & B uml, 2013).
- Make use of mnemonic devices: As you learned earlier in this chapter, mnemonic devices often help us to remember and recall information. There are different types of mnemonic devices, such as the acronym. An acronym is a word formed by the first letter of each of the words you want to remember. For example, even if you live near one, you might have difficulty recalling the names of all five Great Lakes. What if I told you to think of the word Homes? HOMES is an acronym that represents Huron, Ontario, Michigan, Erie, and Superior: the five Great Lakes. Another type of mnemonic device is an acrostic: you make a phrase of all the first letters of the words. For example, if you are taking a math test and you are having difficulty remembering the order of operations, recalling the following sentence will help you: Please Excuse My Dear Aunt Sally, because the order of mathematical operations is Parentheses, Exponents, Multiplication, Division, Addition, Subtraction. There also are jingles, which are rhyming tunes that contain key words related to the concept, such as i before e, except after c.

Summary

There are many ways to combat the inevitable failures of our memory system. Some common strategies that can be used in everyday situations include mnemonic devices, rehearsal, self-referencing, and adequate sleep. These same strategies also can help you to study more effectively.

Review Questions

OpenStax learning exercise

Q. When you are learning how to play the piano, the statement Every good boy does fine can help you remember the notes E, G, B, D, and F for the lines of the treble clef. This is an example of a (an) _____.

A. C

OpenStax learning exercise

Q. According to a study by Yogo and Fujihara (2008), if you want to improve your short-term memory, you should spend time writing about _____.

A. B

OpenStax learning exercise

Q. The self-referencing effect refers to _____.

A. A

OpenStax learning exercise

Q. Memory aids that help organize information for encoding are _____.

A. A

Critical Thinking Questions

OpenStax learning exercise

Q. What is the self-reference effect, and how can it help you study more effectively?

A. The self-reference effect is the tendency an individual to have better memory for information that relates to oneself than information that is not personally relevant. You can use the self-reference effect to relate the material to something you have already learned for another class, or think how you can apply the concepts to your life. When you do this, you are building a web of retrieval cues that will help you access the material when you want to remember it.

OpenStax learning exercise

Q. You and your roommate spent all of last night studying for your psychology test. You think you know the material; however, you suggest that you study again the next morning an hour prior to the test. Your roommate asks you to explain why you think this is a good idea. What do you tell them?

A. You remind her about Ebbinghaus s forgetting curve: the information you learn drops off rapidly with time. Even if you think you know the material, you should study it again right before test time to increase the likelihood the information will remain in your memory. Overlearning can help prevent storage decay.

Personal Application Questions

OpenStax learning exercise

Q. Create a mnemonic device to help you remember a term or concept from this chapter.

OpenStax learning exercise

Q. What is an effective study technique that you have used? How is it similar to/different from the strategies suggested in this chapter?

CHAPTER 12 / CAC.901

Emotion, motivation, and regulation

Lecture: Emotion is information within a relationship

Emotions help people notice what matters, prepare for action, and communicate with others. A child s anger may signal perceived unfairness; fear may signal threat; sadness may signal loss or disconnection. These are possibilities, not a fixed dictionary. The same emotion can serve different functions, and the same behavior can arise from different emotions. Counselors should explore meaning rather than assume it.

Emotional development includes recognizing feelings, understanding their causes, expressing them in socially accessible ways, and regulating their intensity and duration. These capacities grow through biology, learning, language, and relationships. A child is not born with a complete set of self-regulation skills. Adults often provide co-regulation before the child can regulate independently, and support may still be needed during difficult periods later in life.

Co-regulation is not control

Co-regulation occurs when a calm, responsive relationship helps a child manage arousal. The adult may reduce demands, offer predictable language, provide space, or help the child identify a next step. It is not forcing the child to stop feeling. Nor is it requiring immediate eye contact, compliance, or verbal explanation during high distress.

A counselor can model a slower pace and validate the emotion while setting a boundary. You are very angry, and I will keep us safe communicates both acceptance and protection. The adult s own regulation matters. A dysregulated adult may escalate a situation even while using correct words. Reflective practice should therefore examine the counselor s responses as part of the interaction.

Children differ in sensory sensitivity, temperament, communication, and experiences of threat. A strategy that calms one child may overwhelm another. Quiet breathing may help some children, while others find attention to bodily sensations uncomfortable. Offer choices and observe effects. Trauma-sensitive practice avoids making a particular regulation technique a test of cooperation.

Naming feelings without imposing them

Emotion vocabulary can help children communicate, but labels should be offered tentatively. I wonder if that felt embarrassing invites correction. You were jealous closes the question. Visual tools can support communication, but allow the child to identify mixed feelings or reject the available options. Do not treat a selected face as a precise measure of internal experience.

Cultural and family practices shape which emotions are expressed openly and how. Some children learn that distress should be communicated through bodily complaints or requests for closeness rather than direct emotion words. Explore the child s language and context. A different expression style is not automatically alexithymia, denial, or poor emotional development.

Motivation and developmental fit

Motivation is influenced by interest, value, expected success, belonging, and autonomy. A child who avoids a task may not lack motivation in general. The task may feel impossible, humiliating, or disconnected from meaningful goals. Counselors should ask what the child is trying to protect or achieve. Avoid using unmotivated as an explanation that ends inquiry.

Autonomy support means offering meaningful choice and explaining reasons, not removing all structure. Children often benefit from clear expectations combined with influence over how they participate. A young child might choose between two ways to begin a task; an adolescent might help set the agenda and decide which goal matters first. Support should fit development and the real limits of the situation.

Figure 12.1 shows a cycle in which a situation is appraised, emotion and bodily arousal arise, an action follows, and the response of others shapes the next appraisal. The cycle can support learning or maintain distress. Intervention may target any part of it, but the selected point should follow the formulation.

Application: anger after a game

A fictional nine-year-old, David, overturns a board game after losing. An adult might immediately impose a long punishment. A developmental assessment asks what losing meant, whether the rules were understood, how frustration was noticed, and what happened afterward. David may need help tolerating uncertainty, practicing a pause, and repairing the disruption.

The plan can include a shorter game, a visible pause signal, and rehearsal of a recovery statement. The adult remains warm and firm about safety. Later, when David is calm, they review what worked. The measure of progress is not the absence of anger. It is safer expression, quicker recovery, and increased participation.

Caregivers should not be told that every upset requires a therapeutic conversation. Sometimes a child needs rest, food, space, or a simpler demand. Emotional development occurs in ordinary routines as well as formal sessions. The counselor s task is to help adults notice which response fits the moment.

Doctoral seminar task

Design a co-regulation plan for two children with different communication and sensory needs. Include early signs of distress, adult responses, choices offered to the child, and a review of what helped. Identify how the plan would avoid coercion and how it would address safety.

Then observe your own response to a challenging fictional scenario. What assumptions might lead you to escalate? What information would help you slow down? Emotional competence includes the professional s capacity to notice and revise their own contribution to the interaction.

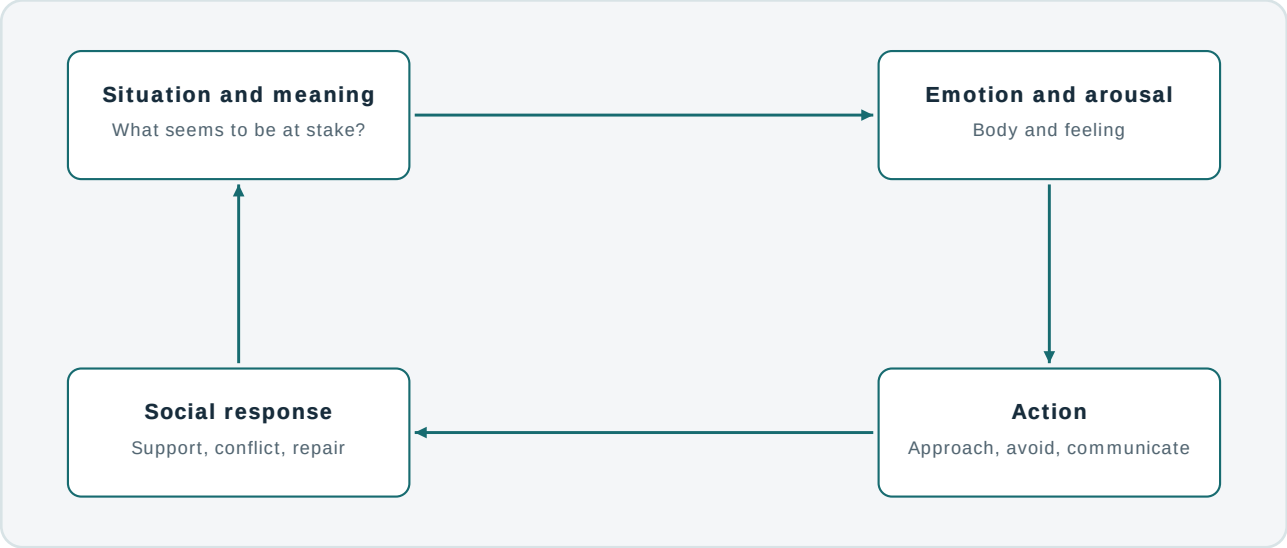


Figure 12.1. Emotion and response form a cycle. Appraisal, arousal, action, and other people s responses influence one another. Co-regulation and environmental change can alter the cycle. The diagram does not imply that every emotion follows the same sequence. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

Motivation

OpenStax source module m82225 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82225/index.cnxml

Why do we do the things we do? What motivations underlie our behaviors? Motivation describes the wants or needs that direct behavior toward a goal. In addition to biological motives, motivations can be intrinsic (arising from internal factors) or extrinsic (arising from external factors) (the accompanying source-figure description). Intrinsically motivated behaviors are performed because of the sense of personal satisfaction that they bring, while extrinsically motivated behaviors are performed in order to receive something from others.

Source illustration description (image not reproduced): An illustration shows a person s upper torso. Inside the person's head are the words intrinsic motivation (from within) and three bullet points: autonomy, mastery, purpose. Outside the person's outline are the words extrinsic motivation (from outside) and three bullet points: compensation, punishment, and reward. Intrinsic motivation comes from within the individual, while extrinsic motivation comes from outside the individual.

Think about why you are pursuing an education. Are you here because you enjoy learning and want to pursue an education to make yourself a more well-rounded individual? If so, then you are intrinsically motivated. However, if you are here because you want to get a college degree to make yourself more marketable for a high-paying career or to satisfy the demands of your parents, then your motivation is more extrinsic in nature.

In reality, our motivations are often a mix of both intrinsic and extrinsic factors, but the nature of the mix of these factors might change over time (often in ways that seem counter-intuitive). There is an old adage: Choose a job that you love, and you will never have to work a day in your life, meaning that if you enjoy your occupation, work doesn t seem like . . . well, work. Some research suggests that this isn t necessarily the case (Daniel & Esser, 1980; Deci, 1972; Deci, Koestner, & Ryan, 1999). According to this research, receiving some sort of extrinsic reinforcement (i.e., getting paid) for engaging in behaviors that we enjoy leads to those behaviors being thought of as work no longer providing that same enjoyment. As a result, we might spend less time engaging in these reclassified behaviors in the absence of any extrinsic reinforcement. For example, Odessa loves baking, so in her free time, she bakes for fun. Oftentimes, after stocking shelves at her grocery store job, she often whips up pastries in the evenings because she enjoys baking. When a coworker in the store s bakery department leaves his job, Odessa applies for his position and gets transferred to the bakery department. Although she enjoys what she does in her new job, after a few months, she no longer has much desire to concoct tasty treats in her free time. Baking has become work in a way that changes her motivation to do it (the accompanying source-figure description). What Odessa has experienced is called the overjustification effect intrinsic motivation is diminished when extrinsic motivation is given. This can lead to extinguishing the intrinsic motivation and creating a dependence on extrinsic rewards for continued performance (Deci et al., 1999).

Source illustration description (image not reproduced): A photograph shows several chefs preparing food together in a kitchen. Research suggests that when something we love to do, like icing cakes, becomes our job, our intrinsic and extrinsic motivations to do it may change. (credit: Agust n Ruiz)

Other studies suggest that intrinsic motivation may not be so vulnerable to the effects of extrinsic reinforcements, and in fact, reinforcements such as verbal praise might actually increase intrinsic motivation (Arnold, 1976; Cameron & Pierce, 1994). In that case, Odessa s motivation to bake in her free time might remain high if, for example, customers regularly compliment her baking or cake decorating skills.

These apparent discrepancies in the researchers findings may be understood by considering several factors. For one, physical reinforcement (such as money) and verbal reinforcement (such as praise) may affect an individual in very different ways. In fact, tangible rewards (i.e., money) tend to have more negative effects on intrinsic motivation than do intangible rewards (i.e., praise). Furthermore, the expectation of the extrinsic motivator by an individual is crucial: If the person expects to receive an extrinsic reward, then intrinsic motivation for the task tends to be reduced. If, however, there is no such expectation, and the extrinsic motivation is presented as a surprise, then intrinsic motivation for the task tends to persist (Deci et al., 1999).

In addition, culture may influence motivation. For example, in collectivistic cultures, it is common to do things for your family members because the emphasis is on the group and what is best for the entire group, rather than what is best for any one individual (Nisbett, Peng, Choi, & Norenzayan, 2001). This focus on others provides a broader perspective that takes into account both situational and cultural influences on behavior; thus, a more nuanced explanation of the causes of others behavior becomes more likely. (You will learn more about collectivistic and individualistic cultures when you learn about social psychology.)

In educational settings, students are more likely to experience intrinsic motivation to learn when they feel a sense of belonging and respect in the classroom. This internalization can be enhanced if the evaluative aspects of the classroom are de-emphasized and if students feel that they exercise some control over the learning environment. Furthermore, providing students with activities that are challenging, yet doable, along with a rationale for engaging in various learning activities can enhance intrinsic motivation for those tasks (Niemiec & Ryan, 2009). Consider Hakim, a first-year law student with two courses this semester: Family Law and Criminal Law. The Family Law professor has a rather intimidating classroom: He likes to put students on the spot with tough questions, which often leaves students feeling belittled or embarrassed. Grades are based exclusively on quizzes and exams, and the instructor posts results of each test on the classroom door. In contrast, the Criminal Law professor facilitates classroom discussions and respectful debates in small groups. The majority of the course grade is not exam-based, but centers on a student-designed research project on a crime issue of the student s choice. Research suggests that Hakim will be less intrinsically motivated in his Family Law course, where students are intimidated in the classroom setting, and there is an emphasis on teacher-driven evaluations. Hakim is likely to experience a higher level of intrinsic motivation in his Criminal Law course, where the class setting encourages inclusive collaboration and a respect for ideas, and where students have more influence over their learning activities.

Theories About Motivation

William James (1842 1910) was an important contributor to early research into motivation, and he is often referred to as the father of psychology in the United States. James theorized that behavior was driven by a number of instincts, which aid survival (the accompanying source-figure description). From a biological perspective, an instinct is a species-specific pattern of behavior that is not learned. There was, however, considerable controversy among James and his contemporaries over the exact definition of instinct. James proposed several dozen special human instincts, but many of his contemporaries had their own lists that differed. A mother s protection of her baby, the urge to lick sugar, and hunting prey were among the human behaviors proposed as true instincts during James s era. This view that human behavior is driven by instincts received a fair amount of criticism because of the undeniable role of learning in shaping all sorts of human behavior. In fact, as early as the 1900s, some instinctive behaviors were experimentally demonstrated to result from associative learning (recall when you learned about Watson s conditioning of fear response in Little Albert) (Faris, 1921).

Source illustration description (image not reproduced): Photograph A shows William James. Photograph B shows a person breastfeeding a baby. (a) William James proposed the instinct theory of motivation, asserting that behavior is driven by instincts. (b) In humans, instincts may include behaviors such as an infant s rooting for a nipple and sucking. (credit b: modification of work by "Mothering Touch"/Flickr)

Another early theory of motivation proposed that the maintenance of homeostasis is particularly important in directing behavior. You may recall from your earlier reading that homeostasis is the tendency to maintain a balance, or optimal level, within a biological system. In a body system, a control center (which is often part of the brain) receives input from receptors (which are often complexes of neurons). The control center directs effectors (which may be other neurons) to correct any imbalance detected by the control center.

According to the drive theory of motivation, deviations from homeostasis create physiological needs. These needs result in psychological drive states that direct behavior to meet the need and, ultimately, bring the system back to homeostasis. For example, if it s been a while since you ate, your blood sugar levels will drop below normal. This low blood sugar will induce a physiological need and a corresponding drive state (i.e., hunger) that will direct you to seek out and consume food (the accompanying source-figure description). Eating will eliminate the hunger, and, ultimately, your blood sugar levels will return to normal. Interestingly, drive theory also emphasizes the role that habits play in the type of behavioral response in which we engage. A habit is a pattern of behavior in which we regularly engage. Once we have engaged in a behavior that successfully reduces a drive, we are more likely to engage in that behavior whenever faced with that drive in the future (Graham & Weiner, 1996).

Source illustration description (image not reproduced): Photograph left shows a child eating watermelon. Photograph center shows a young person eating sushi. Photograph right shows an elderly person eating food. Hunger and subsequent eating are the result of complex physiological processes that maintain homeostasis. (credit "left": modification of work by "Gracie and Viv"/Flickr; credit "center": modification of work by Steven Depolo; credit "right": modification of work by Monica Renata)

Extensions of drive theory take into account levels of arousal as potential motivators. As you recall from your study of learning, these theories assert that there is an optimal level of arousal that we all try to maintain (the accompanying source-figure description). If we are underaroused, we become bored and will seek out some sort of stimulation. On the other hand, if we are overaroused, we will engage in behaviors to reduce our arousal (Berlyne, 1960). Most students have experienced this need to maintain optimal levels of arousal over the course of their academic career. Think about how much stress students experience toward the end of spring semester. They feel overwhelmed with seemingly endless exams, papers, and major assignments that must be completed on time. They probably yearn for the rest and relaxation that awaits them over the extended summer break. However, once they finish the semester, it doesn t take too long before they begin to feel bored. Generally, by the time the next semester is beginning in the fall, many students are quite happy to return to school. This is an example of how arousal theory works.

Source illustration description (image not reproduced): A line graph has an x-axis labeled arousal level with an arrow indicating low to high and a y-axis labeled performance quality with an arrow indicating low to high. A curve charts optimal arousal. Where arousal level and performance quality are both low, the curve is low and labeled boredom or apathy. Where arousal level is medium and performance quality is medium, the curve peaks and is labeled optimal level. Where the arousal level is high and the performance quality is low, the curve is low and is labeled high anxiety. The concept of optimal arousal in relation to performance on a task is depicted here. Performance is maximized at the optimal level of arousal, and it tapers off during under- and overarousal.

So what is the optimal level of arousal? What level leads to the best performance? Research shows that moderate arousal is generally best; when arousal is very high or very low, performance tends to suffer (Yerkes & Dodson, 1908). Think of your arousal level regarding taking an exam for this class. If your level is very low, such as boredom and apathy, your performance will likely suffer. Similarly, a very high level, such as extreme anxiety, can be paralyzing and hinder performance. Consider the example of a softball team facing a tournament. They are favored to win their first game by a large margin, so they go into the game with a lower level of arousal and get beat by a less skilled team.

But optimal arousal level is more complex than a simple answer that the middle level is always best. Researchers Robert Yerkes (pronounced Yerk-EES) and John Dodson discovered that the optimal arousal level depends on the complexity and difficulty of the task to be performed (the accompanying source-figure description). This relationship is known as Yerkes-Dodson law, which holds that a simple task is performed best when arousal levels are relatively high and complex tasks are best performed when arousal levels are lower.

Source illustration description (image not reproduced): A line graph has an x-axis labeled arousal level with an arrow indicating low to high and a y-axis labeled performance quality with an arrow indicating low to high. Two curves charts optimal arousal, one for difficult tasks and the other for easy tasks. The optimal level for easy tasks is reached with slightly higher arousal levels than for difficult tasks. Task performance is best when arousal levels are in a middle range, with difficult tasks best performed under lower levels of arousal and simple tasks best performed under higher levels of arousal.

Self-efficacy and Social Motives

Self-efficacy is an individual s belief in their own capability to complete a task, which may include a previous successful completion of the exact task or a similar task. Albert Bandura (1994) theorized that an individual s sense of self-efficacy plays a pivotal role in motivating behavior. Bandura argues that motivation derives from expectations that we have about the consequences of our behaviors, and ultimately, it is the appreciation of our capacity to engage in a given behavior that will determine what we do and the future goals that we set for ourselves. For example, if you have a sincere belief in your ability to achieve at the highest level, you are more likely to take on challenging tasks and to not let setbacks dissuade you from seeing the task through to the end.

A number of theorists have focused their research on understanding social motives (McAdams & Constantian, 1983; McClelland & Liberman, 1949; Murray et al., 1938). Among the motives they describe are needs for achievement, affiliation, and intimacy. It is the need for achievement that drives accomplishment and performance. The need for affiliation encourages positive interactions with others, and the need for intimacy causes us to seek deep, meaningful relationships. Henry Murray et al. (1938) categorized these needs into domains. For example, the need for achievement and recognition falls under the domain of ambition. Dominance and aggression were recognized as needs under the domain of human power, and play was a recognized need in the domain of interpersonal affection.

Maslow s Hierarchy of Needs

While the theories of motivation described earlier relate to basic biological drives, individual characteristics, or social contexts, Abraham Maslow (1943) proposed a hierarchy of needs that spans the spectrum of motives ranging from the biological to the individual to the social. These needs are often depicted as a pyramid (the accompanying source-figure description).

Source illustration description (image not reproduced): A triangle is divided vertically into five sections with corresponding labels inside and outside of the triangle for each section. From top to bottom, the triangle's sections are

labeled: self-actualization corresponds to Inner fulfillment esteem corresponds to Self-worth, accomplishment, confidence ; social corresponds to Family, friendship, intimacy, belonging security corresponds to Safety, employment, assets ; physiological corresponds to Food, water, shelter, warmth. Maslow s hierarchy of needs is illustrated here. In some versions of the pyramid, cognitive and aesthetic needs are also included between esteem and self-actualization. Others include another tier at the top of the pyramid for self-transcendence.

At the base of the pyramid are all of the physiological needs that are necessary for survival. These are followed by basic needs for security and safety, the need to be loved and to have a sense of belonging, and the need to have self-worth and confidence. The top tier of the pyramid is self-actualization, which is a need that essentially equates to achieving one s full potential, and it can only be realized when needs lower on the pyramid have been met. To Maslow and humanistic theorists, self-actualization reflects the humanistic emphasis on positive aspects of human nature. Maslow suggested that this is an ongoing, life-long process and that only a small percentage of people actually achieve a self-actualized state (Francis & Kritsonis, 2006; Maslow, 1943).

According to Maslow (1943), one must satisfy lower-level needs before addressing those needs that occur higher in the pyramid. So, for example, if someone is struggling to find enough food to meet their nutritional requirements, it is quite unlikely that they would spend an inordinate amount of time thinking about whether others viewed them as a good person or not. Instead, all of their energies would be geared toward finding something to eat. However, it should be pointed out that Maslow s theory has been criticized for its subjective nature and its inability to account for phenomena that occur in the real world (Leonard, 1982). Other research has more recently addressed that late in life, Maslow proposed a self-transcendence level above self-actualization to represent striving for meaning and purpose beyond the concerns of oneself (Koltko-Rivera, 2006). For example, people sometimes make self-sacrifices in order to make a political statement or in an attempt to improve the conditions of others. Mohandas K. Gandhi, a world-renowned advocate for independence through nonviolent protest, on several occasions went on hunger strikes to protest a particular situation. People may starve themselves or otherwise put themselves in danger displaying higher-level motives beyond their own needs.

Summary

Motivation to engage in a given behavior can come from internal and/or external factors. Multiple theories have been put forward regarding motivation. More biologically oriented theories deal with the ways that instincts and the need to maintain bodily homeostasis motivate behavior. Bandura postulated that our sense of self-efficacy motivates behaviors, and there are a number of theories that focus on a variety of social motives. Abraham Maslow s hierarchy of needs is a model that shows the relationship among multiple motives that range from lower-level physiological needs to the very high level of self-actualization.

Review Questions

OpenStax learning exercise

Q. Need for _____ refers to maintaining positive relationships with others.

A. B

OpenStax learning exercise

Q. _____ proposed the hierarchy of needs.

A. C

OpenStax learning exercise

Q. _____ is an individual s belief in her capability to complete some task.

A. D

OpenStax learning exercise

Q. Carl mows the yard of his elderly neighbor each week for $20. What type of motivation is this?

A. A

Critical Thinking Questions

OpenStax learning exercise

Q. How might someone espousing an arousal theory of motivation explain visiting an amusement park?

A. The idea of optimal levels of arousal is similar to a drive theory of motivation. Presumably, we all seek to maintain some intermediate level of arousal. If we are underaroused, we are bored. If we are overaroused, we experience stress. The rides at an amusement park would provide higher arousal (however, we would hope that these rides don t actually pose significant threats to personal safety that would lead to a state of panic) to push us toward our own optimal level of arousal. Individuals at the park would choose different rides based on their specific arousal thresholds; for example, one person might find a simple water ride optimally arousing and an extreme roller coaster overarousing, while others would find the extreme roller coaster optimally arousing.

OpenStax learning exercise

Q. Schools often use concrete rewards to increase adaptive behaviors. How might this be a disadvantage for students intrinsically motivated to learn? What are educational implications of the potential for concrete rewards to diminish intrinsic motivation for a given task?

A. We would expect to see a shift from learning for the sake of learning to learning to earn some reward. This would undermine the foundation upon which traditional institutions of higher education are built. For a student motivated by extrinsic rewards, dependence on those may pose issues later in life (post-school) when there are not typically extrinsic rewards for learning.

Personal Application Question

OpenStax learning exercise

Q. Can you think of recent examples of how Maslow s hierarchy of needs might have affected your behavior in some way?

Emotion

OpenStax source module m82229 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82229/index.cnxml

As we move through our daily lives, we experience a variety of emotions. An emotion is a subjective state of being that we often describe as our feelings. Emotions result from the combination of subjective experience, expression, cognitive appraisal, and physiological responses (Levenson, Carstensen, Friesen, & Ekman, 1991). However, as discussed later in the chapter, the exact order in which the components occur is not clear, and some parts may happen at the same time. An emotion often begins

with a subjective (individual) experience, which is a stimulus. Often the stimulus is external, but it does not have to be from the outside world. For example, it might be that one thinks about war and becomes sad, even though they never experienced war. Emotional expression refers to the way one displays an emotion and includes nonverbal and verbal behaviors (Gross, 1999). One also performs a cognitive appraisal in which a person tries to determine the way they will be impacted by a situation (Roseman & Smith, 2001). In addition, emotions include physiological responses, such as possible changes in heart rate, sweating, etc. (Soussignan, 2002).

The words emotion and mood are sometimes used interchangeably, but psychologists use these words to refer to two different things. Typically, the word emotion indicates a subjective, affective state that is relatively intense and that occurs in response to something we experience (the accompanying source-figure description). Emotions are often thought to be consciously experienced and intentional. Mood, on the other hand, refers to a prolonged, less intense, affective state that does not occur in response to something we experience. Mood states may not be consciously recognized and do not carry the intentionality that is associated with emotion (Beedie, Terry, Lane, & Devonport, 2011). Here we will focus on emotion, and you will learn more about mood in the chapter that covers psychological disorders.

Source illustration description (image not reproduced): Photograph A shows a toddler laughing. Photograph B shows the same toddler crying. Toddlers can cycle through emotions quickly, being (a) extremely happy one moment and (b) extremely sad the next. (credit a: modification of work by Kerry Ceszyk; credit b: modification of work by Kerry Ceszyk)

We can be at the heights of joy or in the depths of despair. We might feel angry when we are betrayed, fear when we are threatened, and surprised when something unexpected happens. This section will outline some of the most well-known theories explaining our emotional experience and provide insight into the biological bases of emotion. This section closes with a discussion of the ubiquitous nature of facial expressions of emotion and our abilities to recognize those expressions in others.

Theories of Emotion

Our emotional states are combinations of physiological arousal, psychological appraisal, and subjective experiences. Together, these are the components of emotion, and our experiences, backgrounds, and cultures inform our emotions. Therefore, different people may have different emotional experiences even when faced with similar circumstances. Over time, several different theories of emotion, shown in the accompanying source-figure description, have been proposed to explain how the various components of emotion interact with one another.

Source illustration description (image not reproduced): A diagram shows a photograph of a snake on the left and a photograph of a frightened person on the right, with an arrow labeled time. Beneath the photos are flow diagrams of four theories of emotion. In the James-Lange theory, a box labeled arousal (snake) leads to a box labeled heart pounding, sweating, which leads to a box labeled fear (emotion). In the Cannon-Bard theory, a box labeled arousal (snake) splits into two boxes labeled heart pounding, sweating, and fear (emotion). In the Schachter-Singer Two-Factor theory, a box labeled arousal (snake) leads to two boxes labeled heart pounding, sweating and cognitive label (I m scared) which then lead to a single box labeled fear (emotion). In the Lazarus Cognitive-mediational theory, a box labeled arousal (snake) leads to a box labeled appraisal, which leads to a box labeled fear/heart pounding, sweating. This figure illustrates the major assertions of the James-Lange, Cannon-Bard, and Schachter-Singer two-factor theories of emotion. (credit "snake": modification of work by "tableatny"/Flickr; credit "face": modification of work by Cory Zanker)

The James-Lange theory of emotion asserts that emotions arise from physiological arousal. Recall what you have learned about the sympathetic nervous system and our fight or flight response when threatened. If you were to encounter some threat in your environment, like a venomous snake in your backyard, your sympathetic nervous system would initiate significant physiological arousal, which would make your heart race and increase your respiration rate. According to the James-Lange theory of emotion, you would only experience a feeling of fear after this physiological arousal had taken place. Furthermore, different arousal patterns would be associated with different feelings.

Other theorists, however, doubted that the physiological arousal that occurs with different types of emotions is distinct enough to result in the wide variety of emotions that we experience. Thus, the Cannon-Bard theory of emotion was developed. According to this view, physiological arousal and emotional experience occur simultaneously, yet independently (Lang, 1994). So, when you see the venomous snake, you feel fear at exactly the same time that your body mounts its fight or flight response. This emotional reaction would be separate and independent of the physiological arousal, even though they co-occur.

Does smiling make you happy? Alternatively, does being happy make you smile? The facial feedback hypothesis proposes that your facial expression can actually affect your emotional experience (Adelman & Zajonc, 1989; Boiger & Mesquita, 2012; Buck, 1980; Capella, 1993; Soussignan, 2001). Research investigating the facial feedback hypothesis suggested that suppression of facial expression of emotion lowered the intensity of some emotions experienced by participants (Davis, Senghas, & Ochsner, 2009). Havas, Glenberg, Gutowski, Lucarelli, and Davidson (2010) used Botox injections to paralyze facial muscles and limit facial expressions, including frowning, and they found that depressed people reported less depression after their frowning muscles were paralyzed. Other research found that the intensities of facial expressions affected the emotional reactions (Soussignan, 2002). In other words, if something insignificant occurs and you smile as if you just won lottery, you will actually be happier about the little thing than you would be if you only had a tiny smile. Conversely, if you walk around frowning all the time, it might cause you to have less positive emotions than you would if you had smiled. Interestingly, Soussignan (2002) also reported physiological arousal differences associated with the intensities of one type of smile.

G. Mara on Posadillo was a Spanish physician who studied the psychological effects of adrenaline to create a model for the experience of emotion. Mara on's model preceded Schachter's two-factor or arousal-cognition theory of emotion (Cornelius, 1991). The Schachter-Singer two-factor theory of emotion is another variation on theories of emotions that takes into account both physiological arousal and the emotional experience. According to this theory, emotions are composed of two factors: physiological and cognitive. In other words, physiological arousal is interpreted in context to produce the emotional experience. In revisiting our example involving the venomous snake in your backyard, the two-factor theory maintains that the snake elicits sympathetic nervous system activation that is labeled as fear given the context, and our experience is that of fear. If you had labeled your sympathetic nervous system activation as joy, you would have experienced joy. The Schachter-Singer two-factor theory depends on labeling the physiological experience, which is a type of cognitive appraisal.

Magda Arnold was the first theorist to offer an exploration of the meaning of appraisal, and to present an outline of what the appraisal process might be and how it relates to emotion (Roseman & Smith, 2001). The key idea of appraisal theory is that you have thoughts (a cognitive appraisal) before you experience an emotion, and the emotion you experience depends on the thoughts you had (Frijda, 1988; Lazarus, 1991). If you think something is positive, you will have more positive emotions about it than if your appraisal was negative, and the opposite is true. Appraisal theory explains the way two people can have two completely different emotions regarding the same event. For example, suppose your psychology instructor selected you to lecture on emotion; you might see that as positive, because it represents an opportunity to be the center of attention, and you would experience happiness. However, if you dislike speaking in public, you could have a negative appraisal and experience discomfort.

Schachter and Singer believed that physiological arousal is very similar across the different types of emotions that we experience, and therefore, the cognitive appraisal of the situation is critical to the actual emotion experienced. In fact, it might be possible to misattribute arousal to an emotional experience if the circumstances were right (Schachter & Singer, 1962). They performed a clever experiment to test their idea. A group of men participating in the experiment were randomly assigned to one of several groups. Some of the participants received injections of epinephrine that caused bodily changes that mimicked the fight-or-flight response of the sympathetic nervous system; however, only some of these men were told to expect these reactions as side effects of the injection. The other men that received injections of epinephrine were told either that the injection would have no side effects or

that it would result in a side effect unrelated to a sympathetic response, such as itching feet or headache. After receiving these injections, participants waited in a room with someone else they thought was another subject in the research project. In reality, the other person was a confederate (someone working on behalf of) of the researcher. The confederate engaged in scripted displays of euphoric or angry behavior (Schachter & Singer, 1962).

When those participants who were told that they should expect to feel symptoms of physiological arousal were asked about any emotional changes that they had experienced related to either euphoria or anger (depending on the way the confederate behaved), they reported none. However, the men who weren t expecting physiological arousal as a function of the injection were more likely to report that they experienced euphoria or anger as a function of their assigned confederate s behavior. While everyone who received an injection of epinephrine experienced the same physiological arousal, only those who were not expecting the arousal used context to interpret the arousal as a change in emotional state (Schachter & Singer, 1962).

Strong emotional responses are associated with strong physiological arousal, which caused some theorists to suggest that the signs of physiological arousal, including increased heart rate, respiration rate, and sweating, might be used to determine whether someone is telling the truth or not. The assumption is that most of us would show signs of physiological arousal if we were being dishonest with someone. A polygraph, or lie detector test, measures the physiological arousal of an individual responding to a series of questions. Someone trained in reading these tests would look for answers to questions that are associated with increased levels of arousal as potential signs that the respondent may have been dishonest on those answers. While polygraphs are still commonly used, their validity and accuracy are highly questionable because there is no evidence that lying is associated with any particular pattern of physiological arousal (Saxe & Ben-Shakhar, 1999).

The relationship between our experiencing of emotions and our cognitive processing of them, and the order in which these occur, remains a topic of research and debate. Lazarus (1991) developed the cognitive-mediational theory that asserts our emotions are determined by our appraisal of the stimulus. This appraisal mediates between the stimulus and the emotional response, and it is immediate and often unconscious. In contrast to the Schachter-Singer model, the appraisal precedes a cognitive label. You will learn more about Lazarus s appraisal concept when you study stress, health, and lifestyle. However, there are other views of emotions that also emphasize the cognitive processes.

Return to the example of being asked to lecture by your professor. Even if you do not enjoy speaking in public, you probably could manage to do it. You would purposefully control your emotions, which would allow you to speak, but we constantly regulate our emotions, and much of our emotion regulation occurs without us actively thinking about it. Mauss and her colleagues studied automatic emotion regulation (AER), which refers to the non-deliberate control of emotions. It is simply not reacting with your emotions, and AER can affect all aspects of emotional processes. AER can influence the things you attend to, your appraisal, your choice to engage in an emotional experience, and your behaviors after an emotion is experienced (Mauss, Bunge, & Gross, 2007; Mauss, Levenson, McCarter, Wilhelm, & Gross, 2005). AER is similar to other automatic cognitive processes in which sensations activate knowledge structures that affect functioning. These knowledge structures can include concepts, schemas, or scripts.

The idea of AER is that people develop an automatic process that works like a script or schema, and the process does not require deliberate thought to regulate emotions. AER works like riding a bicycle. Once you develop the process, you just do it without thinking about it. AER can be adaptive or maladaptive and has important health implications (Hopp, Troy, & Mauss, 2011). Adaptive AER leads to better health outcomes than maladaptive AER, primarily due to experiencing or mitigating stressors better than people with maladaptive AERs (Hopp, Troy, & Mauss, 2011). Alternatively, maladaptive AERs may be critical for maintaining some psychological disorders (Hopp, Troy, & Mauss, 2011). Mauss and her colleagues found that strategies could reduce negative emotions, which in turn should increase psychological health (Mauss, Cook, Cheng, & Gross, 2007; Mauss, Cook, & Gross, 2007; Shallcross,

Troy, Boland, & Mauss, 2010; Troy, Shallcross, & Mauss, 2013; Troy, Wilhelm, Shallcross, & Mauss, 2010). Mauss has also suggested there are problems with the way emotions are measured, but she believes most of the aspects of emotions that are typically measured are useful (Mauss, et al., 2005; Mauss & Robinson, 2009). However, another way of considering emotions challenges our entire understanding of emotions.

After about three decades of interdisciplinary research, Barrett argued that we do not understand emotions. She proposed that emotions were not built into your brain at birth, but rather they were constructed based on your experiences. Emotions in the constructivist theory are predictions that construct your experience of the world. In chapter 7 you learned that concepts are categories or groupings of linguistic information, images, ideas, or memories, such as life experiences. Barrett extended that to include emotions as concepts that are predictions (Barrett, 2017). Two identical physiological states can result in different emotional states depending on your predictions. For example, your brain predicting a churning stomach in a bakery could lead to you constructing hunger. However, your brain predicting a churning stomach while you were waiting for medical test results could lead your brain to construct worry. Thus, you can construct two different emotions from the same physiological sensations. Rather than emotions being something over which you have no control, you can control and influence your emotions.

Two other prominent views arise from the work of Robert Zajonc and Joseph LeDoux. Zajonc asserted that some emotions occur separately from or prior to our cognitive interpretation of them, such as feeling fear in response to an unexpected loud sound (Zajonc, 1998). He also believed in what we might casually refer to as a gut feeling that we can experience an instantaneous and unexplainable like or dislike for someone or something (Zajonc, 1980). LeDoux also views some emotions as requiring no cognition: some emotions completely bypass contextual interpretation. His research into the neuroscience of emotion has demonstrated the amygdala s primary role in fear (Cunha, Monfils, & LeDoux, 2010; LeDoux 1996, 2002). A fear stimulus is processed by the brain through one of two paths: from the thalamus (where it is perceived) directly to the amygdala or from the thalamus through the cortex and then to the amygdala. The first path is quick, while the second enables more processing about details of the stimulus. In the following section, we will look more closely at the neuroscience of emotional response.

The Biology of Emotions

Earlier, you learned about the limbic system, which is the area of the brain involved in emotion and memory (the accompanying source-figure description). The limbic system includes the hypothalamus, thalamus, amygdala, and the hippocampus. The hypothalamus plays a role in the activation of the sympathetic nervous system that is a part of any given emotional reaction. The thalamus serves as a sensory relay center whose neurons project to both the amygdala and the higher cortical regions for further processing. The amygdala plays a role in processing emotional information and sending that information on (Fossati, 2012).The hippocampus integrates emotional experience with cognition (Femen a, G mez-Gal n, Lindskog, & Magara, 2012).

Source illustration description (image not reproduced): An illustration of the brain labels the locations of the thalamus, hypothalamus, amygdala, and hippocampus. The limbic system, which includes the hypothalamus, thalamus, amygdala, and the hippocampus, is involved in mediating emotional response and memory.

Amygdala

The amygdala has received a great deal of attention from researchers interested in understanding the biological basis for emotions, especially fear and anxiety (Blackford & Pine, 2012; Goosens & Maren, 2002; Maren, Phan, & Liberzon, 2013). The amygdala is composed of various subnuclei, including the basolateral complex and the central nucleus (the accompanying source-figure description). The basolateral complex has dense connections with a variety of sensory areas of the brain. It is critical for classical conditioning and for attaching emotional value to learning processes and memory. The central nucleus plays a role in attention, and it has connections with the hypothalamus and various brainstem areas to regulate the autonomic nervous and endocrine systems activity (Pessoa, 2010).

Source illustration description (image not reproduced): An illustration of the brain labels the locations of the basolateral complex and central nucleus within the amygdala. The anatomy of the basolateral complex and central nucleus of the amygdala are illustrated in this diagram.

Animal research has demonstrated that there is increased activation of the amygdala in rat pups that have odor cues paired with electrical shock when their mother is absent. This leads to an aversion to the odor cue that suggests the rats learned to fear the odor cue. Interestingly, when the mother was present, the rats actually showed a preference for the odor cue despite its association with an electrical shock. This preference was associated with no increases in amygdala activation. This suggests a differential effect on the amygdala by the context (the presence or absence of the mother) determined whether the pups learned to fear the odor or to be attracted to it (Moriceau & Sullivan, 2006).

Raineki, Cort s, Belnoue, and Sullivan (2012) demonstrated that, in rats, negative early life experiences could alter the function of the amygdala and result in adolescent patterns of behavior that mimic human mood disorders. In this study, rat pups received either abusive or normal treatment during postnatal days 8 12. There were two forms of abusive treatment. The first form of abusive treatment had an insufficient bedding condition. The mother rat had insufficient bedding material in her cage to build a proper nest that resulted in her spending more time away from her pups trying to construct a nest and less times nursing her pups. The second form of abusive treatment had an associative learning task that involved pairing odors and an electrical stimulus in the absence of the mother, as described above. The control group was in a cage with sufficient bedding and was left undisturbed with their mothers during the same time period. The rat pups that experienced abuse were much more likely to exhibit depressive-like symptoms during adolescence when compared to controls. These depressive-like behaviors were associated with increased activation of the amygdala.

Human research also suggests a relationship between the amygdala and psychological disorders of mood or anxiety. Changes in amygdala structure and function have been demonstrated in adolescents who are either at-risk or have been diagnosed with various mood and/or anxiety disorders (Miguel-Hidalgo, 2013; Qin et al., 2013). It has also been suggested that functional differences in the amygdala could serve as a biomarker to differentiate individuals suffering from bipolar disorder from those suffering from major depressive disorder (Fournier, Keener, Almeida, Kronhaus, & Phillips, 2013).

Hippocampus

As mentioned earlier, the hippocampus is also involved in emotional processing. Like the amygdala, research has demonstrated that hippocampal structure and function are linked to a variety of mood and anxiety disorders. Individuals suffering from posttraumatic stress disorder (PTSD) show marked reductions in the volume of several parts of the hippocampus, which may result from decreased levels of neurogenesis and dendritic branching (the generation of new neurons and the generation of new dendrites in existing neurons, respectively) (Wang et al., 2010). While it is impossible to make a causal claim with correlational research like this, studies have demonstrated behavioral improvements and hippocampal volume increases following either pharmacological or cognitive-behavioral therapy in individuals suffering from PTSD (Bremner & Vermetten, 2004; Levy-Gigi, Szab , Kelemen, & K ri, 2013).

Facial Expression and Recognition of Emotions

Culture can impact the way in which people display emotion. A cultural display rule is one of a collection of culturally specific standards that govern the types and frequencies of displays of emotions that are acceptable (Malatesta & Haviland, 1982). Therefore, people from varying cultural backgrounds can have very different cultural display rules of emotion. For example, research has shown that individuals from the United States express negative emotions like fear, anger, and disgust both alone and in the presence of others, while Japanese individuals only do so while alone (Matsumoto, 1990). Furthermore, individuals from cultures that tend to emphasize social cohesion are more likely to engage in suppression of emotional reaction so they can evaluate which response is most appropriate in a given context (Matsumoto, Yoo, & Nakagawa, 2008).

Other distinct cultural characteristics might be involved in emotionality. For instance, there may be gender differences involved in emotional processing. While research into gender differences in emotional display is equivocal, there is some evidence that people of different genders may differ in regulation of emotions (McRae, Ochsner, Mauss, Gabrieli, & Gross, 2008).

Paul Ekman (1972) researched a New Guinea man who was living in a preliterate culture using stone implements, and which was isolated and had never seen any outsiders before. Ekman asked the man to show what his facial expression would be if: (1) friends visited, (2) his child had just died, (3) he was about to fight, (4) he stepped on a smelly dead pig. After Ekman s return from New Guinea, he researched facial expressions for more than four decades. Despite different emotional display rules, our ability to recognize and produce facial expressions of emotion appears to be universal. In fact, even congenitally blind individuals produce the same facial expression of emotions, despite their never having the opportunity to observe these facial displays of emotion in other people. This would seem to suggest that the pattern of activity in facial muscles involved in generating emotional expressions is universal, and indeed, this idea was suggested in the late 19th century in Charles Darwin s book The Expression of Emotions in Man and Animals (1872). In fact, there is substantial evidence for seven universal emotions that are each associated with distinct facial expressions. These include: happiness, surprise, sadness, fright, disgust, contempt, and anger (the accompanying source-figure description) (Ekman & Keltner, 1997).

Source illustration description (image not reproduced): Each of seven photographs includes a person demonstrating a different facial expression: happiness, surprise, sadness, fright, disgust, contempt, and anger. The seven universal facial expressions of emotion are shown. (credit: modification of work by Cory Zanker)

Of course, emotion is not only displayed through facial expression. We also use the tone of our voices, various behaviors, and body language to communicate information about our emotional states. Body language is the expression of emotion in terms of body position or movement. Research suggests that we are quite sensitive to the emotional information communicated through body language, even if we re not consciously aware of it (de Gelder, 2006; Tamietto et al., 2009).

Connect the Concepts

Autism spectrum disorder (ASD) is a set of neurodevelopmental disorders characterized by repetitive behaviors and communication and social problems. Children who have autism spectrum disorders have difficulty recognizing the emotional states of others, and research has shown that this may stem from an inability to distinguish various nonverbal expressions of emotion (i.e., facial expressions) from one another (Hobson, 1986). In addition, there is evidence to suggest that autistic individuals also have difficulty expressing emotion through tone of voice and by producing facial expressions (Macdonald et al., 1989). Difficulties with emotional recognition and expression may contribute to the behaviors that characterize autism; therefore, various therapeutic approaches have been explored to address these difficulties. Various educational curricula, cognitive-behavioral therapies, and pharmacological therapies have shown some promise in helping autistic individuals process emotionally relevant information (Bauminger, 2002; Golan & Baron-Cohen, 2006; Guastella et al., 2010).

Emotion regulation describes how people respond to situations and experiences by modifying their emotional experiences and expressions. Covert emotion regulation strategies are those that occur within the individual, while overt strategies involve others or actions (such as seeking advice or consuming alcohol). Aldao and Dixon (2014)

studied the relationship between overt emotional regulation strategies and psychopathology. They researched how 218 undergraduate students reported their use of covert and overt strategies and their reported symptoms associated with selected mental disorders, and found that overt emotional regulation strategies were better predictors of psychopathology than covert strategies. Another study examined the relationship between pregaming (the act of drinking heavily before a social event) and two emotion regulation strategies to understand how these might contribute to alcohol-related problems; results suggested a relationship but a complicated one (Pederson, 2016). Further research is needed in these areas to better understand patterns of adaptive and maladaptive emotion regulation (Aldao & Dixon-Gordon, 2014).

Summary

Emotions are subjective experiences that consist of physiological arousal and cognitive appraisal. Various theories have been put forward to explain our emotional experiences. The James-Lange theory asserts that emotions arise as a function of physiological arousal. The Cannon-Bard theory maintains that emotional experience occurs simultaneous to and independent of physiological arousal. The Schachter-Singer two-factor theory suggests that physiological arousal receives cognitive labels as a function of the relevant context and that these two factors together result in an emotional experience.

The limbic system is the brain s emotional circuit, which includes the amygdala and the hippocampus. Both of these structures are implicated in playing a role in normal emotional processing as well as in psychological mood and anxiety disorders. Increased amygdala activity is associated with learning to fear, and it is seen in individuals who are at risk for or suffering from mood disorders. The volume of the hippocampus has been shown to be reduced in individuals suffering from posttraumatic stress disorder.

The ability to produce and recognize facial expressions of emotions seems to be universal regardless of cultural background. However, there are cultural display rules which influence how often and under what circumstances various emotions can be expressed. Tone of voice and body language also serve as a means by which we communicate information about our emotional states.

Review Questions

OpenStax learning exercise

Q. Individuals suffering from posttraumatic stress disorder have been shown to have reduced volumes of the _____.

A. B

OpenStax learning exercise

Q. According to the _____ theory of emotion, emotional experiences arise from physiological arousal.

A. A

OpenStax learning exercise

Q. Which of the following is not one of the seven universal emotions described in this chapter?

A. C

OpenStax learning exercise

Q. Which of the following theories of emotion would suggest that polygraphs should be quite accurate at differentiating one emotion from another?

A. B

Critical Thinking Questions

OpenStax learning exercise

Q. Imagine you find a venomous snake crawling up your leg just after taking a drug that prevented sympathetic nervous system activation. What would the James-Lange theory predict about your experience?

A. The James-Lange theory would predict that I would not feel fear because I haven t had the physiological arousal necessary to induce that emotional state.

OpenStax learning exercise

Q. Why can we not make causal claims regarding the relationship between the volume of the hippocampus and PTSD?

A. The research that exists is correlational in nature. It could be the case that reduced hippocampal volume predisposes people to develop PTSD or the decreased volume could result from PTSD. Causal claims can only be made when performing an experiment.

Personal Application Question

OpenStax learning exercise

Q. Think about times in your life when you have been absolutely elated (e.g., perhaps your school s basketball team just won a closely contested ballgame for the national championship) and very fearful (e.g., you are about to give a speech in your public speaking class to a roomful of 100 strangers). How would you describe how your arousal manifested itself physically? Were there marked differences in physiological arousal associated with each emotional state?

CHAPTER 13 / CAC.901

Self, personality, and culture

Lecture: The self develops with other people

Children build ideas about who they are through experience and reflection. They notice what they can do, how others respond, which groups they belong to, and what they value. The self is not a hidden object that simply becomes visible with age. It is shaped by relationships, language, culture, opportunity, and historical context.

A child s self-description may initially focus on observable actions and possessions. With development, descriptions can become more psychological and differentiated. Yet this is not a simple ladder from primitive to mature. Adults also describe themselves through roles, relationships, and concrete activities. The important question is how a person organizes self-knowledge and how that organization affects participation and well-being.

Self-esteem is not the whole self

Self-esteem concerns evaluation of oneself, but it does not capture every important aspect of self-development. A child may feel worthy yet recognize a learning difficulty. Another may perform well but feel valuable only when successful. Counselors should listen for conditions of worth: I matter if I am useful, I must never need help, or Mistakes mean I am bad.

Praise should be honest and specific. Empty global praise can feel unreliable, especially to a child with repeated experiences of failure. Describe effort, strategy, improvement, and contribution without claiming that effort guarantees success. A child also needs access to teaching and resources. Self-belief cannot compensate indefinitely for inaccessible instruction or discrimination.

Shame and guilt are not identical. Guilt can concern a specific action and support repair. Shame often concerns the whole self and may lead to hiding or withdrawal. These distinctions are not absolute, but they help counselors respond carefully. A child who has harmed another needs accountability without being reduced to the harmful act.

Identity, culture, and belonging

Identity includes personal commitments and group memberships. Family, language, religion, ethnicity, gender, disability, and community can all contribute. Young people may experience belonging in several ways and may face pressure to simplify complex identities. Counselors should avoid treating one identity as the master explanation for every concern.

Experiences of recognition and exclusion matter. A child whose language is mocked at school may become reluctant to speak, not because of an inherent communication deficit but because participation has become costly. A young person exploring identity may need a safe space to ask questions without immediate labeling. Respect does not require the counselor to claim expertise in every community; it requires humility, careful listening, and attention to power.

Figure 13.1 maps sources of self-understanding and the feedback between personal action and social response. It can be used to identify protective relationships and contexts in which the child s identity is constrained. The goal is not to create a perfectly positive self-story. It is to support a coherent, flexible, and dignified account that can include difficulty without totalizing it.

Personality descriptions and developmental caution

Personality frameworks can organize differences in typical patterns of feeling and behaving. They should not be used to confine a child to a fixed type. Behavior varies across contexts and changes with development. A child described as shy may be outgoing with trusted people. A child described as oppositional may be responding to repeated experiences of unfairness.

Trait language can be useful when it describes patterns with appropriate evidence, but it becomes harmful when it explains the pattern by repeating the label. He avoids because he is avoidant adds little. Ask when avoidance occurs, what it achieves, what the child fears, and where confident participation is already present.

Application: the perfect student

A fictional thirteen-year-old, Ife, achieves highly but becomes distressed by minor mistakes. Adults praise her as the responsible one. A developmental formulation considers temperament, family expectations, school reward systems, and Ife s belief that mistakes threaten belonging. The intervention should not simply tell her to care less.

Work can include identifying all-or-nothing standards, practicing manageable imperfection, and helping adults respond to effort and repair rather than flawless performance. Explore whether Ife has responsibilities that leave little room for rest. The aim is flexible commitment, not reduced ambition. Her own goals should guide the work.

The counselor should monitor unintended messages. If sessions focus only on performance management, they may reinforce the idea that Ife s value lies in achievement. Include interests, relationships, pleasure, and the right to need help. Self-development is broader than becoming more efficient.

Doctoral seminar task

Write a developmental self-formulation for a fictional child. Include self-descriptions, important relationships, group memberships, conditions of worth, contexts of confidence, and experiences of exclusion. Identify which statements come from the child and which come from adults.

Then design an intervention that changes both the child s reflective resources and at least one social response. Explain how you would know whether the child experiences greater flexibility rather than merely complying with a new adult expectation. The child s account is an outcome, not an optional addition.

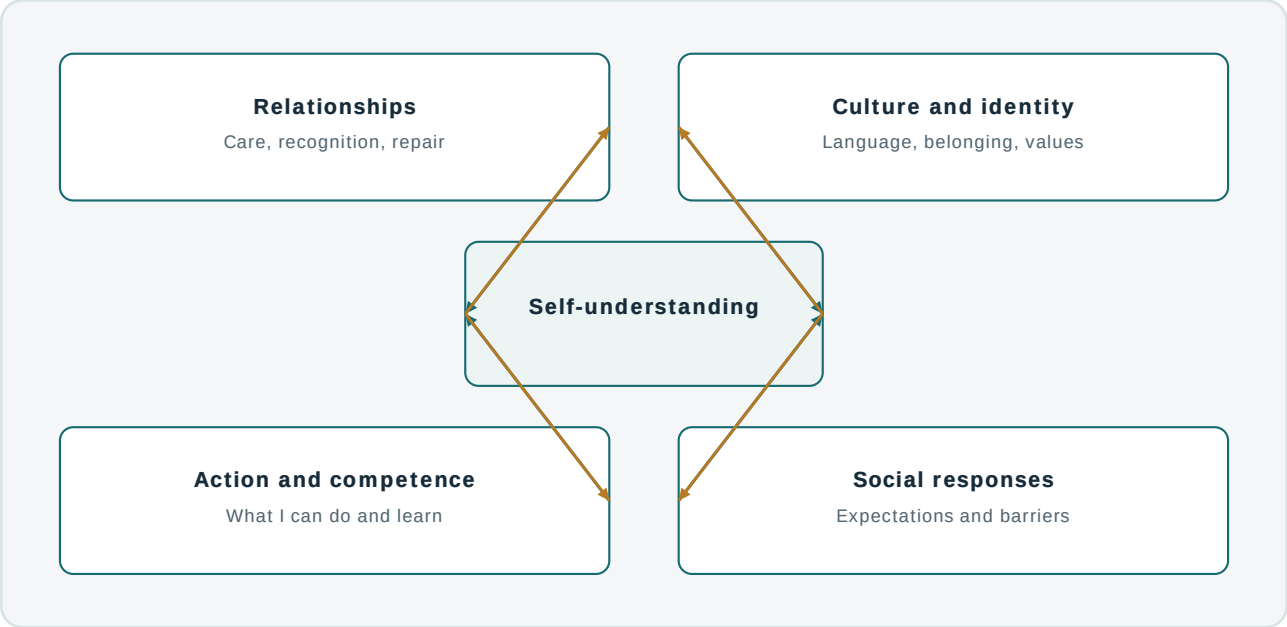


Figure 13.1. Self-understanding develops in context. The self is shaped through action, relationships, group membership, and social recognition. A formulation should include contexts of confidence as well as experiences that impose narrow conditions of worth. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

What Is Personality?

OpenStax source module m82233 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82233/index.cnxml

Personality refers to the long-standing traits and patterns that propel individuals to consistently think, feel, and behave in specific ways. Our personality is what makes us unique individuals. Each person has an idiosyncratic pattern of enduring, long-term characteristics and a manner in which they interact with other individuals and the world around them. Our personalities are thought to be long term, stable, and not easily changed. The word personality comes from the Latin word persona. In the ancient world, a persona was a mask worn by an actor. While we tend to think of a mask as being worn to conceal one s identity, the theatrical mask was originally used to either represent or project a specific personality trait of a character (the accompanying source-figure description).

Source illustration description (image not reproduced): Three masks are arranged side by side. The masks are almost identical, but with slightly different facial expressions resulting from the masks being at different angles. The first mask is tilted downward and has downcast eyes. The second mask is shown straight on and is directing its gaze slightly higher than the first. The third mask is tilted upwards so its gaze is directed more upward. Happy, sad, impatient, shy, fearful, curious, helpful. What characteristics describe your personality?

Historical Perspectives

The concept of personality has been studied for at least 2,000 years, beginning with Hippocrates in 370 BCE (Fazeli, 2012). Hippocrates theorized that personality traits and human behaviors are based on four separate temperaments associated with four fluids (humors) of the body: choleric temperament (yellow bile from the liver), melancholic temperament (black bile from the kidneys), sanguine temperament (red blood from the heart), and phlegmatic temperament (white phlegm from the lungs) (Clark & Watson, 2008; Eysenck & Eysenck, 1985; Lecci & Magnavita, 2013; Noga, 2007). Centuries later, the influential Greek physician and philosopher Galen built on Hippocrates s theory, suggesting that both diseases and personality differences could be explained by imbalances in the humors and that each person exhibits one of the four temperaments. For example, the choleric person is passionate, ambitious, and bold; the melancholic person is reserved, anxious, and unhappy; the sanguine person is joyful, eager, and optimistic; and the phlegmatic person is calm, reliable, and thoughtful (Clark & Watson, 2008; Stelmack & Stalikas, 1991). Galen s theory was prevalent for over 1,000 years and continued to be popular through the Middle Ages.

In 1780, Franz Gall, a German physician, proposed that the distances between bumps on the skull reveal a person s personality traits, character, and mental abilities (the accompanying source-figure description). According to Gall, measuring these distances revealed the sizes of the brain areas underneath, providing information that could be used to determine whether a person was friendly, prideful, murderous, kind, good with languages, and so on. Initially, phrenology was very popular; however, it was soon discredited for lack of empirical support and has long been relegated to the status of pseudoscience (Fancher, 1979).

Source illustration description (image not reproduced): Photograph A shows the cover of the American Phrenological Journal circa 1848. Across the top it reads: American Phrenological Journal. Below that it says Know thyself. Below that is a picture of a human head facing left, with many pictures comprising the area where the brain is. Below the person s ear it says Home truths for home consumption. The lines below that read: 1848, Vol. X, March, No. 3, O.S. Fowler, Editor, Phrenology, Physiology, Physiognomy, Magnetism, New York, Fowlers and Wells, Phrenological cabinet, 131 Nassau-Street, and Terms $1 a year, invariably in advance. Ten cts. a Number. Photograph B shows a printed cartoon of a person in a chair with another person behind. There are three other people in the room, and the wall is decorated with various skulls. Below the picture it reads: Drawn on Stone by E.H, and The Phrenologist. The pseudoscience of measuring the areas of a person s skull is known as phrenology. (a) Gall developed a chart that depicted which areas of the skull corresponded to particular personality traits or characteristics (Hothersall, 1995). (b) An 1825 lithograph depicts Gall examining the skull of a young woman. (credit b: modification of work by Wellcome Library, London)

In the centuries after Galen, other researchers contributed to the development of his four primary temperament types, most prominently Immanuel Kant (in the 18th century) and psychologist Wilhelm Wundt (in the 19th century) (Eysenck, 2009; Stelmack & Stalikas, 1991; Wundt, 1874/1886) (the accompanying source-figure description). Kant agreed with Galen that everyone could be sorted into one of the four temperaments and that there was no overlap between the four categories (Eysenck, 2009). He developed a list of traits that could be used to describe the personality of a person from each of the four temperaments. However, Wundt suggested that a better description of personality could be achieved using two major axes: emotional/nonemotional and changeable/unchangeable. The first axis separated strong from weak emotions (the melancholic and choleric temperaments from the phlegmatic and sanguine). The second axis divided the changeable temperaments (choleric and sanguine) from the unchangeable ones (melancholic and phlegmatic) (Eysenck, 2009).

Source illustration description (image not reproduced): A circle is divided vertically and horizontally into four sections by lines with arrows at the ends. Clockwise from the top, the arrows are labeled Strong Emotions, Changeable Temperaments, Weak Emotions, and Unchangeable Temperaments. The arcs around the perimeter of the circle, clockwise beginning with the top right segment are labeled Choleric, Sanguine, Phlegmatic, and Melancholic. The sections inside each arc contain descriptive words. Inside the Choleric arc are the words excitable, egocentric, exhibitionist, impulsive, histrionic, and active. Inside the Sanguine arc are the words playful, easygoing, sociable, carefree, hopeful, and contented. Inside the Phlegmatic arc are the words reasonable, principled, controlled, persistent, steadfast, and calm. Inside the Melancholic arc are the words anxious, worried, unhappy, suspicious, serious, and thoughtful. Developed from Galen s theory of the four temperaments, Kant proposed trait words to describe each temperament. Wundt later suggested the arrangement of the traits on two major axes.

Sigmund Freud s psychodynamic perspective of personality was the first comprehensive theory of personality, explaining a wide variety of both normal and abnormal behaviors. According to Freud, unconscious drives influenced by sex and aggression, along with childhood sexuality, are the forces that influence our personality. Freud attracted many followers who modified his ideas to create new theories about personality. These theorists, referred to as neo-Freudians, generally agreed with Freud that childhood experiences matter, but they reduced the emphasis on sex and focused more on the social environment and effects of culture on personality. The perspective of personality proposed by Freud and his followers was the dominant theory of personality for the first half of the 20th century.

Other major theories then emerged, including the learning, humanistic, biological, evolutionary, trait, and cultural perspectives. In this chapter, we will explore these various perspectives on personality in depth.

Summary

Personality has been studied for over 2,000 years, beginning with Hippocrates. More recent theories of personality have been proposed, including Freud s psychodynamic perspective, which holds that personality is formed through early childhood experiences. Other perspectives then emerged in reaction to the psychodynamic perspective, including the learning, humanistic, biological, trait, and cultural perspectives.

Review Questions

OpenStax learning exercise

Q. Personality is thought to be _____.

A. D

OpenStax learning exercise

Q. The long-standing traits and patterns that propel individuals to consistently think, feel, and behave in specific ways are known as _____.

A. D

OpenStax learning exercise

Q. _____ is credited with the first comprehensive theory of personality.

A. D

OpenStax learning exercise

Q. An early science that tried to correlate personality with measurements of parts of a person s skull is known as _____.

A. A

Critical Thinking Questions

OpenStax learning exercise

Q. What makes a personal quality part of someone s personality?

A. The particular quality or trait must be part of an enduring behavior pattern, so that it is a consistent or predictable quality.

Personal Application Questions

OpenStax learning exercise

Q. How would you describe your own personality? Do you think that friends and family would describe you in much the same way? Why or why not?

OpenStax learning exercise

Q. How would you describe your personality in an online dating profile?

OpenStax learning exercise

Q. What are some of your positive and negative personality qualities? How do you think these qualities will affect your choice of career?

Neo-Freudians: Adler, Erikson, Jung, and Horney

OpenStax source module m82235 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82235/index.cnxml

Freud attracted many followers who modified his ideas to create new theories about personality. These theorists, referred to as neo-Freudians, generally agreed with Freud that childhood experiences matter, but deemphasized sex, focusing more on the social environment and effects of culture on personality. Four notable neo-Freudians include Alfred Adler, Erik Erikson, Carl Jung (pronounced Yoong), and Karen Horney (pronounced HORN-eye).

Alfred Adler

Alfred Adler, a colleague of Freud s and the first president of the Vienna Psychoanalytical Society (Freud s inner circle of colleagues), was the first major theorist to break away from Freud (the accompanying source-figure description). He subsequently founded a school of psychology called individual psychology, which focuses on our drive to compensate for feelings of inferiority. Adler (1937, 1956) proposed the concept of the inferiority complex. An inferiority complex refers to a person s feelings that they lack worth and don t measure up to the standards of others or of society. Adler s ideas about inferiority represent a major difference between his thinking and Freud s. Freud believed that we are motivated by sexual and aggressive urges, but Adler (1930, 1961) believed that feelings of inferiority in childhood are what drive people to attempt to gain superiority and that this striving is the force behind all of our thoughts, emotions, and behaviors.

Source illustration description (image not reproduced): An illustration shows Alfred Adler. Alfred Adler proposed the concept of the inferiority complex.

Adler also believed in the importance of social connections, seeing childhood development emerging through social development rather than the sexual stages Freud outlined. Adler noted the inter-relatedness of humanity and the need to work together for the betterment of all. He said, The happiness of mankind lies in working together, in living as if each individual had set himself the task of contributing to the common welfare (Adler, 1964, p. 255) with the main goal of psychology being to recognize the equal rights and equality of others (Adler, 1961, p. 691).

With these ideas, Adler identified three fundamental social tasks that all of us must experience: occupational tasks (careers), societal tasks (friendship), and love tasks (finding an intimate partner for a long-term relationship). Rather than focus on sexual or aggressive motives for behavior as Freud did, Adler focused on social motives. He also emphasized conscious rather than unconscious motivation, since he believed that the three fundamental social tasks are explicitly known and pursued. That is not to say that Adler did not also believe in unconscious processes he did but he felt that conscious processes were more important.

One of Adler s major contributions to personality psychology was the idea that our birth order shapes our personality. He proposed that older siblings, who start out as the focus of their parents attention but must share that attention once a new child joins the family, compensate by becoming overachievers. The youngest children, according to Adler, may be spoiled, leaving the middle child with the opportunity to minimize the negative dynamics of the youngest and oldest children. Despite popular attention, research has not conclusively confirmed Adler s hypotheses about birth order.

Erik Erikson

As an art school dropout with an uncertain future, young Erik Erikson met Freud s daughter, Anna Freud, while he was tutoring the children of an American couple undergoing psychoanalysis in Vienna. It was Anna Freud who encouraged Erikson to study psychoanalysis. Erikson received his diploma from the Vienna Psychoanalytic Institute in 1933, and as Nazism spread across Europe, he fled the country and immigrated to the United States that same year. As you learned when you studied lifespan development, Erikson later proposed a psychosocial theory of development, suggesting that an individual s personality develops throughout the lifespan a departure from Freud s view that personality is fixed in early life. In his theory, Erikson emphasized the social relationships that are important at each stage of personality development, in contrast to Freud s emphasis on sex. Erikson identified eight stages, each of which represents a conflict or developmental task (the accompanying source-figure description). The development of a healthy personality and a sense of competence depend on the successful completion of each task.

Source comparison table

Erikson s Psychosocial Stages of Development			
Stage	Age (years)	Developmental Task	Description
1	0 1	Trust vs. mistrust	Trust (or mistrust) that basic needs, such as nourishment and affection, will be met
2	1 3	Autonomy vs. shame/doubt	Sense of independence in many tasks develops
3	3 6	Initiative vs. guilt	Take initiative on some activities, may develop guilt when success not met or boundaries overstepped
4	7 11	Industry vs. inferiority	Develop self-confidence in abilities when competent or sense of inferiority when not
5	12 18	Identity vs. confusion	Experiment with and develop identity and roles
6	19 29	Intimacy vs. isolation	Establish intimacy and relationships with others
7	30 64	Generativity vs. stagnation	Contribute to society and be part of a family
8	65	Integrity vs. despair	Assess and make sense of life and meaning of contributions

Carl Jung

Carl Jung (the accompanying source-figure description) was a Swiss psychiatrist and prot g of Freud, who later split off from Freud and developed his own theory, which he called analytical psychology. The focus of analytical psychology is on working to balance opposing forces of conscious and unconscious thought, and experience within one s personality. According to Jung, this work is a continuous learning process mainly occurring in the second half of life of becoming aware of unconscious elements and integrating them into consciousness.

Source illustration description (image not reproduced): A photograph shows Carl Jung. Carl Jung was interested in exploring the collective unconscious.

Jung s split from Freud was based on two major disagreements. First, Jung, like Adler and Erikson, did not accept that sexual drive was the primary motivator in a person s mental life. Second, although Jung agreed with Freud s concept of a personal unconscious, he thought it to be incomplete. In addition to the personal unconscious, Jung focused on the collective unconscious.

The collective unconscious is a universal version of the personal unconscious, holding mental patterns, or memory traces, which are common to all of us (Jung, 1928). These ancestral memories, which Jung called archetypes, are represented by universal themes in various cultures, as expressed through literature, art, and dreams (Jung). Jung said that these themes reflect common experiences of people the world over, such as facing death, becoming independent, and striving for mastery. Jung (1964) believed that through biology, each person is handed down the same themes and that the same types of symbols such as the hero, the maiden, the sage, and the trickster are present in the folklore and fairy tales of every culture. In Jung s view, the task of integrating these unconscious archetypal aspects of the self is part of the self-realization process in the second half of life. With this orientation toward self-realization, Jung parted ways with Freud s belief that personality is determined solely by past events and anticipated the humanistic movement with its emphasis on self-actualization and orientation toward the future.

Jung also proposed two attitudes or approaches toward life: extroversion and introversion (Jung, 1923) (the accompanying source-figure description). These ideas are considered Jung s most important contributions to the field of personality psychology, as almost all models of personality now include these concepts. If you are an extrovert, then you are a person who is energized by being outgoing and socially oriented: You derive your energy from being around others. If you are an introvert, then you are a person who may be quiet and reserved, or you may be social, but your energy is derived from your inner psychic activity. Jung believed a balance between extroversion and introversion best served the goal of self-realization.

Source comparison table

Introverts and Extroverts	
Introvert	Extrovert
Energized by being alone	Energized by being with others
Avoids attention	Seeks attention
Speaks slowly and softly	Speaks quickly and loudly
Thinks before speaking	Thinks out loud
Stays on one topic	Jumps from topic to topic
Prefers written communication	Prefers verbal communication
Pays attention easily	Distractible

Introverts and Extroverts	
Cautious	Acts first, thinks later

Another concept proposed by Jung was the persona, which he referred to as a mask that we adopt. According to Jung, we consciously create this persona; however, it is derived from both our conscious experiences and our collective unconscious. What is the purpose of the persona? Jung believed that it is a compromise between who we really are (our true self) and what society expects us to be. We hide those parts of ourselves that are not aligned with society s expectations.

Connect the Concepts

Jung proposed that human responses to archetypes are similar to instinctual responses in animals. One criticism of Jung is that there is no evidence that archetypes are biologically based or similar to animal instincts (Roesler, 2012). Jung formulated his ideas about 100 years ago, and great advances have been made in the field of genetics since that time. We ve found that human babies are born with certain capacities, including the ability to acquire language. However, we ve also found that symbolic information (such as archetypes) is not encoded on the genome and that babies cannot decode symbolism, refuting the idea of a biological basis to archetypes. Rather than being seen as purely biological, more recent research suggests that archetypes emerge directly from our experiences and are reflections of linguistic or cultural characteristics (Young-Eisendrath, 1995). Today, most Jungian scholars believe that the collective unconscious and archetypes are based on both innate and environmental influences, with the differences being in the role and degree of each (Sotirova-Kohli et al., 2013).

Karen Horney

Karen Horney was one of the first women trained as a Freudian psychoanalyst. During the Great Depression, Horney moved from Germany to the United States, and subsequently moved away from Freud s teachings. Like Jung, Horney believed that each individual has the potential for self-realization and that the goal of psychoanalysis should be moving toward a healthy self rather than exploring early childhood patterns of dysfunction. Horney also disagreed with the Freudian idea that girls have penis envy and are jealous of male biological features. According to Horney, any jealousy is most likely culturally based, due to the greater privileges that men often have, meaning that the differences between men s and women s personalities are culturally based, not biologically based. She further suggested that men have womb envy, because they cannot give birth.

Horney s theories focused on the role of unconscious anxiety. She suggested that normal growth can be blocked by basic anxiety stemming from needs not being met, such as childhood experiences of loneliness and/or isolation. How do children learn to handle this anxiety? Horney suggested three styles of coping (the accompanying source-figure description). The first coping style, moving toward people, relies on affiliation and dependence. These children become dependent on their parents and other caregivers in an effort to receive attention and affection, which provides relief from anxiety (Burger, 2008). When these children grow up, they tend to use this same coping strategy to deal with relationships, expressing an intense need for love and acceptance (Burger, 2008). The second coping style, moving against people, relies on aggression and assertiveness. Children with this coping style find that fighting is the best way to deal with an unhappy home situation, and they deal with their feelings of insecurity by bullying other children (Burger, 2008). As adults, people with this coping style tend to lash out with hurtful comments and exploit others (Burger, 2008). The third coping style, moving away from people, centers on detachment and isolation. These children handle their anxiety by withdrawing from the world. They need privacy and tend to be self-sufficient. When these children are adults, they continue to avoid such things as love and friendship, and they also tend to gravitate toward careers that require little interaction with others (Burger, 2008).

Source comparison table

Horney s Coping Styles		
Coping Style	Description	Example
Moving toward people	Affiliation and dependence	Child seeking positive attention and affection from parent; adult needing love
Moving against people	Aggression and manipulation	Child fighting or bullying other children; adult who is abrasive and verbally hurtful, or who exploits others
Moving away from people	Detachment and isolation	Child withdrawn from the world and isolated; adult loner

Horney believed these three styles are ways in which people typically cope with day-to-day problems; however, the three coping styles can become neurotic strategies if they are used rigidly and compulsively, leading a person to become alienated from others.

Summary

The neo-Freudians were psychologists whose work followed from Freud s. They generally agreed with Freud that childhood experiences matter, but they decreased the emphasis on sex and focused more on the social environment and effects of culture on personality. Some of the notable neo-Freudians are Alfred Adler, Carl Jung, Erik Erikson, and Karen Horney. The neo-Freudian approaches have been criticized, because they tend to be philosophical rather than based on sound scientific research. For example, Jung s conclusions about the existence of the collective unconscious are based on myths, legends, dreams, and art. In addition, as with Freud s psychoanalytic theory, the neo-Freudians based much of their theories of personality on information from their patients.

Review Questions

OpenStax learning exercise

Q. The universal bank of ideas, images, and concepts that have been passed down through the generations from our ancestors refers to _____.

A. C

Critical Thinking Questions

OpenStax learning exercise

Q. Describe the difference between extroverts and introverts in terms of what is energizing to each.

A. Extroverts are energized by social engagement. Introverts are recharged by solitary time.

OpenStax learning exercise

Q. Discuss Horney s perspective on Freud s concept of penis envy.

A. Horney disagreed with the Freudian idea that women had penis envy and were jealous of a man s biological features. Horney discussed that the jealousy was more likely culturally based, due to the greater privileges that males often have, and that differences between men and women s personalities were cultural, not biologically based. Horney also suggested that men may have womb envy, because men cannot give birth.

Personal Application Questions

OpenStax learning exercise

Q. What is your birth order? Do you agree or disagree with Adler s description of your personality based on his birth order theory, as described in the Link to Learning? Provide examples for support.

OpenStax learning exercise

Q. Would you describe yourself as an extrovert or an introvert? Does this vary based on the situation? Provide examples to support your points.

OpenStax learning exercise

Q. Select an epic story that is popular in contemporary society (such as Harry Potter or Star Wars) and explain it terms of Jung s concept of archetypes.

Trait Theorists

OpenStax source module m82239 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82239/index.cnxml

Trait theorists believe personality can be understood via the approach that all people have certain traits, or characteristic ways of behaving. Do you tend to be sociable or shy? Passive or aggressive? Optimistic or pessimistic? Moody or even-tempered? Early trait theorists tried to describe all human personality traits. For example, one trait theorist, Gordon Allport (Allport & Odbert, 1936), found 4,500 words in the English language that could describe people. He organized these personality traits into three categories: cardinal traits, central traits, and secondary traits. A cardinal trait is one that dominates your entire personality, and hence your life such as Ebenezer Scrooge s greed and Mother Theresa s altruism. Cardinal traits are not very common: Few people have personalities dominated by a single trait. Instead, our personalities typically are composed of multiple traits. Central traits are those that make up our personalities (such as loyal, kind, agreeable, friendly, sneaky, wild, and grouchy). Secondary traits are those that are not quite as obvious or as consistent as central traits. They are present under specific circumstances and include preferences and attitudes. For example, one person gets angry when people try to tickle him; another can only sleep on the left side of the bed; and yet another always orders her salad dressing on the side. And you although not normally an anxious person feel nervous before making a speech in front of your English class.

In an effort to make the list of traits more manageable, Raymond Cattell (1946, 1957) narrowed down the list to about 171 traits. However, saying that a trait is either present or absent does not accurately reflect a person s uniqueness, because all of our personalities are actually made up of the same traits; we differ only in the degree to which each trait is expressed. Cattell (1957) identified 16 factors or dimensions of personality: warmth, reasoning, emotional stability, dominance, liveliness, rule-consciousness, social boldness, sensitivity, vigilance, abstractedness, privateness, apprehension, openness to change, self-reliance, perfectionism, and tension (the accompanying source-figure description). He developed a personality assessment based on these 16 factors, called the 16PF. Instead of a trait being present or absent, each dimension is scored over a continuum, from high to low. For example, your level of warmth describes how warm, caring, and nice to others you are. If you score low on this index, you tend to be more distant and cold. A high score on this index signifies you are supportive and comforting.

Source comparison table

Personality Factors Measured by the 16PF Questionnaire		
Factor	Low Score	High Score
Warmth	Reserved, detached	Outgoing, supportive
Intellect	Concrete thinker	Analytical
Emotional stability	Moody, irritable	Stable, calm
Aggressiveness	Docile, submissive	Controlling, dominant
Liveliness	Somber, prudent	Adventurous, spontaneous
Dutifulness	Unreliable	Conscientious
Social assertiveness	Shy, restrained	Uninhibited, bold
Sensitivity	Tough-minded	Sensitive, caring
Paranoia	Trusting	Suspicious
Abstractness	Conventional	Imaginative
Introversion	Open, straightforward	Private, shrewd
Anxiety	Confident	Apprehensive
Openmindedness	Closeminded, traditional	Curious, experimental
Independence	Outgoing, social	Self-sufficient
Perfectionism	Disorganized, casual	Organized, precise
Tension	Relaxed	Stressed

Psychologists Hans and Sybil Eysenck were personality theorists (the accompanying source-figure description) who focused on temperament, the inborn, genetically based personality differences that you studied earlier in the chapter. They believed personality is largely governed by biology. The Eysencks (Eysenck, 1990, 1992; Eysenck & Eysenck, 1963) viewed people as having two specific personality dimensions: extroversion/introversion and neuroticism/stability.

Source illustration description (image not reproduced): A photograph shows Hans and Sybil Eysenck together. Hans and Sybil Eysenck believed that our personality traits are influenced by our genetic inheritance. (credit: "Sirswindon"/Wikimedia Commons)

According to their theory, people high on the trait of extroversion are sociable and outgoing, and readily connect with others, whereas people high on the trait of introversion have a higher need to be alone, engage in solitary behaviors, and limit their interactions with others. In the neuroticism/stability dimension, people high on neuroticism tend to be anxious; they tend to have an overactive sympathetic nervous system and, even with low stress, their bodies and emotional state tend to go into a flight-or-fight reaction. In contrast, people high on stability tend to need more stimulation to activate their flight-or-fight reaction and are considered more emotionally stable. Based on these two dimensions, the Eysencks theory divides people into four quadrants. These quadrants are sometimes compared with the four temperaments described by the Greeks: melancholic, choleric, phlegmatic, and sanguine (the accompanying source-figure description).

Source illustration description (image not reproduced): A circle is divided vertically and horizontally into four sections by lines with arrows at the ends. Clockwise from the top, the arrows are labeled Unstable Emotions (Neurotic), Extroverted Personality, Stable Emotions, and Introverted Personality. The arcs around the perimeter of the

circle, clockwise beginning with the top right segment are labeled Choleric, Sanguine, Phlegmatic, and Melancholic. The sections inside each arc contain descriptive words. Inside the Choleric arc are the words touchy, restless, aggressive, excitable, impulsive, and active. Inside the Sanguine arc are the words sociable, talkative, responsive, easygoing, lively, and carefree. Inside the Phlegmatic arc are the words passive, thoughtful, peaceful, controlled, reliable, and calm. Inside the Melancholic arc are the words moody, anxious, rigid, pessimistic, unsociable, and quiet. The Eysencks described two factors to account for variations in our personalities: extroversion/introversion and emotional stability/instability.

Later, the Eysencks added a third dimension: psychoticism versus superego control (Eysenck, Eysenck & Barrett, 1985). In this dimension, people who are high on psychoticism tend to be independent thinkers, cold, nonconformists, impulsive, antisocial, and hostile, whereas people who are high on superego control tend to have high impulse control they are more altruistic, empathetic, cooperative, and conventional (Eysenck, Eysenck & Barrett, 1985).

While Cattell s 16 factors may be too broad, the Eysenck s two-factor system has been criticized for being too narrow. Another personality theory, called the Five Factor Model, effectively hits a middle ground, with its five factors referred to as the Big Five personality factors. It is the most popular theory in personality psychology today and the most accurate approximation of the basic personality dimensions (Funder, 2001). The five factors are openness to experience, conscientiousness, extroversion, agreeableness, and neuroticism (the accompanying source-figure description). A helpful way to remember the factors is by using the mnemonic OCEAN.

In the Five Factor Model, each person has each factor, but they occur along a spectrum. Openness to experience is characterized by imagination, feelings, actions, and ideas. People who score high on this factor tend to be curious and have a wide range of interests. Conscientiousness is characterized by competence, self-discipline, thoughtfulness, and achievement-striving (goal-directed behavior). People who score high on this factor are hardworking and dependable. Numerous studies have found a positive correlation between conscientiousness and academic success (Akomolafe, 2013; Chamorro-Premuzic & Furnham, 2008; Conrad & Patry, 2012; Noftle & Robins, 2007; Wagerman & Funder, 2007). Extroversion is characterized by sociability, assertiveness, excitement-seeking, and emotional expression. People who score high on this factor are usually described as outgoing and warm. Not surprisingly, people who score high on both extroversion and openness are more likely to participate in adventure and risky sports due to their curious and excitement-seeking nature (Tok, 2011). The fourth factor is agreeableness, which is the tendency to be pleasant, cooperative, trustworthy, and good-natured. People who score low on agreeableness tend to be described as rude and uncooperative, yet one recent study reported that men who scored low on this factor actually earned more money than men who were considered more agreeable (Judge, Livingston, & Hurst, 2012). The last of the Big Five factors is neuroticism, which is the tendency to experience negative emotions. People high on neuroticism tend to experience emotional instability and are characterized as angry, impulsive, and hostile. Watson and Clark (1984) found that people reporting high levels of neuroticism also tend to report feeling anxious and unhappy. In contrast, people who score low in neuroticism tend to be calm and even-tempered.

Source illustration description (image not reproduced): A diagram includes five vertically stacked arrows, which point to the left and right. A dimension's first letter, name, and description are included inside of each arrow. A box to the left of each arrow includes factors associated with a low score for that arrow's dimension. A box to the right of each arrow includes factors associated with a high score for that arrow's dimension. The top arrow includes the factor openness, which is described with the words, imagination, feelings, actions, and ideas. The box to the left of that arrow includes the words, practical, conventional, and prefers routine, while the box to the right of that arrow includes the words, curious, wide range of interests, and independent. The next arrow includes the factor conscientiousness, which is described with the words, competence, self-discipline, thoughtfulness, and goal-driven. The box to the left of that arrow includes the words, impulsive, careless, and disorganized, while the box to the right of that arrow includes the words, hardworking, dependable, and organized. The next arrow includes the factor extroversion, which is described with the words, sociability, assertiveness, and emotional expression. The box to the left of that arrow includes the words, quiet, reserved, and withdrawn, while the box to the right of that arrow includes the words, outgoing, warm, and seeks adventure. The next arrow includes the factor agreeableness, which is described with the words, cooperative, trustworthy, and good-natured. The box to the left of that arrow includes the words, critical, uncooperative, and suspicious, while the box to the right of

that arrow includes the words, helpful, trusting, and empathetic. The next arrow includes the factor neuroticism, which is described as tendency toward unstable emotions. The box to the left of that arrow includes the words, calm, even-tempered, and secure, while the box to the right of that arrow includes the words, anxious, unhappy, and prone to negative emotions. In the Five Factor Model, each person has five factors, each scored on a continuum from high to low. In the center column, notice that the first letter of each factor spells the mnemonic OCEAN.

The Big Five personality factors each represent a range between two extremes. In reality, most of us tend to lie somewhere midway along the continuum of each factor, rather than at polar ends. It s important to note that the Big Five factors are relatively stable over our lifespan, with some tendency for the factors to increase or decrease slightly. Researchers have found that conscientiousness increases through young adulthood into middle age, as we become better able to manage our personal relationships and careers (Donnellan & Lucas, 2008). Agreeableness also increases with age, peaking between 50 to 70 years (Terracciano, McCrae, Brant, & Costa, 2005). Neuroticism and extroversion tend to decline slightly with age (Donnellan & Lucas; Terracciano et al.). Additionally, The Big Five factors have been shown to exist across ethnicities, cultures, and ages, and may have substantial biological and genetic components (Jang, Livesley, & Vernon, 1996; Jang et al., 2006; McCrae & Costa, 1997; Schmitt et al., 2007).

Another model of personality traits is the HEXACO model. HEXACO is an acronym for six broad traits: honesty-humility, emotionality, extraversion, agreeableness, conscientiousness, and openness to experience (Anglim & O Connor, 2018). the accompanying source-figure description provides a brief overview of each trait.

Source comparison table

The HEXACO Traits	
Trait	Example Aspects of Trait
(H) Honesty-humility	Sincerity, modesty, faithfulness
(E) Emotionality	Sentimentality, anxiety, sensitivity
(X) Extraversion	Sociability, talkativeness, boldness
(A) Agreeableness	Patience, tolerance, gentleness
(C) Conscientiousness	Organization, thoroughness, precision
(O) Openness	Creativity, inquisitiveness, innovativeness

Summary

Trait theorists attempt to explain our personality by identifying our stable characteristics and ways of behaving. They have identified important dimensions of personality. The Five Factor Model is the most widely accepted theory today. The five factors are openness, conscientiousness, extroversion, agreeableness, and neuroticism. These factors occur along a continuum.

Review Questions

OpenStax learning exercise

Q. According to the Eysencks theory, people who score high on neuroticism tend to be _____.

A. D

Critical Thinking Questions

OpenStax learning exercise

Q. How stable are the Big Five factors over one s lifespan?

A. The Big Five traits are relatively stable over our lifespan with a tendency for the traits to increase or decrease slightly. Researchers have found that conscientiousness increases through young adulthood into middle age, as we become better able to manage our personal relationships and careers. Agreeableness also increases with age, peaking between 50 to 70 years. However, neuroticism and extroversion tend to decline slightly with age.

OpenStax learning exercise

Q. Compare the personality of someone who scores high on agreeableness to someone who scores low on agreeableness.

A. A person with a high score on agreeableness is typically pleasant, cooperative, trustworthy and good-natured. People who score low on agreeableness tend to be described as rude and uncooperative. They may be difficult with which to work.

Personal Application Questions

OpenStax learning exercise

Q. Review the Big Five personality factors shown in the accompanying source-figure description. On which areas would you expect you d score high? In which areas does the low score more accurately describe you?

Cultural Understandings of Personality

OpenStax source module m82240 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82240/index.cnxml

As you have learned in this chapter, personality is shaped by both genetic and environmental factors. The culture in which you live is one of the most important environmental factors that shapes your personality (Triandis & Suh, 2002). The term culture refers to all of the beliefs, customs, art, and traditions of a particular society. Culture is transmitted to people through language as well as through the modeling of culturally acceptable and nonacceptable behaviors that are either rewarded or punished (Triandis & Suh, 2002). With these ideas in mind, personality psychologists have become interested in the role of culture in understanding personality. They ask whether personality traits are the same across cultures or if there are variations. It appears that there are both universal and culture-specific aspects that account for variation in people s personalities.

Why might it be important to consider cultural influences on personality? Western ideas about personality may not be applicable to other cultures (Benet-Martinez & Oishi, 2008). In fact, there is evidence that the strength of personality traits varies across cultures. Let s take a look at some of the Big Five factors (conscientiousness, neuroticism, openness, and extroversion) across cultures. As you will learn when you study social psychology, Asian cultures are more collectivist, and people in these cultures tend to be less extroverted. People in Central and South American cultures tend to score higher on openness to experience, whereas Europeans score higher on neuroticism (Benet-Martinez & Karakitapoglu-Aygun, 2003).

According to a study by Rentfrow and colleagues, there also seem to be regional personality differences within the United States (the accompanying source-figure description). Researchers analyzed responses

from over 1.5 million individuals in the United States and found that there are three distinct regional personality clusters: Cluster 1, which is in the Upper Midwest and Deep South, is dominated by people who fall into the friendly and conventional personality; Cluster 2, which includes the West, is dominated by people who are more relaxed, emotionally stable, calm, and creative; and Cluster 3, which includes the Northeast, has more people who are stressed, irritable, and depressed. People who live in Clusters 2 and 3 are also generally more open (Rentfrow et al., 2013).

Source illustration description (image not reproduced): A map of the United States is shown. Above it is the label Personality Clusters in the Continental United States. Below it is a legend which defines areas in the map as either, Cluster 1: friendly, conventional; Cluster 2: relaxed, creative; or Cluster 3: temperamental, uninhibited. Cluster 1occurs mainly in the center of the country. Cluster 2 occurs mainly on the west side of the country. Cluster 3 occurs mainly in the North-East region of the country and also in Texas. These are generalizations; there are several states which are comprised of a combination of two different clusters. Researchers found three distinct regional personality clusters in the United States. People tend to be friendly and conventional in the Upper Midwest and Deep South; relaxed, emotionally stable, and creative in the West; and stressed, irritable, and depressed in the Northeast (Rentfrow et al., 2013).

One explanation for the regional differences is selective migration (Rentfrow et al., 2013). Selective migration is the concept that people choose to move to places that are compatible with their personalities and needs. For example, a person high on the agreeable scale would likely want to live near family and friends, and would choose to settle or remain in such an area. In contrast, someone high on openness would prefer to settle in a place that is recognized as diverse and innovative (such as California). Further, Rentfrow, Jost, Gosling, & Potter (2009) noted an overlap between geographical regions and personality characteristics that goes beyond the often-used explanations of religion, racial diversity, and education. Their research suggests that the psychological profile of a region is closely related to that of its residents. They found that levels of openness and conscientiousness in a state may predict voting patterns, indicating that there are correlations between geographic regions and personality differences between liberals and conservatives relating to political views, levels of economic vitality, and entrepreneurial rates.

Personality in Individualist and Collectivist Cultures

Individualist cultures and collectivist cultures place emphasis on different basic values. People who live in individualist cultures tend to believe that independence, competition, and personal achievement are important. Individuals in Western nations such as the United States, England, and Australia score high on individualism (Oyserman, Coon, & Kemmelmier, 2002). People who live in collectivist cultures value social harmony, respectfulness, and group needs over individual needs. Individuals who live in countries in Asia, Africa, and South America score high on collectivism (Hofstede, 2001; Triandis, 1995). These values influence personality. For example, Yang (2006) found that people in individualist cultures displayed more personally oriented personality traits, whereas people in collectivist cultures displayed more socially oriented personality traits. Frewer and Bleus (1991) conducted a study of the Eysenk Personality Inventory in a collectivist culture using Papua New Guinean university students. They found that the results of the personality inventory were only relevant when analyzed within the context of a collectivist society. Similarly, Dana (1986) suggested that personality assessment services for Native Americans are often provided without a proper recognition of culture-specific responses and a tribe-specific frame of reference. Assessors need to have more than a general knowledge of history, tribal differences, contemporary culture on reservations, and levels of acculturation in order to interpret psychological test responses with a minimal bias.

Approaches to Studying Personality in a Cultural Context

There are three approaches that can be used to study personality in a cultural context, the cultural-comparative approach; the indigenous approach; and the combined approach, which incorporates elements of both views. Since ideas about personality have a Western basis, the cultural-comparative approach seeks to test Western ideas about personality in other cultures to determine whether they can be generalized and if they have cultural validity (Cheung van de Vijver, & Leong, 2011). For example, recall from the previous section on the trait perspective that researchers used the cultural-comparative approach to test the universality of McCrae and Costa s Five Factor Model. They found applicability in numerous cultures around the world, with the Big Five factors being stable in many cultures (McCrae & Costa, 1997; McCrae et al., 2005). The indigenous approach came about in reaction to the dominance of Western approaches to the study of personality in non-Western settings (Cheung et al., 2011). Because Western-based personality assessments cannot fully capture the personality constructs of other cultures, the indigenous model has led to the development of personality assessment instruments that are based on constructs relevant to the culture being studied (Cheung et al., 2011). The third approach to cross-cultural studies of personality is the combined approach, which serves as a bridge between Western and indigenous psychology as a way of understanding both universal and cultural variations in personality (Cheung et al., 2011).

Summary

The culture in which you live is one of the most important environmental factors that shapes your personality. Western ideas about personality may not be applicable to other cultures. In fact, there is evidence that the strength of personality traits varies across cultures. Individualist cultures and collectivist cultures place emphasis on different basic values. People who live in individualist cultures tend to believe that independence, competition, and personal achievement are important. People who live in collectivist cultures value social harmony, respectfulness, and group needs over individual needs. There are three approaches that can be used to study personality in a cultural context: the cultural-comparative approach, the indigenous approach, and the combined approach, which incorporates both elements of both views.

Review Questions

OpenStax learning exercise

Q. The United States is considered a _____ culture.

A. B

OpenStax learning exercise

Q. The concept that people choose to move to places that are compatible with their personalities and needs is known as _____.

A. A

Critical Thinking Questions

OpenStax learning exercise

Q. Why might it be important to consider cultural influences on personality?

A. Since culture influences one s personality, then Western ideas about personality may not be applicable to people of other cultures. In addition, Western-based measures of personality assessment may not be valid when used to collect data on people from other cultures.

Personal Application Questions

OpenStax learning exercise

Q. According to the work of Rentfrow and colleagues, personalities are not randomly distributed. Instead they fit into distinct geographic clusters. Based on where you live, do you agree or disagree with the traits associated with yourself and the residents of your area of the country? Why or why not?

CHAPTER 14 / CAC.901

Social cognition, self-presentation, and peer influence

Lecture: Peer life is a developmental context

Peer relationships are not a distraction from serious development. They are settings in which children practice cooperation, negotiation, status, intimacy, conflict, and repair. A child s behavior may differ across friendships and groups because each setting has different expectations. Counselors should examine the social world rather than treating peer difficulty as a trait carried unchanged from room to room.

Social cognition includes how people interpret intentions, remember social events, and predict others responses. These interpretations can be accurate or biased. A child who expects rejection may notice ambiguous behavior as hostile, withdraw, and miss opportunities for connection. Yet the expectation may have developed from real exclusion. The intervention should address both interpretation and the environment that makes the expectation plausible.

Belonging and status

Groups distribute attention and status. Some children are widely liked; others have one stable friendship that provides substantial protection. Popularity and friendship quality are not the same. A counselor should not define success as becoming prominent in the largest group. Ask whether the child has reliable support, can participate safely, and experiences respect.

Peer influence is not uniformly harmful. Friends can encourage learning, help-seeking, kindness, and resistance to mistreatment. Adults sometimes focus only on negative influence because it is more visible. A balanced assessment identifies positive peer resources as well as pressure. It also recognizes that a young person may remain in a difficult group because alternatives are unavailable or unsafe.

Self-presentation changes across settings. A young person may emphasize competence at school, humor with friends, and responsibility at home. This does not necessarily indicate a false self. It can reflect social understanding. Concern arises when presentation requires chronic concealment of distress, identity, or need, or when the cost of maintaining it becomes harmful.

Interpreting conflict

Conflict can teach negotiation when it occurs within safe relationships and is followed by repair. It becomes more concerning when it involves repeated harm, power imbalance, humiliation, or exclusion. Do not label every disagreement bullying, and do not dismiss bullying as ordinary conflict. The pattern and power dynamics matter.

A developmental response asks what each child understands, what the group rewards, and what adults do. A child who harms others may need help with perspective-taking, emotion regulation, and repair, but consequences should not be replaced by a purely therapeutic story. Safety for the targeted child comes first. Mediation can be inappropriate when it pressures a harmed child to negotiate with someone who holds power over them.

Figure 14.1 maps a social event through interpretation, emotion, action, peer response, and later expectation. It includes a separate route for changing the group environment. Individual

skill-building and environmental change can support each other. A child should not be made responsible for surviving an unsafe social system through better coping alone.

Digital contexts

Digital communication can extend friendship and provide community, especially for young people who feel isolated locally. It can also increase exposure to comparison, exclusion, or persistent harassment. Avoid treating online life as unreal. A message seen by a large peer group can have immediate emotional consequences.

Assess what happens online, who is involved, how long it continues, and what support is available. Do not assume that confiscating a device is always protective; it may remove supportive relationships and discourage future disclosure. Safety planning should be collaborative and proportionate, with attention to evidence preservation and local safeguarding requirements when needed.

Application: the silent lunch table

A fictional twelve-year-old, Ada, stops sitting with her usual group. She says nothing specific happened. A counselor explores changes in group routines, public jokes, online messages, and the cost of being seen with certain peers. Ada eventually describes repeated exclusion that adults have not noticed.

The plan includes a safe adult contact, a place where Ada can spend lunch without stigma, and action on the group behavior. It also supports Ada in identifying trusted relationships without requiring immediate confrontation. The school monitors whether exclusion decreases rather than measuring only Ada s confidence. The problem is relational and institutional, not simply a lack of social skill.

Doctoral seminar task

Map a fictional young person s peer ecology. Include close friendships, group status, adult supervision, digital spaces, and places of safety. Identify one positive influence and one pressure. Then propose an intervention at the individual level and one at the group or institutional level.

Define success from the young person s perspective and from a safeguarding perspective. Explain how you would handle disagreement between those views. The task develops an understanding of peer life as a context with resources and risks, not merely a source of bad decisions.

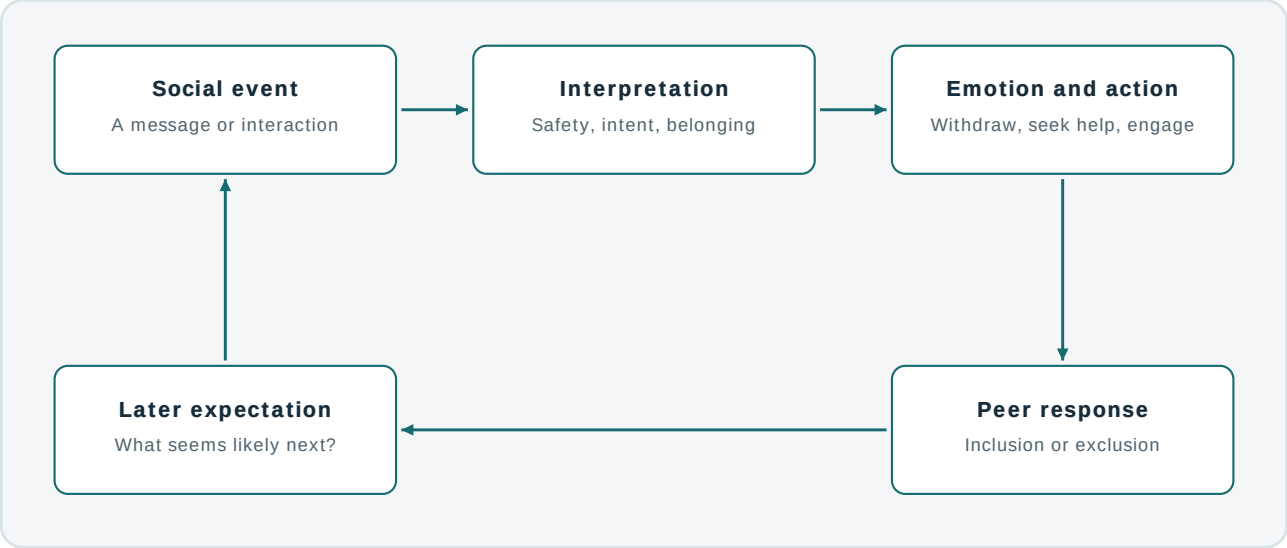


Figure 14.1. Peer experience shapes the next expectation. Interpretation and peer responses can form a feedback loop. Real exclusion must be addressed directly; changing a child s interpretation is not a substitute for changing an unsafe environment. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

What Is Social Psychology?

OpenStax source module m82273 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82273/index.cnxml

Social psychology examines how people affect one another, and it looks at the power of the situation. According to the American Psychological Association (n.d.), social psychologists "are interested in all aspects of personality and social interaction, exploring the influence of interpersonal and group relationships on human behavior." Throughout this chapter, we will examine how the presence of other individuals and groups of people impacts a person's behaviors, thoughts, and feelings. Essentially, people will change their behavior to align with the social situation at hand. If we are in a new situation or are unsure how to behave, we will take our cues from other individuals.

The field of social psychology studies topics at both the intra- and interpersonal levels. Intrapersonal topics (those that pertain to the individual) include emotions and attitudes, the self, and social cognition (the ways in which we think about ourselves and others). Interpersonal topics (those that pertain to dyads and groups) include helping behavior (the accompanying source-figure description), aggression, prejudice and discrimination, attraction and close relationships, and group processes and intergroup relationships.

Source illustration description (image not reproduced): A photograph shows several people pushing a car up an incline. Social psychology deals with all kinds of interactions between people, spanning a wide range of how we connect: from moments of confrontation to moments of working together and helping others, as shown here. (credit: Sgt. Derec Pierson, U.S. Army)

Social psychologists focus on how people conceptualize and interpret situations and how these interpretations influence their thoughts, feelings, and behaviors (Ross & Nisbett, 1991). Thus, social psychology studies individuals in a social context and how situational variables interact to influence behavior. In this chapter, we discuss the intrapersonal processes of self-presentation, cognitive dissonance and attitude change, and the interpersonal processes of conformity and obedience, aggression and altruism, and, finally, love and attraction.

Situational and Dispositional Influences on Behavior

Behavior is a product of both the situation (e.g., cultural influences, social roles, and the presence of bystanders) and of the person (e.g., personality characteristics). Subfields of psychology tend to focus on one influence or behavior over others. Situationism is the view that our behavior and actions are determined by our immediate environment and surroundings. In contrast, dispositionism holds that our behavior is determined by internal factors (Heider, 1958). An internal factor is an attribute of a person and includes personality traits and temperament. Social psychologists have tended to take the situationist perspective, whereas personality psychologists have promoted the dispositionist perspective. Modern approaches to social psychology, however, take both the situation and the individual into account when studying human behavior (Fiske, Gilbert, & Lindzey, 2010). In fact, the field of social-personality psychology has emerged to study the complex interaction of internal and situational factors that affect human behavior (Mischel, 1977; Richard, Bond, & Stokes-Zoota, 2003).

Fundamental Attribution Error

In the United States, the predominant culture tends to favor a dispositional approach in explaining human behavior. Why do you think this is? We tend to think that people are in control of their own behaviors, and, therefore, any behavior change must be due to something internal, such as their personality, habits, or temperament. According to some social psychologists, people tend to overemphasize internal factors as explanations or attributions for the behavior of other people. They tend to assume that the behavior of another person is a trait of that person, and to underestimate the power of the situation on the behavior of others. They tend to fail to recognize when the behavior of another is due to situational variables, and thus to the person s state. This erroneous assumption is called the fundamental attribution error (Ross, 1977; Riggio & Garcia, 2009). To better understand, imagine this scenario: Jamie returns home from work, and opens the front door to a happy greeting from spouse Morgan who inquires how the day has been. Instead of returning the spouse s kind greeting, Jamie yells, Leave me alone! Why did Jamie yell? How would someone committing the fundamental attribution error explain Jamie s behavior? The most common response is that Jamie is a mean, angry, or unfriendly person (traits). This is an internal or dispositional explanation. However, imagine that Jamie was just laid off from work due to company downsizing. Would your explanation for Jamie s behavior change? Your revised explanation might be that Jamie was frustrated and disappointed about being laid off and was therefore in a bad mood (state). This is now an external or situational explanation for Jamie s behavior.

The fundamental attribution error is so powerful that people often overlook obvious situational influences on behavior. A classic example was demonstrated in a series of experiments known as the quizmaster study (Ross, Amabile, & Steinmetz, 1977). Student participants were randomly assigned to play the role of a questioner (the quizmaster) or a contestant in a quiz game. Questioners developed difficult questions to which they knew the answers, and they presented these questions to the contestants. The contestants answered the questions correctly only 4 out of 10 times (the accompanying source-figure description). After the task, the questioners and contestants were asked to rate their own general knowledge compared to the average student. Questioners did not rate their general knowledge higher than the contestants, but the contestants rated the questioners intelligence higher than their own. In a second study, observers of the interaction also rated the questioner as having more general knowledge than the contestant. The obvious influence on performance is the situation. The questioners wrote the questions, so of course they had an advantage. Both the contestants and observers made an internal attribution for the performance. They concluded that the questioners must be more intelligent than the contestants.

Source illustration description (image not reproduced): A photograph shows the game show Jeopardy. In the quizmaster study, people tended to disregard the influence of the situation and wrongly concluded that a questioner s knowledge was greater than their own. (credit: Steve Jurvetson)

The halo effect refers to the tendency to let the overall impression of an individual color the way in which we feel about their character. For instance, we might assume that people who are physically attractive are more likely to be good people than less attractive individuals. Another example of how the halo effect might manifest would involve assuming that someone whom we perceive to be outgoing or friendly has a better moral character than someone who is not.

As demonstrated in the examples above, the fundamental attribution error is considered a powerful influence in how we explain the behaviors of others. However, it should be noted that some researchers have suggested that the fundamental attribution error may not be as powerful as it is often portrayed. In fact, a recent review of more than 173 published studies suggests that several factors (e.g., high levels of idiosyncrasy of the character and how well hypothetical events are explained) play a role in determining just how influential the fundamental attribution error is (Malle, 2006).

Is the Fundamental Attribution Error a Universal Phenomenon?

You may be able to think of examples of the fundamental attribution error in your life. Do people in all cultures commit the fundamental attribution error? Research suggests that they do not. People from an individualistic culture, that is, a culture that focuses on individual achievement and autonomy, have the greatest tendency to commit the fundamental attribution error. Individualistic cultures, which tend to be found in western countries such as the United States, Canada, and the United Kingdom, promote a focus on the individual. Therefore, a person s disposition is thought to be the primary explanation for her behavior. In contrast, people from a collectivistic culture, that is, a culture that focuses on communal relationships with others, such as family, friends, and community (the accompanying source-figure description), are less likely to commit the fundamental attribution error (Markus & Kitayama, 1991; Triandis, 2001).

Source illustration description (image not reproduced): Three photographs show three groups of people: a family preparing a meal, a group of men sitting on a porch, and a group of women playing mahjong. People from collectivistic cultures, such as some Asian cultures, are more likely to emphasize relationships with others than to focus primarily on the individual. Activities such as (a) preparing a meal, (b) hanging out, and (c) playing a game engage people in a group. (credit a: modification of work by Arian Zwegers; credit b: modification of work by "conbon33"/Flickr; credit c: modification of work by Anja Disseldorp)

Why do you think this is the case? Collectivistic cultures, which tend to be found in east Asian countries and in Latin American and African countries, focus on the group more than on the individual (Nisbett, Peng, Choi, & Norenzayan, 2001). This focus on others provides a broader perspective that takes into account both situational and cultural influences on behavior; thus, a more nuanced explanation of the causes of others behavior becomes more likely. the accompanying source-figure description summarizes compares individualistic and collectivist cultures.

Source comparison table

Characteristics of Individualistic and Collectivistic Cultures	
Individualistic Culture	Collectivistic Culture
Achievement oriented	Relationship oriented
Focus on autonomy	Focus on group harmony
Dispositional perspective	Situational perspective
Independent	Interdependent
Analytic thinking style	Holistic thinking style

Masuda and Nisbett (2001) demonstrated that the kinds of information that people attend to when viewing visual stimuli (e.g., an aquarium scene) can differ significantly depending on whether the observer comes from a collectivistic versus an individualistic culture. Japanese participants were much more likely to recognize objects that were presented when they occurred in the same context in which they were originally viewed. Manipulating the context in which object recall occurred had no such impact on American participants. Other researchers have shown similar differences across cultures. For example, Zhang, Fung, Stanley, Isaacowitz, and Zhang (2014) demonstrated differences in the ways that holistic thinking might develop between Chinese and American participants, and Ramesh and Gelfand (2010) demonstrated that job turnover rates are more related to the fit between a person and the organization in which they work in an Indian sample, but the fit between the person and their specific job was more predictive of turnover in an American sample.

Actor-Observer Bias

Returning to our earlier example, Jamie was laid off, but an observer would not know. So a na ve observer would tend to attribute Jamie s hostile behavior to Jamie s disposition rather than to the true, situational cause. Why do you think we underestimate the influence of the situation on the behaviors of others? One reason is that we often don t have all the information we need to make a situational explanation for another person s behavior. The only information we might have is what is observable. Due to this lack of information we have a tendency to assume the behavior is due to a dispositional, or internal, factor. When it comes to explaining our own behaviors, however, we have much more information available to us. If you came home from school or work angry and yelled at your dog or a loved one, what would your explanation be? You might say you were very tired or feeling unwell and needed quiet time a situational explanation. The actor-observer bias is the phenomenon of attributing other people s behavior to internal factors (fundamental attribution error) while attributing our own behavior to situational forces (Jones & Nisbett, 1971; Nisbett, Caputo, Legant, & Marecek, 1973; Choi & Nisbett, 1998). As actors of behavior, we have more information available to explain our own behavior. However as observers, we have less information available; therefore, we tend to default to a dispositionist perspective.

One study on the actor-observer bias investigated reasons male participants gave for why they liked their girlfriend (Nisbett et al., 1973). When asked why participants liked their own girlfriend, participants focused on internal, dispositional qualities of their girlfriends (for example, her pleasant personality). The participants explanations rarely included causes internal to themselves, such as dispositional traits (for example, I need companionship.). In contrast, when speculating why a male friend likes his girlfriend, participants were equally likely to give dispositional and external explanations. This supports the idea that actors tend to provide few internal explanations but many situational explanations for their own behavior. In contrast, observers tend to provide more dispositional explanations for a friend s behavior (the accompanying source-figure description).

Source illustration description (image not reproduced): A bar graph compares own reasons for liking girlfriend to friend s reasons for liking girlfriend. In the former, situational traits are about twice as high as dispositional traits, while in the latter, situational and dispositional traits are nearly equal. Actor-observer bias is evident when subjects explain their own reasons for liking a girlfriend versus their impressions of others reasons for liking a girlfriend.

Self-Serving Bias

We can understand self-serving bias by digging more deeply into attribution, a belief about the cause of a result. One model of attribution proposes three main dimensions: locus of control (internal versus external), stability (stable versus unstable), and controllability (controllable versus uncontrollable). In this context, stability refers to the extent in which the circumstances that result in a given outcome are changeable. The circumstances are considered stable if they are unlikely to change. Controllability refers to the extent to which the circumstances that are associated with a given outcome can be controlled. Obviously, those things that we have the power to control would be labeled controllable (Weiner, 1979).

Following an outcome, self-serving biases are those attributions that enable us to see ourselves in a favorable light (for example, making internal attributions for success and external attributions for failures). When you do well at a task, for example acing an exam, it is in your best interest to make a dispositional attribution for your behavior (I m smart,) instead of a situational one (The exam was easy,). Self-serving bias is the tendency to explain our successes as due to dispositional (internal) characteristics, but to explain our failures as due to situational (external) factors. Again, this is culture dependent. This bias serves to protect self-esteem. You can imagine that if people always made situational attributions for their behavior, they would never be able to take credit and feel good about their accomplishments.

Consider the example of how we explain our favorite sports team s wins. Research shows that we make internal, stable, and controllable attributions for our team s victory (the accompanying source-figure description) (Grove, Hanrahan, & McInman, 1991). For example, we might tell ourselves that our team is talented (internal), consistently works hard (stable), and uses effective strategies (controllable). In contrast, we are more likely to make external, unstable, and uncontrollable attributions when our favorite team loses. For example, we might tell ourselves that the other team has more experienced players or that the referees were unfair (external), the other team played at home (unstable), and the cold weather affected our team s performance (uncontrollable).

Source illustration description (image not reproduced): A photograph shows a hockey team. We tend to believe that our team wins because it s better, but loses for reasons it cannot control (Roesch & Amirkham, 1997). (credit: "TheAHL"/Flickr)

Just-World Hypothesis

One consequence of westerners tendency to provide dispositional explanations for behavior is victim blame (Jost & Major, 2001). When people experience bad fortune, others tend to assume that they somehow are responsible for their own fate. A common ideology, or worldview, in the United States is the just-world hypothesis. The just-world hypothesis is the belief that people get the outcomes they deserve (Lerner & Miller, 1978). In order to maintain the belief that the world is a fair place, people tend to think that good people experience positive outcomes, and bad people experience negative outcomes (Jost, Banaji, & Nosek, 2004; Jost & Major, 2001). The ability to think of the world as a fair place, where people get what they deserve, allows us to feel that the world is predictable and that we have some control over our life outcomes (Jost et al., 2004; Jost & Major, 2001). For example, if you want to experience positive outcomes, you just need to work hard to get ahead in life.

Can you think of a negative consequence of the just-world hypothesis? One negative consequence is people s tendency to blame poor individuals for their plight. What common explanations are given for why people live in poverty? Have you heard statements such as, The poor are lazy and just don t want to work or Poor people just want to live off the government ? What types of explanations are these, dispositional or situational? These dispositional explanations are clear examples of the fundamental attribution error. Blaming poor people for their poverty ignores situational factors that impact them, such as high unemployment rates, recession, poor educational opportunities, and the familial cycle of poverty (the accompanying source-figure description). Other research shows that people who hold just-world beliefs have negative attitudes toward people who are unemployed and people living with AIDS (Sutton & Douglas, 2005). In the United States and other countries, victims of sexual assault may find themselves blamed for their abuse. Victim advocacy groups, such as Domestic Violence Ended (DOVE), attend court in support of victims to ensure that blame is directed at the perpetrators of sexual violence, not the victims.

Source illustration description (image not reproduced): A photograph shows a homeless person and a dog sitting on a sidewalk with a sign reading, homeless, broke, and hungry. People who hold just-world beliefs tend to blame the people in poverty for their circumstances, ignoring situational and cultural causes of poverty. (credit: Adrian Miles)

Summary

Social psychology is the subfield of psychology that studies the power of the situation to influence individuals thoughts, feelings, and behaviors. Psychologists categorize the causes of human behavior as those due to internal factors, such as personality, or those due to external factors, such as cultural and other social influences. Behavior is better explained, however, by using both approaches. Lay people tend to over-rely on dispositional explanations for behavior and ignore the power of situational influences, a perspective called the fundamental attribution error. People from individualistic cultures are more likely to display this bias versus people from collectivistic cultures. Our explanations for our own and others behaviors can be biased due to not having enough information about others motivations for behaviors and by providing explanations that bolster our self-esteem.

Review Questions

OpenStax learning exercise

Q. As a field, social psychology focuses on _____ in predicting human behavior.

A. D

OpenStax learning exercise

Q. Making internal attributions for your successes and making external attributions for your failures is an example of _____.

A. C

OpenStax learning exercise

Q. Collectivistic cultures are to _____ as individualistic cultures are to _____.

A. B

OpenStax learning exercise

Q. According to the actor-observer bias, we have more information about _____.

A. B

Critical Thinking Questions

OpenStax learning exercise

Q. Compare and contrast situational influences and dispositional influences and give an example of each. Explain how situational influences and dispositional influences might explain inappropriate behavior.

A. A situationism view is that our behaviors are determined by the situation for example, a person who is late for work claims that heavy traffic caused the delay. A dispositional view is that our behaviors are determined by personality traits for example, a driver in a road rage incident claims the driver who cut her off is an aggressive person. Thus, a situational view tends to provide an excuse for inappropriate behavior, and a dispositional view tends to lay blame for inappropriate behavior.

OpenStax learning exercise

Q. Provide an example of how people from individualistic and collectivistic cultures would differ in explaining why they won an important sporting event.

A. People from individualistic cultures would tend to attribute athletic success to individual hard work and ability. People from collectivistic cultures would tend attribute athletic success to the team working together and the support and encouragement of the coach.

Personal Application Questions

OpenStax learning exercise

Q. Provide a personal example of an experience in which your behavior was influenced by the power of the situation.

OpenStax learning exercise

Q. Think of an example in the media of a sports figure player or coach who gives a self-serving attribution for winning or losing. Examples might include accusing the referee of incorrect calls, in the case of losing, or citing their own hard work and talent, in the case of winning.

Self-presentation

OpenStax source module m82247 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82247/index.cnxml

As you ve learned, social psychology is the study of how people affect one another s thoughts, feelings, and behaviors. We have discussed situational perspectives and social psychology s emphasis on the ways in which a person s environment, including culture and other social influences, affect behavior. In this section, we examine situational forces that have a strong influence on human behavior including social roles, social norms, and scripts. We discuss how humans use the social environment as a source of information, or cues, on how to behave. Situational influences on our behavior have important consequences, such as whether we will help a stranger in an emergency or how we would behave in an unfamiliar environment.

Social Roles

One major social determinant of human behavior is our social roles. A social role is a pattern of behavior that is expected of a person in a given setting or group (Hare, 2003). Each one of us has several social roles. You may be, at the same time, a student, a parent, an aspiring teacher, a son or daughter, a spouse, and a lifeguard. How do these social roles influence your behavior? Social roles are defined by culturally shared knowledge. That is, nearly everyone in a given culture knows what behavior is expected of a person in a given role. For example, what is the social role for a student? If you look around a college classroom you will likely see students engaging in studious behavior, taking notes, listening to the professor, reading the textbook, and sitting quietly at their desks (the accompanying source-figure description). Of course you may see students deviating from the expected studious behavior such as texting on their phones or using Facebook on their laptops, but in all cases, the students that you observe are attending class a part of the social role of students.

Source illustration description (image not reproduced): A photograph shows students in a classroom. Being a student is just one of the many social roles you have. (credit: modification of work by Rural Institute /Flickr)

Social roles, and our related behavior, can vary across different settings. How do you behave when you are engaging in the role of a child attending a family function? Now imagine how you behave when you

are engaged in the role of employee at your workplace. It is very likely that your behavior will be different. Perhaps you are more relaxed and outgoing with your family, making jokes and doing silly things. But at your workplace you might speak more professionally, and although you may be friendly, you are also serious and focused on getting the work completed. These are examples of how our social roles influence and often dictate our behavior to the extent that identity and personality can vary with context (that is, in different social groups) (Malloy, Albright, Kenny, Agatstein & Winquist, 1997).

Social Norms

As discussed previously, social roles are defined by a culture s shared knowledge of what is expected behavior of an individual in a specific role. This shared knowledge comes from social norms. A social norm is a group s expectation of what is appropriate and acceptable behavior for its members how they are supposed to behave and think (Deutsch & Gerard, 1955; Berkowitz, 2004). How are we expected to act? What are we expected to talk about? What are we expected to wear? In our discussion of social roles we noted that colleges have social norms for students behavior in the role of student and workplaces have social norms for employees behaviors in the role of employee. Social norms are everywhere including in families, gangs, and on social media outlets. What are some social norms on Instagram?

Connect the Concepts

My 11-year-old daughter, Janelle, recently told me she needed shorts and shirts for the summer, and that she wanted me to take her to a store at the mall that is popular with preteens and teens to buy them. I have noticed that many girls have clothes from that store, so I tried teasing her. I said, All the shirts say Aero on the front. If you are wearing a shirt like that and you have a substitute teacher, and the other girls are all wearing that type of shirt, won t the substitute teacher think you are all named Aero ?

My daughter replied, in typical 11-year-old fashion, Mom, you are not funny. Can we please go shopping?

I tried a different tactic. I asked Janelle if having clothing from that particular store will make her popular. She replied, No, it will not make me popular. It is what the popular kids wear. It will make me feel happier. How can a label or name brand make someone feel happier? Think back to what you ve learned about lifespan development. What is it about pre-teens and young teens that make them want to fit in (the accompanying source-figure description)? Does this change over time? Think back to your high school experience, or look around your college campus. What is the main name brand clothing you see? What messages do we get from the media about how to fit in?

Source illustration description (image not reproduced): A photograph shows a group of young people dressed similarly. Young people struggle to become independent at the same time they are desperately trying to fit in with their peers. (credit: Monica Arellano-Ongpin)

Scripts

Because of social roles, people tend to know what behavior is expected of them in specific, familiar settings. A script is a person s knowledge about the sequence of events expected in a specific setting (Schank & Abelson, 1977). How do you act on the first day of school, when you walk into an elevator, or are at a restaurant? For example, at a restaurant in the United States, if we want the server s attention, we try to make eye contact. In Brazil, you would make the sound psst to get the server s attention. You can see the cultural differences in scripts. To an American, saying psst to a server might seem rude, yet to a Brazilian, trying to make eye contact might not seem an effective strategy. Scripts are important sources of information to guide behavior in given situations. Can you imagine being in an unfamiliar situation and not having a script for how to behave? This could be uncomfortable and confusing. How could you find out about social norms in an unfamiliar culture?

Zimbardo s Stanford Prison Experiment

The famous Stanford prison experiment, conducted by social psychologist Philip Zimbardo and his colleagues at Stanford University, demonstrated the power of social roles, social norms, and scripts. In the summer of 1971, an advertisement was placed in a California newspaper asking for male volunteers to participate in a study about the psychological effects of prison life. More than 70 men volunteered, and these volunteers then underwent psychological testing to eliminate candidates who had underlying psychiatric issues, medical issues, or a history of crime or drug abuse. The pool of volunteers was whittled down to 24 healthy male college students. Each student was paid $15 per day (equivalent to about $80 today) and was randomly assigned to play the role of either a prisoner or a guard in the study. Based on what you have learned about research methods, why is it important that participants were randomly assigned?

A mock prison was constructed in the basement of the psychology building at Stanford. Participants assigned to play the role of prisoners were arrested at their homes by Palo Alto police officers, booked at a police station, and subsequently taken to the mock prison. The experiment was scheduled to run for several weeks. To the surprise of the researchers, both the prisoners and guards assumed their roles with zeal. On the second day of the experiment, the guards forced the prisoners to strip, took their beds, and isolated the ringleaders using solitary confinement. In a relatively short time, the guards came to harass the prisoners in an increasingly sadistic manner, through a complete lack of privacy, lack of basic comforts such as mattresses to sleep on, and through degrading chores and late-night counts.

The prisoners, in turn, began to show signs of severe anxiety and hopelessness they began tolerating the guards abuse. Even the Stanford professor who designed the study and was the head researcher, Philip Zimbardo, found himself acting as if the prison was real and his role, as prison supervisor, was real as well. After only six days, the experiment had to be ended due to the participants deteriorating behavior. Zimbardo explained,

At this point it became clear that we had to end the study. We had created an overwhelmingly powerful situation a situation in which prisoners were withdrawing and behaving in pathological ways, and in which some of the guards were behaving sadistically. Even the good guards felt helpless to intervene, and none of the guards quit while the study was in progress. Indeed, it should be noted that no guard ever came late for his shift, called in sick, left early, or demanded extra pay for overtime work. (Zimbardo, 2013)

The Stanford Prison Experiment has been used as a memorable demonstration of the incredible power that social roles, norms, and scripts have in affecting human behavior. However, multiple aspects of the study have been subject to criticism since its inception. The nature of these criticisms range from ethical concerns to issues of generalizability (Bartels, Milovich, & Moussier, 2016; Griggs, 2014; Le Texier, 2019). One criticism is that the way students were recruited for the experiment may have impacted the outcome (Carnahan & McFarland, 2007). Another criticism questions the conclusions that can be drawn from the study. Zimbardo appears to have provided specific guidelines of the types of behaviors that were expected of the guards (Zimbardo, 2007). Subsequent research suggests that such guidelines likely created an expectation of the types of behavior that Zimbardo reported observing in the Stanford Prison Experiment (Bartels, 2019), and that given these expectations, the guards simply acted as they thought they were expected to act. It has also been problematic that attempts to replicate aspects of the study have not been successful. For example, when no guidelines were presented to the guards, researchers documented different outcomes than those observed by Zimbardo. (Reicher & Haslam, 2006).

The Stanford Prison Experiment has some parallels with the abuse of prisoners of war by U.S. Army troops and CIA personnel at the Abu Ghraib prison in 2003 and 2004 during the Iraq War. The offenses at Abu Ghraib were documented by photographs of the abuse, some taken by the abusers themselves (the accompanying source-figure description).

Source illustration description (image not reproduced): A photograph shows a person standing on a box with arms held out. The person is covered in shawl-like attire and a full hood that covers the face completely. Iraqi prisoners of war were abused by their American captors in Abu Ghraib prison, during the second Iraq war. (credit: United States Department of Defense)

Summary

Human behavior is largely influenced by our social roles, norms, and scripts. In order to know how to act in a given situation, we have shared cultural knowledge of how to behave depending on our role in society. Social norms dictate the behavior that is appropriate or inappropriate for each role. Each social role has scripts that help humans learn the sequence of appropriate behaviors in a given setting. The famous Stanford prison experiment is an example of how the power of the situation can dictate the social roles, norms, and scripts we follow in a given situation, even if this behavior is contrary to our typical behavior.

Review Questions

OpenStax learning exercise

Q. A(n) _____ is a set of group expectations for appropriate thoughts and behaviors of its members.

A. B

OpenStax learning exercise

Q. On his first day of soccer practice, Jose suits up in a t-shirt, shorts, and cleats and runs out to the field to join his teammates. Jose s behavior is reflective of _____.

A. A

OpenStax learning exercise

Q. When it comes to buying clothes, teenagers often follow social norms; this is likely motivated by _____.

A. C

OpenStax learning exercise

Q. In the Stanford prison experiment, even the lead researcher succumbed to his role as a prison supervisor. This is an example of the power of _____ influencing behavior.

A. D

Critical Thinking Questions

OpenStax learning exercise

Q. Why didn t the good guards in the Stanford prison experiment object to other guards abusive behavior? Were the student prisoners simply weak people? Why didn t they object to being abused?

A. The good guards were fulfilling their social roles and they did not object to other guards abusive behavior because of the power of the situation. In addition, the prison supervisor s behavior sanctioned the guards negative treatment of prisoners. The prisoners were not weak people; they were recruited because they were healthy, mentally stable adults. The power of their social role influenced them to engage in subservient prisoner behavior. The script for prisoners is to accept

abusive behavior from authority figures, especially for punishment, when they do not follow the rules.

OpenStax learning exercise

Q. Describe how social roles, social norms, and scripts were evident in the Stanford prison experiment. How can this experiment be applied to everyday life? Are there any more recent examples where people started fulfilling a role and became abusive?

A. Social roles were in play as each participant acted out behaviors appropriate to his role as prisoner, guard, or supervisor. Scripts determined the specific behaviors the guards and prisoners displayed, such as humiliation and passivity. The social norms of a prison environment sanctions abuse of prisoners since they have lost many of their human rights and became the property of the government. This experiment can be applied to other situations in which social norms, roles, and scripts dictate our behavior, such as in mob behavior. A more recent example of similar behavior was the abuse of prisoners by American soldiers who were working as prison guards at the Abu Ghraib prison in Iraq.

Personal Application Questions

OpenStax learning exercise

Q. Try attending a religious service very different from your own and see how you feel and behave without knowing the appropriate script. Or, try attending an important, personal event that you have never attended before, such as a bar mitzvah (a coming-of-age ritual in Jewish culture), a quincea era (in some Latin American cultures a party is given to a girl who is turning 15 years old), a wedding, a funeral, or a sporting event new to you, such as horse racing or bull riding. Observe and record your feelings and behaviors in this unfamiliar setting for which you lack the appropriate script. Do you silently observe the action, or do you ask another person for help interpreting the behaviors of people at the event? Describe in what ways your behavior would change if you were to attend a similar event in the future?

OpenStax learning exercise

Q. Name and describe at least three social roles you have adopted for yourself. Why did you adopt these roles? What are some roles that are expected of you, but that you try to resist?

Conformity, Compliance, and Obedience

OpenStax source module m82249 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82249/index.cnxml

In this section, we discuss additional ways in which people influence others. The topics of conformity, social influence, obedience, and group processes demonstrate the power of the social situation to change our thoughts, feelings, and behaviors. We begin this section with a discussion of a famous social psychology experiment that demonstrated how susceptible humans are to outside social pressures.

Conformity

Solomon Asch conducted several experiments in the 1950s to determine how people are affected by the thoughts and behaviors of other people. In one study, a group of participants was shown a series of printed line segments of different lengths: a, b, and c (the accompanying source-figure description). Participants were then shown a fourth line segment: x. They were asked to identify which line segment from the first group (a, b, or c) most closely resembled the fourth line segment in length.

Source illustration description (image not reproduced): A drawing has two boxes: in the first is a line labeled x and in the second are three lines of different lengths from each other, labeled a, b, and c. These line segments illustrate the judgment task in Asch s conformity study. Which line on the right a, b, or c is the same length as line x on the left?

Each group of participants had only one true, na ve subject. The remaining members of the group were confederates of the researcher. A confederate is a person who is aware of the experiment and works for the researcher. Confederates are used to manipulate social situations as part of the research design, and the true, na ve participants believe that confederates are, like them, uninformed participants in the experiment. In Asch s study, the confederates identified a line segment that was obviously shorter than the target line a wrong answer. The na ve participant then had to identify aloud the line segment that best matched the target line segment.

How often do you think the true participant aligned with the confederates response? That is, how often do you think the group influenced the participant, and the participant gave the wrong answer? Asch (1955) found that 76% of participants conformed to group pressure at least once by indicating the incorrect line. Conformity is the change in a person s behavior to go along with the group, even if he does not agree with the group. Why would people give the wrong answer? What factors would increase or decrease someone giving in or conforming to group pressure?

The Asch effect is the influence of the group majority on an individual s judgment.

What factors make a person more likely to yield to group pressure? Research shows that the size of the majority, the presence of another dissenter, and the public or relatively private nature of responses are key influences on conformity.

- The size of the majority: The greater the number of people in the majority, the more likely an individual will conform. There is, however, an upper limit: a point where adding more members does not increase conformity. In Asch s study, conformity increased with the number of people in the majority up to seven individuals. At numbers beyond seven, conformity leveled off and decreased slightly (Asch, 1955).
- The presence of another dissenter: If there is at least one dissenter, conformity rates drop to near zero (Asch, 1955).
- The public or private nature of the responses: When responses are made publicly (in front of others), conformity is more likely; however, when responses are made privately (e.g., writing down the response), conformity is less likely (Deutsch & Gerard, 1955).

The finding that conformity is more likely to occur when responses are public than when they are private is the reason government elections require voting in secret, so we are not coerced by others (the accompanying source-figure description). The Asch effect can be easily seen in children when they have to publicly vote for something. For example, if the teacher asks whether the children would rather have extra recess, no homework, or candy, once a few children vote, the rest will comply and go with the majority. In a different classroom, the majority might vote differently, and most of the children would comply with that majority. When someone s vote changes if it is made in public versus private, this is known as compliance. Compliance can be a form of conformity. Compliance is going along with a request or demand, even if you do not agree with the request. In Asch s studies, the participants complied by giving the wrong answers, but privately did not accept that the obvious wrong answers were correct.

Source illustration description (image not reproduced): A photograph shows a row of curtained voting booths; two are occupied by people. Voting for government officials in the United States is private to reduce the pressure of conformity. (credit: Nicole Klauss)

Now that you have learned about the Asch line experiments, why do you think the participants conformed? The correct answer to the line segment question was obvious, and it was an easy task. Researchers have categorized the motivation to conform into two types: normative social influence and informational social influence (Deutsch & Gerard, 1955).

In normative social influence, people conform to the group norm to fit in, to feel good, and to be accepted by the group. However, with informational social influence, people conform because they believe the group is competent and has the correct information, particularly when the task or situation is ambiguous. What type of social influence was operating in the Asch conformity studies? Since the line judgment task was unambiguous, participants did not need to rely on the group for information. Instead, participants complied to fit in and avoid ridicule, an instance of normative social influence.

An example of informational social influence may be what to do in an emergency situation. Imagine that you are in a movie theater watching a film and what seems to be smoke comes in the theater from under the emergency exit door. You are not certain that it is smoke it might be a special effect for the movie, such as a fog machine. When you are uncertain you will tend to look at the behavior of others in the theater. If other people show concern and get up to leave, you are likely to do the same. However, if others seem unconcerned, you are likely to stay put and continue watching the movie (the accompanying source-figure description).

Source illustration description (image not reproduced): Photograph A shows people seated in an auditorium. Photograph B shows a person crowd surfing. People in crowds tend to take cues from others and act accordingly. (a) An audience is listening to a lecture and people are relatively quiet, still, and attentive to the speaker on the stage. (b) An audience is at a rock concert where people are dancing, singing, and possibly engaging in activities like crowd surfing. (credit a: modification of work by Matt Brown; credit b: modification of work by Christian Holm r)

How would you have behaved if you were a participant in Asch s study? Many students say they would not conform, that the study is outdated, and that people nowadays are more independent. To some extent this may be true. Research suggests that overall rates of conformity may have reduced since the time of Asch s research. Furthermore, efforts to replicate Asch s study have made it clear that many factors determine how likely it is that someone will demonstrate conformity to the group. These factors include the participant s age, gender, and socio-cultural background (Bond & Smith, 1996; Larsen, 1990; Walker & Andrade, 1996).

Stanley Milgram s Experiment

Conformity is one effect of the influence of others on our thoughts, feelings, and behaviors. Another form of social influence is obedience to authority. Obedience is the change of an individual s behavior to comply with a demand by an authority figure. People often comply with the request because they are concerned about a consequence if they do not comply. To demonstrate this phenomenon, we review another classic social psychology experiment.

Stanley Milgram was a social psychology professor at Yale who was influenced by the trial of Adolf Eichmann, a Nazi war criminal. Eichmann s defense for the atrocities he committed was that he was just following orders. Milgram (1963) wanted to test the validity of this defense, so he designed an experiment and initially recruited 40 men for his experiment. The volunteer participants were led to believe that they were participating in a study to improve learning and memory. The participants were told that they were to teach other students (learners) correct answers to a series of test items. The participants were shown how to use a device that they were told delivered electric shocks of different intensities to the learners. The participants were told to shock the learners if they gave a wrong answer to a test item that the shock would help them to learn. The participants believed they gave the learners shocks, which increased in 15-volt increments, all the way up to 450 volts. The participants did not know that the learners were confederates and that the confederates did not actually receive shocks.

In response to a string of incorrect answers from the learners, the participants obediently and repeatedly shocked them. The confederate learners cried out for help, begged the participant teachers to stop, and even complained of heart trouble. Yet, when the researcher told the participant-teachers to continue the shock, 65% of the participants continued the shock to the maximum voltage and to the point that the learner became unresponsive (the accompanying source-figure description). What makes someone obey authority to the point of potentially causing serious harm to another person?

Source illustration description (image not reproduced): A graph shows the voltage of shock given on the x-axis, and the percentage of participants who delivered voltage on the y-axis. All or nearly all participants delivered slight to moderate shock (15 135 volts); with strong to very strong shock (135 255 volts), the participation percentage dropped to about 80%; with intense to extremely intense shock (255 375 volts), the participation percentage dropped to about 65%; the participation percentage remained at about 65% for severe shock (375 435 volts) and XXX (435 450 volts). The Milgram experiment showed the surprising degree to which people obey authority. Two out of three (65%) participants continued to administer shocks to an unresponsive learner.

Several variations of the original Milgram experiment were conducted to test the boundaries of obedience. When certain features of the situation were changed, participants were less likely to continue to deliver shocks (Milgram, 1965). For example, when the setting of the experiment was moved to an off-campus office building, the percentage of participants who delivered the highest shock dropped to 48%. When the learner was in the same room as the teacher, the highest shock rate dropped to 40%. When the teachers and learners hands were touching, the highest shock rate dropped to 30%. When the researcher gave the orders by phone, the rate dropped to 23%. These variations show that when the humanity of the person being shocked was increased, obedience decreased. Similarly, when the authority of the experimenter decreased, so did obedience.

This case is still very applicable today. What does a person do if an authority figure orders something done? What if the person believes it is incorrect, or worse, unethical? In a study by Martin and Bull (2008), midwives privately filled out a questionnaire regarding best practices and expectations in delivering a baby. Then, a more senior midwife and supervisor asked the junior midwives to do something they had previously stated they were opposed to. Most of the junior midwives were obedient to authority, going against their own beliefs. Burger (2009) partially replicated this study. He found among a multicultural sample of women and men that their levels of obedience matched Milgram's research. Doli ski et al. (2017) performed a replication of Burger's work in Poland and controlled for the gender of both participants and learners, and once again, results that were consistent with Milgram's original work were observed.

Groupthink

When in group settings, we are often influenced by the thoughts, feelings, and behaviors of people around us. Whether it is due to normative or informational social influence, groups have power to influence individuals. Another phenomenon of group conformity is groupthink. Groupthink is the modification of the opinions of members of a group to align with what they believe is the group consensus (Janis, 1972). In group situations, the group often takes action that individuals would not perform outside the group setting because groups make more extreme decisions than individuals do. Moreover, groupthink can hinder opposing trains of thought. This elimination of diverse opinions contributes to faulty decision by the group.

Dig Deeper

There have been several instances of groupthink in the U.S. government. One example occurred when the United States led a small coalition of nations to invade Iraq in March 2003. This invasion occurred because a small group of advisors and former President George W. Bush were convinced that Iraq represented a significant terrorism threat with a large stockpile of weapons of mass destruction at its disposal. Although some of these individuals may have had some doubts about the credibility of the information available to them at the time, in the end, the group arrived at a consensus that Iraq had weapons of mass destruction and represented a significant threat to national security. It later came to light that Iraq did not have weapons of mass destruction, but not until the invasion was well underway. As a result, 6000 American soldiers were killed and many more civilians died. How did the Bush administration arrive at their conclusions? View this video of Colin Powell, 10 years after his famous United Nations speech, discussing the information he had at the time (https://www.openstax.org/l/GroupThink) that his decisions were based on. ("CNN Official Interview: Colin Powell now regrets UN speech about WMDs," 2010).

Do you see evidence of groupthink?

Why does groupthink occur? There are several causes of groupthink, which makes it preventable. When the group is highly cohesive, or has a strong sense of connection, maintaining group harmony may become more important to the group than making sound decisions. If the group leader is directive and makes his opinions known, this may discourage group members from disagreeing with the leader. If the group is isolated from hearing alternative or new viewpoints, groupthink may be more likely. How do you know when groupthink is occurring?

There are several symptoms of groupthink including the following:

- perceiving the group as invulnerable or invincible believing it can do no wrong
- believing the group is morally correct
- self-censorship by group members, such as withholding information to avoid disrupting the group consensus
- the quashing of dissenting group members opinions
- the shielding of the group leader from dissenting views
- perceiving an illusion of unanimity among group members
- holding stereotypes or negative attitudes toward the out-group or others with differing viewpoints (Janis, 1972)

Given the causes and symptoms of groupthink, how can it be avoided? There are several strategies that can improve group decision making including seeking outside opinions, voting in private, having the leader withhold position statements until all group members have voiced their views, conducting research on all viewpoints, weighing the costs and benefits of all options, and developing a contingency plan (Janis, 1972; Mitchell & Eckstein, 2009).

Group Polarization

Another phenomenon that occurs within group settings is group polarization. Group polarization (Teger & Pruitt, 1967) is the strengthening of an original group attitude after the discussion of views within a group. That is, if a group initially favors a viewpoint, after discussion the group consensus is likely a stronger endorsement of the viewpoint. Conversely, if the group was initially opposed to a viewpoint, group discussion would likely lead to stronger opposition. Group polarization explains many actions taken by groups that would not be undertaken by individuals. Group polarization can be observed at political conventions, when platforms of the party are supported by individuals who, when not in a group, would decline to support them. Recently, some theorists have argued that group polarization may be partly responsible for the extreme political partisanship that seems ubiquitous in modern society. Given that people can self-select media outlets that are most consistent with their own political views, they are less likely to encounter opposing viewpoints. Over time, this leads to a strengthening of their own perspective and of hostile attitudes and behaviors towards those with different political ideals. Remarkably, political polarization leads to open levels of discrimination that are on par with, or perhaps exceed, racial discrimination (Iyengar & Westwood, 2015). A more everyday example is a group s discussion of how attractive someone is. Does your opinion change if you find someone attractive, but your friends do not agree? If your friends vociferously agree, might you then find this person even more attractive?

Social traps refer to situations that arise when individuals or groups of individuals behave in ways that are not in their best interest and that may have negative, long-term consequences. However, once established, a social trap is very difficult to escape. For example, following World War II, the United States and the former Soviet Union engaged in a nuclear arms race. While the presence of nuclear weapons is not in either party's best interest, once the arms race began, each country felt the need to continue producing nuclear weapons to protect itself from the other.

Social Loafing

Imagine you were just assigned a group project with other students whom you barely know. Everyone in your group will get the same grade. Are you the type who will do most of the work, even though the final grade will be shared? Or are you more likely to do less work because you know others will pick up the slack? Social loafing involves a reduction in individual output on tasks where contributions are pooled. Because each individual's efforts are not evaluated, individuals can become less motivated to perform well. Karau and Williams (1993) and Simms and Nichols (2014) reviewed the research on social loafing and discerned when it was least likely to happen. The researchers noted that social loafing could be alleviated if, among other situations, individuals knew their work would be assessed by a manager (in a workplace setting) or instructor (in a classroom setting), or if a manager or instructor required group members to complete self-evaluations.

The likelihood of social loafing in student work groups increases as the size of the group increases (Shepperd & Taylor, 1999). According to Kamau and Williams (1993), college students were the population most likely to engage in social loafing. Their study also found that women and participants from collectivistic cultures were less likely to engage in social loafing, explaining that their group orientation may account for this.

College students could work around social loafing or free-riding by suggesting to their professors use of a flocking method to form groups. Harding (2018) compared groups of students who had self-selected into groups for class to those who had been formed by flocking, which involves assigning students to groups who have similar schedules and motivations. Not only did she find that students reported less free riding, but that they also did better in the group assignments compared to those whose groups were self-selected.

Interestingly, the opposite of social loafing occurs when the task is complex and difficult (Bond & Titus, 1983; Geen, 1989). In a group setting, such as the student work group, if your individual performance cannot be evaluated, there is less pressure for you to do well, and thus less anxiety or physiological

arousal (Latan , Williams, & Harkens, 1979). This puts you in a relaxed state in which you can perform your best, if you choose (Zajonc, 1965). If the task is a difficult one, many people feel motivated and believe that their group needs their input to do well on a challenging project (Jackson & Williams, 1985).

Deindividuation

Another way that being part of a group can affect behavior is exhibited in instances in which deindividuation occurs. Deindividuation refers to situations in which a person may feel a sense of anonymity and therefore a reduction in accountability and sense of self when among others. Deindividuation is often pointed to in cases in which mob or riot-like behaviors occur (Zimbardo, 1969), but research on the subject and the role that deindividuation plays in such behaviors has resulted in inconsistent results (as discussed in Granstr m, Guv , Hylander, & Rosander, 2009).

the accompanying source-figure description summarizes the types of social influence you have learned about in this chapter.

Source comparison table

Types of Social Influence	
Type of Social Influence	Description
Conformity	Changing your behavior to go along with the group even if you do not agree with the group
Compliance	Going along with a request or demand
Normative social influence	Conformity to a group norm to fit in, feel good, and be accepted by the group
Informational social influence	Conformity to a group norm prompted by the belief that the group is competent and has the correct information
Obedience	Changing your behavior to please an authority figure or to avoid aversive consequences
Groupthink	Tendency to prioritize group cohesion over critical thinking that might lead to poor decision making; more likely to occur when there is perceived unanimity among the group
Group polarization	Strengthening of the original group attitude after discussing views within a group
Social loafing	Exertion of less effort by a person working in a group because individual performance cannot be evaluated separately from the group, thus causing performance decline on easy tasks
Deindividuation	Group situation in which a person may feel a sense of anonymity and a resulting reduction in accountability and sense of self

Summary

The power of the situation can lead people to conform, or go along with the group, even in the face of inaccurate information. Conformity to group norms is driven by two motivations, the desire to fit in and be liked and the desire to be accurate and gain information from the group. Authority figures also have influence over our behaviors, and many people become obedient and follow orders even if the orders are contrary to their personal values. Conformity to group pressures can also result in groupthink, or the faulty decision-making process that results from cohesive group members trying to maintain group harmony. Group situations can improve human behavior through facilitating performance on easy tasks, but inhibiting performance on difficult tasks. The presence of others can also lead to social loafing when individual efforts cannot be evaluated.

Review Questions

OpenStax learning exercise

Q. In the Asch experiment, participants conformed due to _____ social influence.

A. B

OpenStax learning exercise

Q. Under what conditions will informational social influence be more likely?

A. D

OpenStax learning exercise

Q. Social loafing occurs when _____.

A. C

OpenStax learning exercise

Q. If group members modify their opinions to align with a perceived group consensus, then _____ has occurred.

A. C

Critical Thinking Questions

OpenStax learning exercise

Q. Describe how seeking outside opinions can prevent groupthink.

A. Outsiders can serve as a quality control by offering diverse views and views that may differ from the leader s opinion. The outsider can also remove the illusion of invincibility by having the group s action held up to outside scrutiny. An outsider may offer additional information and uncover information that group members withheld.

OpenStax learning exercise

Q. Explain why the following situation is not an example of discrimination: A teacher seats students wearing short sleeves on the left half of the room and students wearing long sleeves on the right half of the room.

A. In social loafing individual performance cannot be evaluated; however, in social facilitation individual performance can be evaluated. Social loafing and social facilitation both occur for easy or well-known tasks and when individuals are relaxed.

Personal Application Questions

OpenStax learning exercise

Q. Conduct a conformity study the next time you are in an elevator. After you enter the elevator, stand with your back toward the door. See if others conform to your behavior. Watch this video (https://www.youtube.com/watch?v=dDAbdMv14Is) for a candid camera demonstration of this phenomenon. Did your results turn out as expected?

OpenStax learning exercise

Q. Most students adamantly state that they would never have turned up the voltage in the Milligram experiment. Do you think you would have refused to shock the learner? Looking at your own past behavior, what evidence suggests that you would go along with the order to increase the voltage?

CHAPTER 15 / CAC.901

Belonging, prejudice, aggression, and helping

Lecture: Belonging, harm, and the courage to help

Children learn who belongs through ordinary interactions. They notice which languages are respected, which bodies are treated as normal, which families are represented, and whose distress receives attention. Prejudice is not only an individual attitude. It can be built into routines, expectations, and access to opportunity. A developmental counselor should understand both interpersonal behavior and the conditions that make harm more likely.

Social identities help people understand themselves and connect with others. Problems arise when categories become rigid hierarchies or when a child is treated as a stereotype rather than a person. Counselors should avoid color-evasive statements such as I do not see difference when difference shapes experience. Seeing difference respectfully is not the same as judging someone by it.

Prejudice and institutional practice

A school may intend inclusion while maintaining practices that exclude. A costly activity may prevent participation. A discipline rule may be applied unevenly. A counseling service may be available only during hours when caregivers cannot attend. These examples do not require an individual adult to express hostility. Structural patterns can produce unequal outcomes through ordinary operation.

Assessment should examine patterns rather than rely only on stated values. Who is referred, who receives support, who is excluded, and who is missing from the data? Disaggregate information carefully while protecting privacy. A small number of cases can identify a concern, but it should not be overinterpreted as a precise group effect. Combine numbers with the experiences of children and families.

Aggression is not one behavior

Aggression can be physical, verbal, relational, or digital. It can be reactive to perceived threat or used to gain a goal. These forms can overlap. A child who lashes out under extreme distress may need regulation and safety support; a child who organizes exclusion may need accountability and changes in group incentives. The response should follow the pattern rather than a generic label.

Moral disengagement describes ways people distance themselves from the harm they cause. Everyone does it, They deserved it, and It was only a joke can reduce responsibility. Counselors can help young people notice these explanations without humiliating them into defensiveness. Repair requires recognition of impact, not merely a forced apology.

Empathy is useful but not a complete solution. A child may understand another s feelings and still harm them. Group norms, status, fear, and perceived permission matter. Interventions should address the setting and consequences as well as individual perspective-taking.

Helping and bystander action

Prosocial behavior develops through relationships, modeling, opportunity, and a sense of responsibility. Children are more likely to help when they know what to do and believe that action is safe. Adults should teach practical options: interrupt gently, seek help, support the targeted child afterward, or document a concern according to local procedures. Do not require a child to confront a dangerous situation alone.

Figure 15.1 shows several routes from witnessing harm to action or inaction. It includes perceived safety, responsibility, skill, group norms, and adult response. This prevents a simple judgment that children who do not intervene lack kindness. A school can increase helping by making reporting accessible and responding reliably.

Application: a joke that keeps returning

A fictional class repeatedly mocks a student s accent. Some participants call it a joke; the targeted student has become quieter. A counselor and school team should not decide the issue by asking whether the group intended harm. Examine repetition, impact, power, and the student s ability to stop the interaction.

The response may include direct support for the targeted student, clear expectations and accountability for those involved, and work with the class on language and belonging. The school should review whether adults have normalized the behavior. Progress includes reduced harassment and increased participation, not merely completion of an assembly.

The targeted student should not be required to educate the class or publicly explain their identity. Confidentiality and choice matter. Adults carry responsibility for changing the environment. The counselor can support the student s recovery while the institution addresses the conditions that produced harm.

Doctoral seminar task

Audit a fictional school practice for inclusion. Identify its intended purpose, who benefits, who faces barriers, and what evidence would reveal unequal impact. Propose a change and a way to monitor unintended consequences. Include children s perspectives without making them responsible for fixing the institution.

Then design a bystander-support plan that offers safe options for different ages and levels of risk. Explain how adults will respond when a child reports harm. A plan that encourages disclosure without reliable adult action can increase danger and reduce trust.

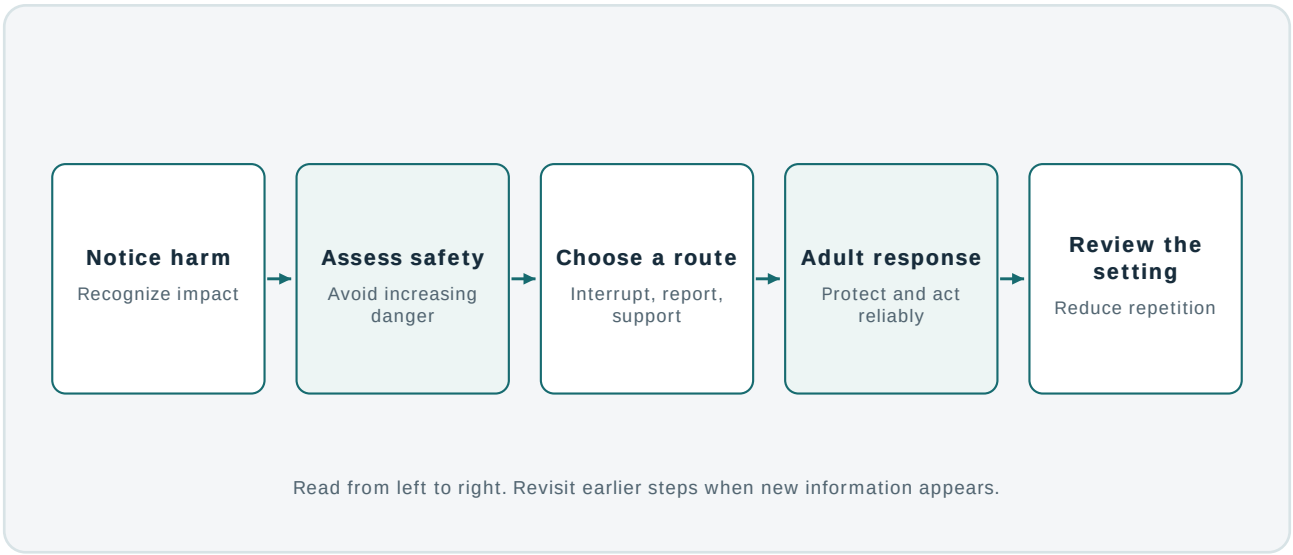


Figure 15.1. Helping requires a safe route to action. Witnessing harm does not automatically lead to helping. Safety, responsibility, practical skills, group norms, and reliable adult responses influence whether a child can act. Support afterward is also a form of help. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

Prejudice and Discrimination

OpenStax source module m82250 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82250/index.cnxml

Human conflict can result in crime, war, and mass murder, such as genocide. Prejudice and discrimination often are root causes of human conflict, which explains how strangers come to hate one another to the extreme of causing others harm. Prejudice and discrimination affect everyone. In this section we will examine the definitions of prejudice and discrimination, examples of these concepts, and causes of these biases.

Source illustration description (image not reproduced): Photograph A shows a sign written in German. Photograph B shows a man drinking at a drinking fountain. Photograph C shows two people holding signs with hate messages. Prejudice and discrimination occur across the globe. (a) A 1939 sign in German-occupied Poland warns No Entrance for Poles! (b) An African-American male drinks from a designated colored water fountain in Oklahoma in 1939 during the era of racial segregation as a practice of discrimination. (c) Members of the Westboro Baptist Church, widely identified as a hate group, engage in discrimination based on religion and sexual orientation. (credit b: modification of work by United States Farm Security Administration; credit c: modification of work by JCWilmore /Wikimedia Commons)

Understanding Prejudice and Discrimination

As we discussed in the opening story of Trayvon Martin, humans are very diverse and although we share many similarities, we also have many differences. The social groups we belong to help form our identities (Tajfel, 1974). These differences may be difficult for some people to reconcile, which may lead to prejudice toward people who are different. Prejudice is a negative attitude and feeling toward an individual based solely on one s membership in a particular social group (Allport, 1954; Brown, 2010). Prejudice is common against people who are members of an unfamiliar cultural group. Thus, certain types of education, contact, interactions, and building relationships with members of different cultural groups can reduce the tendency toward prejudice. In fact, simply imagining interacting with members of different cultural groups might affect prejudice. Indeed, when experimental participants were asked to imagine themselves positively interacting with someone from a different group, this led to an increased positive attitude toward the other group and an increase in positive traits associated with the other group. Furthermore, imagined social interaction can reduce anxiety associated with inter-group interactions (Crisp & Turner, 2009). What are some examples of social groups that you belong to that contribute to your identity? Social groups can include gender, race, ethnicity, nationality, social class, religion, sexual orientation, profession, and many more. And, as is true for social roles, you can simultaneously be a member of more than one social group. An example of prejudice is having a negative attitude toward people who are not born in the United States. Although people holding this prejudiced attitude do not know all people who were not born in the United States, they dislike them due to their status as foreigners.

Can you think of a prejudiced attitude you have held toward a group of people? How did your prejudice develop? Prejudice often begins in the form of a stereotype that is, a specific belief or assumption about individuals based solely on their membership in a group, regardless of their individual

characteristics. Stereotypes become overgeneralized and applied to all members of a group. For example, someone holding prejudiced attitudes toward older adults, may believe that older adults are slow and incompetent (Cuddy, Norton, & Fiske, 2005; Nelson, 2004). We cannot possibly know each individual person of advanced age to know that all older adults are slow and incompetent. Therefore, this negative belief is overgeneralized to all members of the group, even though many of the individual group members may in fact be spry and intelligent.

Another example of a well-known stereotype involves beliefs about racial differences among athletes. As Hodge, Burden, Robinson, and Bennett (2008) point out, Black athletes are often believed to be more athletic, yet less intelligent, than their White counterparts. These beliefs persist despite a number of high profile examples to the contrary. Sadly, such beliefs often influence how these athletes are treated by others and how they view themselves and their own capabilities. Whether or not you agree with a stereotype, stereotypes are generally well-known within a given culture (Devine, 1989).

Sometimes people will act on their prejudiced attitudes toward a group of people, and this behavior is known as discrimination. Discrimination is negative action toward an individual as a result of one s membership in a particular group (Allport, 1954; Dovidio & Gaertner, 2004). As a result of holding negative beliefs (stereotypes) and negative attitudes (prejudice) about a particular group, people often treat the target of prejudice poorly, such as excluding older adults from their circle of friends. In this way, discrimination is a form of oppression. An example of a psychologist experiencing gender discrimination is found in the life and studies of Mary Whiton Calkins. Calkins was given special permission to attend graduate seminars at Harvard (at that time in the late 1880s, Harvard did not accept women) and at one point was the sole student of the famous psychologist William James. She passed all the requirements needed for a PhD and was described by psychologist Hugo M nsterberg as one of the strongest professors of psychology in this country. However, Harvard refused to grant Calkins a PhD because she was a woman (Harvard University, 2019). the accompanying source-figure description summarizes the characteristics of stereotypes, prejudice, and discrimination. Have you ever been the target of discrimination? If so, how did this negative treatment make you feel?

Source comparison table

Connecting Stereotypes, Prejudice, and Discrimination			
Item	Function	Connection	Example
Stereotype	Cognitive; thoughts about people	Overgeneralized beliefs about people may lead to prejudice.	Yankees fans are arrogant and obnoxious.
Prejudice	Affective; feelings about people, both positive and negative	Feelings may influence treatment of others, leading to discrimination.	I hate Yankees fans; they make me angry.
Discrimination	Behavior; positive or negative treatment of others	Holding stereotypes and harboring prejudice may lead to excluding, avoiding, and biased treatment of group members.	I would never hire nor become friends with a person if I knew they were a Yankees fan.

So far, we ve discussed stereotypes, prejudice, and discrimination as negative thoughts, feelings, and behaviors because these are typically the most problematic. However, it is important to also point out that people can hold positive thoughts, feelings, and behaviors toward individuals based on group membership; for example, they would show preferential treatment for people who are like themselves that is, who share the same gender, race, or favorite sports team.

Prejudice and Discrimination

When we meet strangers we automatically process three pieces of information about them: their race, gender, and age (Ito & Urland, 2003). Why are these aspects of an unfamiliar person so important? Why don t we instead notice whether their eyes are friendly, whether they are smiling, their height, the type of clothes they are wearing? Although these secondary characteristics are important in forming a first impression of a stranger, the social categories of race, gender, and age provide a wealth of information about an individual. This information, however, often is based on stereotypes. We may have different expectations of strangers depending on their race, gender, and age. What stereotypes and prejudices do you hold about people who are from a race, gender, and age group different from your own?

Racism

Racism is prejudice and discrimination against an individual based solely on one s membership in a specific racial group (such as toward African Americans, Asian Americans, Latinos, Native Americans, European Americans). What are some stereotypes of various racial or ethnic groups? Research suggests cultural stereotypes for Asian Americans include cold, sly, and intelligent; for Latinos, cold and unintelligent; for European Americans, cold and intelligent; and for African Americans, aggressive, athletic, and more likely to be law breakers (Devine & Elliot, 1995; Fiske, Cuddy, Glick, & Xu, 2002; Sommers & Ellsworth, 2000; Dixon & Linz, 2000).

Racism exists for many racial and ethnic groups. For example, Black people are significantly more likely to have their vehicles searched during traffic stops than White people, particularly when Black people are driving in predominately White neighborhoods, a phenomenon often termed DWB or driving while Black (Rojek, Rosenfeld, & Decker, 2012).

Mexican Americans and other Latino groups also are targets of racism from the police and other members of the community. For example, when purchasing items with a personal check, Latino shoppers are more likely than White shoppers to be asked to show formal identification (Dovidio et al., 2010).

In one case of alleged harassment by the police, several East Haven, Connecticut, police officers were arrested on federal charges due to reportedly continued harassment and brutalization of Latinos. When the accusations came out, the mayor of East Haven was asked, What are you doing for the Latino community today? The Mayor responded, I might have tacos when I go home, I m not quite sure yet (East Haven Mayor, 2012). This statement undermines the important issue of racial profiling and police harassment of Latinos, while belittling Latino culture by emphasizing an interest in a food product stereotypically associated with Latinos.

Racism is prevalent toward many other groups in the United States including Native Americans, Arab Americans, Jewish Americans, and Asian Americans. Have you experienced or witnessed racism toward any of these racial or ethnic groups? Are you aware of racism in your community?

One reason modern forms of racism, and prejudice in general, are hard to detect is related to the dual attitudes model (Wilson, Lindsey, & Schooler, 2000). Humans have two forms of attitudes: explicit attitudes, which are conscious and controllable, and implicit attitudes, which are unconscious and uncontrollable (Devine, 1989; Olson & Fazio, 2003). Because holding egalitarian views is socially desirable (Plant & Devine, 1998), most people do not show extreme racial bias or other prejudices on measures of their explicit attitudes. However, measures of implicit attitudes often show evidence of mild to strong racial bias or other prejudices (Greenwald, McGee, & Schwartz, 1998; Olson & Fazio, 2003).

Sexism

Sexism is prejudice and discrimination toward individuals based on their sex. Typically, sexism takes the form of men holding biases against women, but anyone can show sexism toward their own or the other sex. Like racism, sexism may be subtle and difficult to detect. Common forms of sexism in modern society include gender role expectations, such as expecting women to be the caretakers of the household. Sexism also includes people s expectations for how members of a gender group should behave. For example, women are expected to be friendly, passive, and nurturing, and when women behave in an unfriendly, assertive, or neglectful manner they often are disliked for violating their gender role (Rudman, 1998). Research by Laurie Rudman (1998) finds that when female job applicants self-promote, they are likely to be viewed as competent, but they may be disliked and are less likely to be hired because they violated gender expectations for modesty. In this way, women are sometimes blamed for wage and employment inequality, but they would be viewed negatively for trying to overcome them, a phenomenon sometimes referred to as the double bind. Sexism can exist on a societal level such as in hiring, employment opportunities, and education. Women are less likely to be hired or promoted in male-dominated professions such as engineering, aviation, and construction (the accompanying source-figure description) (Blau, Ferber, & Winkler, 2010; Ceci & Williams, 2011). Have you ever experienced or witnessed sexism? Think about your family members jobs or careers. Why do you think there are differences in the jobs women and men have, such as more women nurses but more male surgeons (Betz, 2008)?

Source illustration description (image not reproduced): A photograph shows a female soldier and male soldiers standing beside a large gun. Women now have many jobs previously closed to them, though they still face challenges in men-dominated occupations. (credit: "The National Guard"/Flickr)

Ageism

People often form judgments and hold expectations about people based on their age. These judgments and expectations can lead to ageism, or prejudice and discrimination toward individuals based solely on their age. Think of expectations you hold for older adults. How could someone s expectations influence the feelings they hold toward individuals from older age groups? Ageism is widespread in U.S. culture (Nosek, 2005), and a common ageist attitude toward older adults is that they are incompetent, physically weak, and slow (Greenberg, Schimel, & Martens, 2002) and some people consider older adults less attractive. Chang, Kannoth, Levy, Wang, Lee, and Levy (2020) reported on relationships between ageism and health outcomes over a 40-year-plus period from countries around the world. Across 11 health domains, people over 50 were likely to experience ageism most often in the form of being denied access to health services and work opportunities. Some cultures, however, including some Asian, Latino, and African American cultures, both outside and within the United States afford older adults respect and honor.

Typically, ageism occurs against older adults, but ageism also can occur toward younger adults. What expectations do you hold toward younger people? Does society expect younger adults to be immature and irresponsible? Are younger generations seen as having it too easy or having weaker characters than older generations? Raymer, Reed, Spiegel, and Purvanova (2017) examined ageism against younger workers. They found that older workers endorsed negative stereotypes of younger workers, believing that they had more work deficit characteristics (including perceptions of incompetence). How might these forms of ageism affect a younger and older adult who are applying for a sales clerk position?

Homophobia and Transphobia

Another form of prejudice is homophobia: an umbrella term referring to prejudice and discrimination of individuals based solely on their sexual orientation, which is often applied to bisexual, lesbian, gay, and other non-heterosexual people. Transphobia is the hatred or fear of those who are perceived to break or blur stereotypical gender roles, often expressed as stereotyping, discrimination, harassment and/or violence. Like ageism, homophobia is a widespread prejudice in U.S. society that is tolerated by many people (Herek & McLemore, 2013; Nosek, 2005). Negative feelings often result in discrimination, such as the exclusion of lesbian, gay, bisexual, transgender, and queer (LGBTQ+) people from social groups and the avoidance of LGBTQ+ neighbors and co-workers. This discrimination also extends to employers deliberately declining to hire qualified LGBTQ+ job applicants, a practice officially outlawed by a 2020 Surpreme Court decision but which remains a significant issue. Have you experienced or witnessed homophobia? If so, what stereotypes, prejudiced attitudes, and discrimination were evident?

Dig Deeper

Some people are quite passionate in their hatred for nonheterosexuals in our society. In some cases, people have been tortured and/or murdered simply because they were not straight. This passionate response has led some researchers to question what motives might exist for homophobic people. Adams, Wright, & Lohr (1996) conducted a study investigating this issue and their results were quite an eye-opener.

In this experiment, male college students were given a scale that assessed how homophobic they were; those with extreme scores were recruited to participate in the experiment. In the end, 64 men agreed to participate and were split into 2 groups: homophobic men and nonhomophobic men. Both groups of men were fitted with a penile plethysmograph, an instrument that measures changes in blood flow to the penis and serves as an objective measurement of sexual arousal.

All men were shown segments of sexually explicit videos. One of these videos involved a sexual interaction between a man and a woman (straight clip). One video displayed two females engaged in a sexual interaction (lesbian clip), and the final video displayed two men engaged in a sexual interaction (gay clip). Changes in penile tumescence (a measure of physiological genital arousal) were recorded during all three clips, and a subjective measurement of sexual arousal was also obtained. While both groups of men became sexually aroused to the straight and lesbian video clips, only those men who were identified as homophobic showed sexual arousal to the gay male video clip. While all men reported that their erections indicated arousal for the straight and lesbian clips, the homophobic men indicated that they were not sexually aroused (despite their erections) to the gay clips. Adams et al. (1996) suggest that these findings may indicate that homophobia is related to gay arousal that the homophobic individuals either deny or are unaware.

Why Do Prejudice and Discrimination Exist?

Prejudice and discrimination persist in society due to social learning and conformity to social norms. Children learn prejudiced attitudes and beliefs from society: their parents, teachers, friends, the media, and other sources of socialization, such as social media (O Keeffe & Clarke-Pearson, 2011). If certain types of prejudice and discrimination are acceptable in a society, there may be normative pressures to conform and share those prejudiced beliefs, attitudes, and behaviors. For example, public and private schools are still somewhat segregated by social class. Historically, only children from wealthy families could afford to attend private schools, whereas children from middle- and low-income families typically attended public schools. If a child from a low-income family received a merit scholarship to attend a private school, how might the child be treated by classmates? Can you recall a time when you held prejudiced attitudes or beliefs or acted in a discriminatory manner because your group of friends expected you to?

Stereotypes and Self-Fulfilling Prophecy

When we hold a stereotype about a person, we have expectations that they will fulfill that stereotype. A self-fulfilling prophecy is an expectation held by a person that alters their behavior in a way that tends to make it true. When we hold stereotypes about a person, we tend to treat the person according to our expectations. This treatment can influence the person to act according to our stereotypic expectations, thus confirming our stereotypic beliefs. Research by Rosenthal and Jacobson (1968) found that disadvantaged students whose teachers expected them to perform well had higher grades than disadvantaged students whose teachers expected them to do poorly.

Consider this example of cause and effect in a self-fulfilling prophecy: If an employer expects an openly gay job applicant to be incompetent, the potential employer might treat the applicant negatively during the interview by engaging in less conversation, making little eye contact, and generally behaving coldly toward the applicant (Hebl, Foster, Mannix, & Dovidio, 2002). In turn, the job applicant will perceive that the potential employer dislikes him, and he will respond by giving shorter responses to interview questions, making less eye contact, and generally disengaging from the interview. After the interview, the employer will reflect on the applicant s behavior, which seemed cold and distant, and the employer will conclude, based on the applicant s poor performance during the interview, that the applicant was in fact incompetent. Thus, the employer s stereotype gay people are incompetent and do not make good employees is reinforced. Do you think this job applicant is likely to be hired? Treating individuals according to stereotypic beliefs can lead to prejudice and discrimination.

Another dynamic that can reinforce stereotypes is confirmation bias. When interacting with the target of our prejudice, we tend to pay attention to information that is consistent with our stereotypic expectations and ignore information that is inconsistent with our expectations. In this process, known as confirmation bias, we seek out information that supports our stereotypes and ignore information that is inconsistent with our stereotypes (Wason & Johnson-Laird, 1972). In the job interview example, the employer may not have noticed that the job applicant was friendly and engaging, and that he provided competent responses to the interview questions in the beginning of the interview. Instead, the employer focused on the job applicant s performance in the later part of the interview, after the applicant changed his demeanor and behavior to match the interviewer s negative treatment. Have you ever fallen prey to the self-fulfilling prophecy or confirmation bias, either as the source or target of such bias? How might we stop the cycle of the self-fulfilling prophecy?

In-Groups and Out-Groups

As discussed previously in this section, we all belong to a gender, race, age, and social economic group. These groups provide a powerful source of our identity and self-esteem (Tajfel & Turner, 1979). These groups serve as our in-groups. An in-group is a group that we identify with or see ourselves as belonging to. A group that we don t belong to, or an out-group, is a group that we view as fundamentally different from us. For example, if you are a woman, your gender in-group includes all women, and your gender out-group includes all men (the accompanying source-figure description). People often view gender groups as being fundamentally different from each other in personality traits, characteristics, social roles, and interests. Because we often feel a strong sense of belonging and emotional connection to our in-groups, we develop in-group bias: a preference for our own group over other groups. This in-group bias can result in prejudice and discrimination because the out-group is perceived as different and is less preferred than our in-group.

Source illustration description (image not reproduced): A photograph shows children climbing on playground equipment. These children are very young, but they are already aware of their gender in-group and out-group. (credit: modification of work by "Reiner Kraft"/Flickr)

Despite the group dynamics that seem only to push groups toward conflict, there are forces that promote reconciliation between groups: the expression of empathy, of acknowledgment of past suffering on both sides, and the halt of destructive behaviors.

One function of prejudice is to help us feel good about ourselves and maintain a positive self-concept. This need to feel good about ourselves extends to our in-groups: We want to feel good and protect our in-groups. We seek to resolve threats individually and at the in-group level. This often happens by blaming an out-group for the problem. Scapegoating is the act of blaming an out-group when the in-group experiences frustration or is blocked from obtaining a goal (Allport, 1954).

Summary

As diverse individuals, humans can experience conflict when interacting with people who are different from each other. Prejudice, or negative feelings and evaluations, is common when people are from a different social group (i.e., out-group). Negative attitudes toward out-groups can lead to discrimination. Prejudice and discrimination against others can be based on gender, race, ethnicity, social class, sexual orientation, or a variety of other social identities. In-group s who feel threatened may blame the out-groups for their plight, thus using the out-group as a scapegoat for their frustration.

Review Questions

OpenStax learning exercise

Q. Prejudice is to _____ as discrimination is to _____.

A. A

OpenStax learning exercise

Q. Which of the following is not a type of prejudice?

A. D

OpenStax learning exercise

Q. _____ occurs when the out-group is blamed for the in-group s frustration.

A. C

OpenStax learning exercise

Q. When we seek out information that supports our stereotypes we are engaged in _____.

A. B

Critical Thinking Questions

OpenStax learning exercise

Q. Some people seem more willing to openly display prejudice regarding sexual orientation than prejudice regarding race and gender. Descbribe why this might be.

A. In the United States, many people believe that sexual orientation is a choice, and there is some debate in the research literature as to the extent sexual orientation is biological or influenced by social factors. Because race and gender are not chosen, many Americans believe it is unfair to negatively judge women or racial minority groups for a characteristic that is determined by genetics. In addition, many people in the United States practice religions that believe homosexuality is wrong.

OpenStax learning exercise

Q. When people blame a scapegoat, how do you think they choose evidence to support the blame?

A. One way in which they might do this is to selectively attend to information that would bolster their argument. Furthermore, they may actively seek out information to confirm their assertions.

Personal Application Questions

OpenStax learning exercise

Q. Give an example when you felt that someone was prejudiced against you. What do you think caused this attitude? Did this person display any discrimination behaviors and, if so, how?

OpenStax learning exercise

Q. Give an example when you felt prejudiced against someone else. How did you discriminate against them? Why do you think you did this?

Aggression

OpenStax source module m82251 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82251/index.cnxml

Throughout this chapter we have discussed how people interact and influence one another s thoughts, feelings, and behaviors in both positive and negative ways. People can work together to achieve great things, such as helping each other in difficult times: consider how educators and healthcare workers went to great lengths to help others during the COVID-19 pandemic. People also can do great harm to one another, such as conforming to group norms that are immoral and obeying authority to the point of murder: consider the mass conformity of Nazis during WWII. In this section we will discuss a negative side of human behavior aggression.

A number of researchers have explored ways to reduce prejudice. One of the earliest was a study by Sherif et al. (1961) known as the Robbers Cave experiment. They found that when two opposing groups at a camp worked together toward a common goal, prejudicial attitudes between the groups decreased (Gaertner, Dovidio, Banker, Houlette, Johnson, & McGlynn, 2000). Focusing on superordinate goals was the key to attitude change in the research. Another study examined the jigsaw classroom, a technique designed by Aronson and Bridgeman in an effort to increase success in desegregated classrooms. In this technique, students work on an assignment in groups inclusive of various races and abilities. They are assigned tasks within their group, then collaborate with peers from other groups who were assigned the same task, and then report back to their original group. Walker and Crogan (1998) noted that the jigsaw classroom reduced potential for prejudice in Australia, as diverse students worked together on projects needing all of the pieces to succeed. This research suggests that anything that can allow individuals to work together toward common goals can decrease prejudicial attitudes. Obviously, the application of such strategies in real-world settings would enhance opportunities for conflict resolution.

Aggression

Humans engage in aggression when they seek to cause harm or pain to another person. Aggression takes two forms depending on one s motives: hostile or instrumental. Hostile aggression is motivated by feelings of anger with intent to cause pain; a fight in a bar with a stranger is an example of hostile aggression. In contrast, instrumental aggression is motivated by achieving a goal and does not necessarily involve intent to cause pain (Berkowitz, 1993); a contract killer who murders for hire displays instrumental aggression.

There are many different theories as to why aggression exists. Some researchers argue that aggression serves an evolutionary function (Buss, 2004). Men are more likely than women to show aggression

(Wilson & Daly, 1985). From the perspective of evolutionary psychology, human male aggression, like that in nonhuman primates, likely serves to display dominance over other males, both to protect a mate and to perpetuate the male s genes (the accompanying source-figure description). Sexual jealousy is part of male aggression; males endeavor to make sure their mates are not copulating with other males, thus ensuring their own paternity of the female s offspring. Although aggression provides an obvious evolutionary advantage for men, women also engage in aggression. Women typically display more indirect forms of aggression, with their aggression serving as a means to an end (Dodge & Schwartz, 1997). For example, women may express their aggression covertly by communication that impairs the social standing of another person. Another theory that explains one of the functions of human aggression is frustration aggression theory (Dollard, Doob, Miller, Mowrer, & Sears, 1939). This theory states that when humans are prevented from achieving an important goal, they become frustrated and aggressive.

Source illustration description (image not reproduced): A photograph shows two monkeys face to face. Human males and nonhuman male primates endeavor to gain and display dominance over other males, as demonstrated in the behavior of these monkeys. (credit: Arcadiu /Flickr)

Bullying

Another form of aggression is bullying. As you learn in your study of child development, socializing and playing with other children is beneficial for children s psychological development. However, as you may have experienced as a child, not all play behavior has positive outcomes. Some children are aggressive and want to play roughly. Other children are selfish and do not want to share toys. One form of negative social interactions among children that has become a national concern is bullying. Bullying is repeated negative treatment of another person, often an adolescent, over time (Olweus, 1993). A one-time incident in which one child hits another child on the playground would not be considered bullying: Bullying is repeated behavior. The negative treatment typical in bullying is the attempt to inflict harm, injury, or humiliation, and bullying can include physical or verbal attacks. However, bullying doesn t have to be physical or verbal, it can be psychological. Research finds gender differences in how girls and boys bully others (American Psychological Association, 2010; Olweus, 1993). Boys tend to engage in direct, physical aggression such as physically harming others. Girls tend to engage in indirect, social forms of aggression such as spreading rumors, ignoring, or socially isolating others. Based on what you have learned about child development and social roles, why do you think boys and girls display different types of bullying behavior?

Bullying involves three parties: the bully, the victim, and witnesses or bystanders. The act of bullying involves an imbalance of power with the bully holding more power physically, emotionally, and/or socially over the victim. The experience of bullying can be positive for the bully, who may enjoy a boost to self-esteem. However, there are several negative consequences of bullying for the victim, and also for the bystanders. How do you think bullying negatively impacts adolescents? Being the victim of bullying is associated with decreased mental health, including experiencing anxiety and depression (APA, 2010). Victims of bullying may underperform in schoolwork (Bowen, 2011). Bullying also can result in the victim dying by suicide (APA, 2010). How might bullying negatively affect witnesses?

Although there is not one single personality profile for who becomes a bully and who becomes a victim of bullying (APA, 2010), researchers have identified some patterns in children who are at a greater risk of being bullied (Olweus, 1993): Children who are emotionally reactive are at a greater risk for being bullied. Bullies may be attracted to children who get upset easily because the bully can quickly get an emotional reaction from them. Children who are different from others are likely to be targeted for bullying. Children who are overweight, cognitively impaired, or racially or ethnically different from their peer group may be at higher risk. Gay, lesbian, bisexual, and transgender teens are at very high risk of being bullied and hurt due to their sexual orientation.

Cyberbullying

With widely available mobile technology and social networking media, a new form of bullying has emerged: cyberbullying. Cyberbullying, like bullying, is repeated behavior that is intended to cause psychological or emotional harm to another person. What is unique about cyberbullying is that it is typically covert, concealed, done in private, and the bully can remain anonymous. This anonymity gives the bully power, and the victim may feel helpless, unable to escape the harassment, and unable to retaliate (Spears, Slee, Owens, & Johnson, 2009). About one in three middle and high school students report that they have experienced cyberbullying (Patchin, 2016; Patchin, 2019), with some studies indicating that nearly three in five teens have experienced some type of online abusive behavior such as non-repeated name-calling or being sent unsolicited links or images (Anderson, 2018).

Cyberbullying can take many forms, including harassing a victim by spreading rumors, creating a website defaming the victim, threatening the victim, or teasing the victim (Spears et al., 2009). Overall, LGBTQ youth are targeted at a higher rate than heterosexual and cisgender youth, and members of minority populations overall are more likely to be cyberbullying victims (Hinjuda & Patchin, 2020). In terms of gender, experiences vary in terms of both the prevelance and types of cyberbullying experienced and perpetrated. In cyberbullying, it is more common for girls to be the bullies and victims. Interestingly, girls who become cyberbullies often have been the victims of cyberbullying at one time (Vandebosch & Van Cleemput, 2009). Girls were more likely to say someone spread rumors about them online while boys were more likely to say that someone threatened to hurt them online (Patchin, 2019). The effects of cyberbullying are just as harmful as traditional bullying and include the victim feeling frustration, anger, sadness, helplessness, powerlessness, and fear. Victims will also experience lower self-esteem (Hoff & Mitchell, 2009; Spears et al., 2009). Furthermore, recent research suggests that both cyberbullying victims and perpetrators are more likely to experience suicidal ideation, and they are more likely to attempt suicide than individuals who have no experience with cyberbullying (John, 2019). Cyberbullying is a form of aggression and, depending on the messages and methods, can be considered harassment, stalking, or assault all subject to prosecution. Finally, while much of the concern and research regarding cyberbullying centers on adolescents, adults (particularly college students) are frequent victims and perpetrators. What features of technology make cyberbullying easier and perhaps more accessible to young adults? What can parents, teachers, and social networking websites do to prevent cyberbullying?

Source illustration description (image not reproduced): A photograph shows a young person looking at a handheld electronic device. Cyberbullying can create a feeling of powerlessness and inescapability because victimization is not constrained to physical locations. Many victims report being harassed across multiple platforms or formats at the same time. (credit: Steven Depolo)

The Bystander Effect

The discussion of bullying highlights the problem of witnesses not intervening to help a victim. Researchers Latan and Darley (1968) described a phenomenon called the bystander effect. The bystander effect is a phenomenon in which a witness or bystander does not volunteer to help a victim or person in distress. Instead, they just watch what is happening. Social psychologists hold that we make these decisions based on the social situation, not our own personality variables. The impetus behind the bystander effect was the murder of a young woman named Kitty Genovese in 1964. The story of her tragic death took on a life of its own when it was reported that none of her neighbors helped her or called the police when she was being attacked. However, Kassin (2017) noted that her killer was apprehended due to neighbors who called the police when they saw him committing a burglary days later. Not only did bystanders indeed intervene in her murder (one man who shouted at the killer, a woman who said she called the police, and a friend who comforted her in her last moments), but other bystanders intervened in the capture of the murderer. Social psychologists claim that diffusion of responsibility is the likely explanation. Diffusion of responsibility is the tendency for no one in a group to help because the responsibility to help is spread throughout the group (Bandura, 1999). Because there were many witnesses to the attack on Genovese, as evidenced by the number of lit apartment windows in the building, individuals assumed someone else must have already called the police. The responsibility to call the police was diffused across the number of witnesses to the crime. Have you ever passed an accident on the freeway and assumed that a victim or certainly another motorist has already reported the accident? In general, the greater the number of bystanders, the less likely any one person will help.

Summary

Aggression is seeking to cause another person harm or pain. Hostile aggression is motivated by feelings of anger with intent to cause pain, and instrumental aggression is motivated by achieving a goal and does not necessarily involve intent to cause pain Bullying is an international public health concern that largely affects the adolescent population. Bullying is repeated behaviors that are intended to inflict harm on the victim and can take the form of physical, psychological, emotional, or social abuse. Bullying has negative mental health consequences for youth including suicide. Cyberbullying is a newer form of bullying that takes place in an online environment where bullies can remain anonymous and victims are helpless to address the harassment. Despite the social norm of helping others in need, when there are many bystanders witnessing an emergency, diffusion of responsibility will lead to a lower likelihood of any one person helping.

Review Questions

OpenStax learning exercise

Q. Typically, bullying from boys is to _____ as bullying from girls is to _____.

A. B

OpenStax learning exercise

Q. Which of the following adolescents is least likely to be targeted for bullying?

A. D

OpenStax learning exercise

Q. The bystander effect likely occurs due to _____.

A. C

Critical Thinking Questions

OpenStax learning exercise

Q. Compare and contrast hostile and instrumental aggression.

A. Hostile aggression is intentional with the purpose to inflict pain. Hostile aggression is often motivated by anger. In contrast, instrumental aggression is not motivated by anger or the intention to cause pain. Instrumental aggression serves as a means to reach a goal. In a sense it is a more practical or functional form of aggression, whereas hostile aggression is more emotion-driven and less functional and rational.

OpenStax learning exercise

Q. What evidence discussed in the previous section suggests that cyberbullying is difficult to detect and prevent?

A. Cyberbullying is difficult to prevent because there are so many forms of media that adolescents use and are exposed to. The Internet is virtually everywhere: computers, phones, tablets, TVs, gaming systems, and so on. Parents likely do not monitor all of their children s use of the Internet, thus their children could be exposed to cyberbullying without their knowledge. Cyberbullying is difficult to detect because it can be done anonymously. Cyberbullies can use pseudonyms and can attack victims in untraceable ways, such as hacking into Facebook accounts or making Twitter posts on their behalf.

Personal Application Questions

OpenStax learning exercise

Q. Have you ever experienced or witnessed bullying or cyberbullying? How did it make you feel? What did you do about it? After reading this section would you have done anything differently?

OpenStax learning exercise

Q. The next time you see someone needing help, observe your surroundings. Look to see if the bystander effect is in action and take measures to make sure the person gets help. If you aren t able to help, notify an adult or authority figure that can.

Prosocial Behavior

OpenStax source module m82252 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82252/index.cnxml

You ve learned about many of the negative behaviors of social psychology, but the field also studies many positive social interactions and behaviors. What makes people like each other? With whom are we friends? Whom do we date? Researchers have documented several features of the situation that influence whether we form relationships with others. There are also universal traits that humans find attractive in others. In this section we discuss conditions that make forming relationships more likely, what we look for in friendships and romantic relationships, the different types of love, and a theory explaining how our relationships are formed, maintained, and terminated.

Prosocial Behavior and Altruism

Do you voluntarily help others? Voluntary behavior with the intent to help other people is called prosocial behavior. Why do people help other people? Is personal benefit such as feeling good about oneself the only reason people help one another? Research suggests there are many other reasons. Altruism is people s desire to help others even if the costs outweigh the benefits of helping. In fact, people acting in altruistic ways may disregard the personal costs associated with helping (the accompanying source-figure description). For example, news accounts of the 9/11 terrorist attacks on the World Trade Center in New York reported an employee in the first tower helped his co-workers make it to the exit stairwell. After helping a co-worker to safety he went back in the burning building to help additional co-workers. In this case the costs of helping were great, and the hero lost his life in the destruction (Stewart, 2002).

Source illustration description (image not reproduced): A photograph shows two people covered in dust; one appears to be helping the other. The events of 9/11 unleashed an enormous show of altruism and heroism on the parts of first responders and many ordinary people. (credit: Don Halasy)

Some researchers suggest that altruism operates on empathy. Empathy is the capacity to understand another person s perspective, to feel what they feel. An empathetic person makes an emotional connection with others and feels compelled to help (Batson, 1991). Other researchers argue that altruism is a form of selfless helping that is not motivated by benefits or feeling good about oneself. Certainly, after helping, people feel good about themselves, but some researchers argue that this is a consequence of altruism, not a cause. Other researchers argue that helping is always self-serving because our egos are involved, and we receive benefits from helping (Cialdini, Brown, Lewis, Luce, & Neuberg 1997). It is challenging to determine experimentally the true motivation for helping, whether it is largely self-serving (egoism) or selfless (altruism). Thus, a debate on whether pure altruism exists continues.

Forming Relationships

What do you think is the single most influential factor in determining with whom you become friends and whom you form romantic relationships? You might be surprised to learn that the answer is simple: the people with whom you have the most contact. This most important factor is proximity. You are more likely to be friends with people you have regular contact with. For example, there are decades of research that shows that you are more likely to become friends with people who live in your dorm, your apartment building, or your immediate neighborhood than with people who live farther away (Festinger, Schachler, & Back, 1950). It is simply easier to form relationships with people you see often because you have the opportunity to get to know them.

Similarity is another factor that influences who we form relationships with. We are more likely to become friends or lovers with someone who is similar to us in background, attitudes, and lifestyle. In fact, there is no evidence that opposites attract. Rather, we are attracted to people who are most like us (the accompanying source-figure description) (McPherson, Smith-Lovin, & Cook, 2001). Why do you think we are attracted to people who are similar to us? Sharing things in common will certainly make it easy to get along with others and form connections. When you and another person share similar music taste, hobbies, food preferences, and so on, deciding what to do with your time together might be easy. Homophily is the tendency for people to form social networks, including friendships, marriage, business relationships, and many other types of relationships, with others who are similar (McPherson et al., 2001).

Source illustration description (image not reproduced): A photograph shows a bride and groom in a wedding ceremony. People tend to be attracted to similar people. Many couples share a cultural background. This can be quite obvious in a ceremony such as a wedding, and more subtle (but no less significant) in the day-to-day workings of a relationship. (credit: modification of work by Shiraz Chanawala)

But, homophily limits our exposure to diversity (McPherson et al., 2001). By forming relationships only with people who are similar to us, we will have homogenous groups and will not be exposed to different points of view. In other words, because we are likely to spend time with those who are most like ourselves, we will have limited exposure to those who are different than ourselves, including people of different races, ethnicities, social-economic status, and life situations.

Once we form relationships with people, we desire reciprocity. Reciprocity is the give and take in relationships. We contribute to relationships, but we expect to receive benefits as well. That is, we want our relationships to be a two way street. We are more likely to like and engage with people who like us back. Self-disclosure is part of the two way street. Self-disclosure is the sharing of personal information (Laurenceau, Barrett, & Pietromonaco, 1998). We form more intimate connections with people with whom we disclose important information about ourselves. Indeed, self-disclosure is a characteristic of healthy intimate relationships, as long as the information disclosed is consistent with our own views (Cozby, 1973).

Attraction

We have discussed how proximity and similarity lead to the formation of relationships, and that reciprocity and self-disclosure are important for relationship maintenance. But, what features of a person do we find attractive? We don t form relationships with everyone that lives or works near us, so how is it that we decide which specific individuals we will select as friends and lovers?

Researchers have documented several characteristics that humans find attractive. First we look for friends and lovers who are physically attractive. People differ in what they consider attractive, and attractiveness is culturally influenced. Research, however, suggests that some universally attractive features in women include large eyes, high cheekbones, a narrow jaw line, a slender build (Buss, 1989), and a lower waist-to-hip ratio (Singh, 1993). For men, attractive traits include being tall, having broad shoulders, and a narrow waist (Buss, 1989). Both men and women with high levels of facial and body symmetry are generally considered more attractive than asymmetric individuals (Fink, Neave, Manning, & Grammer, 2006; Penton-Voak et al., 2001; Rikowski & Grammer, 1999). Social traits that people find attractive in potential female mates include warmth, affection, and social skills; in males, the attractive traits include achievement, leadership qualities, and job skills (Regan & Berscheid, 1997). Although humans want mates who are physically attractive, this does not mean that we look for the most attractive person possible. In fact, this observation has led some to propose what is known as the matching hypothesis which asserts that people tend to pick someone they view as their equal in physical attractiveness and social desirability (Taylor, Fiore, Mendelsohn, & Cheshire, 2011). For example, you and most people you know likely would say that a very attractive movie star is out of your league. So, even if you had proximity to that person, you likely would not ask them out on a date because you believe you likely would be rejected. People weigh a potential partner s attractiveness against the likelihood of success with that person. If you think you are particularly unattractive (even if you are not), you likely will seek partners that are fairly unattractive (that is, unattractive in physical appearance or in behavior).

Sternberg s Triangular Theory of Love

We typically love the people with whom we form relationships, but the type of love we have for our family, friends, and lovers differs. Robert Sternberg (1986) proposed that there are three components of love: intimacy, passion, and commitment. These three components form a triangle that defines multiple types of love: this is known as Sternberg s triangular theory of love (the accompanying source-figure description). Intimacy is the sharing of details and intimate thoughts and emotions. Passion is the physical attraction the flame in the fire. Commitment is standing by the person the in sickness and health part of the relationship.

Source illustration description (image not reproduced): Diagram shows a triangle. The interior of the triangle is labeled, Consummate love; intimacy + passion + commitment. The peak of the triangle is labeled, Liking; intimacy. The left side of the triangle is labeled, Romantic love; passion + intimacy. The right side of the triangle is labeled,

Companionate love; intimacy + commitment. The bottom left corner of the triangle is labeled, Infatuation; passion. The bottom side of the triangle is labeled, Fatuous love; passion + commitment. The bottom right corner of the triangle is labeled, Empty love; commitment. According to Sternberg s triangular theory of love, seven types of love can be described from combinations of three components: intimacy, passion, and commitment. (credit: modification of work by Lnesa /Wikimedia Commons)

Sternberg (1986) states that a healthy relationship will have all three components of love intimacy, passion, and commitment which is described as consummate love (the accompanying source-figure description). However, different aspects of love might be more prevalent at different life stages. Other forms of love include liking, which is defined as having intimacy but no passion or commitment. Infatuation is the presence of passion without intimacy or commitment. Empty love is having commitment without intimacy or passion. Companionate love, which is characteristic of close friendships and family relationships, consists of intimacy and commitment but no passion. Romantic love is defined by having passion and intimacy, but no commitment. Finally, fatuous love is defined by having passion and commitment, but no intimacy, such as a long term sexual love affair. Can you describe other examples of relationships that fit these different types of love?

Source illustration description (image not reproduced): Two boys touch foreheads and noses. According to Sternberg, consummate love describes a healthy relationship containing intimacy, passion, and commitment. (credit: Carloxito/Wikimedia)

Social Exchange Theory

We have discussed why we form relationships, what attracts us to others, and different types of love. But what determines whether we are satisfied with and stay in a relationship? One theory that provides an explanation is social exchange theory. According to social exchange theory, we act as na ve economists in keeping a tally of the ratio of costs and benefits of forming and maintaining a relationship with others (the accompanying source-figure description) (Rusbult & Van Lange, 2003).

Source illustration description (image not reproduced): An illustration shows a balance scale, with one side labeled positives or benefits appearing heavier than the other side, which is labeled negatives or costs. Acting like na ve economists, people may keep track of the costs and benefits of maintaining a relationship. Typically, only those relationships in which the benefits outweigh the costs will be maintained.

People are motivated to maximize the benefits of social exchanges, or relationships, and minimize the costs. People prefer to have more benefits than costs, or to have nearly equal costs and benefits, but most people are dissatisfied if their social exchanges create more costs than benefits. Let s discuss an example. If you have ever decided to commit to a romantic relationship, you probably considered the advantages and disadvantages of your decision. What are the benefits of being in a committed romantic relationship? You may have considered having companionship, intimacy, and passion, but also being comfortable with a person you know well. What are the costs of being in a committed romantic relationship? You may think that over time boredom from being with only one person may set in; moreover, it may be expensive to share activities such as attending movies and going to dinner. However, the benefits of dating your romantic partner presumably outweigh the costs, or you wouldn t continue the relationship.

Summary

Altruism is a pure form of helping others out of empathy, which can be contrasted with egoistic motivations for helping. Forming relationships with others is a necessity for social beings. We typically form relationships with people who are close to us in proximity and people with whom we share similarities. We expect reciprocity and self-disclosure in our relationships. We also want to form relationships with people who are physically attractive, though standards for attractiveness vary by culture and gender. There are many types of love that are determined by various combinations of intimacy, passion, and commitment; consummate love, which is the ideal form of love, contains all three components. When determining satisfaction and whether to maintain a relationship, individuals often use a social exchange approach and weigh the costs and benefits of forming and maintaining a relationship.

Review Questions

OpenStax learning exercise

Q. Altruism is a form of prosocial behavior that is motivated by _____.

A. B

OpenStax learning exercise

Q. After moving to a new apartment building, research suggests that Sam will be most likely to become friends with _____.

A. A

OpenStax learning exercise

Q. What trait do both men and women tend to look for in a romantic partner?

A. D

OpenStax learning exercise

Q. According to the triangular theory of love, what type of love is defined by passion and intimacy but no commitment?

A. C

OpenStax learning exercise

Q. According to social exchange theory, humans want to maximize the _____ and minimize the _____ in relationships.

A. B

Critical Thinking Questions

OpenStax learning exercise

Q. Describe what influences whether relationships will be formed.

A. Proximity is a major situational factor in relationship formation; people who have frequent contact are more likely to form relationships. Whether or not individuals will form a relationship is based on non-situational factors such as similarity, reciprocity, self-disclosure, and physical attractiveness. In

relationships, people seek reciprocity (i.e., a give and take in costs and benefits), self-disclosure of intimate information, and physically attractive partners.

OpenStax learning exercise

Q. The evolutionary theory argues that humans are motivated to perpetuate their genes and reproduce. Using an evolutionary perspective, describe traits in men and women that humans find attractive.

A. Traits that promote reproduction in females warmth, affection, and social skills; women with these traits are presumably better able to care for children. Traits that are desired in males include achievement, leadership qualities, and job skills; men with these traits are thought to be better able to financially provide for their families.

Personal Application Questions

OpenStax learning exercise

Q. Think about your recent friendships and romantic relationship(s). What factors do you think influenced the development of these relationships? What attracted you to becoming friends or romantic partners?

OpenStax learning exercise

Q. Have you ever used a social exchange theory approach to determine how satisfied you were in a relationship, either a friendship or romantic relationship? Have you ever had the costs outweigh the benefits of a relationship? If so, how did you address this imbalance?

CHAPTER 16 / CAC.901

Stress, coping, and adaptation

Lecture: Resilience is a process, not a prize

Resilience is often described as a quality that some children possess and others lack. A more useful developmental account treats it as a process involving a child, relationships, institutions, and circumstances over time. A child may function well in one domain while struggling in another. A young person may maintain school attendance while experiencing severe private distress. Resilience should not be inferred from the absence of visible inconvenience to adults.

Masten s account of ordinary magic draws attention to common adaptive systems, including relationships, learning, motivation, and opportunities for competence. The phrase does not mean that adaptation is easy or that adversity is harmless. It means that positive development often depends on ordinary human resources rather than extraordinary heroism. Protecting those resources is a practical task for families, schools, and services.

Risk is not a identity

A risk factor is associated with a greater likelihood of an outcome under particular conditions. It is not a certainty, and it is not a complete explanation. Several risks may accumulate, but their effects depend on timing, interaction, and available support. Avoid turning a list of risks into a score that defines a child s future.

Protective factors should also be described carefully. A supportive adult can matter, but one relationship cannot neutralize every structural danger. A child s problem-solving skill can help, but it should not be used to excuse inaccessible services. Resilience-informed practice asks which systems need strengthening, not merely how much adversity a child can tolerate.

Cultural and community resources may be overlooked by standard checklists. Extended family, faith communities, language networks, and collective practices can provide meaning and support. Ask what is meaningful and reliable for this child. Do not assume that every community resource is safe or equally accessible; assess rather than romanticize.

Stress and coping

Coping includes efforts to manage demands. Some strategies address the problem, some regulate emotion, and some seek support. No strategy is universally best. Problem-solving is useful when a situation can be changed; acceptance and support may be more useful when it cannot. Avoidance may provide short-term relief but maintain difficulty when it prevents necessary engagement. The function and flexibility of coping matter more than a simple good-versus-bad label.

Children learn coping through modeling and practice. A counselor can teach a strategy, but daily conditions determine whether it can be used. A breathing exercise will not remove bullying. A planning sheet will not create electricity or transport. Combine individual skills with action on the environment, and evaluate whether the child experiences the plan as helpful.

Figure 16.1 represents resilience as a changing system: adversity, child processes, relationships, institutional supports, and outcomes influence one another over time. It includes feedback from outcomes to later opportunities. A success can increase confidence and access; a setback can restrict participation. Intervention may interrupt a negative sequence by changing support at several points.

Recovery is not always a straight line

A child may improve and then become distressed again during a transition or reminder. This does not erase earlier progress. Review what changed, whether supports faded too quickly, and whether new demands require additional help. Avoid describing recurrence as failure. At the same time, do not assume that every recurrence is normal; reassess safety, treatment fit, and emerging needs.

Outcome measurement should include the child s priorities. Attendance, symptom scores, and academic performance can be useful, but they do not capture every meaningful change. A child may value being able to ask for help, sleep without fear, or spend time with a friend. Combine measures thoughtfully and explain why each is included.

Application: praise that becomes pressure

A fictional sixteen-year-old, Amina, is described by adults as resilient because she cares for siblings and maintains high grades. She privately reports exhaustion and panic. The praise has made it difficult to ask for help. A developmental formulation recognizes competence and overload together.

The plan should reduce avoidable demands, identify reliable adult support, and address Amina s own goals. It should not remove every responsibility without consultation or assume that her caregiving is inherently harmful. Ask what she values, what feels obligatory, and what support would make her life more sustainable. The school and family may need to change expectations.

This case warns against using resilience as a moral demand. A young person should not have to remain impressive to deserve care. Services should be available before a crisis and should not depend on proving exceptional strength.

Doctoral seminar task

Develop a resilience formulation that includes adversity, resources, outcomes in at least two domains, and change over time. Identify one resource that is real but insufficient. Propose an intervention that strengthens support without placing the entire burden on the child.

Write a short critique of a program that measures resilience only by school attendance. What might the measure miss? Which additional perspectives and indicators would improve the evaluation? The task connects developmental theory with ethical outcome measurement.

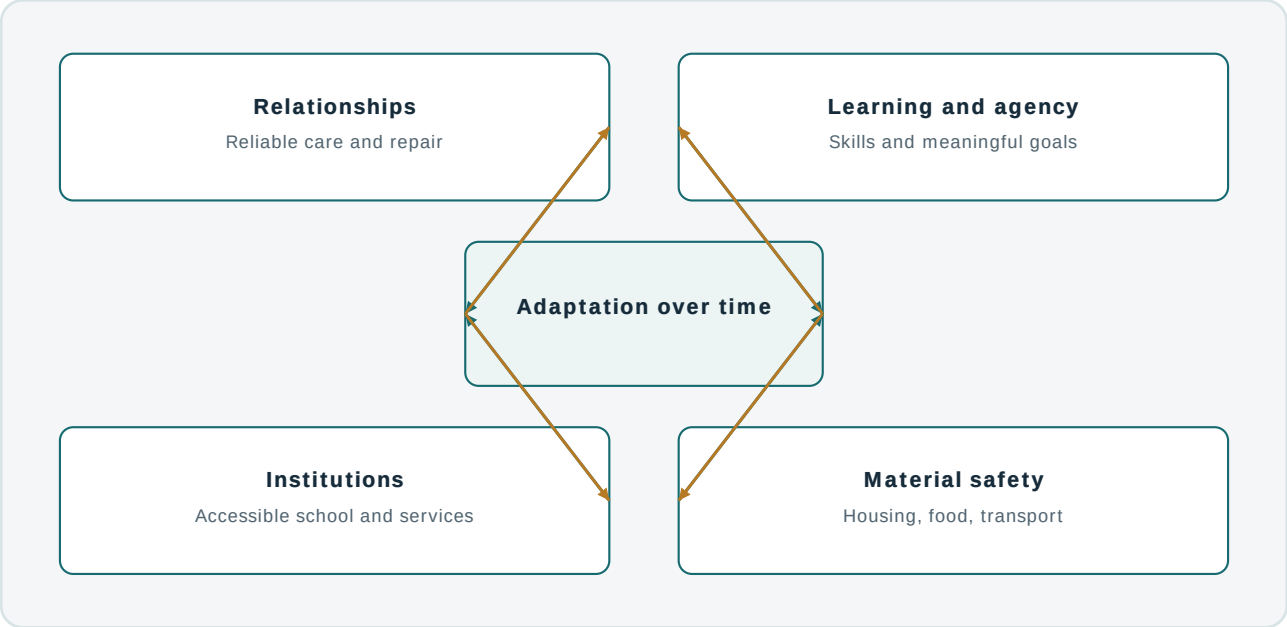


Figure 16.1. Resilience is supported by systems. Positive adaptation depends on interacting resources and conditions over time. A child s skills matter, but relationships, institutional access, and material safety also shape what is possible. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

What Is Stress?

OpenStax source module m82259 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82259/index.cnxml

The term stress as it relates to the human condition first emerged in scientific literature in the 1930s, but it did not enter the popular vernacular until the 1970s (Lyon, 2012). Today, we often use the term loosely in describing a variety of unpleasant feeling states; for example, we often say we are stressed out when we feel frustrated, angry, conflicted, overwhelmed, or fatigued. Despite the widespread use of the term, stress is a fairly vague concept that is difficult to define with precision.

Researchers have had a difficult time agreeing on an acceptable definition of stress. Some have conceptualized stress as a demanding or threatening event or situation (e.g., a high-stress job, overcrowding, and long commutes to work). Such conceptualizations are known as stimulus-based definitions because they characterize stress as a stimulus that causes certain reactions. Stimulus-based definitions of stress are problematic, however, because they fail to recognize that people differ in how they view and react to challenging life events and situations. For example, a conscientious student who has studied diligently all semester would likely experience less stress during final exams week than would a less responsible, unprepared student.

Others have conceptualized stress in ways that emphasize the physiological responses that occur when faced with demanding or threatening situations (e.g., increased arousal). These conceptualizations are referred to as response-based definitions because they describe stress as a response to environmental conditions. For example, the endocrinologist Hans Selye, a famous stress researcher, once defined stress as the response of the body to any demand, whether it is caused by, or results in, pleasant or unpleasant conditions (Selye, 1976, p. 74). Selye s definition of stress is response-based in that it conceptualizes stress chiefly in terms of the body s physiological reaction to any demand that is placed on it. Neither stimulus-based nor response-based definitions provide a complete definition of stress. Many of the physiological reactions that occur when faced with demanding situations (e.g., accelerated heart rate) can also occur in response to things that most people would not consider to be genuinely stressful, such as receiving unanticipated good news: an unexpected promotion or raise.

A useful way to conceptualize stress is to view it as a process whereby an individual perceives and responds to events that they appraise as overwhelming or threatening to their well-being (Lazarus & Folkman, 1984). A critical element of this definition is that it emphasizes the importance of how we appraise that is, judge demanding or threatening events (often referred to as stressors); these appraisals, in turn, influence our reactions to such events. Two kinds of appraisals of a stressor are especially important in this regard: primary and secondary appraisals. A primary appraisal involves judgment about the degree of potential harm or threat to well-being that a stressor might entail. A stressor would likely be appraised as a threat if one anticipates that it could lead to some kind of harm, loss, or other negative consequence; conversely, a stressor would likely be appraised as a challenge if one believes that it carries the potential for gain or personal growth. For example, an employee who is promoted to a leadership position would likely perceive the promotion as a much greater threat if they

believed the promotion would lead to excessive work demands than if they viewed it as an opportunity to gain new skills and grow professionally. Similarly, a college student on the cusp of graduation may face the change as a threat or a challenge (the accompanying source-figure description).

Source illustration description (image not reproduced): A photo shows a smiling person wearing a graduation cap and gown. Graduating from college and entering the workforce can be viewed as either a threat (loss of financial support) or a challenge (opportunity for independence and growth). (credit: Timothy Zanker)

The perception of a threat triggers a secondary appraisal: judgment of the options available to cope with a stressor, as well as perceptions of how effective such options will be (Lyon, 2012) (the accompanying source-figure description). As you may recall from what you learned about self-efficacy, an individual s belief in their ability to complete a task is important (Bandura, 1994). A threat tends to be viewed as less catastrophic if one believes something can be done about it (Lazarus & Folkman, 1984). Imagine that two middle-aged people, Robin and Madhuri, perform breast self-examinations one morning and each notices a lump on the lower region of their left breast. Although both view the breast lump as a potential threat (primary appraisal), their secondary appraisals differ considerably. In considering the breast lump, some of the thoughts racing through Robin s mind are, Oh my God, I could have breast cancer! What if the cancer has spread to the rest of my body and I cannot recover? What if I have to go through chemotherapy? I ve heard that experience is awful! What if I have to quit my job? My partner and I won t have enough money to pay the mortgage. Oh, this is just horrible I can t deal with it! On the other hand, Madhuri thinks, Hmm, this may not be good. Although most times these things turn out to be benign, I need to have it checked out. If it turns out to be breast cancer, there are doctors who can take care of it because the medical technology today is quite advanced. I ll have a lot of different options, and I ll be just fine. Clearly, Robin and Madhuri have different outlooks on what might turn out to be a very serious situation: Robin seems to think that little could be done about it, whereas Madhuri believes that, worst case scenario, a number of options that are likely to be effective would be available. As such, Robin would clearly experience greater stress than would Madhuri.

Source illustration description (image not reproduced): A concept map begins with a box titled Stressor at the top with an arrow underneath that leads to a box labeled Primary appraisal: challenge or threat? Below Primary appraisal: challenge or threat? is a line leading to the word challenge on the left side and threat on the right side. Below the word challenge is a box labeled Potential for gain or growth. There are no additional lines, arrows, or boxes under Potential for gain or growth. Below the word threat, there is a box labeled May lead to harm, loss, or negative consequences. Underneath the box, there is an arrow leading to another box labeled Secondary appraisal: potential options and how effective? The box has a line underneath that leads to the words effective option on the left side and ineffective/no option on the right side. Below the words effective option, there is an arrow leading to a box labeled Low threat. Below the words ineffective/no option, there is an arrow leading to a box labeled High threat. When encountering a stressor, a person judges its potential threat (primary appraisal) and then determines if effective options are available to manage the situation. Stress is likely to result if a stressor is perceived as extremely threatening or threatening with few or no effective coping options available.

To be sure, some stressors are inherently more stressful than others in that they are more threatening and leave less potential for variation in cognitive appraisals (e.g., objective threats to one s health or safety). Nevertheless, appraisal will still play a role in augmenting or diminishing our reactions to such events (Everly & Lating, 2002).

If a person appraises an event as harmful and believes that the demands imposed by the event exceed the available resources to manage or adapt to it, the person will subjectively experience a state of stress. In contrast, if one does not appraise the same event as harmful or threatening, she is unlikely to experience stress. According to this definition, environmental events trigger stress reactions by the way they are interpreted and the meanings they are assigned. In short, stress is largely in the eye of the beholder: it s not so much what happens to you as it is how you respond (Selye, 1976).

Good Stress?

Although stress carries a negative connotation, at times it may be of some benefit. Stress can motivate us to do things in our best interests, such as study for exams, visit the doctor regularly, exercise, and perform to the best of our ability at work. Indeed, Selye (1974) pointed out that not all stress is harmful. He argued that stress can sometimes be a positive, motivating force that can improve the quality of our lives. This kind of stress, which Selye called eustress (from the Greek eu = good), is a good kind of stress associated with positive feelings, optimal health, and performance. A moderate amount of stress can be beneficial in challenging situations. For example, athletes may be motivated and energized by pregame stress, and students may experience similar beneficial stress before a major exam. Indeed, research shows that moderate stress can enhance both immediate and delayed recall of educational material. Participants in one study who memorized a scientific text passage showed improved memory of the passage immediately after exposure to a mild stressor as well as one day following exposure to the stressor (Hupbach & Fieman, 2012).

Increasing one s level of stress will cause performance to change in a predictable way. As shown in the accompanying source-figure description, as stress increases, so do performance and general well-being (eustress); when stress levels reach an optimal level (the highest point of the curve), performance reaches its peak. A person at this stress level is colloquially at the top of their game, meaning they feel fully energized, focused, and can work with minimal effort and maximum efficiency. But when stress exceeds this optimal level, it is no longer a positive force it becomes excessive and debilitating, or what Selye termed distress (from the Latin dis = bad). People who reach this level of stress feel burned out; they are fatigued, exhausted, and their performance begins to decline. If the stress remains excessive, health may begin to erode as well (Everly & Lating, 2002). A good example of distress is severe test anxiety. When students are feeling very stressed about a test, negative emotions combined with physical symptoms may make concentration difficult, thereby negatively affecting test scores.

Source illustration description (image not reproduced): A graph features a bell curve that has a line going through the middle labeled Optimal level. The curve is labeled eustress on the left side and distress on the right side. The x-axis is labeled Stress level and moves from low to high, and the y-axis is labeled Performance level and moves from low to high. The graph shows that stress levels increase with performance levels and that once stress levels reach optimal level, they move from eustress to distress. As the stress level increases from low to moderate, so does performance (eustress). At the optimal level (the peak of the curve), performance has reached its peak. If stress exceeds the optimal level, it will reach the distress region, where it will become excessive and debilitating, and performance will decline (Everly & Lating, 2002).

The Prevalence of Stress

Stress is everywhere and, as shown in the accompanying source-figure description, it has been on the rise over the last several years. Each of us is acquainted with stress some are more familiar than others. In many ways, stress feels like a load you just can t carry a feeling you experience when, for example, you have to drive somewhere in a blizzard, when you wake up late the morning of an important job interview, when you run out of money before the next pay period, and before taking an important exam for which you realize you are not fully prepared.

Source illustration description (image not reproduced): A pie chart is labeled Change in Stress Levels Over Past 5 Years and split into three sections. The largest section is labeled Increased and accounts for 44% of the pie chart. The second largest section is labeled Stayed the same and accounts for 31% of the pie chart. The smallest section is labeled Decreased and accounts for 25% of the pie chart. Nearly half of U.S. adults indicated that their stress levels have increased over the last five years (Neelakantan, 2013).

Stress is an experience that evokes a variety of responses, including those that are physiological (e.g., accelerated heart rate, headaches, or gastrointestinal problems), cognitive (e.g., difficulty concentrating or making decisions), and behavioral (e.g., drinking alcohol, smoking, or taking actions directed at eliminating the cause of the stress). Although stress can be positive at times, it can have deleterious health implications, contributing to the onset and progression of a variety of physical illnesses and diseases (Cohen & Herbert, 1996).

The scientific study of how stress and other psychological factors impact health falls within the realm of health psychology, a subfield of psychology devoted to understanding the importance of psychological influences on health, illness, and how people respond when they become ill (Taylor, 1999). Health psychology emerged as a discipline in the 1970s, a time during which there was increasing awareness of the role behavioral and lifestyle factors play in the development of illnesses and diseases (Straub, 2007). In addition to studying the connection between stress and illness, health psychologists investigate issues such as why people make certain lifestyle choices (e.g., smoking or eating unhealthy food despite knowing the potential adverse health implications of such behaviors). Health psychologists also design and investigate the effectiveness of interventions aimed at changing unhealthy behaviors. Perhaps one of the more fundamental tasks of health psychologists is to identify which groups of people are especially at risk for negative health outcomes, based on psychological or behavioral factors. For example, measuring differences in stress levels among demographic groups and how these levels change over time can help identify populations who may have an increased risk for illness or disease.

the accompanying source-figure description depicts the results of three national surveys in which several thousand individuals from different demographic groups completed a brief stress questionnaire; the surveys were administered in 1983, 2006, and 2009 (Cohen & Janicki-Deverts, 2012). All three surveys demonstrated higher stress in women than in men. Unemployed individuals reported high levels of stress in all three surveys, as did those with less education and income; retired persons reported the lowest stress levels. However, from 2006 to 2009 the greatest increase in stress levels occurred among men, Hispanic people aged 45 64, college graduates, and those with full-time employment. One interpretation of these findings is that concerns surrounding the 2008 2009 economic downturn (e.g., threat of or actual job loss and substantial loss of retirement savings) may have been especially stressful to college-educated employed men with limited time remaining in their working careers.

Source illustration description (image not reproduced): Graphs a through f show mean stress scores in 1983, 2006, and 2009, and how they have been impacted by different factors. Graph a shows the relationship between mean stress score and sex. The mean stress score for men steadily increased from 12 in 1983 to a little over 14 in 2006 to a little over 15 in 2009. The mean stress score for women increased rapidly from a little under 13 in 1983 to 16 in 2006 and remained the same in 2009. The graph indicates that the mean stress score for women is higher than the mean stress score for men overall. Graph b shows the relationship between mean stress score and age. The mean stress scores for people under 25 years old increased from a little over 14 in 1983 to a little over 18 in 2006, and then decreased to 17 in 2009. The mean stress scores for people 25 to 34 years old increased from a little under 14 in 1983 to 18 in 2006, then decreased to a little over 16 in 2009. The mean stress scores for people 35 44 years old increased from 13 in 1983 to a little under 17 in 2006, then decreased to a little over 16 in 2009. The mean stress scores for people 45 54 years old from a little under 13 in 1983 to 15 in 2006, then increased to a little under 17 in 2009. The mean stress scores for people 55 64 years old steadily increased from 12 in 1983 to a little over 13 in 2006 to a little over 14 in 2009. The mean stress scores for people 65 years old or older decreased from 12 in 1983 to a little under 11 in 2006, then slightly increased to 11 in 2009. Graph c shows the relationship between mean stress score and race. The mean stress scores for White people steadily increased from a little under 13 in 1983 to 15 in 2006 to a little over 15 in 2009. The mean stress scores for Black people increased from a little over 15 in 1983 to a little over 16 in 2006, then slightly decreased to a little over 15 in 2009. The mean stress scores for Hispanic people steadily increased from 14 in 1983 to a little under 16 in 2006 to 17 in 2009. The mean stress score for people classified as Other increased from 14 in 1983 to a little over 17 in 2006 where it remained. Graph d shows the relationship between mean stress scores and education. The mean stress scores for those with less than a high school education steadily increased from a little over 14 in 1983 to a little over 17 in 2006 to 19 in 2009. The mean stress scores for those with a high school education increased from 12 in 1983 to a little over 16 in 2006 and remained the same in 2009. The mean stress scores for those with some college education increased from 12 in 1983 to a little over 15 in 2006, then slightly increased to a little under 16 in 2009. The mean stress scores for those with a bachelor s degree steadily increased from 12 in 1983 to a little over 13 in 2006 to 15 in 2009. The mean stress scores for those with advanced degrees also steadily increased, from a little over 11 in 1983 to 13 in 2006 to a little under 15 in 2009. Graph e shows the relationship between mean stress scores and employment status. The mean stress scores for those with full time employment status steadily increased from a little over 12 in 1983 to 15 in 2006 to 16 in 2009. The mean stress scores for those with part time employment status increased from 14 in 1983 to 16 in 2006, then decreased to 15 in 2009.The mean stress scores for those who were unemployed rapidly increased from a little over 16 in 1983 to 20 in 2006, then decreased back to a little over 16 in 2009. The mean stress scores for those who were retired remained lower than the other groups, remaining at a little under 12 in 1983 and 2006, then slightly increasing to a little over 12 in 2009. Graph f shows the relationship between the mean stress score and income in U.S. dollars. The mean stress scores for those with an income of $25,000 or lower

steadily increased from a little over 15 in 1983 to 17 in 2006 to a little under 18 in 2009. The mean stress scores for those with an income of $25,001 to $35,000 steadily increased from 14 in 1983 to 16 in 2006 to a little under 17 in 2009. The mean stress scores for those with an income of $35,001 $50,000 steadily increased from a little under 13 in 1983 to a little over 15 in 2006 to a little over 16 in 2009. The mean stress scores for those with an income of $50,001 $75,000 increased rapidly from 12 in 1983 to a little under 15 in 2006, then slightly increased to a little over 15 in 2009. The mean stress scores for those with an income of $75,001 or more steadily increased from 12 in 1983 to a little under 13 in 2006 to a little over 14 in 2009. The charts above, adapted from Cohen & Janicki-Deverts (2012), depict the mean stress level scores among different demographic groups during the years 1983, 2006, and 2009. Across categories of sex, age, race, education level, employment status, and income, stress levels generally show a marked increase over this quarter-century time span.

Early Contributions to the Study of Stress

As previously stated, scientific interest in stress goes back nearly a century. One of the early pioneers in the study of stress was Walter Cannon, an eminent American physiologist at Harvard Medical School (the accompanying source-figure description). In the early part of the 20th century, Cannon was the first to identify the body s physiological reactions to stress.

Source illustration description (image not reproduced): A photo of Walter Cannon is shown. Harvard physiologist Walter Cannon first articulated and named the fight-or-flight response, the nervous system s sympathetic response to a significant stressor.

Cannon and the Fight-or-Flight Response

Imagine that you are hiking in the beautiful mountains of Colorado on a warm and sunny spring day. At one point during your hike, a large, frightening-looking black bear appears from behind a stand of trees and sits about 50 yards from you. The bear notices you, sits up, and begins to lumber in your direction. In addition to thinking, This is definitely not good, a constellation of physiological reactions begins to take place inside you. Prompted by a deluge of epinephrine (adrenaline) and norepinephrine (noradrenaline) from your adrenal glands, your pupils begin to dilate. Your heart starts to pound and speeds up, you begin to breathe heavily and perspire, you get butterflies in your stomach, and your muscles become tense, preparing you to take some kind of direct action. Cannon proposed that this reaction, which he called the fight-or-flight response, occurs when a person experiences very strong emotions especially those associated with a perceived threat (Cannon, 1932). During the fight-or-flight response, the body is rapidly aroused by activation of both the sympathetic nervous system and the endocrine system (the accompanying source-figure description). This arousal helps prepare the person to either fight or flee from a perceived threat.

Source illustration description (image not reproduced): A figure shows the basic outline of a human body and indicates the body s various responses to fight or flight, including: pupils dilate, heart rate increases, muscles tense and may tremble, respiration quickens, bronchial tubes dilate, and perspiration begins. Fight or flight is a physiological response to a stressor.

According to Cannon, the fight-or-flight response is a built-in mechanism that assists in maintaining homeostasis an internal environment in which physiological variables such as blood pressure, respiration, digestion, and temperature are stabilized at levels optimal for survival. Thus, Cannon viewed the fight-or-flight response as adaptive because it enables people to adjust internally and externally to threats in their environment, allowing them to continue to be alive and overcome the threat.

Selye and the General Adaptation Syndrome

Another important early contributor to the stress field was Hans Selye, mentioned earlier. He would eventually become one of the world s foremost experts in the study of stress (the accompanying source-figure description). As a young assistant in the biochemistry department at McGill University in the 1930s, Selye was engaged in research involving sex hormones in rats. Although he was unable to find an answer for what he was initially researching, he incidentally discovered that when exposed to prolonged negative stimulation (stressors) such as extreme cold, surgical injury, excessive muscular exercise, and shock the rats showed signs of adrenal enlargement, thymus and lymph node shrinkage, and stomach ulceration. Selye realized that these responses were triggered by a coordinated series of physiological reactions that unfold over time during continued exposure to a stressor. These physiological reactions were nonspecific, which means that regardless of the type of stressor, the same pattern of reactions would occur. What Selye discovered was the general adaptation syndrome, the body s nonspecific physiological response to stress.

Source illustration description (image not reproduced): A stamp featuring Hans Selye is shown. Hans Selye specialized in research about stress. In 2009, his native Hungary honored his work with this stamp, released in conjunction with the 2nd annual World Conference on Stress.

The general adaptation syndrome, shown in the accompanying source-figure description, consists of three stages: (1) alarm reaction, (2) stage of resistance, and (3) stage of exhaustion (Selye, 1936; 1976). Alarm reaction describes the body s immediate reaction upon facing a threatening situation or emergency, and it is roughly analogous to the fight-or-flight response described by Cannon. During an alarm reaction, you are alerted to a stressor, and your body alarms you with a cascade of physiological reactions that provide you with the energy to manage the situation. A person who wakes up in the middle of the night to discover her house is on fire, for example, is experiencing an alarm reaction.

Source illustration description (image not reproduced): A graph shows the three stages of Selye s general adaption syndrome: alarm reaction, resistance, and exhaustion. The x-axis represents time while the y-axis represents stress levels. The x-axis is labeled Time and the y-axis is labeled Stress resistance. The graph shows that an increase in time and stress ultimately leads to exhaustion. The three stages of Selye s general adaptation syndrome are shown in this graph. Prolonged stress ultimately results in exhaustion.

If exposure to a stressor is prolonged, the organism will enter the stage of resistance. During this stage, the initial shock of alarm reaction has worn off and the body has adapted to the stressor. Nevertheless, the body also remains on alert and is prepared to respond as it did during the alarm reaction, although with less intensity. For example, suppose a child who went missing is still missing 72 hours later. Although the parents would obviously remain extremely disturbed, the magnitude of physiological reactions would likely have diminished over the 72 intervening hours due to some adaptation to this event.

If exposure to a stressor continues over a longer period of time, the stage of exhaustion ensues. At this stage, the person is no longer able to adapt to the stressor: the body s ability to resist becomes depleted as physical wear takes its toll on the body s tissues and organs. As a result, illness, disease, and other permanent damage to the body even death may occur. If a missing child still remained missing after three months, the long-term stress associated with this situation may cause a parent to literally faint with exhaustion at some point or even to develop a serious and irreversible illness.

In short, Selye s general adaptation syndrome suggests that stressors tax the body via a three-phase process an initial jolt, subsequent readjustment, and a later depletion of all physical resources that ultimately lays the groundwork for serious health problems and even death. It should be pointed out, however, that this model is a response-based conceptualization of stress, focusing exclusively on the body s physical responses while largely ignoring psychological factors such as appraisal and interpretation of threats. Nevertheless, Selye s model has had an enormous impact on the field of stress because it offers a general explanation for how stress can lead to physical damage and, thus, disease. As we shall discuss later, prolonged or repeated stress has been implicated in development of a number of disorders such as hypertension and coronary artery disease.

The Physiological Basis of Stress

What goes on inside our bodies when we experience stress? The physiological mechanisms of stress are extremely complex, but they generally involve the work of two systems the sympathetic nervous system and the hypothalamic-pituitary-adrenal (HPA) axis. When a person first perceives something as stressful (Selye s alarm reaction), the sympathetic nervous system triggers arousal via the release of adrenaline from the adrenal glands. Release of these hormones activates the fight-or-flight responses to stress, such as accelerated heart rate and respiration. At the same time, the HPA axis, which is primarily endocrine in nature, becomes especially active, although it works much more slowly than the sympathetic nervous system. In response to stress, the hypothalamus (one of the limbic structures in the brain) releases corticotropin-releasing factor, a hormone that causes the pituitary gland to release adrenocorticotropic hormone (ACTH) (the accompanying source-figure description). The ACTH then activates the adrenal glands to secrete a number of hormones into the bloodstream; an important one is cortisol, which can affect virtually every organ within the body. Cortisol is commonly known as a stress hormone and helps provide that boost of energy when we first encounter a stressor, preparing us to run away or fight. However, sustained elevated levels of cortisol weaken the immune system.

Source illustration description (image not reproduced): A figure shows an outline of the human body that indicates various parties of the body related to the hypothalamic-pituitary-adrenal axis. The hypothalamus, pituitary gland, and adrenal glands are labeled. There is an arrow from hypothalamus to pituitary gland and another arrow from pituitary gland to adrenal glands. These arrows represent the flow between these organs. This diagram shows the functioning of the hypothalamic-pituitary-adrenal (HPA) axis. The hypothalamus activates the pituitary gland, which in turn activates the adrenal glands, increasing their secretion of cortisol.

In short bursts, this process can have some favorable effects, such as providing extra energy, improving immune system functioning temporarily, and decreasing pain sensitivity. However, extended release of cortisol as would happen with prolonged or chronic stress often comes at a high price. High levels of cortisol have been shown to produce a number of harmful effects. For example, increases in cortisol can significantly weaken our immune system (Glaser & Kiecolt-Glaser, 2005), and high levels are frequently observed among depressed individuals (Geoffroy, Hertzman, Li, & Power, 2013). In summary, a stressful event causes a variety of physiological reactions that activate the adrenal glands, which in turn release epinephrine, norepinephrine, and cortisol. These hormones affect a number of bodily processes in ways that prepare the stressed person to take direct action, but also in ways that may heighten the potential for illness.

When stress is extreme or chronic, it can have profoundly negative consequences. For example, stress often contributes to the development of certain psychological disorders, including post-traumatic stress disorder, major depressive disorder, and other serious psychiatric conditions. Additionally, we noted earlier that stress is linked to the development and progression of a variety of physical illnesses and diseases. For example, researchers in one study found that people injured during the September 11, 2001, World Trade Center disaster or who developed post-traumatic stress symptoms afterward later suffered significantly elevated rates of heart disease (Jordan, Miller-Archie, Cone, Morabia, & Stellman, 2011). Another investigation yielded that self-reported stress symptoms among aging and retired Finnish food industry workers were associated with morbidity 11 years later. This study also predicted the onset of musculoskeletal, nervous system, and endocrine and metabolic disorders (Salonen, Arola, Nyg rd, & Huhtala, 2008). Another study reported that male South Korean manufacturing employees who reported high levels of work-related stress were more likely to catch the common cold over the next several months than were those employees who reported lower work-related stress levels (Park et al., 2011). Later, you will explore the mechanisms through which stress can produce physical illness and disease.

Summary

Stress is a process whereby an individual perceives and responds to events appraised as overwhelming or threatening to one s well-being. The scientific study of how stress and emotional factors impact health and well-being is called health psychology, a field devoted to studying the general impact of psychological factors on health. The body s primary physiological response during stress, the fight-or-flight response, was first identified in the early 20th century by Walter Cannon. The fight-or-flight response involves the coordinated activity of both the sympathetic nervous system and the hypothalamic-pituitary-adrenal (HPA) axis. Hans Selye, a noted endocrinologist, referred to these physiological reactions to stress as part of general adaptation syndrome, which occurs in three stages: alarm reaction (fight-or-flight reactions begin), resistance (the body begins to adapt to continuing stress), and exhaustion (adaptive energy is depleted, and stress begins to take a physical toll).

Review Questions

OpenStax learning exercise

Q. Negative effects of stress are most likely to be experienced when an event is perceived as _____.

A. D

OpenStax learning exercise

Q. Between 2006 and 2009, the greatest increases in stress levels were found to occur among _____.

A. B

OpenStax learning exercise

Q. At which stage of Selye s general adaptation syndrome is a person especially vulnerable to illness?

A. A

OpenStax learning exercise

Q. During an encounter judged as stressful, cortisol is released by the _____.

A. D

Critical Thinking Questions

OpenStax learning exercise

Q. Provide an example (other than the one described earlier) of a situation or event that could be appraised as either threatening or challenging.

A. Answers will vary. One example is divorce. People may perceive a divorce as a threat if they believe it will result in loneliness, change of lifestyle (due to loss of additional income), or humiliation in the eyes of their family. However, divorce may be perceived as a challenge if they view it as an opportunity to find somebody more compatible, and if they consider the process of finding a new partner a pleasant one, perhaps involving mystery and excitement.

OpenStax learning exercise

Q. Provide an example of a stressful situation that may cause a person to become seriously ill. How would Selye s general adaptation syndrome explain this occurrence?

A. Answers will vary. One example is when somebody s spouse dies or is unexpectedly diagnosed with a fatal disease. In both cases, the stress experienced by the surviving spouse would be intense, continuous, and according the general adaptation syndrome would eventually increase vulnerability to illness or disease (exhaustion stage).

Personal Application Question

OpenStax learning exercise

Q. Think of a time in which you and others you know (family members, friends, and classmates) experienced an event that some viewed as threatening and others viewed as challenging. What were some of the differences in the reactions of those who experienced the event as threatening compared to those who viewed the event as challenging? Why do you think there were differences in how these individuals judged the same event?

Stressors

OpenStax source module m82260 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82260/index.cnxml

For an individual to experience stress, they must first encounter a potential stressor. In general, stressors can be placed into one of two broad categories: chronic and acute. Chronic stressors include events that persist over an extended period of time, such as caring for a parent with dementia, long-term unemployment, or imprisonment. Acute stressors involve brief focal events that sometimes continue to be experienced as overwhelming well after the event has ended, such as falling on an icy sidewalk and breaking your leg (Cohen, Janicki-Deverts, & Miller, 2007). Whether chronic or acute, potential stressors come in many shapes and sizes. They can include major traumatic events, significant life changes, daily hassles, as well as other situations in which a person is regularly exposed to threat, challenge, or danger.

Traumatic Events

Some stressors involve traumatic events or situations in which a person is exposed to actual or threatened death or serious injury. Stressors in this category include exposure to military combat, threatened or actual physical assaults (e.g., physical attacks, sexual assault, robbery, childhood abuse), terrorist attacks, natural disasters (e.g., earthquakes, floods, hurricanes), and automobile accidents. Men, non-White people, and individuals in lower socioeconomic status (SES) groups report experiencing a greater number of traumatic events than do women, White people, and individuals in higher SES groups (Hatch & Dohrenwend, 2007). Some individuals who are exposed to stressors of extreme magnitude develop post-traumatic stress disorder (PTSD): a chronic stress reaction characterized by experiences and behaviors that may include intrusive and painful memories of the stressor event, jumpiness, persistent negative emotional states, detachment from others, angry outbursts, and avoidance of reminders of the event (American Psychiatric Association [APA], 2013).

Life Changes

Most stressors that we encounter are not nearly as intense as the ones described above. Many potential stressors we face involve events or situations that require us to make changes in our ongoing lives and require time as we adjust to those changes. Examples include death of a close family member, marriage, divorce, and moving (the accompanying source-figure description).

Source illustration description (image not reproduced): A photo shows a person next to the back of a moving truck unloading furniture. Some fairly typical life events, such as moving, can be significant stressors. Even when the move is intentional and positive, the amount of resulting change in daily life can cause stress. (credit: "Jellaluna"/Flickr)

In the 1960s, psychiatrists Thomas Holmes and Richard Rahe wanted to examine the link between life stressors and physical illness, based on the hypothesis that life events requiring significant changes in a person s normal life routines are stressful, whether these events are desirable or undesirable. They developed the Social Readjustment Rating Scale (SRRS), consisting of 43 life events that require varying degrees of personal readjustment (Holmes & Rahe, 1967). Many life events that most people would consider pleasant (e.g., holidays, retirement, marriage) are among those listed on the SRRS; these are examples of eustress. Holmes and Rahe also proposed that life events can add up over time, and that experiencing a cluster of stressful events increases one s risk of developing physical illnesses.

In developing their scale, Holmes and Rahe asked 394 participants to provide a numerical estimate for each of the 43 items; each estimate corresponded to how much readjustment participants felt each event would require. These estimates resulted in mean value scores for each event often called life change units (LCUs) (Rahe, McKeen, & Arthur, 1967). The numerical scores ranged from 11 to 100, representing the perceived magnitude of life change each event entails. Death of a spouse ranked highest on the scale with 100 LCUs, and divorce ranked second highest with 73 LCUs. In addition, personal injury or illness, marriage, and job termination also ranked highly on the scale with 53, 50, and 47 LCUs, respectively. Conversely, change in residence (20 LCUs), change in eating habits (15 LCUs), and vacation (13 LCUs) ranked low on the scale (the accompanying source-figure description). Minor violations of the law ranked the lowest with 11 LCUs. To complete the scale, participants checked yes for events experienced within the last 12 months. LCUs for each checked item are totaled for a score quantifying the amount of life change. Agreement on the amount of adjustment required by the various life events on the SRRS is highly consistent, even cross-culturally (Holmes & Masuda, 1974).

Source comparison table

Some Stressors on the Social Readjustment Rating Scale (Holmes & Rahe, 1967)	
Life event	Life change units
Death of a close family member	63
Personal injury or illness	53
Dismissal from work	47
Change in financial state	38
Change to different line of work	36
Outstanding personal achievement	28
Beginning or ending school	26
Change in living conditions	25
Change in working hours or conditions	20
Change in residence	20
Change in schools	20
Change in social activities	18
Change in sleeping habits	16
Change in eating habits	15
Minor violation of the law	11

Extensive research has demonstrated that accumulating a high number of life change units within a brief period of time (one or two years) is related to a wide range of physical illnesses (even accidents and athletic injuries) and mental health problems (Monat & Lazarus, 1991; Scully, Tosi, & Banning, 2000). In an early demonstration, researchers obtained LCU scores for U.S. and Norwegian Navy personnel who were about to embark on a six-month voyage. A later examination of medical records revealed positive (but small) correlations between LCU scores prior to the voyage and subsequent illness symptoms during the ensuing six-month journey (Rahe, 1974). In addition, people tend to experience more physical symptoms, such as backache, upset stomach, diarrhea, and acne, on specific days in which self-reported LCU values are considerably higher than normal, such as the day of a family member s wedding (Holmes & Holmes, 1970).

The Social Readjustment Rating Scale (SRRS) provides researchers a simple, easy-to-administer way of assessing the amount of stress in people s lives, and it has been used in hundreds of studies (Thoits, 2010). Despite its widespread use, the scale has been subject to criticism. First, many of the items on the SRRS are vague; for example, death of a close friend could involve the death of a long-absent childhood friend that requires little social readjustment (Dohrenwend, 2006). In addition, some have challenged its assumption that undesirable life events are no more stressful than desirable ones (Derogatis & Coons, 1993). However, most of the available evidence suggests that, at least as far as mental health is concerned, undesirable or negative events are more strongly associated with poor outcomes (such as depression) than are desirable, positive events (Hatch & Dohrenwend, 2007). Perhaps the most serious criticism is that the scale does not take into consideration respondents appraisals of the life events it contains. As you recall, appraisal of a stressor is a key element in the conceptualization and overall experience of stress. Being fired from work may be devastating to some but a welcome opportunity to obtain a better job for others. The SRRS remains one of the most well-known instruments in the study of stress, and it is a useful tool for identifying potential stress-related health outcomes (Scully et al., 2000).

Connect the Concepts

The Holmes and Rahe Social Readjustment Rating Scale (SRRS) uses the correlational research method to identify the connection between stress and health. That is, respondents LCU scores are correlated with the number or frequency of self-reported symptoms indicating health problems. These correlations are typically positive as LCU scores increase, the number of symptoms increase. Consider all the thousands of studies that have used this scale to correlate stress and illness symptoms: If you were to assign an average correlation coefficient to this body of research, what would be your best guess? How strong do you think the correlation coefficient would be? Why can t the SRRS show a causal relationship between stress and illness? If it were possible to show causation, do you think stress causes illness or illness causes stress?

Hassles

Potential stressors do not always involve major life events. Daily hassles the minor irritations and annoyances that are part of our everyday lives (e.g., rush hour traffic, lost keys, obnoxious coworkers, inclement weather, arguments with friends or family) can build on one another and leave us just as stressed as life change events (the accompanying source-figure description) (Kanner, Coyne, Schaefer, & Lazarus, 1981).

Source illustration description (image not reproduced): Photograph A shows heavy traffic going both ways on a scenic road. Photograph B shows a crowded bus with people sitting in the seats and standing in the aisles. Daily commutes, whether (a) on the road or (b) via public transportation, can be hassles that contribute to our feelings of everyday stress. (credit a: modification of work by Jeff Turner; credit b: modification of work by "epSos.de"/Flickr)

Researchers have demonstrated that the frequency of daily hassles is actually a better predictor of both physical and psychological health than are life change units. In a well-known study of San Francisco residents, the frequency of daily hassles was found to be more strongly associated with physical health problems than were life change events (DeLongis, Coyne, Dakof, Folkman, & Lazarus, 1982). In

addition, daily minor hassles, especially interpersonal conflicts, often lead to negative and distressed mood states (Bolger, DeLongis, Kessler, & Schilling, 1989). Cyber hassles that occur on social media may represent a modern and evolving source of stress. In one investigation, social media stress was tied to loss of sleep in adolescents, presumably because ruminating about social media caused a physiological stress response that increased arousal (van der Schuur, Baumgartner, & Sumter, 2018). Clearly, daily hassles can add up and take a toll on us both emotionally and physically.

OCCUPATION-RELATED Stressors

Stressors can include situations in which one is frequently exposed to challenging and unpleasant events, such as difficult, demanding, or unsafe working conditions. Although most jobs and occupations can at times be demanding, some are clearly more stressful than others (the accompanying source-figure description). For example, most people would likely agree that a firefighter s work is inherently more stressful than that of a florist. Equally likely, most would agree that jobs containing various unpleasant elements, such as those requiring exposure to loud noise (heavy equipment operator), constant harassment and threats of physical violence (prison guard), perpetual frustration (bus driver in a major city), or those mandating that an employee work alternating day and night shifts (hotel desk clerk), are much more demanding and thus, more stressful than those that do not contain such elements. the accompanying source-figure description lists several occupations and some of the specific stressors associated with those occupations (Sulsky & Smith, 2005).

Source illustration description (image not reproduced): Photograph A shows uniformed police officers marching with synchronized arms swinging. Photograph B shows firefighters fighting a fire. (a) Police officers and (b) firefighters hold high stress occupations. (credit a: modification of work by Australian Civil-Military Centre; credit b: modification of work by Andrew Magill)

Source comparison table

Occupations and Their Related Stressors	
Occupation	Stressors Specific to Occupation
Police officer	physical dangers, excessive paperwork, dealing with court system, tense interactions, life-and-death decision making
Firefighter	uncertainty over whether a serious fire or hazard awaits after an alarm, potential for extreme physical danger
Social worker	little positive feedback from jobs or from the public, unsafe work environments, frustration in dealing with bureaucracy, excessive paperwork, sense of personal responsibility for clients, work overload
Teacher	Excessive paperwork, lack of adequate supplies or facilities, work overload, lack of positive feedback, threat of physical violence, lack of support from parents and administrators
Nurse	Work overload, heavy physical work, patient concerns (dealing with death and medical concerns), interpersonal problems with other medical staff (especially physicians)
Emergency medical worker	Unpredictable and extreme nature of the job, inexperience

Occupations and Their Related Stressors	
Clerical and secretarial work	Few opportunities for advancement, unsupportive supervisors, work overload, lack of perceived control
Managerial work	Work overload, conflict and ambiguity in defining the managerial role, difficult work relationships

Although the specific stressors for these occupations are diverse, they seem to share some common denominators such as heavy workload and uncertainty about and lack of control over certain aspects of a job. Chronic occupational stress contributes to job strain, a work situation that combines excessive job demands and workload with little discretion in decision making or job control (Karasek & Theorell, 1990). Clearly, many occupations other than the ones listed in the accompanying source-figure description involve at least a moderate amount of job strain in that they often involve heavy workloads and little job control (e.g., inability to decide when to take breaks). Such jobs are often low-status and include those of factory workers, postal clerks, supermarket cashiers, taxi drivers, and short-order cooks. Job strain can have adverse consequences on both physical and mental health; it has been shown to be associated with increased risk of hypertension (Schnall & Landsbergis, 1994), heart attacks (Theorell et al., 1998), recurrence of heart disease after a first heart attack (Aboa- boul et al., 2007), significant weight loss or gain (Kivim ki et al., 2006), and major depressive disorder (Stansfeld, Shipley, Head, & Fuhrer, 2012). A longitudinal study of over 10,000 British civil servants reported that workers under 50 years old who earlier had reported high job strain were 68% more likely to later develop heart disease than were those workers under 50 years old who reported little job strain (Chandola et al., 2008).

Some people who are exposed to chronically stressful work conditions can experience job burnout, which is a general sense of emotional exhaustion and cynicism in relation to one s job (Maslach & Jackson, 1981). Job burnout occurs frequently among those in human service jobs (e.g., social workers, teachers, therapists, and police officers). Job burnout consists of three dimensions. The first dimension is exhaustion a sense that one s emotional resources are drained or that one is at the end of their rope and has nothing more to give at a psychological level. Second, job burnout is characterized by depersonalization: a sense of emotional detachment between the worker and the recipients of their services, often resulting in callous, cynical, or indifferent attitudes toward these individuals. Third, job burnout is characterized by diminished personal accomplishment, which is the tendency to evaluate one s work negatively by, for example, experiencing dissatisfaction with one s job-related accomplishments or feeling as though one has categorically failed to influence others lives through one s work.

Job strain appears to be one of the greatest risk factors leading to job burnout, which is most commonly observed in workers who are older (ages 55 64), unmarried, and whose jobs involve manual labor. Heavy alcohol consumption, physical inactivity, being overweight, and having a physical or lifetime mental disorder are also associated with job burnout (Ahola, et al., 2006). In addition, depression often co-occurs with job burnout. One large-scale study of over 3,000 Finnish employees reported that half of the participants with severe job burnout had some form of depressive disorder (Ahola et al., 2005). Job burnout is often precipitated by feelings of having invested considerable energy, effort, and time into one s work while receiving little in return (e.g., little respect or support from others or low pay) (Tatris, Peeters, Le Blanc, Schreurs, & Schaufeli, 2001).

As an illustration, consider Tyre, a nursing assistant who worked in a nursing home. Tyre worked long hours for little pay in a difficult facility. Tyre's supervisor was domineering, unpleasant, and unsupportive, as well as disrespectful of Tyre's personal time, frequently informing them at the last minute they must work several additional hours after their shift ended or report to work on weekends. Tyre had very little autonomy at work. They had little input in day-to-day duties and how to perform them, and was not permitted to take breaks unless explicitly told by their supervisor. Tyre did not feel as though their hard work was appreciated, either by supervisory staff or by the residents of the home.

Tyre was very unhappy over the low pay, and felt that many of the residents treated them disrespectfully.

After several years, Tyre began to hate their job. Tyre dreaded going to work in the morning, and gradually developed a callous, hostile attitude toward many of the residents. Eventually, they began to feel they could no longer help the nursing home residents. Tyre s absenteeism from work increased, and one day they decided that they had had enough and quit. Tyre now has a job in sales, vowing never to work in nursing again.

Finally, our close relationships with friends and family particularly the negative aspects of these relationships can be a potent source of stress. Negative aspects of close relationships can include conflicts such as disagreements or arguments, lack of emotional support or confiding, and lack of reciprocity. All of these can be overwhelming, threatening to the relationship, and thus stressful. Such stressors can take a toll both emotionally and physically. A longitudinal investigation of over 9,000 British civil servants found that those who at one point had reported the highest levels of negative interactions in their closest relationship were 34% more likely to experience serious heart problems (fatal or nonfatal heart attacks) over a 13 15 year period, compared to those who experienced the lowest levels of negative interaction (De Vogli, Chandola & Marmot, 2007).

Summary

Stressors can be chronic (long term) or acute (short term), and can include traumatic events, significant life changes, daily hassles, and situations in which people are frequently exposed to challenging and unpleasant events. Many potential stressors include events or situations that require us to make changes in our lives, such as a divorce or moving to a new residence. Thomas Holmes and Richard Rahe developed the Social Readjustment Rating Scale (SRRS) to measure stress by assigning a number of life change units to life events that typically require some adjustment, including positive events. Although the SRRS has been criticized on a number of grounds, extensive research has shown that the accumulation of many LCUs is associated with increased risk of illness. Many potential stressors also include daily hassles, which are minor irritations and annoyances that can build up over time. In addition, jobs that are especially demanding, offer little control over one s working environment, or involve unfavorable working conditions can lead to job strain, thereby setting the stage for job burnout.

Review Questions

OpenStax learning exercise

Q. According to the Holmes and Rahe scale, which life event requires the greatest amount of readjustment?

A. D

OpenStax learning exercise

Q. While waiting to pay for his weekly groceries at the supermarket, Paul had to wait about 20 minutes in a long line at the checkout because only one cashier was on duty. When he was finally ready to pay, his debit card was declined because he did not have enough money left in his checking account. Because he had left his credit cards at home, he had to place the groceries back into the cart and head home to retrieve a credit card. While driving back to his home, traffic was backed up two miles due to an accident. These events that Paul had to endure are best characterized as _____.

A. C

OpenStax learning exercise

Q. What is one of the major criticisms of the Social Readjustment Rating Scale?

A. C

OpenStax learning exercise

Q. Which of the following is not a dimension of job burnout?

A. B

Critical Thinking Questions

OpenStax learning exercise

Q. Review the items on the Social Readjustment Rating Scale. Select one of the items and discuss how it might bring about distress and eustress.

A. Answers will vary. For example, many people look forward to celebrating the Christmas holiday, but it can be stressful in that it requires some degree of readjustment. Getting together with family may bring eustress, while the schedule and travel demands of may bring distress. Giving gifts to others and seeing their enjoyment may bring eustress, but the financial burden associated with buying presents could produce distress. Each of these things requires making some minor adjustments to one s life, and thus is considered somewhat stressful.

OpenStax learning exercise

Q. Job burnout tends to be high in people who work in human service jobs. Considering the three dimensions of job burnout, explain how various job aspects unique to being a police officer might lead to job burnout in that line of work.

A. Answers will vary. Many calls that police officers make can be emotionally draining (e.g., tragic deaths, suicides, and children who live in squalid conditions), which might eventually lead to feelings of exhaustion that one can no longer deal with such things. Depersonalization may occur if a police officer works in an environment in which she feels disrespected and unappreciated, which may lead to cynical and callous feelings toward the public. Constant disrespect from others may diminish a police officer s sense of personal accomplishment.

Personal Application Question

OpenStax learning exercise

Q. Suppose you want to design a study to examine the relationship between stress and illness, but you cannot use the Social Readjustment Rating Scale. How would you go about measuring stress? How would you measure illness? What would you need to do in order to tell if there is a cause-effect relationship between stress and illness?

Regulation of Stress

OpenStax source module m82262 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82262/index.cnxml

As we learned in the previous section, stress especially if it is chronic takes a toll on our bodies and can have enormously negative health implications. When we experience events in our lives that we appraise as stressful, it is essential that we use effective coping strategies to manage our stress. Coping refers to mental and behavioral efforts that we use to deal with problems relating to stress.

Coping Styles

Lazarus and Folkman (1984) distinguished two fundamental kinds of coping: problem-focused coping and emotion-focused coping. In problem-focused coping, one attempts to manage or alter the problem that is causing one to experience stress (i.e., the stressor). Problem-focused coping strategies are similar to strategies used in everyday problem-solving: they typically involve identifying the problem, considering possible solutions, weighing the costs and benefits of these solutions, and then selecting an alternative (Lazarus & Folkman, 1984). As an example, suppose Bradford receives a midterm notice that he is failing statistics class. If Bradford adopts a problem-focused coping approach to managing his stress, he would be proactive in trying to alleviate the source of the stress. He might contact his professor to discuss what must be done to raise his grade, he might also decide to set aside two hours daily to study statistics assignments, and he may seek tutoring assistance. A problem-focused approach to managing stress means we actively try to do things to address the problem.

Emotion-focused coping, in contrast, consists of efforts to change or reduce the negative emotions associated with stress. These efforts may include avoiding, minimizing, or distancing oneself from the problem, or positive comparisons with others (I m not as bad off as she is), or seeking something positive in a negative event (Now that I ve been fired, I can sleep in for a few days). In some cases, emotion-focused coping strategies involve reappraisal, whereby the stressor is construed differently (and somewhat self-deceptively) without changing its objective level of threat (Lazarus & Folkman, 1984). For example, a person sentenced to federal prison who thinks, This will give me a great chance to network with others, is using reappraisal. If Bradford adopted an emotion-focused approach to managing his midterm deficiency stress, he might watch a comedy movie, play video games, or spend hours on social media to take his mind off the situation. In a certain sense, emotion-focused coping can be thought of as treating the symptoms rather than the actual cause.

While many stressors elicit both kinds of coping strategies, problem-focused coping is more likely to occur when encountering stressors we perceive as controllable, while emotion-focused coping is more likely to predominate when faced with stressors that we believe we are powerless to change (Folkman & Lazarus, 1980). Clearly, emotion-focused coping is more effective in dealing with uncontrollable stressors. For example, the stress you experience when a loved one dies can be overwhelming. You are simply powerless to change the situation as there is nothing you can do to bring this person back. The most helpful coping response is emotion-focused coping aimed at minimizing the pain of the grieving period.

Fortunately, most stressors we encounter can be modified and are, to varying degrees, controllable. A person who cannot stand her job can quit and look for work elsewhere; a middle-aged divorcee can find another potential partner; the freshman who fails an exam can study harder next time, and a breast lump does not necessarily mean that one is fated to die of breast cancer.

Control and Stress

The desire and ability to predict events, make decisions, and affect outcomes that is, to enact control in our lives is a basic tenet of human behavior (Everly & Lating, 2002). Albert Bandura (1997) stated that the intensity and chronicity of human stress is governed largely by perceived control over the demands of one s life (p. 262). As cogently described in his statement, our reaction to potential stressors depends to a large extent on how much control we feel we have over such things. Perceived control is our beliefs about our personal capacity to exert influence over and shape outcomes, and it has major implications for our health and happiness (Infurna & Gerstorf, 2014). Extensive research has demonstrated that perceptions of personal control are associated with a variety of favorable outcomes, such as better physical and mental health and greater psychological well-being (Diehl & Hay, 2010). Greater personal control is also associated with lower reactivity to stressors in daily life. For example, researchers in one investigation found that higher levels of perceived control at one point in time were later associated with lower emotional and physical reactivity to interpersonal stressors (Neupert, Almeida, & Charles, 2007). Further, a daily diary study with 34 older widows found that their stress and anxiety levels were significantly reduced on days during which the widows felt greater perceived control (Ong, Bergeman, & Bisconti, 2005).

Dig Deeper

When we lack a sense of control over the events in our lives, particularly when those events are threatening, harmful, or noxious, the psychological consequences can be profound. In one of the better illustrations of this concept, psychologist Martin Seligman conducted a series of classic experiments in the 1960s (Seligman & Maier, 1967) in which dogs were placed in a chamber where they received electric shocks from which they could not escape. Later, when these dogs were given the opportunity to escape the shocks by jumping across a partition, most failed to even try; they seemed to just give up and passively accept any shocks the experimenters chose to administer. In comparison, dogs who were previously allowed to escape the shocks tended to jump the partition and escape the pain (the accompanying source-figure description).

Source illustration description (image not reproduced): An illustration shows a dog about to jump over a partition separating an area of a floor delivering shocks from an area that doesn t deliver shocks. Seligman s learned helplessness experiments with dogs used an apparatus that measured when the animals would move from a floor delivering shocks to one without.

Seligman believed that the dogs who failed to try to escape the later shocks were demonstrating learned helplessness: They had acquired a belief that they were powerless to do anything about the stimulation they were receiving. Seligman also believed that the passivity and lack of initiative these dogs demonstrated was similar to that observed in human depression. Therefore, Seligman speculated that learned helplessness might be an important cause of depression in humans: Humans who experience negative life events that they believe they are unable to control may become helpless. As a result, they give up trying to change the situation and some may become depressed and show lack of initiative in future situations in which they can control the outcomes (Seligman, Maier, & Geer, 1968). In an application Seligman never proposed, learned helplessness was later used as a methodology in the torture of prisoners by U.S. military and intelligence personnel following the 2001 attacks on the World Trade Center. The psychologists who designed the torture program, James E. Mitchell and Bruce Jesson, theorized that detainees who were subjected to uncontrollable afflictions would eventually become passive and compliant, making them more likely to reveal information to their interrogators. There is little evidence that the program achieved worthwhile results. It is now widely regarded as unethical and unjustified. This example emphasizes the need to consistently consider the ethics of research studies and their applications (Konnikova, 2015).

Seligman and colleagues later reformulated the original learned helplessness model of depression (Abramson, Seligman, & Teasdale, 1978). In their reformulation, they emphasized attributions (i.e., a mental explanation for why something occurred) that fostered a sense of learned helplessness. For example, suppose a coworker shows up late to work; your belief as to what caused the coworker s tardiness would be an attribution (e.g., too much traffic, slept too late, or just doesn t care about being on time).

The reformulated version of Seligman s study holds that the attributions made for negative life events contribute to depression. Consider the example of a student who performs poorly on a midterm exam. This model suggests that the student will make three kinds of attributions for this outcome: internal vs. external (believing the outcome was caused by his own personal inadequacies or by environmental factors), stable vs. unstable (believing the cause can

be changed or is permanent), and global vs. specific (believing the outcome is a sign of inadequacy in most everything versus just this area). Assume that the student makes an internal (I m just not smart), stable (Nothing can be done to change the fact that I m not smart) and global (This is another example of how lousy I am at everything) attribution for the poor performance. The reformulated theory predicts that the student would perceive a lack of control over this stressful event and thus be especially prone to developing depression. Indeed, research has demonstrated that people who have a tendency to make internal, global, and stable attributions for bad outcomes tend to develop symptoms of depression when faced with negative life experiences (Peterson & Seligman, 1984). Fortunately, attribution habits can be changed through practice. Training in healthy attribution habits has been shown to make people less vulnerable to depression (Konnikova, 2015).

Seligman s learned helplessness model has emerged over the years as a leading theoretical explanation for the onset of major depressive disorder. When you study psychological disorders, you will learn more about the latest reformulation of this model now called hopelessness theory.

People who report higher levels of perceived control view their health as controllable, thereby making it more likely that they will better manage their health and engage in behaviors conducive to good health (Bandura, 2004). Not surprisingly, greater perceived control has been linked to lower risk of physical health problems, including declines in physical functioning (Infurna, Gerstorf, Ram, Schupp, & Wagner, 2011), heart attacks (Rosengren et al., 2004), and both cardiovascular disease incidence (St rmer, Hasselbach, & Amelang, 2006) and mortality from cardiac disease (Surtees et al., 2010). In addition, longitudinal studies of British civil servants have found that those in low-status jobs (e.g., clerical and office support staff) in which the degree of control over the job is minimal are considerably more likely to develop heart disease than those with high-status jobs or considerable control over their jobs (Marmot, Bosma, Hemingway, & Stansfeld, 1997).

The link between perceived control and health may provide an explanation for the frequently observed relationship between social class and health outcomes (Kraus, Piff, Mendoza-Denton, Rheinschmidt, & Keltner, 2012). In general, research has found that more affluent individuals experience better health partly because they tend to believe that they can personally control and manage their reactions to life s stressors (Johnson & Krueger, 2006). Perhaps buoyed by the perceived level of control, individuals of higher social class may be prone to overestimating the degree of influence they have over particular outcomes. For example, those of higher social class tend to believe that their votes have greater sway on election outcomes than do those of lower social class, which may explain higher rates of voting in more affluent communities (Krosnick, 1990). Other research has found that a sense of perceived control can protect less affluent individuals from poorer health, depression, and reduced life-satisfaction all of which tend to accompany lower social standing (Lachman & Weaver, 1998).

Taken together, findings from these and many other studies clearly suggest that perceptions of control and coping abilities are important in managing and coping with the stressors we encounter throughout life.

Social Support

The need to form and maintain strong, stable relationships with others is a powerful, pervasive, and fundamental human motive (Baumeister & Leary, 1995). Building strong interpersonal relationships with others helps us establish a network of close, caring individuals who can provide social support in times of distress, sorrow, and fear. Social support can be thought of as the soothing impact of friends, family, and acquaintances (Baron & Kerr, 2003). Social support can take many forms, including advice, guidance, encouragement, acceptance, emotional comfort, and tangible assistance (such as financial help). Thus, other people can be very comforting to us when we are faced with a wide range of life stressors, and they can be extremely helpful in our efforts to manage these challenges. Even in nonhuman animals, species mates can offer social support during times of stress. For example, elephants seem to be able to sense when other elephants are stressed and will often comfort them with physical contact such as a trunk touch or an empathetic vocal response (Krumboltz, 2014).

Scientific interest in the importance of social support first emerged in the 1970s when health researchers developed an interest in the health consequences of being socially integrated (Stroebe &

Stroebe, 1996). Interest was further fueled by longitudinal studies showing that social connectedness reduced mortality. In one classic study, nearly 7,000 Alameda County, California, residents were followed over 9 years. Those who had previously indicated that they lacked social and community ties were more likely to die during the follow-up period than those with more extensive social networks. Compared to those with the most social contacts, isolated men and women were, respectively, 2.3 and 2.8 times more likely to die. These trends persisted even after controlling for a variety of health-related variables, such as smoking, alcohol consumption, self-reported health at the beginning of the study, and physical activity (Berkman & Syme, 1979).

Since the time of that study, social support has emerged as one of the well-documented psychosocial factors affecting health outcomes (Uchino, 2009). A statistical review of 148 studies conducted between 1982 and 2007 involving over 300,000 participants concluded that individuals with stronger social relationships have a 50% greater likelihood of survival compared to those with weak or insufficient social relationships (Holt-Lunstad, Smith, & Layton, 2010). According to the researchers, the magnitude of the effect of social support observed in this study is comparable with quitting smoking and exceeded many well-known risk factors for mortality, such as obesity and physical inactivity (the accompanying source-figure description).

Source illustration description (image not reproduced): Photograph A shows four people walking along the beach with the sun setting in the distance. Photograph B shows a close relationship between three people by the water. Close relationships with others, whether (a) a group of friends or (b) a family circle, provide more than happiness and fulfillment they can help foster good health. (credit a: modification of work by "Damian Gadal_Flickr"/Flickr; credit b: modification of work by Christian Haugen)

A number of large-scale studies have found that individuals with low levels of social support are at greater risk of mortality, especially from cardiovascular disorders (Brummett et al., 2001). Further, higher levels of social supported have been linked to better survival rates following breast cancer (Falagas et al., 2007) and infectious diseases, especially HIV infection (Lee & Rotheram-Borus, 2001). In fact, a person with high levels of social support is less likely to contract a common cold. In one study, 334 participants completed questionnaires assessing their sociability; these individuals were subsequently exposed to a virus that causes a common cold and monitored for several weeks to see who became ill. Results showed that increased sociability was linearly associated with a decreased probability of developing a cold (Cohen, Doyle, Turner, Alper, & Skoner, 2003).

For many of us, friends are a vital source of social support. But what if you find yourself in a situation in which you have few friends and companions? Many students who leave home to attend and live at college experience drastic reductions in their social support, which makes them vulnerable to anxiety, depression, and loneliness. Social media can sometimes be useful in navigating these transitions (Raney & Troop Gordon, 2012) but might also cause increases in loneliness (Hunt, Marx, Lipson, & Young, 2018). For this reason, many colleges have designed first-year programs, such as peer mentoring (Raymond & Shepard, 2018), that can help students build new social networks. For some people, our families especially our parents are a major source of social support.

Social support appears to work by boosting the immune system, especially among people who are experiencing stress (Uchino, Vaughn, Carlisle, & Birmingham, 2012). In a pioneering study, spouses of cancer patients who reported high levels of social support showed indications of better immune functioning on two out of three immune functioning measures, compared to spouses who were below the median on reported social support (Baron, Cutrona, Hicklin, Russell, & Lubaroff, 1990). Studies of other populations have produced similar results, including those of spousal caregivers of dementia sufferers, medical students, elderly adults, and cancer patients (Cohen & Herbert, 1996; Kiecolt-Glaser, McGuire, Robles, & Glaser, 2002).

In addition, social support has been shown to reduce blood pressure for people performing stressful tasks, such as giving a speech or performing mental arithmetic (Lepore, 1998). In these kinds of studies, participants are usually asked to perform a stressful task either alone, with a stranger present (who may be either supportive or unsupportive), or with a friend present. Those tested with a friend

present generally exhibit lower blood pressure than those tested alone or with a stranger (Fontana, Diegnan, Villeneuve, & Lepore, 1999). In one study, 112 female participants who performed stressful mental arithmetic exhibited lower blood pressure when they received support from a friend rather than a stranger, but only if the friend was a male (Phillips, Gallagher, & Carroll, 2009). Although these findings are somewhat difficult to interpret, the authors mention that it is possible that females feel less supported and more evaluated by other females, particularly females whose opinions they value.

Taken together, the findings above suggest one of the reasons social support is connected to favorable health outcomes is because it has several beneficial physiological effects in stressful situations. However, it is also important to consider the possibility that social support may lead to better health behaviors, such as a healthy diet, exercising, smoking cessation, and cooperation with medical regimens (Uchino, 2009).

Dig Deeper

Being the recipient of prejudice and discrimination is associated with a number of negative outcomes. Many studies have shown how discrimination is a significant stressor for marginalized groups (Pascoe & Smart Richman, 2009). Discrimination negatively impacts both physical and mental health for individuals in stigmatized groups. As you ll learn when you study social psychology, various social identities (such as gender, age, religion, sexuality, ethnicity) often lead people to simultaneously be exposed to multiple forms of discrimination, which can have even stronger negative effects on mental and physical health (Vines, Ward, Cordoba, & Black, 2017). For example, the amplified levels of discrimination faced by Latinx transgender women may have related effects, leading to high stress levels and poor mental and physical health outcomes.

Perceived control and the general adaptation syndrome help explain the process by which discrimination affects mental and physical health. Discrimination can be conceptualized as an uncontrollable, persistent, and unpredictable stressor. When a discriminatory event occurs, the target of the event initially experiences an acute stress response (alarm stage). This acute reaction alone does not typically have a great impact on health. However, discrimination tends to be a chronic stressor. As people in marginalized groups experience repeated discrimination, they develop a heightened reactivity as their bodies prepare to act quickly (resistance stage). This long-term accumulation of stress responses can eventually lead to increases in negative emotion and wear on physical health (exhaustion stage). This explains why a history of perceived discrimination is associated with a host of mental and physical health problems including depression, cardiovascular disease, and cancer (Pascoe & Smart Richman, 2009).

Protecting stigmatized groups from the negative impact of discrimination-induced stress may involve reducing the incidence of discriminatory behaviors in conjunction with protective strategies that reduce the impact of discriminatory events when they occur. Civil rights legislation has protected some stigmatized groups by making discrimination a prosecutable offense in many social contexts. However, some groups (e.g., transgender people) often lack important legal recourse when discrimination occurs. Moreover, most modern discrimination comes in subtle forms that fall below the radar of the law. For example, discrimination may be experienced as selective inhospitality toward people of specific races or ethnicities, but little is done in response since it would be easy to attribute the behavior to other causes. Although some cultural changes are increasingly helping people to recognize and control subtle discrimination, such shifts may take a long time.

Similar to other stressors, buffers like social support and healthy coping strategies appear to be effective in lowering the impact of perceived discrimination. For example, one study (Ajrouch, Reisine, Lim, Sohn, & Ismail, 2010) showed that discrimination predicted high psychological distress among African American mothers living in Detroit. However, the women who had readily available emotional support from friends and family experienced less distress than those with fewer social resources. While coping strategies and social support may buffer the effects of discrimination, they fail to erase all of the negative impacts. Vigilant antidiscrimination efforts, including the development of legal protections for vulnerable groups, are needed to reduce discrimination, stress, and the resulting physical and mental health effects.

Stress Reduction Techniques

Beyond having a sense of control and establishing social support networks, there are numerous other means by which we can manage stress (the accompanying source-figure description). A common technique people use to combat stress is exercise (Salmon, 2001). It is well-established that exercise, both of long (aerobic) and short (anaerobic) duration, is beneficial for both physical and mental health (Everly & Lating, 2002). There is considerable evidence that physically fit individuals are more resistant to the adverse effects of stress and recover more quickly from stress than less physically fit individuals (Cotton, 1990). In a study of more than 500 Swiss police officers and emergency service personnel, increased physical fitness was associated with reduced stress, and regular exercise was reported to protect against stress-related health problems (Gerber, Kellman, Hartman, & P hse, 2010).

Source illustration description (image not reproduced): Photograph A shows an exercise room with several treadmills, elliptical machines, and stationary bikes. There are people exercising with multiple televisions hanging from the ceiling in front of them. Photograph B shows a person meditating next to a tree. Photograph C shows two people sitting across from each other at a table, each in front of a monitor. The person in the foreground has straps around the head holding up wires or devices. Stress reduction techniques may include (a) exercise, (b) meditation and relaxation, or (c) biofeedback. (credit a: modification of work by UNE Photos /Flickr; credit b: modification of work by Caleb Roenigk; credit c: modification of work by Dr. Carmen Russoniello)

One reason exercise may be beneficial is because it might buffer some of the deleterious physiological mechanisms of stress. One study found rats that exercised for six weeks showed a decrease in hypothalamic-pituitary-adrenal responsiveness to mild stressors (Campeau et al., 2010). In high-stress humans, exercise has been shown to prevent telomere shortening, which may explain the common observation of a youthful appearance among those who exercise regularly (Puterman et al., 2010). Further, exercise in later adulthood appears to minimize the detrimental effects of stress on the hippocampus and memory (Head, Singh, & Bugg, 2012). Among cancer survivors, exercise has been shown to reduce anxiety (Speck, Courneya, Masse, Duval, & Schmitz, 2010) and depressive symptoms (Craft, VanIterson, Helenowski, Rademaker, & Courneya, 2012). Clearly, exercise is a highly effective tool for regulating stress.

In the 1970s, Herbert Benson, a cardiologist, developed a stress reduction method called the relaxation response technique (Greenberg, 2006). The relaxation response technique combines relaxation with transcendental meditation, and consists of four components (Stein, 2001): sitting upright on a comfortable chair with feet on the ground and body in a relaxed position, being in a quiet environment with eyes closed, repeating a word or a phrase a mantra to oneself, such as alert mind, calm body, passively allowing the mind to focus on pleasant thoughts, such as nature or the warmth of your blood nourishing your body.

The relaxation response approach is conceptualized as a general approach to stress reduction that reduces sympathetic arousal, and it has been used effectively to treat people with high blood pressure (Benson & Proctor, 1994).

Another technique to combat stress, biofeedback, was developed by Gary Schwartz at Harvard University in the early 1970s. Biofeedback is a technique that uses electronic equipment to accurately measure a person s neuromuscular and autonomic activity feedback is provided in the form of visual or auditory signals. The main assumption of this approach is that providing somebody biofeedback will enable the individual to develop strategies that help gain some level of voluntary control over what are normally involuntary bodily processes (Schwartz & Schwartz, 1995). A number of different bodily measures have been used in biofeedback research, including facial muscle movement, brain activity, and skin temperature, and it has been applied successfully with individuals experiencing tension headaches, high blood pressure, asthma, and phobias (Stein, 2001).

Summary

When faced with stress, people must attempt to manage or cope with it. In general, there are two basic forms of coping: problem-focused coping and emotion-focused coping. Those who use problem-focused coping strategies tend to cope better with stress because these strategies address the source of stress rather than the resulting symptoms. To a large extent, perceived control greatly impacts reaction to stressors and is associated with greater physical and mental well-being. Social support has been demonstrated to be a highly effective buffer against the adverse effects of stress. Extensive research has shown that social support has beneficial physiological effects for people, and it seems to influence immune functioning. However, the beneficial effects of social support may be related to its influence on promoting healthy behaviors.

Review Questions

OpenStax learning exercise

Q. Emotion-focused coping would likely be a better method than problem-focused coping for dealing with which of the following stressors?

A. A

OpenStax learning exercise

Q. Studies of British civil servants have found that those in the lowest status jobs are much more likely to develop heart disease than those who have high status jobs. These findings attest to the importance of _____ in dealing with stress.

A. C

OpenStax learning exercise

Q. Relative to those with low levels of social support, individuals with high levels of social support _____.

A. D

OpenStax learning exercise

Q. The concept of learned helplessness was formulated by Seligman to explain the _____.

A. A

Critical Thinking Questions

OpenStax learning exercise

Q. Although problem-focused coping seems to be a more effective strategy when dealing with stressors, do you think there are any kinds of stressful situations in which emotion-focused coping might be a better strategy?

A. Emotion-focused coping would likely be a better coping strategy in situations in which a stressor is uncontrollable, or in which nothing could otherwise be done about it, such as a fatal illness.

OpenStax learning exercise

Q. Describe how social support can affect health both directly and indirectly.

A. Social support seems to have a direct effect on immune system functioning. Social support can affect health indirectly by influencing health-related behaviors, such as exercise and eating properly.

Personal Application Question

OpenStax learning exercise

Q. Try to think of an example in which you coped with a particular stressor by using problem-focused coping. What was the stressor? What did your problem-focused efforts involve? Were they effective?

CHAPTER 17 / CAC.901

Developmental pathways to distress and disorder

Lecture: Symptoms belong to a developmental story

Diagnostic categories can help professionals communicate, identify patterns, and access evidence-based support. They can also obscure context when used too quickly. A developmental approach asks how a difficulty began, how it changes across settings and time, and what it means for participation. The diagnosis may be one part of the formulation, not the whole account.

Children s distress can appear through behavior, bodily complaints, withdrawal, irritability, or changes in learning. These expressions are influenced by development and culture. A young child may not describe worry in abstract terms; a teenager may hide distress behind anger or silence. Counselors should not require adult-style symptom language before taking suffering seriously.

Equifinality and multifinality

Equifinality means that different pathways can lead to a similar outcome. Several children may show school avoidance for different reasons, including anxiety, bullying, learning difficulty, health problems, or family responsibilities. Multifinality means that a similar beginning or exposure can lead to different outcomes. Not every child exposed to adversity develops the same difficulty.

These concepts discourage one-cause thinking. They also show why a label should not replace assessment of mechanism. Two children with the same diagnosis may need different supports. A plan based only on category may miss the process maintaining the problem.

Comorbidity, or the presence of more than one condition, should be considered without assuming that every difficulty is a separate disorder. Sleep loss, stress, and learning demands can produce overlapping patterns. A careful timeline helps identify what appeared first and what changes together. Use qualified diagnostic assessment when indicated and remain within professional scope.

Functioning across settings

A child s behavior may vary because settings differ in demands, predictability, relationships, and cultural expectations. Variation does not prove that the difficulty is unreal. A child may hold distress together at school and release it at home, or function well with a trusted teacher and struggle in a chaotic environment. Gather specific examples rather than asking only whether a problem occurs everywhere.

Impairment should be understood in context. A child who cannot participate because a school is inaccessible experiences a mismatch between needs and environment. Support may require changing the environment as well as building skills. Avoid locating all impairment inside the child when institutional barriers are central.

Figure 17.1 shows several pathways toward a common presenting concern and several possible outcomes from a shared starting condition. It is a conceptual diagram, not a predictive model. Use it to generate questions about timing, mechanisms, and supports.

Safety and proportionate response

Assessment of distress includes attention to safety. Ask directly and calmly about self-harm or suicidal thoughts when concerns are present, using appropriate training and local procedures. Do not rely on a checklist score alone. Immediate danger requires urgent action and appropriate emergency or crisis services. A book cannot replace individualized risk assessment.

Safety work should preserve dignity and involve caregivers or other responsible adults when appropriate, while recognizing that disclosure can create additional risk in some families. Follow law and professional guidance, seek consultation, and document reasoning. Do not promise absolute confidentiality. Explain what will be shared, with whom, and why, as far as safely possible.

Application: anxiety or inaccessible teaching?

A fictional ten-year-old, Emeka, becomes restless during written work and asks to leave. Several adults suggest anxiety. A developmental assessment finds that he enjoys discussion, understands the topic orally, and becomes distressed when required to write at length. Learning assessment, classroom observation, and Emeka s account help clarify the pattern.

Anxiety may be present, but it may be connected to repeated failure rather than existing independently. The plan can address accessible instruction and appropriate learning support while also helping Emeka manage distress. Treating only the anxiety without changing the writing demand may leave the pathway intact. Treating only the learning need without attending to fear may limit participation.

The counselor should avoid presenting this as a choice between a real disorder and a social explanation. Difficulties can become layered. A developmental formulation explains how they interact and identifies where change is most feasible.

Doctoral seminar task

Create two different pathways leading to the same presenting concern. For each, identify onset, maintaining processes, protective resources, and an intervention target. Then describe one shared exposure that could lead to different outcomes and explain why.

Add a safety decision point to each pathway. Identify what would require urgent action, consultation, or information sharing. The exercise integrates developmental reasoning with the responsibility to act when a child may be unsafe.

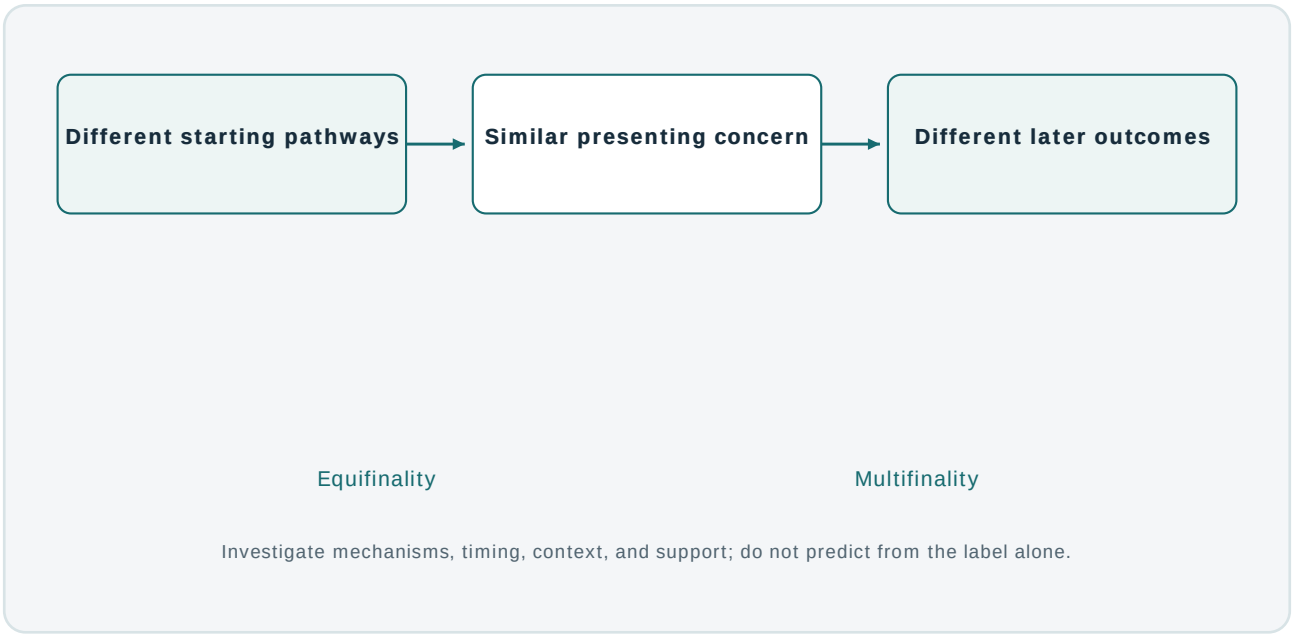


Figure 17.1. Similar concerns can have different pathways. Equifinality means that different processes can lead to a similar presenting concern. Multifinality means that a shared starting condition can lead to different outcomes. The arrows are hypotheses to investigate, not predictions for a particular child. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

What Are Psychological Disorders?

OpenStax source module m82265 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82265/index.cnxml

According to the American Psychiatric Association, a psychological disorder, or mental disorder, is a syndrome characterized by clinically significant disturbance in an individual's cognition, emotion regulation, or behavior that reflects a dysfunction in the psychological, biological, or developmental processes underlying mental functioning. Mental disorders are usually associated with significant distress in social, occupational, or other important activities (2013). Psychopathology is the study of psychological disorders, including their symptoms, etiology (i.e., their causes), and treatment. The term psychopathology can also refer to the manifestation of a psychological disorder. Although consensus can be difficult, it is extremely important for mental health professionals to agree on what kinds of thoughts, feelings, and behaviors are truly abnormal in the sense that they genuinely indicate the presence of psychopathology. Certain patterns of behavior and inner experience can easily be labeled as abnormal and clearly signify some kind of psychological disturbance. The person who washes their hands 40 times per day and the person who claims to hear the voices of demons exhibit behaviors and inner experiences that most would regard as abnormal: beliefs and behaviors that suggest the existence of a psychological disorder. But, consider the nervousness a young man feels when talking to an attractive person or the loneliness and longing for home a first-year student experiences during her first semester of college these feelings may not be regularly present, but they fall in the range of normal. So, what kinds of thoughts, feelings, and behaviors represent a true psychological disorder? Psychologists work to distinguish psychological disorders from inner experiences and behaviors that are merely situational, idiosyncratic, or unconventional.

Definition of a Psychological Disorder

Perhaps the simplest approach to conceptualizing psychological disorders is to label behaviors, thoughts, and inner experiences that are atypical, distressful, dysfunctional, and sometimes even dangerous, as signs of a disorder. For example, if you ask a classmate for a date and you are rejected, you probably would feel a little dejected. Such feelings would be normal. If you felt extremely depressed so much so that you lost interest in activities, had difficulty eating or sleeping, felt utterly worthless, and contemplated suicide your feelings would be atypical, would deviate from the norm, and could signify the presence of a psychological disorder. Just because something is atypical, however, does not necessarily mean it is disordered.

For example, only about 4% of people in the United States have red hair, so red hair is considered an atypical characteristic (the accompanying source-figure description), but it is not considered disordered, it s just unusual. And it is less unusual in Scotland, where approximately 13% of the population has red hair (DNA Project Aims, 2012). As you will learn, some disorders, although not exactly typical, are far from atypical, and the rates in which they appear in the population are surprisingly high.

Source illustration description (image not reproduced): Photograph A shows Isla Fischer. Photograph B shows Prince Harry. Photograph C shows Marcia Cross. Red hair is considered unusual, but not abnormal. (a) Isla Fischer, (b) Prince Harry, and (c) Marcia Cross are three natural redheads. (credit a: modification of work by Richard Goldschmidt; credit b: modification of work by Glyn Lowe; credit c: modification of work by Kirk Weaver)

If we can agree that merely being atypical is an insufficient criterion for a having a psychological disorder, is it reasonable to consider behavior or inner experiences that differ from widely expected cultural values or expectations as disordered? Using this criterion, a person who walks around a subway platform wearing a heavy winter coat in July while screaming obscenities at strangers may be considered as exhibiting symptoms of a psychological disorder. Their actions and clothes violate socially accepted rules governing appropriate dress and behavior; these characteristics are atypical.

Cultural Expectations

Violating cultural expectations is not, in and of itself, a satisfactory means of identifying the presence of a psychological disorder. Since behavior varies from one culture to another, what may be expected and considered appropriate in one culture may not be viewed as such in other cultures. For example, returning a stranger s smile is expected in the United States because a pervasive social norm dictates that we reciprocate friendly gestures. A person who refuses to acknowledge such gestures might be considered socially awkward perhaps even disordered for violating this expectation. However, such expectations are not universally shared. Cultural expectations in Japan involve showing reserve, restraint, and a concern for maintaining privacy around strangers. Japanese people are generally unresponsive to smiles from strangers (Patterson et al., 2007). Eye contact provides another example. In the United States and Europe, eye contact with others typically signifies honesty and attention. However, most Latin-American, Asian, and African cultures interpret direct eye contact as rude, confrontational, and aggressive (Pazain, 2010). Thus, someone who makes eye contact with you could be considered appropriate and respectful or brazen and offensive, depending on your culture (the accompanying source-figure description).

Source illustration description (image not reproduced): A photograph shows two people making eye contact during a conversation.c Eye contact is one of many social gestures that vary from culture to culture. (credit: Joi Ito)

Hallucinations (seeing or hearing things that are not physically present) in Western societies is a violation of cultural expectations, and a person who reports such inner experiences is readily labeled as psychologically disordered. In other cultures, visions that, for example, pertain to future events may be regarded as normal experiences that are positively valued (Bourguignon, 1970). Finally, it is important to recognize that cultural norms change over time: what might be considered typical in a society at one time may no longer be viewed this way later, similar to how fashion trends from one era may elicit quizzical looks decades later imagine how a headband, legwarmers, and the big hair of the 1980s would go over on your campus today.

Dig Deeper

In the 1950s and 1960s, the concept of mental illness was widely criticized. One of the major criticisms focused on the notion that mental illness was a myth that justifies psychiatric intervention in socially disapproved behavior (Wakefield, 1992). Thomas Szasz (1960), a noted psychiatrist, was perhaps the biggest proponent of this view. Szasz argued that the notion of mental illness was invented by society (and the mental health establishment) to stigmatize and subjugate people whose behavior violates accepted social and legal norms. Indeed, Szasz suggested that what appear to be symptoms of mental illness are more appropriately characterized as problems in living (Szasz, 1960).

In his 1961 book, The Myth of Mental Illness: Foundations of a Theory of Personal Conduct, Szasz expressed his disdain for the concept of mental illness and for the field of psychiatry in general (Oliver, 2006). The basis for Szasz s attack was his contention that detectable abnormalities in bodily structures and functions (e.g., infections and organ damage or dysfunction) represent the defining features of genuine illness or disease, and because symptoms of purported mental illness are not accompanied by such detectable abnormalities, so-called psychological disorders are not disorders at all. Szasz (1961/2010) proclaimed that disease or illness can only

affect the body; hence, there can be no mental illness (p. 267).

Today, we recognize the extreme level of psychological suffering experienced by people with psychological disorders: the painful thoughts and feelings they experience, the disordered behavior they demonstrate, and the levels of distress and impairment they exhibit. This makes it very difficult to deny the reality of mental illness.

However controversial Szasz s views and those of his supporters might have been, they have influenced the mental health community and society in several ways. First, lay people, politicians, and professionals now often refer to mental illness as mental health problems, implicitly acknowledging the problems in living perspective Szasz described (Buchanan-Barker & Barker, 2009). Also influential was Szasz s view of sexual orientation. Szasz was perhaps the first psychiatrist to openly challenge the idea that same-sex attraction represented a form of mental illness or disease (Szasz, 1965), which helped pave the way for the social and civil rights that gay and lesbian people now have (Barker, 2010). His work also inspired legal changes that protect the rights of people in psychiatric institutions and allow such individuals a greater degree of influence and responsibility over their lives (Buchanan-Barker & Barker, 2009).

Harmful Dysfunction

If none of the criterion discussed so far is adequate by itself to define the presence of a psychological disorder, how can a disorder be conceptualized? Many efforts have been made to identify the specific dimensions of psychological disorders, yet none is entirely satisfactory. No universal definition of psychological disorder exists that can apply to all situations in which a disorder is thought to be present (Zachar & Kendler, 2007). However, one of the more influential conceptualizations was proposed by Wakefield (1992), who defined psychological disorder as a harmful dysfunction. Wakefield argued that natural internal mechanisms that is, psychological processes honed by evolution, such as cognition, perception, and learning have important functions, such as enabling us to experience the world the way others do and to engage in rational thought, problem solving, and communication. For example, learning allows us to associate a fear with a potential danger in such a way that the intensity of fear is roughly equal to the degree of actual danger. Dysfunction occurs when an internal mechanism breaks down and can no longer perform its normal function. But, the presence of a dysfunction by itself does not determine a disorder. The dysfunction must be harmful in that it leads to negative consequences for the individual or for others, as judged by the standards of the individual s culture. The harm may include significant internal anguish (e.g., high levels of anxiety or depression) or problems in day-to-day living (e.g., in one s social or work life).

To illustrate, Janet has an extreme fear of spiders. Janet s fear might be considered a dysfunction in that it signals that the internal mechanism of learning is not working correctly (i.e., a faulty process prevents Janet from appropriately associating the magnitude of fear with the actual threat posed by spiders). Janet s fear of spiders has a significant negative influence on daily life: she avoids all situations in which she suspects spiders to be present (e.g., the basement or a friend s home), and she quit her job last month because she saw a spider in the restroom at work and is now unemployed. According to the harmful dysfunction model, Janet s condition would signify a disorder because (a) there is a dysfunction in an internal mechanism, and (b) the dysfunction has resulted in harmful consequences. Similar to how the symptoms of physical illness reflect dysfunctions in biological processes, the symptoms of psychological disorders presumably reflect dysfunctions in mental processes. The internal mechanism component of this model is especially appealing because it implies that disorders may occur through a breakdown of biological functions that govern various psychological processes, thus supporting contemporary neurobiological models of psychological disorders (Fabrega, 2007).

The American Psychiatric Association (APA) Definition

Many of the features of the harmful dysfunction model are incorporated in a formal definition of psychological disorder developed by the American Psychiatric Association (APA). According to the APA (2013), a psychological disorder is a condition that is said to consist of the following:

- There are significant disturbances in thoughts, feelings, and behaviors. A person must experience inner states (e.g., thoughts and/or feelings) and exhibit behaviors that are clearly disturbed that is,

unusual, but in a negative, self-defeating way. Often, such disturbances are troubling to those around the individual who experiences them. For example, an individual who is uncontrollably preoccupied by thoughts of germs spends hours each day bathing, has inner experiences, and displays behaviors that most would consider atypical and negative (disturbed) and that would likely be troubling to family members.

- The disturbances reflect some kind of biological, psychological, or developmental dysfunction. Disturbed patterns of inner experiences and behaviors should reflect some flaw (dysfunction) in the internal biological, psychological, and developmental mechanisms that lead to normal, healthy psychological functioning. For example, the hallucinations observed in schizophrenia could be a sign of brain abnormalities.
- The disturbances lead to significant distress or disability in one s life. A person s inner experiences and behaviors are considered to reflect a psychological disorder if they cause the person considerable distress, or greatly impair their ability to function as a normal individual (often referred to as functional impairment, or occupational and social impairment). As an illustration, a person s fear of social situations might be so distressing that it causes the person to avoid all social situations (e.g., preventing that person from being able to attend class or apply for a job).
- The disturbances do not reflect expected or culturally approved responses to certain events. Disturbances in thoughts, feelings, and behaviors must be socially unacceptable responses to certain events that often happen in life. For example, it is perfectly natural (and expected) that a person would experience great sadness and might wish to be left alone following the death of a close family member. Because such reactions are in some ways culturally expected, the individual would not be assumed to signify a mental disorder.

Some believe that there is no essential criterion or set of criteria that can definitively distinguish all cases of disorder from nondisorder (Lilienfeld & Marino, 1999). In truth, no single approach to defining a psychological disorder is adequate by itself, nor is there universal agreement on where the boundary is between disordered and not disordered. From time to time we all experience anxiety, unwanted thoughts, and moments of sadness; our behavior at other times may not make much sense to ourselves or to others. These inner experiences and behaviors can vary in their intensity, but are only considered disordered when they are highly disturbing to us and/or others, suggest a dysfunction in normal mental functioning, and are associated with significant distress or disability in social or occupational activities.

Summary

Psychological disorders are conditions characterized by abnormal thoughts, feelings, and behaviors. Although challenging, it is essential for psychologists and mental health professionals to agree on what kinds of inner experiences and behaviors constitute the presence of a psychological disorder. Inner experiences and behaviors that are atypical or violate social norms could signify the presence of a disorder; however, each of these criteria alone is inadequate. Harmful dysfunction describes the view that psychological disorders result from the inability of an internal mechanism to perform its natural function. Many of the features of harmful dysfunction conceptualization have been incorporated in the APA s formal definition of psychological disorders. According to this definition, the presence of a psychological disorder is signaled by significant disturbances in thoughts, feelings, and behaviors; these disturbances must reflect some kind of dysfunction (biological, psychological, or developmental), must cause significant impairment in one s life, and must not reflect culturally expected reactions to certain life events.

Review Questions

OpenStax learning exercise

Q. In the harmful dysfunction definition of psychological disorders, dysfunction involves _____.

A. A

OpenStax learning exercise

Q. Patterns of inner experience and behavior are thought to reflect the presence of a psychological disorder if they _____.

A. B

Critical Thinking Question

OpenStax learning exercise

Q. Discuss why thoughts, feelings, or behaviors that are merely atypical or unusual would not necessarily signify the presence of a psychological disorder. Provide an example.

A. Just because something is atypical or unusual does not mean it is disordered. A person may experience atypical inner experiences or exhibit unusual behaviors, but she would not be considered disordered if they are not distressing, disturbing, or reflecting a dysfunction. For example, a classmate might stay up all night studying before exams; although atypical, this behavior is unlikely to possess any of the other criteria for psychological disorder mentioned previously.

Personal Application Question

OpenStax learning exercise

Q. Identify a behavior that is considered unusual or abnormal in your own culture; however, it would be considered normal and expected in another culture.

Diagnosing and Classifying Psychological Disorders

OpenStax source module m82266 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82266/index.cnxml

A first step in the study of psychological disorders is carefully and systematically discerning significant signs and symptoms. How do mental health professionals ascertain whether or not a person s inner states and behaviors truly represent a psychological disorder? Arriving at a proper diagnosis that is, appropriately identifying and labeling a set of defined symptoms is absolutely crucial. This process enables professionals to use a common language with others in the field and aids in communication about the disorder with the patient, colleagues and the public. A proper diagnosis is an essential element to guide proper and successful treatment. For these reasons, classification systems that organize psychological disorders systematically are necessary.

The Diagnostic and Statistical Manual of Mental Disorders (DSM)

Although a number of classification systems have been developed over time, the one that is used by most mental health professionals in the United States is the Diagnostic and Statistical Manual of Mental Disorders (DSM-5), published by the American Psychiatric Association (2013). (Note that the American Psychiatric Association differs from the American Psychological Association; both are abbreviated APA.) The first edition of the DSM, published in 1952, classified psychological disorders according to a format developed by the U.S. Army during World War II (Clegg, 2012). In the years since, the DSM has undergone numerous revisions and editions. The most recent edition, published in 2013, is the DSM-5 (APA, 2013). The DSM-5 includes many categories of disorders (e.g., anxiety disorders, depressive disorders, and dissociative disorders). Each disorder is described in detail, including an overview of the disorder (diagnostic features), specific symptoms required for diagnosis (diagnostic criteria), prevalence information (what percent of the population is thought to be afflicted with the disorder), and risk factors associated with the disorder. the accompanying source-figure description shows lifetime prevalence rates the percentage of people in a population who develop a disorder in their lifetime of various psychological disorders among U.S. adults. These data were based on a national sample of 9,282 U.S. residents (National Comorbidity Survey, 2007).

Source illustration description (image not reproduced): A bar graph has an x-axis labeled DSM disorder and a y-axis labeled Lifetime prevalence rates. For each disorder, a prevalence rate is given for total population, females, and males. The approximate data shown is: major depressive disorder 17% total, 20% females, 13% males; alcohol abuse 13% total, 7% females, 20% males; specific phobia 13% total, 16% females, 8% males; social anxiety disorder 12% total, 13% females, 11% males; drug abuse 8% total, 5% females, 12% males; posttraumatic stress disorder 7% total, 10% females, 3% males; generalized anxiety disorder 6% total, 7% females, 4% males; panic disorder 5% total, 6% females, 3% males; obsessive-compulsive disorder 3% total, 3% females, 2% males; dysthymia 3% total, 3% females, 2% males. The graph shows the breakdown of psychological disorders, comparing the percentage prevalence among adult males and adult females in the United States. Because the data is from 2007, the categories shown here are from the DSM-IV, which has been supplanted by the DSM-5. Most categories remain the same; however, alcohol abuse now falls under a broader Alcohol Use Disorder category.

The DSM-5 also provides information about comorbidity; the co-occurrence of two disorders. For example, the DSM-5 mentions that 41% of people with obsessive-compulsive disorder (OCD) also meet the diagnostic criteria for major depressive disorder (the accompanying source-figure description). Problematic substance use is highly comorbid with other mental illnesses; 6 out of 10 people who have a substance use disorder also suffer from another form of mental illness (National Institute on Drug Abuse [NIDA], 2007).

Source illustration description (image not reproduced): A Venn-diagram shows two overlapping circles. One circle is titled Obsessive-Compulsive Disorder and the other is titled Major Depressive Disorder. The area in which these two circles overlap includes forty-one percent of each circle. This area is titled Comorbidity 41%. Obsessive-compulsive disorder and major depressive disorder frequently occur in the same person.

Connect the Concepts

As you ve learned in the text, comorbidity refers to situations in which an individual suffers from more than one disorder, and often the symptoms of each can interact in negative ways. Co-occurrence and comorbidity of psychological disorders are quite common, and some of the most pervasive comorbidities involve substance use disorders that co-occur with psychological disorders. Indeed, some estimates suggest that around a quarter of people who suffer from the most severe cases of mental illness exhibit substance use disorder as well. Conversely, around 10 percent of individuals seeking treatment for substance use disorder have serious mental illnesses. Observations such as these have important implications for treatment options that are available. When people with a mental illness are also habitual drug users, their symptoms can be exacerbated and resistant to treatment. Furthermore, it is not always clear whether the symptoms are due to drug use, the mental illness, or a combination of the two. Therefore, it is recommended that behavior is observed in situations in which the individual has ceased using drugs and is no longer experiencing withdrawal from the drug in order to make the most accurate diagnosis (NIDA, 2018).

Obviously, substance use disorders are not the only possible comorbidities. In fact, some of the most common psychological disorders tend to co-occur. For instance, more than half of individuals who have a primary diagnosis

of depressive disorder are estimated to exhibit some sort of anxiety disorder. The reverse is also true for those diagnosed with a primary diagnosis of an anxiety disorder. Further, anxiety disorders and major depression have a high rate of comorbidity with several other psychological disorders (Al-Asadi, Klein, & Meyer, 2015).

The DSM has changed considerably in the half-century since it was originally published. The first two editions of the DSM, for example, listed homosexuality as a disorder; however, in 1973, the APA voted to remove it from the manual (Silverstein, 2009). While the DSM-III did not list homosexuality as a disorder, it introduced a new diagnosis, ego-dystonic homosexuality, which emphasized same-sex arousal that the patient viewed as interfering with desired heterosexual relationships and causing distress for the individual. This new diagnosis was considered by many as a compromise to appease those who viewed homosexuality as a mental illness. Other professionals questioned how appropriate it was to have a separate diagnosis that described the content of an individual's distress. In 1986, the diagnosis was removed from the DSM-III-R (Herek, 2012). Additionally, beginning with the DSM-III in 1980, mental disorders have been described in much greater detail, and the number of diagnosable conditions has grown steadily, as has the size of the manual itself. DSM-I included 106 diagnoses and was 130 total pages, whereas DSM-III included more than 2 times as many diagnoses (265) and was nearly seven times its size (886 total pages) (Mayes & Horowitz, 2005). Although DSM-5 is longer than DSM-IV, the volume includes only 237 disorders, a decrease from the 297 disorders that were listed in DSM-IV. The latest edition, DSM-5, includes revisions in the organization and naming of categories and in the diagnostic criteria for various disorders (Regier, Kuhl, & Kupfer, 2012), while emphasizing careful consideration of the importance of gender and cultural difference in the expression of various symptoms (Fisher, 2010).

Some believe that establishing new diagnoses might overpathologize the human condition by turning common human problems into mental illnesses (The Associated Press, 2013). Indeed, the finding that nearly half of all Americans will meet the criteria for a DSM disorder at some point in their life (Kessler et al., 2005) likely fuels much of this skepticism. The DSM-5 is also criticized on the grounds that its diagnostic criteria have been loosened, thereby threatening to turn our current diagnostic inflation into diagnostic hyperinflation (Frances, 2012, para. 22). For example, DSM-IV specified that the symptoms of major depressive disorder must not be attributable to normal bereavement (loss of a loved one). The DSM-5, however, has removed this bereavement exclusion, essentially meaning that grief and sadness after a loved one s death can constitute major depressive disorder.

The International Classification of Diseases

A second classification system, the International Classification of Diseases (ICD), is also widely recognized. Published by the World Health Organization (WHO), the ICD was developed in Europe shortly after World War II and, like the DSM, has been revised several times. The categories of psychological disorders in both the DSM and ICD are similar, as are the criteria for specific disorders; however, some differences exist. Although the ICD is used for clinical purposes, this tool is also used to examine the general health of populations and to monitor the prevalence of diseases and other health problems internationally (WHO, 2013). The ICD is in its 10th edition (ICD-10); however, efforts are now underway to develop a new edition (ICD-11) that, in conjunction with the changes in DSM-5, will help harmonize the two classification systems as much as possible (APA, 2013).

A study that compared the use of the two classification systems found that worldwide the ICD is more frequently used for clinical diagnosis, whereas the DSM is more valued for research (Mezzich, 2002). Most research findings concerning the etiology and treatment of psychological disorders are based on criteria set forth in the DSM (Oltmanns & Castonguay, 2013). The DSM also includes more explicit disorder criteria, along with an extensive and helpful explanatory text (Regier et al., 2012). The DSM is the classification system of choice among U.S. mental health professionals, and this chapter is based on the DSM paradigm.

The Compassionate View of Psychological Disorders

As these disorders are outlined, please bear two things in mind. First, remember that psychological disorders represent extremes of inner experience and behavior. If, while reading about these disorders, you feel that these descriptions begin to personally characterize you, do not worry this moment of enlightenment probably means nothing more than you have had a range of experiences and emotions. Each of us experiences episodes of sadness, anxiety, and preoccupation with certain thoughts times when we do not quite feel ourselves. These episodes should not be considered problematic unless the accompanying thoughts and behaviors become extreme and have a disruptive effect on one s life. Second, understand that people with psychological disorders are far more than just embodiments of their disorders. We do not use terms such as schizophrenics, depressives, or phobics because they are labels that objectify people who have these conditions, thus promoting biased and disparaging assumptions about them. It is important to remember that a psychological disorder is not what a person is; it is something that a person has through no fault of their own. As is the case with cancer or diabetes, those with psychological disorders suffer debilitating, often painful conditions that are not of their own choosing. These individuals deserve to be viewed and treated with compassion, understanding, and dignity.

Summary

The diagnosis and classification of psychological disorders is essential in studying and treating psychopathology. The classification system used by most U.S. professionals is the DSM-5. The first edition of the DSM was published in 1952, and has undergone numerous revisions. The 5th and most recent edition, the DSM-5, was published in 2013. The diagnostic manual includes a total of 237 specific diagnosable disorders, each described in detail, including its symptoms, prevalence, risk factors, and comorbidity. Over time, the number of diagnosable conditions listed in the DSM has grown steadily, prompting criticism from some. Nevertheless, the diagnostic criteria in the DSM are more explicit than that of any other system, which makes the DSM system highly desirable for both clinical diagnosis and research.

Review Questions

OpenStax learning exercise

Q. The letters in the abbreviation DSM-5 stand for _____.

A. D

OpenStax learning exercise

Q. A study based on over 9,000 U. S. residents found that the most prevalent disorder was _____.

A. A

Critical Thinking Questions

OpenStax learning exercise

Q. Describe the DSM-5. What is it, what kind of information does it contain, and why is it important to the study and treatment of psychological disorders?

A. The DSM-5 is the classification system of psychological disorders preferred by most U.S. mental health professionals, and it is published by the American Psychiatric Association (APA). It consists of broad categories of disorders and specific disorders that fall within each category. Each disorder has an explicit description of its symptoms, as well as information concerning prevalence, risk factors,

and comorbidity. The DSM-5 provides a common language that enables mental health professionals to communicate effectively about sets of symptoms.

OpenStax learning exercise

Q. The International Classification of Diseases (ICD) and the DSM differ in various ways. What are some of the differences in these two classification systems?

A. The ICD is used primarily for making clinical diagnoses and more broadly for examining the general health of populations and monitoring the international prevalence of diseases and other health problems. While the DSM is also used for diagnostic purposes, it is also highly valued as a research tool. For example, much of the data regarding the etiology and treatment of psychological disorders are based on diagnostic criteria set forth in the DSM.

Perspectives on Psychological Disorders

OpenStax source module m82267 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82267/index.cnxml

Scientists, mental health professionals, and cultural healers may adopt different perspectives in attempting to understand or explain the underlying mechanisms that contribute to the development of a psychological disorder. The specific perspective used in explaining a psychological disorder is extremely important. Each perspective explains psychological disorders, their causes or etiology, and effective treatments from a different viewpoint. Different perspectives provide alternate ways for how to think about the nature of psychopathology.

Supernatural Perspectives of Psychological Disorders

For centuries, psychological disorders were viewed from a supernatural perspective: attributed to a force beyond scientific understanding. Those afflicted were thought to be practitioners of black magic or possessed by spirits (the accompanying source-figure description) (Maher & Maher, 1985). For example, convents throughout Europe in the 16th and 17th centuries reported hundreds of nuns falling into a state of frenzy in which the afflicted foamed at the mouth, screamed and convulsed, sexually propositioned priests, and confessed to having carnal relations with devils or Christ. Although, today, these cases would suggest serious mental illness; at the time, these events were routinely explained as possession by devilish forces (Waller, 2009a). Similarly, grievous fits by young girls are believed to have precipitated the witch panic in New England late in the 17th century (Demos, 1983). Such beliefs in supernatural causes of mental illness are still held in some societies today; for example, beliefs that supernatural forces cause mental illness are common in some cultures in modern-day Nigeria (Aghukwa, 2012).

Source illustration description (image not reproduced): The Extraction of the Stone of Madness is shown. In The Extraction of the Stone of Madness, a 15th century painting by Hieronymus Bosch, a practitioner is using a tool to extract an object (the supposed stone of madness) from the head of an afflicted person.

Dig Deeper

Between the 11th and 17th centuries, a curious epidemic swept across Western Europe. Groups of people would suddenly begin to dance with wild abandon. This compulsion to dance referred to as dancing mania sometimes gripped thousands of people at a time (the accompanying source-figure description). Historical accounts indicate that those afflicted would sometimes dance with bruised and bloody feet for days or weeks, screaming of terrible visions and begging priests and monks to save their souls (Waller, 2009b). What caused dancing mania is not known, but several explanations have been proposed, including spider venom and ergot poisoning (Dancing Mania, 2011).

Source illustration description (image not reproduced): A painting shows a group of pilgrims dancing in a way that appears inconsistent and aimless. Although the cause of dancing mania, depicted in this painting, was unclear, the behavior was attributed to supernatural forces.

Historian John Waller (2009a, 2009b) has provided a comprehensive and convincing explanation of dancing mania that suggests the phenomenon was attributable to a combination of three factors: psychological distress, social contagion, and belief in supernatural forces. Waller argued that various disasters of the time (such as famine, plagues, and floods) produced high levels of psychological distress that could increase the likelihood of succumbing to an involuntary trance state. Waller indicated that anthropological studies and accounts of possession rituals show that people are more likely to enter a trance state if they expect it to happen, and that entranced individuals behave in a ritualistic manner, their thoughts and behavior shaped by the spiritual beliefs of their culture. Thus, during periods of extreme physical and mental distress, all it took were a few people believing themselves to have been afflicted with a dancing curse to slip into a spontaneous trance and then act out the part of one who is cursed by dancing for days on end.

Biological Perspectives of Psychological Disorders

The biological perspective views psychological disorders as linked to biological phenomena, such as genetic factors, chemical imbalances, and brain abnormalities; it has gained considerable attention and acceptance in recent decades (Wyatt & Midkiff, 2006). Evidence from many sources indicates that most psychological disorders have a genetic component; in fact, there is little dispute that some disorders are largely due to genetic factors. The graph in the accompanying source-figure description shows heritability estimates for schizophrenia.

Source illustration description (image not reproduced): A bar graph has an x-axis labeled Percent risk of developing schizophrenia and a y-axis labeled relationship to person with schizophrenia. A series of relationships are correlated with the percentage risk, shown with brackets indicating the generic relationship. The general population has a 1% risk. First cousins have 2% risk; they share 12.5% of genes. The next relationships are uncles/aunts, nephews/nieces, grandchildren, and half-siblings; they share 25% of genes and the risk ranges from about 3 6%. The next relationships are parents, siblings, children, and fraternal twins; they share 50% of genes and the risks are about 6, 9, 13, and 17%, respectively. Identical twins share 100% of genes and have about a 48% risk. A person s risk of developing schizophrenia increases if a relative has schizophrenia. The closer the genetic relationship, the higher the risk.

Findings such as these have led many of today s researchers to search for specific genes and genetic mutations that contribute to mental disorders. Also, sophisticated neural imaging technology in recent decades has revealed how abnormalities in brain structure and function might be directly involved in many disorders, and advances in our understanding of neurotransmitters and hormones have yielded insights into their possible connections. The biological perspective is currently thriving in the study of psychological disorders.

The Diathesis-Stress Model of Psychological Disorders

Despite advances in understanding the biological basis of psychological disorders, the psychosocial perspective is still very important. This perspective emphasizes the importance of learning, stress, faulty and self-defeating thinking patterns, and environmental factors. Perhaps the best way to think about psychological disorders, then, is to view them as originating from a combination of biological and psychological processes. Many develop not from a single cause, but from a delicate fusion between partly biological and partly psychosocial factors.

The diathesis-stress model (Zuckerman, 1999) integrates biological and psychosocial factors to predict the likelihood of a disorder. This diathesis-stress model suggests that people with an underlying predisposition for a disorder (i.e., a diathesis) are more likely than others to develop a disorder when faced with adverse environmental or psychological events (i.e., stress), such as childhood maltreatment, negative life events, trauma, and so on. A diathesis is not always a biological vulnerability to an illness; some diatheses may be psychological (e.g., a tendency to think about life events in a pessimistic, self-defeating way).

The key assumption of the diathesis-stress model is that both factors, diathesis and stress, are necessary in the development of a disorder. Different models explore the relationship between the two factors: the level of stress needed to produce the disorder is inversely proportional to the level of diathesis.

Summary

Psychopathology is very complex, involving a plethora of etiological theories and perspectives. For centuries, psychological disorders were viewed primarily from a supernatural perspective and thought to arise from divine forces or possession from spirits. Some cultures continue to hold this supernatural belief. Today, many who study psychopathology view mental illness from a biological perspective, whereby psychological disorders are thought to result largely from faulty biological processes. Indeed, scientific advances over the last several decades have provided a better understanding of the genetic, neurological, hormonal, and biochemical bases of psychopathology. The psychological perspective, in contrast, emphasizes the importance of psychological factors (e.g., stress and thoughts) and environmental factors in the development of psychological disorders. A contemporary, promising approach is to view disorders as originating from an integration of biological and psychosocial factors. The diathesis-stress model suggests that people with an underlying diathesis, or vulnerability, for a psychological disorder are more likely than those without the diathesis to develop the disorder when faced with stressful events.

Review Questions

OpenStax learning exercise

Q. The diathesis-stress model presumes that psychopathology results from _____.

A. A

OpenStax learning exercise

Q. Dr. Anastasia believes that major depressive disorder is caused by an over-secretion of cortisol. His view on the cause of major depressive disorder reflects a _____ perspective.

A. C

Critical Thinking Question

OpenStax learning exercise

Q. Why is the perspective one uses in explaining a psychological disorder important?

A. The perspective one uses in explaining a psychological disorder consists of assumptions that will guide how to best study and understand the nature of a disorder, including its causes, and how to most effectively treat the disorder.

Personal Application Question

OpenStax learning exercise

Q. Even today, some believe that certain occurrences have supernatural causes. Think of an event, recent or historical, for which others have provided supernatural explanation.

Anxiety Disorders

OpenStax source module m82268 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82268/index.cnxml

Everybody experiences anxiety from time to time. Although anxiety is closely related to fear, the two states possess important differences. Fear involves an instantaneous reaction to an imminent threat, whereas anxiety involves apprehension, avoidance, and cautiousness regarding a potential threat, danger, or other negative event (Craske, 1999). While anxiety is unpleasant to most people, it is important to our health, safety, and well-being. Anxiety motivates us to take actions such as preparing for exams, watching our weight, showing up to work on time that enable us to avert potential future problems. Anxiety also motivates us to avoid certain things such as running up debts and engaging in illegal activities that could lead to future trouble. Most individuals level and duration of anxiety approximates the magnitude of the potential threat they face. For example, suppose a student who came to the U.S. as a Dreamer (someone whose parents didn t lawfully immigrate) is concerned about the possibility of being unable to continue in the university program or of losing access to academic financial aid, due to changes and litigation around the Deferred Action for Childhood Arrivals (DACA) program. This person likely would experience anxiety of greater intensity and duration than would a 21-year-old junior who entered college as a birthright citizen. Some people experience anxiety that is excessive, persistent, and greatly out of proportion to the actual threat; if one s anxiety has a disruptive influence on one s life, this is a strong indicator that the individual is experiencing an anxiety disorder.

Anxiety disorders are characterized by excessive and persistent fear and anxiety, and by related disturbances in behavior (APA, 2013). Although anxiety is universally experienced, anxiety disorders cause considerable distress. As a group, anxiety disorders are common: approximately 25% 30% of the U.S. population meets the criteria for at least one anxiety disorder during their lifetime (Kessler et al., 2005). Also, these disorders appear to be much more common in women than they are in men; within a 12-month period, around 23% of women and 14% of men will experience at least one anxiety disorder (National Comorbidity Survey, 2007). Anxiety disorders are the most frequently occurring class of mental disorders and are often comorbid with each other and with other mental disorders (Kessler, Ruscio, Shear, & Wittchen, 2009).

Specific Phobia

Phobia is a Greek word that means fear. A person diagnosed with a specific phobia (formerly known as simple phobia) experiences excessive, distressing, and persistent fear or anxiety about a specific object or situation (such as animals, enclosed spaces, elevators, or flying) (APA, 2013). Even though people realize their level of fear and anxiety in relation to the phobic stimulus is irrational, some people with a specific phobia may go to great lengths to avoid the phobic stimulus (the object or situation that triggers the fear and anxiety). Typically, the fear and anxiety a phobic stimulus elicits is disruptive to the person s life. For example, a man with a phobia of flying might refuse to accept a job that requires frequent air travel, thus negatively affecting his career. Clinicians who have worked with people who have specific phobias have encountered many kinds of phobias, some of which are shown in the accompanying source-figure description.

Source comparison table

Specific Phobias	
Phobia	Feared Object or Situation
Acrophobia	heights
Aerophobia	flying
Arachnophobia	spiders

Specific Phobias	
Claustrophobia	enclosed spaces
Cynophobia	dogs
Hematophobia	blood
Ophidiophobia	snakes
Taphophobia	being buried alive
Trypanophobia	injections
Xenophobia	strangers

Specific phobias are common; in the United States, around 12.5% of the population will meet the criteria for a specific phobia at some point in their lifetime (Kessler et al., 2005). One type of phobia, agoraphobia, is listed in the DSM-5 as a separate anxiety disorder. Agoraphobia, which literally means fear of the marketplace, is characterized by intense fear, anxiety, and avoidance of situations in which it might be difficult to escape or receive help if one experiences symptoms of a panic attack (a state of extreme anxiety that we will discuss shortly). These situations include public transportation, open spaces (parking lots), enclosed spaces (stores), crowds, or being outside the home alone (APA, 2013). About 1.4% of Americans experience agoraphobia during their lifetime (Kessler et al., 2005).

Acquisition of Phobias Through Learning

Many theories suggest that phobias develop through learning. Rachman (1977) proposed that phobias can be acquired through three major learning pathways. The first pathway is through classical conditioning. As you may recall, classical conditioning is a form of learning in which a previously neutral stimulus is paired with an unconditioned stimulus (UCS) that reflexively elicits an unconditioned response (UCR), eliciting the same response through its association with the unconditioned stimulus. The response is called a conditioned response (CR). For example, a child who has been bitten by a dog may come to fear dogs because of a past association with pain. In this case, the dog bite is the UCS and the fear it elicits is the UCR. Because a dog was associated with the bite, any dog may come to serve as a conditioned stimulus, thereby eliciting fear; the fear the child experiences around dogs, then, becomes a CR.

The second pathway of phobia acquisition is through vicarious learning, such as modeling. For example, a child who observes their cousin react fearfully to spiders may later express the same fears, even though spiders have never presented any danger to them. This phenomenon has been observed in both humans and nonhuman primates (Olsson & Phelps, 2007). A study of laboratory-reared monkeys readily acquired a fear of snakes after observing wild-reared monkeys react fearfully to snakes (Mineka & Cook, 1993).

The third pathway is through verbal transmission or information. For example, a child whose parents, siblings, friends, and classmates constantly tell them how disgusting and dangerous snakes are may come to acquire a fear of snakes.

Interestingly, people are more likely to develop phobias of things that do not represent much actual danger to themselves, such as animals and heights, and are less likely to develop phobias toward things that present legitimate danger in contemporary society, such as motorcycles and weapons (hman & Mineka, 2001). Why might this be so? One theory suggests that the human brain is evolutionarily predisposed to more readily associate certain objects or situations with fear (Seligman, 1971). This theory argues that throughout our evolutionary history, our ancestors associated certain stimuli (e.g., snakes, spiders, heights, and thunder) with potential danger. As time progressed, the mind has become adapted to more readily develop fears of these things than of others. Experimental evidence has

consistently demonstrated that conditioned fears develop more readily to fear-relevant stimuli (images of snakes and spiders) than to fear-irrelevant stimuli (images of flowers and berries) (hman & Mineka, 2001). Such prepared learning has also been shown to occur in monkeys. In one study (Cook & Mineka, 1989), monkeys watched videotapes of model monkeys reacting fearfully to either fear-relevant stimuli (toy snakes or a toy crocodile) or fear-irrelevant stimuli (flowers or a toy rabbit). The observer monkeys developed fears of the fear-relevant stimuli but not the fear-irrelevant stimuli.

Social Anxiety Disorder

Social anxiety disorder (formerly called social phobia) is characterized by extreme and persistent fear or anxiety and avoidance of social situations in which the person could potentially be evaluated negatively by others (APA, 2013). As with specific phobias, social anxiety disorder is common in the United States; a little over 12% of all Americans experience social anxiety disorder during their lifetime (Kessler et al., 2005).

The heart of the fear and anxiety in social anxiety disorder is the person s concern that they may act in a humiliating or embarrassing way, such as appearing foolish, showing symptoms of anxiety (blushing), or doing or saying something that might lead to rejection (such as offending others). The kinds of social situations in which individuals with social anxiety disorder usually have problems include public speaking, having a conversation, meeting strangers, eating in restaurants, and, in some cases, using public restrooms. Although many people become anxious in social situations like public speaking, the fear, anxiety, and avoidance experienced in social anxiety disorder are highly distressing and lead to serious impairments in life. Adults with this disorder are more likely to experience lower educational attainment and lower earnings (Katzelnick et al., 2001), perform more poorly at work and are more likely to be unemployed (Moitra, Beard, Weisberg, & Keller, 2011), and report greater dissatisfaction with their family lives, friends, leisure activities, and income (Stein & Kean, 2000).

When people with social anxiety disorder are unable to avoid situations that provoke anxiety, they typically perform safety behaviors: mental or behavioral acts that reduce anxiety in social situations by reducing the chance of negative social outcomes. Safety behaviors include avoiding eye contact, rehearsing sentences before speaking, talking only briefly, and not talking about oneself (Alden & Bieling, 1998). Other examples of safety behaviors include the following (Marker, 2013):

- assuming roles in social situations that minimize interaction with others (e.g., taking pictures, setting up equipment, or helping prepare food)
- asking people many questions to keep the focus off of oneself
- selecting a position to avoid scrutiny or contact with others (sitting in the back of the room)
- wearing bland, neutral clothes to avoid drawing attention to oneself
- avoiding substances or activities that might cause anxiety symptoms (such as caffeine, warm clothing, and physical exercise)

Although these behaviors are intended to prevent the person with social anxiety disorder from doing something awkward that might draw criticism, these actions usually exacerbate the problem because they do not allow the individual to disconfirm their negative beliefs, often eliciting rejection and other negative reactions from others (Alden & Bieling, 1998).

People with social anxiety disorder may resort to self-medication, such as drinking alcohol, as a means to avert the anxiety symptoms they experience in social situations (Battista & Kocovski, 2010). The use of alcohol when faced with such situations may become negatively reinforcing: encouraging individuals with social anxiety disorder to turn to the substance whenever they experience anxiety symptoms. The tendency to use alcohol as a coping mechanism for social anxiety, however, can come with a hefty price tag: a number of large scale studies have reported a high rate of comorbidity between social anxiety disorder and alcohol use disorder (Morris, Stewart, & Ham, 2005).

As with specific phobias, it is highly probable that the fears inherent to social anxiety disorder can develop through conditioning experiences. For example, a child who is subjected to early unpleasant social experiences (e.g., bullying at school) may develop negative social images of herself that become activated later in anxiety-provoking situations (Hackmann, Clark, & McManus, 2000). Indeed, one study reported that 92% of a sample of adults with social anxiety disorder reported a history of severe teasing in childhood, compared to only 35% of a sample of adults with panic disorder (McCabe, Antony, Summerfeldt, Liss, & Swinson, 2003).

One of the most well-established risk factors for developing social anxiety disorder is behavioral inhibition (Clauss & Blackford, 2012). Behavioral inhibition is thought to be an inherited trait, and it is characterized by a consistent tendency to show fear and restraint when presented with unfamiliar people or situations (Kagan, Reznick, & Snidman, 1988). Behavioral inhibition is displayed very early in life; behaviorally inhibited toddlers and children respond with great caution and restraint in unfamiliar situations, and they are often timid, fearful, and shy around unfamiliar people (Fox, Henderson, Marshall, Nichols, & Ghera, 2005). A recent statistical review of studies demonstrated that behavioral inhibition was associated with more than a sevenfold increase in the risk of development of social anxiety disorder, demonstrating that behavioral inhibition is a major risk factor for the disorder (Clauss & Blackford, 2012).

Panic Disorder

Imagine that you are at the mall one day with your friends and suddenly and inexplicably you begin sweating and trembling, your heart starts pounding, you have trouble breathing, and you start to feel dizzy and nauseous. This episode lasts for 10 minutes and is terrifying because you start to think that you are going to die. When you visit your doctor the following morning and describe what happened, she tells you that you have experienced a panic attack (the accompanying source-figure description). If you experience another one of these episodes two weeks later and worry for a month or more that similar episodes will occur in the future, it is likely that you have developed panic disorder.

Source illustration description (image not reproduced): A diagram shows an outline of a person s upper body. Within this outline, some of the major organs appear. The brain is labeled, Feeling dizzy, unsteady, lightheaded. The heart is labeled, Chest pain, palpitations and/or accelerated heart rate. The lungs are labeled, Shortness of breath. The stomach is labeled, Nausea or abdominal distress. Some of the physical manifestations of a panic attack are shown. People may also experience sweating, trembling, feelings of faintness, or a fear of losing control, among other symptoms.

People with panic disorder experience recurrent (more than one) and unexpected panic attacks, along with at least one month of persistent concern about additional panic attacks, worry over the consequences of the attacks, or self-defeating changes in behavior related to the attacks (e.g., avoidance of exercise or unfamiliar situations) (APA, 2013). As is the case with other anxiety disorders, the panic attacks cannot result from the physiological effects of drugs and other substances, a medical condition, or another mental disorder. A panic attack is defined as a period of extreme fear or discomfort that develops abruptly and reaches a peak within 10 minutes. Its symptoms include accelerated heart rate, sweating, trembling, choking sensations, hot flashes or chills, dizziness or lightheadedness, fears of losing control or going crazy, and fears of dying (APA, 2013). Sometimes panic attacks are expected, occurring in response to specific environmental triggers (such as being in a tunnel); other times, these episodes are unexpected and emerge randomly (such as when relaxing). According to the DSM-5, the person must experience unexpected panic attacks to qualify for a diagnosis of panic disorder.

Experiencing a panic attack is often terrifying. Rather than recognizing the symptoms of a panic attack merely as signs of intense anxiety, individuals with panic disorder often misinterpret them as a sign that something is intensely wrong internally (thinking, for example, that the pounding heart represents an impending heart attack). Panic attacks can occasionally precipitate trips to the emergency room because several symptoms of panic attacks are, in fact, similar to those associated with heart problems (e.g., palpitations, racing pulse, and a pounding sensation in the chest) (Root, 2000). Unsurprisingly,

those with panic disorder fear future attacks and may become preoccupied with modifying their behavior in an effort to avoid future panic attacks. For this reason, panic disorder is often characterized as fear of fear (Goldstein & Chambless, 1978).

Panic attacks themselves are not mental disorders. Indeed, around 23% of Americans experience isolated panic attacks in their lives without meeting the criteria for panic disorder (Kessler et al., 2006), indicating that panic attacks are fairly common. Panic disorder is, of course, much less common, afflicting 4.7% of Americans during their lifetime (Kessler et al., 2005). Many people with panic disorder develop agoraphobia, which is marked by fear and avoidance of situations in which escape might be difficult or help might not be available if one were to develop symptoms of a panic attack. People with panic disorder often experience a comorbid disorder, such as other anxiety disorders or major depressive disorder (APA, 2013).

Researchers are not entirely sure what causes panic disorder. Children are at a higher risk of developing panic disorder if their parents have the disorder (Biederman et al., 2001), and family and twins studies indicate that the heritability of panic disorder is around 43% (Hettema, Neale, & Kendler, 2001). The exact genes and gene functions involved in this disorder, however, are not well-understood (APA, 2013). Neurobiological theories of panic disorder suggest that a region of the brain called the locus coeruleus may play a role in this disorder. Located in the brainstem, the locus coeruleus is the brain s major source of norepinephrine, a neurotransmitter that triggers the body s fight-or-flight response. Activation of the locus coeruleus is associated with anxiety and fear, and research with nonhuman primates has shown that stimulating the locus coeruleus either electrically or through drugs produces panic-like symptoms (Charney et al., 1990). Such findings have led to the theory that panic disorder may be caused by abnormal norepinephrine activity in the locus coeruleus (Bremner, Krystal, Southwick, & Charney, 1996).

Conditioning theories of panic disorder propose that panic attacks are classical conditioning responses to subtle bodily sensations resembling those normally occurring when one is anxious or frightened (Bouton, Mineka, & Barlow, 2001). For example, consider a child who has asthma. An acute asthma attack produces sensations, such as shortness of breath, coughing, and chest tightness, that typically elicit fear and anxiety. Later, when the child experiences subtle symptoms that resemble the frightening symptoms of earlier asthma attacks (such as shortness of breath after climbing stairs), they may become anxious, fearful, and then experience a panic attack. In this situation, the subtle symptoms would represent a conditioned stimulus, and the panic attack would be a conditioned response. The finding that panic disorder is nearly three times as frequent among people with asthma as it is among people without asthma (Weiser, 2007) supports the possibility that panic disorder has the potential to develop through classical conditioning.

Cognitive factors may play an integral part in panic disorder. Generally, cognitive theories (Clark, 1996) argue that those with panic disorder are prone to interpret ordinary bodily sensations catastrophically, and these fearful interpretations set the stage for panic attacks. For example, a person might detect bodily changes that are routinely triggered by innocuous events such as getting up from a seated position (dizziness), exercising (increased heart rate, shortness of breath), or drinking a large cup of coffee (increased heart rate, trembling). The individual interprets these subtle bodily changes catastrophically (Maybe I m having a heart attack!). Such interpretations create fear and anxiety, which trigger additional physical symptoms; subsequently, the person experiences a panic attack. Support of this contention rests with findings that people with more severe catastrophic thoughts about sensations have more frequent and severe panic attacks, and among those with panic disorder, reducing catastrophic cognitions about their sensations is as effective as medication in reducing panic attacks (Good & Hinton, 2009).

Generalized Anxiety Disorder

Alex was always worried about many things. He worried that his children would drown when they played at the beach. Each time he left the house, he worried that an electrical short circuit would start a fire in his home. He worried that his husband would lose his job at the prestigious law firm. He worried that his daughter s minor staph infection could turn into a massive life-threatening condition. These and other worries constantly weighed heavily on Alex s mind, so much so that they made it difficult for him to make decisions and often left him feeling tense, irritable, and worn out. One night, Alex s husband was to drive their son home from a soccer game. However, his husband stayed after the game and talked with some of the other parents, resulting in him arriving home 45 minutes late. Alex had tried to call his cell phone three or four times, but he could not get through because the soccer field did not have a signal. Extremely worried, Alex eventually called the police, convinced that his husband and son had not arrived home because they had been in a terrible car accident.

Alex suffers from generalized anxiety disorder: a relatively continuous state of excessive, uncontrollable, and pointless worry and apprehension. People with generalized anxiety disorder often worry about routine, everyday things, even though their concerns are unjustified (the accompanying source-figure description). For example, an individual may worry about her health and finances, the health of family members, the safety of her children, or minor matters (e.g., being late for an appointment) without having any legitimate reason for doing so (APA, 2013). A diagnosis of generalized anxiety disorder requires that the diffuse worrying and apprehension characteristic of this disorder what Sigmund Freud referred to as free-floating anxiety is not part of another disorder, occurs more days than not for at least six months, and is accompanied by any three of the following symptoms: restlessness, difficulty concentrating, being easily fatigued, muscle tension, irritability, and sleep difficulties.

Source illustration description (image not reproduced): AA photograph shows a woman biting her fingernails. Worry is a defining feature of generalized anxiety disorder. (credit: Freddie Pe a)

About 5.7% of the U.S. population will develop symptoms of generalized anxiety disorder during their lifetime (Kessler et al., 2005), and females are 2 times as likely as males to experience the disorder (APA, 2013). Generalized anxiety disorder is highly comorbid with mood disorders and other anxiety disorders (Noyes, 2001), and it tends to be chronic. Also, generalized anxiety disorder appears to increase the risk for heart attacks and strokes, especially in people with preexisting heart conditions (Martens et al., 2010).

Although there have been few investigations aimed at determining the heritability of generalized anxiety disorder, a summary of available family and twin studies suggests that genetic factors play a modest role in the disorder (Hettema et al., 2001). Cognitive theories of generalized anxiety disorder suggest that worry represents a mental strategy to avoid more powerful negative emotions (Aikins & Craske, 2001), perhaps stemming from earlier unpleasant or traumatic experiences. Indeed, one longitudinal study found that childhood maltreatment was strongly related to the development of this disorder during adulthood (Moffitt et al., 2007); worrying might distract people from remembering painful childhood experiences.

Summary

Anxiety disorders are a group of disorders in which a person experiences excessive, persistent, and distressing fear and anxiety that interferes with normal functioning. Anxiety disorders include specific phobia: a specific unrealistic fear; social anxiety disorder: extreme fear and avoidance of social situations; panic disorder: suddenly overwhelmed by panic even though there is no apparent reason to be frightened; agoraphobia: an intense fear and avoidance of situations in which it might be difficult to escape; and generalized anxiety disorder: a relatively continuous state of tension, apprehension, and dread.

Review Questions

OpenStax learning exercise

Q. In which of the following anxiety disorders is the person in a continuous state of excessive, pointless worry and apprehension?

A. B

OpenStax learning exercise

Q. Which of the following would constitute a safety behavior?

A. C

Critical Thinking Question

OpenStax learning exercise

Q. Describe how cognitive theories of the etiology of anxiety disorders differ from learning theories.

A. Learning theories suggest that some anxiety disorders, especially specific phobia, can develop through a number of learning mechanisms. These mechanisms can include classical and operant conditioning, modeling, or vicarious learning. Cognitive theories, in contrast, assume that some anxiety disorder, especially panic disorder, develop through cognitive misinterpretations of anxiety and other symptoms.

Posttraumatic Stress Disorder

OpenStax source module m82274 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82274/index.cnxml

Extremely stressful or traumatic events, such as combat, natural disasters, and crimes, place the people who experience them at an increased risk for developing psychological disorders such as posttraumatic stress disorder (PTSD). Throughout much of the 20th century, this disorder was called shell shock and combat neurosis because its symptoms were observed in soldiers who had engaged in wartime combat. By the late 1970s it had become clear that women who had experienced sexual traumas (e.g., rape, domestic battery, and incest) often experienced the same set of symptoms as did soldiers (Herman, 1997). The term posttraumatic stress disorder was developed given that these symptoms could happen to anyone who experienced psychological trauma.

A Broader Definition of PTSD

PTSD was listed among the anxiety disorders in previous DSM editions. In DSM-5, it is now listed among a group called Trauma-and-Stressor-Related Disorders. For a person to be diagnosed with PTSD, they must be exposed to, witness, or experience the details of a traumatic experience (e.g., a first responder), one that involves actual or threatened death, serious injury, or sexual violence (APA, 2013, p. 271). These experiences can include such events as combat, threatened or actual physical attack, sexual assault, natural disasters, terrorist attacks, and automobile accidents. This criterion makes PTSD the only disorder listed in the DSM in which a cause (extreme trauma) is explicitly specified.

Symptoms of PTSD include intrusive and distressing memories of the event, flashbacks (states that can last from a few seconds to several days, during which the individual relives the event and behaves as if the event were occurring at that moment [APA, 2013]), avoidance of stimuli connected to the event,

persistently negative emotional states (e.g., fear, anger, guilt, and shame), feelings of detachment from others, irritability, proneness toward outbursts, and an exaggerated startle response (jumpiness). For PTSD to be diagnosed, these symptoms must occur for at least one month.

Roughly 7% of adults in the United States, including 9.7% of women and 3.6% of men, experience PTSD in their lifetime (National Comorbidity Survey, 2007), with higher rates among people exposed to mass trauma and people whose jobs involve duty-related trauma exposure (e.g., police officers, firefighters, and emergency medical personnel) (APA, 2013). Nearly 21% of residents of areas affected by Hurricane Katrina suffered from PTSD one year following the hurricane (Kessler et al., 2008), and 12.6% of Manhattan residents were observed as having PTSD 2 3 years after the 9/11 terrorist attacks (DiGrande et al., 2008).

Risk Factors For PTSD

Of course, not everyone who experiences a traumatic event will go on to develop PTSD; several factors strongly predict the development of PTSD: trauma experience, greater trauma severity, lack of immediate social support, and more subsequent life stress (Brewin, Andrews, & Valentine, 2000). Traumatic events that involve harm by others (e.g., combat, rape, and sexual molestation) carry greater risk than do other traumas (e.g., natural disasters) (Kessler, Sonnega, Bromet, Hughes, & Nelson, 1995). Women are more likely to have been traumatized because of sexual trauma, childhood neglect, and childhood physical abuse. Men are more likely to have been traumatized by natural disaster, life-threatening accident, and physical violence, either witnessed or directed at them. Adolescent boys are more likely to experience accident, physical assault, and witness death/injury; adolescent girls are more likely to experience rape/sexual assault, intimate partner violence, or unexpected death or injury of a loved one. Assaultive violence and witnessing trauma to others is more prevalent among non-White people when compared to White people. African American males are more likely to be exposed to and victims of violence than males of other races (Kilpatrick, Badour, & Resnick, 2017). A 2012 study found that 27% of corrections officers reported experiencing symptoms of PTSD in the past 30 days. Rates were higher for males (31%) than females (22%) (Spinaris, Denhof, & Kellaway, 2012). A study conducted by Jaegers et al (2019) found that 53.4% of jail correctional officers screened positively for PTSD. PTSD is more prevalent in prison populations than in the general public, with prevalence estimates of 6% in male prisoners and 21% in female prisoners (Facer-Irwin et al, 2019). Factors that increase the risk of PTSD include female gender, low socioeconomic status, low intelligence, personal history of mental disorders, history of childhood adversity (abuse or other trauma during childhood), and family history of mental disorders (Brewin et al., 2000). Personality characteristics such as neuroticism and somatization (the tendency to experience physical symptoms when one encounters stress) have been shown to elevate the risk of PTSD (Bramsen, Dirkzwager, & van der Ploeg, 2000). People who experience childhood adversity and/or traumatic experiences during adulthood are at significantly higher risk of developing PTSD if they possess one or two short versions of a gene that regulates the neurotransmitter serotonin (Xie et al., 2009). This suggests a possible diathesis-stress interpretation of PTSD: its development is influenced by the interaction of psychosocial and biological factors.

Support For Sufferers of PTSD

Research has shown that social support following a traumatic event can reduce the likelihood of PTSD (Ozer, Best, Lipsey, & Weiss, 2003). Social support is often defined as the comfort, advice, and assistance received from relatives, friends, and neighbors. Social support can help individuals cope during difficult times by allowing them to discuss feelings and experiences and providing a sense of being loved and appreciated. A 14-year study of 1,377 people who had served in the Vietnam War found that those who perceived less social support when they came home were more likely to develop PTSD than were those who perceived greater support (the accompanying source-figure description). In addition, those who became involved in the community were less likely to develop PTSD, and they were more likely to experience a remission of PTSD than were those who were less involved (Koenen, Stellman, Stellman, & Sommer, 2003).

Source illustration description (image not reproduced): A photograph shows a person looking at the Vietnam Traveling Memorial Wall. PTSD was first recognized in soldiers who had engaged in combat. Research has shown that strong social support decreases the risk of PTSD. This person stands at the Vietnam Traveling Memorial Wall. (credit: Kevin Stanchfield)

Learning and the Development of PTSD

PTSD learning models suggest that some symptoms are developed and maintained through classical conditioning. The traumatic event may act as an unconditioned stimulus that elicits an unconditioned response characterized by extreme fear and anxiety. Cognitive, emotional, physiological, and environmental cues accompanying or related to the event are conditioned stimuli. These traumatic reminders evoke conditioned responses (extreme fear and anxiety) similar to those caused by the event itself (Nader, 2001). A person who witnessed or experienced a plane crash and who developed PTSD may display excessive hypervigilance and distress when planes fly overhead; this behavior constitutes a conditioned response to the traumatic reminder (conditioned stimulus of the sight and sound of an airplane). Differences in how conditionable individuals are help to explain differences in the development and maintenance of PTSD symptoms (Pittman, 1988). Conditioning studies demonstrate facilitated acquisition of conditioned responses and delayed extinction of conditioned responses in people with PTSD (Orr et al., 2000).

Cognitive factors are important in the development and maintenance of PTSD. One model suggests that two key processes are crucial: disturbances in memory for the event, and negative appraisals of the trauma and its aftermath (Ehlers & Clark, 2000). According to this theory, some people who experience traumas do not form coherent memories of the trauma; memories of the traumatic event are poorly encoded and, thus, are fragmented, disorganized, and lacking in detail. Therefore, these individuals are unable remember the event in a way that gives it meaning and context. A rape victim who cannot coherently remember the event may remember only bits and pieces (e.g., the attacker repeating a word or insult); because the victim was unable to develop a fully integrated memory, the fragmentary memory tends to stand out. Although unable to retrieve a complete memory of the event, the victim may be haunted by intrusive fragments involuntarily triggered by stimuli associated with the event (e.g., memories of the attacker s comments when encountering a person who resembles the attacker). This interpretation fits previously discussed material concerning PTSD and conditioning. The model also proposes that negative appraisals of the event (I deserved to be raped because I m stupid) may lead to dysfunctional behavioral strategies (e.g., avoiding social activities where men are likely to be present) that maintain PTSD symptoms by preventing both a change in the nature of the memory and a change in the problematic appraisals.

Summary

Posttraumatic stress disorder (PTSD) was described through much of the 20th century and was referred to as shell shock and combat neurosis in the belief that its symptoms were thought to emerge from the stress of active combat. Today, PTSD is defined as a disorder in which the experience of a traumatic or profoundly stressful event, such as combat, sexual assault, or natural disaster, produces a constellation of symptoms that must last for one month or more. These symptoms include intrusive and distressing memories of the event, flashbacks, avoidance of stimuli or situations that are connected to the event, persistently negative emotional states, feeling detached from others, irritability, proneness toward outbursts, and a tendency to be easily startled. Not everyone who experiences a traumatic event will develop PTSD; a variety of risk factors associated with its development have been identified.

Review Questions

OpenStax learning exercise

Q. Symptoms of PTSD include all of the following except _____.

A. D

OpenStax learning exercise

Q. Which of the following elevates the risk for developing PTSD?

A. A

Critical Thinking Question

OpenStax learning exercise

Q. List some of the risk factors associated with the development of PTSD following a traumatic event.

A. Risk factors associated with PTSD include gender (female), low socioeconomic status, low intelligence, personal and family history of mental illness, and childhood abuse or trauma. Personality factors, including neuroticism and somatization, may also serve as risk factors. Also, certain versions of a gene that regulates serotonin may constitute a diathesis.

CHAPTER 18 / CAC.901

Neurodevelopmental differences and support

Lecture: Neurodevelopmental differences require more than a label

Neurodevelopmental differences can affect attention, communication, learning, movement, sensory experience, and adaptation to demands. They are not failures of character. A child may work very hard while still finding a task difficult. At the same time, a neurodevelopmental diagnosis does not remove agency, preferences, or the capacity to learn. The counselor s task is to understand the person and the fit between needs and environment.

ADHD and autism are common examples, but they are not interchangeable. Assessment requires appropriate professional competence, developmental history, and evidence about functioning. A brief observation in a quiet office may not capture daily demands. A caregiver s report may be essential, but the young person s experience should also be central. Avoid diagnosing from a single behavior such as poor eye contact, high activity, or intense interest.

Function before assumptions

Ask what the child is trying to do and what makes it difficult. A child may leave a group because noise is overwhelming, because instructions are unclear, or because the activity has no predictable ending. The observable behavior is similar, but the support differs. Examine sensory conditions, communication, task structure, and social expectations.

Supports can include visual schedules, clear instructions, movement opportunities, alternative communication, reduced sensory load, and explicit teaching of routines. Select supports with the child where possible. Some aids that adults consider helpful may feel stigmatizing or uncomfortable. The aim is meaningful participation, not making the child appear typical.

Neurodiversity perspectives emphasize variation, dignity, and the removal of unnecessary barriers. They do not require ignoring disability, distress, or the need for treatment. A balanced approach can value identity while providing support for communication, health, learning, and safety. Avoid presenting acceptance and intervention as opposites.

Intersection with culture and service access

Families may encounter different explanations for a child s behavior. Some experience blame, stigma, or pressure to conceal difficulty. Others distrust services because previous institutions treated them unfairly. Ask about the family s understanding and goals without assuming that disagreement reflects denial. Explain assessment clearly, use appropriate language support, and identify practical barriers.

Diagnostic practices can reflect unequal access. A child who is quiet may be overlooked; a child from a marginalized group may be interpreted through a behavior-disorder lens. Review referral patterns and seek multiple sources of evidence. Equity does not mean giving every child the same assessment regardless of need. It means making valid assessment and useful support accessible.

Figure 18.1 maps the fit between a child s profile, task demands, environmental supports, and participation. It shows that intervention can change skills, demands, or supports. The diagnosis informs the conversation but does not determine a single plan.

Application: a busy classroom

A fictional eight-year-old, Halima, frequently misses instructions and becomes upset during transitions. A developmental assessment considers attention, hearing, language, sleep, classroom noise, and the way transitions are announced. It also asks what happens in a quieter setting and which supports already help.

If ADHD is identified through appropriate assessment, the plan should still be individualized. Halima may benefit from a visual sequence, a predictable warning before transitions, and instructions checked privately. Her preferences matter. Monitor whether the supports improve participation and reduce distress, not merely whether she sits still.

The counselor can help teachers distinguish support from lowered expectations. A clear route to a task can allow a child to demonstrate competence. Review supports regularly and involve Halima in decisions about what is useful. Avoid making access dependent on perfect behavior.

Doctoral seminar task

Design an assessment plan for a fictional child whose activity level concerns adults. Include developmental history, health considerations, functioning across settings, the child s perspective, and possible alternative explanations. Identify which questions require another professional s expertise.

Then design two support plans for the same diagnosis under different environmental conditions. Explain why identical labels do not imply identical interventions. Include a measure of participation and a measure of the child s experience. This keeps the work person-centered rather than category-centered.

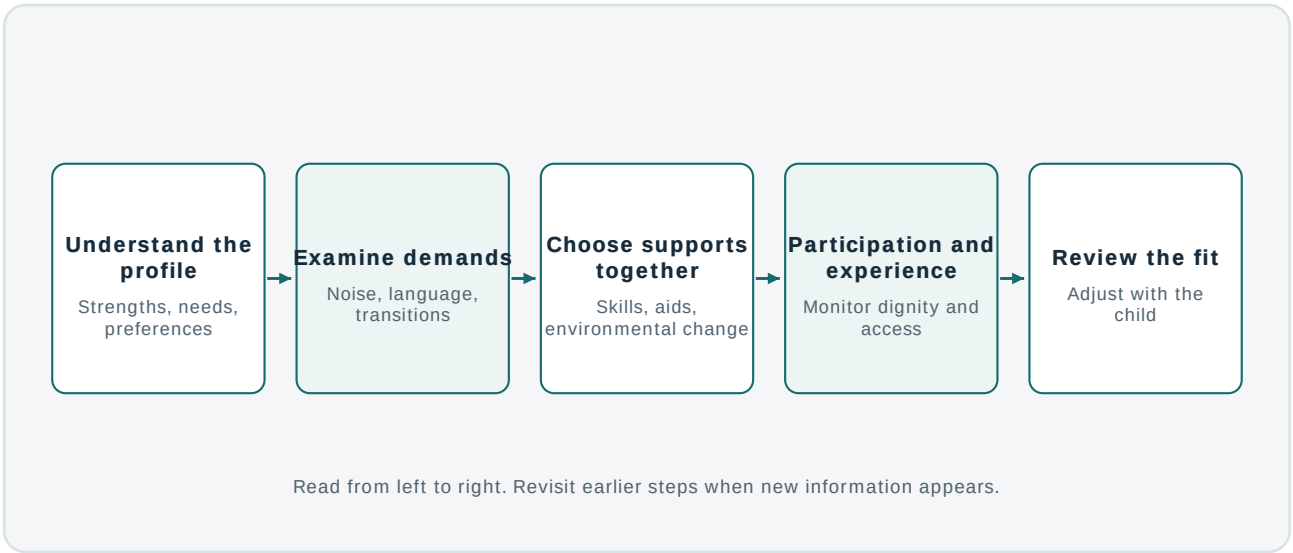


Figure 18.1. Support improves the person environment fit. Participation can improve through teaching skills, changing demands, and providing accessible supports. A diagnosis informs this process but does not prescribe one identical plan for every child. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

Disorders in Childhood

OpenStax source module m82278 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82278/index.cnxml

Most of the disorders we have discussed so far are typically diagnosed in adulthood, although they can and sometimes do occur during childhood. However, there are a group of conditions that, when present, are diagnosed early in childhood, often before the time a child enters school. These conditions are listed in the DSM-5 as neurodevelopmental disorders, and they involve developmental problems in personal, social, academic, and intellectual functioning (APA, 2013). While they are often diagnosed in childhood, many people live with them throughout adulthood. In this section, we will discuss two such disorders: attention deficit/ hyperactivity disorder and autism.

Attention Deficit/Hyperactivity Disorder

Diego is always active, from the time he wakes up in the morning until the time he goes to bed at night. His mother reports that he came out the womb kicking and screaming, and he has not stopped moving since. He has a sweet disposition, but always seems to be in trouble with his teachers, parents, and after-school program counselors. He seems to accidentally break things; he lost his jacket three times last winter, and he never seems to sit still. His teachers believe he is a smart child, but he never finishes anything he starts and is so impulsive that he does not seem to learn much in school.

Diego likely has attention deficit/hyperactivity disorder (ADHD). The symptoms of this disorder were first described by Hans Hoffman in the 1920s. While taking care of his son while his wife was in the hospital giving birth to a second child, Hoffman noticed that the boy had trouble concentrating on his homework, had a short attention span, and had to repeatedly go over easy homework to learn the material (Jellinek & Herzog, 1999). Later, it was discovered that many hyperactive children those who are fidgety, restless, socially disruptive, and have trouble with impulse control also display short attention spans, problems with concentration, and distractibility. By the 1970s, it had become clear that many children who display attention problems often also exhibit signs of hyperactivity. In recognition of such findings, the DSM-III (published in 1980) included a new disorder: attention deficit disorder with and without hyperactivity, now known as attention deficit/hyperactivity disorder (ADHD).

A child with ADHD shows a constant pattern of inattention and/or hyperactive and impulsive behavior that interferes with normal functioning (APA, 2013). Some of the signs of inattention include great difficulty with and avoidance of tasks that require sustained attention (such as conversations or reading), failure to follow instructions (often resulting in failure to complete school work and other duties), disorganization (difficulty keeping things in order, poor time management, sloppy and messy work), lack of attention to detail, becoming easily distracted, and forgetfulness. Hyperactivity is characterized by excessive movement, and includes fidgeting or squirming, leaving one s seat in situations when remaining seated is expected, having trouble sitting still (e.g., in a restaurant), running about and climbing on things, blurting out responses before another person s question or statement has been completed, difficulty waiting one s turn for something, and interrupting and intruding on others.

Frequently, the hyperactive child comes across as noisy and boisterous. The child s behavior is hasty, impulsive, and seems to occur without much forethought; these characteristics may explain why adolescents and young adults diagnosed with ADHD receive more traffic tickets and have more automobile accidents than do others (Thompson, Molina, Pelham, & Gnagy, 2007).

ADHD occurs in about 8% of children (Danielson et al, 2016), and studies estimate that for about 60% of these people, ADHD continues into adulthood (Sibley et al 2016). On the average, boys are 3 times more likely to have ADHD than are girls; however, such findings might reflect the greater propensity of boys to engage in aggressive and antisocial behavior and thus incur a greater likelihood of being referred to psychological clinics (Barkley, 2006). Children with ADHD face severe academic and social challenges. Compared to their non-ADHD counterparts, children with ADHD have lower grades and standardized test scores and higher rates of expulsion, grade retention, and dropping out (Loe & Feldman, 2007). they also are less well-liked and more often rejected by their peers (Hoza et al., 2005).

A recent study found that nearly 81% of those whose ADHD persisted into adulthood had experienced at least one other comorbid disorder, compared to 47% of those whose ADHD did not persist (Barbaresi et al., 2013).

Life Problems from ADHD

Children with ADHD face considerably worse long-term outcomes than do those children who do not have ADHD. Adults diagnosed with ADHD in childhood, but not treated for ADHD, have been reported to have poor outcomes in a wide range of areas of life, including social function, education, criminality, alcohol use, substance use, and occupational outcomes (Arnold et al, 2015). In one investigation, 135 adults who had been identified as having ADHD symptoms in the 1970s were contacted decades later and interviewed (Klein et al., 2012). Compared to a control sample of 136 participants who had never been diagnosed with ADHD, those who were diagnosed as children: had worse educational attainment (more likely to have dropped out of high school and less likely to have earned a bachelor s degree); had lower socioeconomic status; held less prestigious occupational positions; were more likely to be unemployed; made considerably less in salary; scored worse on a measure of occupational functioning (indicating, for example, lower job satisfaction, poorer work relationships, and more firings); scored worse on a measure of social functioning (indicating, for example, fewer friendships and less involvement in social activities); were more likely to be divorced; and were more likely to have non-alcohol-related substance abuse problems. (Klein et al., 2012)

Longitudinal studies also show that children diagnosed with ADHD are at higher risk for substance abuse problems. One study reported that childhood ADHD predicted later drinking problems, daily smoking, and use of marijuana and other illicit drugs (Molina & Pelham, 2003). The risk of substance abuse problems appears to be even greater for those with ADHD who also exhibit antisocial tendencies (Marshal & Molina, 2006).

Causes of ADHD

Family and twin studies indicate that genetics play a significant role in the development of ADHD. Burt (2009), in a review of 26 studies, reported that the median rate of concordance for identical twins was .66 (one study reported a rate of .90), whereas the median concordance rate for fraternal twins was .20. This study also found that the median concordance rate for unrelated (adoptive) siblings was .09; although this number is small, it is greater than 0, thus suggesting that the environment may have at least some influence. Another review of studies concluded that the heritability of inattention and hyperactivity were 71% and 73%, respectively (Nikolas & Burt, 2010).

The specific genes involved in ADHD are thought to include at least two that are important in the regulation of the neurotransmitter dopamine (Gizer, Ficks, & Waldman, 2009), suggesting that dopamine may be important in ADHD. Indeed, medications used in the treatment of ADHD, such as methylphenidate (Ritalin) and amphetamine with dextroamphetamine (Adderall), have stimulant qualities and elevate dopamine activity. People with ADHD show less dopamine activity in key regions of the brain, especially those associated with motivation and reward (Volkow et al., 2009), which

provides support to the theory that dopamine deficits may be a vital factor in the development this disorder (Swanson et al., 2007).

Brain imaging studies have shown that children with ADHD exhibit abnormalities in their frontal lobes, an area in which dopamine is in abundance. Compared to children without ADHD, those with ADHD appear to have smaller frontal lobe volume, and they show less frontal lobe activation when performing mental tasks. Recall that one of the functions of the frontal lobes is to inhibit our behavior. Thus, abnormalities in this region may go a long way toward explaining the hyperactive, uncontrolled behavior of ADHD.

By the 1970s, many had become aware of the connection between nutritional factors and childhood behavior. At the time, much of the public believed that hyperactivity was caused by sugar and food additives, such as artificial coloring and flavoring. Undoubtedly, part of the appeal of this hypothesis was that it provided a simple explanation of (and treatment for) behavioral problems in children. A statistical review of 16 studies, however, concluded that sugar consumption has no effect at all on the behavioral and cognitive performance of children (Wolraich, Wilson, & White, 1995). Additionally, although food additives have been shown to increase hyperactivity in non-ADHD children, the effect is rather small (McCann et al., 2007). Numerous studies, however, have shown a significant relationship between exposure to nicotine in cigarette smoke during the prenatal period and ADHD (Linnet et al., 2003). Maternal smoking during pregnancy is associated with the development of more severe symptoms of the disorder (Thakur et al., 2013).

Is ADHD caused by poor parenting? No. Remember, the genetics studies discussed above suggested that the family environment does not seem to play much of a role in the development of this disorder; if it did, we would expect the concordance rates to be higher for fraternal twins and adoptive siblings than has been demonstrated. All things considered, the evidence seems to point to the conclusion that ADHD is triggered more by genetic and neurological factors and less by social or environmental ones.

Dig Deeper

Many people believe that the rates of ADHD have increased in recent years, and there is evidence to support this contention. In a recent study, investigators found that the parent-reported prevalence of ADHD among children (4 17 years old) in the United States increased by 22% during a 4-year period, from 7.8% in 2003 to 9.5% in 2007 (CDC, 2010). Over time this increase in parent-reported ADHD was observed in all sociodemographic groups and was reflected by substantial increases in 12 states (Indiana, North Carolina, and Colorado were the top three). The increases were greatest for older teens (ages 15 17), multiracial and Hispanic children, and children with a primary language other than English. Another investigation found that from 1998 2000 through 2007 2009 the parent-reported prevalence of ADHD increased among U.S. children between the ages of 5 17 years old, from 6.9% to 9.0% (Akinbami, Liu, Pastor, & Reuben, 2011).

A major weakness of both studies was that children were not actually given a formal diagnosis. Instead, parents were simply asked whether or not a doctor or other health-care provider had ever told them their child had ADHD; the reported prevalence rates thus may have been affected by the accuracy of parental memory. Nevertheless, the findings from these studies raise important questions concerning what appears to be a demonstrable rise in the prevalence of ADHD. Although the reasons underlying this apparent increase in the rates of ADHD over time are poorly understood and, at best, speculative, several explanations are viable: ADHD may be over-diagnosed by doctors who are too quick to medicate children as a behavior treatment. There is greater awareness of ADHD now than in the past. Nearly everyone has heard of ADHD, and most parents and teachers are aware of its key symptoms. Thus, parents may be quick to take their children to a doctor if they believe their child possesses these symptoms, or teachers may be more likely now than in the past to notice the symptoms and refer the child for evaluation. The use of computers, video games, iPhones, and other electronic devices has become pervasive among children in the early 21st century, and these devices could potentially shorten children s attentions spans. Thus, what might seem like inattention to some parents and teachers could simply reflect exposure to too much technology. ADHD diagnostic criteria have changed over time.

Autism Spectrum Disorder

A seminal paper published in 1943 by psychiatrist Leo Kanner described an unusual neurodevelopmental condition he observed in a group of children. He called this condition early infantile autism, and it was characterized mainly by an inability to form close emotional ties with others, speech and language abnormalities, repetitive behaviors, and an intolerance of minor changes in the environment and in normal routines (Bregman, 2005). What the DSM-5 refers to as autism spectrum disorder today, is a direct extension of Kanner s work.

Autism spectrum disorder is probably the most misunderstood of the neurodevelopmental disorders. Children with this disorder show signs of significant disturbances in three main areas: (a) deficits in social interaction, (b) deficits in communication, and (c) repetitive patterns of behavior or interests. These disturbances appear early in life and cause serious impairments in functioning (APA, 2013). The child with autism spectrum disorder might exhibit deficits in social interaction by not initiating conversations with other children or turning their head away when spoken to. Typically, these children do not make eye contact with others and seem to prefer playing alone rather than with others. In some cases, it is almost as though these individuals live in a personal and isolated social world others are simply not privy to or able to penetrate. Communication deficits can range from a complete lack of speech, to one word responses (e.g., saying Yes or No when replying to questions or statements that require additional elaboration), to echoed speech (e.g., parroting what another person says, either immediately or several hours or even days later), to difficulty maintaining a conversation because of an inability to reciprocate others comments. These deficits can also include problems in using and understanding nonverbal cues (e.g., facial expressions, gestures, and postures) that facilitate normal communication.

Repetitive patterns of behavior or interests can be exhibited a number of ways. The child might engage in stereotyped, repetitive movements (rocking, head-banging, or repeatedly dropping an object and then picking it up), or they might show great distress at small changes in routine or the environment. In some cases, the person with autism spectrum disorder might show highly restricted and fixated interests that appear to be abnormal in their intensity. For instance, the person might learn and memorize every detail about something even though doing so serves no apparent purpose. Importantly, autism spectrum disorder is not the same thing as intellectual disability, although these two conditions are often comorbid. The DSM-5 specifies that the symptoms of autism spectrum disorder are not caused or explained by intellectual disability.

Life Problems From Autism Spectrum Disorder

Autism spectrum disorder is referred to in everyday language as autism; in fact, the disorder was termed autistic disorder in earlier editions of the DSM, and its diagnostic criteria were much narrower than those of autism spectrum disorder. The qualifier spectrum in autism spectrum disorder is used to indicate that individuals with the disorder can show a range, or spectrum, of symptoms that vary in their magnitude and severity: some severe, others less severe. The previous edition of the DSM included a diagnosis of Asperger s disorder, generally recognized as a less severe form of autistic disorder; individuals diagnosed with Asperger s disorder were described as having average or high intelligence and a strong vocabulary, but exhibiting impairments in social interaction and social communication, such as talking only about their special interests (Wing, Gould, & Gillberg, 2011). However, because research has failed to demonstrate that Asperger s disorder differs qualitatively from autistic disorder, the DSM-5 does not include it, which is prompting concerns among some parents that their children may no longer be eligible for special services (Asperger s Syndrome Dropped, 2012). Some individuals with autism spectrum disorder, particularly those with better language and intellectual skills, can live and work independently as adults. However, most do not because the symptoms remain sufficient to cause serious impairment in many realms of life (APA, 2013).

Current estimates from the Center for Disease Control and Prevention s Autism and Developmental Disabilities Monitoring Network indicate that 1 in 59 children in the United States has autism spectrum disorder; the disorder is 4 times more common among boys (1 in 38) than in girls (1 in 152) (Baio et al,

2018). Rates of autistic spectrum disorder have increased dramatically since the 1980s. For example, California saw an increase of 273% in reported cases from 1987 through 1998 (Byrd, 2002); between 2000 and 2008, the rate of autism diagnoses in the United States increased 78% (CDC, 2012). Although it is difficult to interpret this increase, it is possible that the rise in prevalence is the result of the broadening of the diagnosis, increased efforts to identify cases in the community, and greater awareness and acceptance of the diagnosis. In addition, mental health professionals are now more knowledgeable about autism spectrum disorder and are better equipped to make the diagnosis, even in subtle cases (Novella, 2008).

Causes of Autism Spectrum Disorder

The exact causes of autism spectrum disorder remain unknown despite massive research efforts over the last two decades (Meek, Lemery-Chalfant, Jahromi, & Valiente, 2013). Autism appears to be strongly influenced by genetics, as identical twins show concordance rates of 60% 90%, whereas concordance rates for fraternal twins and siblings are 5% 10% (Autism Genome Project Consortium, 2007). Many different genes and gene mutations have been implicated in autism (Meek et al., 2013). Among the genes involved are those important in the formation of synaptic circuits that facilitate communication between different areas of the brain (Gauthier et al., 2011). A number of environmental factors are also thought to be associated with increased risk for autism spectrum disorder, at least in part, because they contribute to new mutations. These factors include exposure to pollutants, such as plant emissions and mercury, urban versus rural residence, and vitamin D deficiency (Kinney, Barch, Chayka, Napoleon, & Munir, 2009).

Child Vaccinations and Autism Spectrum Disorder

In the late 1990s, a prestigious medical journal published an article purportedly showing that autism is triggered by the MMR (measles, mumps, and rubella) vaccine. These findings were very controversial and drew a great deal of attention, sparking an international forum on whether children should be vaccinated. In a shocking turn of events, some years later the article was retracted by the journal that had published it after evidence of fraud and unethical practices on the part of the lead researcher. Despite the retraction, the reporting in popular media led to concerns about a possible link between vaccines and autism persisting. A recent survey of parents, for example, found that roughly a third of respondents expressed such a concern (Kennedy, LaVail, Nowak, Basket, & Landry, 2011); and perhaps fearing that their children would develop autism, more than 10% of parents of young children refuse or delay vaccinations (Dempsey et al., 2011). Some parents of children with autism mounted a campaign against scientists who refuted the vaccine-autism link. Even politicians and several well-known celebrities weighed in; for example, actress Jenny McCarthy (who believed that a vaccination caused her son s autism) co-authored a book on the matter. However, there is no scientific evidence that a link exists between autism and vaccinations (Hughes, 2007). Indeed, a recent study compared the vaccination histories of 256 children with autism spectrum disorder with that of 752 control children across three time periods during their first two years of life (birth to 3 months, birth to 7 months, and birth to 2 years) (DeStefano, Price, & Weintraub, 2013). At the time of the study, the children were between 6 and 13 years old, and their prior vaccination records were obtained. Because vaccines contain immunogens (substances that fight infections), the investigators examined medical records to see how many immunogens children received to determine if those children who received more immunogens were at greater risk for developing autism spectrum disorder. The results of this study, a portion of which are shown in the accompanying source-figure description, clearly demonstrate that the quantity of immunogens from vaccines received during the first two years of life were not at all related to the development of autism spectrum disorder. There is not a relationship between vaccinations and autism spectrum disorders.

Source illustration description (image not reproduced): A graph has an x-axis labeled total cumulative immunogens and a y-axis with percentage numbers. For children aged 0 3 months, the data is approximately as follows: 0 25 immunogens are about 48% for ASD cases and 41% for controls, 26 50 immunogens are 5% for ASD cases and 6% for controls, and for 3000 3258 immunogens45% for ASD cases and 50% for controls. For children aged 0 7months, the data is approximately as follows: 26 50 immunogens are about 20% for ASD cases and 18% for controls, 51 75

immunogens are 25% for ASD cases and 22% for controls, 3000 3258 immunogens are 45% for ASD cases and 52% for controls, 6000 6258 immunogens are 10% for ASD cases and 8% for controls, and for 9000 9258 immunogens 33% for ASD cases and 40% for controls. For children aged 0 24 months, the data is approximately as follows: 151 175 immunogens are about 25% for ASD cases and 25% for controls, 176 200 immunogens are 18% for ASD cases and 13% for controls, 9000 9528 immunogens are 17% for ASD cases and 20% for controls, and for 12000 12258 immunogens 25% for ASD cases and25% for controls. In terms of their exposure to immunogens in vaccines, overall, there is not a significant difference between children with autism spectrum disorder and their age-matched controls without the disorder (DeStefano et al., 2013).

Why does concern over vaccines and autism spectrum disorder persist? Since the proliferation of the Internet in the 1990s, parents have been constantly bombarded with online information that can become magnified and take on a life of its own. The enormous volume of electronic information pertaining to autism spectrum disorder, combined with how difficult it can be to grasp complex scientific concepts, can make separating good research from bad challenging (Downs, 2008). Notably, the study that fueled the controversy reported that 8 out of 12 children according to their parents developed symptoms consistent with autism spectrum disorder shortly after receiving a vaccination. To conclude that vaccines cause autism spectrum disorder on this basis, as many did, is clearly incorrect for a number of reasons, not the least of which is because correlation does not imply causation, as you ve learned.

Additionally, as was the case with diet and ADHD in the 1970s, the notion that autism spectrum disorder is caused by vaccinations is appealing to some because it provides a simple explanation for this condition. Like all disorders, however, there are no simple explanations for autism spectrum disorder. Although the research discussed above has shed some light on its causes, science is still a long way from complete understanding of the disorder.

Summary

Neurodevelopmental disorders are a group of disorders that are typically diagnosed during childhood and are characterized by developmental deficits in personal, social, academic, and intellectual realms; these disorders include attention deficit/hyperactivity disorder (ADHD) and autism spectrum disorder. ADHD is characterized by a pervasive pattern of inattention and/or hyperactive and impulsive behavior that interferes with normal functioning. Genetic and neurobiological factors contribute to the development of ADHD, which can persist well into adulthood and is often associated with poor long-term outcomes. The major features of autism spectrum disorder include deficits in social interaction and communication and repetitive movements or interests. As with ADHD, genetic factors appear to play a prominent role in the development of autism spectrum disorder; exposure to environmental pollutants such as mercury have also been linked to the development of this disorder. Although it is believed by some that autism is triggered by the MMR vaccination, evidence does not support this claim.

Review Questions

OpenStax learning exercise

Q. Which of the following is not a primary characteristic of ADHD?

A. C

OpenStax learning exercise

Q. One of the primary characteristics of autism spectrum disorder is _____.

A. B

Critical Thinking Questions

OpenStax learning exercise

Q. Compare the factors that are important in the development of ADHD with those that are important in the development of autism spectrum disorder.

A. Genetic factors appear to play a major role in the development of both ADHD and autism spectrum disorder: studies show higher rates of concordance among identical twins than among fraternal twins for both disorders. In ADHD, genes that regulate dopamine have been implicated; in autism spectrum disorder, de novo genetic mutations appear to be important. Imaging studies suggest that abnormalities in the frontal lobes may be important in the development of ADHD. Parenting practices are not connected to the development of either disorder. Although environmental toxins are generally unimportant in the development of ADHD, exposure to cigarette smoke during the prenatal period has been linked to the development of the disorder; a number of environmental factors are thought to be associated with an increased risk for autism spectrum disorder: exposure to pollutants, an urban versus rural residence, and vitamin D deficiency. Although some people continue to believe that MMR vaccinations can cause autism spectrum disorder (due to an influential paper that was later retracted), there is no scientific evidence that supports this assertion.

Personal Application Question

OpenStax learning exercise

Q. Discuss the characteristics of autism spectrum disorder with a few of your friends or members of your family (choose friends or family members who know little about the disorder) and ask them if they think the cause is due to vaccinations. If they indicate that they believe this to be true, why do you think this might be the case? What would be your response?

CHAPTER 19 / CAC.901

Developmentally informed intervention

Lecture: Intervention should fit the developmental mechanism

An intervention is a planned effort to change a process that contributes to difficulty. It should follow a formulation, not merely a diagnosis or a popular technique. Developmentally informed intervention considers the child s capacities, relationships, goals, and environment. It also considers whether the proposed change is feasible for the people expected to carry it out.

Begin with a shared understanding of the problem. A caregiver may want immediate compliance, a teacher may want uninterrupted instruction, and the child may want the teasing to stop. These goals are not automatically compatible. The counselor should make them explicit, protect safety, and identify a goal that respects the child s welfare. Agreement does not require pretending that every perspective is equally accurate.

Match method to communication and capacity

Talk-based interventions can be adapted with play, drawing, stories, movement, visual schedules, and caregiver participation. Adaptation should preserve the active purpose of the intervention rather than merely make the session entertaining. If the goal is to practice a coping sequence, the activity should provide a meaningful opportunity to rehearse it.

Evidence for an intervention in one age group does not automatically establish the same effect in another. Developmental differences affect understanding, engagement, and generalization. Review the evidence, explain uncertainty, and monitor outcomes. Do not claim that a technique is evidence-based merely because it resembles a named therapy.

Caregiver involvement can support practice and change the environment, but it should be planned carefully. Some caregivers need coaching; others need practical resources or relief from strain. In situations of abuse or coercive control, joint sessions may be unsafe. Assess rather than assuming that family involvement is always protective.

Generalization is part of the plan

A skill practiced in a calm office may not transfer automatically to a noisy classroom or a stressful home. Plan for use in real settings. Identify prompts, practice opportunities, and supportive adults. Fade assistance thoughtfully, and review whether the child can use the skill when it matters.

Figure 19.1 shows the intervention cycle: formulate, agree on goals, select methods, practice, monitor, and revise. The cycle is not a rigid sequence. New information may require returning to formulation. A lack of progress should prompt investigation of fit, implementation, barriers, and the original hypothesis rather than automatic blame.

Ethics and scope

Use methods within training and professional scope. Seek supervision or consultation when a case exceeds competence, and refer when needed without abandoning the family. Explain the intervention, alternatives, expected benefits, possible discomfort, and limits of evidence. Obtain appropriate permission and the child s assent, following local requirements.

Confidentiality should be discussed with caregivers and the child in understandable terms. Share information needed for coordinated care with appropriate authorization and attention to safety. Avoid sending full session notes when a brief functional summary is sufficient. Professional collaboration does not remove the duty to minimize unnecessary disclosure.

Medication may be part of care for some conditions. Counselors should understand their role in observation, communication, and support without prescribing or changing medication unless separately qualified and legally authorized. Report concerns to the prescriber and family through appropriate channels. Do not interpret every behavior change as a medication effect without considering context.

Application: adapting exposure without coercion

A fictional eleven-year-old, Titi, avoids entering the classroom after repeated panic symptoms. A developmental plan first assesses safety, bullying, health, and learning demands. If gradual exposure is appropriate, it is explained collaboratively and paired with support. The aim is not to force compliance while ignoring danger.

The team identifies manageable steps, a way to signal distress, and a review routine. Titi helps choose meaningful goals. The counselor monitors anxiety, attendance, participation, and the child s sense of control. If the classroom remains unsafe or inaccessible, the plan changes. Exposure should not be used to adapt a child to preventable harm.

Caregivers and teachers receive consistent guidance without being given private details that are unnecessary for their role. Progress is reviewed against the original formulation. If avoidance persists, the team asks what was missed rather than increasing pressure automatically.

Doctoral seminar task

Take a developmental formulation from an earlier chapter and design an intervention with a clear mechanism. Specify the active ingredients, developmental adaptations, participants, setting, and expected barriers. Include an outcome that matters to the child and a measure of implementation.

Then write a revision rule. What result would lead you to change the plan, seek consultation, or reconsider the formulation? An intervention is ethically stronger when it includes a way to learn from failure rather than protecting the method from criticism.

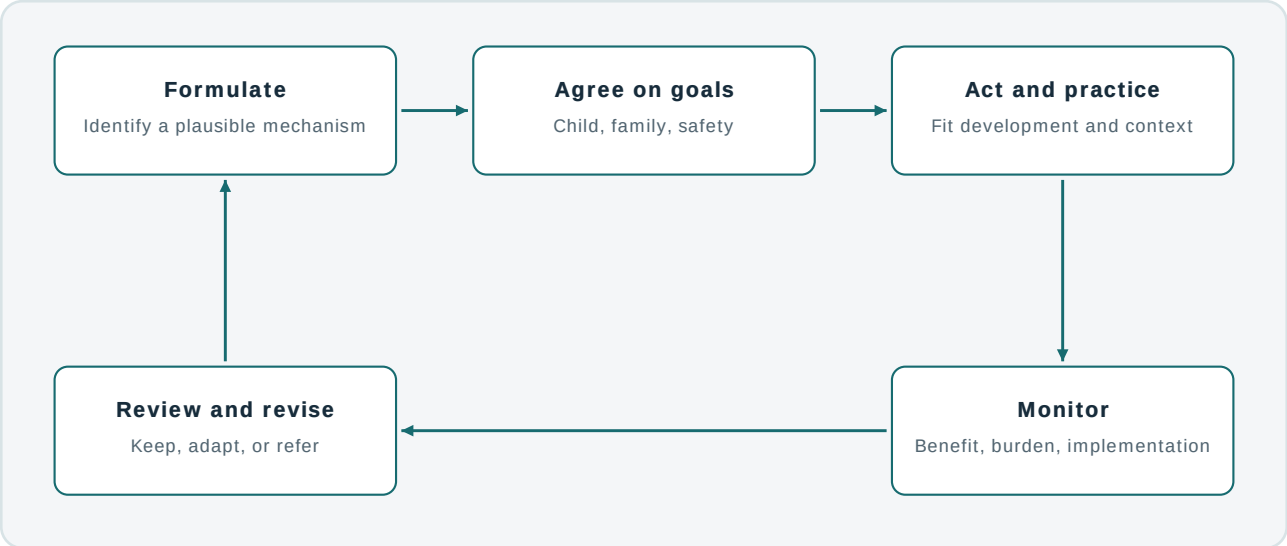


Figure 19.1. Intervention is a learning cycle. A formulation guides a proportionate plan. Practice and monitoring reveal whether the proposed mechanism is relevant and whether the intervention can be used in daily life. New information may require revision of the original formulation. Original course diagram.

Foundational Readings

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

These readings provide introductory foundations, not current diagnostic or legal authority. Read examples and statistical claims critically. Consult the chapter lecture, current professional guidance, and the source record when making real decisions.

Types of Treatment

OpenStax source module m82282 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82282/index.cnxml

One of the goals of therapy is to help a person stop repeating and reenacting destructive patterns and to start looking for better solutions to difficult situations. This goal is reflected in the following poem:

Autobiography in Five Short Chapters by Portia Nelson (1993)

Chapter One I walk down the street. There is a deep hole in the sidewalk. I fall in. I am lost. . . . I am helpless. It isn't my fault. It takes forever to find a way out.

Chapter Two I walk down the same street. There is a deep hole in the sidewalk. I pretend I don't see it. I fall in again. I can't believe I am in this same place. But, it isn't my fault. It still takes a long time to get out.

Chapter Three I walk down the same street. There is a deep hole in the sidewalk. I see it is there. I still fall in . . . it's a habit . . . but, my eyes are open. I know where I am. It is my fault. I get out immediately.

Chapter Four I walk down the same street. There is a deep hole in the sidewalk. I walk around it.

Chapter Five I walk down another street.

Two types of therapy are psychotherapy and biomedical therapy. Both types of treatment help people with mental health issues, such as depression, anxiety, and schizophrenia. Psychotherapy is a psychological treatment that employs various methods to help someone overcome personal problems, or to attain personal growth. In modern practice, it has evolved into what is known as psychodynamic therapy, which will be discussed later. Biomedical therapy involves medication and/or medical procedures to treat psychological disorders. First, we will explore the various psychotherapeutic orientations outlined in the accompanying source-figure description (many of these orientations were discussed in the Introduction chapter).

Source comparison table

Various Psychotherapy Techniques		
Type	Description	Example
Psychodynamic psychotherapy	Talk therapy based on belief that the unconscious and childhood conflicts impact behavior	Patient talks about their past
Play therapy	Psychoanalytical therapy wherein interaction with toys is used instead of talk; used in child therapy	Patient (child) acts out family scenes with dolls

Various Psychotherapy Techniques		
Behavior therapy	Principles of learning applied to change undesirable behaviors	Patient learns to overcome fear of elevators through several stages of relaxation techniques
Cognitive therapy	Awareness of cognitive process helps patients eliminate thought patterns that lead to distress	Patient learns not to overgeneralize failure based on single failure
Cognitive-behavioral therapy	Work to change cognitive distortions and self-defeating behaviors	Patient learns to identify self-defeating behaviors to overcome an eating disorder
Humanistic therapy	Increase self-awareness and acceptance through focus on conscious thoughts	Patient learns to articulate thoughts that keep them from achieving their goals

Psychotherapy Techniques: Psychoanalysis

Psychoanalysis was developed by Sigmund Freud and was the first form of psychotherapy. It was the dominant therapeutic technique in the early 20th century, but it has since waned significantly in popularity. Freud believed most of our psychological problems are the result of repressed impulses and trauma experienced in childhood, and he believed psychoanalysis would help uncover long-buried feelings. In a psychoanalyst s office, you might see a patient lying on a couch speaking of dreams or childhood memories, and the therapist using various Freudian methods such as free association and dream analysis (the accompanying source-figure description). In free association, the patient relaxes and then says whatever comes to mind at the moment. However, Freud felt that the ego would at times try to block, or repress, unacceptable urges or painful conflicts during free association. Consequently, a patient would demonstrate resistance to recalling these thoughts or situations. In dream analysis, a therapist interprets the underlying meaning of dreams.

Psychoanalysis is a therapy approach that typically takes years. Over the course of time, the patient reveals a great deal about himself to the therapist. Freud suggested that during this patient-therapist relationship, the patient comes to develop strong feelings for the therapist maybe positive feelings, maybe negative feelings. Freud called this transference: the patient transfers all the positive or negative emotions associated with the patient s other relationships to the psychoanalyst. For example, Crystal is seeing a psychoanalyst. During the years of therapy, she comes to see her therapist as a father figure. She transfers her feelings about her father onto her therapist, perhaps in an effort to gain the love and attention she did not receive from her own father.

Source illustration description (image not reproduced): This photograph shows what Freud s famous psychoanalytic couch looked like. The couch is draped in tapestries and pillows, and the room is decorated with sculptures, books and pictures on the wall. This is the famous couch in Freud s consulting room. Patients were instructed to lie comfortably on the couch and to face away from Freud in order to feel less inhibited and to help them focus. Today, psychotherapy patients are not likely to lie on a couch; instead they are more likely to sit facing the therapist (Prochaska & Norcross, 2010). (credit: Robert Huffstutter)

Today, Freud s psychoanalytical perspective has been expanded upon by the developments of subsequent theories and methodologies: the psychodynamic perspective. This approach to therapy remains centered on the role of people s internal drives and forces, but treatment is less intensive than Freud s original model.

Psychotherapy: Play Therapy

Play therapy is often used with children since they are not likely to sit on a couch and recall their dreams or engage in traditional talk therapy. This technique uses a therapeutic process of play to help clients prevent or resolve psychosocial difficulties and achieve optimal growth (O Connor, 2000, p. 7). The idea is that children play out their hopes, fantasies, and traumas while using dolls, stuffed animals, and sandbox figurines (the accompanying source-figure description). Play therapy can also be used to help a therapist make a diagnosis. The therapist observes how the child interacts with toys (e.g., dolls, animals, and home settings) in an effort to understand the roots of the child s disturbed behavior. Play therapy can be nondirective or directive. In nondirective play therapy, children are encouraged to work through their problems by playing freely while the therapist observes (LeBlanc & Ritchie, 2001). In directive play therapy, the therapist provides more structure and guidance in the play session by suggesting topics, asking questions, and even playing with the child (Harter, 1977).

Source illustration description (image not reproduced): This photograph shows a person playing with objects in a small box filled with sand. The person is organizing these objects and small play figures in a form of treatment called sandplay. This type of play therapy is known as sandplay or sandtray therapy. Children can set up a three-dimensional world using various figures and objects that correspond to their inner state (Kalff, 1991). (credit: Kristina Walter)

Psychotherapy: Behavior Therapy

In psychoanalysis, therapists help their patients look into their past to uncover repressed feelings. In behavior therapy, a therapist employs principles of learning to help clients change undesirable behaviors rather than digging deeply into one s unconscious. Therapists with this orientation believe that dysfunctional behaviors, like phobias and bedwetting, can be changed by teaching clients new, more constructive behaviors. Behavior therapy employs both classical and operant conditioning techniques to change behavior.

One type of behavior therapy utilizes classical conditioning techniques. Therapists using these techniques believe that dysfunctional behaviors are conditioned responses. Applying the conditioning principles developed by Ivan Pavlov, these therapists seek to recondition their clients and thus change their behavior. Emmie is eight years old, and frequently wets her bed at night. She s been invited to several sleepovers, but she won t go because of her problem. Using a type of conditioning therapy, Emmie begins to sleep on a liquid-sensitive bed pad that is hooked to an alarm. When moisture touches the pad, it sets off the alarm, waking up Emmie. When this process is repeated enough times, Emmie develops an association between urinary relaxation and waking up, and this stops the bedwetting. Emmie has now gone three weeks without wetting her bed and is looking forward to her first sleepover this weekend.

One commonly used classical conditioning therapeutic technique is counterconditioning: a client learns a new response to a stimulus that has previously elicited an undesirable behavior. Two counterconditioning techniques are aversive conditioning and exposure therapy. Aversive conditioning uses an unpleasant stimulus to stop an undesirable behavior. Therapists apply this technique to eliminate addictive behaviors, such as smoking, nail biting, and drinking. In aversion therapy, clients will typically engage in a specific behavior (such as nail biting) and at the same time are exposed to something unpleasant, such as a mild electric shock or a bad taste. After repeated associations between the unpleasant stimulus and the behavior, the client can learn to stop the unwanted behavior.

Aversion therapy has been used effectively for years in the treatment of alcoholism (Davidson, 1974; Elkins, 1991; Streeton & Whelan, 2001). One common way this occurs is through a chemically based substance known as Antabuse. When a person takes Antabuse and then consumes alcohol, uncomfortable side effects result including nausea, vomiting, increased heart rate, heart palpitations, severe headache, and shortness of breath. Antabuse is repeatedly paired with alcohol until the client associates alcohol with unpleasant feelings, which decreases the client s desire to consume alcohol. Antabuse creates a conditioned aversion to alcohol because it replaces the original pleasure response with an unpleasant one.

In exposure therapy, a therapist seeks to treat clients fears or anxiety by presenting them with the object or situation that causes their problem, with the idea that they will eventually get used to it. This can be done via reality, imagination, or virtual reality. Exposure therapy was first reported in 1924 by Mary Cover Jones, who is considered the mother of behavior therapy. Jones worked with a boy named Peter who was afraid of rabbits. Her goal was to replace Peter s fear of rabbits with a conditioned response of relaxation, which is a response that is incompatible with fear (the accompanying source-figure description). How did she do it? Jones began by placing a caged rabbit on the other side of a room with Peter while he ate his afternoon snack. Over the course of several days, Jones moved the rabbit closer and closer to where Peter was seated with his snack. After two months of being exposed to the rabbit while relaxing with his snack, Peter was able to hold the rabbit and pet it while eating (Jones, 1924).

Source illustration description (image not reproduced): This figure, titled Exposure Therapy, illustrates the exposure therapy strategy of Mary Cover Jones to rid a person of the fear of rabbits. The first of four levels depicts an image of a person and a rabbit with an equals sign between them. Under the rabbit reads conditioned stimulus (CS), and under the person reads fear of rabbits. The second level depicts an image of milk and cookies, labeled unconditioned stimulus (US), and on the other side of an equals sign there is a picture of the same person labeled unconditioned response (UR). The third level shows the milk and cookies, labeled unconditioned stimulus (US), and rabbit, labeled conditioned stimulus (CS), to the left and right of a plus sign, with the person on the other side of an equals sign. The label unconditioned response (UR) is below the person. The final level shows the person and the rabbit separated by an equals sign. This time the rabbit is labeled conditioned stimulus (CS) and the person is labeled conditioned response (CR). Exposure therapy seeks to change the response to a conditioned stimulus (CS). An unconditioned stimulus is presented over and over just after the presentation of the conditioned stimulus. This figure shows conditioning as conducted in Mary Cover Jones 1924 study.

Thirty years later, Joseph Wolpe (1958) refined Jones s techniques, giving us systematic desensitization. Using this method, a person creates a hierarchy of anxiety, ranging from the least-anxiety-producing stimulus to the feared object. These activities are paired with relaxation techniques, which are taught to the client beforehand and used during the graduated exposures. The idea is that a person cannot be nervous and relaxed at the same time. Therefore, if they can learn to relax when they are facing environmental stimuli that make them nervous or fearful, they can eventually eliminate the unwanted fear response. (the accompanying source-figure description).

Source illustration description (image not reproduced): Figure (a): A close-up picture of a very large spider on a person s arm is shown. The person is using its other hand to hold up two of the spider s legs. Figure (b): A chart has two columns: Behavior and Fear Rating. The first behavior is look at a photo of a spider with a fear rating of 15. The second behavior is look at a real spider through clear glass with a fear rating of 35. The third behavior is hold a box that contains a spider with a fear rating of 55. The fourth behavior is watch a spider crawl across the floor with a fear rating of 65. The fifth behavior is watch a spider crawl across your desk with a fear rating of 75. The sixth behavior is hold a spider with gloved hands with a fear rating of 90. The seventh behavior is hold a spider with bare hands with a fear rating of 100. This person suffers from arachnophobia (fear of spiders). Through exposure therapy they are learning how to face their fear in a controlled, therapeutic setting. (credit: GollyGforce Living My Worst Nightmare /Flickr)

For example, Jayden is terrified of elevators. Nothing bad has ever happened to him on an elevator, but he s so afraid of elevators that he will always take the stairs. That wasn t a problem when Jayden worked on the second floor of an office building, but now he has a new job on the 29th floor of a skyscraper in downtown Los Angeles. Jayden knows he can t climb 29 flights of stairs in order to get to work each day, so he decided to see a behavior therapist for help. The therapist asks Jayden to first construct a hierarchy of elevator-related situations that elicit fear and anxiety. They range from situations of mild anxiety such as being nervous around the other people in the elevator, to the fear of getting an arm caught in the door, to panic-provoking situations such as getting trapped or the cable snapping. Next, the therapist uses progressive relaxation. They teach Jayden how to relax each of his muscle groups so that he achieves a drowsy, relaxed, and comfortable state of mind. Once he s in this state, the therapist asks Jayden to imagine a mildly anxiety-provoking situation. Jayden is standing in front of the elevator thinking about pressing the call button.

If this scenario causes Jayden anxiety, he lifts his finger. The therapist would then tell Jayden to forget the scene and return to his relaxed state. They repeat this scenario over and over until Jayden can

imagine himself pressing the call button without anxiety. Over time the therapist and Jayden use progressive relaxation and imagination to proceed through all of the situations on Jayden s hierarchy until he becomes desensitized to each one. After this, Jayden and the therapist begin to practice what he only previously envisioned in therapy, gradually going from pressing the button to actually riding an elevator. The goal is that Jayden will soon be able to take the elevator all the way up to the 29th floor of his office without feeling any anxiety.

Sometimes, it s too impractical, expensive, or embarrassing to re-create anxiety- producing situations, so a therapist might employ virtual reality exposure therapy by using a simulation to help conquer fears. Virtual reality exposure therapy has been used effectively to treat numerous anxiety disorders such as the fear of public speaking, claustrophobia (fear of enclosed spaces), aviophobia (fear of flying), and post-traumatic stress disorder (PTSD), a trauma and stressor-related disorder (Gerardi, Cukor, Difede, Rizzo, & Rothbaum, 2010).

Some behavior therapies employ operant conditioning. Recall what you learned about operant conditioning: We have a tendency to repeat behaviors that are reinforced. What happens to behaviors that are not reinforced? They become extinguished. These principles, defined by Skinner as operant conditioning, can be applied to help people with a wide range of psychological problems.

One popular operant conditioning intervention is called the token economy. This involves a controlled setting where individuals are reinforced for desirable behaviors with tokens, such as a poker chip, that can be exchanged for items or privileges. Token economies are often used in psychiatric hospitals to increase patient cooperation and activity levels. Patients are rewarded with tokens when they engage in positive behaviors (e.g., making their beds, brushing their teeth, coming to the cafeteria on time, and socializing with other patients). They can later exchange the tokens for extra TV time, private rooms, visits to the canteen, and so on (Dickerson, Tenhula, & Green-Paden, 2005).

Psychotherapy: Cognitive Therapy

Cognitive therapy is a form of psychotherapy that focuses on how a person s thoughts lead to feelings of distress. The idea behind cognitive therapy is that how you think determines how you feel and act. Cognitive therapists help their clients change dysfunctional thoughts in order to relieve distress. They help a client see how they misinterpret a situation (cognitive distortion). For example, a client may overgeneralize. Because a person failed one test in Psychology 101, they feel stupid and worthless. These thoughts then cause their mood to worsen. Therapists also help clients recognize when they blow things out of proportion. Because Ray failed his Psychology 101 test, he has concluded that he s going to fail the entire course and probably flunk out of college altogether. These errors in thinking have contributed to Ray s feelings of distress. His therapist will help him challenge these irrational beliefs, focus on their illogical basis, and correct them with more logical and rational thoughts and beliefs.

Cognitive therapy was developed by psychiatrist Aaron Beck in the 1960s. His initial focus was on depression and how a client s self-defeating attitude served to maintain a depression despite positive factors in her life (Beck, Rush, Shaw, & Emery, 1979) (the accompanying source-figure description). Through questioning, a cognitive therapist can help a client recognize dysfunctional ideas, challenge catastrophizing thoughts about themselves and their situations, and find a more positive way to view things (Beck, 2011).

Source illustration description (image not reproduced): This graphic depicts two three-box flowcharts showing reactions to failing a test. The first flowchart flows from Failed test to Internal beliefs: I m worthless and stupid to Depression. The second flowchart flows from Failed test to Internal beliefs: I m smart, but I didn t study for this test. I can do better. to No depression. Your emotional reactions are the result of your thoughts about the situation rather than the situation itself. For instance, if you consistently interpret events and emotions around the themes of loss and defeat, then you are likely to be depressed. Through therapy, you can learn more logical ways to interpret situations.

Psychotherapy: Cognitive-Behavioral Therapy

Cognitive-behavioral therapists focus much more on present issues than on a patient s childhood or past, as in other forms of psychotherapy. One of the first forms of cognitive-behavioral therapy was rational emotive therapy (RET), which was founded by Albert Ellis and grew out of his dislike of Freudian psychoanalysis (Daniel, n.d.). Behaviorists such as Joseph Wolpe also influenced Ellis s therapeutic approach (National Association of Cognitive-Behavioral Therapists, 2009).

Cognitive-behavioral therapy (CBT) helps clients examine how their thoughts affect their behavior. It aims to change cognitive distortions and self-defeating behaviors. In essence, this approach is designed to change the way people think as well as how they act. It is similar to cognitive therapy in that CBT attempts to make individuals aware of their irrational and negative thoughts and helps people replace them with new, more positive ways of thinking. It is also similar to behavior therapies in that CBT teaches people how to practice and engage in more positive and healthy approaches to daily situations. In total, hundreds of studies have shown the effectiveness of cognitive-behavioral therapy in the treatment of numerous psychological disorders such as depression, PTSD, anxiety disorders, eating disorders, bipolar disorder, and substance abuse (Beck Institute for Cognitive Behavior Therapy, n.d.). For example, CBT has been found to be effective in decreasing levels of hopelessness and suicidal thoughts in previously suicidal teenagers (Alavi, Sharifi, Ghanizadeh, & Dehbozorgi, 2013). Cognitive-behavioral therapy has also been effective in reducing PTSD in specific populations, such as transit workers (Lowinger & Rombom, 2012).

Cognitive-behavioral therapy aims to change cognitive distortions and self-defeating behaviors using techniques like the ABC model. With this model, there is an Action (sometimes called an activating event), the Belief about the event, and the Consequences of this belief. Let s say Jon and Joe both go to a party. Jon and Joe each have met an interesting person at the party and spend a few hours chatting with them. At the end of the party, Jon and Joe ask to exchange phone numbers with the person they've been talking to, and the request is refused. Both Jon and Joe are surprised, as they thought things were going well. What can Jon and Joe tell themselves about why the person was not interested? Let s say Jon tells himself he is a loser, or is ugly, or has no game. Jon then gets depressed and decides not to go to another party, which starts a cycle that keeps him depressed. Joe tells himself that he had bad breath, goes out and buys a new toothbrush, goes to another party, and meets someone new.

Jon s belief about what happened results in a consequence of further depression, whereas Joe s belief does not. Jon is internalizing the attribution or reason for the rebuffs, which triggers his depression. On the other hand, Joe is externalizing the cause, so his thinking does not contribute to feelings of depression. Cognitive-behavioral therapy examines specific maladaptive and automatic thoughts and cognitive distortions. Some examples of cognitive distortions are all-or-nothing thinking, overgeneralization, and jumping to conclusions. In overgeneralization, someone takes a small situation and makes it huge for example, instead of saying, This particular person was not interested in me, the man says, I am ugly, a loser, and no one is ever going to be interested in me.

All or nothing thinking, which is a common type of cognitive distortion for people suffering from depression, reflects extremes. In other words, everything is black or white. After being turned down for a date, Jon begins to think, No woman will ever go out with me. I m going to be alone forever. He begins to feel anxious and sad as he contemplates his future.

The third kind of distortion involves jumping to conclusions assuming that people are thinking negatively about you or reacting negatively to you, even though there is no evidence. Consider the example of Savannah and Hillaire, who recently met at a party. They have a lot in common, and Savannah thinks they could become friends. She calls Hillaire to invite her for coffee. Since Hillaire doesn t answer, Savannah leaves her a message. Several days go by and Savannah never hears back from her potential new friend. Maybe Hillaire never received the message because she lost her phone or she is too busy to return the phone call. But if Savannah believes that Hillaire didn t like Savannah or didn t want to be her friend, she is demonstrating the cognitive distortion of jumping to conclusions.

How effective is CBT? One client said this about his cognitive-behavioral therapy:

I have had many painful episodes of depression in my life, and this has had a negative effect on my career and has put considerable strain on my friends and family. The treatments I have received, such as taking antidepressants and psychodynamic counseling, have helped [me] to cope with the symptoms and to get some insights into the roots of my problems. CBT has been by far the most useful approach I have found in tackling these mood problems. It has raised my awareness of how my thoughts impact on my moods. How the way I think about myself, about others and about the world can lead me into depression. It is a practical approach, which does not dwell so much on childhood experiences, whilst acknowledging that it was then that these patterns were learned. It looks at what is happening now, and gives tools to manage these moods on a daily basis. (Martin, 2007, n.p.)

Psychotherapy: Humanistic Therapy

Humanistic psychology focuses on helping people achieve their potential. So it makes sense that the goal of humanistic therapy is to help people become more self-aware and accepting of themselves. In contrast to psychoanalysis, humanistic therapists focus on conscious rather than unconscious thoughts. They also emphasize the patient s present and future, as opposed to exploring the patient s past.

Psychologist Carl Rogers developed a therapeutic orientation known as Rogerian, or client-centered therapy. Note the change from patients to clients. Rogers (1951) felt that the term patient suggested the person seeking help was sick and looking for a cure. Since this is a form of nondirective therapy, a therapeutic approach in which the therapist does not give advice or provide interpretations but helps the person to identify conflicts and understand feelings, Rogers (1951) emphasized the importance of the person taking control of his own life to overcome life s challenges.

In client-centered therapy, the therapist uses the technique of active listening. In active listening, the therapist acknowledges, restates, and clarifies what the client expresses. Therapists also practice what Rogers called unconditional positive regard, which involves not judging clients and simply accepting them for who they are. Rogers (1951) also felt that therapists should demonstrate genuineness, empathy, and acceptance toward their clients because this helps people become more accepting of themselves, which results in personal growth.

Evaluating Various Forms of Psychotherapy

How can we assess the effectiveness of psychotherapy? Is one technique more effective than another? For anyone considering therapy, these are important questions. According to the American Psychological Association, three factors work together to produce successful treatment. The first is the use of evidence-based treatment that is deemed appropriate for your particular issue. The second important factor is the clinical expertise of the psychologist or therapist. The third factor is your own characteristics, values, preferences, and culture. Many people begin psychotherapy feeling like their problem will never be resolved; however, psychotherapy helps people see that they can do things to make their situation better. Psychotherapy can help reduce a person s anxiety, depression, and maladaptive behaviors. Through psychotherapy, individuals can learn to engage in healthy behaviors designed to help them better express emotions, improve relationships, think more positively, and perform more effectively at work or school.

Many studies have explored the effectiveness of psychotherapy. For example, one large-scale study that examined 16 meta-analyses of CBT reported that it was equally effective or more effective than other therapies in treating PTSD, generalized anxiety disorder, depression, and social phobia (Butlera, Chapmanb, Formanc, & Becka, 2006). Another study found that CBT was as effective at treating depression (43% success rate) as prescription medication (50% success rate) compared to the placebo rate of 25% (DeRubeis et al., 2005). Another meta-analysis found that psychodynamic therapy was also as effective at treating these types of psychological issues as CBT (Shedler, 2010). However, no studies have found one psychotherapeutic approach more effective than another (Abbass, Kisely, & Kroenke, 2006; Chorpita et al., 2011), nor have they shown any relationship between a client s treatment

outcome and the level of the clinician s training or experience (Wampold, 2007). Regardless of which type of psychotherapy an individual chooses, one critical factor that determines the success of treatment is the person s relationship with the psychologist or therapist.

Biomedical Therapies

Individuals can be prescribed biologically based treatments or psychotropic medications that are used to treat mental disorders. While these are often used in combination with psychotherapy, they also are taken by individuals not in therapy. This is known as biomedical therapy. Medications used to treat psychological disorders are called psychotropic medications and are prescribed by medical doctors, including psychiatrists. In Louisiana and New Mexico, psychologists are able to prescribe some types of these medications (American Psychological Association, 2014).

Different types and classes of medications are prescribed for different disorders. An individual with depression might be given an antidepressant, an individual with bipolar disorder might be given a mood stabilizer, and an individual with schizophrenia might be given an antipsychotic. These medications treat the symptoms of a psychological disorder by altering the levels or effects of neurotransmitters. For example, each type of antidepressant affects a different neurotransmitter, such as SSRI (selective serotonin reuptake inhibitor) antidepressants that increase the level of the neurotransmitter serotonin, and SNRI (serotonin-norepinephrine reuptake inhibitor) antidepressants that increase the levels of both serotonin and norepinephrine. They can help people feel better so that they can function on a daily basis, but they do not cure the disorder. Some people may only need to take a psychotropic medication for a short period of time. Others with severe disorders like bipolar disorder or schizophrenia may need to take psychotropic medication for a long time.

Psychotropic medications are a popular treatment option for many types of disorders, and research suggests that they are most effective when combined with psychotherapy. This is especially true for the most common mental disorders, such as depressive and anxiety disorders (Cuijpers et al, 2014). When considering adding medication as a treatment option, individuals should know that some psychotropic medications have very concerning side effects. the accompanying source-figure description shows the commonly prescribed types of medications, how they are used, and some of the potential side effects that may occur.

Source comparison table

Some Commonly Prescribed Psychotropic Medications				
Type of Medication	Used to Treat	Brand Names of Commonly Prescribed Medications	How They Work	Side Effects
Antipsychotics (developed in the 1950s)	Schizophrenia and other types of severe thought disorders	Haldol, Mellaril, Prolixin, Thorazine	Treat positive psychotic symptoms such as auditory and visual hallucinations, delusions, and paranoia by blocking the neurotransmitter dopamine	Long-term use can lead to tardive dyskinesia, involuntary movements of the arms, legs, tongue and facial muscles, resulting in Parkinson s-like tremors

Some Commonly Prescribed Psychotropic Medications				
Atypical Antipsychotics (developed in the late 1980s)	Schizophrenia and other types of severe thought disorders	Abilify, Risperdal, Clozaril	Treat the negative symptoms of schizophrenia, such as withdrawal and apathy, by targeting both dopamine and serotonin receptors; newer medications may treat both positive and negative symptoms	Can increase the risk of obesity and diabetes as well as elevate cholesterol levels; constipation, dry mouth, blurred vision, drowsiness, and dizziness
Anti-depressants	Depression and increasingly for anxiety	Paxil, Prozac, Zoloft (selective serotonin reuptake inhibitors, [SSRIs]); Tofranil and Elavil (tricyclics)	Alter levels of neurotransmitters such as serotonin and norepinephrine	SSRIs: headache, nausea, weight gain, drowsiness, reduced sex drive Tricyclics: dry mouth, constipation, blurred vision, drowsiness, reduced sex drive, increased risk of suicide
Anti-anxiety agents	Anxiety and agitation that occur in OCD, PTSD, panic disorder, and social phobia	Xanax, Valium, Ativan (Benzodiazepines) Buspar (non-Benzodiazepine)	Depress central nervous system activity	Drowsiness, dizziness, headache, fatigue, lightheadedness
Mood Stabilizers	Bipolar disorder	Lithium, Depakote, Lamictal, Tegretol	Treat episodes of mania as well as depression	Excessive thirst, irregular heartbeat, itching/rash, swelling (face, mouth, and extremities), nausea, loss of appetite
Stimulants	ADHD	Adderall, Ritalin	Improve ability to focus on a task and maintain attention	Decreased appetite, difficulty sleeping, stomachache, headache

Another biologically based treatment that continues to be used, although infrequently, is electroconvulsive therapy (ECT) (formerly known by its unscientific name as electroshock therapy). It involves using an electrical current to induce seizures to help alleviate the effects of severe depression. The exact mechanism is unknown, although it does help alleviate symptoms for people with severe depression who have not responded to traditional drug therapy (Pagnin, de Queiroz, Pini, & Cassano, 2004). About 85% of people treated with ECT improve (Reti, n.d.). However, the memory loss associated with repeated administrations has led to it being implemented as a last resort (Donahue, 2000; Prudic, Peyser, & Sackeim, 2000). A more recent alternative is transcranial magnetic stimulation (TMS), a procedure approved by the FDA in 2008 that uses magnetic fields to stimulate nerve cells in the brain to improve depression symptoms; it is used when other treatments have not worked (Mayo Clinic, 2012).

Dig Deeper

A buzzword in therapy today is evidence-based practice. However, it s not a novel concept but one that has been used in medicine for at least two decades. Evidence-based practice is used to reduce errors in treatment selection by making clinical decisions based on research (Sackett & Rosenberg, 1995). In any case, evidence-based treatment is on the rise in the field of psychology. So what is it, and why does it matter? In an effort to determine which treatment methodologies are evidenced-based, professional organizations such as the American Psychological Association (APA) have recommended that specific psychological treatments be used to treat certain psychological disorders (Chambless & Ollendick, 2001). According to the APA (2005), Evidence-based practice in psychology (EBPP) is the integration of the best available research with clinical expertise in the context of patient characteristics, culture, and preferences (p. 1).

The foundational idea behind evidence based treatment is that best practices are determined by research evidence that has been compiled by comparing various forms of treatment (Charman & Barkham, 2005). These treatments are then operationalized and placed in treatment manuals trained therapists follow these manuals. The benefits are that evidence-based treatment can reduce variability between therapists to ensure that a specific approach is delivered with integrity (Charman & Barkham, 2005). Therefore, clients have a higher chance of receiving therapeutic interventions that are effective at treating their specific disorder. While EBPP is based on randomized control trials, critics of EBPP reject it stating that the results of trials cannot be applied to individuals and instead determinations regarding treatment should be based on a therapist s judgment (Mullen & Streiner, 2004).

Summary

Psychoanalysis was developed by Sigmund Freud. Freud s theory is that a person s psychological problems are the result of repressed impulses or childhood trauma. The goal of the therapist is to help a person uncover buried feelings by using techniques such as free association and dream analysis.

Play therapy is a psychodynamic therapy technique often used with children. The idea is that children play out their hopes, fantasies, and traumas, using dolls, stuffed animals, and sandbox figurines.

In behavior therapy, a therapist employs principles of learning from classical and operant conditioning to help clients change undesirable behaviors. Counterconditioning is a commonly used therapeutic technique in which a client learns a new response to a stimulus that has previously elicited an undesirable behavior via classical conditioning. Principles of operant conditioning can be applied to help people deal with a wide range of psychological problems. Token economy is an example of a popular operant conditioning technique.

Cognitive therapy is a technique that focuses on how thoughts lead to feelings of distress. The idea behind cognitive therapy is that how you think determines how you feel and act. Cognitive therapists help clients change dysfunctional thoughts in order to relieve distress. Cognitive-behavioral therapy explores how our thoughts affect our behavior. Cognitive-behavioral therapy aims to change cognitive distortions and self-defeating behaviors.

Humanistic therapy focuses on helping people achieve their potential. One form of humanistic therapy developed by Carl Rogers is known as client-centered or Rogerian therapy. Client-centered therapists use the techniques of active listening, unconditional positive regard, genuineness, and empathy to help clients become more accepting of themselves.

Often in combination with psychotherapy, people can be prescribed biologically based treatments such as psychotropic medications and/or other medical procedures such as electro-convulsive therapy.

Review Questions

OpenStax learning exercise

Q. The idea behind _____ is that how you think determines how you feel and act.

A. A

OpenStax learning exercise

Q. Mood stabilizers, such as lithium, are used to treat _____.

A. C

OpenStax learning exercise

Q. Clay is in a therapy session. The therapist asks him to relax and say whatever comes to his mind at the moment. This therapist is using _____, which is a technique of _____.

A. D

Critical Thinking Question

OpenStax learning exercise

Q. Imagine that you are a psychiatrist. Your patient, Pat, comes to you with the following symptoms: anxiety and feelings of sadness. Which therapeutic approach would you recommend and why?

A. I would recommend psychodynamic talk therapy or cognitive therapy to help the person see how her thoughts and behaviors are having negative effects.

Personal Application Question

OpenStax learning exercise

Q. If you were to choose a therapist practicing one of the techniques presented in this section, which kind of therapist would you choose and why?

Treatment Modalities

OpenStax source module m82283 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82283/index.cnxml

Once a person seeks treatment, whether voluntarily or involuntarily, they have an intake done to assess their clinical needs. An intake is the therapist s first meeting with the client. The therapist gathers specific information to address the client s immediate needs, such as the presenting problem, the client s support system, and insurance status. The therapist informs the client about confidentiality, fees, and what to expect in treatment. Confidentiality means the therapist cannot disclose confidential communications to any third party unless mandated or permitted by law to do so. During the intake, the therapist and client will work together to discuss treatment goals. Then a treatment plan will be formulated, usually with specific measurable objectives. Also, the therapist and client will discuss how treatment success will be measured and the estimated length of treatment. There are several different modalities of treatment (the accompanying source-figure description): Individual therapy, family therapy, couples therapy, and group therapy are the most common.

Source illustration description (image not reproduced): Two photographs are shown. Photograph A depicts two people in conversation. Photograph B depicts a large group of people sitting in a circle on the beach. Therapy may occur (a) one-on-one between a therapist and client, or (b) in a group setting. (credit a: modification of work by Connor Ashleigh, AusAID/Department of Foreign Affairs and Trade)

Individual Therapy

In individual therapy, also known as individual psychotherapy or individual counseling, the client and clinician meet one-on-one (usually from 45 minutes to 1 hour). These meetings typically occur weekly or every other week, and sessions are conducted in a confidential and caring environment (the accompanying source-figure description). The clinician will work with clients to help them explore their feelings, work through life challenges, identify aspects of themselves and their lives that they wish to change, and set goals to help them work towards these changes. A client might see a clinician for only a few sessions, or the client may attend individual therapy sessions for a year or longer. The amount of time spent in therapy depends on the needs of the client as well as her personal goals.

Source illustration description (image not reproduced): A photograph depicting a woman in a therapy session with her therapist is shown. In an individual therapy session, a client works one-on-one with a trained therapist. (credit: Alan Cleaver)

Group Therapy

In group therapy, a clinician meets together with several clients with similar problems (the accompanying source-figure description). When children are placed in group therapy, it is particularly important to match clients for age and problems. One benefit of group therapy is that it can help decrease a client s shame and isolation about a problem while offering needed support, both from the therapist and other members of the group (American Psychological Association, 2014). A nine-year-old sexual abuse victim, for example, may feel very embarrassed and ashamed. If he is placed in a group with other sexually abused boys, he will realize that he is not alone. A child struggling with poor social skills would likely benefit from a group with a specific curriculum to foster special skills. A woman suffering from post-partum depression could feel less guilty and more supported by being in a group with similar women.

Group therapy also has some specific limitations. Members of the group may be afraid to speak in front of other people because sharing secrets and problems with complete strangers can be stressful and overwhelming. There may be personality clashes and arguments among group members. There could also be concerns about confidentiality: Someone from the group might share what another participant said to people outside of the group.

Source illustration description (image not reproduced): A group of people arranged in a circle having a conversation is shown. In group therapy, usually 5 10 people meet with a trained therapist to discuss a common issue such as divorce, grief, an eating disorder, substance abuse, or anger management. (credit: Cory Zanker)

Another benefit of group therapy is that members can confront each other about their patterns. For those with some types of problems, such as sexual abusers, group therapy is the recommended treatment. Group treatment for this population is considered to have several benefits:

Group treatment is more economical than individual, couples, or family therapy. Sexual abusers often feel more comfortable admitting and discussing their offenses in a treatment group where others are modeling openness. Clients often accept feedback about their behavior more willingly from other group members than from therapists. Finally, clients can practice social skills in group treatment settings. (McGrath, Cumming, Burchard, Zeoli, & Ellerby, 2009)

Groups that have a strong educational component are called psycho-educational groups. For example, a group for children whose parents have cancer might discuss in depth what cancer is, types of treatment for cancer, and the side effects of treatments, such as hair loss. Often, group therapy sessions with children take place in school. They are led by a school counselor, a school psychologist, or a school social worker. Groups might focus on test anxiety, social isolation, self-esteem, bullying, or school failure (Shechtman, 2002). Whether the group is held in school or in a clinician s office, group therapy has been found to be effective with children facing numerous kinds of challenges (Shechtman, 2002).

During a group session, the entire group could reflect on an individual s problem or difficulties, and others might disclose what they have done in that situation. When a clinician is facilitating a group, the

focus is always on making sure that everyone benefits and participates in the group and that no one person is the focus of the entire session. Groups can be organized in various ways: some have an overarching theme or purpose, some are time-limited, some have open membership that allows people to come and go, and some are closed. Some groups are structured with planned activities and goals, while others are unstructured: There is no specific plan, and group members themselves decide how the group will spend its time and on what goals it will focus. This can become a complex and emotionally charged process, but it is also an opportunity for personal growth (Page & Berkow, 1994).

Couples Therapy

Couples therapy involves two people in an intimate relationship who are having difficulties and are trying to resolve them (the accompanying source-figure description). The couple may be dating, partnered, engaged, or married. The primary therapeutic orientation used in couples counseling is cognitive-behavioral therapy (Rathus & Sanderson, 1999). Couples meet with a therapist to discuss conflicts and/or aspects of their relationship that they want to change. The therapist helps them see how their individual backgrounds, beliefs, and actions are affecting their relationship. Often, a therapist tries to help the couple resolve these problems, as well as implement strategies that will lead to a healthier and happier relationship, such as how to listen, how to argue, and how to express feelings. However, sometimes, after working with a therapist, a couple will realize that they are too incompatible and will decide to separate. Some couples seek therapy to work out their problems, while others attend therapy to determine whether staying together is the best solution. Counseling couples in a high-conflict and volatile relationship can be difficult. In fact, psychologists Peter Pearson and Ellyn Bader, who founded the Couples Institute in Palo Alto, California, have compared the experience of the clinician in couples therapy to be like piloting a helicopter in a hurricane (Weil, 2012, para. 7).

Source illustration description (image not reproduced): A photograph shows two people talking to a third person. In couples counseling, a therapist helps people work on their relationship. (credit: Cory Zanker)

Family Therapy

Family therapy is a special form of group therapy, consisting of one or more families. Although there are many theoretical orientations in family therapy, one of the most predominant is the systems approach. The family is viewed as an organized system, and each individual within the family is a contributing member who creates and maintains processes within the system that shape behavior (Minuchin, 1985). Each member of the family influences and is influenced by the others. The goal of this approach is to enhance the growth of each family member as well as that of the family as a whole.

Often, dysfunctional patterns of communication that develop between family members can lead to conflict. A family with this dynamic might wish to attend therapy together rather than individually. In many cases, one member of the family has problems that detrimentally affect everyone. For example, a mother s depression, teen daughter s eating disorder, or father s alcohol dependence could affect all members of the family. The therapist would work with all members of the family to help them cope with the issue, and to encourage resolution and growth in the case of the individual family member with the problem.

With family therapy, the nuclear family (i.e., parents and children) or the nuclear family plus whoever lives in the household (e.g., grandparent) come into treatment. Family therapists work with the whole family unit to heal the family. There are several different types of family therapy. In structural family therapy, the therapist examines and discusses the boundaries and structure of the family: who makes the rules, who sleeps in the bed with whom, how decisions are made, and what are the boundaries within the family. In some families, the parents do not work together to make rules, or one parent may undermine the other, leading the children to act out. The therapist helps them resolve these issues and learn to communicate more effectively.

In strategic family therapy, the goal is to address specific problems within the family that can be dealt with in a relatively short amount of time. Typically, the therapist would guide what happens in the therapy session and design a detailed approach to resolving each member s problem (Madanes, 1991).

Summary

There are several modalities of treatment: individual therapy, group therapy, couples therapy, and family therapy are the most common. In an individual therapy session, a client works one-on-one with a trained therapist. In group therapy, usually 5 10 people meet with a trained group therapist to discuss a common issue (e.g., divorce, grief, eating disorders, substance abuse, or anger management). Couples therapy involves two people in an intimate relationship who are having difficulties and are trying to resolve them. The couple may be dating, partnered, engaged, or married. The therapist helps them resolve their problems as well as implement strategies that will lead to a healthier and happier relationship. Family therapy is a special form of group therapy. The therapy group is made up of one or more families. The goal of this approach is to enhance the growth of each individual family member and the family as a whole.

Review Questions

OpenStax learning exercise

Q. A treatment modality in which 5 10 people with the same issue or concern meet together with a trained clinician is known as _____.

A. C

OpenStax learning exercise

Q. What happens during an intake?

A. A

Critical Thinking Question

OpenStax learning exercise

Q. Compare and contrast individual and group therapies.

A. In an individual therapy session, a client works one-on-one with a trained therapist. In group therapy, usually 5 10 people meet with a trained group therapist to discuss a common issue, such as divorce, grief, eating disorder, substance abuse, or anger management.

Personal Application

OpenStax learning exercise

Q. Your best friend tells you that she is concerned about her cousin. The cousin a teenage girl is constantly coming home after her curfew, and your friend suspects that she has been drinking. What treatment modality would you recommend to your friend and why?

The Sociocultural Model and Therapy Utilization

OpenStax source module m82285 https://github.com/cnx-user-books/cnxbook-psychology-2e/blob/f5c9cf50686749d28cc5a594e0fdc487398f8307/modules/m82285/index.cnxml

The sociocultural perspective looks at you, your behaviors, and your symptoms in the context of your culture and background. For example, Jos is an 18-year-old Hispanic male from a traditional family. Jos comes to treatment because of depression. During the intake session, he reveals that he is gay and is nervous about telling his family. He also discloses that he is concerned because his religious background has taught him that being gay is wrong. How does his religious and cultural background affect him? How might his cultural background affect how his family reacts if Jos were to tell them he is gay?

Mental health professionals must develop cultural competence (the accompanying source-figure description), which means they must understand and address issues of race, culture, and ethnicity. They must also develop strategies to effectively address the needs of various populations for which Eurocentric therapies have limited application (Sue, 2004). For example, a counselor whose treatment focuses on individual decision making may be ineffective at helping a Chinese client with a collectivist approach to problem solving (Sue, 2004).

Multicultural counseling and therapy aims to offer both a helping role and process that uses modalities and defines goals consistent with the life experiences and cultural values of clients. It strives to recognize client identities to include individual, group, and universal dimensions, advocate the use of universal and culture-specific strategies and roles in the healing process, and balances the importance of individualism and collectivism in the assessment, diagnosis, and treatment of client and client systems (Sue, 2001).

This therapeutic perspective integrates the impact of cultural and social norms, starting at the beginning of treatment. Therapists who use this perspective work with clients to obtain and integrate information about their cultural patterns into a unique treatment approach based on their particular situation (Stewart, Simmons, & Habibpour, 2012). Sociocultural therapy can include individual, group, family, and couples treatment modalities.

Source illustration description (image not reproduced): A photo montage composed of eight photographs arranged in two parallel rows of four. From the top-left-hand-side, the photos are as follows: a person with a bicycle standing in a rice paddy, three children, three elderly people sitting along a rock wall, four cooks standing around a table, a classroom of students, a group of people seated at a covered outdoor table, two children wearing robes, and two people being held up by other people during a wedding ceremony. How do your cultural and religious beliefs affect your attitude toward mental health treatment? (credit top-left : modification of work by Staffan Scherz; credit top-left-middle : modification of work by Alejandra Quintero Sinisterra; credit top-right-middle : modification of work by Pedro Ribeiro Sim es; credit top-right : modification of work by Agustin Ruiz; credit bottom-left : modification of work by Czech Provincial Reconstruction Team; credit bottom-left-middle : modification of work by Arian Zwegers; credit bottom-right-middle : modification of work by Wonderlane /Flickr; credit bottom-right : modification of work by Shiraz Chanawala)

Barriers to Treatment

Statistically, ethnic minorities tend to utilize mental health services less frequently than White, middle-class Americans (Alegr a et al., 2008; Richman, Kohn-Wood, & Williams, 2007). Why is this so? Perhaps the reason has to do with access and availability of mental health services. Ethnic minorities and individuals of low socioeconomic status (SES) report that barriers to services include lack of insurance, transportation, and time (Thomas & Snowden, 2002). However, researchers have found that even when income levels and insurance variables are taken into account, ethnic minorities are far less likely to seek out and utilize mental health services. And when access to mental health services is comparable across ethnic and racial groups, differences in service utilization remain (Richman et al., 2007).

In a study involving thousands of women, it was found that the prevalence rate of anorexia was similar across different races, but that bulimia nervosa was more prevalent among Hispanic and African American women when compared with non-Hispanic White people (Marques et al., 2011). Although they have similar or higher rates of eating disorders, Hispanic and African American women with these disorders tend to seek and engage in treatment far less than White women. These findings suggest

ethnic disparities in access to care, as well as clinical and referral practices that may prevent Hispanic and African American women from receiving care, which could include lack of bilingual treatment, stigma, fear of not being understood, family privacy, and lack of education about eating disorders.

Perceptions and attitudes toward mental health services may also contribute to this imbalance. A recent study at King s College, London, found many complex reasons why people do not seek treatment: self-sufficiency and not seeing the need for help, not seeing therapy as effective, concerns about confidentiality, and the many effects of stigma and shame (Clement et al., 2014). And in another study, African Americans exhibiting depression were less willing to seek treatment due to fear of possible psychiatric hospitalization as well as fear of the treatment itself (Sussman, Robins, & Earls, 1987). Instead of mental health treatment, many African Americans prefer to be self-reliant or use spiritual practices (Snowden, 2001; Belgrave & Allison, 2010). For example, it has been found that the Black church plays a significant role as an alternative to mental health services by providing prevention and treatment-type programs designed to enhance the psychological and physical well-being of its members (Blank, Mahmood, Fox, & Guterbock, 2002).

Additionally, people belonging to ethnic groups that already report concerns about prejudice and discrimination are less likely to seek services for a mental illness because they view it as an additional stigma (Gary, 2005; Townes, Cunningham, & Chavez-Korell, 2009; Scott, McCoy, Munson, Snowden, & McMillen, 2011). For example, in one recent study of 462 older Korean Americans (over the age of 60) many participants reported suffering from depressive symptoms. However, 71% indicated they thought depression was a sign of personal weakness, and 14% reported that having a mentally ill family member would bring shame to the family (Jang, Chiriboga, & Okazaki, 2009).

Language differences are a further barrier to treatment. In the previous study on Korean Americans attitudes toward mental health services, it was found that there were no Korean-speaking mental health professionals where the study was conducted (Orlando and Tampa, Florida) (Jang et al., 2009). Because of the growing number of people from ethnically diverse backgrounds, there is a need for therapists and psychologists to develop knowledge and skills to become culturally competent (Ahmed, Wilson, Henriksen, & Jones, 2011). Those providing therapy must approach the process from the context of the unique culture of each client (Sue & Sue, 2007).

Dig Deeper

In the United States, about one in six children and one in five adults experiences a mental health disorder, but fewer than half of these people receive professional support for their disorder (Whitney & Peterson, 2019). Access to qualified mental health professionals is not universal or equitable, but it has improved to the point that more people could receive help if they sought it. Why then, do so many people go without support, therapy, or treatment?

It seems that the public has a negative perception of people with mental health disorders. According to researchers from Indiana University, the University of Virginia, and Columbia University, interviews with over 1,300 U.S. adults show that they believe children with depression are prone to violence and that if a child receives treatment for a psychological disorder, then that child is more likely to be rejected by peers at school.

Bernice Pescosolido, author of the study, asserts that this is a misconception. And it is not limited to perceptions of mental health issues in children: adults living with mental health issues may face even more scrutiny when sharing their condition or seeking support. Stigmatization of psychological disorders is one of the main reasons why people do not get the help they need when they are having difficulties. Pescosolido and her colleagues caution that this stigma surrounding mental illness, based on misconceptions rather than facts, can be devastating to emotional and social well-being.

Fortunately, we are starting to see discussions related to the destigmatization of mental illness and an increase in public education and awareness. Dozens of leaders have contributed to the conversation, including athletes like Naomi Osaka, Simone Biles, Michael Phelps, Kevin Love, and Dak Prescott, as well as artists such as Adele, Bruce Springsteen, Ariana Grande, Big Sean, and Bebe Rexha. Mental health awareness is stronger within workplaces, educational settings, and communities overall. However, stigma remains, particularly regarding mental health issues that are frequently misunderstood.

I've become accustomed to the reaction when I tell people where I'm from.

Eleven months later, it's as consistent as it was back in January.

Just yesterday, inquiring as to the availability of a rental house this holiday season, the gentleman taking my information paused to ask, Newtown, CT? Isn't that where that...that thing happened?

A recent encounter in the Massachusetts Berkshires, however, took me by surprise.

It was in a small, charming art gallery. The proprietor, a woman who looked to be in her 60s, asked where we were from. My response usually depends on my present mood and readiness for the inevitable dialogue. Sometimes it's simply, Connecticut. This time, I replied, Newtown, CT.

The woman's demeanor abruptly shifted from one of amiable graciousness to one of visible agitation.

Oh my god, she said wide eyed and open mouthed. Did you know her?

. . . .

Her? I inquired

That woman, she replied with disdain, that woman that raised that monster.

That woman's name was Nancy Lanza. Her son, Adam, killed her with a rifle blast to the head before heading out to kill 20 children and six educators at Sandy Hook Elementary School in Newtown, CT last December 14th.

When Nelba Marquez Greene, whose beautiful 6-year-old daughter, Ana, was killed by Adam Lanza, was recently asked how she felt about that woman, this was her reply:

She's a victim herself. And it's time in America that we start looking at mental illness with compassion, and helping people who need it.

This was a family that needed help, an individual that needed help and didn't get it. And what better can come of this, of this time in America, than if we can get help to people who really need it? (pars. 1 7, 10 15)

Join the effort by encouraging and supporting those around you to seek help if they need it. To learn more, visit the National Alliance on Mental Illness (NAMI) website (http://www.nami.org/).

Summary

The sociocultural perspective looks at you, your behaviors, and your symptoms in the context of your culture and background. Clinicians using this approach integrate cultural and religious beliefs into the therapeutic process. Research has shown that ethnic minorities are less likely to access mental health services than their White middle-class American counterparts. Barriers to treatment include lack of insurance, transportation, and time; cultural views that mental illness is a stigma; fears about treatment; and language barriers. Supporting mental health treatment involves speaking and listening openly about mental health, avoiding assumptions, being conscious about language, and encouraging others to get help when needed.

Review Questions

OpenStax learning exercise

Q. The sociocultural perspective looks at you, your behaviors, and your symptoms in the context of your _____.

A. C

OpenStax learning exercise

Q. Which of the following was not listed as a barrier to mental health treatment?

A. D

Critical Thinking Question

OpenStax learning exercise

Q. Lashawn is a 24-year-old African American female. For years she has been struggling with bulimia. She knows she has a problem, but she is not willing to seek mental health services. What are some reasons why she may be hesitant to get help?

A. One reason may be that her culture views having a mental illness as a stigma. Additionally, perhaps she doesn t have insurance and is worried about the cost of therapy. She could also be afraid that a White counselor would not understand her cultural background, so she would feel uncomfortable sharing things. Also, she may believe she is self-reliant and tell herself that she s a strong woman who can fix this problem on her own without the help of a therapist.

Personal Application Question

OpenStax learning exercise

Q. What is your attitude toward mental health treatment? Would you seek treatment if you were experiencing symptoms or having trouble functioning in your life? Why or why not? In what ways do you think your cultural and/or religious beliefs influence your attitude toward psychological intervention?

CHAPTER 20 / CAC.901

Integration, reflection, and doctoral practice

Lecture: Integrating evidence without losing the child

Doctoral practice requires more than collecting techniques. It requires the ability to integrate research, clinical observation, ethical reasoning, and the perspectives of the child and family. Integration is an active judgment. Evidence may be incomplete, measures may conflict, and the most studied intervention may not fit the local setting. The counselor must reason transparently rather than hide uncertainty behind authority.

Begin with the question that matters. Is the task to understand a developmental pattern, select an intervention, evaluate a service, or decide whether urgent action is needed? Different questions require different evidence. A randomized trial may estimate an average intervention effect, while a developmental interview may reveal why a child cannot use the intervention. Neither replaces the other.

A disciplined integration sequence

First, describe the concern without explanation. Second, locate it in time and context. Third, generate competing hypotheses. Fourth, examine the quality and relevance of evidence. Fifth, consider the child s goals, rights, and preferences. Sixth, select a proportionate action. Seventh, monitor outcomes and unintended effects. Finally, revise the account when new information appears.

This sequence is not a mechanical formula. Urgent safety concerns may require immediate action before a complete formulation is possible. Even then, document what is known, what is uncertain, and what will happen next. A decision can be necessary without being final.

Figure 20.1 brings together the book s central domains: biological development, cognition, emotion, relationships, culture, institutions, and time. The arrows indicate possible influence rather than fixed causation. The child is located within the system, not outside it as a passive outcome. Use the map to identify where information is missing and where change is possible.

Evidence-based practice and local knowledge

Research evidence should be examined for design, sample, measurement, effect size, uncertainty, and relevance. A statistically significant average difference may be small or may not apply to a particular child. A null result may reflect limited power or poor implementation rather than proof that an approach cannot help. Look beyond the abstract and consider what was actually delivered.

Local knowledge includes the child s account, family experience, school conditions, and service realities. It should not be treated as automatically superior to research, nor dismissed because it is not formal. A caregiver may notice a pattern that a standardized measure misses. A child may explain a barrier that changes the meaning of nonattendance. Integrate these sources while remaining open to correction.

Cultural responsiveness is part of validity. An intervention that cannot be understood, accessed, or safely used is unlikely to help as intended. Adaptation should distinguish surface features from active mechanisms. Changing the language or examples may improve access; removing the core

practice may change the intervention. Document adaptations and evaluate rather than assuming equivalence.

Integrated case: anxiety, school refusal, and family conflict

Consider a fictional thirteen-year-old, Mayowa, who has missed increasing amounts of school. Mayowa reports panic before tests and describes peer exclusion. A caregiver reports morning arguments and fears that absence will become permanent. Teachers describe declining work and inconsistent attendance. A younger sibling has health needs that draw family attention.

A single explanation would be premature. Anxiety, bullying, learning demands, family strain, and reinforcement through relief may all contribute. Begin with immediate safety and a collaborative account of the absence pattern. Ask Mayowa about the worst part of the school day, not only whether school is frightening. Examine whether the exclusion is ongoing and whether academic demands are accessible.

Coordinate with the school to address safety and provide a predictable entry plan. Support caregivers in responding calmly while maintaining realistic expectations. Consider appropriate health assessment and qualified evaluation of anxiety or learning needs. If gradual return is used, pair it with environmental change rather than treating attendance as the only outcome.

Monitor several indicators: attendance, distress, participation, peer safety, learning access, and Mayowa s sense of agency. Review family conflict as an outcome and a possible maintaining process. If attendance improves while panic remains severe, the plan is incomplete. If Mayowa feels safer but cannot access learning, additional educational support is needed.

The case demonstrates why developmental psychology is integrative. Biology may influence arousal; cognition shapes threat appraisal; learning can maintain avoidance; relationships can buffer or intensify distress; institutions determine whether return is safe. Time reveals whether the pattern is changing and whether intervention alters the pathway.

Doctoral reflective practice

Reflection should examine the counselor s contribution. What did you notice first, and what did you overlook? Did the child s behavior evoke urgency, frustration, or a desire to rescue? Did a theory make you more confident than the evidence justified? Did you share decision-making in a way the child could understand? Reflection is not self-criticism without structure; it is a method for improving judgment.

Use supervision and consultation appropriately. Protect identifying information, clarify roles, and remain accountable for decisions made within your scope. Record disagreement and the reasoning used to resolve it. A consultation does not remove responsibility, but it can reveal assumptions and improve safety.

Research integrity follows the same habits. Distinguish planned analyses from exploratory observations, report limitations, and avoid presenting a fictional case as empirical evidence. When evaluating your own service, seek appropriate ethics review and permission. Do not turn routine care into research without the required protections.

Final applied task

Prepare a developmental formulation of a fictional child using no more than two pages. Include a timeline, competing hypotheses, evidence sources, cultural and structural context, the child s goals, safety considerations, and an intervention plan. Add a monitoring table with a review date and a rule for revision.

Then write a one-page reflection on the limits of your formulation. Identify the information you would most want next and the professional expertise that could help. Finally, explain the case to a caregiver in simple English without labels that blame or predictions that exceed the evidence.

The purpose of this course is not to make developmental complexity disappear. It is to help you work with it responsibly. A child is changing, relationships are changing, and settings are changing. Good practice follows those changes with curiosity, evidence, ethical care, and a willingness to revise. That is the foundation for the more specialized courses that follow.

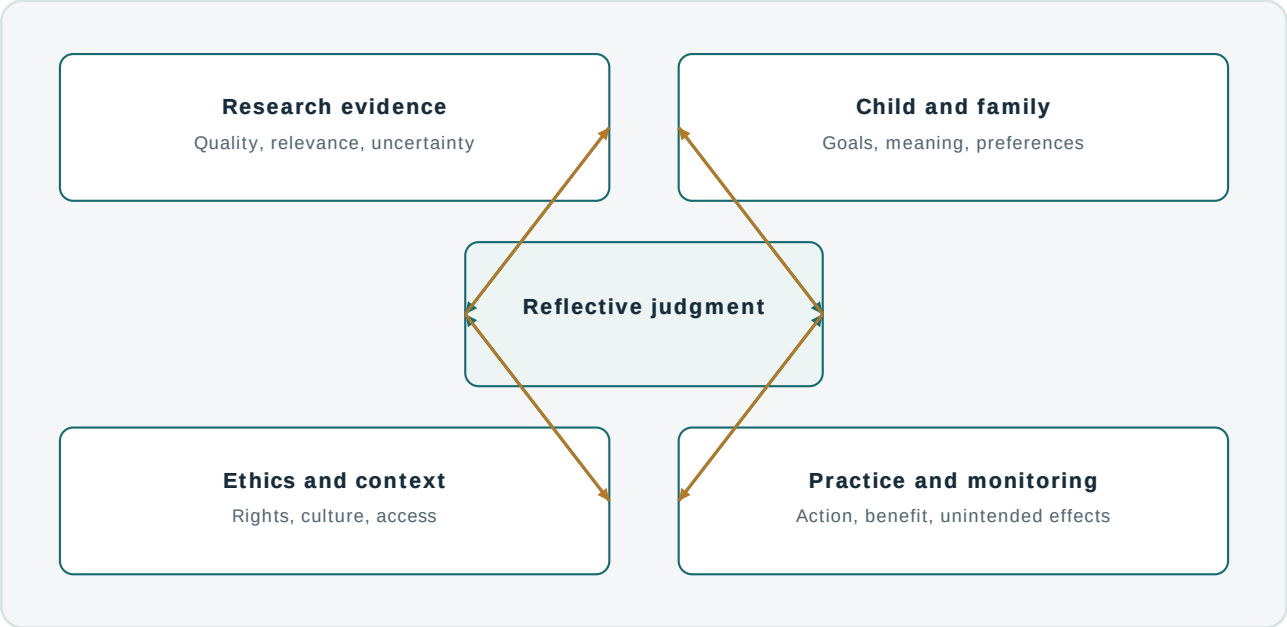


Figure 20.1. Doctoral integration keeps the child in view. Evidence, developmental theory, lived experience, ethics, and outcome monitoring inform one another. Professional judgment remains accountable to the child s welfare and to new information. Original course diagram.

Applied Developmental Workbook

The exercises below extend the course lectures. They are practice in reasoning, not tests of a real child and not instructions to conduct research without approval. Use fictional information or properly de-identified material only within an authorized learning setting. Do not place a child s identifying information in a public seminar document.

Exercise 1: Build a timeline before choosing a label

A school reports that a nine-year-old has become inattentive. Create a timeline with four columns: date or period, observed behavior, setting and demands, and changes in relationships or health. Include at least one period before the concern began. Add the child s account in a separate column so it is not confused with adult interpretation.

Mark each statement as an observation, a report from another person, or a hypothesis. Appears anxious is an interpretation unless the observable signs are specified. Asked to leave the room three times during written work is more precise. Do not assume that a behavior beginning after a school transition was caused only by that transition. The timeline organizes questions; it does not prove causation.

For review, ask a colleague to identify the strongest description and the weakest inference. Discuss what additional information would change your formulation. A good answer includes periods of successful functioning, not only difficulty. End with two competing explanations and a proportionate next step for each.

Exercise 2: Write an assent conversation

Imagine a study inviting ten-year-olds to complete a weekly feelings questionnaire and discuss a school coping program. Write an explanation of no more than 250 words. Include what will happen, how long it takes, who will see the information, the limits of privacy, and the right to decline or stop. Avoid saying that the program will certainly help.

Now write three questions to check understanding without making the child repeat a memorized script. For example: What could you do if you did not want to answer a question? Record what the study team would do if the child says yes but later appears distressed. Explain how permission from a legally authorized representative differs from the child s assent.

Review the conversation for pressure. Would a child believe that refusal affects grades, counseling access, or a teacher s opinion? Propose a recruitment process that reduces that pressure. Check local law and ethics requirements rather than treating this exercise as legal authorization.

Exercise 3: Separate two questions in repeated data

Use a fictional dataset with five adolescents, four weekly observations of sleep duration, and four weekly distress ratings. You do not need to calculate a model. First describe how you would examine whether adolescents who generally sleep more differ in average distress. Then describe how you would examine whether a particular adolescent reports more distress than usual in weeks with less sleep than usual.

Create two small graphs. One compares person averages. The other centers each adolescent around their own average and shows weekly deviations. Explain why the graphs answer different questions. Identify a factor, such as examination pressure, that could influence both sleep and distress.

For discussion, write one sentence that overgeneralizes from the between-person pattern to an individual intervention. Rewrite it as a bounded statement. Then identify what an intervention trial or carefully monitored individual plan could add. The goal is not to reject group research; it is to use it at the right level.

Exercise 4: Compare explanations without collecting every technique

Choose a fictional child who avoids group work. Write four brief formulations: cognitive, learning-based, relational, and ecological. Each must identify a mechanism, a possible observation, and an intervention implication. Do not use a theorist s name as a substitute for explanation.

Now add information that the child reports being mocked during presentations. Which formulation changes? Which environmental action becomes necessary before expecting the child to practice participation? Identify one technique that would be inappropriate if used alone.

Review your final plan for coherence. It should explain why the chosen actions belong together and which outcome would show that they are helping. Include the child s sense of safety as well as visible participation. A quiet child is not necessarily a successfully supported child.

Exercise 5: Adapt a task without changing the child s worth

A seven-year-old is asked to explain a recent conflict with a friend. The adult s original question is long and contains several assumptions. Rewrite the invitation using short sentences and one idea at a time. Offer a drawing or sequence cards as an optional communication aid.

List what the adapted method can reveal and what it cannot establish. A drawing may support conversation, but it does not provide a fixed symbolic code for the child s experience. Write two neutral follow-up questions and two questions that would be too leading. Explain the difference.

Finally, describe how you would involve a caregiver without requiring the child s private account to be repeated in full. Identify information that is necessary for support and information that can remain private under applicable rules.

Exercise 6: Challenge a biological certainty claim

An advertisement says that a child s attention problem can be identified from one brain measure and corrected by a proprietary training package. Write a list of questions about the measure s validity, the population studied, the comparison condition, the size and uncertainty of effects, and possible conflicts of interest.

Separate three claims: the measure is associated with attention in a research sample; the measure can diagnose an individual; the training causes meaningful improvement in daily life. Explain why evidence for the first does not automatically establish the other two.

Write a short response to a caregiver who is interested in the package. Be respectful and avoid ridicule. Acknowledge the desire for help, explain the uncertainty, and suggest a proportionate assessment pathway. Identify when medical or specialist consultation would be appropriate.

Exercise 7: Design support for a real decision moment

A fifteen-year-old understands a safety rule but sometimes breaks it when peers are watching. Design a plan that addresses both the decision moment and the surrounding conditions. Include a pause cue, a practicable exit strategy, a trusted adult, and a change in the environment where feasible.

Do not assume that role-play in a quiet office will transfer automatically. Specify where and how the young person can practice, what might make practice embarrassing, and how the plan can preserve dignity. Ask the young person to identify the part that seems least realistic.

Choose two indicators of progress. One may concern behavior, but the other should concern safety, agency, or the ability to seek support. Set a review date and a rule for revising the plan. Avoid explaining the entire concern with a claim about an immature brain.

Exercise 8: Test a behavioral hypothesis ethically

A child leaves the table when a difficult task begins. Adults respond by removing the task. Write an operational definition of the behavior and identify setting events that might affect it. Generate an escape hypothesis and a communication hypothesis; explain how they may overlap.

Design a support that teaches a respectful help request and makes the task more accessible. Specify what adults will do when the request is used. Include a baseline, a review period, and a measure of task participation. Do not use deprivation, humiliation, or escalating punishment as a convenience.

Now imagine that the behavior decreases but the child remains distressed. Explain why the plan may still be incomplete. Review the child s experience, the task s difficulty, and whether the behavior was a reasonable response to an inaccessible demand. Ethical evaluation includes more than compliance.

Exercise 9: Analyze hidden task demands

Choose a school assignment that requires planning over several days. Break it into steps and identify the language, memory, organization, and emotional demands of each step. Mark where a child could understand the goal but struggle with the route.

Design a visual support that preserves the learning objective. Include a way for the child to choose among acceptable approaches and a plan for reducing assistance. Do not turn the support into permanent adult completion of the task.

Write two outcome measures: successful task progress and the child s ability to use the planning process. Explain how you would distinguish improved access from simply making the assignment less demanding. Include a review of whether the support is acceptable in the actual classroom.

Exercise 10: Audit a score-based decision

A school proposes a fixed placement based on one group-administered score. Create an audit checklist covering the referral question, test norms, language, administration, measurement error, daily functioning, and the consequences of placement. Identify which missing information would prevent a confident decision.

Write a caregiver explanation that does not call the score the child s identity. Include strengths, needs, uncertainty, and a review plan. Then write a team recommendation that identifies what support should be provided while further information is gathered.

Discuss why rejecting all standardized measures and accepting every score without question are both poor responses. The goal is evidence that fits the decision, not loyalty to or rejection of testing in general.

Exercise 11: Protect an account from suggestion

Review a fictional conversation in which an adult repeatedly asks a child whether a named person caused a particular event. Mark the adult s assumptions and the child s spontaneous statements. Rewrite the opening using a neutral invitation and a calm response to silence.

Separate the counselor s responsibilities from those of an investigative interviewer. Identify the immediate safety actions that cannot wait for a complete narrative. Write a documentation note using exact quotation, observed behavior, and interpretation in separate sections.

Explain why a detailed account is not automatically complete and a sparse account is not automatically false. Avoid using this exercise to rehearse a real allegation outside authorized professional channels. The purpose is careful communication and role clarity.

Exercise 12: Plan co-regulation for two different children

One child becomes quiet and withdraws when distressed; another becomes loud and active. Design an initial response for each without assuming that the outward behavior reveals the whole emotional state. Include adult tone, space, choices, communication supports, and safety boundaries.

Identify a common regulation technique that might be uncomfortable for one child. Offer an alternative and a way to review what helped. Do not require breathing, eye contact, or touch as proof of cooperation.

Write a short caregiver script that validates emotion and sets a boundary. Then describe how the adult can reflect afterward on their own contribution to escalation or recovery. The plan should increase usable support rather than demand a particular performance of calm.

Exercise 13: Identify conditions of worth

A child says, I am only the good one when I do not need help. Map possible sources of this belief across family expectations, school rewards, peer comparison, and the child s own interpretation. Avoid assuming that any one source is responsible.

Design a response that changes the child s reflective resources and at least one adult response. Include honest feedback that does not depend on flawless performance. Identify an activity in which the child can experience belonging without proving competence.

Choose an outcome that reflects flexibility rather than higher achievement alone. Explain how the child s account will be included. Review whether the intervention unintentionally turns self-acceptance into another performance target.

Exercise 14: Map peer resources and risks

Create a peer-ecology map for a fictional adolescent. Include close friends, group status, trusted adults, digital spaces, and places where the young person feels safe. Mark supportive connections and pressures without treating every online relationship as either dangerous or therapeutic.

The young person reports exclusion but does not want immediate confrontation. Develop a plan that protects safety, preserves supportive relationships, and changes the school environment. Identify why removing the device or forcing a joint meeting could be harmful.

Define success from both the young person s perspective and the school s safeguarding responsibilities. Explain how disagreement will be handled and when urgent action overrides a preference for delay. The map should guide collaborative reasoning, not surveillance.

Exercise 15: Make a helping pathway safe

A class repeatedly mocks a student s language. Design a bystander-support plan with several routes: seek an adult, support the targeted student afterward, interrupt only when safe, and report through an accessible channel. Do not require a child to confront a dangerous group alone.

Specify what adults must do when harm is reported. Include protection from retaliation, a review of group patterns, and support for the targeted student that does not require public explanation of identity. Identify why an assembly alone may be insufficient.

Write indicators for reduced harm and increased participation. Include a check for unintended consequences, such as students being punished for reporting. A school s response is part of the intervention, not merely the background.

Exercise 16: Redefine resilience beyond attendance

A sixteen-year-old maintains grades and caregiving responsibilities but reports exhaustion. Write a formulation that recognizes competence and overload together. Identify resources that are real but insufficient, and demands that could be shared or reduced.

Critique an evaluation that calls the young person resilient only because attendance is high. Add measures of distress, rest, support, and personally meaningful participation. Avoid treating help-seeking as a loss of resilience.

Design a review meeting in which the young person can disagree with adult praise. Identify a change that the family or school will make, not only a coping skill the young person must practice. Resilience work should expand support rather than increase the burden to appear strong.

Exercise 17: Build two pathways to the same concern

Two children refuse school. One faces bullying; the other struggles with inaccessible written work and has developed panic around failure. Create separate timelines and formulations. Identify where the presenting behavior is similar and where the mechanisms differ.

For each case, specify an environmental action, a child-focused support, and a safety decision point. Explain why attendance alone is an incomplete outcome. Include learning access, distress, and the child s sense of control.

Now imagine that both plans produce only partial improvement. Write a revision process that examines implementation, new information, and the original hypotheses. Avoid blaming the child for a plan that did not address the relevant pathway.

Exercise 18: Individualize support after a diagnosis

A child receives an ADHD diagnosis. Design two support plans for different classroom environments. One classroom is noisy and uses long verbal instructions; the other provides predictable routines but little movement opportunity. Explain why the same label does not produce the same support plan.

Include the child s preferences, a measure of participation, and a measure of experience. Identify which supports change the environment and which teach skills. Do not make access to support conditional on perfect behavior.

Write a question for an appropriate health professional and a separate question for the school team. Keep role boundaries clear. The diagnosis can inform care without becoming the entire description of the child.

Exercise 19: Write an intervention revision rule

Choose a fictional intervention and identify its proposed mechanism. Specify the active practice, developmental adaptation, setting, and people involved. Then define implementation: what must actually happen for the intervention to have a fair opportunity to work?

Write one indicator of benefit, one possible burden, and one sign that the formulation may be wrong. Set a review date. State what would lead to adaptation, consultation, referral, or stopping the approach.

Explain how the child will participate in the review. A plan that measures only adult satisfaction may miss discomfort or lack of meaningful change. The revision rule protects the child from continued investment in an approach that is not helping.

Exercise 20: Complete an integrated formulation and reflection

Select one fictional case from the workbook. Write a two-page formulation containing a timeline, strengths, competing hypotheses, evidence sources, cultural and structural context, the child s goals, safety considerations, and a proportionate plan. Include a monitoring table and a review date.

Write a one-page caregiver explanation in simple English. Avoid blame, diagnostic certainty beyond the evidence, and predictions about the child s entire future. Explain what is known, what remains uncertain, and what will happen next.

Finally, write a professional reflection. Identify an assumption you made, information you initially overlooked, and a way supervision or consultation could improve the decision. Distinguish routine care from research and identify when ethics approval would be required. The final product should demonstrate accountable judgment rather than the number of theories or techniques you can name.

Portfolio review guide

A completed portfolio should show progression from description to explanation, from explanation to action, and from action to review. The reader should be able to see where each claim came from and how it could be revised. Strong work includes the child s perspective, structural conditions, measurement limits, and a realistic account of implementation.

Use four review questions. First, is the reasoning clear enough that another professional can follow it? Second, does the evidence support the strength of the conclusion? Third, does the plan protect dignity, safety, and meaningful participation? Fourth, is there a credible way to learn that the plan is not working?

A portfolio is not a clinical record unless it is created within an authorized care system under appropriate rules. It is not research evidence merely because it contains numbers. It is a structured opportunity to practice the habits that responsible developmental counseling requires.

Working Glossary

Accommodation. Changing an existing idea or mental pattern when new information does not fit it. In assessment, the word can also mean changing how a task is accessed without changing the intended construct; context determines the meaning.

Adaptation. Adjustment to conditions or demands. In developmental work, adaptation should be judged in context and across more than one domain, not inferred only from compliance.

Adolescence. A developmental period involving biological, cognitive, social, and contextual changes. Its boundaries and expectations vary across cultures and institutions.

Antecedent. An event or condition that occurs before a behavior. It may help explain when the behavior is more likely, but temporal order alone does not establish its function.

Assent. A child s affirmative agreement to participate when they cannot provide the full legally authorized consent. It should be informed, voluntary, and appropriate to the child s understanding and applicable rules.

Assimilation. Interpreting new information through an existing idea or mental pattern. It is often discussed alongside accommodation in accounts of cognitive development.

Attachment. A developing pattern of emotional connection and organized behavior in a close caregiving relationship. It should not be reduced to a single moment of distress or a simple label for a child.

Attrition. Loss of participants during a study. It can reduce power and introduce bias if the people who leave differ systematically from those who remain.

Baseline. Information about functioning before an intervention or comparison period. A single baseline observation may not represent a stable pattern.

Between-person difference. Variation among individuals. It answers a different question from variation within one individual over time.

Bias. A systematic influence that can distort measurement, selection, interpretation, or decisions. Bias is not removed merely by collecting a large amount of data.

Case formulation. A reasoned account of a person s concerns, strengths, context, and possible maintaining processes. It guides action and should be revised when new evidence appears.

Causal inference. A conclusion that one factor contributes to producing an outcome. It requires stronger evidence than association alone and depends on design and assumptions.

Cohort. A group that moves through time under partly shared historical conditions. Cohort membership should not be treated as a complete explanation of every member s experience.

Confidence interval. A range produced by a statistical procedure that reflects uncertainty under its assumptions. It should not be interpreted as a guarantee about an individual child.

Confidentiality. The professional duty to protect information shared in a care or research relationship. Its limits should be explained and applicable law and ethical standards followed.

Confound. A factor associated with both an exposure and an outcome that can help explain an observed association. It is one reason correlation does not establish causation.

Consequence. An event following a behavior. Its effect on future behavior must be examined rather than assumed from the adult s intention.

Construct. The concept a measure or theory is intended to represent, such as belonging or attention. A score is an indicator of a construct, not the construct itself.

Co-regulation. Support through a relationship that helps a person manage arousal and respond to demands. It is not the same as controlling or suppressing emotion.

Correlation. A pattern of association between variables. It does not by itself show that one variable causes the other.

Critical period. A limited interval during which particular experience may be especially necessary for a developmental process. Claims about critical periods should be domain-specific and evidence-based.

Cross-sectional design. A design comparing people at different ages or conditions at one time. It can be informative but does not directly observe individual change.

Developmental cascade. A process in which functioning in one domain influences later functioning in another domain or system. A proposed cascade requires evidence about timing and mechanism.

Developmental plasticity. The capacity for development to be modified. Plasticity is conditional and constrained, not unlimited change through effort.

Developmental trajectory. A pattern of stability and change over time. It describes a path and should not be treated as an inevitable destination.

Dynamic assessment. An approach that examines performance and response to guided support. Its uses and limits differ from standardized testing for a fixed decision.

Ecological perspective. An approach that locates development within connected settings, relationships, institutions, and social conditions.

Effect size. A quantitative description of the magnitude of a difference or association. Its practical meaning depends on the outcome, context, and uncertainty.

Equifinality. The possibility that different developmental pathways lead to a similar outcome.

Executive functions. Processes involved in organizing and regulating goal-directed activity, including planning, inhibition, shifting, and monitoring. Definitions vary across research traditions.

Functional behavior assessment. A process for developing hypotheses about the conditions and consequences that maintain a behavior. It should inform support rather than merely label the child.

Generalization. Use of a learned skill across settings, people, or conditions. It usually needs planning and should not be assumed from success in one session.

Heritability. A population-level estimate concerning variation associated with genetic differences in a particular population and environment. It does not describe a fixed proportion of an individual s trait.

Implementation fidelity. The degree to which a program is delivered as intended. It should be considered alongside appropriate adaptation and actual access.

Informed permission. Authorization from a parent or legally authorized representative for a child s research participation, subject to applicable law and ethics review.

Longitudinal design. A design that measures the same people repeatedly over time. It can describe change and temporal order but does not automatically establish causality.

Measurement invariance. The extent to which a measure represents the same construct across groups or occasions. It matters when scores are compared over development or context.

Multifinality. The possibility that a similar starting condition or exposure is followed by different outcomes.

Multimethod assessment. Use of more than one method or information source to answer an assessment question. Disagreement should be interpreted, not automatically averaged away.

Negative reinforcement. An increase in behavior associated with removal or avoidance of an aversive condition. It is not the same as punishment.

Neurodiversity. A perspective that recognizes variation in neurodevelopment while allowing attention to disability, support needs, identity, and distress.

Norm-referenced score. A score interpreted relative to a reference sample. Its usefulness depends on the relevance and quality of that sample and the intended decision.

Operational definition. A clear statement of how a variable or behavior is identified and measured. It supports consistent observation and communication.

Plasticity constraint. A biological, historical, or contextual limit on the change that is possible under particular conditions.

Positive reinforcement. An increase in behavior associated with the addition of a consequence. Whether something functions as reinforcement is shown by the pattern, not by its name.

Protective factor. A resource or condition associated with better adaptation, especially under adversity. Its effect depends on context and should not be treated as a guarantee.

Random assignment. Allocation of participants to conditions by a chance process. It can reduce systematic selection differences in an experiment.

Random sampling. Selection of participants from a population by a chance process. It concerns who enters a study and is distinct from random assignment.

Reliability. The consistency or precision of measurement. A reliable measure is not necessarily valid for every interpretation or decision.

Resilience. A dynamic process of positive adaptation in relation to adversity, supported by interacting personal, relational, and environmental resources.

Risk factor. A characteristic or condition associated with increased likelihood of an outcome. It is probabilistic, not a diagnosis or destiny.

Scaffolding. Temporary support that helps a person perform a task or use a skill, with adjustment as competence grows.

Sensitive period. A time when experience may have a stronger influence on a particular developmental process. It is usually less absolute than a critical period.

Setting event. A broader condition that changes the likelihood or meaning of an immediate behavior sequence, such as fatigue or prior conflict.

Social validity. The acceptability and meaningfulness of goals, procedures, and outcomes to the people affected. Definitions vary, so the intended use should be stated.

Standardization. Consistent procedures for administering and scoring a measure. It supports comparability but does not by itself establish validity.

Transactional model. An account of development in which the person and context influence one another over time.

Trauma-informed practice. Care that attends to safety, trust, choice, collaboration, and the possibility of trauma-related distress without assuming that every difficulty has the same cause.

Validity. Evidence supporting an interpretation or use of a measure or decision. It is not a permanent property that a test possesses in every context.

Within-person change. Variation within the same individual across time or situations. It should be distinguished from differences among individuals.

Zone of proximal development. A concept associated with Vygotsky concerning the difference between what a learner can do independently and what becomes possible with appropriate assistance.

References and Source Notes

Supplemental sources checked for this edition

The following sources support the new lectures and editorial framework. Verification means that the cited bibliographic record was located; it does not mean that every claim in a source has been independently replicated.

American Counseling Association. (2014). 2014 ACA code of ethics. https://www.counseling.org/resources/aca-code-of-ethics.pdf

Use in this volume: Professional ethics, minors, assent, confidentiality, and scope. Applicable law and later professional guidance must also be checked.

Baltes, P. B. (1987). Theoretical propositions of life-span developmental psychology: On the dynamics between growth and decline. Developmental Psychology, 23(5), 611 626. https://doi.org/10.1037/0012-1649.23.5.611

Use in this volume: Lifespan development, multidirectionality, plasticity, and contextual influences.

Curran, P. J., & Bauer, D. J. (2011). The disaggregation of within-person and between-person effects in longitudinal models of change. Annual Review of Psychology, 62, 583 619. https://doi.org/10.1146/annurev-psych-093008-100356

Use in this volume: Separating within-person and between-person questions in longitudinal research.

Hamaker, E. L., Kuiper, R. M., & Grasman, R. P. P. P. (2015). A critique of the cross-lagged panel model. Psychological Methods, 20(1), 102 116. https://doi.org/10.1037/a0038889

Use in this volume: Limitations of traditional cross-lagged models and within-person interpretation.

Masten, A. S. (2001). Ordinary magic: Resilience processes in development. American Psychologist, 56(3), 227 238. https://doi.org/10.1037/0003-066X.56.3.227

Use in this volume: Resilience as a process supported by ordinary adaptive systems.

Masten, A. S., & Cicchetti, D. (2010). Developmental cascades. Development and Psychopathology, 22(3), 491 495. https://doi.org/10.1017/S0954579410000222

Use in this volume: Cross-domain developmental influence over time.

Sameroff, A. (2010). A unified theory of development: A dialectic integration of nature and nurture. Child Development, 81(1), 6 22. https://doi.org/10.1111/j.1467-8624.2009.01378.x

Use in this volume: Transactional integration of person and context over development.

Spielman, R. M., Jenkins, W. J., & Lovett, M. D. (2020). Psychology 2e. OpenStax, Rice University. https://openstax.org/books/psychology-2e/pages/preface

Use in this volume: Openly licensed foundational readings adapted in this course volume; specific source modules and snapshot are listed in the attribution register.

Inherited OpenStax reference lists

The entries below are retained from the source textbook to make its reading trail visible. They have not all been independently reverified for this edition. Some URLs, editions, and terminology are historical. Their inclusion is not a current clinical endorsement. These entries are excluded from the substantive word count.

Psychological research

American Cancer Society. (n.d.). History of the cancer prevention studies. http://www.cancer.org/research/researchtopreventcancer/history-cancer-prevention-study

American Psychological Association. (2009). Publication Manual of the American Psychological Association (6th ed.). Author.

American Psychological Association. (n.d.). Research with animals in psychology. https://www.apa.org/research/responsible/research-animals.pdf

Arnett, J. (2008). The neglected 95%: Why American psychology needs to become less American. American Psychologist, 63(7), 602 614.

Aschwanden, C. (2018, December 6). Psychology s replication crisis has made the field better. FiveThirtyEight. https://fivethirtyeight.com/features/psychologys-replication-crisis-has-made-the-field-better/

Barnett, W. S. (2011). Effectiveness of early educational intervention. Science, 333(6045), 975 978. https://science.sciencemag.org/content/333/6045/975

Barton, B. A., Eldridge, A. L., Thompson, D., Affenito, S. G., Striegel-Moore, R. H., Franko, D. L., . . . Crockett, S. J. (2005). The relationship of breakfast and cereal consumption to nutrient intake and body mass index: The national heart, lung, and blood institute growth and health study. Journal of the American Dietetic Association, 105(9), 1383 1389. http://dx.doi.org/10.1016/j.jada.2005.06.003

Blann, L.E. (2005). Early intervention for children and families: With special needs. MCN: The American Journal of Maternal Child Nursing, 30, 263 67.

Chwalisz, K., Diener, E., & Gallagher, D. (1988). Autonomic arousal feedback and emotional experience: Evidence from the spinal cord injured. Journal of Personality and Social Psychology, 54, 820 828.

Dominus, S. (2011, May 25). Could conjoined twins share a mind? New York Times Sunday Magazine. http://www.nytimes.com/2011/05/29/magazine/could-conjoined-twins-share-a-mind.html?_r=5&hp&

Egnor, M. (2017, November 24). What the craniopagus twins teach us about the mind and the brain. Evolution News & Science Today. https://evolutionnews.org/2017/11/what-the-craniopagus-twins-teach-us-about-the-mind-and-the-brain/

Ethnicity and Health in America Series: Featured Psychologists (n.d.). American Psychological Association. https://www.apa.org/pi/oema/resources/ethnicity-health/psychologists/sumner-prosser

Fanger, S. M., Frankel, L. A., & Hazen, N. (2012). Peer exclusion in preschool children s play: Naturalistic observations in a playground setting. Merrill-Palmer Quarterly, 58, 224 254.

Fiedler, K. (2004). Illusory correlation. In R. F. Pohl (Ed.), Cognitive illusions: A handbook on fallacies and biases in thinking, judgment and memory (pp. 97 114). Psychology Press.

Frantzen, L. B., Trevi o, R. P., Echon, R. M., Garcia-Dominic, O., & DiMarco, N. (2013). Association between frequency of ready-to-eat cereal consumption, nutrient intakes, and body mass index in fourth- to sixth-grade low-income minority children. Journal of the Academy of Nutrition and Dietetics, 113(4), 511 519.

Godoy, J. C., & Brussino, S. (2010). Psychology in Argentina. The Corsini Encyclopedia of Psychology. John Wiley & Sons.

Gunamudian David Boaz. (n.d.). World Library. http://www.worldlibrary.org/articles/Gunamudian_David_Boaz

Harper, J. (2013, July 5). Ice cream and crime: Where cold cuisine and hot disputes intersect. The Times-Picaune. http://www.nola.com/crime/index.ssf/2013/07/ice_cream_and_crime_where_hot.html

Jenkins, W. J., Ruppel, S. E., Kizer, J. B., Yehl, J. L., & Griffin, J. L. (2012). An examination of post 9-11 attitudes towards Arab Americans. North American Journal of Psychology, 14, 77 84.

Jones, J. M. (2013, May 13). Same-sex marriage support solidifies above 50% in U.S. Gallup Politics. http://www.gallup.com/poll/162398/sex-marriage-support-solidifies-above.aspx

Kobrin, J. L., Patterson, B. F., Shaw, E. J., Mattern, K. D., & Barbuti, S. M. (2008). Validity of the SAT for predicting first-year college grade point average (Research Report No. 2008-5). https://research.collegeboard.org/sites/default/files/publications/2012/7/researchreport-2008-5-validity-sat-predicting-first-year-college-grade-point-average.pdf

Lewin, T. (2014, March 5). A new SAT aims to realign with schoolwork. New York Times. http://www.nytimes.com/2014/03/06/education/major-changes-in-sat-announced-by-college-board.html.

Lowcock, E. C., Cotterchio, M., Anderson, L. N., Boucher, B. A., & El-Sohemy, A. (2013). High coffee intake, but not caffeine, is associated with reduced estrogen receptor negative and postmenopausal breast cancer risk with no effect modification by CYP1A2 genotype. Nutrition and Cancer, 65(3), 398 409. doi:10.1080/01635581.2013.768348

Lowry, M., Dean, K., & Manders, K. (2010). The link between sleep quantity and academic performance for the college student. Sentience: The University of Minnesota Undergraduate Journal of Psychology, 3(Spring), 16 19. http://www.psych.umn.edu/sentience/files/SENTIENCE_Vol3.pdf

Margaret Floy Washburn, PhD. (n.d.). American Psychological Association. https://www.apa.org/about/governance/president/bio-margaret-washburn.

Mary Whiton Calkins. (n.d.). American Psychological Association. https://www.apa.org/about/governance/president/bio-mary-whiton-calkins

Massimini, M., & Peterson, M. (2009). Information and communication technology: Affects of U.S. college students. Cyberpsychology: Journal of Psychosocial Research on Cyberspace, 3(1).

McKie, R. (2010, June 26). Chimps with everything: Jane Goodall s 50 years in the jungle. The Guardian. http://www.theguardian.com/science/2010/jun/27/jane-goodall-chimps-afric a-interview

Narendra Nath Sen Gupta. (n.d.). Veethi: The face of India. https://www.veethi.com/india-people/narendra_nath_sen_gupta-profile-4893 -16.htm

Neil, A. L., & Christensen, H. (2009). Efficacy and effectiveness of school-based prevention and early intervention programs for anxiety. Clinical Psychology Review, 29, 208 15.

Nobel Prize-winning scientist Frances Arnold retracts paper. (2020, January 3). BBC News. https://www.bbc.com/news/world-us-canada-50989423

Offit, P. (2008). Autism's false prophets: Bad science, risky medicine, and the search for a cure. Columbia University Press.

Patel, M., Lee, A. D., Clemmons, N. S., Redd, S. B., Poser, S., Blog, D., Gasta aduy, P. A. (2019, October 11). National update on measles cases and outbreaks United States, January 1 October 1, 2019. Morbidity and Mortality Weekly Report, 68, 893 96.

Perkins, H. W., Haines, M. P., & Rice, R. (2005). Misperceiving the college drinking norm and related problems: A nationwide study of exposure to prevention information, perceived norms and student alcohol misuse. J. Stud. Alcohol, 66(4), 470 478.

Peters-Scheffer, N., Didden, R., Korzilius, H., & Sturmey, P. (2011). A meta-analytic study on the effectiveness of comprehensive ABA-based early intervention programs for children with Autism Spectrum Disorders. Research in Autism Spectrum Disorders, 5, 60 69.

Rimer, S. (2008, September 21). College panel calls for less focus on SATs. The New York Times. http://www.nytimes.com/2008/09/22/education/22admissions.html?_r=0

Rothstein, J. M. (2004). College performance predictions and the SAT. Journal of Econometrics, 121, 297 317.

Rotton, J., & Kelly, I. W. (1985). Much ado about the full moon: A meta-analysis of lunar-lunacy research. Psychological Bulletin, 97(2), 286 306. doi:10.1037/0033-2909.97.2.286

Santelices, M. V., & Wilson, M. (2010). Unfair treatment? The case of Freedle, the SAT, and the standardization approach to differential item functioning. Harvard Education Review, 80, 106 134.

Sears, D. O. (1986). College sophomores in the laboratory: Influences of a narrow data base on social psychology s view of human nature. Journal of Personality and Social Psychology, 51, 515 530.

Shaw, C. M., & Tan, S. A. (2015). Integration of mobile technology in educational materials improves participation: Creation of a novel smartphone application for resident education. Journal of Surgical Education, 72(4), 670 73. https://www.sciencedirect.com/science/article/abs/pii/S1931720415000318

Shrout, P. E., & Rodgers, J. L. (2018). Psychology, science, and knowledge construction: Broadening perspectives from the replication crisis. Annual Review of Psychology, 69, 487-510.

Strauss, V. (2019, March 19). Is it finally time to get rid of the SAT and ACT college admissions tests? The Washington Post. https://www.washingtonpost.com/education/2019/03/19/is-it-finally-time-g et-rid-sat-act-college-admissions-tests/

Tuskegee University. (n.d.). About the USPHS Syphilis Study. http://www.tuskegee.edu/about_us/centers_of_excellence/bioethics_center/ about_the_usphs_syphilis_study.aspx.

Women and Minorities in Psychology. (n.d.). IResearch. http://psychology.iresearchnet.com/history-of-psychology/women-and-minorities/

Biopsychology

Anderson, P. J., & Leuzzi, V. (2010). White matter pathology in phenylketonuria. Molecular Genetics and Metabolism, 99, S3 S9.

Arnst, C. (2003, November). Commentary: Getting rational about health-care rationing. Bloomberg Businessweek Magazine. http://www.businessweek.com/stories/2003-11-16/commentary-getting-rational-about-health-care-rationing

Azevedo, F. A., Carvalho, L. R., Grinberg, L. T., Farfel, J. M., Ferretti, R. E., Leite, R. E., ... & Herculano Houzel, S. (2009). Equal numbers of neuronal and nonneuronal cells make the human brain an isometrically scaled up primate brain. Journal of Comparative Neurology, 513(5), 532 541.

Banich, M. T., & Heller, W. (1998). Evolving perspectives on lateralization of function. Current Directions in Psychological Science, 7(1), 1 2.

Berridge, K. C., & Robinson, T. E. (1998). What is the role of dopamine in reward: Hedonic impact, reward learning, or incentive salience? Brain Research Reviews, 28, 309 369.

Chandola, T., Brunner, E., & Marmot, M. (2006). Chronic stress at work and the metabolic syndrome: A prospective study. BMJ, 332, 521 524.

Comings, D. E., Gonzales, N., Saucier, G., Johnson, J. P., & MacMurray, J. P. (2000). The DRD4 gene and the spiritual transcendence scale of the character temperament index. Psychiatric Genetics, 10, 185 189.

Confer, J. C., Easton, J. A., Fleischman, D. S., Goetz, C. D., Lewis, D. M. G, Perilloux, C., & Buss, D. M. (2010). Evolutionary psychology: Controversies, questions, prospects, and limitations. American Psychologist, 65, 110 126.

Connors, B. W., & Long, M. A. (2004). Electrical synapses in the mammalian brain. Annual Review of Neuroscience, 27, 393 418.

Ehret, G. (2006). Hemisphere dominance of brain function which functions are lateralized and why?. 23 Problems in Systems Neuroscience, 44 61.

Fernandez, A. (2008, October 16). ABC reporter Bob Woodruff's incredible recovery from traumatic brain injury. The Huffington Post. https://www.huffpost.com/entry/abc-reporter-bob-woodruff_b_125863

Gaines, C. (2013, August). An A-Rod suspension would save the Yankees as much as $37.5 million in 2014 alone. Business Insider. http://www.businessinsider.com/an-a-rod-suspension-would-save-the-yankees-as-much-as-375-million-in-2014-2013-8

Gardner, E. L. (2011). Addiction and brain reward and antireward pathways. Advances in Psychosomatic Medicine, 30, 22 60.

Gazzaniga, M. S. (2005). Forty-five years of split-brain research and still going strong. Nature Reviews Neuroscience, 6(8), 653 659.

George, O., Le Moal, M., & Koob, G. F. (2012). Allostasis and addiction: Role of the dopamine and corticotropin-releasing factor systems. Physiology & Behavior, 106, 58 64.

Glaser, R., & Kiecolt-Glaser, J. K. (2005). Stress-induced immune dysfunction: Implications for health. Nature Reviews Immunology, 5, 243 251.

Gong, L., Parikh, S., Rosenthal, P. J., & Greenhouse, B. (2013). Biochemical and immunological mechanisms by which sickle cell trait protects against malaria. Malaria Journal. Advance online publication. doi:10.1186/1475-2875-12-317

Hardt, O., Einarsson, E. ., & Nader, K. (2010). A bridge over troubled water: Reconsolidation as a link between cognitive and neuroscientific memory research traditions. Annual Review of Psychology, 61, 141 167.

Herculano-Houzel, S. (2009). The human brain in numbers: a linearly scaled-up primate brain. Frontiers in Human Neuroscience, 3, 31.

Herculano-Houzel, S. (2012). The remarkable, yet not extraordinary, human brain as a scaled-up primate brain and its associated cost. Proceedings of the National Academy of Sciences, 109(Supplement 1), 10661 10668.

Huttenlocher, P. R. (2000). The neuropathology of phenylketonuria: human and animal studies. European Journal of Pediatrics, 159(2), S102 S106.

Macmillan, M. (1999). The Phineas Gage Information Page. http://www.uakron.edu/gage

March, J. S., Silva, S., Petrycki, S., Curry, J., Wells, K., Fairbank, J., Severe, J. (2007). The treatment for adolescents with depression study (TADS): Long-term effectiveness and safety outcomes. Arch Gen Psychiatry, 64, 1132 1143.

Miguel, P. M., Pereira, L. O., Silveira, P. P., & Meaney, M. J. (2019). Early environmental influences on the development of children's brain structure and function. Developmental Medicine and Child Neurology, 61, 1127 33.

Mustanski, B. S., DuPree, M. G., Nievergelt, C. M., Bocklandt, S., Schork, N. J., & Hamer, D. H. (2005). A genome wide scan of male sexual orientation. Human Genetics, 116, 272 278.

National Institute on Drug Abuse. (2001, July). Anabolic steroid abuse: What are the health consequences of steroid abuse? National Institutes of Health. http://www.drugabuse.gov/publications/research-reports/anabolic-steroid-abuse/what-are-health-consequences-steroid-abuse

Short, A. K., and Baram, T. Z. (2019). Early-life adversity and neurological disease: Age-old questions and novel answers. Nature Reviews Neurology, 15, 657 69.

Squire, L. R. (2009). The legacy of patient H. M. for neuroscience. Neuron, 61, 6 9.

Tienari, P., Wynne, L. C., Sorri, A., et al. (2004). Genotype environment interaction in schizophrenia spectrum disorder: long-term follow-up study of Finnish adoptees. British Journal of Psychiatry, 184, 216 222.

University of Utah Genetic Science Learning Center. (n.d.). What are genetic disorders? http://learn.genetics.utah.edu/content/disorders/whataregd/

Yudell, M., Roberts, D., DeSalle, R., & Tishkoff, S. (2016). Taking race out of human genetics. Science, 351(6273), 564 565.

Learning

Anderson, C. A., & Gentile, D. A. (2008). Media violence, aggression, and public policy. In E. Borgida & S. Fiske (Eds.), Beyond common sense: Psychological science in the courtroom (p. 322). Blackwell.

Anderson, C. A., Bushman, B. J., Donnerstein, E., Hummer, T. A., & Warburton, W. (2015). SPSSI research summary on media violence. Analyses of Social Issues and Public Policy (ASAP), 15(1), 4 19. https://doi.org/10.1111/asap.12093

Anderson, C. A., Shibuya, A., Ihori, N., Swing, E. L., Bushman, B. J., Sakamoto, A., Rothstein, H.R., & Saleem, M. (2010). Violent video game effects on aggression, empathy, and prosocial behavior in Eastern and Western countries: A meta-analytic review. Psychological Bulletin, 136(2), 151 173. HYPERLINK "https://psycnet.apa.org/doi/10.1037/a0018251"https://doi.org/10.1037/a0018251

Bandura, A., Ross, D., & Ross, S. A. (1961). Transmission of aggression through imitation of aggressive models. Journal of Abnormal and Social Psychology, 63, 575 582.

Bushman, B. J., Anderson, C. A., Donnerstein, E., Hummer, T. A., & Warburton, W. A. (2016). Reply to comments on SPSSI research summary on media violence by Cupit (2016), Gentile (2016), Glackin and Gray (2016), Gollwitzer (2016), and Krah (2016). Analyses of Social Issues and Public Policy (ASAP), 16(1), 443 450. https://doi.org/10.1111/asap.12123

Cangi, K., & Daly, M. (2013). The effects of token economies on the occurrence of appropriate and inappropriate behaviors by children with autism in a social skills setting. West Chester University: Journal of Undergraduate Research. http://www.wcupa.edu/UndergraduateResearch/journal/documents/cangi_S2012.pdf

Carlson, L., Holscher, C., Shipley, T., & Conroy Dalton, R. (2010). Getting lost in buildings. Current Directions in Psychological Science, 19(5), 284 289.

Chance, P. (2009). Learning and behavior (6th ed.). Wadsworth, Cengage Learning.

Chase, H. W., & Clark, L. (2010). Gambling severity predicts midbrain response to near-miss outcomes. The Journal of Neuroscience, 30(18), 6180 6187. https://www.jneurosci.org/content/30/18/6180.short

Cialdini, R. B. (2008). Influence: Science and practice (5th ed.). Pearson Education.

DeAngelis, T. (2010). Little Albert regains his identity. Monitor on Psychology, 41(1), 10.

Ferguson, C. J. (2011). Video games and youth violence: A prospective analysis in adolescents. Journal of Youth and Adolescence, 40(4), 377 391. https://link.springer.com/article/10.1007/s10964-010-9610-x

Fryer, R. G., Jr. (2010, April). Financial incentives and student achievement: Evidence from randomized trials. National Bureau of Economic Research [NBER] Working Paper, No. 15898. http://www.nber.org/papers/w15898

Garcia, J., & Koelling, R. A. (1966). Relation of cue to consequence in avoidance learning. Psychonomic Science, 4, 123 124.

Garcia, J., & Rusiniak, K. W. (1980). What the nose learns from the mouth. In D. M ller-Schwarze & R. M. Silverstein (Eds.), Chemical signals: Vertebrates and aquatic invertebrates (pp. 141 156). Plenum Press.

Gentile, D. A. (2016). The evolution of scientific skepticism in the media violence debate. Analyses of Social Issues and Public Policy, 16, 429 434.

Gershoff, E. T. (2002). Corporal punishment by parents and associated child behaviors and experiences: A meta-analytic and theoretical review. Psychological Bulletin, 128(4), 539 579. doi:10.1037//0033-2909.128.4.539

Gershoff, E.T., Grogan-Kaylor, A., Lansford, J. E., Chang, L., Zelli, A., Deater-Deckard, K., & Dodge, K. A. (2010). Parent discipline practices in an international sample: Associations with child behaviors and moderation by perceived normativeness. Child Development, 81(2), 487 502.

Hickock, G. (2010). The role of mirror neurons in speech and language processing. Brain and Language, 112, 1 2.

Holmes, S. (1993). Food avoidance in patients undergoing cancer chemotherapy. Support Care Cancer, 1(6), 326 330.

Hunt, M. (2007). The story of psychology. Doubleday.

Huston, A. C., Donnerstein, E., Fairchild, H., Feshbach, N. D., Katz, P. A., Murray, J. P., . . . Zuckerman, D. (1992). Big world, small screen: The role of television in American society. University of Nebraska Press.

Hutton, J. L., Baracos, V. E., & Wismer, W. V. (2007). Chemosensory dysfunction is a primary factor in the evolution of declining nutritional status and quality of life with patients with advanced cancer. Journal of Pain Symptom Management, 33(2), 156 165.

Jacobsen, P. B., Bovbjerg, D. H., Schwartz, M. D., Andrykowski, M. A., Futterman, A. D., Gilewski, T., . . . Redd, W. H. (1993). Formation of food aversions in cancer patients receiving repeated infusions of chemotherapy. Behaviour Research and Therapy, 31(8), 739 748.

Kirsch, SJ (2010). Media and youth: A developmental perspective. Wiley Blackwell.

Laskowski, C. S., Dorchak, D. L., Ward, K. M., Christensen, D. R., & Euston, D. R. (2019). Can slot-machine reward schedules induce gambling addiction in rats? Journal of Gambling Studies, 35(3), 887 914. https://www.ncbi.nlm.nih.gov/pubmed/31049772

Lefran ois, G. R. (2012). Theories of human learning: What the professors said (6th ed.). Wadsworth, Cengage Learning.

Miller, L. E., Grabell, A., Thomas, A., Bermann, E., & Graham-Bermann, S. A. (2012). The associations between community violence, television violence, intimate partner violence, parent-child aggression, and aggression in sibling relationships of a sample of preschoolers. Psychology of Violence, 2(2), 165 78. doi:10.1037/a0027254

Murch, W. S., & Clark, L. (2016). Games in the brain: Neural substrates of gambling addiction. The Neuroscientist, 22(5), 534 545. https://www.ncbi.nlm.nih.gov/pubmed/26116634

Murrell, A., Christoff, K. & Henning, K. (2007) Characteristics of domestic violence offenders: associations with childhood exposure to violence. Journal of Family Violence, 22(7), 523-532.

Pavlov, I. P. (1927). Conditioned reflexes: An investigation of the physiological activity of the cerebral cortex (G. V. Anrep, Ed. & Trans.). Oxford University Press.

Potenza, M. N. (2013). Neurobiology of gambling behaviors. Current Opinion in Neurobiology, 23(4), 660 667. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3803105/

Rescorla, R. A., & Wagner, A. R. (1972). A theory of Pavlovian conditioning: Variations in the effectiveness of reinforcement and nonreinforcement. Classical Conditioning II: Current Research and Theory, 2, 64-99.

Rizzolatti, G., Fadiga, L., Fogassi, L., & Gallese, V. (2002). From mirror neurons to imitation: Facts and speculations. In A. N. Meltzoff & W. Prinz (Eds.), The imitative mind: Development, evolution, and brain bases (pp. 247 66). Cambridge University Press.

Rizzolatti, G., Fogassi, L., & Gallese, V. (2006, November). Mirrors in the mind. Scientific American, pp. 54 61.

Skinner, B. F. (1938). The behavior of organisms: An experimental analysis. Appleton-Century-Crofts.

Skinner, B. F. (1953). Science and human behavior. Macmillan.

Skinner, B. F. (1961). Cumulative record: A selection of papers. Appleton-Century-Crofts.

Skinner s utopia: Panacea, or path to hell? (1971, September 20). Time. http://www.wou.edu/~girodm/611/Skinner%27s_utopia.pdf

Skolin, I., Wahlin, Y. B., Broman, D. A., Hursti, U-K. K., Larsson, M. V., & Hernell, O. (2006). Altered food intake and taste perception in children with cancer after start of chemotherapy: Perspectives of children, parents and nurses. Supportive Care in Cancer, 14, 369 78.

Thorndike, E. L. (1911). Animal intelligence: An experimental study of the associative processes in animals. Psychological Monographs, 8.

Tolman, E. C., & Honzik, C. H. (1930). Degrees of hunger, reward, and non-reward, and maze performance in rats. University of California Publications in Psychology, 4, 241 256.

Tolman, E. C., Ritchie, B. F., & Kalish, D. (1946). Studies in spatial learning: II. Place learning versus response learning. Journal of Experimental Psychology, 36, 221 229. doi:10.1037/h0060262

Watson, J. B. & Rayner, R. (1920). Conditioned emotional reactions. Journal of Experimental Psychology, 3, 1 14.

Watson, J. B. (1919). Psychology from the standpoint of a behaviorist. J. B. Lippincott.

Yamamoto, S., Humle, T., & Tanaka, M. (2013). Basis for cumulative cultural evolution in chimpanzees: Social learning of a more efficient tool-use technique. PLoS ONE, 8(1): e55768. doi:10.1371/journal.pone.0055768

Thinking and intelligence

Abler, W. (2013). Sapir, Harris, and Chomsky in the twentieth century. Cognitive Critique, 7, 29 48.

American Association on Intellectual and Developmental Disabilities. (2013). Definition of intellectual disability. http://aaidd.org/intellectual-disability/definition#.UmkR2xD2Bh4

American Psychiatric Association. (2013). In Diagnostic and statistical manual of psychological disorders (5th ed., pp. 34 36). American Psychiatric Association.

Aronson, E. (Ed.). (1995). Social cognition. In The social animal (p. 151). W.H. Freeman and Company.

Atkins v. Virginia, 00-8452 (2002).

Bartels, M., Rietveld, M., Van Baal, G., & Boomsma, D. I. (2002). Genetic and environmental influences on the development of intelligence. Behavior Genetics, 32(4), 237 238.

Bartlett, F. C. (1932). Remembering: A study in experimental and social psychology. Cambridge University Press.

Bayer, J. B., & Campbell, S. W. (2012). Texting while driving on automatic: Considering the frequency-independent side of habit. Computers in Human Behavior, 28, 2083 2090.

Barton, S. M. (2003). Classroom accommodations for students with dyslexia. Learning Disabilities Journal, 13, 10 14.

Berlin, B., & Kay, P. (1969). Basic color terms: Their universality and evolution. University of California Press.

Berninger, V. W. (2008). Defining and differentiating dysgraphia, dyslexia, and language learning disability within a working memory model. In M. Mody & E. R. Silliman (Eds.), Brain, behavior, and learning in language and reading disorders (pp. 103 134). The Guilford Press.

Blossom, M., & Morgan, J. L. (2006). Does the face say what the mouth says? A study of infants sensitivity to visual prosody. In the 30th annual Boston University Conference on Language Development, Somerville, MA.

Boake, C. (2002, May 24). From the Binet-Simon to the Wechsler-Bellevue: Tracing the history of intelligence testing. Journal of Clinical and Experimental Neuropsychology, 24(3), 383 405.

Boroditsky, L. (2001). Does language shape thought? Mandarin and English speakers conceptions of time. Cognitive Psychology, 43, 1 22.

Boroditsky, L. (2011, February). How language shapes thought. Scientific American, 63 65.

Bouchard, T. J., Lykken, D. T., McGue, M., Segal, N. L., & Tellegen, A. (1990). Sources of human psychological differences: The Minnesota Study of Twins Reared Apart. Science, 250, 223 228.

Buck v. Bell, 274 U.S. 200

Cairns Regional Council. (n.d.). Cultural greetings. http://www.cairns.qld.gov.au/__data/assets/pdf_file/0007/8953/CulturalGreetingExercise.pdf

Callero, P. L. (1994). From role-playing to role-using: Understanding role as resource. Social Psychology Quarterly, 57, 228 243.

Cattell, R. (1963). Theory of fluid and crystallized intelligence: A critical experiment. Journal of Educational Psychology, 54(1), 1 22.

Cianciolo, A. T., & Sternberg, R. J. (2004). Intelligence: A brief history. Blackwell Publishing.

Chomsky, N. (1965). Aspects of the theory of syntax. MIT Press

Corballis, M. C., & Suddendorf, T. (2007). Memory, time, and language. In C. Pasternak (Ed.), What makes us human (pp. 17 36). Oneworld Publications.

Cropley, A. (2006). In praise of convergent thinking. Creativity Research Journal, 18(3), 391 404.

Csikszentmihalyi, M., & Csikszentmihalyi, I. (1993). Family influences on the development of giftedness. Ciba Foundation Symposium, 178, 187 206.

Curtiss, S. (1981). Dissociations between language and cognition: Cases and implications. Journal of Autism and Developmental Disorders, 11(1), 15 30.

Cyclopedia of Puzzles. (n.d.) http://www.mathpuzzle.com/loyd/

Dates and Events. (n.d.). Oprah Winfrey timeline. http://www.datesandevents.org/people-timelines/05-oprah-winfrey-timeline .htm

Duncker, K. (1945). On problem-solving (L. S. Lees, Trans.) Psychological Monographs, 58(5), i 113. https://psycnet.apa.org/record/2011-16110-001

Fern ndez, E. M., & Cairns, H. S. (2011). Fundamentals of psycholinguistics. Wiley-Blackwell.

Flanagan, D., & Kaufman, A. (2004). Essentials of WISC-IV assessment. John Wiley and Sons, Inc.

Flynn, J., Shaughnessy, M. F., & Fulgham, S. W. (2012) Interview with Jim Flynn about the Flynn effect. North American Journal of Psychology, 14(1), 25 38.

Fox, M. (2012, November 1). Arthur R. Jensen dies at 89; Set off debate about I.Q. New York Times, p. B15.

Fromkin, V., Krashen, S., Curtiss, S., Rigler, D., & Rigler, M. (1974). The development of language in Genie: A case of language acquisition beyond the critical period. Brain and Language, 1, 81 107.

Gardner, H. (1983). Frames of mind: The theory of multiple intelligences. Basic Books.

German, T. P., & Barrett, H. C. (2005). Functional fixedness in a technologically sparse culture. Psychological Science, 16, 1 5.

Goad, B. (2013, January 25). SSA wants to stop calling people 'mentally retarded. http://thehill.com/blogs/regwatch/pending-regs/279399-ssa-wants-to-stop-calling-people-mentally-retarded

Goldstone, R. L., & Kersten, A. (2003). Concepts and categorization. In A. F. Healy, R. W. Proctor, & I.B. Weiner (Eds.), Handbook of psychology (Volume IV, pp. 599 622). John Wiley & Sons, Inc.

Goleman, D. (1995). Emotional intelligence; Why it can matter more than IQ. Bantam Books.

Gordon, O. E. (1995). Francis Galton (1822 1911). http://www.psych.utah.edu/gordon/Classes/Psy4905Docs/PsychHistory/Cards/Galton.html

Gottfredson, L. S. (2004). Social consequences of group differences in cognitive ability. University of Delaware.

Gresham, F. M., & Witt, J. C. (1997). Utility of intelligence tests for treatment planning, classification, and placement decisions: Recent empirical findings and future directions. School Psychology Quarterly, 12(3), 249 267.

Guilford, J. P. (1967). The nature of human intelligence. McGraw Hill.

Heaton, S. (2004). Making the switch: Unlocking the mystery of the WISC-IV. Case Conference. University of Florida.

Jensen, J. (2011). Phoneme acquisition: Infants and second language learners. The Language Teacher, 35(6), 24 28.

Johnson, J. S., & Newport, E. L. (1989). Critical period effects in second language learning: The influence of maturational state on the acquisition of English as a second language. Cognitive Psychology, 21, 60 99.

Kahneman, D. (2011). Thinking, fast and slow. Farrar, Straus, and Giroux.

Kishyama, M. M., Boyce, W. T., Jimenez, A. M., Perry, L. M., & Knight, R. T. (2009). Socioeconomic disparities affect prefrontal function in children. Journal of Cognitive Neuroscience, 21(6), 1106 1115.

Ko, L. (2016). Unwanted sterilization and eugenics programs in the United States. Independent Lens. http://www.pbs.org/independentlens/blog/unwanted-sterilization-and-eugen ics-programs-in-the-united-states/

Larry P v. Riles, C-71-2270 RFP. (1979).

Lenneberg, E. (1967). Biological foundations of language. Wiley.

Liptak, A. (2008, January 19). Lawyer reveals secret, toppling death sentence. New York Times. http://www.nytimes.com/2008/01/19/us/19death.html?_r=0

Locke, E. A. (2005, April 14). Why emotional intelligence is an invalid concept. Journal of Organizational Behavior, 26, 425 431.

Mayer, J. D., Salovey, P., & Caruso, D. (2004). Emotional intelligence: Theory, findings, and implications, Psychological Inquiry, 15(3), 197 215.

Modgil, S., & Routledge, C. M. (Eds.). (1987). Arthur Jensen: Consensus and controversy. Falmer Press.

Moskowitz, B. A. (1978). The acquisition of language. Scientific American, 239, 92 108. Petitto, L. A., Holowka, S., Sergio, L. E., Levy, B., & Ostry, D. J. (2004). Baby hands that move to the rhythm of language: Hearing babies acquiring sign languages babble silently on the hands. Cognition, 93, 43 73.

Neyfakh, L. (2013, October 7). Why you can t stop checking your phone. http://www.bostonglobe.com/ideas/2013/10/06/why-you-can-stop-checking-yo ur-phone/rrBJzyBGDAr1YlEH5JQDcM/story.html

Parker, J. D., Saklofske, D. H., & Stough, C. (Eds.). (2009). Assessing emotional intelligence: Theory, research, and applications. Springer.

Petitto, L. A., Holowka, S., Sergio, L. E., Levy, B., & Ostry, D. J. (2004). Baby hands that move to the rhythm of language: Hearing babies acquiring sign languages babble silently on the hands. Cognition, 93, 43 73.

Pickens, J. (1994). Full-term and preterm infants perception of face-voice synchrony. Infant Behavior and Development, 17, 447 455.

Pratkanis, A. (1989). The cognitive representation of attitudes. In A. R. Pratkanis, S. J. Breckler, & A. G. Greenwald (Eds.), Attitude structure and function (pp. 71 98). Erlbaum.

Regier, T., & Kay, P. (2009). Language, thought, and color: Whorf was half right. Trends in Cognitive Sciences, 13(10), 439 446.

Riccio, C. A., Gonzales, J. J., & Hynd, G. W. (1994). Attention-deficit Hyperactivity Disorder (ADHD) and learning disabilities. Learning Disability Quarterly, 17, 311 322.

Richardson, K. (2002). What IQ tests test. Theory & Psychology, 12(3), 283 314.

Roberts, D. (2014, May 27). U.S. Supreme Court bars Florida from using IQ score cutoff for executions. The Guardian. http://www.theguardian.com/world/2014/may/27/us-supreme-court-iq-score-c utoff-florida-execution

Rushton, J. P., & Jensen, A. R. (2005). Thirty years of research on race differences in cognitive ability. Psychology, public policy, and law, 11(2), 235 294.

Rymer, R. (1993). Genie: A Scientific Tragedy. Harper Collins.

Sapir, E. (1964). Culture, language, and personality. University of California Press. (Original work published 1941)

Schlinger, H. D. (2003). The myth of intelligence. The Psychological Record, 53(1), 15 32.

Schneider, W. J., & McGrew, K. S. (2018). The Cattell-Horn-Carroll theory of cognitive abilities. In D. P. Flanagan & E. M. McDonough (Eds.), Contemporary intellectual assessment: Theories, tests, and issues (pp. 73 163). The Guilford Press.

Severson, K. (2011, December 9). Thousands sterilized, a state weighs restitution. The New York Times. http://www.nytimes.com/2011/12/10/us/redress-weighed-for-forcedsteriliza tions-in-north-carolina.html?pagewanted=all&_r=0

Singleton, D. M. (1995). Introduction: A critical look at the critical period hypothesis in second language acquisition research. In D.M. Singleton & Z. Lengyel (Eds.), The age factor in second language acquisition: A critical look at the critical period hypothesis in second language acquisition research (pp. 1 29). Multilingual Matters Ltd.

Skinner, B. F. (1957). Verbal behavior. Copley Publishing Group.

Smits-Engelsman, B. C. M., & Van Galen, G. P. (1997). Dysgraphia in children: Lasting psychomotor deficiency or transient developmental delay? Journal of Experimental Child Psychology, 67, 164 184.

Spelke, E. S., & Cortelyou, A. (1981). Perceptual aspects of social knowing: Looking and listening in infancy. In M.E. Lamb & L.R. Sherrod (Eds.), Infant social cognition: Empirical and theoretical considerations (pp. 61 83). Erlbaum.

Steitz, T. (2010). Thomas A. Steitz Biographical. (K. Grandin, Ed.) http://www.nobelprize.org/nobel_prizes/chemistry/laureates/2009/steitz-bio.html

Sternberg, R. J. (1988). The triarchic mind: A new theory of intelligence. Viking-Penguin.

Terman, L. M. (1925). Mental and physical traits of a thousand gifted children (I). Stanford University Press.

Terman, L. M., & Oden, M. H. (1947). The gifted child grows up: 25 years follow-up of a superior group: Genetic studies of genius (Vol. 4). Stanford University Press.

Terman, L. M. (1916). The measurement of intelligence. Houghton-Mifflin.

Tomasello, M., & Rakoczy, H. (2003). What makes human cognition unique? From individual to shared to collective intentionality. Mind & Language, 18(2), 121 147.

Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. Science, 185(4157), 1124 1131.

van Troyer, G. (1994). Linguistic determinism and mutability: The Sapir-Whorf hypothesis and intercultural communication. JALT Journal, 2, 163 178.

Wechsler, D. (1958). The measurement of adult intelligence. Williams & Wilkins.

Wechsler, D. (1981). Manual for the Wechsler Adult Intelligence Scale revised. Psychological Corporation.

Wechsler, D. (2002). WPPSI-R manual. Psychological Corporation.

Werker, J. F., & Lalonde, C. E. (1988). Cross-language speech perception: Initial capabilities and developmental change. Developmental Psychology, 24, 672 683.

Werker, J. F., & Tees, R. C. (1984). Cross-language speech perception: Evidence for perceptual reorganization during the first year of life. Infant Behavior and Development, 7, 49 63.

Whorf, B. L. (1956). Language, thought and relativity. MIT Press.

Williams, R. L., (1970). Danger: Testing and dehumanizing Black children. Clinical Child Psychology Newsletter, 9(1), 5 6.

Zwicker, J. G. (2005). Effectiveness of occupational therapy in remediating handwriting difficulties in primary students: Cognitive versus multisensory interventions. Unpublished master s thesis, University of Victoria, Victoria, British Columbia, Canada). http://dspace.library.uvic.ca:8080/bitstream/handle/1828/49/Zwicker%20thesis.pdf?sequence=1

Memory

Abel, M., & B uml, K.-H. T. (2013). Sleep can reduce proactive interference. Memory, 22(4), 332 339. doi:10.1080/09658211.2013.785570. http://www.psychologie.uni-regensburg.de/Baeuml/papers_in_press/sleepPI.pdf

Adams, J. K. (1957). Laboratory studies of behavior without awareness. Psychological Bulletin, 54, 383 405. https://psycnet.apa.org/record/1959-00483-001

Anderson, J. R., & Reder, L. M. (1999). The fan effect: New results and new theories. Journal of Experimental Psychology: General, 128, 186 197. https://psycnet.apa.org/record/1999-05245-004

Anderson, N. S. (1969). The influence of acoustic similarity on serial recall of letter sequences. Quarterly Journal of Experimental Psychology, 21(3), 248 255.

Anderson, R. C. (1984). Role of the reader's schema in comprehension, learning, and memory. In R. C. Anderson, J. Osborn, & R. J. Tierney (Eds.), Learning to read in American schools: Basal Readers and Content Texts (pp. 243 257). Erlbaum.

Atkinson, R. C., & Shiffrin, R. M. (1968). Human memory: A proposed system and its control processes. In K. W. Spence & J. T. Spence (Eds.), The psychology of learning and motivation: Volume 2 (pp. 89 195). Academic Press.

Baddeley, A. (2004). Your memory: A user's guide. Firefly Books.

Baddeley, A. D. (2000). The episodic buffer: A new component of working memory? Trends in Cognitive Sciences, 4, 417 423. https://www.ncbi.nlm.nih.gov/pubmed/11058819

Baddeley, A. D., & Hitch, G. (1974). Working memory. In G. H. Bower (Ed.), The psychology of learning and motivation: Advances in research and theory (Vol. 8, pp. 47 89). Academic Press.

Bayley, P. J., & Squire, L. R. (2002). Medial temporal lobe amnesia: Gradual acquisition of factual information by nondeclarative memory. Journal of Neuroscience, 22, 5741 5748.

Bellezza, F. S. (1981). Mnemonic devices: Classification, characteristics and criteria. Review of Educational Research, 51, 247 275.

Benjamin N. Cardozo School of Law, Yeshiva University. (2009). Reevaluating lineups: Why witnesses make mistakes and how to reduce the chance of a misidentification. The Innocence Project website: http://www.innocenceproject.org/docs/Eyewitness_ID_Report.pdf

Blockland, A. (1996). Acetylcholine: A neurotransmitter for learning and memory? Brain Research Reviews, 21, 285 300.

Bodie, G. D., Powers, W. G., & Fitch-Hauser, M. (2006). Chunking, priming, and active learning: Toward an innovative approach to teaching communication-related skills. Interactive Learning Environment, 14(2), 119 135.

Bousfield, W. (1935). The occurrence of clustering in the recall of randomly arranged associates. Journal of General Psychology, 49, 229 240.

Bransford, J. D., & McCarrell, N. S. (1974). A sketch of a cognitive approach to comprehension. In W. B. Weimer & D. J. Palermo (Eds.), Cognition and the symbolic processes (pp. 189 229). Lawrence Erlbaum Associates.

Briere, J., & Conte, J. (1993). Self-reported amnesia for abuse in adults molested as children. Journal of Traumatic Stress, 6, 21 31.

Carli, L. (1999). Cognitive reconstruction, hindsight, and reactions to victims and perpetrators. Personality and Social Psychology Bulletin, 25(8), 966 979. doi:10.1177/01461672992511005

Ceci, S. J., & Bruck, M. (1993). Child witness: Translating research into policy. Social Policy Report, 7(3), 1 30.

Ceci, S. J., & Bruck, M. (1995). Jeopardy in the courtroom: A scientific analysis of children s testimony. American Psychological Association.

Cheit, R. E. (2007). The recovered memory project. http://blogs.brown.edu/recoveredmemory/.

Christianson, S. A. (1992). The handbook of emotion and memory: Research and theory. Erlbaum.

Clark, R. E., Zola, S. M., & Squire, L. R. (2000). Impaired recognition memory in rats after damage to the hippocampus. The Journal of Neuroscience, 20(23), 8853 8860.

Collins, A. M., & Loftus, E. F. (1975). A spreading-activation theory of semantic processing. Psychological Review, 82, 407 428. https://psycnet.apa.org/record/1976-03421-001

Corkin, S. (1965). Tactually-guided maze learning in man: Effects of unilateral cortical excisions and bilateral hippocampal lesions. Neuropsychologia, 3, 339 351.

Corkin, S. (1968). Acquisition of motor skill after bilateral medial temporal-lobe excision. Neuropsychologia, 6, 255 264.

Corkin, S., Amaral D. G., Gonz lez, R. G., Johnson, K. A., & Hyman, B. T. (1997). H. M. s medial temporal lobe lesion: Findings from magnetic resonance imaging. Journal of Neuroscience, 17(10), 3964 3979.

Cowan, N. (2010). The magical mystery tour: How is working memory capacity limited, and why? Current Directions in Psychological Science, 19, 51 57. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2864034/

Craik, F. I. M., & Lockhart, R. S. (1972). Levels of processing: A framework for memory research. Journal of Verbal Learning and Verbal Behavior, 11, 671 684.

Craik, F. I. M., Moroz, T. M., Moscovitch, M., Stuss, D. T., Winocur, G., Tulving, E., & Kapur, S. (1999). In search of the self: A positron emission tomography study. Psychological Science, 10(1), 26 34.

Craik, F. I. M., & Tulving, E. (1975). Depth of processing and the retention of words in episodic memory. Journal of Experimental Psychology, 104(3), 268 294.

Craik, F. I. M., & Watkins, M. J. (1973). The role of rehearsal in short-term memory. Journal of Verbal Learning and Verbal Behavior, 12, 599 607.

Green, J. T., & Woodruff-Pak, D. S. (2000). Eyeblink classical conditioning in aging animals. In D. S. Woodruff-Pak & J. E. Steinmetz (Eds.), Eyeblink classical conditioning: Animal models (Vol. 2, pp.155 178). Kluwer Academic.

Greenberg, D. L. (2004). President Bush's false [flashbulb] memory of 9/11/01. Applied. Cognitive Psychology, 18(3), 363 370. doi:10.1002/acp.1016

Devilly, G. J. (2007). If nothing happened why do I still hurt? An update on the memory wars. InPsych, 29(2), 16 18.

Ebbinghaus, H. (1964). Memory: A contribution to experimental psychology (H. A. Ruger & C. E. Bussenius, Trans.). Dover. (Original work published 1885)

Giddan, N. S., & Eriksen, C. W. (1959). Generalization of response biases acquired with and without verbal awareness. Journal of Personality, 27, 104 115. https://psycnet.apa.org/record/1960-03930-001

Goodman, G. S. (2006). Children s eyewitness memory: A modern history and contemporary commentary. Journal of Social Issues, 62, 811 832.

Greenspoon, J. (1955). The reinforcing effect of two spoken sounds on the frequency of two responses. American Journal of Psychology, 68, 409 416. https://psycnet.apa.org/record/1956-04488-001

Hassabis D., & Maguire E. A. (2007). Deconstructing episodic memory with construction. Trends in Cognitive Sciences, 11(7), 299 306.

Jacobs, J. (1887). Experiments on prehension. Mind, 12, 75 79.

Jacoby, L. L. (1983). Perceptual enhancement: Persistent effects of an experience. Journal of Experimental Psychology: Learning, Memory, and Cognition, 9, 21 38. https://www.ncbi.nlm.nih.gov/pubmed/6220114

Jacoby, L. L., & Witherspoon, D. (1982). Remembering without awareness. Canadian Journal of Psychology, 36, 300 324. https://psycnet.apa.org/record/1983-04928-001

Johnson, K. E., & Mervis, C. B. (1997). Effects of varying levels of expertise on the basic level of categorization. Journal of Experimental Psychology, General, 126, 248 277. https://psycnet.apa.org/record/1997-05801-003

Johnson, K. E., & Mervis, C. B. (1998). Impact of intuitive theories on feature recruitment throughout the continuum of expertise. Memory & Cognition, 26, 382 401. https://link.springer.com/article/10.3758/BF03201148

Josselyn, J. A. (2010). Continuing the search for the engram: Examining the mechanism of fear memories. Journal of Psychiatry Neuroscience, 35(4), 221 228.

Kapur, S., Craik, F. I. M., Tulving, E., Wilson, A. A., Houle, S., & Brown, G. M. (1994). Neuroanatomical correlates of encoding in episodic memory: Levels of processing effect. Proceedings of the National Academy of Sciences of the United States of America, 91(6), 208 2011.

Keppel, G., & Underwood, B. J. (1962). Proactive inhibition in short-term retention of single items. Journal of Verbal Learning & Verbal Behavior, 1, 153 161. https://www.sciencedirect.com/science/article/abs/pii/S0022537162800231

Krieckhaus, E. E., & Eriksen, C. W. (1960). A study of awareness and its effects on learning and generalization. Journal of Personality, 28, 503 517. https://psycnet.apa.org/record/1961-04444-001

Lacey, J. L., & Smith, R. L. (1954). Conditioning and generalization of unconscious anxiety. Science, 120, 1045 1052. https://www.ncbi.nlm.nih.gov/pubmed/13216221

Lashley K. S. (1950). In search of the engram. Society of Experimental Biology Symposium, 4: Psychological Mechanisms in Animal Behavior. Cambridge University Press.

Lazarus, R. S., & McCleary, R. (1951). Autonomic discrimination without awareness: A study of subception. Psychological Review, 58, 113 122. https://www.ncbi.nlm.nih.gov/pubmed/14834294

Loftus, E. F., & Palmer, J. C. (1974). Reconstruction of auto-mobile destruction: An example of the interaction between language and memory. Journal of Verbal Learning and Verbal Behavior, 13, 585 589.

MacLeod, C. M., Gopie, N., Hourihan, K. L., Neary, K. R., & Ozubko, J. D. (2010). The production effect: Delineation of a phenomenon. Journal of Experimental Psychology: Learning, Memory, and Cognition, 36(3), 671 685.

Mayford, M., Siegelbaum, S. A., & Kandel, E. R. (2012). Synapses and memory storage. Cold Spring Harbor Perspectives in Biology, Cold Spring Harbor Laboratory Press.

McGaugh, J. L. (2003). Memory and emotion: The making of lasting memories. Columbia University Press.

McLeod, S. A. (2011). Anterograde amnesia. http://www.simplypsychology.org/anterograde-amnesia.html

Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. Psychological Review, 68, 81 87.

Myhrer, T. (2003). Neurotransmitter systems involved in learning and memory in the rat: A meta-analysis based on studies of four behavioral tasks. Brain Research Reviews, 41(2 3), 268 287.

Newseum. (n.d.). G-men and journalists: D. C. sniper. http://www.newseum.org/exhibits-and-theaters/temporary-exhibits/g-men-and-journalists/sniper/

Nickerson, R. S., & Adams, M. J. (1979). Long-term memory for a common object. Cognitive Psychology, 11(3), 287 307.

Olson, M. A., & Fazio, R. H. (2001). Implicit attitude formation through classical conditioning. Psychological Science, 12, 413 417. https://www.ncbi.nlm.nih.gov/pubmed/11554676

Paivio, A. (1986). Mental representations: A dual coding approach. Oxford University Press.

Palmer, C. F., Jones, R. K., Hennessy, B. L., Unze, M. G., & Pick, A. D. (1989). How is a trumpet known? The basic object level concept and perception of musical instruments. American Journal of Psychology, 102, 17 37. https://www.jstor.org/stable/1423114?seq=1

Parker, E. S., Cahill, L., & McGaugh, J. L. (2006). A case of unusual autobiographical remembering. Neurocase, 12, 35 49.

Payne, B. K., Jacoby, L. L., & Lambert, A. J. (2004). Memory monitoring and the control of stereotype distortion. Journal of Experimental Social Psychology, 40, 52 64.

Peterson, L., & Peterson, M. J. (1959). Short-term retention of individual verbal items. Journal of Experimental Psychology, 58(3), 193 198. https://psycnet.apa.org/record/1960-05499-001

Pew Research Center (2011, September 1). Ten years after 9/11: United in remembrance, divided over policies. People Press.

Pipe, M.-E. (1996). Children s eyewitness memory. New Zealand Journal of Psychology, 25(2), 36 43.

Pipe, M.-E., Lamb, M., Orbach, Y., & Esplin, P. W. (2004). Recent research on children s testimony about experienced and witnessed events. Developmental Review, 24, 440 468.

Reber, A. S. (1976). Implicit learning of synthetic languages: The role of instructional set. Journal of Experimental Psychology: Human Learning and Memory, 2, 88 94. https://psycnet.apa.org/doiLanding?doi=10.1037%2F0278-7393.2.1.88

Roediger, H. L., III. (1990). Implicit memory: Retention without remembering. American Psychologist, 45, 1043 1056. https://www.ncbi.nlm.nih.gov/pubmed/2221571

Roediger, H. L., & DeSoto, K. A. (2015). The psychology of reconstructive memory. In J. Wright (Ed.), International Encyclopedia of the Social and Behavioral sciences, 2e. Elsevier.

Roediger, H. L., III, & McDermott, K. B. (2000). Tricks of memory. Current Directions in Psychological Science, 9, 123 127.

Rogers, T. B., Kuiper, N. A., & Kirker, W. S. (1977). Self-reference and the encoding of personal information. Journal of Personal Social Psychology, 35(9), 677 688.

Rosch, E., Mervis, C. B., Gray, W. D., Johnson, D. M., & Boyes-Braem, P. (1976). Basic objects in natural categories. Cognitive Psychology, 8(3), 382 439. https://psycnet.apa.org/record/1976-27194-001

Schacter, D. L. (1987). Implicit memory: History and current status. Journal of Experimental Psychology: Learning, Memory, and Cognition, 13, 501 551. https://pdfs.semanticscholar.org/8c35/06e3c78a987fb5a704bc60611680fcb5365b.pdf

Schacter, D. L. (1992). Priming and multiple memory systems: Perceptual mechanisms of implicit memory. Journal of Cognitive Neuroscience 4, 244 256. https://dash.harvard.edu/bitstream/handle/1/3627272/Schacter_PrimingMultiple.pdf?sequence=2

Schacter, D. (2001). The seven sins of memory: How the mind forgets and remembers. Houghton Mifflin.

Steinmetz, J. E. (1999). A renewed interest in human classical eyeblink conditioning. Psychological Science, 10, 24 25.

Tanaka, J. M., & Taylor, M. (1991). Object categories and expertise: Is the basic level in the eye of the beholder? Cognitive Psychology, 23, 457 482. http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.320.9949&rep=rep1&type=pdf

Tigner, R. B. (1999). Putting memory research to good use. College Teaching, 47(4), 149 152.

Tulving, E. (1972). Episodic and semantic memory. In E. Tulving & W. Dolandson (Eds.), Organization of memory (pp. 381 403). Academic Press.

Tulving, E. (2002, February). Episodic memory: From mind to brain. Annual Review of Psychology, 53, 1 25. doi:10.1146/annurev.psych.53.100901.135114

van Praag, H. (2008). Neurogenesis and exercise: Past and future directions. NeuroMolecular Medicine, 10(2), 128 140.

Wells, G. L., & Quinlivan, D. S. (2009). Suggestive eyewitness identification procedures and the Supreme Court s reliability test in light of eyewitness science: 30 years later. Law and Human Behavior, 33, 1 24. doi:10.1007/s10979-008-9130-3

Wrubel, B. (Writer), & Spiller, M. (Director). (2010). The Old Wagon. In S. Levitan & C. Lloyd (Executive producers), Modern Family. 20th Century Fox Television.

Yang, J., Xu, X., Du, X., Shi, C., & Fang, F. (2011). Effects of unconscious processing on implicit memory for fearful faces. PLoS One, 6(2). https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0014641

Lifespan development

Ainsworth, M. D. S., & Bell, S. M. (1970). Attachment, exploration, and separation: Illustrated by the behavior of one-year-olds in a strange situation. Child Development, 41, 49 67.

Ainsworth, M. D. S., Blehar, M. C., Waters, E., & Wall, S. (1978). Patterns of attachment: A psychological study of the strange situation. Erlbaum.

American Academy of Pediatrics. (2007). The importance of play in promoting healthy child development and maintaining strong parent-child bonds. Pediatrics, 199(1), 182 191.

American Psychological Association. (2019). Ruth Howard, PhD. https://www.apa.org/pi/oema/resources/ethnicity-health/psychologists/ruth-howard

Amsterdam, B. (1972). Mirror image reactions before age two. Developmental Psychobiology, 5, 297 305.

Archer, J. (1992). Ethology and human development. Harvester Wheatsheaf.

Arnett, J. (2000). Emerging adulthood: A theory of development from the late teens through the twenties. American Psychologist, 55(5), 469 480.

Arora, R., Mittal, A., & Kakkar, R. (2015). Alzheimer's disease: Recent advances. Journal of Biochemistry and Molecular Biology Research, 1, 87 104.

Ashley, S. J., Magnuson, S. I., Omnell, L. M., & Clarren, S. K. (1999). Fetal alcohol syndrome: Changes in craniofacial form with age, cognition, and timing of ethanol exposure in the macaque. Teratology, 59(3), 163 172.

Bahr, S. J., & Hoffman, J. P. (2010). Parenting style, religiosity, peers, and adolescent heavy drinking. Journal of Studies on Alcohol and Drugs, 71, 539 543.

Baillargeon, R. (2004). Infants reasoning about hidden objects: Evidence for event-general and event-specific expectations. Developmental Science, 7(4), 391 424.

Baillargeon, R. (1987). Young infants' reasoning about the physical and spatial properties of a hidden object. Cognitive Development, 2(3), 179 200.

Baillargeon, R., Li, J., Gertner, Y., & Wu, D. (2011). How do infants reason about physical events. The Wiley-Blackwell handbook of childhood cognitive development, 2, 11 48.

Barber, B. K. (1994). Cultural, family, and person contexts of parent-adolescent conflict. Journal of Marriage and the Family, 56, 375 386.

Basseches, M. (1984). Dialectical thinking as metasystematic form of cognitive organization. In M. L. Commons, F. A. Richards, & C. Armon (Eds.), Beyond formal operations: Late adolescent and adult cognitive development (pp. 216 238). Praeger.

Baumrind, D. (1971). Current patterns of parental authority. Developmental Psychology, 4(1, Pt. 2), 1 103. doi:10.1037/h0030372

Baumrind, D. (1991). The influence of parenting style on adolescent competence and substance use. Journal of Early Adolescence, 11(1), 56 95.

Bayley, N., & Oden, M. H. (1955). The maintenance of intellectual ability in gifted adults. Journal of Gerontology, 10, 91 107.

Benjamin, L., Henry, K., & McMahon, L. (2005). Inez Beverly Prosser and the education of African Americans. Journal of the History of the Behavioral Sciences, 41, 43 62.

Bjorklund, D. F. (1987). A note on neonatal imitation. Developmental Review, 7, 86 92.

Blossom, M., & Morgan, J.L. (2006). Does the face say what the mouth says? A study of infants sensitivity to visual prosody. In 30th anuual Boston University conference on language development, Somerville, MA.

Bogartz, R. S., Shinskey, J. L., & Schilling, T. (2000). Infancy, 1(4), 403 428.

Bowlby, J. (1969). Attachment and loss: Attachment (Vol. 1). Basic Books.

Bowlby, J. (1988). A secure base: Parent-child attachment and health human development. Basic Books.

Bronfenbrenner, U. (1977). Toward an experimental ecology of human development. American Psychologist, 32, 513 531.

Brumley, R., Enquidanos, S., Jamison, P., Seitz, R., Morgenstern, N., Saito, S., . . . Gonzalez, J. (2007). Increased satisfaction with care and lower costs: Results of a randomized trial of in-home palliative care. Journal of the American Geriatric Society, 55(7), 993 1000.

Brumley, R. D., Enquidanos, S., & Cherin, D. A. (2003). Effectiveness of a home-based palliative care program for end-of-life. Journal of Palliative Medicine, 6(5), 715 724.

Callaghan, T. C., Rochat, P., Lillard, A., Claux, M.L., Odden, H., Itakura, S., . . . Singh, S. (2005). Synchrony in the onset of mental-state reasoning. Psychological Science, 16, 378 384.

Carel, J-C., Lahlou, N., Roger, M., & Chaussain, J. L. (2004). Precocious puberty and statural growth. Human Reproduction Update, 10(2), 135 147.

Carstensen, L. L. (1992). Social and emotional patterns in adulthood: Support for socioemotional selectivity. Psychology and Aging, 7(3), 331 338.

Case, R. (1985). Intellectual development: Birth to Adulthood. Academic.

Casey, B. J., Tottenham, N., Liston, C., & Durston, S. (2005). Imaging the developing brain: What have we learned about cognitive development? TRENDS in Cognitive Sciences, 19(3), 104 110.

Centers for Disease Control and Prevention. (2013). Smoking during pregnancy. http://www.cdc.gov/tobacco/basic_information/health_effects/pregnancy/

Chick, K., Heilman-Houser, R., & Hunter, M. (2002). The impact of child care on gender role development and gender stereotypes. Early Childhood Education Journal, 29(3), 149 154.

Chomsky, N. (1957). Syntactic structures. Mouton.

Clark, K. & Clark, M. (1950). Emotional factors in racial identification and preference in Negro children. Journal of Negro Education, 19, 341 350.

Clements, R. (2004). An investigation of the status of outdoor play. Contemporary Issues in Early Childhood, 5(1), 68 80.

Commons, M. L., & Bresette, L. M. (2006). Illuminating major creative scientific innovators with postformal stages. In C. Hoare (Ed.), Handbook of adult development and learning (pp. 255 280). Oxford University Press.

Connor, S. R., Pyenson, B., Fitch, K., Spence, C., & Iwasaki, K. (2007). Comparing hospice and nonhospice patient survival among patients who die within a three-year window. Journal of Pain and Symptom Management, 33(3), 238 246.

Courage, M. L., & Howe, M. L. (2002). From infant to child: The dynamics of cognitive change in the second year of life. Psychological Bulletin, 128, 250 277.

Darling, N. (1999). Parenting style and its correlates. ERIC database (EDO-PS-99-3) http://ecap.crc.illinois.edu/eecearchive/digests/1999/darlin99.pdf

de Hevia, M. D., & Spelke, E. S. (2010). Number-space mapping in human infants. Psychological Science, 21(5), 653 660.

Dennett, D. (1987). The intentional stance. MIT Press.

Diamond, A. (2009). The interplay of biology and the environment broadly defined. Developmental Psychology, 45(1), 1 8.

Donenberg, G. R., Wilson, H. W., Emerson, E., Bryant, F. B. (2002). Holding the line with a watchful eye: The impact of perceived parental permissiveness and parental monitoring on risky sexual behavior among adolescents in psychiatric care. AIDS Education Prevention, 14(2), 138 157.

Dornbusch, S. M., Ritter, P. L., Leiderman, P. H., Roberts, D. F., & Fraleigh, M. J. (1987). The relation of parenting style to adolescent school performance. Child Development, 58(5), 1244 1257.

Duncan, G. J., & Magnuson, K. A. (2005). Can family socioeconomic resources account for racial and ethnic test score gaps? The Future of Children, 15(1), 35 54.

Erikson, E. H. (1963). Childhood and Society (2nd ed.). Norton.

Erikson, E. H. (1968). Identity: Youth and crisis. Norton.

Ferrer, M., & Fugate, A. (2003). Helping your school-age child develop a healthy self-concept. http://edis.ifas.ufl.edu/fy570#FOOTNOTE_2

Figdor, E., & Kaeser, L. (1998). Concerns mount over punitive approaches to substance abuse among pregnant women. The Guttmacher Report on Public Policy 1(5), 3 5.

Fischer, K. W., Yan, Z., & Stewart, J. (2003). Adult cognitive development: Dynamics in the developmental web. In J. Valsiner & K Connolly (Eds.), Handbook of developmental psychology (pp. 491 516). Sage Publications.

Flannery, D. J., Rowe, D. C., & Gulley, B. L. (1993). Impact of pubertal status, timing, and age on adolescent sexual experience and delinquency. Journal of Adolescent Research, 8, 21 40.

Forstmann, B., Tittgemeyer, M., Wagenmakers, E., Derrfuss, J., Imperati, D., & Brown, S. (2011). The speed-accuracy tradeoff in the elderly brain: A structural model-based approach. Journal of Neuroscience, 31, 17242 17249.

Freud, S. (1909). Analysis of a phobia in a five-year-old boy. In Collected Papers: Volume 111, Case Histories (1949) (pp. 149 289). Hogarth Press.

Galambos, N. L., & Almeida, D. M. (1992). Does parent-adolescent conflict increase in early adolescence? Journal of Marriage and the Family, 54, 737 747.

Galvan, A., Hare, T., Voss, H., Glover, G., & Casey, B. J. (2007). Risk-taking and the adolescent brain: Who is at risk? Developmental Science, 10, F8 F14.

Ganger, J., & Brent, M.R. (2004). Reexamining the vocabulary spurt. Developmental Psychology, 40(4), 621 632.

Garcia-Betances, R., Jimenez-Mixco, V., Arredondo, M., & Cabrera-Umpierrez, M. (2015). Using virtual reality for cognitive training of the elderly. American Journal of Alzheimer's Disease and Other Dementias, 30, 49 54.

Ge, X., Conger, R. D., & Elder, G. H. (2001). Pubertal transition, stressful life events, and the emergence of gender differences in adolescent depressive symptoms. Developmental Psychology, 37, 404 417.

Gervai, J. (2009). Environmental and genetic influences on early attachment. Child and Adolescent Psychiatry and Mental Health, 3, 25.

Gesell, A. (1933). Maturation and the patterning of behavior. In C. Murchison (Ed.), A handbook of child psychology (2nd ed., pp. 209 235). Clark University Press.

Gesell, A. (1939). Biographies of child development. Paul B. Hoeber.

Gesell, A. (1940). The first five years of life. Harper.

Gesell, A., & Ilg, F. L. (1946). The child from five to ten. Harper.

Gilligan, C. (1982). In a different voice: Psychological theory and women's development. Harvard University Press.

Gleitman, L. R., & Newport, E. L. (1995). The invention of language by children: Environmental and biological influences on the acquisition of language. In L. R. Gleitman & M. Liberman (Eds.), An invitation to cognitive science, Vol. 1: Language. (2nd ed.) (pp. 1 24). MIT Press.

Godkin, M., Krant, M., & Doster, N. (1984). The impact of hospice care on families. International Journal of Psychiatry in Medicine, 13, 153 165.

Graber, J. A., Lewinsohn, P. M., Seeley, J. R., & Brooks-Gunn, J. (1997). Is psychopathology associated with the timing of pubertal development? Journal of the Academy of Child and Adolescent Psychiatry, 36, 1768 1776.

Hair, E. C., Moore, K. A., Garrett, S. B., Kinukawa, A., Lippman, L., & Michelson, E. (2005). The parent-adolescent relationship scale. In L. Lippman (Ed.), Conceptualizing and Measuring Indicators of Positive Development: What Do Children Need to Fluorish? (pp. 183 202). Kluwer Academic/Plenum Press.

Hall, G. S. (1904). Adolescence. Appleton.

Hall, S. S. (2004, May). The good egg. Discover, 30 39.

Harlow, H. (1958). The nature of love. American Psychologist, 13, 673 685.

Harris, J. R. (2009). The nurture assumption: Why children turn out the way they do (2nd ed.). Free Press.

Hart, B., & Risley, T. R. (2003). The early catastrophe: The 30 million word gap. American Educator, 27(1), 4 9.

Hatch, E. (1983). Psycholinguistics: A second language perspective. Newbury House.

Hertzog, C., Kramer, A. F., Wilson, R. S., & Lindenberger, U. (2009). Enrichment effects on adult cognitive development. Psychological Science in the Public Interest, 9(1), 1 65.

Hood, R. W., Jr., Spilka, B., Hunsberger, B., & Corsuch, R. (1996). The psychology of religion: An empirical approach (2nd ed.). Guilford.

Huebler, F. (2005, December 14). International education statistics. http://huebler.blogspot.com/2005/12/age-and-level-of-education-in-nigeri a.html

Hutchinson, N. (2011). A geographically informed vision of skills development. Geographical Education, 24, 15.

Iverson, J.M., & Goldin-Meadow, S. (2005). Gesture paves the way for language development. Psychological Science, 16(5), 367 71.

Iyengar, S. S., Wells, R. E., & Schwartz, B. (2006). Doing better but feeling worse: Looking for the best job undermines satisfaction. Psychological Science, 17, 143 150.

John-Steiner, V. & Mahn, H. (1996). Sociocultural approaches to learning and development: A Vygotskian framework. Educational Psychologist, 31, 191 206.

Jos, P. H., Marshall, M. F., & Perlmutter, M. (1995). The Charleston policy on cocaine use during pregnancy: A cautionary tale. The Journal of Law, Medicine & Ethics, 23(2), 120 128.

Kaltiala-Heino, R. A., Rimpela, M., Rissanen, A., & Rantanen, P. (2001). Early puberty and early sexual activity are associated with bulimic-type eating pathology in middle adolescence. Journal of Adolescent Health, 28, 346 352.

Kaplan, H., & Dove, H. (1987). Infant development among the Ach of Eastern Paraguay. Developmental Psychology, 23, 190 198.

Karasik, L. B., Adolph, K. E., Tamis-LeMonda, C. S., & Bornstein, M. H. (2010). WEIRD Walking: Cross-cultural research on motor development. Behavioral & Brain Sciences, 33(2-3), 95 96.

Karnik, S., & Kanekar, A. (2012). Childhood obesity: A global public health crisis. International Journal of Preventive Medicine, 3(1), 1 7.

Kohlberg, L. (1969). Stage and sequence: The cognitive-developmental approach to socialization. In D. A. Goslin (Ed.), Handbook of socialization theory and research (p. 379). Rand McNally.

Kolb, B., & Whishaw, I. Q. (2009). Fundamentals of human neuropsychology. Worth.

K bler-Ross, E. (1969). On death and dying. Macmillan.

Labouvie-Vief, G., & Diehl, M. (1999). Self and personality development. In J. C. Cavanaugh & S. K. Whitbourne (Eds.), Gerontology: An interdisciplinary perspective (pp. 238 268). Oxford University Press.

Larson, E. B., Wang, L., Bowen, J. D., McCormick, W. C., Teri, L., Crane, P., & Kukull, W. (2006). Exercise is associated with reduced risk for incident dementia among persons 65 years of age or older. Annals of Internal Medicine, 144, 73 81.

Lee, V. E., & Burkam, D. T. (2002). Inequality at the starting gate: Social background differences in achievement as children begin school. Economic Policy Institute.

Lobo, I. (2008) Environmental influences on gene expression. Nature Education 1(1), 39.

Loop, E. (2013). Major milestones in cognitive development in early childhood. http://everydaylife.globalpost.com/major-milestones-cognitive-developmen t-early-childhood-4625.html

Maccoby, E. (1980). Social development: Psychological growth and the parent-child relationship. Harcourt Brace Jovanovich.

MacFarlane, A. (1978, February). What a baby knows. Human Nature, 74 81.

Maier, S. E., & West, J. R. (2001). Drinking patterns and alcohol-related birth defects. Alcohol Research & Health, 25(3), 168 174.

Main, M., & Solomon, J. (1990). Procedures for identifying infants as disorganized/disoriented during the Ainsworth Strange Situation. In M. T. Greenberg, D. Cicchetti, & E. M. Cummings (Eds.), Attachment in the Preschool Years (pp. 121 160). University of Chicago Press.

Markus, H. R., Ryff, C. D., Curan, K., & Palmersheim, K. A. (2004). In their own words: Well-being at midlife among high school-educated and college-educated adults. In O. G. Brim, C. D. Ryff, & R. C. Kessler (Eds.), How healthy are we? A national study of well-being at midlife (pp. 273 319). University of Chicago Press.

McIntosh, D. N., Silver, R. C., & Wortman, C. B. (1993). Religion s role in adjustment to a negative life event: Coping with the loss of a child. Journal of Personality and Social Psychology, 65, 812 821.

McMillan, S. C., Small, B. J., Weitzner, M., Schonwetter, R., Tittle, M., Moody, L., & Haley, W. E. (2006). Impact of coping skills intervention with family caregivers of hospice patients with cancer. Cancer, 106(1), 214-222.

Miklikowska, M., Duriez, B., & Soenens, B. (2011). Family roots of empathy-related characteristics: The role of perceived maternal and paternal need support in adolescence. Developmental Psychology, 47(5), 1342 1352.

Mills, M., & Melhuish, E. (1974). Recognition of mother s voice in early infancy. Nature, 252, 123 124.

Mohr, R. D., & Zoghi, C. (2006). Is job enrichment really enriching? (U.S. Bureau of Labor Statistics Working Paper 389). Washington, DC: U.S. Bureau of Labor Statistics. http://www.bls.gov/ore/pdf/ec060010.pdf

Moore, K. A., Guzman, L., Hair, E. C., Lippman, L., & Garrett, S. B. (2004). Parent-teen relationships and interactions: Far more positive than not. Child Trends Research Brief, 2004-25. Child Trends.

National Institutes of Health. (2013). What is prenatal care and why is it important? http://www.nichd.nih.gov/health/topics/pregnancy/conditioninfo/Pages/pre natal-care.aspx

National Institute of Neurological Disorders and Stroke. (2019). Brain basics: The life and death of a neuron. https://www.ninds.nih.gov/Disorders/Patient-CareGiver-Education/Life-and -Death-Neuron

Neblett, E., Smalls, C., Ford, K., Nguyen, H., & Sellers, R. (2009). Racial socialization and racial identity: African American parents' messages about race as precursors to identity. Journal of Youth and Adolescence, 38, 189 203.

Newcombe, N. (2011). What is neoconstructivism? Child Development Perspectives, 5, 157 160.

Nolen-Hoeksema, S., & Larson, J. (1999). Coping with loss. Erlbaum.

Overman, W. H., Bachevalier, J., Turner, M., & Peuster, A. (1992). Object recognition versus object discrimination: Comparison between human infants and infant monkeys. Behavioral Neuroscience, 106, 15 29.

Paloutzian, R. F. (1996). Invitation to the psychology of religion. Allyn & Bacon.

Parent, J., Forehand, R., Merchant, M. J., Edwards, M. C., Conners-Burrow, N. A., Long, N., & Jones, D. J. (2011). The relation of harsh and permissive discipline with child disruptive behaviors: Does child gender make a difference in an at-risk sample? Journal of Family Violence, 26, 527 533.

Piaget, J. (1954). The construction of reality in the child. Basic Books.

Piaget, J. (1930). The child s conception of the world. Harcourt, Brace & World.

Piaget, J. (1932). The moral judgment of the child. Harcourt, Brace & World.

Podewils, L. J., Guallar, E., Kuller, L. H., Fried, L. P., Lopez, O. L., Carlson, M., & Lyketsos, C. G. (2005). Physical activity, APOE genotype, and dementia risk: Findings from the Cardiovascular Health Cognition Study. American Journal of Epidemiology, 161, 639 651.

Pollack, W., & Shuster, T. (2000). Real boys voices. Random House.

Rhodes, R. L., Mitchell, S. L., Miller, S. C., Connor, S. R., & Teno, J. M. (2008). Bereaved family members' evaluation of hospice care: What factors influence overall satisfaction with services? Journal of Pain and Symptom Management, 35, 365 371.

Risley, T. R., & Hart, B. (2006). Promoting early language development. In N. F. Watt, C. Ayoub, R. H. Bradley, J. E. Puma, & W. A. LeBoeuf (Eds.), The crisis in youth mental health: Early intervention programs and policies (Vol. 4, pp. 83 88). Praeger.

Rogers, L. & Meltzoff, A. (2017). Is gender more important and meaningful than race? An analysis of racial and gender identity among Black, White, and mixed-race children. Cultural Diversity and Ethnic Minority Psychology, 23, 323 334.

Rothbaum, R., Weisz, J., Pott, M., Miyake, K., & Morelli, G. (2000). Attachment and culture: Security in the United States and Japan. American Psychologist, 55, 1093 1104.

Ruggiero, G., D'Errico, O., & Iachini, T. (2016). Development of egocentric and allocentric spatial representations from childhood to elderly age. Psychological Research, 80, 259 272.

Russell, S. T., Crockett, L. J., & Chao, R. (Eds.). (2010). Asian American parenting and parent-adolescent relationships. In R. Levesque (Series Ed.), Advancing responsible adolescent development. Springer.

Ryff, C. D., & Singer, B. (2009). Understanding healthy aging: Key components and their integration. In V. L. Bengtson, D. Gans., N. M. Putney, & M. Silverstein. (Eds.), Handbook of theories of aging (2nd ed., pp. 117 144). Springer.

Samarel, N. (1991). Caring for life after death. Hemisphere.

Sanson, A., & Rothbart, M. K. (1995). Child temperament and parenting. In M. Bornstein (Ed.), Applied and practical parenting (Vol. 4, pp. 299 321). Lawrence Erlbaum.

Schechter, C., & Byeb, B. (2007). Preliminary evidence for the impact of mixed-income preschools on low-income children s language growth. Early Childhood Research Quarterly, 22, 137 146.

Shamay-Tsoory, S. G., Tomer, R., & Aharon-Peretz, J. (2005). The neuroanatomical basis of understanding sarcasm and its relationship to social cognition. Neuropsychology, 19(3), 288 300.

Shanahan, L., McHale, S. M., Osgood, D. W., & Crouter, A. C. (2007). Conflict frequency with mothers and fathers from middle childhood to late adolescence: Within and between family comparisons. Developmental Psychology, 43, 539 550.

Siegler, R. S. (2005). Children s thinking (4th ed). Erlbaum.

Siegler, R. S. (2006). Microgenetic analyses of learning. In D. Kuhn & R. S. Siegler (Eds.), Handbook of child psychology: Cognition, perception, and language (6th ed., Vol. 2). Wiley.

Sinnott, J. D. (1998). The development of logic in adulthood: Postformal thought and its applications. Springer.

Small, M. F. (1999). Our babies, ourselves: How biology and culture shape the way we parent. Anchor Books.

Spelke, E.S., & Cortelyou, A. (1981). Perceptual aspects of social knowing: Looking and listening in infancy. In M.E. Lamb & L.R. Sherrod (Eds.), Infant social cognition: Empirical and theoretical considerations (pp. 61 83). Erlbaum.

Spencer, M. B., Dupree, D., & Hartmann, T. (1997). A phenomenological variant of ecological systems theory (PVEST): A self-organization perspective in context. Development and Psychopathology, 9, 817 833.

Steinberg, L., & Morris, A. S. (2001). Adolescent development. Annual Review of Psychology, 52, 83 110.

Sterns, H. L., & Huyck, M. H. (2001). The role of work in midlife. In M. Lachman (Ed.), The handbook of midlife development (pp. 447 486). Wiley.

Steven L. Youngentob, et. al. (2007). Experience-induced fetal plasticity: The effect of gestational ethanol exposure on the behavioral and neurophysiologic olfactory response to ethanol odor in early postnatal and adult rats. Behavioral Neuroscience, 121(6), 1293 1305.

Stork, F. C., & Widdowson, D. A. (1974). Learning about linguistics. Hutchinson Ltd.

Streissguth, A. P., Bookstein, F. L., Barr, H. M., Sampson, P. D., O Malley, K., & Young, J. K. (2004). Risk factors for adverse life outcomes in fetal alcohol syndrome and fetal alcohol effects. Developmental and Behavioral Pediatrics, 25(4), 228 238.

Striegel-Moore, R. H., & Cachelin, F. M. (1999). Body image concerns and disordered eating in adolescent girls: Risk and protective factors. In N. G. Johnson, M. C. Roberts, & J. Worell (Eds.), Beyond appearance: A new look at adolescent girls. American Psychological Association

Tanner, J. M. (1978). Fetus into man: Physical growth from conception to maturity. Harvard University Press.

Temel, J. S., Greer, J. A., Muzikansky, A., Gallagher, E. R., Admane, S., Jackson, V. A. . . . Lynch, T. J. (2010). Early palliative care for patients with metastic non-small-cell lung cancer. New England Journal of Medicine, 363, 733 742.

Thomas, A. (1984). Temperament research: Where we are, where we are going. Merrill-Palmer Quarterly, 30(2), 103 109.

Tran, T. D., & Kelly, S. J. (2003). Critical periods for ethanol-induced cell loss in the hippocampal formation. Neurotoxicology and Teratology, 25(5), 519 528.

Umberson, D., Pudrovska, T., & Reczek, C. (2010). Parenthood, childlessness, and well-being: A life course perspective. Journal of Marriage and the Family, 72(3), 612 629.

United Nations Educational, Scientific and Cultural Organization. (2013, June). UIS Fact Sheet: Schooling for millions of children jeopardized by reductions in aid. UNESCO Institute for Statistics.

Vaillant, G. E. (2002). Aging well. Little Brown & Co.

Van der Graaff, J., Branje, S., De Wied, M., Hawk, S., Van Lier, P., & Meeus, W. (2013). Perspective taking and empathetic concern in adolescence: Gender differences in developmental changes. Developmental Psychology, 50(3), 881.

van Ijzendoorn, M. H., & Sagi-Schwartz, A. (2008). Cross-cultural patterns of attachment: Universal and contextual dimensions. In J. Cassidy & P. R. Shaver (Eds.), Handbook of attachment. Guilford.

Vouloumanos, A., & Werker, J. F. (2004). Tuned to the signal: The privileged status of speech for young infants. Developmental Science, 7, 270 276.

Westermann, G., Mareschal, D., Johnson, M., Sirois, S., Spratling, M., & Thomas, M. (2007). Neuroconstructivism. Developmental Science, 10, 75 83.

WHO Multicentre Growth Reference Study Group. (2006). WHO Child growth standards: Methods and development: Length/height-for-age, weight-for-age, weight-for-length, weight-for-height and body mass index-for-age. World Health Organization.

Winerman, L. (2011). Closing the achievement gap. Monitor of Psychology, 42(8), 36.

Wortman, J. H., & Park, C. L. (2008). Religion and spirituality in adjustment following bereavement: An integrative review. Death Studies

Emotion and motivation

Adelmann, P. K., & Zajonc, R. B. (1989). Facial efference and the experience of emotion. Annual Review of Psychology, 40, 249 280. https://www.annualreviews.org/doi/abs/10.1146/annurev.ps.40.020189.00134
1

Ahima, R. S., & Antwi, D. A. (2008). Brain regulation of appetite and satiety. Endocrinology and Metabolism Clinics of North America, 37, 811 823.

Aldao, A., & Dixon-Gordon, K. L. (2014). Broadening the scope of research on emotion regulation strategies and psychopathology. Cognitive Behaviour Therapy, 43(1), 22 33. https://www.tandfonline.com/doi/abs/10.1080/16506073.2013.816769

Aldao, A., Nolen-Hoeksema, S., Schweizer, S. (2010). Emotion regulation strategies across psychopathology: A meta-analysis. Clinical Psychology Review, 30, 217 237. https://www.ncbi.nlm.nih.gov/pubmed/20015584

Allen, L. S., & Gorski, R. A. (1992). Sexual orientation and the size of the anterior commissure in the human brain. Proceedings of the National Academy of Sciences, USA, 89, 7199 7202.

American Psychiatric Association. (2013). Feeding and eating disorders. http://www.dsm5.org/documents/eating%20disorders%20fact%20sheet.pdf

Arnold, H. J. (1976). Effects of performance feedback and extrinsic reward upon high intrinsic motivation. Organizational Behavior and Human Performance, 17, 275 288.

Bailey, M. J., & Pillard, R. C. (1991). A genetic study of male sexual orientation. Archives of General Psychiatry, 48, 1089 1096.

Baldwin, J. D., & Baldwin, J. I. (1989). The socialization of homosexuality and heterosexuality in a non-western society. Archives of Sexual Behavior, 18, 13 29.

Bancroft, J. (2004). Alfred C. Kinsey and the politics of sex research. Annual Review of Sex Research, 15, 1 39.

Bandura, A. (1994). Self-efficacy. In V. S. Ramachandran (Ed.), Encyclopedia of human behavior (Vol. 4, pp. 71 81). Academic Press.

Barrett, L. F. (2017). The theory of constructed emotion: An active inference account of interoception and categorization. Social Cognitive and Affective Neuroscience, 12, 1 23. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5691871/

Bauminger, N. (2002). The facilitation of social-emotional understanding and social interaction in high-functioning children with autism: Intervention outcomes. Journal of Autism and Developmental Disorders, 32, 283 298.

Becker, J. B., Rudick, C. N., & Jenkins, W. J. (2001). The role of dopamine in the nucleus accumbens and striatum during sexual behavior in the female rat. Journal of Neuroscience, 21, 3236 3241.

Becker, J. M. (2012, April 25). Dr. Robert Spitzer apologizes to gay community for infamous ex-gay study. http://www.truthwinsout.org/news/2012/04/24542/

Beedie, C. J., Terry, P. C., Lane, A. M., & Devonport, T. J. (2011). Differential assessment of emotions and moods: Development and validation of the emotion and mood components of anxiety questionnaire. Personality and Individual Differences, 50, 228 233.

Bell, A. P., Weinberg, M. S., & Hammersmith, S. K. (1981). Sexual preferences: Its development in men and women. Indiana University Press.

Berlyne, D. E. (1960). Toward a theory of exploratory behavior: II. Arousal potential, perceptual curiosity, and learning. In (Series Ed.), Conflict, arousal, and curiosity (pp. 193 227). McGraw-Hill Book Company.

Bhasin, S., Enzlin, P., Coviello, A., & Basson, R. (2007). Sexual dysfunction in men and women with endocrine disorders. The Lancet, 369, 597 611.

Blackford, J. U., & Pine, D. S. (2012). Neural substrates of childhood anxiety disorders: A review of neuroimaging findings. Child and Adolescent Psychiatric Clinics of North America, 21, 501 525.

Boiger, M., & Mesquita, B. (2012). The construction of emotion in interactions, relationships, and cultures. Emotion Review, 4, 221 229. https://journals.sagepub.com/doi/10.1177/1754073912439765

Bremner, J. D., & Vermetten, E. (2004). Neuroanatomical changes associated with pharmacotherapy in posttraumatic stress disorder. Annals of the New York Academy of Sciences, 1032, 154 157.

Buck, R. (1980). Nonverbal behavior and the theory of emotion: The facial feedback hypothesis. Journal of Personality and Social Psychology, 38, 811 824.

Bullough, V. L. (1998). Alfred Kinsey and the Kinsey report: Historical overview and lasting contributions. The Journal of Sex Research, 35, 127 131.

Byne, W., Tobet, S., Mattiace, L. A., Lasco, M. S., Kemether, E., Edgar, M. A., . . . Jones, L. B. (2001). The interstitial nuclei of the human anterior hypothalamus: An investigation of variation with sex, sexual orientation, and HIV status. Hormones and Behavior, 40, 86 92.

Cameron, J., & Pierce, W. D. (1994). Reinforcement, reward, and intrinsic motivation: A meta-analysis. Review of Educational Research, 64, 363 423.

Capella, J. N. (1993). The facial feedback hypothesis in human interaction: Review and speculation. Journal of Language and Social Psychology, 12(12), 13 29. https://journals.sagepub.com/doi/10.1177/0261927X93121002

Carey, B. (2012, May 18). Psychiatry giant sorry for backing gay cure. The New York Times. http://www.nytimes.com/2012/05/19/health/dr-robert-l-spitzer-noted-psych iatrist-apologizes-for-study-on-gay-cure.html?_r=0

Carter, C. S. (1992). Hormonal influences on human sexual behavior. In J. B. Becker, S. M. Breedlove, & D. Crews (Eds.), Behavioral Endocrinology (pp.131 142). MIT Press.

Cassidy, S. B., & Driscoll, D. J. (2009). Prader-Willi syndrome. European Journal of Human Genetics, 17, 3 13.

Centers for Disease Control and Prevention. (2012). Overweight and obesity. http://www.cdc.gov/obesity/index.html

Colapinto, J. (2000). As nature made him: The boy who was raised as a girl. Harper Collins.

Collier, D. A., & Treasure, J. L. (2004). The aetiology of eating disorders. The British Journal of Psychiatry, 185, 363 365.

Conrad, P. (2005). The shifting engines of medicalization. Journal of Health and Social Behavior, 46, 3 14.

Cornelius, R. R. (1991). Gregorio Mara on's two-factor theory of emotion. Personality and Social Psychology Bulletin, 17(1), 65 69. https://psycnet.apa.org/doi/10.1177/0146167291171010

Creighton, S. (2001). Surgery for intersex. Journal of the Royal Society of Medicine, 94, 218 220. https://journals.sagepub.com/doi/10.1177/014107680109400505

Cunha, C., Monfils, M. H., & LeDoux, J. E. (2010). GABA(C) receptors in the lateral amygdala: A possible novel target for the treatment of fear and anxiety disorders? Frontiers in Behavioral Neuroscience, 4, 6.

Daniel, T. L., & Esser, J. K. (1980). Intrinsic motivation as influenced by rewards, task interest, and task structure. Journal of Applied Psychology, 65, 566 573.

Darwin, C. (1872). The expression of emotions in man and animals. Appleton.

Davis, J. I., Senghas, A., & Ochsner, K. N. (2009). How does facial feedback modulate emotional experience? Journal of Research in Personality, 43, 822 829.

Deci, E. L. (1972). Intrinsic motivation, extrinsic reinforcement, and inequity. Journal of Personality and Social Psychology, 22, 113 120.

Deci, E. L., Koestner, R., & Ryan, R. M. (1999). A meta-analytic review of experiments examining the effects of extrinsic rewards on intrinsic motivation. Psychological Bulletin, 125, 627 668.

de Gelder, B. (2006). Towards the neurobiology of emotional body language. Nature Reviews Neuroscience, 7, 242 249.

Drazen, D. L., & Woods, S. C. (2003). Peripheral signals in the control of satiety and hunger. Current Opinion in Clinical Nutrition and Metabolic Care, 5, 621 629.

Druce, M.R., Small, C.J., & Bloom, S.R. (2004). Minireview: Gut peptides regulating satiety. Endocrinology, 145, 2660 2665.

Ekman, P. (1972). Universals and cultural differences in facial expressions of emotions. In J. Cole (Ed.), Nebraska Symposium on Motivation (pp. 207 282). University of Nebraska Press. https://1ammce38pkj41n8xkp1iocwe-wpengine.netdna-ssl.com/wp-content/uploads/2013/07/Universals-And-Cultural-Differences-In-Facial-Expressions-Of.pdf

Ekman, P., & Keltner, D. (1997). Universal facial expressions of emotion: An old controversy and new findings. In U. Segerstr le & P. Moln r (Eds.), Nonverbal communication: Where nature meets culture (pp. 27 46). Lawrence Erlbaum.

Everett, B. J. (1990). Sexual motivation: A neural and behavioural analysis of the mechanisms underlying appetitive and copulatory responses of male rats. Neuroscience and Biobehavioral Reviews, 14, 217 232.

Faris, E. (1921). Are instincts data or hypotheses? American Journal of Sociology, 27, 184 196.

Femen a, T., G mez-Gal n, M., Lindskog, M., & Magara, S. (2012). Dysfunctional hippocampal activity affects emotion and cognition in mood disorders. Brain Research, 1476, 58 70.

Fossati, P. (2012). Neural correlates of emotion processing: From emotional to social brain. European Neuropsychopharmacology, 22, S487 S491.

Fournier, J. C., Keener, M. T., Almeida, J., Kronhaus, D. M., & Phillips, M. L. (2013). Amygdala and whole-brain activity to emotional faces distinguishes major depressive disorder and bipolar disorder. Bipolar Disorders. Advance online publication. doi:10.1111/bdi.12106

Francis, N. H., & Kritsonis, W. A. (2006). A brief analysis of Abraham Maslow s original writing of Self-Actualizing People: A Study of Psychological Health. Doctoral Forum National Journal of Publishing and Mentoring Doctoral Student Research, 3, 1 7.

Frijda, N. H. (1988). The laws of emotion. American Psychologist, 43, 349 358. https://psycnet.apa.org/record/1988-28577-001

Gloy, V. L., Briel, M., Bhatt, D. L., Kashyap, S. R., Schauer, P. R., Mingrone, G., . . . Nordmann, A. J. (2013, October 22). Bariatric surgery versus non-surgical treatment for obesity: A systematic review and meta-analysis of randomized controlled trials. BMJ, 347. doi:http://dx.doi.org/10.1136/bmj.f5934

Golan, O., & Baron-Cohen, S. (2006). Systemizing empathy: Teaching adults with Asperger syndrome or high-functioning autism to recognize complex emotions using interactive multimedia. Development and Psychopathology, 18, 591 617.

Goosens, K. A., & Maren, S. (2002). Long-term potentiation as a substrate for memory: Evidence from studies of amygdaloid plasticity and Pavlovian fear conditioning. Hippocampus, 12, 592 599.

Graham, S., & Weiner, B. (1996). Theories and principles of motivation. In D. C. Berliner & R. C. Calfee (Eds.), Handbook of educational psychology (pp. 63 84). Routledge.

Greary, N. (1990). Pancreatic glucagon signals postprandial satiety. Neuroscience and Biobehavioral Reviews, 14, 323 328.

Gross, J. J. (1999). Emotion regulation: Past, present, and future. Cognition & Emotion, 13, 551 573. https://www.tandfonline.com/doi/abs/10.1080/026999399379186

Guastella, A. J., Einfeld, S. L., Gray, K. M., Rinehart, N. J., Tonge, B. J., Lambert, T. J., & Hickie, I. B. (2010). Intranasal oxytocin improves emotion recognition for youth with autism spectrum disorders. Biological Psychiatry, 67, 692 694.

Hall, J. A., & Kimura, D. (1994). Dermatoglyphic asymmetry and sexual orientation in men. Behavioral Neuroscience, 108(6), 1203 1206.

Hamer, D. H., Hu. S., Magnuson, V. L., Hu, N., & Pattatucci, A. M. (1993). A linkage between DNA markers on the X chromosome and male sexual orientation. Science, 261, 321-327.

Havas, D. A., Glenberg, A. M., Gutowski, K. A., Lucarelli, M. J., & Davidson, R. J. (2010). Cosmetic use of botulinum toxin-A affects processing of emotional language. Psychological Science, 21, 895 900.

Hobson, R. P. (1986). The autistic child s appraisal of expressions of emotion. The Journal of Child Psychology and Psychiatry, 27, 321 342.

Hock, R. R. (2008). Emotion and Motivation. In Forty studies that changed psychology: Explorations into the history of psychological research (6th ed.) (pp. 158 168). Pearson.

Hopp, H., Troy, A. S., & Mauss, I. B. (2011). The unconscious pursuit of emotion regulation: Implications for psychological health. Cognition and Emotion, 25, 532 545. https://www.ocf.berkeley.edu/~eerlab/pdf/papers/2011_Hopp_unconscious_pu rsuit_of_emotion_regulation.pdf

Hu, S. H., Wei, N., Wang, Q. D., Yan, L. Q., Wei, E.Q., Zhang, M. M., . . . Xu, Y. (2008). Patterns of brain activation during visually evoked sexual arousal differ between homosexual and heterosexual men. American Journal of Neuroradiology, 29, 1890 1896.

Hughes, I. A., Houk, C., Ahmed, S. F., Lee, P. A., LWPES Consensus Group, & ESPE Consensus Group. (2006). Consensus statement on management of intersex disorders. Archives of Disease in Childhood, 91, 554 563. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2082839/

Human Rights Campaign. (n.d.). The lies and dangers of efforts to change sexual orientation or gender identity. http://www.hrc.org/resources/entry/the-lies-and-dangers-of-reparative-th erapy

Jenkins, W. J. (2010). Can anyone tell me why I m gay? What research suggests regarding the origins of sexual orientation. North American Journal of Psychology, 12, 279 296.

Jenkins, W. J., & Becker, J. B. (2001). Role of the striatum and nucleus accumbens in paced copulatory behavior in the female rat. Behavioural Brain Research, 121, 19 28.

Kennedy, K. (2017). Coverage in transition: Considerations when expanding employer-provided health coverage to LGBTI employees and beneficiaries. The John Marhsall Institutional Repository, 24(1). https://repository.jmls.edu/cgi/viewcontent.cgi?article=1691&context=fac pubs

Kinsey, A. C., Pomeroy, W. B., & Martin, C. E. (1948). Sexual behavior in the human male. W.B. Saunders Company.

Koltko-Rivera, M. E. (2006). Rediscovering the later version of Maslow s hierarchy of needs: Self-transcendence and opportunities for theory, research, and unification. Review of General Psychology, 10, 302 317.

Konturek, S. J., Pepera, J., Zabielski, K., Konturek, P. C., Pawlick, T., Szlachcic, A., & Hahn. (2003). Brain-gut axis in pancreatic secretion and appetite control. Journal of Physiology and Pharmacology, 54, 293 317.

Lang, P. J. (1994). The varieties of emotional experience: A meditation on James-Lange theory. Psychological Review, 101, 211 221.

Lazarus, R. S. (1991). Cognition and motivation in emotion. American Psychologist, 46, 352 367. https://psycnet.apa.org/record/1991-23402-001

Lazarus, R. S. (1991). Emotion and adaptation. Oxford University Press.

LeDoux, J. E. (1996). The Emotional Brain: The Mysterious Underpinnings of Emotional Life. Simon & Schuster.

LeDoux, J. E. (2002). The synaptic self. Macmillan.

Leonard, G. (1982). The failure of self-actualization theory: A critique of Carl Rogers and Abraham Maslow. Journal of Humanistic Psychology, 22, 56 73.

LeVay, S. (1991). A difference in the hypothalamic structure between heterosexual and homosexual men. Science, 253, 1034 1037.

LeVay, S. (1996). Queer science: The use and abuse of research into homosexuality. The MIT Press.

Levenson, R. W., Carstensen, L. L., Friesen, W. V., & Ekman, P. (1991). Emotion, physiology, and expression in old age. Psychology and Aging, 6, 28 35. https://psycnet.apa.org/record/1991-18170-001

Levy-Gigi, E., Szab , C., Kelemen, O., & K ri, S. (2013). Association among clinical response, hippocampal volume, and FKBP5 gene expression in individuals with posttraumatic stress disorder receiving cognitive behavioral therapy. Biological Psychiatry, 74, 793 800.

Lippa, R. A. (2003). Handedness, sexual orientation, and gender-related personality traits in men and women. Archives of Sexual Behavior, 32, 103 114.

Loehlin, J. C., & McFadden, D. (2003). Otoacoustic emissions, auditory evoked potentials, and traits related to sex and sexual orientation. Archives of Sexual Behavior, 32, 115 127.

Macdonald, H., Rutter, M., Howlin, P., Rios, P., Conteur, A. L., Evered, C., & Folstein, S. (1989). Recognition and expression of emotional cues by autistic and normal adults. Journal of Child Psychology and Psychiatry, 30, 865 877.

Malatesta, C. Z., & Haviland, J. M. (1982). Learning display rules: The socialization of emotion expression in infancy. Child Development, 53, 991 1003.

Maren, S., Phan, K. L., & Liberzon, I. (2013). The contextual brain: Implications for fear conditioning, extinction and psychopathology. Nature Reviews Neuroscience, 14, 417 428.

Martin-Gronert, M. S., & Ozanne, S. E. (2013). Early life programming of obesity. Developmental Period Medicine, 17, 7 12.

Maslow, A. H. (1943). A theory of human motivation. Psychological Review, 50, 370 396.

Matsumoto, D. (1990). Cultural similarities and differences in display rules. Motivation and Emotion, 14, 195 214.

Matsumoto, D., Yoo, S. H., & Nakagawa, S. (2008). Culture, emotion regulation, and adjustment. Journal of Personality and Social Psychology, 94, 925 937.

Mauss, I. B., & Robinson, M. D. (2009). Measures of emotion: A review. Cognition and Emotion, 23, 209 237. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2756702/

Mauss, I. B., Bunge, S. A., & Gross, J. J. (2007). Automatic emotion regulation. Social and Personality Psychology Compass, 1, 146 167. https://www.ocf.berkeley.edu/~eerlab/pdf/papers/2007_Mauss_Automatic%20E motion%20Regulation.pdf

Mauss, I. B., Cook, C. L., & Gross, J. J. (2007). Automatic emotion regulation during anger provocation. Journal of Experimental Social Psychology, 43, 698 711. https://psycnet.apa.org/record/2007-12302-004

Mauss, I. B., Cook, C. L., Cheng, J. Y. J., & Gross, J. J. (2007). Individual differences in cognitive reappraisal: Experiential and physiological responses to an anger provocation. International Journal of Psychophysiology, 66, 116 124. https://www.ncbi.nlm.nih.gov/pubmed/17543404

Mauss, I. B., Levenson, R. W., McCarter, L., Wilhelm, F. H., & Gross, J. J. (2005). The tie that binds? Coherence among emotion experience, behavior, and physiology. Emotion, 5, 175 190. https://www.ncbi.nlm.nih.gov/pubmed/15982083

Mayo Clinic. (2012a). Anorexia nervosa. http://www.mayoclinic.com/health/anorexia/DS00606

Mayo Clinic. (2012b). Bulimia nervosa. http://www.mayoclinic.com/health/bulimia/DS00607

Mayo Clinic. (2013). Gastric bypass surgery. http://www.mayoclinic.com/health/gastric-bypass/MY00825

McAdams, D. P., & Constantian, C. A. (1983). Intimacy and affiliation motives in daily living: An experience sampling analysis. Journal of Personality and Social Psychology, 45, 851 861.

McClelland, D. C., & Liberman, A. M. (1949). The effect of need for achievement on recognition of need-related words. Journal of Personality, 18, 236 251.

McFadden, D., & Champlin, C. A. (2000). Comparisons of auditory evoked potentials in heterosexual, homosexual, and bisexual males and females. Journal of the Association for Research in Otolaryngology, 1, 89 99.

McFadden, D., & Pasanen, E. G. (1998). Comparisons of the auditory systems of heterosexuals and homosexuals: Clicked-evoked otoacoustic emissions. Proceedings of the National Academy of Sciences, USA, 95, 2709 2713.

McRae, K., Ochsner, K. N., Mauss, I. B., Gabrieli, J. J. D., & Gross, J. J. (2008). Gender differences in emotion regulation: An fMRI study of cognitive reappraisal. Group Processes and Intergroup Relations, 11, 143 162.

Miguel-Hidalgo, J. J. (2013). Brain structural and functional changes in adolescents with psychiatric disorders. International Journal of Adolescent Medicine and Health, 25, 245 256.

Money, J. (1962). Cytogenic and psychosexual incongruities with a note on space-form blindness. Paper presented at the 118th meeting of the American Psychiatric Association, Toronto, Canada.

Money, J. (1975). Ablatio penis: Normal male infant sex-reassigned as a girl. Archives of Sexual Behavior, 4, 65 71.

Moriceau, S., & Sullivan, R. M. (2006). Maternal presence serves as a switch between learning fear and attraction in infancy. Nature Neuroscience, 9, 1004 1006.

Murray, H. A., Barrett, W. G., Homburger, E., Langer, W. C., Mekeel, H. S., Morgan, C. D., . . . Wolf, R. E. (1938). Explorations in personality: A clinical and experimental study of fifty men of college age. Oxford University Press.

Niemiec, C. P., & Ryan, R. M. (2009). Autonomy, competence, and relatedness in the classroom: Applying self-determination theory to educational practice. Theory and Research in Education, 7, 133 144.

Nisbett, R. E., Peng, K., Choi, I., & Norenzayan, A. (2001). Culture and systems of thought: Holistic versus analytic cognition. Psychological Review, 108(2), 291 310. https://psycnet.apa.org/record/2001-17194-001

Novin, D., Robinson, K., Culbreth, L. A., & Tordoff, M. G. (1985). Is there a role for the liver in the control of food intake? The American Journal of Clinical Nutrition, 42, 1050 1062.

O Connell, S. (Writer/Producer). (2004). Dr. Money and the boy with no penis. In Horizon. BBC.

Paramaguru, K. (2013, November). Boy, girl, or intersex? Germany adjusts to a third option at birth. Time. http://world.time.com/2013/11/12/boy-girl-or-intersex/

Pedersen, E. R. (2016). Using the solid research base on pregaming to begin intervention development: An epilogue to the special issue on pregaming. Substance Use & Misuse, 51(8), 1067 1073. https://www.tandfonline.com/doi/abs/10.1080/10826084.2016.1187533

Pessoa, L. (2010). Emotion and cognition and the amygdala: From what is it? to what s to be done? Neuropsychologia, 48, 3416 3429.

Pillard, R. C., & Bailey, M. J. (1995). A biologic perspective on sexual orientation. The Psychiatric Clinics of North America, 18(1), 71 84.

Pillard, R. C., & Bailey, M. J. (1998). Sexual orientation has a heritable component. Human Biology, 70, 347 365.

Ponseti, J., Bosinski, H. A., Wolff, S., Peller, M., Jansen, O., Mehdorn, H.M., . . . Siebner, H. R. (2006). A functional endophenotype for sexual orientation in humans. Neuroimage, 33(3), 825 833.

Prader-Willi Syndrome Association. (2012). What is Prader-Willi Syndrome? http://www.pwsausa.org/syndrome/index.htm

Qin, S., Young, C. B., Duan, X., Chen, T., Supekar, K., & Menon, V. (2013). Amygdala subregional structure and intrinsic functional connectivity predicts individual differences in anxiety during early childhood. Biological Psychiatry. Advance online publication. doi:10.1016/j.biopsych.2013.10.006

Rahman, Q., & Wilson, G. D. (2003a). Large sexual-orientation-related differences in performance on mental rotation and judgment of line orientation tasks. Neuropsychology, 17, 25 31.

Rahman, Q., & Wilson, G. D. (2003b). Sexual orientation and the 2nd to 4th finger length ratio: Evidence for organising effects of sex hormones or developmental instability? Psychoneuroendocrinology, 28, 288 303.

Raineki, C., Cort s, M. R., Belnoue, L., & Sullivan, R. M. (2012). Effects of early-life abuse differ across development: Infant social behavior deficits are followed by adolescent depressive-like behaviors mediated by the amygdala. The Journal of Neuroscience, 32, 7758 7765.

Rodriguez-Larralde, A., & Paradisi, I. (2009). Influence of genetic factors on human sexual orientation. Investigacion Clinica, 50, 377 391.

Roseman, I. J., and Smith, C. A. (2001). Appraisal theory: Overview, assumptions, varieties, controversies. In K. R. Scherer, A. Schorr, & T. Johnstone (Eds.), Appraisal processes in emotion: Theory, methods, research, (pp. 3 19). Oxford University Press.

Ross, M. W., & Arrindell, W. A. (1988). Perceived parental rearing patterns of homosexual and heterosexual men. The Journal of Sex Research, 24, 275 281.

Saxe, L., & Ben-Shakhar, G. (1999). Admissibility of polygraph tests: The application of scientific standards post-Daubert. Psychology, Public Policy, and Law, 5, 203 223.

Schachter, S., & Singer, J. E. (1962). Cognitive, social, and physiological determinants of emotional state. Psychological Review, 69, 379 399.

Shallcross, A. J., Troy, A. S., Boland, M., & Mauss, I. B. (2010). Let it be: Accepting negative emotional experiences predicts decreased negative affect and depressive symptoms. Behaviour Research and Therapy, 48, 921 929. https://www.ncbi.nlm.nih.gov/pubmed/20566191

Sherwin, B. B. (1988). A comparative analysis of the role of androgen in human male and female sexual behavior: Behavioral specificity, critical thresholds, and sensitivity. Psychobiology, 16, 416 425.

Smink, F. R. E., van Hoeken, D., & Hoek, H. W. (2012). Epidemiology of eating disorders: Incidence, prevalence, and mortality rates. Current Psychiatry Reports, 14, 406 414.

Soussignan, R. (2002). Duchenne smile, emotional experience, and autonomic reactivity: A test of the facial feedback hypothesis. Emotion, 2, 52 74. https://psycnet.apa.org/record/2002-18343-004

Speakman, J. R., Levitsky, D. A., Allison, D. B., Bray, M. S., de Castro, J. M., Clegg, D. J., . . . Westerterp-Plantenga, M. S. (2011). Set points, settling points and some alternative models: Theoretical options to understand how genes and environment combine to regulate body adiposity. Disease Models & Mechanisms, 4, 733 745.

Swaab, D. F., & Hofman, M. A. (1990). An enlarged suprachiasmatic nucleus in homosexual men. Brain Research, 537, 141 148.

Tamietto, M., Castelli, L., Vighetti, S., Perozzo, P., Geminiani, G., Weiskrantz, L., & de Gelder, B. (2009). Unseen facial and bodily expressions trigger fast emotional reactions. Proceedings of the National Academy of Sciences,USA, 106, 17661 17666.

Tangmunkongvorakul, A., Banwell, C., Carmichael, G., Utomo, I. D., & Sleigh, A. (2010). Sexual identities and lifestyles among non-heterosexual urban Chiang Mai youth: Implications for health. Culture, Health, and Sexuality, 12, 827 841.

Troy, A. S., Shallcross, A. J., & Mauss, I. B. (2013). A person by situation approach to emotion regulation: Cognitive reappraisal can either hurt or help, depending on the context. Psychological Science, 12, 2505 2514. https://www.ncbi.nlm.nih.gov/pubmed/24145331

Troy, A. S., Wilhelm, F. H., Shallcross, A. J., & Mauss, I. B. (2010). Seeing the silver lining: Cognitive reappraisal ability moderates the relationship between stress and depression. Emotion, 10, 783 795. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3278301/

Wang, Z., Neylan, T. C., Mueller, S. G., Lenoci, M., Truran, D., Marmar, C. R., . . . Schuff, N. (2010). Magnetic resonance imaging of hippocampal subfields in posttraumatic stress disorder. Arch Gen Psychiatry, 67(3), 296 303. doi:10.1001/archgenpsychiatry.2009.205

Weinsier, R. L., Nagy, T. R., Hunter, G. R., Darnell, B. E., Hensrud, D. D., & Weiss, H. L. (2000). Do adaptive changes in metabolic rate favor weight regain in weight-reduced individuals? An examination of the set-point theory. The American Journal of Clinical Nutrition, 72, 1088 1094.

Woods, S. C. (2004). Gastrointestinal satiety signals I. An overview of gastrointestinal signals that influence food intake. American Journal of Physiology: Gastrointestinal and Liver Physiology, 286, G7 G13.

Woods, S. C., & D Alessio, D. A. (2008). Central control of body weight and appetite. Journal of Clinical Endocrinology and Metabolism, 93, S37 S50.

Yerkes, R. M., & Dodson, J. D. (1908). The relation of strength of stimulus to rapidity of habit-formation. Journal of Comparative Neurology and Psychology, 18, 459 482. doi:10.1002/cne.920180503

Zajonc, R. B. (1980). Feeling and thinking: Preferences need no inferences. American Psychologist, 35(2), 151 175.

Zajonc, R. B. (1998). Emotions. In D. T. Gilbert & S. T. Fiske (Eds.), Handbook of social psychology (4th ed., Vol. 1, pp. 591 632). McGraw-Hill.

Personality

Adler, A. (1930). Individual psychology. In C. Murchison (Ed.), Psychologies of 1930 (pp. 395 405). Clark University Press.

Adler, A. (1937). A school girl's exaggeration of her own importance. International Journal of Individual Psychology, 3(1), 3 12.

Adler, A. (1956). The individual psychology of Alfred Adler: A systematic presentation in selections from his writings. (C. H. Ansbacher & R. Ansbacher, Eds.). Harper.

Adler, A. (1961). The practice and theory of individual psychology. In T. Shipley (Ed.), Classics in psychology (pp. 687 714). Philosophical Library

Adler, A. (1964). Superiority and social interest. Norton.

Akomolafe, M. J. (2013). Personality characteristics as predictors of academic performance of secondary school students. Mediterranean Journal of Social Sciences, 4(2), 657 664.

Allport, G. W. & Odbert, H. S. (1936). Trait-names: A psycho-lexical study. Psychological Review Company.

Anglim, J., & O Connor, P. (2018). Measurement and research using Big Five, HEXACO, and narrow traits: A primer for researchers and practitioners. Australian Journal of Psychology, 71(1), 16 25. https://aps.onlinelibrary.wiley.com/doi/full/10.1111/ajpy.12202

Aronow, E., Weiss, K. A., & Rezinkoff, M. (2001). A practical guide to the Thematic Apperception Test. Brunner Routledge.

Bandura, A. (1977). Self-efficacy: Toward a unifying theory of behavioral change. Psychological Review, 84, 191 215.

Bandura, A. (1986). Social foundations of thought and action: A social cognitive theory. Prentice Hall.

Bandura, A. (1995). Self-efficacy in changing societies. Cambridge University Press.

Benassi, V. A., Sweeney, P. D., & Dufour, C. L. (1988). Is there a relation between locus of control orientation and depression? Journal of Abnormal Psychology, 97(3), 357.

Ben-Porath, Y., & Tellegen, A. (2008). Minnesota Multiphasic Personality Inventory-2-RF. University of Minnesota Press.

Benet-Mart nez, V. & Karakitapoglu-Aygun, Z. (2003). The interplay of cultural values and personality in predicting life-satisfaction: Comparing Asian- and European-Americans. Journal of Cross-Cultural Psychology, 34, 38 61.

Benet-Mart nez, V., & Oishi, S. (2008). Culture and personality. In O. P. John, R.W. Robins, L. A. Pervin (Eds.), Handbook of personality: Theory and research. Guildford Press.

Beutler, L. E., Nussbaum, P. D., & Meredith, K. E. (1988). Changing personality patterns of police officers. Professional Psychology: Research and Practice, 19(5), 503 507.

Bouchard, T., Jr. (1994). Genes, environment, and personality. Science, 264, 1700 1701.

Bouchard, T., Jr., Lykken, D. T., McGue, M., Segal, N. L., & Tellegen, A. (1990). Sources of human psychological differences: The Minnesota Study of Twins Reared Apart. Science, 250, 223 228.

Burger, J. (2008). Personality (7th ed.). Thompson Higher Education.

Buss, D. M. (2009). How can evolutionary psychology successfully explain personality and individual differences? Perspectives on Psychological Science, 4(4), 359 366. https://labs.la.utexas.edu/buss/files/2015/09/evolution-personality-and-individual-differences-2009.pdf

Carter, S., Champagne, F., Coates, S., Nercessian, E., Pfaff, D., Schecter, D., & Stern, N. B. (2008). Development of temperament symposium. Philoctetes Center, New York.

Cattell, R. B. (1946). The description and measurement of personality. Harcourt, Brace, & World.

Cattell, R. B. (1957). Personality and motivation structure and measurement. World Book.

Chamorro-Premuzic, T., & Furnham, A. (2008). Personality, intelligence, and approaches to learning as predictors of academic performance. Personality and Individual Differences, 44, 1596 1603.

Cheung, F. M., van de Vijver, F. J. R., & Leong, F. T. L. (2011). Toward a new approach to the study of personality in culture. American Psychologist, 66(7), 593 603.

Clark, A. L., & Watson, D. (2008). Temperament: An organizing paradigm for trait psychology. In O. P. John, R. W. Robins, & L. A. Previn (Eds.), Handbook of personality: Theory and research (3rd ed., pp. 265 286). Guilford Press.

Conrad, N. & Party, M.W. (2012). Conscientiousness and academic performance: A Mediational Analysis. International Journal for the Scholarship of Teaching and Learning, 6 (1), 1 14.

Costantino, G. (1982). TEMAS: A new technique for personality research assessment of Hispanic children. Hispanic Research Center, Fordham University Research Bulletin, 5, 3 7.

Cramer, P. (2004). Storytelling, narrative, and the Thematic Apperception Test. Guilford Press.

Dana, R. H. (1986). Personality assessment and Native Americans. Journal of Personality Assessment, 50(3), 480 500.

Donnellan, M. B., & Lucas, R. E. (2008). Age differences in the big five across the life span: Evidence from two national samples. Psychology and Aging, 23(3), 558 566.

Duzant, R. (2005). Differences of emotional tone and story length of African American respondents when administered the Contemporized Themes Concerning Blacks test versus the Thematic Apperception Test. Unpublished doctoral dissertation, The Chicago School of Professional Psychology, Chicago, IL.

Exner, J. E. (2002). The Rorschach: Basic foundations and principles of interpretation (Vol. 1). Wiley.

Eysenck, H. J. (1990). An improvement on personality inventory. Current Contents: Social and Behavioral Sciences, 22(18), 20.

Eysenck, H. J. (1992). Four ways five factors are not basic. Personality and Individual Differences, 13, 667 673.

Eysenck, H. J. (2009). The biological basis of personality (3rd ed.). Transaction Publishers.

Eysenck, S. B. G., & Eysenck, H. J. (1963). The validity of questionnaire and rating assessments of extroversion and neuroticism, and their factorial stability. British Journal of Psychology, 54, 51 62.

Eysenck, H. J., & Eysenck, M. W. (1985). Personality and individual differences: A natural science approach. Plenum Press.

Eysenck, S. B. G., Eysenck, H. J., & Barrett, P. (1985). A revised version of the psychoticism scale. Personality and Individual Differences, 6(1), 21 29.

Fazeli, S. H. (2012). The exploring nature of the assessment instrument of five factors of personality traits in the current studies of personality. Asian Social Science, 8(2), 264 275.

Fancher, R. W. (1979). Pioneers of psychology. Norton.

Freud, S. (1920). Resistance and suppression. A general introduction to psychoanalysis (pp. 248 261). Horace Liveright.

Freud, S. (1923/1949). The ego and the id. Hogarth.

Freud, S. (1931/1968). Female sexuality. In J. Strachey (Ed. &Trans.), The standard edition of the complete psychological works of Sigmund Freud (Vol. 21). Hogarth Press.

Frewer, L., & Bleus A. V. (1991). Personality assessment in a collectivist culture. South Pacific Journal of Psychology, 4, 1 5.

Funder, D. C. (2001). Personality. Annual Review of Psychology, 52, 197 221.

Goldstein, M. L. Projective personality assessment in child custody evaluations. n.d. https://www.drmarkgoldsteinphd.com/projective-personality-assessment-in-child-custody-evaluations

Hofstede, G. (2001). Culture s consequences: Comparing values, behaviors, institutions, and organizations across nations (2nd ed.). Sage.

Holaday, D., Smith, D. A., & Sherry, Alissa. (2010). Sentence completion tests: A review of the literature and results of a survey of members of the society for personality assessment. Journal of Personality Assessment, 74(3), 371 383.

Hothersall, D. (1995). History of psychology. McGraw-Hill.

Hoy, M. (1997). Contemporizing of the Themes Concerning Blacks test (C-TCB). California School of Professional Psychology.

Hoy-Watkins, M., & Jenkins-Moore, V. (2008). The Contemporized-Themes Concerning Blacks Test (C-TCB). In S. R. Jenkins (Ed.), A Handbook of Clinical Scoring Systems for Thematic Apperceptive Techniques (pp. 659 698). Lawrence Erlbaum Associates.

Jang, K. L., Livesley, W. J., & Vernon, P. A. (1996). Heritability of the big five personality dimensions and their facts: A twin study. Journal of Personality, 64(3), 577 591.

Jang, K. L., Livesley, W. J., Ando, J., Yamagata, S., Suzuki, A., Angleitner, A., et al. (2006). Behavioral genetics of the higher-order factors of the Big Five. Personality and Individual Differences, 41, 261 272.

Judge, T. A., Livingston, B. A., & Hurst, C. (2012). Do nice guys-and gals- really finish last? The joint effects of sex and agreeableness on income. Journal of Personality and Social Psychology, 102(2), 390 407.

Jung, C. G. (1923). Psychological types. Harcourt Brace.

Jung, C. G. (1928). Contributions to analytical psychology. Harcourt Brace Jovanovich.

Jung, C. G. (1964). Man and his symbols. Doubleday and Company.

Jung, C., & Kerenyi, C. (1963). Science of mythology. In R. F. C. Hull (Ed. & Trans.), Essays on the myth of the divine child and the mysteries of Eleusis. Harper & Row.

Launer, J. (2005). Anna O. and the talking cure. QJM: An International Journal of Medicine, 98(6), 465 466.

Lecci, L. B. & Magnavita, J. J. (2013). Personality theories: A scientific approach. Bridgepoint Education.

Lefcourt, H. M. (1982). Locus of control: Current trends in theory and research (2nd ed.). Erlbaum.

Likert, R. (1932). A technique for the measurement of attitudes. Archives of Psychology, 140, 1 55.

Lilienfeld, S. O., Wood, J. M., & Garb, H. N. (2000). The scientific status of projective techniques. Psychological Science in the Public Interest, 1(2), 27 66.

Maltby, J., Day, L., & Macaskill, A. (2007). Personality, individual differences and intelligence (3rd ed.). Pearson.

Maslow, A. H. (1970). Motivation and personality. Harper & Row.

Maslow, A. H. (1950). Self-actualizing people: A study of psychological health. In W. Wolff (Ed.), Personality Symposia: Symposium 1 on Values (pp. 11 34). Grune & Stratton.

McCrae, R. R., & Costa, P. T. (1997). Personality trait structure as a human universal. American Psychologist, 52(5), 509 516.

McCrae, R. R., et al. (2005). Universal features of personality traits from the observer s perspective: Data from 50 cultures. Journal of Personality and Social Psychology, 88, 547 561.

Mischel, W. (1993). Introduction to personality (5th ed.). Harcourt Brace Jovanovich.

Mischel, W., Ayduk, O., Berman, M. G., Casey, B. J., Gotlib, I. H., Jonides, J., et al. (2010). Willpower over the life span: Decomposing self-regulation. Social Cognitive and Affective Neuroscience, 6(2), 252 256.

Mischel, W., Ebbesen, E. B., & Raskoff Zeiss, A. (1972). Cognitive and attentional mechanisms of delay in gratification. Journal of Personality and Social Psychology, 21(2), 204 218.

Mischel, W., & Shoda, Y. (1995). A cognitive-affective system theory of personality: Reconceptualizing situations, dispositions, dynamics, and invariance in personality structure. Psychological Review, 102(2), 246 268.

Mischel, W., Shoda, Y., & Rodriguez, M. L. (1989, May 26). Delay of gratification in children. Science, 244, 933-938.

Motley, M. T. (2002). Theory of slips. In E. Erwin (Ed.), The Freud encyclopedia: Theory, therapy, and culture (pp. 530 534). Routledge.

Noftle, E. E., & Robins, R. W. (2007). Personality predictors of academic outcomes: Big Five correlates of GPA and SAT scores. Personality Processes and Individual Differences, 93, 116 130.

Noga, A. (2007). Passions and tempers: A history of the humors. Harper Collins.

Oyserman, D., Coon, H., & Kemmelmier, M. (2002). Rethinking individualism and collectivism: Evaluation of theoretical assumptions and meta-analyses. Psychological Bulletin, 128, 3 72.

Piotrowski, Z. A. (1987). Perceptanalysis: The Rorschach method fundamentally reworked, expanded and systematized. Routledge.

Rentfrow, P. J., Gosling, S. D., Jokela, M., Stillwell, D. J., Kosinski, M., & Potter, J. (2013, October 14). Divided we stand: Three psychological regions of the United States and their political, economic, social, and health correlates. Journal of Personality and Social Psychology, 105(6), 996 1012.

Rentfrow, P. J., Jost, J. T., Gosling, S. D., & Potter, J. (2009). Statewide differences in personality predict voting patterns in 1996 2004 U.S. Presidential elections. In J. T. Jost, A. C. Kay, & H. Thorisdottir (Eds.), Series in political psychology: Social and psychological bases of ideology and system justification (pp. 314 347). Oxford University

Press.

Roesler, C. (2012). Are archetypes transmitted more by culture than biology? Questions arising from conceptualizations of the archetype. Journal of Analytical Psychology, 57(2), 223 246.

Rogers, C. (1980). A way of being. Houghton Mifflin.

Rothbart, M. K. (2011). Becoming who we are: Temperament and personality in development. Guilford Press.

Rothbart, M. K., Ahadi, S. A., & Evans, D. E. (2000). Temperament and personality: Origins and outcomes. Journal of Personality and Social Psychology, 78(1), 122 135.

Rothbart, M. K., & Derryberry, D. (1981). Development of individual differences in temperament. In M. E. Lamb & A. L. Brown (Eds.), Advances in developmental psychology (Vol. 1, pp. 37 86). Erlbaum.

Rothbart, M. K., Sheese, B. E., Rueda, M. R., & Posner, M. I. (2011). Developing mechanisms of self-regulation in early life. Emotion Review, 3(2), 207 213.

Rotter, J. (1966). Generalized expectancies for internal versus external control of reinforcements. Psychological Monographs, 80, 609.

Rotter, J. B., & Rafferty, J. E. (1950). Manual the Rotter Incomplete Sentences Blank College Form. The Psychological Corporation.

Schmitt, D. P., Allik, J., McCrae, R. R., & Benet-Martinez, V. (2007). The geographic distribution of Big Five personality traits: Patterns and profiles of human self-description across 56 nations. Journal of Cross-Cultural Psychology, 38, 173 212.

Scott, J. (2005). Electra after Freud: Myth and culture. Cornell University Press.

Segal, N. L. (2012). Born together-reared apart: The landmark Minnesota Twin Study. Harvard University Press.

Skinner, B. F. (1953). Science and human behavior. The Free Press.

Sotirova-Kohli, M., Opwis, K., Roesler, C., Smith, S. M., Rosen, D. H., Vaid, J., & Djnov, V. (2013). Symbol/meaning paired-associate recall: An archetypal memory advantage? Behavioral Sciences, 3, 541 561. http://www2.cnr.edu/home/araia/Myth_%20Body.pdf

Stelmack, R. M., & Stalikas, A. (1991). Galen and the humour theory of temperament. Personal Individual Difference, 12(3), 255 263.

Terracciano A., McCrae R. R., Brant L. J., Costa P. T., Jr. (2005). Hierarchical linear modeling analyses of the NEO-PI-R scales in the Baltimore Longitudinal Study of Aging. Psychology and Aging, 20, 493 506.

Thomas, A., & Chess, S. (1977). Temperament and development. Brunner/Mazel.

Tok, S. (2011). The big five personality traits and risky sport participation. Social Behavior and Personality: An International Journal, 39(8), 1105 1111.

Triandis, H. C. (1995). Individualism and collectivism. Westview.

Triandis, H. C., & Suh, E. M. (2002). Cultural influences on personality. Annual Review of Psychology, 53, 133 160.

Wagerman, S. A., & Funder, D. C. (2007). Acquaintance reports of personality and academic achievement: A case for conscientiousness. Journal of Research in Personality, 41, 221 229.

Watson, D., & Clark, L. A. (1984). Negative affectivity: The disposition to experience aversive emotional states. Psychological Bulletin, 96, 465 490.

Watts, T. W., Duncan, G. J., & Quan, H. (2018). Revisiting the marshmallow test: A conceptual replication investigating links between early delay of gratification and later outcomes. Psychological Science 29(7), 1159 1177. https://doi.org/10.1177/0956797618761661

Weiner, I. B. (2003). Principles of Rorschach interpretation. Lawrence Erlbaum.

Whyte, C. (1980). An integrated counseling and learning center. In K. V. Lauridsen (Ed.), Examining the scope of learning centers (pp. 33 43). Jossey-Bass.

Whyte, C. (1978). Effective counseling methods for high-risk college freshmen. Measurement and Evaluation in Guidance, 6(4), 198 200.

Whyte, C. B. (1977). High-risk college freshman and locus of control. The Humanist Educator, 16(1), 2 5.

Williams, R. L. (1972). Themes Concerning Blacks: Manual. Williams.

Wundt, W. (1874/1886). Elements du psychologie, physiologique (2ieme tome). [Elements of physiological psychology, Vol. 2]. (E. Rouvier, Trans.). Paris: Ancienne Librairie Germer Bailliere et Cie.

Yang, K. S. (2006). Indigenous personality research: The Chinese case. In U. Kim, K.-S. Yang, & K.-K. Hwang (Eds.), Indigenous and cultural psychology: Understanding people in context (pp. 285 314). Springer.

Young-Eisendrath, P. (1995). Myth and body: Pandora s legacy in a post-modern world. http://www2.cnr.edu/home/araia/Myth_%20Body.pdf

Social psychology

Adams, H. E., Wright, L. W., Jr., & Lohr, B.A. (1996). Is homophobia associated with homosexual arousal? Journal of Abnormal Psychology, 105, 440 445.

Albarrac n, D., & Wyer, R. S. (2001). Elaborative and nonelaborative processing of a behavior-related communication. Personality and Social Psychology Bulletin, 27, 691 705.

Allport, G. W. (1954). The Nature of Human Prejudice. Addison-Wesley.

American Psychological Association. (n.d.) Social psychology studies human interactions. https://www.apa.org/action/science/social/

American Psychological Association. (2010). Bullying: What parents, teachers can do to stop it. http://www.apa.org/news/press/releases/2010/04/bullying.aspx.

Aronson, E., & Mills, J. (1959). The effect of severity of initiation on liking for a group, Journal of Abnormal and Social Psychology, 59, 177 181.

Asch, S. E. (1955). Opinions and social pressure. Scientific American, 193, 31 35.

Bandura, A. (1999). Moral disengagement in the perpetration of inhumanities. Personality and Social Psychology Review, 3(3), 193 209. doi:10.1207/s15327957pspr0303_3.

Bartels, J. (2019). Revisiting the Stanford prison experiment, again: Examining demand characteristics in the guard orientation. The Journal of Social Psychology, 159(6), 780 790.

Bartels, J. M., Milovich, M. M., & Moussier, S. (2016). Coverage of the Stanford prison experiment in introductory psychology courses: A survey of introductory psychology instructors. Teaching of Psychology, 43(2), 136 141.

Batson, C. D. (1991). The altruism question: Toward a social-psychological answer. Erlbaum.

Berkowitz, A. D. (2004). The social norms approach: Theory, research and annotated bibliography. http://www.alanberkowitz.com/articles/social_norms.pdf.

Berkowitz, L. (1993). Aggression: Its causes, consequences, and control. McGraw-Hill.

Betz, N. E. (2008). Women s career development. In F. Denmark & M. Paludi (Eds.), Psychology of women: Handbook of issues and theories (2nd ed., pp. 717 752). Praeger.

Blau, F. D., Ferber, M. A., & Winkler, A. E. (2010). The economics of women, men, and work (6th ed.). Prentice Hall.

Bond, C. F., & Titus, L. J. (1983). Social facilitation: A meta-analysis of 241 studies. Psychological Bulletin, 94, 265 292.

Bond, R., & Smith, P. B. (1996). Culture and conformity: A meta-analysis of studies using Asch s (1952b, 1956) line judgment task. Psychological Bulletin, 119(1), 111 137.

Bowen, L. (2011). Bullying may contribute to lower test scores. Monitor on Psychology, 42(9), 19.

Brown, R. (2010). Prejudice: Its social psychology (2nd ed.). Wiley-Blackwell.

Burger, J. (2009). Replicating Milgram: Would people still obey today? American Psychologist, 64, 1 11. https://www.ncbi.nlm.nih.gov/pubmed/19209958

Buss, D. M. (2004). Evolutionary psychology: The new science of the mind (2nd ed.). Allyn and Bacon.

Buss, D. M. (1989). Sex differences in human mate preferences: Evolutionary hypotheses tested in 37 cultures. Behavioral and Brain Sciences, 12, 1 49.

Carnahan, T., & McFarland, S. (2007). Revisiting the Stanford Prison Experiment: Could participant self-selection have led to the cruelty? Personality and Social Psychology Bulletin, 33(5), 603 614.

Ceci, S. J., & Williams, W. M. (2011). Understanding current causes of women s underrepresentation in science. Proceedings of the National Academy of Sciences, 108, 3157 3162.

Chang, E., Kannoth, S., Levy, S., Wang, S., Lee, J., & Levy, B. (2020). Global reach of ageism on older persons' health: A systematic review. PLoS ONE, 15, 1 24.

Choi, I., & Nisbett R. E. (1998). Situational salience and cultural differences in the correspondence bias and actor-observer bias. Personality and Social Psychology Bulletin, 24(9), 949 960. doi:10.1177/0146167298249003.

Cialdini, R. B. (2001). Harnessing the science of persuasion. Harvard Business Review, 79, 72 81.

Cialdini, R. B., Brown, S. L., Lewis, B. P., Luce, C., & Neuberg, S. L. (1997). Reinterpreting the empathy-altruism relationship: When one into one equals oneness. Journal of Personality and Social Psychology, 73, 481 494.

Colin Powell regrets Iraq war intelligence. (2011). http://www.aljazeera.com/news/americas/2011/09/20119116916873488.html.

Cozby, P. C. (1973). Self-disclosure: A literature review. Psychological Bulletin, 79, 73 91.

Crisp, R. J., & Turner, R. N. (2009). Can imagined interactions produce positive perceptions? Reducing prejudice through simulated social contact. American Psychologist, 64, 231 240.

Crowley, A. E., & Hoyer, W. D. (1994). An integrative framework for understanding two-sided persuasion. Journal of Consumer Research, 20(4), 561 574.

Croyle, R. T., & Cooper, J. (1983). Dissonance arousal: Physiological evidence. Journal of Personality and Social Psychology, 45, 782 791.

Cuddy, A. J., Norton, M. I., & Fiske, S. T. (2005). This old stereotype: The pervasiveness and persistence of the elderly stereotype. Journal of Social Issues, 61, 267 285.

Deutsch, M., & Gerard, H. (1955). A study of normative and informational social influences upon individual judgment. Journal of Abnormal and Social Psychology, 51, 629 636.

Devine, P. G. (1989). Stereotypes and prejudice: Their automatic and controlled components. Journal of Personality and Social Psychology, 56, 5 18.

Devine, P. G., & Elliot, A. J. (1995). Are racial stereotypes really fading? The Princeton trilogy revisited. Personality and Social Psychology Bulletin, 21, 1139 1150.

Dixon, T. L., & Linz D. (2000). Overrepresentation and underrepresentation of African Americans and Latinos as lawbreakers on television news. Journal of Communication, 50(2), 131 154.

Dodge, K. A., & Schwartz, D. (1997). Social information processing mechanisms in aggressive behavior. In D. M. Stoff and J. Breiling (Eds.), Handbook of Antisocial Behavior (pp. 171 180). John Wiley and Sons.

Doli ski, D., Grzyb, T., Folwarczny, M., Grzyba a, P., Krzyszycha, K., Martynowska, K., & Trojanowski, J. (2017). Would you deliver an electric shock in 2015? Obedience in the experimental paradigm developed by Stanley Milgram in the 50 years following the original studies. Social Psychological and Personality Science, 8(8), 927 933.

Dollard, J., Miller, N. E., Doob, L. W., Mowrer, O. H., & Sears, R. R. (1939). Frustration and aggression. Yale University Press.

Dovidio, J. F., & Gaertner, S. L. (2004). On the nature of contemporary prejudice. In P. S. Rothenberg, (Ed.), Race, class, and gender in the United States: An integrated study (6th ed., pp. 132 142). Worth.

Dovidio, J. F., Gluszek, A., John, M. S., Ditlmann, R., & Lagunes, P. (2010). Understanding bias toward Latinos: Discrimination, dimensions of difference, and experience of exclusion. Journal of Social Issues, 66, 59 78.

Eagly, A. H., & Chaiken, S. (1975). An attribution analysis of the effect of communicator characteristics on opinion change: The case of communicator attractiveness. Journal of Personality and Social Psychology, 32, 136 144.

Eagly, A. H., & Chaiken, S. (1993). The psychology of attitudes. Harcourt Brace Jovanovich College.

East Haven mayor suggests he might have tacos to repair relations with Latinos. (2012). https://www.youtube.com/watch?v=PCUwtfqF4wU.

Ehrlinger, J., Gilovich, T., & Ross, L. (2005). Peering into the bias blind spot: People's assessments of bias in themselves and others. Personality and Social Psychology Bulletin, 31, 680 692.

Festinger, L. (1957). A theory of cognitive dissonance. Stanford University Press.

Festinger, L., & Maccoby, N. (1964). On resistance to persuasive communications. The Journal of Abnormal and Social Psychology, 68, 359 366.

Festinger, L., Schachler, S., & Back, K. W. (1950). Social pressures in informal groups: A study of human factors in housing. Harper.

Fink, B., Neave, N., Manning, J. T., & Grammer, K. (2006). Facial symmetry and judgments of attractiveness, health and personality. Personality and Individual Differences, 41, 491 499.

Fiske, S. T., Cuddy, A. J., Glick, P., & Xu, J. (2002). A model of (often mixed) stereotype content: Competence and warmth respectively follow from perceived status and competition. Journal of Personality and Social Psychology, 82(6), 878 902.

Fiske, S. T., Gilbert, D. T., & Lindzey, G. (2010). Handbook of social psychology (5th ed.). Wiley.

Freedman, J. L., & Fraser, S. C. (1966). Compliance without pressure: The foot-in-the-door technique. Journal of Personality and Social Psychology, 4, 195 202.

Gaertner, S., Dovidio, J., Banker, B., Houlette, M., Johnson, K., & McGlynn, E. (2000). Reducing intergroup conflict: From superordinate goals to decategorization, recategorization, and mutual differentiation. Group Dynamics: Theory, Research, and Practice, 4, 98 114. https://psycnet.apa.org/doiLanding?doi=10.1037%2F1089-2699.4.1.98

Geen, R. G. (1989). Alternative conceptions of social facilitation. In P. B. Paulus (Ed.), Psychology of group influence (2nd ed., pp. 15 51). Lawrence Erlbaum.

Granstr m, K., Guv , G., Hylander, I., & Rosander, M. (2009). Riots and disturbances: How riots start and how order is secured. Link ping University Electronic Press.

Greenberg, J., Schimel, J., & Martens, A. (2002). Ageism: Denying the face of the future. In T. D. Nelson (Ed.), Ageism: Stereotyping and prejudice against older persons (pp. 27 48). MIT Press.

Greenwald, A. G., McGhee, D. E., & Schwartz, J. L. (1998). Measuring individual differences in implicit cognition: The implicit association test. Journal of Personality and Social Psychology, 74, 1464 1480.

Greenwald, A. G., & Ronis, D. L. (1978). Twenty years of cognitive dissonance: Case study of the evolution of a theory. Psychological Review, 85, 53 57.

Griggs, R. A. (2014). Coverage of the Stanford Prison Experiment in introductory psychology textbooks. Teaching of Psychology, 41(3), 195 203.

Grove, J. R., Hanrahan, S. J., & McInman, A. (1991). Success/failure bias in attributions across involvement categories in sport. Personality and Social Psychology Bulletin, 17(1), 93 97.

Gupta, P. B., & Lord, K. R. (1998). Product placement in movies: The effect of prominence and mode on recall. Journal of Current Issues and Research in Advertising, 20, 47 59.

Harding, L. (2018). Students of a feather "flocked" together: A group assignment method for reducing free-riding and improving group and individual learning outcomes. Journal of Marketing Education, 40, 117 127. https://journals.sagepub.com/doi/10.1177/0273475317708588

Hare, A. P. (2003). Roles, relationships, and groups in organizations: Some conclusions and recommendations. Small Group Research, 34, 123 154.

Harvard University, Department of Psychology. (2019). Mary Whiton Calkins. https://psychology.fas.harvard.edu/people/mary-whiton-calkins

Haugtvedt, C. P., & Wegener, D. T. (1994). Message order effects in persuasion: An attitude strength perspective. Journal of Consumer Research, 21, 205 218.

Hebl, M. R., Foster, J. B., Mannix, L. M., & Dovidio, J. F. (2002). Formal and interpersonal discrimination: A field study of bias toward homosexual applicants. Personality and Social Psychology Bulletin, 28(6), 815 825.

Heckert, T. M., Latier, A., Ringwald-Burton, A., & Drazen, C. (2006). Relations among student effort, perceived class difficulty appropriateness, and student evaluations of teaching: Is it possible to "buy" better evaluations through lenient grading? College Student Journal, 40(3), 588.

Herek, G. M., & McLemore, K. A. (2013). Sexual prejudice. Annual Review of Psychology, 64, 309 33. doi:10.1146/annurev-psych-113011-143826.

Heider, F. (1958). The psychology of interpersonal relations. Wiley.

Hinduja, S. & Patchin, J. W. (2011). Cyberbullying research summary: Bullying, cyberbullying, and sexual orientation. Cyberbullying Research Center. http://www.cyberbullying.us/cyberbullying_sexual_orientation_fact_sheet.pdf.

Hodge, S. R., Burden, J. W., Jr., Robinson, L. E., & Bennett, R. A., III. (2008). Theorizing on the stereotyping of Black male student-athletes. Journal for the Study of Sports and Athletes in Education, 2, 203 226.

Hoff, D. L., & Mitchell, S. N. (2009). Cyberbullying: Causes, effects, and remedies. Journal of Education, 47, 652 665.

Hovland, C. I., Janis, I. L. and Kelley, H. H. (1953). Communications and persuasion: Psychological studies in opinion change. Yale University Press.

Hovland, C.I., Weiss, W. (1951, Winter). The influence of source credibility on communication effectiveness. Public Opinion Quarterly, 15(4), 635 650.

Igou, E. R., & Bless, H. (2003). Inferring the importance of arguments: Order effects and conversational rules. Journal of Experimental Social Psychology, 39, 91 99.

Ito, T. A., & Urland, G. R., (2003). Race and gender on the brain: Electrocortical measures of attention to race and gender of multiply categorizable individuals. Journal of Personality & Social Psychology, 85, 616 626.

Iyengar, S., & Westwood, S. J. (2015). Fear and loathing across party lines: New evidence on group polarization. American Journal of Political Science, 59(3), 690 707.

Jackson, J. M., & Williams, K. D. (1985). Social loafing on difficult tasks: Working collectively can improve performance. Journal of Personality and Social Psychology, 49, 937 942.

Janis, I. L. (1972). Victims of groupthink. Houghton Mifflin.

Jones, E. E., & Nisbett, R. E. (1971). The actor and the observer: Divergent perceptions of the causes of behavior. General Learning Press.

Jost, J. T., Banaji, M. R., & Nosek, B. A. (2004). A decade of system justification theory: Accumulated evidence of conscious and unconscious bolstering of the status quo. Political Psychology, 25, 881 919.

Jost, J. T., & Major, B. (Eds.). (2001). The psychology of legitimacy: Emerging perspectives on ideology, justice, and intergroup relations. Cambridge University Press.

Karau, S. J., & Williams, K. D. (1993). Social loafing: A meta-analytic review and theoretical integration. Journal of Personality and Social Psychology, 65, 681 706.

Kassin, S. (2017). The killing of Kitty Genovese: What else does this case tell us? Perspectives on Psychological Science, 12, 374 381. https://journals.sagepub.com/doi/abs/10.1177/1745691616679465

Krosnick, J. A., & Alwin, D. F. (1989). Aging and susceptibility to attitude change. Journal of Personality and Social Psychology, 57, 416 425.

Kumkale, G. T., & Albarrac n, D. (2004). The sleeper effect in persuasion: A meta-analytic review. Psychological Bulletin, 130(1), 143 172. doi:10.1037/0033-2909.130.1.143.

Larsen, K. S. (1990). The Asch conformity experiment: Replication and transhistorical comparisons. Journal of Social Behavior & Personality, 5(4), 163 168.

Latan , B., & Darley, J. M. (1968). Group inhibition of bystander intervention in emergencies. Journal of Personality and Social Psychology, 10, 215 221.

Latan , B., Williams, K. and Harkins, S. G. (1979). Many hands make light the work: The causes and consequences of social loafing. Journal of Personality and Social Psychology, 37, 822 832.

Laurenceau, J.-P., Barrett, L. F., & Pietromonaco, P. R. (1998). Intimacy as an interpersonal process: The importance of self-disclosure, partner disclosure, and perceived partner responsiveness in interpersonal exchanges. Journal of Personality and Social Psychology, 74(5), 1238 1251. doi:10.1037/0022-3514.74.5.1238.

Le Texier, T. (2019). Debunking the Stanford Prison Experiment. American Psychologist. 74(7), 823 839.

Lerner, M. J., & Miller, D. T. (1978). Just world research and the attribution process: Looking back and ahead. Psychological Bulletin, 85, 1030 1051.

Lumsdaine, A. A., & Janis, I. L. (1953). Resistance to counterpropaganda produced by one-sided and two-sided propaganda presentations. Public Opinion Quarterly, 17, 311 318.

Malle, B. F. (2006). The actor observer asymmetry in attribution: A (surprising) meta-analysis. Psychological Bulletin, 132(6), 895 919. doi:10.1037/0033-2909.132.6.895.

Malloy, T. E., Albright, L., Kenny, D. A., Agatstein, F., & Winquist, L. (1997). Interpersonal perception and metaperception in non-overlapping social groups. Journal of Personality and Social Psychology, 72, 390 398.

Markus, H. R., & Kitayama, S. (1991). Culture and the self: Implications for cognition, emotion, and motivation. Psychological Review, 98, 224 253.

Martin, C. H., & Bull, P. (2008). Obedience and conformity in clinical practice. British Journal of Midwifery, 16(8), 504 509.

Masuda, T., & Nisbett, R. E. (2001). Attending holistically versus analytically: Comparing the context sensitivity of Japanese and Americans. Journal of Personality and Social Psychology, 81(5), 922 934. https://psycnet.apa.org/doiLanding?doi=10.1037%2F0022-3514.81.5.922

McPherson, M., Smith-Lovin, L., & Cook, J. M. (2001). Birds of a feather: Homophily in social networks. Annual Review of Sociology, 27, pp. 415 444. doi:10.1146/annurev.soc.27.1.415.

Milgram, S. (1963). Behavioral study of obedience. Journal of Abnormal and Social Psychology, 67, 371 378.

Milgram, S. (1965). Some conditions of obedience and disobedience to authority. Human Relations, 18, 57 76.

Miller, D. T., & Ross, M. (1975). Self-serving biases in the attribution of causality: Fact or fiction? Psychological Bulletin, 82, 213 225.

Miller, N., & Campbell, D. T. (1959). Recency and primacy in persuasion as a function of the timing of speeches and measurements. The Journal of Abnormal and Social Psychology, 59, 1 9.

Mischel, W. (1977). The interaction of person and situation. Personality at the crossroads: Current issues in interactional psychology, 333, 352.

Mitchell, D. H., & Eckstein, D. (2009). Jury dynamics and decision-making: A prescription for groupthink. International Journal of Academic Research, 1(1), 163 169.

Nelson, T. (Ed.). (2004). Ageism: Stereotyping and prejudice against older persons. The MIT Press.

Nisbett, R. E., Caputo, C., Legant, P., & Marecek, J. (1973). Behavior as seen by the actor and as seen by the observer. Journal of Personality and Social Psychology, 27, 154 164.

Nisbett, R. E., Peng, K., Choi, I., & Norenzayan, A. (2001). Culture and systems of thought: Holistic versus analytic cognition. Psychological Review, 108, 291 310.

Nosek, B. A. (2005). Moderators of the relationship between implicit and explicit evaluation. Journal of Experimental Psychology: General, 134(4), 565 584.

O'Keeffe, G. S., & Clarke-Pearson, K. (2011). The impact of social media on children, adolescents, and families. Pediatrics, (127)4, 800 4. doi:10.1542/peds.2011-0054.

Olson, M. A., & Fazio, R. H. (2003). Relations between implicit measures of prejudice what are we measuring? Psychological Science, 14, 636 639.

Olweus, D. (1993). Bullying at school: What we know and what we can do. Wiley-Blackwell.

Penton-Voak, I. S., Jones, B. C., Little, A. C., Baker, S., Tiddeman, B., Burt, D. M., & Perrett, D. I. (2001). Symmetry, sexual dimorphism in facial proportions and male facial attractiveness. Proceedings of the Royal Society B: Biological Sciences, 268, 1617 1623.

Petty, R. E., & Cacioppo, J. T. (1986). The elaboration likelihood model of persuasion. In Communication and persuasion: Central and peripheral routes to attitude change (pp. 1 24). Springer. doi:10.1007/978-1-4612-4964-1.

Petty, R. E., Wegener, D. T., & Fabrigar, L. R. (1997). Attitudes and attitude change. Annual Review of Psychology, 48, 609 647.

Pliner, P., Hart, H., Kohl, J., & Saari, D. (1974). Compliance without pressure: Some further data on the foot-in-the-door technique. Journal of Experimental Social Psychology, 10, 17 22.

Plant, E. A., & Devine, P. G. (1998). Internal and external motivation to respond without prejudice. Journal of Personality and Social Psychology, 75, 811 832.

Ramesh, A., & Gelfand, M. J. (2010). Will they stay or will they go? The role of job embeddedness in predicting turnover in individualistic and collectivistic cultures. Journal of Applied Psychology, 95(5), 807.

Raymer, M., Reid, M., Spiegel, M., & Purvanova, R. (2017). An examination of generational stereotypes as a path towards reverse ageism. The Psychologist-Manager Journal, 20, 148 175.

Regan, P. C., & Berscheid, E. (1997). Gender differences in characteristics desired in a potential sexual and marriage partner. Journal of Psychology & Human Sexuality, 9, 25 37.

Reicher, S., & Haslam, S. A. (2006). Rethinking the psychology of tyranny: The BBC prison study. British Journal of Social Psychology, 45(1), 1 40.

Rhodes, N., & Wood, W. (1992). Self-esteem and intelligence affect influenceability: The mediating role of message reception. Psychological Bulletin, 111, 156 171.

Richard, F. D., Bond, C. F., Jr., & Stokes-Zoota, J. J. (2003). One hundred years of social psychology quantitatively described. Review of General Psychology, 7(4), 331 363. doi:10.1037/1089-2680.7.4.331.

Riggio, H. R., & Garcia, A. L. (2009). The power of situations: Jonestown and the fundamental attribution error. Teaching of Psychology, 36(2), 108 112. doi:10.1080/00986280902739636.

Rikowski, A., & Grammer, K. (1999). Human body odour, symmetry and attractiveness. Proceedings of the Royal Society B: Biological Sciences, 266(1422), 869 874. doi:10.1098/rspb.1999.0717.

Roesch, S. C., & Amirkham, J. H. (1997). Boundary conditions for self-serving attributions: Another look at the sports pages. Journal of Applied Social Psychology, 27, 245 261.

Rojek, J., Rosenfeld, R., & Decker, S. (2012). Policing race: The racial stratification of searches in police traffic stops. Criminology, 50, 993 1024.

Rosenberg, M. J., & Hovland, C. I. (1960). Cognitive, affective and behavioral components of attitudes. In Attitude organization and change: An analysis of consistency among attitude components (pp. 1 14). Yale University Press.

Rosenthal, R., & Jacobson, L. F. (1968). Teacher expectations for the disadvantaged. Scientific American, 218, 19 23.

Ross, L. (1977). The intuitive psychologist and his shortcomings: Distortions in the attribution process. Advances in Experimental Social Psychology, 10, 173 220.

Ross, L., Amabile, T. M., & Steinmetz, J. L. (1977). Social roles, social control, and biases in social-perception processes. Journal of Personality and Social Psychology, 35, 485 494.

Ross, L., & Nisbett, R. E. (1991). The person and the situation: Perspectives of social psychology. McGraw-Hill.

Rudman, L. A. (1998). Self-promotion as a risk factor for women: The costs and benefits of counterstereotypical impression management. Journal of Personality and Social Psychology, 74(3), 629 645.

Rusbult, C. E., & Van Lange, P. A. (2003). Interdependence, interaction, and relationships. Annual Review of Psychology, 54, 351 575.

Schank, R. C., Abelson, R. (1977). Scripts, plans, goals, and understanding: An inquiry into human knowledge. Lawrence Erlbaum Associates.

Shepperd, J. A., & Taylor, K. M. (1999). Social loafing and expectancy-value theory. Personality and Social Psychology Bulletin, 25, 1147 1158.

Sherif, M., Harvey, O. J., White, B. J., Hood, W. R., & Sherif, C. W. (1961). Intergroup conflict and cooperation: The Robbers Cave experiment. The University Book Exchange. http://livros01.livrosgratis.com.br/ps000162.pdf

Simms, A., & Nichols, T. (2014). Social loafing: A review of the literature. Journal of Management Policy and Practice, 15, 58 67.

Singh, D. (1993). Adaptive significance of female physical attractiveness: Role of waist-to-hip ratio. Journal of Personality and Social Psychology, 65, 293 307.

Sommers, S. R., & Ellsworth, P. C. (2000). Race in the courtroom: Perceptions of guilt and dispositional attributions. Personality and Social Psychology Bulletin, 26, 1367 1379.

Spears, B., Slee, P., Owens, L., & Johnson, B. (2009). Behind the scenes and screens: Insights into the human dimension of covert and cyberbullying. Journal of Psychology, 217(4), 189 196. doi:10.1027/0044-3409.217.4.189.

Sternberg, R. J. (1986). A triangular theory of love. Psychological Review, 93, 119 135.

Stewart, J. B. (2002). Heart of a soldier. Simon and Schuster.

Sutton, R.M. and Douglas, K.M. (2005). Justice for all, or just for me? More support for self-other differences in just world beliefs. Personality and Individual Differences, 9(3). pp. 637-645. ISSN 0191-8869.

Tajfel, H. (1974). Social identity and intergroup behaviour. Social Science Information, 13(2), 65 93.

Tajfel, H., & Turner, J. C. (1979). An integrative theory of intergroup conflict. In W. G. Austin & S. Worchel (Eds.), The social psychology of intergroup relations (pp. 33 48). Brooks-Cole.

Tavris, C., & Aronson, E. (2008). Mistakes were made (but not by me): Why we justify foolish beliefs, bad decisions, and hurtful acts. Houghton Mifflin Harcourt.

Taylor, L. S., Fiore, A. T., Mendelsohn, G. A., & Cheshire, C. (2011). Out of my league : A real-world test of the matching hypothesis. Personality and Social Psychology Bulletin, 37(7), 942 954. doi:10.1177/0146167211409947.

Teger, A. I., & Pruitt, D. G. (1967). Components of group risk taking. Journal of Experimental Social Psychology, 3, 189 205.

Triandis, H. C. (2001). Individualism-collectivism and personality. Journal of Personality, 69, 907 924.

van Veen, V., Krug, M. K., Schooler, J. W., & Carter, C. S. (2009). Neural activity predicts attitude change in cognitive dissonance. Nature Neuroscience, 12, 1469 1474.

Vandebosch, H., & Van Cleemput, K. (2009). Cyberbullying among youngsters: Profiles of bullies and victims. New media & Society, 11(8), 1349 1371. doi:10.1177/1461444809341263.

Walker, I., & Crogan, M. (1998). Academic performance, prejudice, and the jigsaw classroom: New pieces to the puzzle. Journal of Community and Applied Social Psychology, 8, 381 393.

Walker, M. B., & Andrade, M. G. (1996). Conformity in the Asch task as a function of age. The Journal of Social Psychology, 136, 367 372.

Walster, E., & Festinger, L. (1962). The effectiveness of overheard persuasive communications. Journal of Abnormal and Social Psychology, 65, 395 402.

Wason, P. C., & Johnson-Laird, P. N. (1972). The psychology of deduction: Structure and content. Harvard University Press.

Weiner, B. (1979). A theory of motivation for some classroom experiences. Journal of Educational Psychology, 71(1), 3 25.

Wilson, M., & Daly, M. (1985). Competitiveness, risk taking, and violence: The young male syndrome. Ethology and Sociobiology, 6, 59 73.

Wilson, T. D., Lindsey, S., & Schooler, T. Y. (2000). A model of dual attitudes. Psychological Review, 107, 101 126.

Zajonc, R. B. (1965). Social facilitation. Science, 149(3681), 269 274. doi:10.1126/science.149.3681.269

Zhang, X., Fung, H. H., Stanley, J. T., Isaacowitz, D. M., & Zhang, Qi. (2014). Thinking more holistically as we grow older? Results from different tasks in two cultures. Culture and Brain, 2, 109 121. https://link.springer.com/article/10.1007%2Fs40167-014-0018-4

Zimbardo, P. G. (1969). The human choice: Individuation, reason, and order versus deindividuation, impulse, and chaos. In Nebraska symposium on motivation. University of Nebraska press.

Zimbardo, P. G. (2007). The Lucifer effect: Understanding how good people turn evil. Random House.

Zimbardo, P. G. (2013). An end to the experiment. http://www.prisonexp.org/psychology/37.

Stress, lifestyle, and health

Aboa- boul , C., Brisson, C., Maunsell, E., M sse, B., Bourbonnais, R., V zina, M., . . . Dagenais, G. R. (2007). Job strain and risk for acute recurrent coronary heart disease events. Journal of the American Medical Association, 298, 1652 1660.

Abramson, L. Y., Seligman, M. E. P., & Teasdale, J. D. (1978). Learned helplessness in humans: Critique and reformulation. Journal of Abnormal Psychology, 87, 49 74.

Ader, R. & Cohen, N. (2001). Conditioning and immunity. In R. Ader, D. L. Felten & N. Cohen (Eds.), Psychoneuroimmunology (3rd ed., pp. 3 34). Academic Press.

Ader, R., & Cohen, N. (1975). Behaviorally conditioned immunosuppression. Psychosomatic Medicine, 37, 333 340.

Ahola, K., Honkonen, T., Isomets , E., Kalimo, R., Nykyri, E., Aromaa, A., & L nnqvist, J. (2005). The relationship between job-related burnout and depressive disorders Results from the Finnish Health 2000 study. Journal of Affective Disorders, 88, 55 62.

Ahola, K., Honkonen, T., Kivam ki, M., Virtanen, M., Isomets , E., Aromaa, A., & L nnqvist, J. (2006). Contribution of burnout to the association between job strain and depression: The Health 2000 study. Journal of Occupational and Environmental Medicine, 48, 1023 1030.

Ajrouch, K. J., Reisine, S., Lim, S., Sohn, W., & Ismail, A. (2010). Perceived everyday discrimination and psychological distress: Does social support matter? Ethnicity & Health, 15(4), 417 434. https://www.tandfonline.com/doi/abs/10.1080/13557858.2010.484050

American Academy of Neurology. (2014). Headache. https://patients.aan.com/disorders/index.cfm?event=view&disorder_id=936

American Heart Association. (2012a). What is angina? http://www.heart.org/idc/groups/heart-public/@wcm/@hcm/documents/downloadable/ucm_300287.pdf

American Heart Association. (2012b). Why blood pressure matters. http://www.heart.org/HEARTORG/Conditions/HighBloodPressure/WhyBloodPressureMatters/Why-Blood-Pressure-Matters_UCM_002051_Article.jsp

American Heart Association. (2014, February 24). Depression as a risk factor for poor prognosis among patients with acute coronary syndrome: Systematic review and recommendations: A scientific statement from the American Heart Association. Circulation. http://circ.ahajournals.org/content/early/2014/02/24/CIR.0000000000000019.full.pdf+html

American Lung Association. (2010). Asthma. http://www.lung.org/assets/documents/publications/solddc-chapters/asthma.pdf

American Psychiatric Association. (2013). Diagnostic and statistical manual of mental disorders (5th ed.). Author.

Bandura, A. (1997). Self-efficacy: The exercise of control. Freeman.

Bandura, A. (2004). Health promotion by social cognitive means. Health Education & Behavior, 31, 143 164.

Barefoot, J. C., & Schroll, M. S. (1996). Symptoms of depression, acute myocardial infarction, and total mortality in a community sample. Circulation, 93, 1976 1980.

Baron, R. S., & Kerr, N. L. (2003). Group process, group decision, group action (2nd ed.). Open University Press.

Baron, R. S., Cutrona, C. E., Hicklin, D., Russell, D. W., & Lubaroff, D. M. (1990). Social support and immune function among spouses of cancer patients. Journal of Personality and Social Psychology, 59, 344 352.

Baumeister, R. F., & Leary, M. R. (1995). The need to belong: Desire for interpersonal attachments as a fundamental human motivation. Psychological Bulletin, 117, 497 529.

Benson, H., & Proctor, W. (1994). Beyond the relaxation response: How to harness the healing power of your personal beliefs. Berkley Publishing Group.

Berkman, L. F., & Syme, L. (1979). Social networks, host resistance, and mortality: A nine-year follow-up study of Alameda County residents. American Journal of Epidemiology, 109, 186 204.

Blackburn, E. H., & Epel, E. S. (2012). Telomeres and adversity: Too toxic to ignore. Nature, 490(7419), 169 171.

Boehm, J. K., & Kubzansky, L. D. (2012). The heart s content: The association between positive psychological well-being and cardiovascular health. Psychological Bulletin, 138, 655 691.

Bolger, N., DeLongis, A., Kessler, R. C., & Schilling, E. A. (1989). Effects of daily stress on negative mood. Journal of Personality and Social Psychology, 57, 808 818.

Boyle, S. H., Michalek, J. E., & Suarez, E. C. (2006). Covariation of psychological attributes and incident coronary heart disease in U.S. Air Force veterans of the Vietnam War. Psychosomatic Medicine, 68, 844 850.

Brickman, P., & Campbell, D. T. (1971). Hedonic relativism and planning the good society. In M. H. Appley (Ed.), Adaptation level theory: A symposium (pp. 287 302). Academic Press.

Brickman, P., Coats, D., & Janoff-Bulman, R. (1978). Lottery winners and accident victims: Is happiness relative? Journal of Personality and Social Psychology, 36, 917 927.

Brummett, B. H., Barefoot, J. C., Siegler, I. C., Clapp-Channing, N. E., Lytle, B. L., Bosworth, H. B., . . . Mark, D. B. (2001). Characteristics of socially isolated patients with coronary artery disease who are at elevated risk for mortality. Psychosomatic Medicine, 63, 267 272.

Caceres, C., & Burns, J. W. (1997). Cardiovascular reactivity to psychological stress may enhance subsequent pain sensitivity. Pain, 69, 237 244.

Campeau, S., Nyhuis, T. J., Sasse, S. K., Kryskow, E. M., Herlihy, L., Masini, C. V., . . . Day, H. E. W. (2010). Hypothalamic pituitary adrenal axis responses to low-intensity stressors are reduced after voluntary wheel running in rats. Journal of Neuroendocrinology, 22, 872 888.

Cannon, W. B. (1932). The wisdom of the body. W. W. Norton.

Carroll, J. (2007). Most Americans very satisfied with their personal lives. Gallup. http://www.gallup.com/poll/103483/Most-Americans-Very-Satisfied-Their-Personal-Lives.aspx

Cathcart, S., Petkov, J., & Pritchard, D. (2008). Effects of induced stress on experimental pain sensitivity in chronic tension-type headache sufferers. European Journal of Neurology, 15, 552 558.

Centers for Disease Control and Prevention (CDC). (2006). You can control your asthma: A guide to understanding asthma and its triggers. http://www.cdc.gov/asthma/pdfs/asthma_brochure.pdf

Centers for Disease Control and Prevention (CDC). (2011). Million hearts: Strategies to reduce the prevalence of leading cardiovascular disease risk factors United States, 2011. Morbidity and Mortality Weekly Report [MMWR], 60(36), 1248 1251. http://www.cdc.gov/mmwr/pdf/wk/mm6036.pdf

Centers for Disease Control and Prevention (CDC). (2013a). Asthma s impact on the nation: Data from the CDC National Asthma Control Program. http://www.cdc.gov/asthma/impacts_nation/AsthmaFactSheet.pdf

Centers for Disease Control and Prevention (CDC). (2013b). Breathing easier. http://www.cdc.gov/asthma/pdfs/breathing_easier_brochure.pdf

Center for Investigating Health Minds. (2013). About. http://www.investigatinghealthyminds.org/cihmCenter.html

Chandola, T., Britton, A., Brunner, E., Hemingway, H., Malik, M., Kumari, M., . . . Marmot, M. (2008). Work stress and coronary heart disease: What are the mechanisms? European Heart Journal, 29, 640 648.

Chang, E. C. (2001). Introduction: Optimism and pessimism and moving beyond the most fundamental questions. In E. C. Chang (Ed.), Optimism and pessimism: Implications for theory, research, and practice (pp. 3 12). American Psychological Association.

Chang, P. P., Ford, D. E., Meoni, L. A., Wang, N. Y., & Klag, M. J. (2002). Anger in young and subsequent premature cardiovascular disease. Archives of Internal Medicine, 162, 901 906.

Chida, Y., & Steptoe, A. (2009). The association of anger and hostility with future coronary heart disease: A meta-analytic review or prospective evidence. Journal of the American College of Cardiology, 53, 936 946.

Cohen, S., Frank, E., Doyle, W. J., Skoner, D. P., Rabin, B. S., & Gwaltney, J. M. J. (1998). Types of stressors that increase susceptibility to the common cold in healthy adults. Health Psychology, 17, 214 223.

Cohen, S., & Herbert, T. B. (1996). Health psychology: Psychological factors and physical disease from the perspective of human psychoneuroimmunology. Annual Review of Psychology, 47, 113 142.

Cohen, S., & Janicki-Deverts, D. (2012). Who s stressed? Distributions of psychological stress in the United States in probability samples in 1993, 2006, and 2009. Journal of Applied Social Psychology, 42, 1320 1334.

Cohen, S., Janicki-Deverts, D., & Miller, G. E. (2007). Psychological distress and disease. Journal of the American Medical Association, 98, 1685 1687.

Cohen, S., Doyle, W. J., Turner, R., Alper, C. M., & Skoner, D. P. (2003). Sociability and susceptibility to the common cold. Psychological Science, 14, 389 395.

Cohrs, J. C., Christie, D. J., White, M. P., & Das, C. (2013). Contributions of positive psychology to peace: Toward global well-being and resilience. American Psychologist, 68, 590 600.

Compton, W. C. (2005). An introduction to positive psychology. Thomson Wadsworth.

Cotton, D. H. G. (1990). Stress management: An integrated approach to therapy. Brunner/Mazel.

Craft, L. L., VanIterson, E. H., Helenowski, I. B., Rademaker, A. W., & Courneya, K. S. (2012). Exercise effects on depressive symptoms in cancer survivors: A systematic review and meta-analysis. Cancer Epidemiology, Biomarkers & Prevention, 21, 3 19.

Csikszentmihalyi, M. (1997). Finding flow. Basic Books.

Csikszentmihalyi, M. (1999). If we are so rich, why aren t we happy? American Psychologist, 54, 821 827.

D Amato, G., Liccardi, G., Cecchi, L., Pellegrino, F., & D Amato, M. (2010). Facebook: A new trigger for asthma? The Lancet, 376, 1740.

Davidson, K. W., Mostofsky, E., & Whang, W. (2010). Don t worry: be happy: Positive affect and reduced 10-year incident coronary heart disease: The Canadian Nova Scotia Health Survey. European Heart Journal, 31, 1065 1070.

de Kluizenaar, Y., Gansevoort, R. T., Miedema, H. M. E., & de Jong, P. E. (2007). Hypertension and road traffic noise exposure. Journal of Occupational and Environmental Medicine, 49, 484 492.

De Vogli, R., Chandola, T., & Marmot, M. G. (2007). Negative aspects of close relationships and heart disease. Archives of Internal Medicine, 167, 1951 1957.

DeLongis, A., Coyne, J. C., Dakof, G., Folkman, S., & Lazarus, R. S. (1982). Relationship of daily hassles, uplifts, and major life events to health status. Health Psychology, 1, 119 136.

Derogatis, L. R., & Coons, H. L. (1993). Self-report measures of stress. In L. Goldberger & S. Breznitz (Eds.), Handbook of stress: Theoretical and clinical aspects (2nd ed., pp. 200 233). Free Press.

Diehl, M., & Hay, E. L. (2010). Risk and resilience factors in coping with daily stress in adulthood: The role of age, self-concept incoherence, and personal control. Developmental Psychology, 46, 1132 1146.

Diener, E. (2012). New findings and future directions for subjective well-being research. American Psychologist, 67, 590 597.

Diener, E. (2013). The remarkable changes in the science of subjective well-being. Perspectives on Psychological Science, 8, 663 666.

Diener, E., & Biswas-Diener, R. (2002). Will money increase subjective well-being? A literature review and guide to needed research. Social Indicators Research, 57, 119 169.

Diener, E., Diener, M., & Diener, C. (1995). Factors predicting the subjective well-being of nations. Journal of Personality and Social Psychology, 69, 851 864.

Diener, E., Lucas, R., & Scollon, C. N. (2006). Beyond the hedonic treadmill: Revising the adaptation theory of well-being. American Psychologist, 61, 305 314.

Diener, E., Ng, W., Harter, J., & Arora, R. (2010). Wealth and happiness across the world: Material prosperity predicts life evaluation, whereas psychosocial prosperity predicts positive feelings. Journal of Personality and Social Psychology, 99, 52 61.

Diener, E., Oishi, S., & Ryan, K. L. (2013). Universals and cultural differences in the causes and structure of happiness: A multilevel review. In Mental Well-Being (pp. 153 176). Springer Netherlands.

Diener, E., Suh, E. M., Lucas, R. E., & Smith, H. L. (1999). Subjective well-being: Three decades of progress. Psychological Bulletin, 125, 276 302.

Diener, E., Tay, L., & Myers, D. (2011). The religion paradox: If religion makes people happy, why are so many dropping out? Journal of Personality and Social Psychology, 101, 1278 1290.

Diener, E., Tay, L., & Oishi, S. (2013). Rising income and the subjective well-being of nations. Journal of Personality and Social Psychology, 104, 267 276.

Diener, E., Wolsic, B., & Fujita, F. (1995). Physical attractiveness and subjective well-being. Journal of Personality and Social Psychology, 69, 120 129.

Dohrenwend, B. P. (2006). Inventorying stressful life events as risk factors for psychopathology: Toward resolution of the problem of intracategory variability. Psychological Bulletin, 132, 477 495.

Entringer, S., Epel, E. S., Kumsta, R., Lin, J., Hellhammer, D. H., Blackburn, E., W st, S., & Wadhwa, P. D. (2011). Stress exposure in intrauterine life is associated with shorter telomere length in young adulthood. Proceedings of the National Academy of Sciences, USA, 108, E513 E518.

Epel, E. S., Blackburn, E. H., Lin, J., Dhabhar, F. S., Adler, N. E., Morrow, J. D., & Cawthon, R. M. (2004). Accelerated telomere shortening in response to life stress. Proceedings of the National Academy of Sciences, USA, 101, 17312 17315.

Everly, G. S., & Lating, J. M. (2002). A clinical guide to the treatment of the human stress response (2nd ed.). Kluwer Academic/Plenum Publishing.

Falagas, M. E., Zarkadoulia, E. A., Ioannidou, E. N., Peppas, G., Christodoulou, C., & Rafailidis, P. I. (2007). The effect of psychosocial factors on breast cancer outcome: A systematic review. Breast Cancer Research, 9:R44. http://breast-cancer-research.com/content/pdf/bcr1744.pdf

Folkman, S., & Lazarus, R. S. (1980). An analysis of coping in a middle-aged community sample. Journal of Health and Social Behavior, 21, 219 239.

Fontana, A. M., Diegnan, T., Villeneuve, A., & Lepore, S. J. (1999). Nonevaluative social support reduces cardiovascular reactivity in young women during acutely stressful performance situations. Journal of Behavioral Medicine, 22, 75 91.

Friedman, H. S., & Booth-Kewley, S. (1987). The disease-prone personality : A meta-analytic view of the construct. American Psychologist, 42, 539 555.

Friedman, M. (1977). Type A behavior pattern: Some of its pathophysiological components. Bulletin of the New York Academy of Medicine, 53, 593 604.

Friedman, M., & Rosenman, R. (1974). Type A behavior and your heart. Alfred A. Knopf.

Friedman, M., & Rosenman, R. H. (1959). Association of specific overt behavior pattern with blood and cardiovascular findings blood cholesterol level, blood clotting time, incidence of arcus senilis, and clinical coronary artery disease. Journal of the American Medical Association, 169(12), 1286 1296.

Fujita, F., & Diener, E. (2005). Life satisfaction set point: Stability and change. Journal of Personality and Social Psychology, 88, 158 164.

Fulmer, C. A., Gelfand, M. J., Kruglanski, A., Kim-Prieto, C., Diener, E., Pierro, A., & Higgins, E. T. (2010). On feeling right in cultural contexts: How person-culture match affects self-esteem and subjective well-being. Psychological Science, 21, 1563 1569.

Geoffroy, M. C., Hertzman, C., Li, L., & Power, C. (2013). Prospective association of morning salivary cortisol with depressive symptoms in mid-life: A life-course study. PLoS ONE, 8(11), 1 9.

Gerber, M., Kellman, M., Hartman, T., & P hse, U. (2010). Do exercise and fitness buffer against stress among Swiss police and emergency response service officers? Psychology of Sport and Exercise, 11, 286 294.

Glaser, R., Kiecolt-Glaser, J. K., Marucha, P. T., MacCallum, R. C., Laskowski, B. F., & Malarkey, W. B. (1999). Stress-related changes in proinflammatory cytokine production in wounds. Archives of General Psychiatry, 56, 450 456.

Glassman, A. H. (2007). Depression and cardiovascular comorbidity. Dialogues in Clinical Neuroscience, 9, 9 17.

Glassman, A. H., & Shapiro, P. A. (1998). Depression and the course of coronary artery disease. American Journal of Psychiatry, 155, 4 11.

Greenberg, J. S. (2006). Comprehensive stress management (9th ed.). McGraw-Hill.

Hackney, C. H., & Sanders, G. S. (2003). Religiosity and mental health: A meta-analysis of recent studies. Journal for the Scientific Study of Religion, 42, 43 55.

Hafeez, H., Zeshan, M., Tahir, M. A., Jahan, N., & Naveed, S. (2017). Health care disparities among lesbian, gay, bisexual, and transgender youth: A literature review. Cureus, 9(4). https://www.ncbi.nlm.nih.gov/pubmed/28638747

Hansen, T. (2012). Parenthood and happiness: A review of folk theories versus empirical evidence. Social Indicators Research, 108, 29 64.

Hare, D. L., Toukhsati, S. R., Johansson, P., & Jaarsma, T. (2013). Depression and cardiovascular disease: A clinical review. European Heart Journal. Advance online publication. doi:10.1093/eurheartj/eht462

Hatch, S. L., & Dohrenwend, B. P. (2007). Distribution of traumatic and other stressful life events by race/ethnicity, gender, SES, and age: A review of the research. American Journal of Community Psychology, 40, 313 332.

Haynes, S. G., Feinleib, M., & Kannel, W. B. (1980). The relationship of psychosocial factors to coronary disease in the Framingham study: III. Eight-year incidence of coronary heart disease. American Journal of Epidemiology, 111, 37 58.

Head, D., Singh, T., & Bugg, J. M. (2012). The moderating role of exercise on stress-related effects on the hippocampus and memory in later adulthood. Neuropsychology, 26, 133 143.

Helliwell, J., Layard, R., & Sachs, J. (Eds.). (2013). World happiness report 2013. United Nations Sustainable Development Solutions Network: http://unsdsn.org/wp-content/uploads/2014/02/WorldHappinessReport2013_online.pdf

Helson, H. (1964). Current trends and issues in adaptation-level theory. American Psychologist, 19, 26 38.

Holmes, T. H., & Masuda, M. (1974). Life change and illness susceptibility. In B. S. Dohrenwend & B. P. Dohrenwend (Eds.), Stressful life events: Their nature and effects (pp. 45 72). John Wiley & Sons.

Holmes, T. H., & Rahe, R. H. (1967). The social readjustment rating scale. Journal of Psychosomatic Research, 11, 213 218.

Holmes, T. S., & Holmes, T. H. (1970). Short-term intrusions into the life style routine. Journal of Psychosomatic Research, 14, 121 132.

Holt-Lunstad, J., Smith, T. B., & Layton, J. B. (2010). Social relationships and mortality risk: A meta-analytic review. PLoS Medicine, 7(7), e1000316.

Hunt, M. G., Marx, R., Lipson, C., & Young, J. (2018). No more FOMO: Limiting social media decreases loneliness and depression. Journal and Social and Clinical Psychology, 37(10), 751 768. https://guilfordjournals.com/doi/abs/10.1521/jscp.2018.37.10.751

Hupbach, A., & Fieman, R. (2012). Moderate stress enhances immediate and delayed retrieval of educationally relevant material in healthy young men. Behavioral Neuroscience, 126, 819 825.

Infurna, F. J., & Gerstorf, D. (2014). Perceived control relates to better functional health and lower cardio-metabolic risk: The mediating role of physical activity. Health Psychology, 33, 85 94.

Infurna, F. J., Gerstorf, D., Ram, N., Schupp, J., & Wagner, G. G. (2011). Long-term antecedents and outcomes of perceived control. Psychology and Aging, 26, 559 575.

Johnson, W., & Krueger, R. F. (2006). How money buys happiness: Genetic and environmental processes linking finances and life satisfaction. Journal of Personality and Social Psychology, 90, 680 691.

Jonas, B. S., & Lando, J. F. (2000). Negative affect as a prospective risk factor for hypertension. Psychosomatic Medicine, 62, 188 196.

Jordan, H. T., Miller-Archie, S. A., Cone, J. E., Morabia, A., & Stellman, S. D. (2011). Heart disease among those exposed to the September 11, 2001 World Trade Center disaster: Results from the World Trade Center Health Registry. Preventive Medicine: An International Journal Devoted to Practice and Theory, 53, 370 376.

Kahneman, D. (2011). Thinking fast and slow. Farrar, Straus, & Giroux.

Kahneman, D., & Deaton, A. (2010). High income improves evaluation of life, but not emotional well-being. Proceedings of the National Academy of Sciences, USA, 107, 16489 16493.

Kanner, A. D., Coyne, J. C., Schaefer, C., & Lazarus, R. S. (1981). Comparison of two modes of stress measurement: Daily hassles and uplifts versus major life events. Journal of Behavioral Medicine, 4, 1 39.

Karasek, R., & Theorell, T. (1990). Healthy work: Stress, productivity, and the reconstruction of working life. Basic Books.

Kiecolt-Glaser, J. K. (2009). Psychoneuroimmunology: Psychology s gateway to the biomedical future. Perspectives on Psychological Science, 4, 367 369.

Kiecolt-Glaser, J. K., Glaser, R., Gravenstein, S., Malarkey, W. B., & Sheridan, J., (1996). Chronic stress alters the immune response to influenza virus vaccine in older adults. Proceedings of the National Academy of Sciences, USA, 93, 3043 3047.

Kiecolt-Glaser, J. K., McGuire, L., Robles, T. F., & Glaser, R. (2002). Psychoneuroimmunology and psychosomatic medicine: Back to the future. Psychosomatic Medicine, 64, 15 28.

Kiecolt-Glaser, J. K., McGuire, L., Robles, T. F., & Glaser, R. (2002).Psychoneuroimmunology: Psychological influences on immune function and health. Journal of Consulting and Clinical Psychology, 70, 537 547.

Kilpel inen, M., Koskenvuo, M., Helenius, H., & Terho, E. O. (2002). Stressful life events promote the manifestation of asthma and atopic diseases. Clinical and Experimental Allergy, 32, 256 263.

Kivim ki, M., Head, J., Ferrie, J. E., Shipley, M. J., Brunner, E., Vahtera, J., & Marmot, M. G. (2006). Work stress, weight gain and weight loss. Evidence for bidirectional effects of body mass index in the Whitehall II study. International Journal of Obesity, 30, 982 987.

Klinnert, M. D., Nelson, H. S., Price, M. R., Adinoff, A. D., Leung, M., & Mrazek, D. A. (2001). Onset and persistence of childhood asthma: Predictors from infancy. Pediatrics, 108, E69.

Konnikova, M. (2015, January 14). Trying to cure depression, but inspiring torture. The New Yorker. https://www.newyorker.com/science/maria-konnikova/theory-psychology-just ified-torture?verso=true

Kraus, M. W., Piff, P. K., Mendoza-Denton, R., Rheinschmidt, M. L., & Keltner, D. (2012). Social class, solipsism, and contextualism: How the rich are different from the poor. Psychological Review, 119, 546 572.

Krosnick, J. A. (1990). Thinking about politics: Comparisons of experts and novices. Guilford.

Krumboltz, M. (2014, February 18). Just like us? Elephants comfort each other when they re stressed out. Yahoo News. http://news.yahoo.com/elephants-know-a-thing-or-two-about-empathy-202224 477.html

Lachman, M. E., & Weaver, S. L. (1998). The sense of control as a moderator of social class differences in health and well-being. Journal of Personality and Social Psychology, 74, 763 773.

Lavner, J. A., Karney, B. R., & Bradbury, T. N. (2013). Newlyweds optimistic forecasts of their marriage: For better or for worse? Journal of Family Psychology, 27, 531 540.

Lazarus, R. P., & Folkman, S. (1984). Stress, appraisal, and coping. Springer.

Lee, M., & Rotheram-Borus, M. J. (2001). Challenges associated with increased survival among parents living with HIV. American Journal of Public Health, 91, 1303 1309.

Lehrer, P. M., Isenberg, S., & Hochron, S. M. (1993). Asthma and emotion: A review. Journal of Asthma, 30, 5 21.

Lepore, S. J. (1998). Problems and prospects for the social support-reactivity hypothesis. Annals of Behavioral Medicine, 20, 257 269.

Lichtman, J. H., Bigger, T., Blumenthal, J. A., Frasure-Smith, N., Kaufmann, P. G., Lesp rance, F., . . . Froelicher, E. S. (2008). Depression and coronary heart disease: Recommendations for screening, referral, and treatment: A science advisory from the American Heart Association Prevention Committee of the Council on Cardiovascular Nursing, Council on Clinical Cardiology, Council on Epidemiology and Prevention, and Interdisciplinary Council on Quality of Care and Outcomes Research. Circulation, 118, 1768 1775.

Loerbroks, A., Apfelbacher, C. J., Thayer, J. F., Debling, D., & St rmer, T. (2009). Neuroticism, extraversion, stressful life events and asthma: A cohort study of middle-aged adults. Allergy, 64, 1444 1450.

Logan H., Lutgendorf, S., Rainville, P., Sheffield, D., Iverson, K., & Lubaroff, D. (2001). Effects of stress and relaxation on capsaicin-induced pain. The Journal of Pain, 2, 160 170.

Lutter, M. (2007). Book review: Winning a lottery brings no happiness. Journal of Happiness Studies, 8, 155 160.

Lyon, B. L. (2012). Stress, coping, and health. In V. H. Rice (Ed.), Handbook of stress, coping, and health: Implications for nursing research, theory, and practice (2nd ed., pp. 2 20). Sage.

Lyubomirsky, S. (2001). Why are some people happier than others? The role of cognitive and motivational processes in well-being. American Psychologist, 56, 239 249.

Lyubomirsky, S., King, L., & Diener, E. (2005). The benefits of frequent positive affect: Does happiness lead to success? Psychological Bulletin, 131, 803 855.

Maier, S. F., Watkins, L. R., & Fleshner, M. (1994). Psychoneuroimmunology: The interface between behavior, brain, and immunity. American Journal of Psychology 49(12), 1004 1017.

Malzberg, B. (1937). Mortality among patients with involution melancholia. American Journal of Psychiatry, 93, 1231 1238.

Manuel, J. I. (2018). Racial/ethnic and gender disparities in health care use and access. Health Services Research 53(3), 1407 1429. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5980371/

Marmot, M. G., Bosma, H., Hemingway, H., & Stansfeld, S. (1997). Contribution of job control and other risk factors to social variations in coronary heart disease incidence. The Lancet, 350, 235 239.

Marmot, M. G. & Sapolsky, R. (2014). Of baboons and men: Social circumstances, biology, and the social gradient in health. In M. Weinstein, & M. A. Lane (Eds.), Sociality, hierarchy, health: comparative biodemography, A collection of papers (pp. 365 388). National Academies Press (US).

Martin, M. W. (2012). Happiness and the good life. Oxford University Press.

Maslach, C., & Jackson, S. E. (1981). The measurement of experienced burnout. Journal of Occupational Behavior, 2, 99 113.

McCarthy, J. (2020, January 10). Happiness not quite as widespread as usual in the U.S. Gallup. https://news.gallup.com/poll/276503/happiness-not-quite-widespread-usual .aspx

McEwan, B. (1998). Protective and damaging effects of stress mediators. New England Journal of Medicine, 338(3), 171 179.

McIntosh, J. (2014, July 28) What are headaches? What causes headaches? Medical News Today. http://www.medicalnewstoday.com/articles/73936.php

MedicineNet. (2013). Headaches. http://www.medicinenet.com/tension_headache/article.htm#what_causes_tens ion_headaches

Monat, A., & Lazarus, R. S. (1991). Stress and coping: An anthology (3rd ed.). Columbia University Press.

Myers, D. G. (2000). The funds, friends, and faith of happy people. American Psychologist, 55, 56 67.

Myers, T. C., Wittrock, D. A., & Foreman, G. W., (1998). Appraisal of subjective stress in individuals with tension-type headache: The influence of baseline measures. Journal of Behavioral Medicine, 21, 469 484.

Mykletun, A., Bjerkeset, O., Dewey, M., Prince, M., Overland, S., & Stewart, R. (2007). Anxiety, depression, and cause-speci c mortality: The HUNT study. Psychosomatic Medicine, 69, 323 331.

Myrtek, M. (2001). Meta-analyses of prospective studies on coronary heart disease, type A personality, and hostility. International Journal of Cardiology, 79, 245 251.

Nabi, H., Kivimaki, M., De Vogli, R., Marmot, M. G., & Singh-Manoux, A. (2008). Positive and negative affect and risk of coronary heart disease: Whitehall II prospective cohort study. British Medical Journal, 337, a118.

National Institute of Arthritis and Musculoskeletal and Skin Diseases (NIAMS). (2012). Understanding autoimmune diseases. http://www.niams.nih.gov/Health_Info/Autoimmune/understanding_autoimmune .pdf

Nealey-Moore, J. B., Smith, T. W., Uchino, B. N., Hawkins, M. W., & Olson-Cerny, C. (2007). Cardiovascular reactivity during positive and negative marital interactions. Journal of Behavioral Medicine, 30, 505 519.

Neelakantan, S. (2013). Mind over myocardium. Nature, 493, S16 S17.

Neupert, S. D., Almeida, D. M., & Charles, S. T. (2007). Age differences in reactivity to daily stressors: The role of personal control. Journal of Gerontology: Psychological Sciences, 62B, P216 P225.

Nusair, M., Al-dadah, A., & Kumar, A. (2012). The tale of mind and heart: Psychiatric disorders and coronary heart disease. Missouri Medicine, 109, 199 203.

Office on Women s Health, U.S. Department of Health and Human Services. (2009). Heart disease: Frequently asked questions. http://www.womenshealth.gov/publications/our-publications/fact-sheet/hea rt-disease.pdf

Ong, A. D., Bergeman, C. S., & Bisconti, T. L. (2005). Unique effects of daily perceived control on anxiety symptomatology during conjugal bereavement. Personality and Individual Differences, 38, 1057 1067.

sby, U., Brandt, L., Correia, N., Ekbom, A., & Spar n, P. (2001). Excess mortality in bipolar and unipolar depression in Sweden. Archives of General Psychiatry, 58, 844 850.

Park, S. G., Kim, H. C., Min, J. Y., Hwang, S. H., Park, Y. S., & Min, K. B. (2011). A prospective study of work stressors and the common cold. Occupational Medicine, 61, 53 56.

Pascoe, E. A., & Smart Richman, L. (2009). Perceived discrimination and health: A meta-analytic review. Psychological Bulletin, 135(4), 531 554. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2747726/

Peterson, C., & Seligman, M. E. P. (1984). Causal explanations as a risk factor for depression: Theory and evidence. Psychological Review, 91, 347 374.

Peterson, C., & Steen, T. A. (2002). Optimistic explanatory style. In C. R. Snyder & S. J. Lopez (Eds.), Handbook of positive psychology (pp. 244 256). Oxford University Press.

Phillips, A. C. (2011). Blunted as well as exaggerated cardiovascular reactivity to stress is associated with negative health outcomes. Japanese Psychological Research, 53, 177 192.

Phillips, A. C., Gallagher, S., & Carroll, D. (2009). Social support, social intimacy, and cardiovascular reactions to acute psychological stress. Annals of Behavioral Medicine, 37, 38 45.

Pinquart, M., & S rensen, S. (2000). Influence of socioeconomic status, social network, and competence on subjective well-being in later life. A meta-analysis. Psychology and Aging, 15, 187 224.

Ploubidis, G. B., & Grundy, E. (2009). Personality and all cause mortality: Evidence for indirect links. Personality and Individual differences, 47, 203 208.

Powell, J. (1996). AIDS and HIV-related diseases: An educational guide for professionals and the public. Insight Books.

Pressman, S. D., & Cohen, S. (2005). Does positive affect influence health? Psychological Bulletin, 131, 925 971.

Puterman, E., Lin, J., Blackburn, E., O Donovan, A., Adler, N., & Epel, E. (2010). The power of exercise: Buffering the effect of chronic stress on telomere length. PLoS ONE, 5(5), e10837.

Quoidbach, J., Dunn, E. W., Petrides, K. V., & Mikolajczak, M. (2010). Money giveth, money taketh away: The dual effect of wealth on happiness. Psychological Science, 21, 759 763.

Rahe, R. H. (1974). The pathway between subjects recent life changes and their near-future illness reports: Representative results and methodological issues. In B. S. Dohrenwend & B. P. Dohrenwend (Eds.), Stressful life events: Their nature and effects (pp. 73 86). Wiley & Sons.

Rahe, R. H., McKeen, J. D., & Arthur, R. J. (1967). A longitudinal study of life change and illness patterns. Journal of Psychosomatic Research, 10, 355 366.

Raney, J. D., & Troop-Gordon, W. (2012). Computer-mediated communication with distant friends: Relations with adjustment during students first semester in college. Journal of Educational Psychology, 104, 848 861.

Rasmussen, H. N., & Wallio, S. C. (2008). The health benefits of optimism. In S. J. Lopez (Ed.), Positive psychology: Exploring the best in people (pp. 131 149). Praeger Publishers.

Rasmussen, H. N., Scheier, M. F., & Greenhouse, J. B. (2009). Optimism and physical health: A meta-analytic review. Annals of Behavioral Medicine, 37, 239 256.

Raymond, J. M., & Sheppard, K. (2018). Effects of peer mentoring on nursing students' perceived stress, sense of belonging, self-efficacy and loneliness. Journal of Nursing Education and Practice, 8(1), 16 23. https://pdfs.semanticscholar.org/ce60/8eca933ab45f3d608f9b5a1a505014dd2cb7.pdf

Ritz, T., Steptoe, A., Bobb, C., Harris, A. H. S., & Edwards, M. (2006). The asthma trigger inventory: Validation of a questionnaire for perceived triggers of asthma. Psychosomatic Medicine, 68, 956 965.

Rosengren, A., Hawken, S., Ounpuu, S., Sliwa, K., Zubaid, M., Almahmeed, W. A., . . . Yusuf, S. (2004). Association of psychosocial risk factors with risk of acute myocardial infarction in 11,119 cases and 13,648 controls from 52 countries (the INTERHEART study): Case-control study. The Lancet, 364, 953 962.

Rosenman, R. H., Brand, R. J., Jenkins, C. D., Friedman, M., Straus, R., & Wurm, M. (1975). Coronary heart disease in the Western Collaborative Group Study: Final follow-up experience of 8.5 years. Journal of the American Medical Association, 223, 872 877.

Rottenberg, J., Yaroslavsky, I., Carney, R. M., Freedland, K. E., George, C. J., Baki, I., Kovacs, M. (2014). The association between major depressive disorder and risk factors for cardiovascular disease in adolescence. Psychosomatic Medicine, 76, 122 127.

Salmon, P. (2001). Effects of physical exercise on anxiety, depression, and sensitivity to stress: A unifying theory. Clinical Psychology Review, 21, 33 61.

Saito, K., Kim, J. I., Maekawa, K., Ikeda, Y., & Yokoyama, M. (1997). The great Hanshin-Awaji earthquake aggravates blood pressure control in treated hypertensive patients. American Journal of Hypertension, 10, 217 221.

Salonen, P., Arola, H., Nyg rd, C., & Huhtala, H. (2008). Long-term associations of stress and chronic diseases in ageing and retired employees. Psychology, Health, and Medicine, 13, 55 62.

Sapolsky, R. M. (1998). Why zebras don t get ulcers: An updated guide to stress, stress-related disease, and coping. Freeman.

Sapolsky, R. M. (2004). Organismal stress and telomeric aging. An unexpected connection. Proceedings of the National Academy of Sciences, USA, 101, 17323 17324.

Schnall, P. L., & Landsbergis, P. A. (1994). Job strain and cardiovascular disease. Annual Review of Public Health, 15, 381 411.

Schwartz, B. S., Stewart, W. F., Simon, D., & Lipton, R. B. (1998). Epidemiology of tension-type headache. Journal of the American Medical Association, 279, 381 383.

Schwartz, N. M., & Schwartz, M. S. (1995). Definitions of biofeedback and applied physiology. In M. S. Schwartz & F. Andrasik (Eds.), Biofeedback: A practitioners guide (pp. 32 42). Guilford.

Scully, J. A., Tosi, H., & Banning, K. (2000). Life event checklists: Revisiting the Social Readjustment Rating Scale after 30 years. Educational and Psychological Measurement, 60, 864 876.

Segerstrom, S. C., & Miller, G. E. (2004). Psychological stress and the human immune system: A meta-analytic study of 30 years of inquiry. Psychological Bulletin, 130, 601 630.

Seligman, M. E., & Maier, S. F. (1967). Failure to escape traumatic shock. Journal of Experimental Psychology, 74, 1 9.

Seligman, M. E., Maier, S. F., & Geer, J. H. (1968). Alleviation of learned helplessness in the dog. Journal of Abnormal Psychology, 3, 256 262.

Seligman, M. E. P. (2002). Authentic happiness: Using the new positive psychology to realize your potential for lasting fulfillment. Free Press.

Seligman, M. E. P., Steen, T. A., Park, N., & Peterson, C. (2005). Positive psychology progress: Empirical validation of interventions. American Psychologist, 60, 410 421.

Seligman, M. P., & Csikszentmihalyi, M. (2000). Positive psychology: An introduction. American Psychologist, 55, 5 14.

Selye, H. (1936). A syndrome produced by diverse nocuous agents. Nature, 138, 32 33.

Selye, H. (1974). Stress without distress. Lippencott.

Selye, H. (1976). The stress of life (Rev. ed.). McGraw-Hill.

Shalev, I., Moffitt, T. E., Sugden, K., Williams, B., Houts, R. M., Danese, A., . . . Caspi, A. (2013). Exposure to violence during childhood is associated with telomere erosion from 5 to 10 years of age: A longitudinal study. Molecular Psychiatry, 18, 576 581.

Shapiro, P. A. (2005). Heart disease. In J. L. Levenson (Ed.), Textbook of psychosomatic medicine (pp. 423 444). American Psychiatric Publishing.

Sims, M., Diez-Roux, A. V., Dudley, A., Gebreab, S., Wyatt, S. B., Bruce, M. A., . . . Taylor, H. A. (2012). Perceived discrimination and hypertension among African Americans in the Jackson Heart Study. American Journal of Public Health, 102(2 Suppl.), S258 S265.

Smyth, J. M., Soefer, M. H., Hurewitz, A., Kliment, A., & Stone, A. A. (1999). Daily psychosocial factors predict levels and diurnal cycles of asthma symptomatology and peak flow. Journal of Behavioral Medicine, 22, 179 193.

Sodergren, S. C., & Hyland, M. H. (1999). Expectancy and asthma. In I. Kirsch (Ed.), How expectancies shape experience (pp. 197 212). American Psychological Association.

Speck, R. M., Courneya, K. S., Masse, L. C., Duval, S., & Schmitz, K. H. (2010). An update of controlled physical activity trials in cancer survivors: A systematic review and meta-analysis. Journal of Cancer Survivorship, 4, 87 100.

Stansfeld, S. A., Shipley, M. J., Head, J., & Fuhrer, R. (2012). Repeated job strain and the risk of depression: Longitudinal analyses from the Whitehall II study. American Journal of Public Health, 102, 2360 2366.

Stein, F. (2001). Occupational stress, relaxation therapies, exercise, and biofeedback. Work: Journal of Prevention, Assessment, and Rehabilitation, 17, 235 246.

Steptoe, A., O Donnell, K., Marmot, M., & Wardle, J. (2008). Positive affect and psychosocial processes related to health. British Journal of Psychology, 99, 211 227.

Stovner, L. J., Hagen, K., Jensen, R., Katsarava, Z., Lipson, R., Scher, A., . . . Zwart, J. (2007). The global burden of headache: A documentation of headache prevalence and disability worldwide. Cephalalgia, 27, 193 210.

Straub, R. O. (2007). Health psychology: A biopsychosocial approach (2nd ed.). Worth.

Stroebe, W., & Stroebe, M. (1996). The social psychology of social support. In E. T. Higgins & A. W. Kruglanski (Eds.), Social psychology: Handbook of basic principles (pp. 597 621). Guilford.

St rmer, T., Hasselbach, P., & Amelang, M. (2006). Personality, lifestyle, and risk of cardiovascular disease and cancer: Follow-up of population based cohort. British Medical Journal, 332, 1359 1362.

Suls, J., & Bunde, J. (2005). Anger, anxiety, and depression as risk factors for cardiovascular disease: The problems and implications of overlapping affective dispositions. Psychological Bulletin, 131, 260 300.

Sulsky, L., & Smith, C. (2005). Work stress. Thomson Wadsworth.

Surtees, P. G., Wainwright, N. W. J., Luben, R., Wareham, N. J., Bingham, S. A., & Khaw, K.-T. (2010). Mastery is associated with cardiovascular disease mortality in men and women at apparently low risk. Health Psychology, 29, 412 420.

Tatris, T. W., Peeters, M. C. W., Le Blanc, P. M., Schreurs, P. J. G., & Schaufeli, W. B. (2001). From inequity to burnout: The role of job stress. Journal of Occupational Health Psychology, 6, 303 323.

Taylor, S. E. (1999). Health psychology (4th ed.). McGraw-Hill.

Theorell, T., Tsutsumi, A., Hallquist, J., Reuterwall, C., Hogstedt, C., Fredlund, P., . . . Johnson, J. V. (1998). Decision latitude, job strain, and myocardial infarction: A study of working men in Stockholm. American Journal of Public Health, 88, 382 388.

Thoits, P. A. (2010). Stress and health: Major findings and policy implications. Journal of Health and Social Behavior, 51(1 Suppl.), S41 S53.

Trudel, X., Brisson, C., & Milot, A. (2010). Job strain and masked hypertension. Psychosomatic Medicine, 72, 786 793.

Trueba, A. F., & Ritz, T. (2013). Stress, asthma, and respiratory infections: Pathways involving airway immunology and microbal endocrinology. Brain, Behavior and Immunity, 29, 11 27.

Uchino, B. N. (2009). Understanding the links between social support and physical health: A life-span perspective with emphasis on the separability of perceived and received support. Perspectives on Psychological Science, 4, 236 255.

Uchino, B. N., Vaughn, A. A., Carlisle, M., & Birmingham, W. (2012). Social support and immunity. In S. C. Segerstrom (Ed.), The Oxford handbook of psychoneuroimmunology (pp. 214 233). Oxford University Press.

Ukestad, L. K., and Wittrock, D. A. (1996). Pain perception and coping in recurrent headache. Health Psychology, 15, 65 68.

van der Schurr, W. A., Baumgartner, S. E., & Sumter, S. R. (2018). Social media use, social media stress, and sleep: Examining cross-sectional and longitudinal relationships in adolescents. Health Communication, 34(5), 552 559. https://www.tandfonline.com/doi/full/10.1080/10410236.2017.1422101

Vella, E. J., Kamarck, T. W., Flory, J. D., & Manuck, S. (2012). Hostile mood and social strain during daily life: A test of the transactional model. Annals of Behavioral Medicine, 44, 341 352.

Vines, A., Ward, J. B., Cordoba, E., & Black, K. Z. (2017). Perceived racial/ethnic discrimination and mental health: A review and future directions for social epidemiology. Current Epidemiology Reports, 4(2), 156 165. https://www.ncbi.nlm.nih.gov/pubmed/28920011

von Leupoldt, A., Ehnes, F., & Dahme, B. (2006). Emotions and respiratory function in asthma: A comparison of findings in everyday life and laboratory. British Journal of Health Psychology, 11, 185 198.

Watson, D., Clark, L. A., & Tellegen, A. (1988). Development and validation of brief measures of positive and negative affect: The PANAS scales. Journal of Personality and Social Psychology, 54, 1063 1070.

Weeke, A. (1979). Causes of death in manic-depressive. In M. Shou & M. Stromgren (Eds.), Origin, prevention, and treatment of affective disorders (pp. 289 299). Academic Press.

Whang, W., Kubzansky, L. D., Kawachi, I., Rexrod, K. M., Kroenke, C. H., Glynn, R. J., . . . Albert, C. M. (2009). Depression and risk of sudden cardiac death and coronary heart disease in women: Results from the Nurses Health Study. Journal of the American College of Cardiology, 53, 950 958.

Wilson, T. D., & Gilbert, D. T. (2003). Affective forecasting. Advances in Experimental Social Psychology, 35, 345 411.

World Health Organization (WHO). (2013). A global brief on hypertension: Silent killer, global public health crisis. http://apps.who.int/iris/bitstream/10665/79059/1/WHO_DCO_WHD_2013.2_eng. pdf?ua=1

Wright, R. J., Rodriguez, M., & Cohen, S. (1998). Review of psychosocial stress and asthma: An integrated biopsychosocial approach. Thorax, 53, 1066 1074.

Wulsin, L. R., & Singal, B. M. (2003). Do depressive symptoms increase the risk for the onset of coronary disease? A systematic quantitative review. Psychosomatic Medicine, 65, 201 210.

Zacharie, R. (2009). Psychoneuroimmunology: A bio-psycho-social approach to health and disease. Scandinavian Journal of Psychology, 50, 645 651.

Psychological disorders

Abela, J. R., & Hankin, B. L. (2011). Rumination as a vulnerability factor to depression during the transition from early to middle adolescence: A multiwave longitudinal study. Journal of Abnormal Psychology, 120, 259 271.

Abramowitz, J. S., & Siqueland, L. (2013). Obsessive-compulsive disorder. In L. G. Castonguay & T. F. Oltmanns (Eds.), Psychopathology: From science to clinical practice (pp. 143 171). Guilford Press.

Abramson, L. Y., Metalsky, G. I., & Alloy, L. B. (1989). Hopelessness depression: A theory- based subtype of depression. Psychological Review, 96, 358 372.

Afifi, T. O., Mather, A., Boman, J., Fleisher, W., Enns, M. W., MacMillan, H., & Sareen, J. (2010). Childhood adversity and personality disorder: Results from a nationally representative population-based survey. Journal of Psychiatric Research, 45, 814 822.

Agerbo, E., Nordentoft, M., & Mortensen, P. B. (2002). Familial, psychiatric, and socioeconomic risk factors for suicide in young people: Nested case-control study. British Medical Journal, 325, 74 77.

Aghukwa, C. N. (2012). Care seeking and beliefs about the cause of mental illness among Nigerian psychiatric patients and their families. Psychiatric Services, 63, 616 618.

Aikins, D. E., & Craske, M. G. (2001). Cognitive theories of generalized anxiety disorder. Psychiatric Clinics of North America, 24, 57 74.

Akinbami, L. J., Liu, X., Pastor, P., & Reuben, C. A. (2011, August). Attention deficit hyperactivity disorder among children aged 5 17 years in the United States, 1998 2009 (NCHS data brief No. 70). National Center for Health Statistics. http://www.cdc.gov/nchs/data/databriefs/db70.pdf

Al-Asadi, A. M., Klein, B., & Meyer, D. (2015). Multiple comorbidities of 21 psychological disorders and relationships with psychosocial variables: A study of the online assessment and diagnostic system within a web-based population. Journal of Medical Internet Research, 17(2), e55. https://www.ncbi.nlm.nih.gov/pubmed/25803420

Alden, L. E., & Bieling, P. (1998). Interpersonal consequences in the pursuit of safety. Behaviour Research and Therapy, 36, 53 64.

Alegria, A. A., Blanco, C., Petry, N. M., Skodol, A. E., Liu, S. M., & Grant, B. (2013). Sex differences in antisocial personality disorder: Results from the National Epidemiological Survey on Alcohol and Related Conditions. Personality Disorders: Theory, Research, and Treatment, 4, 214 222.

American Psychological Association. (n.d.) Forensic psychology. https://www.apa.org/ed/graduate/specialize/forensic

Andreasen, N. C. (1987). The diagnosis of schizophrenia. Schizophrenia Bulletin, 13, 9 22.

Andr asson, S., Allbeck, P., Engstr m, A., & Rydberg, U. (1987). Cannabis and schizophrenia: A longitudinal study of Swedish conscripts. Lancet, 330, 1483 1486.

Asberg, M., Thor n, P., Tr skman, L., Bertilsson, L., & Ringberger, V. (1976). Serotonin depression a biochemical subgroup within the affective disorders? Science, 191(4226), 478 480. doi:10.1126/science.1246632

Asperger s syndrome dropped from psychiatrists handbook the DSM: DSM-5, latest revision of Diagnostic and Statistical Manual, merges Asperger's with autism and widens dyslexia category. (2012, December 1). The Guardian. http://www.theguardian.com/society/2012/dec/02/aspergers-syndrome-droppe d-psychiatric-dsm

Autism Genome Project Consortium. (2007). Mapping autism risk loci using genetic linkage and chromosomal rearrangements. Nature Genetics, 39, 319 328.

Baes, C. V. W., Tofoli, S. M. C., Martins, C. M. S., & Juruena, M. F. (2012). Assessment of the hypothalamic pituitary adrenal axis activity: Glucocorticoid receptor and mineralocorticoid receptor function in depression with early life stress a systematic review. Acta Neuropsychiatrica, 24, 4 15.

Baio, J., Wiggins, L., Christensen, D. L., et al. (2018). Prevalence of Autism Spectrum Disorder among children aged 8 years Autism and Developmental Disabilities Monitoring Network, 11 sites, United States, 2014. Surveillance Summaries, 67(6), 1 23. https://www.cdc.gov/mmwr/volumes/67/ss/ss6706a1.htm?s_cid=ss6706a1_w#sug gestedcitation

Baker, L. A., Bezdjian, S., & Raine, A. (2006). Behavioral genetics: The science of antisocial behavior. Law and Contemporary Problems, 69, 7 46.

Barbaresi, W. J., Colligan, R. C., Weaver, A. L., Voigt, R. G., Killian, J. M., & Katusic, S. K. (2013). Mortality, ADHD, and psychosocial adversity in adults with childhood ADHD: A prospective study. Pediatrics, 131, 637 644.

Barker, P. (2010). The legacy of Thomas Szasz. http://www.centerforindependentthought.org/SzaszLegacy.html

Barkley, R. A. (2006). Attention-deficit hyperactivity disorder: A handbook for diagnosis and treatment. Guilford Press.

Barkley, R. A., Fischer, M., Smallish, L., & Fletcher, K. (2002). The persistence of attention-deficit/hyperactivity disorder into young adulthood as a function of reporting source and definition of disorder. Journal of Abnormal Psychology, 111, 279 289.

Battista, S. R., & Kocovski, N. L. (2010). Exploring the effect of alcohol on post-event processing specific to a social event. Cognitive Behaviour Therapy, 39, 1 10.

Baumeister, A. A., & Hawkins, M. F. (2004). The serotonin hypothesis of schizophrenia: A historical case study on the heuristic value of theory in clinical neuroscience. Journal of the History of the Neurosciences, 13, 277 291.

Beck, A. T. (1976). Cognitive therapy and the emotional disorders. International Universities Press.

Beck, A. T. (2008). The evolution of the cognitive model of depression and its neurobiological correlates. American Journal of Psychiatry, 165, 969 977.

Berman, A. L. (2009). School-based suicide prevention: Research advances and practice implications. School Psychology Review, 38, 233 238.

Bettelheim, B. (1967). The empty fortress: Infantile autism and the birth of the self. Free Press.

Beucke, J. C., Sepulcre, J., Talukdar, T., Linnman, C., Zschenderlein, K., Endrass, T., . . . Kathman, N. (2013). Abnormally high degree connectivity of the orbitofrontal cortex in obsessive-compulsive disorder. JAMA Psychiatry, 70, 619 629.

Biederman, J., Faraone, S. V., Hirshfeld-Becker, D. R., Friedman, D., Robin, J. A., & Rosenbaum, J. F. (2001). Patterns of psychopathology and dysfunction in high-risk children of parents with panic disorder and major depression. American Journal of Psychiatry, 158, 49 57.

Boland, R. J., & Keller, M. B. (2009). Course and outcome of depression. In I. H. Gotlib & C. L. Hammen (Eds.), Handbook of depression (pp. 23 43). Guilford Press.

Bolton, D., Rijsdijk, F., O Connor, T. G., Perrin, S., & Eley, T. C. (2007). Obsessive-compulsive disorder, tics and anxiety in 6-year-old twins. Psychological Medicine, 37, 39 48.

Bourguignon, E. (1970). Hallucinations and trance: An anthropologist's perspective. In W. Keup (Ed.), Origins and mechanisms of hallucination (pp. 183 190). Plenum Press.

Bouton, M. E., Mineka, S., & Barlow, D. H. (2001). A modern learning theory perspective on the etiology of panic disorder. Psychological Review, 108, 4 32.

Bramsen, I., Dirkzwager, A. J. E., & van der Ploeg, H. M. (2000). Predeployment personality traits and exposure to trauma as predictors of posttraumatic stress symptoms: A prospective study of former peacekeepers. American Journal of Psychiatry, 157, 1115 1119.

Bregman, J. D. (2005). Definitions and characteristics of the spectrum. In D. Zager (Ed.), Autism spectrum disorders: Identification, education, and treatment (3rd ed., pp. 3 46). Erlbaum.

Brewin, C. R., Andrews, B., & Valentine, J. D. (2000). Meta-analysis of risk factors for posttraumatic stress disorder in trauma-exposed adults. Journal of Consulting and Clinical Psychology, 68, 748 756.

Bremner, J. D., Krystal, J. H., Southwick, S. M., & Charney, D. S. (1996). Noradrenergic mechanisms in stress and anxiety: I. preclinical studies. Synapse, 23, 28 38.

Brown, A. S., Begg, M. D., Gravenstein, S., Schaefer, C. A., Wyatt, R. J., Breshnahan, M., . . . Susser, E. S. (2004). Serologic evidence of prenatal influenza in the etiology of schizophrenia. Archives of General Psychiatry, 61, 774 780.

Brent, D. A., & Bridge, J. (2003). Firearms availability and suicide: A review of the literature. American Behavioral Scientist, 46, 1192 1210.

Brown, G. W., & Harris, T. O. (1989). Depression. In G. W. Brown and T. O. Harris (Eds.), Life events and illness (pp. 49 93). Guilford Press.

Brown, G. W., Ban, M., Craig, T. J. K., Harris, T. O., Herbert, J., & Uher, R. (2013). Serotonin transporter length polymorphism, childhood maltreatment, and chronic depression: A specific gene-environment interaction. Depression and Anxiety, 30, 5 13.

Buchanan-Barker, P., Barker, P. (2009, February). The convenient myth of Thomas Szasz. Journal of Psychiatric and Mental Health Nursing, 16(1): 87 95. doi:10.1111/j.1365-2850.2008.01310.x

Buchsbaum, M. S., Nuechterlein, K. H., Haier, R. J., Wu, J., Sicotte, N., Hazlett, E., . . . Guich, S. (1990). Glucose metabolic rate in normal and schizophrenics during the continuous performance test assessed by positron emission tomography. British Journal of Psychiatry, 156, 216 227.

Burt, S. A. (2009). Rethinking environmental contributions to child and adolescent psychopathology: A meta-analysis of shared environmental influences. Psychological Bulletin, 135, 608 637.

Butler, L. D., & Nolen-Hoeksema, S. (1994). Gender differences in responses to depressed mood in a college sample. Sex Roles, 30, 331 346.

Byrd, R. (2002, October 17). Report to the legislature on the principal findings from the epidemiology of autism in California: A comprehensive pilot study. http://www.dds.ca.gov/Autism/MindReport.cfm

Cadoret, R., Yates, W., Ed., T., Woodworth, G., & Stewart, M. (1995). Genetic environmental interactions in the genesis of aggressivity and conduct disorders. Archives of General Psychiatry, 52, 916 924.

Cannon, M., Jones, P. B., & Murray, R. M. (2002). Obstetric complications and schizophrenia: Historical and meta-analytic review. American Journal of Psychiatry, 159, 1080 1092.

Casadio, P., Fernandes, C., Murray, R. M., & Di Forti, M. (2011). Cannabis use in young people: The risk for schizophrenia. Neuroscience and Biobehavioral Reviews, 35, 1779 1787.

Carde a, E., & Gleaves, D. H. (2006). Dissociative disorders. In M. Hersen, S. M. Turner, & D. C. Beidel (Eds.), Adult psychopathology and diagnosis (pp. 473 503). John Wiley & Sons.

Caspi, A., Sugden, K., Moffitt, T. E., Taylor, A., Craig, I. W., Harrington, H., Poulton, R. (2003). Influence of life stress on depression: Moderation by a polymorphism in the 5-HTT gene. Science, 301(5631), 386 389.

Centers for Disease Control and Prevention. (2010, November 12). Increasing prevalence of parent-reported attention-deficit/hyperactivity disorder among children, United States, 2003 2007. Morbidity and Mortality Weekly Report, 59(44), 1439 1443.

Centers for Disease Control and Prevention. (2012). Suicide: Facts at a glance. http://www.cdc.gov/ViolencePrevention/pdf/Suicide_DataSheet-a.pdf

Centers for Disease Control and Prevention. (2012, March 30). Prevalence of autism spectrum disorders autism and developmental disabilities monitoring network, 14 sites, United States, 2008. Morbidity and Mortality Weekly Report: Surveillance Summaries, 61(3), 1 19. http://www.cdc.gov/mmwr/pdf/ss/ss6103.pdf

Centers for Disease Control and Prevention. (2013a). Definitions: Self-directed violence. http://www.cdc.gov/violenceprevention/suicide/definitions.html.

Centers for Disease Control and Prevention. (2013b). National suicide statistics at a glance: Trends in suicide rates among both sexes, by age group, United States, 1991 2009. http://www.cdc.gov/violenceprevention/suicide/statistics/trends02.html

Charney, D. S., Woods, S. W., Nagy, L. M., Southwick, S. M., Krystal, J. H., & Heninger, G. R. (1990). Noradrenergic function in panic disorder. Journal of Clinical Psychiatry, 51, 5 11.

Clark, D. M. (1996). Panic disorder: From theory to therapy. In R. M. Rapee (Ed.), Current controversies in the anxiety disorders (pp. 318 344). Guilford Press.

Clauss, J. A., & Blackford, J. U. (2012). Behavioral inhibition and risk for developing social anxiety disorder: A meta-analytic study. Journal of the American Academy of Child and Adolescent Psychiatry, 51(10), 1066 1075.

Clegg, J. W. (2012). Teaching about mental health and illness through the history of the DSM. History of Psychology, 15, 364 370.

Compton, W. M., Conway, K. P., Stinson, F. S., Colliver, J. D., & Grant, B. F. (2005). Prevalence, correlates, and comorbidity of DSM-IV antisocial personality syndromes and alcohol and specific drug use disorders in the United States: Results from the national epidemiologic survey on alcohol and related conditions. Journal of Clinical Psychiatry, 66, 677 685.

Cook, M., & Mineka, S. (1989). Observational conditioning of fear to fear-relevant versus fear-irrelevant stimuli in rhesus monkeys. Journal of Abnormal Psychology, 98, 448 459.

Craske, M. G. (1999). Anxiety disorders: Psychological approaches to theory and treatment. Westview Press.

Crego, C., & Widiger, T. A. (2014). Psychopathy and the DSM. Journal of Personality. http://www.sakkyndig.com/psykologi/artvit/crego2014.pdf

Crosby, A. E., Ortega, L., & Melanson, C. (2011). Self-directed violence surveillance: Uniform definitions and recommended data elements, version 1.0. http://www.cdc.gov/violenceprevention/pdf/self-directed-violence-a.pdf

Dalenberg, C. J., Brand, B. L., Gleaves, D. H., Dorahy, M. J., Loewenstein, R. J., Carde a, E., . . . Spiegel, D. (2012). Evaluation of the evidence for the trauma and fantasy models of dissociation. Psychological Bulletin, 138, 550 588.

Dancing mania. (2011, July 2). Sometimes interesting: Weird, forgotten, and sometimes interesting things. https://sometimes-interesting.com/2011/07/02/dancing-mania/

Davidson, R. J., Pizzagalli, D. A., & Nitschke, J. B. (2009). Representation and regulation of emotional depression: Perspectives from cognitive neuroscience. In I. H. Gotlib & C. L. Hammen (Eds.), Handbook of depression (pp. 218 248). Guilford Press.

Davis, K. L., Kahn, R. S., Ko., G., & Davidson, M. (1991). Dopamine in schizophrenia: A review and reconceptualization. American Journal of Psychiatry, 148, 1474 1486.

Demos, J. (1983). Entertaining Satan: Witchcraft and the culture of early New England. Oxford University Press.

Dempsey, A. F., Schaffer, S., Singer, D., Butchart, A., Davis, M., & Freed, G. L. (2011). Alternative vaccination schedule preferences among parents of young children. Pediatrics, 128, 848 856.

DeStefano, F., Price, C. S., & Weintraub, E. S. (2013). Increasing exposures to antibody-stimulating proteins and polysaccharides in vaccines is not associated with risk of autism. The Journal of Pediatrics, 163, 561 567.

DiGrande, L., Perrin, M. A., Thorpe, L. E., Thalji, L., Murphy, J., Wu, D., . . . Brackbill, R. M. (2008). Posttraumatic stress symptoms, PTSD, and risk factors among lower Manhattan residents 2 3 years after the September 11, 2001 terrorist attacks. Journal of Traumatic Stress, 21, 264 273.

DNA project aims to count Scots redheads. (2012, November 7). British Broadcast Corporation [BBC]. http://www.bbc.com/news/uk-scotland-20237511

Downs, M. (2008, March 31). Autism-vaccine link: Evidence doesn t dispel doubts. www.webmd.com/brain/autism/searching-for-answers/vaccines-autism

Dozois, D. J. A., & Beck, A. T. (2008). Cognitive schemas, beliefs and assumptions. In K. S. Dobson & D. J. A. Dozois (Eds.), Risk factors in depression (pp. 121 143). Academic Press.

Drevets, W. C., Bogers, W. U., & Raichle, M. E. (2002). Functional anatomical correlates of antidepressant drug treatment assessed using PET measures of regional glucose metabolism. European Neuropsychopharmacology, 12, 527 544.

Ehlers, A., & Clark, D. M. (2000). A cognitive model of posttraumatic stress disorder. Behaviour Research and Therapy, 38, 319 345.

Etzersdorfer, E., Voracek, M., & Sonneck, G. (2004). A dose-response relationship between imitational suicides and newspaper distribution. Archives of Suicide Research, 8, 137 145.

Fabrega, H. (2007). How psychiatric conditions were made. Psychiatry, 70, 130 153.

Facer-Irwin, E., Blackwood, N. J., Bird, A., Dickson, H., McGlade, D., Alves-Costa, F., & MacManus, D. (2019, Sept 26). PTSD in prison settings: A systematic review and meta-analysis of comorbid mental disorders and problematic behaviours. PLOS ONE, 14(9). https://journals.plos.org/plosone/article?id=10.1371/journal.pone.022240 7

Fitzgerald, P. B., Laird, A. R., Maller, J., & Daskalakis, Z. J. (2008). A meta-analytic study of changes in brain activation in depression. Human Brain Mapping, 29, 683 695.

Fields, T. (2010). Postpartum depression effects on early interactions, parenting, and safety practices: A review. Infant Behavior and Development, 33, 1 6.

Fisher, C. (2010, February 11). DSM-5 development process included emphasis on gender and cultural sensitivity. http://www.bmedreport.com/archives/9359

Fleischman, A., Bertolote, J. M., Belfer, M., & Beautrais, A. (2005). Completed suicide and psychiatric diagnoses in young people: A critical examination of the evidence. American Journal of Orthopsychiatry, 75, 676 683.

Foote, B., Smolin, Y., Kaplan, M., Legatt, M. E., & Lipschitz, D. (2006). Prevalence of dissociative disorders in psychiatric outpatients. American Journal of Psychiatry, 163, 623 629.

Fox, N. A., Henderson, H. A., Marshall, P. J., Nichols, K. E., & Ghera, M. M. (2005). Behavioral inhibition: Linking biology and behavior within a developmental framework. Annual Review of Psychology, 56, 235 262.

Frances, A. (2012, December 2). DSM 5 is guide not bible ignore its ten worst changes. http://www.psychologytoday.com/blog/dsm5-in-distress/201212/dsm-5-is-gui de-not-bible-ignore-its-ten-worst-changes

Freeman, A., Stone, M., Martin, D., & Reinecke, M. (2005). A review of borderline personality disorder. In A. Freeman, M. Stone, D. Martin, & M. Reinecke (Eds.), Comparative treatments for borderline personality disorder (pp. 1 20). Springer.

Fung, M. T., Raine, A., Loeber, R., Lynam, D. R., Steinhauer, S. R., Venables, P. H., & Stouthamer-Loeber, M. (2005). Reduced electrodermal activity in psychopathy-prone adolescents. Journal of Abnormal Psychology, 114, 187 196.

Fusar-Poli, P., Borgwardt, S., Bechdolf, A., Addington, J., Riecher-R ssler, A., Schultze-Lutter, F., . . . Yung, A. (2013). The psychosis high-risk state: A comprehensive state-of-the-art review. Archives of General Psychiatry, 70, 107 120.

Gauthier, J., Siddiqui, T. J., Huashan, P., Yokomaku, D., Hamdan, F. F., Champagne, N., . . . Rouleau, G.A. (2011). Truncating mutations in NRXN2 and NRXN1 in autism spectrum disorders and schizophrenia. Human Genetics, 130, 563 573.

Gizer, I. R., Ficks, C., & Waldman, I. D. (2009). Candidate gene studies of ADHD: A meta-analytic review. Human Genetics, 126, 51 90.

Goldstein, A. J., & Chambless, D. L. (1978). A reanalysis of agoraphobia. Behavior Therapy, 9, 47 59.

Goldstein, J. M., Buka, S. L., Seidman, L. J., & Tsuang, M. T. (2010). Specificity of familial transmission of schizophrenia psychosis spectrum and affective psychoses in the New England family study s high-risk design. Archives of General Psychiatry, 67, 458 467.

Good, B. J., & Hinton, D. E. (2009). Panic disorder in cross-cultural and historical perspective. In D. E. Hinton & B. J. Good (Eds.), Culture and panic disorder (pp. 1 28). Stanford University Press.

Goodman, S. H., & Brand, S. R. (2009). Depression and early adverse experiences. In I. H. Gotlib & C. L. Hammen (Eds.), Handbook of depression (pp. 249 274). Guilford Press.

Gotlib, I. H., & Joormann, J. (2010). Cognition and depression: Current status and future directions. Annual Review of Clinical Psychology, 6, 285 312.

Gottesman, I. I. (2001). Psychopathology through a life span-genetic prism. American Psychologist, 56, 867 878.

Graybiel, A. N., & Rauch, S. L. (2000). Toward a neurobiology of obsessive-compulsive disorder. Neuron, 28, 343 347.

Green, M. F. (2001). Schizophrenia revealed: From neurons to social interactions. W. W. Norton.

Hackmann, A., Clark, D. M., & McManus, F. (2000). Recurrent images and early memories in social phobia. Behaviour Research and Therapy, 38, 601 610.

Halligan, S. L., Herbert, J., Goodyer, I., & Murray, L. (2007). Disturbances in morning cortisol secretion in association with maternal postnatal depression predict subsequent depressive symptomatology in adolescents. Biological Psychiatry, 62, 40 46.

Hare, R. D. (1965). Temporal gradient of fear arousal in psychopaths. Journal of Abnormal Psychology, 70, 442 445.

Hasin, D. S., Fenton, M. C., & Weissman, M. M. (2011). Epidemiology of depressive disorders. In M. T. Tsuang, M. Tohen, & P. Jones (Eds.), Textbook of psychiatric epidemiology (pp. 289 309). John Wiley & Sons.

Herek, G. M. (2012). Facts about homosexuality and mental health. Sexual orientation: Science, education, and policy. https://psychology.ucdavis.edu/rainbow/html/facts_mental_health.html

Herman, J. (1997). Trauma and recovery: The aftermath of violence from domestic abuse to political terror. Basic Books.

Herrenkohl, T. I., Maguin, E., Hill, K. G., Hawkins, J. D., Abbott, R. D., & Catalano, R. (2000). Developmental risk factors for youth violence. Journal of Adolescent Health, 26, 176 186.

Heston, L. L. (1966). Psychiatric disorders in foster home reared children of schizophrenic mothers. British Journal of Psychiatry, 112, 819 825.

Hettema, J. M., Neale, M. C., & Kendler, K. S. (2001). A review and meta-analysis of the genetic epidemiology of anxiety disorders. The American Journal of Psychiatry, 158, 1568 1578.

Holsboer, F., & Ising, M. (2010). Stress hormone regulation: Biological role and translation into therapy. Annual Review of Psychology, 61, 81 109.

Hopwood, C. J., Schade, N., Krueger, R. F., Wright, A. G. C., & Markon, K. E. (2012). Connecting DSM-5 personality traits and pathological beliefs: Toward a unifying model. Journal of Psychopathology and Behavioral Assessment, 35(2). https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3833658/

Howes, O. D., & Kapur, S. (2009). The dopamine hypothesis of schizophrenia: Version III The final common pathway. Schizophrenia Bulletin, 35, 549 562.

Hoza, B., Mrug, S., Gerdes, A. C., Hinshaw, S. P., Bukowski, W. M., Gold, J. A., . . . Arnold, L. E. (2005). What aspects of peer relationships are impaired in children with ADHD? Journal of Consulting and Clinical Psychology, 73, 411 423.

Hughes, V. (2007). Mercury rising. Nature Medicine, 13, 896 897.

Jaegers, L. A., Matthieu, M. M., Vaughn, M. G., Werth, P., Katz I. M., & Ahmad, S. O. (2019, June). Posttraumatic stress disorder and job burnout among jail officers. Journal of Occupational and Environmental Medicine, 61(6), 505 510. https://journals.lww.com/joem/Citation/2019/06000/Posttraumatic_Stress_Disorder_and_Job_Burnout.9.aspx

Jellinek, M. S., & Herzog, D. B. (1999). The child. In A. M. Nicholi, Jr. (Ed.), The Harvard guide to psychiatry (pp. 585 610). The Belknap Press of Harvard University.

Johnson, J. G., Cohen, P., Kasen, S., & Brook, J. S. (2006). Dissociative disorders among adults in the community, impaired functioning, and axis I and II comorbidity. Journal of Psychiatric Research, 40, 131 140.

Joormann, J. (2009). Cognitive aspects of depression. In I. H. Gotlib & C. L. Hammen (Eds.), Handbook of depression (pp. 298 321). Guilford Press.

Joyce, P. R., McKenzie, J. M., Luty, S. E., Mulder, R. T., Carter, J. D., Sullivan, P. F., & Cloninger, C. R. (2003). Temperament, childhood environment, and psychopathology as risk factors for avoidant and borderline personality disorders. Australian and New Zealand Journal of Psychiatry, 37, 756 764.

Judd, L. L. (2012). Dimensional paradigm of the long-term course of unipolar major depressive disorder. Depression and Anxiety, 29, 167 171.

Kagan, J., Reznick, J. S., & Snidman, N. (1988). Biological bases of childhood shyness. Science, 240, 167 171.

Katzelnick, D. J., Kobak, K. A., DeLeire, T., Henk, H. J., Greist, J. H., Davidson, J. R. T., . . . Helstad, C. P. (2001). Impact of generalized social anxiety disorder in managed care. The American Journal of Psychiatry, 158, 1999 2007.

Kendler, K. S., Hettema, J. M., Butera, F., Gardner, C. O., & Prescott, C. A. (2003). Life event dimensions of loss, humiliation, entrapment, and danger in the prediction of onsets of major depression and generalized anxiety. Archives of General Psychiatry, 60, 789 796.

Kennedy, A., LaVail, K., Nowak, G., Basket, M., & Landry, S. (2011). Confidence about vaccines in the United States: Understanding parents perceptions. Health Affairs, 30, 1151 1159.

Kessler, R. C. (1997). The effects of stressful life events on depression. Annual Review of Psychology, 48, 191 214.

Kessler, R. C. (2003). Epidemiology of women and depression. Journal of Affective Disorders, 74, 5 13.

Kessler, R. C., Berglund, P., Demler, O., Jin, R., Merikangas, K. P., & Walters, E. F. (2005). Lifetime prevalence and age-of-onset distributions of DSM-IV disorders in the National Comorbidity Survey Replication. Archives of General Psychiatry, 62, 593 602.

Kessler, R. C., Chiu, W. T., Jin, R., Ruscio, A. M., Shear, K., & Walters, E. (2006). The epidemiology of panic attacks, panic disorder, and agoraphobia in the National Comorbidity Survey Replication. Archives of General Psychiatry, 63, 415 424.

Kessler, R. C., Galea, S., Gruber, M. J., Sampson, N. A., Ursano, R. J., & Wessely, S. (2008). Trends in mental illness and suicidality after Hurricane Katrina. Molecular Psychiatry, 13, 374 384.

Kessler, R. C., Ruscio, A. M., Shear, K., & Wittchen, H. U. (2009). Epidemiology of anxiety disorders. In M. B. Stein & T. Steckler (Eds.), Behavioral neurobiology of anxiety and its treatment (pp. 21 35). Springer.

Kessler, R. C. Sonnega, A., Bromet, E., Hughes, M., & Nelson, C. B. (1995). Posttraumatic stress disorder in the National Comorbidity Survey. Archives of General Psychiatry, 52, 1048 1060.

Kessler, R. C., & Wang, P. S. (2009). Epidemiology of depression. In I. H. Gotlib & C. L. Hammen (Eds.), Handbook of depression (pp. 5 22). Guilford Press.

Khashan, A. S., Abel, K. M., McNamee, R., Pedersen, M. G., Webb, R., Baker, P., . . . Mortensen, P. B. (2008). Higher risk of offspring schizophrenia following antenatal maternal exposure to severe adverse life events. Archives of General Psychiatry, 65, 146 152.

Kilpatrick, D. G., Badour, C. L., & Resnick, H. S. (2017). Trauma and posttraumatic stress disorder prevalence and sociodemographic characteristics. In S. N. Gold (Ed.), APA handbook of trauma psychology: Foundations in knowledge: Volume 1 (pp. 63 85). American Psychological Association.

Kinney, D. K., Barch, D. H., Chayka, B., Napoleon, S., & Munir, K. M. (2009). Environmental risk factors for autism: Do they help or cause de novo genetic mutations that contribute to the disorder? Medical Hypotheses, 74, 102 106.

Kleim, B., Gonzalo, D., & Ehlers, A. (2011). The Depressive Attributions Questionnaire (DAQ): Development of a short self-report measure of depressogenic attributions. Journal of Psychopathology and Behavioral Assessment, 33, 375 385.

Klein, R. G., Mannuzza, S., Olazagasti, M. A. R., Roizen, E., Hutchison, J. A., Lashua, E. C., & Castellanos, F. X. (2012). Clinical and functional outcome of childhood attention-deficit/hyperactivity disorder 33 years later. Archives of General Psychiatry, 69, 1295 1303.

Koenen, K. C., Stellman, J. M., Stellman, S. D., & Sommer, J. F. (2003). Risk factors for course of posttraumatic stress disorder among Vietnam veterans: A 14-year follow-up of American Legionnaires. Journal of Consulting and Clinical Psychology, 71, 980 986.

Kopell, B. H., & Greenberg, B. D. (2008). Anatomy and physiology of the basal ganglia: Implications for DBS in psychiatry. Neuroscience and Biobehavioral Reviews, 32, 408 422.

Large, M., Sharma, S., Compton, M. T., Slade, T., & Nielssen, O. (2011). Cannabis use and earlier onset of psychosis: A systematic meta-analysis. Archives of General Psychiatry, 68, 555 561.

Lasalvia, A., Zoppei, S., Van Bortel, T., Bonetto, C., Cristofalo, D., Wahlbeck, K., Thornicroft, G. (2013). Global pattern of experienced and anticipated discrimination reported by people with major depressive disorder: A cross-sectional survey. The Lancet, 381, 55 62.

Lawrie, S. M., & Abukmeil, S. S. (1998). Brain abnormality in schizophrenia: A systematic and quantitative review of volumetric magnetic resonance imaging studies. British Journal of Psychiatry, 172, 110 120.

LeMoult, J., Castonguay, L. G., Joormann, J., & McAleavey, A. (2013). Depression. In L. G. Castonguay & T. F. Oltmanns (Eds.), Psychopathology: From science to clinical practice (pp. 17 61). Guilford Press.

Lezenweger, M. F., Lane, M. C., Loranger, A. W., & Kessler, R. C. (2007). DSM-IV personality disorders in the National Comorbidity Survey Replication. Biological Psychiatry, 62, 553 564.

Lilienfeld, S. O., & Marino, L. (1999). Essentialism revisited: Evolutionary theory and the concept of mental disorder. Journal of Abnormal Psychology, 108, 400 411.

Linnet, K. M., Dalsgaard, S., Obel, C., Wisborg, K., Henriksen, T. B., Rodriquez, A., . . . Jarvelin, M. R. (2003). Maternal lifestyle factors in pregnancy risk of attention deficit hyperactivity disorder and associated behaviors: A review of current evidence. The American Journal of Psychiatry, 160, 1028 1040.

Livesley, J. (2008). Toward a genetically-informed model of borderline personality disorder. Journal of Personality Disorders, 22, 42 71.

Livesley, J., & Jang, K. L. (2008). The behavioral genetics of personality disorders. Annual Review of Clinical Psychology, 4, 247 274.

Loe, I. M., & Feldman, H. M. (2007). Academic and educational outcomes of children with ADHD. Journal of Pediatric Psychology, 32, 643 654.

Luxton, D. D., June, J. D., & Fairall, J. M. (2012, May). Social media and suicide: A public health perspective. American Journal of Public Health, 102(S2), S195 S200. doi:10.2105/AJPH.2011.300608

Mackin, P., & Young, A. H. (2004, May 1). The role of cortisol and depression: Exploring new opportunities for treatments. Psychiatric Times. http://www.psychiatrictimes.com/articles/role-cortisol-and-depression-exploring-new-opportunities-treatments

Maher, W. B., & Maher, B. A. (1985). Psychopathology: I. from ancient times to the eighteenth century. In G. A. Kimble & K. Schlesinger (Eds.), Topics in the history of psychology: Volume 2 (pp. 251 294). Erlbaum.

Mann, J. J., (2003). Neurobiology of suicidal behavior. Nature Reviews Neuroscience, 4, 819 828.

Marker, C. D. (2013, March 3). Safety behaviors in social anxiety: Playing it safe in social anxiety. http://www.psychologytoday.com/blog/face-your-fear/201303/safety-behaviors-in-social-anxiety

Martens, E. J., de Jonge, P., Na, B., Cohen, B. E., Lett, H., & Whooley, M. A. (2010). Scared to death? Generalized anxiety disorder and cardiovascular events in patients with stable coronary heart disease. Archives of General Psychiatry, 67, 750 758.

Mataix-Cols, D., Frost, R. O., Pertusa, A., Clark, L. A., Saxena, S., Leckman, J. F., . . . Wilhelm, S. (2010). Hoarding disorder: A new diagnosis for DSM-V? Depression and Anxiety, 27, 556 572.

Mayes, R., & Horowitz, A. V. (2005). DSM-III and the revolution in the classification of mental illness. Journal of the History of the Behavioral Sciences, 41, 249 267.

Mazure, C. M. (1998). Life stressors as risk factors in depression. Clinical Psychology: Science and Practice, 5, 291 313.

Marshal, M. P., & Molina, B. S. G. (2006). Antisocial behaviors moderate the deviant peer pathway to substance use in children with ADHD. Journal of Clinical Child and Adolescent Psychology, 35, 216 226.

McCabe, K. (2010, January 24). Teen s suicide prompts a look at bullying. Boston Globe. http://www.boston.com

McCabe, R. E., Antony, M. M., Summerfeldt, L. J., Liss, A., & Swinson, R. P. (2003). Preliminary examination of the relationship between anxiety disorders in adults and self-reported history of teasing or bullying experiences. Cognitive Behaviour Therapy, 32, 187 193.

McCann, D., Barrett, A., Cooper, A., Crumpler, D., Dalen, L., Grimshaw, K., . . . Stevenson, J. (2007). Food additives and hyperactive behaviour in 3-year-old and 8/9-year-old children in the community: A randomised, double-blinded, placebo-controlled trial. The Lancet, 370(9598), 1560 1567.

McEwen, B. S. (2005). Glucocorticoids, depression, and mood disorders: Structural remodeling in the brain. Metabolism: Clinical and Experimental, 54, 20 23.

McNally, R. J. (2003). Remembering trauma. Harvard University Press.

Meek, S. E., Lemery-Chalfant, K., Jahromi, L. D., & Valiente, C. (2013). A review of gene-environment correlations and their implications for autism: A conceptual model. Psychological Review, 120, 497 521.

Mental health myths and facts. (2017, August 29). MentalHealth.gov. https://www.mentalhealth.gov/basics/mental-health-myths-facts

Merikangas, K. R., & Tohen, M. (2011). Epidemiology of bipolar disorder in adults and children. In M. T. Tsuang, M. Tohen, & P. Jones (Eds.), Textbook of psychiatric epidemiology (pp. 329 342). John Wiley & Sons.

Merikangas, K. R., Jin, R., He, J. P., Kessler, R. C., Lee, S., Sampson, N. A., Zarkov, Z. (2011). Prevalence and correlates of bipolar spectrum disorder in the World Mental Health Survey Initiative. Archives of General Psychiatry, 68, 241 251.

Mezzich, J. E. (2002). International surveys on the use of ICD-10 and related diagnostic systems. Psychopathology, 35, 72 75.

Michaud, K., Matheson, K., Kelly, O., & Anisman, H. (2008). Impact of stressors in a natural context on release of cortisol in healthy adult humans: A meta-analysis. Stress, 11, 177 197.

Mineka, S., & Cook, M. (1993). Mechanisms involved in the observational conditioning of fear. Journal of Experimental Psychology: General, 122, 23 38.

Moffitt, T. E., Caspi, A., Harrington, H., Milne, B. J., Melchior, M., Goldberg, D., & Poulton, R. (2007). Generalized anxiety disorder and depression: Childhood risk factors in a birth cohort followed to age 32. Psychological Medicine, 37, 441 452.

Moitra, E., Beard, C., Weisberg, R. B., & Keller, M. B. (2011). Occupational impairment and social anxiety disorder in a sample of primary care patients. Journal of Affective Disorders, 130, 209 212.

Molina, B. S. G., & Pelham, W. E. (2003). Childhood predictors of adolescent substance abuse in a longitudinal study of children with ADHD. Journal of Abnormal Psychology, 112, 497 507.

Moore, T. H., Zammit, S., Lingford-Hughes, A., Barnes, T. R., Jones, P. B., Burke, M., & Lewis, G. (2007). Cannabis use and risk of psychotic or affective mental health outcomes. Lancet, 370, 319 328.

Morris, E. P., Stewart, S. H., & Ham, L. S. (2005). The relationship between social anxiety disorder and alcohol use disorders: A critical review. Clinical Psychology Review, 25, 734 760.

Mowrer, O. H. (1960). Learning theory and behavior. John Wiley & Sons.

Nader, K. (2001). Treatment methods for childhood trauma. In J. P. Wilson, M. J. Friedman, & J. D. Lindy (Eds.), Treating psychological trauma and PTSD (pp. 278 334). Guilford Press.

Nanni, V., Uher, R., & Danese, A. (2012). Childhood maltreatment predicts unfavorable course of illness and treatment outcome in depression: A meta-analysis. American Journal of Psychiatry, 169, 141 151.

Nathan, D. (2011). Sybil exposed: The extraordinary story behind the famous multiple personality case. Free Press.

National Comorbidity Survey. (2007). NCS-R lifetime prevalence estimates. http://www.hcp.med.harvard.edu/ncs/index.php

National Institute on Drug Abuse (NIDA). (2007, October). Comorbid drug use and mental illness: A research update from the National Institute on Drug Abuse. http://www.drugabuse.gov/sites/default/files/comorbid.pdf

National Institute on Drug Abuse (NIDA). (2018, February). Common comorbidities with substance use disorders. https://www.drugabuse.gov/node/pdf/1155/common-comorbidities-with-substa nce-use-disorders

Nelson, A. J. (2014, Nov. 8). Teen was insane when he killed child, judge rules. Omaha World-Herald. https://www.omaha.com/eedition/sunrise/articles/teen-was-insane-when-he-killed-child-judge-rules/article_d680d09b-be8e-55c3-992f-22da28e60bf8.html

Nestadt, G., Samuels, J., Riddle, M., Bienvenu, J., Liang, K. Y., LaBuda, M., . . . Hoehn-Saric, R. (2000). A family study of obsessive-compulsive disorder. Archives of General Psychiatry, 57, 358 363.

Newman, C. F. (2004). Suicidality. In S. L. Johnson & R. L. Leahy (Eds.), Psychological treatment of bipolar disorder (pp. 265 285). Guilford Press.

Nikolas, M. A., & Burt, S. A. (2010). Genetic and environmental influences on ADHD symptom dimensions of inattention and hyperactivity: A meta-analysis. Journal of Abnormal Psychology, 119, 1 17.

Nolen-Hoeksema, S. (1987). Sex differences in unipolar depression: Evidence and theory. Psychological Bulletin, 101, 259 282.

Nolen-Hoeksema, S. (1991). Responses to depression and their effects on the duration of depressive episodes. Journal of Abnormal Psychology, 100, 569 582.

Nolen-Hoeksema, S. & Hilt, L. M. (2009). Gender differences in depression. In I. H. Gotlib & C. L. Hammen (Eds.), Handbook of depression (pp. 386 404). Guilford Press.

Nolen-Hoeksema, S., Larson, J., & Grayson, C. (1999). Explaining the gender difference in depressive symptoms. Journal of Personality and Social Psychology, 77, 1061 1072.

Norberg, M. M., Calamari, J. E., Cohen, R. J., & Riemann, B. C. (2008). Quality of life in obsessive-compulsive disorder: An evaluation of impairment and a preliminary analysis of the ameliorating effects of treatment. Depression and Anxiety, 25, 248 259.

Novella, S. (2008, April 16). The increase in autism diagnoses: Two hypotheses. http://www.sciencebasedmedicine.org/the-increase-in-autism-diagnoses-two -hypotheses/

Novick, D. M., Swartz, H. A., & Frank, E. (2010). Suicide attempts in bipolar I and bipolar II disorder: A review and meta-analysis of the evidence. Bipolar Disorders, 12, 1 9.

Noyes, R. (2001). Comorbidity in generalized anxiety disorder. Psychiatric Clinics of North America, 24, 41 55.

O Connor, R. C., Smyth, R., Ferguson, E., Ryan, C., & Williams, J. M. G. (2013). Psychological processes and repeat suicidal behavior: A four-year prospective study. Journal of Consulting and Clinical Psychology. Advance online publication. doi:10.1037/a0033751

hman, A., & Mineka, S. (2001). Fears, phobias, and preparedness: Toward an evolved module of fear and fear learning. Psychological Review, 108, 483 552.

Oliver, J. (2006, Summer). The myth of Thomas Szasz. The New Atlantis, 13. http://www.thenewatlantis.com/docLib/TNA13-Oliver.pdf

Olsson, A., & Phelps, E. A. (2007). Social learning of fear. Nature Neuroscience, 10, 1095 1102.

Oltmanns, T. F., & Castonguay, L. G. (2013). General issues in understanding and treating psychopathology. In L. G. Castonguay & T. F. Oltmanns (Eds.), Psychopathology: From science to clinical practice (pp. 1 16). Guilford Press.

Orr, S. P., Metzger, L. J., Lasko, N. B., Macklin, M. L., Peri, T., & Pitman, R. K. (2000). De novo conditioning in trauma-exposed individuals with and without posttraumatic stress disorder. Journal of Abnormal Psychology, 109, 290 298.

Owens, D., Horrocks, J., & House, A. (2002). Fatal and non-fatal repetition of self-harm: Systematic review. British Journal of Psychiatry, 181, 193 199.

Ozer, E. J., Best, S. R., Lipsey, T. L., & Weiss, D. S. (2003). Predictors of posttraumatic stress disorder and symptoms in adults: A meta-analysis. Psychological Bulletin, 129, 52 73.

Parker-Pope, T. (2013, May 2). Suicide rates rise sharply in U.S. The New York Times. http://www.nytimes.com.

Patterson, M. L., Iizuka, Y., Tubbs, M. E., Ansel, J., Tsutsumi, M., & Anson, J. (2007). Passing encounters east and west: Comparing Japanese and American pedestrian interactions. Journal of Nonverbal Behavior, 31, 155 166.

Pauls, D. L. (2010). The genetics of obsessive-compulsive disorder: A review. Dialogues in Clinical Neuroscience, 12, 149 163.

Paykel, E. S. (2003). Life events and affective disorders. Acta Psychiatrica Scandinavica, 108(S418), 61 66.

Pazain, M. (2010, December 2). To look or not to look? Eye contact differences in different cultures. http://www.examiner.com/article/to-look-or-not-to-look-eye-contact-diffe rences-different-cultures

Phan, K. L., Fitzgerald, D. A., Nathan, P. J., Moore, G. J., Uhde, T. W., & Tancer, M. E. (2005). Neural substrates for voluntary suppression of negative affect: A functional magnetic resonance imaging study. Biological Psychiatry, 57, 210 219.

Phillips, D. P. (1974). The influence of suggestion on suicide: Substantive and theoretical implications of the Werther Effect. American Sociological Review, 39, 340 354.

Phillips, K. (2005). The broken mirror: Understanding and treating body dysmorphic disorder. Oxford University Press.

Piper, A., & Merskey, H. (2004). The persistence of folly: A critical examination of dissociative identity disorder: Part I: The excesses of an improbable concept. Canadian Journal of Psychiatry, 49, 592 600.

Pittman, R. K. (1988). Post-traumatic stress disorder, conditioning, and network theory. Psychiatric Annals, 18, 182 189.

Pompili, M., Serafini, G., Innamorati, M., M ller-Leimk hler, A. M., Guipponi, G., Girardi, P., Tatarelli, R., & Lester, D. (2010). The hypothalamic-pituitary-adrenal axis and serotonin abnormalities: A selective overview of the implications of suicide prevention. European Archives of Psychiatry and Clinical Neuroscience, 260, 583 600.

Pope, H. G., Jr., Barry, S. B., Bodkin, A., & Hudson, J. I. (2006). Tracking scientific interest in the dissociative disorders: A study of scientific publication output 1984 2003. Psychotherapy and Psychosomatics, 75, 19 24.

Pope, H. G., Jr., Hudson, J. I., Bodkin, J. A., & Oliva, P. S. (1998). Questionable validity of dissociative amnesia in trauma victims: Evidence from prospective studies. British Journal of Psychiatry, 172, 210 215.

Pope, H. G., Jr., Poliakoff, M. B., Parker, M. P., Boynes, M., & Hudson, J. I. (2006). Is dissociative amnesia a culture-bound syndrome? Findings from a survey of historical literature. Psychological Medicine, 37, 225 233.

Postolache, T. T., Mortensen, P. B., Tonelli, L. H., Jiao, X., Frangakis, C., Soriano, J. J., & Qin, P. (2010). Seasonal spring peaks of suicide in victims with and without prior history of hospitalization for mood disorders. Journal of Affective Disorders, 121, 88 93.

Putnam, F.W., Guroff, J, J., Silberman, E. K., Barban, L., & Post, R. M. (1986). The clinical phenomenology of multiple personality disorder: A review of 100 recent cases. Journal of Clinical Psychiatry, 47, 285 293.

Rachman, S. (1977). The conditioning theory of fear acquisition: A critical examination. Behaviour Theory and Research, 15, 375 387.

Regier, D. A., Kuhl, E. A., & Kupfer, D. A. (2012). DSM-5: Classification and criteria changes. World Psychiatry, 12, 92 98.

Rhee, S. H., & Waldman, I. D. (2002). Genetic and environmental influences on antisocial behavior: A meta-analysis of twin and adoption studies. Psychological Bulletin, 128, 490 529.

Robinson, M. S., & Alloy, L. B. (2003). Negative cognitive styles and stress-reactive rumination interact to predict depression: A prospective study. Cognitive Therapy and Research, 27, 275 292.

Roche, T. (2002, March 18). Andrea Yates: More to the story. Time. http://content.time.com/time/nation/article/0,8599,218445,00.html.

Root, B. A. (2000). Understanding panic and other anxiety disorders. University Press of Mississippi.

Ross, C. A., Miller, S. D., Reagor, P., Bjornson, L., Fraser, G. A., & Anderson, G. (1990). Structured interview data on 102 cases of multiple personality disorder from four centers. The American Journal of Psychiatry, 147, 596 601.

Rothschild, A. J. (1999). Mood disorders. In A. M. Nicholi, Jr. (Ed.), The Harvard guide to psychiatry (pp. 281 307). The Belknap Press of Harvard University.

Ruder, T. D., Hatch, G. M., Ampanozi, G., Thali, M. J., & Fischer, N. (2011). Suicide announcement on Facebook. Crisis, 35, 280 282.

Ruscio, A. M., Stein, D. J., Chiu, W. T., & Kessler, R. C. (2010). The epidemiology of obsessive-compulsive disorder in the National Comorbidity Survey Replication. Molecular Psychiatry, 15, 53 63.

Rushworth, M. F., Noonan, M. P., Boorman, E. D., Walton, M. E., & Behrens, T. E. (2011). Frontal cortex and reward-guided learning and decision-making. Neuron, 70, 1054 1069.

Rotge, J. Y., Guehl, D., Dilharreguy, B., Cuny, E., Tignol, J., Biolac, B., . . . Aouizerate, B. (2008). Provocation of obsessive-compulsive symptoms: A quantitative voxel-based meta-analysis of functional neuroimaging studies. Journal of Psychiatry and Neuroscience, 33, 405 412.

Saxena, S., Bota, R. G., & Brody, A. L. (2001). Brain-behavior relationships in obsessive- compulsive disorder. Seminars in Clinical Neuropsychiatry, 6, 82 101.

Schwartz, T. (1981). The hillside strangler: A murderer s mind. New American Library.

Seligman, M. E. P. (1971). Phobias and preparedness. Behavioral Therapy, 2, 307 320.

Shih, R. A., Belmonte, P. L., & Zandi, P. P. (2004). A review of the evidence from family, twin, and adoption studies for a genetic contribution to adult psychiatric disorders. International Review of Psychiatry, 16, 260 283.

Siegle, G. J., Thompson, W., Carter, C. S., Steinhauer, S. R., & Thase, M. E. (2007). Increased amygdala and decreased dorsolateral prefrontal BOLD responses in unipolar depression: Related and independent features. Biological

Psychiatry, 61, 198 209.

Silverstein, C. (2009). The implications of removing homosexuality from the DSM as a mental disorder. Archives of Sexual Behavior, 38, 161 163.

Simon, D., Kaufmann, C., M sch, K., Kischkel, E., & Kathmann, N. (2010). Fronto-striato-limbic hyperactivation in obsessive-compulsive disorder during individually tailored symptom provocation. Psychophysiology, 47(4), 728 738. doi:10.1111/j.1469-8986.2010.00980.x

Snyder, S. H. (1976). The dopamine hypothesis of schizophrenia: Focus on the dopamine receptor. The American Journal of Psychiatry, 133, 197 202.

Spinaris, C. G., Denhof, M. D., & Kellaway, J. A. (2012). Posttraumatic stress disorder in United States corrections professionals: Prevalence and impact on health and functioning. Desert Waters Correctional Outreach. http://desertwaters.com/wp-content/uploads/2013/09/PTSD_Prev_in_Correcti ons_09-03-131.pdf

Stack, S. (2000). Media impacts on suicide: A quantitative review of 243 findings. Social Science Quarterly, 81, 957 971.

Stanley, B., Molcho, A., Stanley, M., Winchel, R., Gameroff, M. J., Parson, B., & Mann, J. J. (2000). Association of aggressive behavior with altered serotonergic function in patients who are not suicidal. American Journal of Psychiatry, 157, 609 614.

Stein, M. B., & Kean, Y. M. (2000). Disability and quality of life in social phobia: Epidemiological findings. The American Journal of Psychiatry, 157, 1606 1613.

Steinmetz, J. E., Tracy, J. A., & Green, J. T. (2001). Classical eyeblink conditioning: Clinical models and applications. Integrative Physiological and Behavioral Science, 36, 220 238.

Stuppy-Sullivan, A., & Baskin-Sommers, A. (2019). Evaluating dysfunction in cognition and reward among offenders with antisocial personality disorder. Personality Disorders Theory, Research, and Treatment. https://modlab.yale.edu/sites/default/files/files/StuppySullivanBaskinSo mmers_reward_2019.pdf

Surguladze, S., Brammer, M. J., Keedwell, P., Giampietro, V., Young, A. W., Travis, M. J., . . . Phillips, M. L. (2005). A differential pattern of neural response toward sad versus happy facial expressions in major depressive disorder. Biological Psychiatry, 57, 201 209.

Szasz, T. S. (1960). The myth of mental illness. American Psychologist, 15, 113 118.

Szasz, T. S. (2010). The myth of mental illness: Foundations of a theory of personal conduct. HarperCollins (Original work published 1961)

Szasz, T. S. (1965). Legal and moral aspects of homosexuality. In J. Marmor (Ed.), Sexual inversion: The multiple roots of homosexuality (pp. 124 139). Basic Books.

Swanson, J. M., Kinsbourne, M., Nigg, J., Lanphear, B., Stephanatos, G., Volkow, N., . . . Wadhwa, P. D. (2007). Etiologic subtypes of attention-deficit/hyperactivity disorder: Brain imaging, molecular genetic and environmental factors and the dopamine hypothesis. Neuropsychology Review, 17, 39 59.

Thakur, G. A., Sengupta, S. M., Grizenko, N., Schmitz, N., Pag , V., & Joober, R. (2013). Maternal smoking during pregnancy and ADHD: A comprehensive clinical and neurocognitive characterization. Nicotine and Tobacco Research, 15, 149 157.

Thase, M. E. (2009). Neurobiological aspects of depression. In I. H. Gotlib & C. L. Hammen (Eds.), Handbook of depression (pp. 187 217). Guilford Press.

The Associated Press. (2013, May 15). New psychiatric manual, DSM-5, faces criticism for turning normal human problems into mental illness. http://www.nydailynews.com/life-style/health/shrinks-critics-face-new-ps ychiatric-manual-article-1.1344935

Thompson, A., Molina, B. S. G., Pelham, W., & Gnagy, E. M. (2007). Risky driving in adolescents and young adults with childhood ADHD. Journal of Pediatric Psychology, 32, 745 759.

Thornicroft, G. (1990). Cannabis and psychosis: Is there epidemiological evidence for an association? British Journal of Psychiatry, 157, 25 33.

Tienari, P., Wynne, L. C., Sorri, A., Lahti, I., Lasky, K., Moring, J., . . . Wahlberg, K. (2004). Genotype-environment interaction in schizophrenia spectrum disorder. British Journal of Psychiatry, 184, 216 222.

Trezza V., Cuomo, V., & Vanderschuren, L. J. (2008). Cannabis and the developing brain: Insights from behavior. European Journal of Pharmacology, 585, 441 452.

Tsuang, M. T., Farone, S. V., & Green, A. I. (1999). Schizophrenia and other psychotic disorders. In A. M. Nicholi, Jr. (Ed.), The Harvard guide to psychiatry (pp. 240 280). The Belknap Press of Harvard University Press.

van Praag, H. M. (2005). Can stress cause depression?. The World Journal of Biological Psychiatry, 6(S2), 5 22.

Victor, T. A., Furey, M. L., Fromm, S. J., hman, A., & Drevets, W. C. (2010). Relationship between amygdala responses to masked faces and mood state and treatment in major depressive disorder. Archives of General Psychiatry, 67, 1128 1138.

Volkow N. D., Fowler J. S., Logan J., Alexoff D., Zhu W., Telang F., . . . Apelskog-Torres K. (2009). Effects of modafinil on dopamine and dopamine transporters in the male human brain: clinical implications. Journal of the American Medical Association, 301, 1148 1154.

Wakefield, J. C. (1992). The concept of mental disorder: On the boundary between biological facts and social values. American Psychologist, 47, 373 388.

Waller, J. (2009a). Looking back: Dancing plagues and mass hysteria. The Psychologist, 22(7), 644 647.

Waller, J. (2009b, February 21). A forgotten plague: Making sense of dancing mania. The Lancet, 373(9664), 624 625. doi:10.1016/S0140-6736(09)60386-X

Weiser, E. B. (2007). The prevalence of anxiety disorders among adults with Asthma: A meta-analytic review. Journal of Clinical Psychology in Medical Settings, 14, 297 307.

White, C. N., Gunderson, J. G., Zanarani, M. C., & Hudson, J. I. (2003). Family studies of borderline personality disorder: A review. Harvard Review of Psychiatry, 11, 8 19.

Whooley, M. A. (2006). Depression and cardiovascular disease: Healing the broken-hearted. Journal of the American Medical Association, 295, 2874 2881.

Wilcox, H. C., Conner, K. R., & Caine, E. D. (2004). Association of alcohol and drug use disorders and completed suicide: An empirical review of cohort studies. Drug and Alcohol Dependence, 76, S11 S19.

Wing, L., Gould, J., & Gillberg, C. (2011). Autism spectrum disorders in the DSM-V: Better or worse than the DSM IV? Research in Developmental Disabilities, 32, 768 773.

Wisner, K. L., Sit, D. K. Y., McShea, M. C., Rizzo, D. M., Zoretich, R. A., Hughes, C. L., Hanusa, B. H. (2013). Onset timing, thoughts of self-harm, and diagnoses in postpartum women with screen-positive depression findings. JAMA Psychiatry, 70, 490 498.

Wolraich, M. L., Wilson, D. B., & White, J. W. (1995). The effect of sugar on behavior or cognition in children. Journal of the American Medical Association, 274, 1617 1621.

World Health Organization (WHO). (2013). International classification of diseases (ICD). http://www.who.int/classifications/icd/en/

Wyatt, W. J., & Midkiff, D. M. (2006). Biological psychiatry: A practice in search of a science. Behavior and Social Issues, 15, 132 151.

Xie, P., Kranzler, H. R., Poling, J., Stein, M. B., Anton, R. F., Brady, K., Gelernter, J. (2009). Interactive effect of stressful life events and the serotonin transporter 5-HTTLPR genotype on posttraumatic stress disorder diagnosis in 2 independent populations. Archives of General Psychology, 66, 1201 1209.

Zachar, P., & Kendler, K. S. (2007). Psychiatric disorders: A conceptual taxonomy. The American Journal of Psychiatry, 16, 557 565.

Zuckerman, M. (1999). Vulnerability to psychopathology: A biosocial model. American Psychological Association.

Therapy and treatment

Abbass, A., Kisely, S., & Kroenke, K. (2006). Short-term psychodynamic psychotherapy for somatic disorders: Systematic review and meta-analysis of clinical trials. Psychotherapy and Psychosomatics, 78, 265 274.

Ahmed, S., Wilson, K. B., Henriksen, R. C., & Jones, J. W. (2011). What does it mean to be a culturally competent counselor? Journal for Social Action in Counseling and Psychology, 3(1), 17 28.

Alavi, A., Sharifi, B., Ghanizadeh, A., & Dehbozorgi, G. (2013). Effectiveness of cognitive-behavioral therapy in decreasing suicidal ideation and hopelessness of the adolescents with previous suicidal attempts. Iranian Journal of Pediatrics, 23(4), 467 472.

Alegr a, M., Chatterji, P., Wells, K., Cao, Z., Chen, C. N., Takeuchi, D., . . . Meng, X. L. (2008). Disparity in depression treatment among racial and ethnic minority populations in the United States. Psychiatric Services, 59(11), 1264 1272.

American Psychological Association. (2005). Policy statement on evidence-based practice in psychology. http://www.apapracticecentral.org/ce/courses/ebpstatement.pdf

American Psychological Association. (2014). Can psychologists prescribe medications for their patients? http://www.apa.org/news/press/releases/2004/05/louisiana-rx.aspx

American Psychological Association. (2014). Psychotherapy: Understanding group therapy. http://www.apa.org/helpcenter/group-therapy.aspx

Beck, A. T., Rush, A. J., Shaw, B. F., & Emery, G. (1979). Cognitive therapy of depression. The Guilford Press.

Beck Institute for Cognitive Behavior Therapy. (n.d.). History of cognitive therapy. http://www.beckinstitute.org/history-of-cbt/

Beck, J. S. (2011). Cognitive behavior therapy: Basics and beyond (2nd ed.). The Guilford Press.

Belgrave, F., & Allison, K. (2010). African-American psychology: From Africa to America (2nd ed.). Sage Publications.

Bertrand, K., Richer, I., Brunelle, N., Beaudoin, I., Lemieux, A., & M nard, J-M. (2013). Substance abuse treatment for adolescents: How are family factors related to substance use change? Journal of Psychoactive Drugs, 45(1), 28 38.

Blank, M. B., Mahmood, M., Fox, J. C., & Guterbock, T. (2002). Alternative mental health services: The role of the Black church in the South. American Journal of Public Health, 92, 1668 1672.

Blumberg, J. (2007, October 24). A brief history of the Salem witch trials. Smithsonian.com. http://www.smithsonianmag.com/history-archaeology/brief-salem.html?c=y&p age=2

Butlera, A. C., Chapmanb, J. E., Formanc, E. M., & Becka, A. T. (2006). The empirical status of cognitive-behavioral therapy: A review of meta-analyses. Clinical Psychology Review, 26,17 31.

Center for Substance Abuse Treatment. (2005). Substance Abuse Treatment: Group Therapy. Treatment Improvement Protocol (TIP) Series 41. DHHS Publication No. (SMA) 05-3991. Substance Abuse and Mental Health Services Administration.

Centers for Disease Control and Prevention. (2014). Suicide prevention: Youth suicide. http://www.cdc.gov/violenceprevention/pub/youth_suicide.html

Chambless, D. L., & Ollendick, T. H. (2001). Empirically supported psychological interventions: Controversies and evidence. Annual Review of Psychology, 52, 685 716.

Charman, D., & Barkham, M. (2005). Psychological treatments: Evidence-based practice and practice-based evidence. InPsych Highlights. www.psychology.org.au/publications/inpsych/treatments

Chorpita, B. F., Daleiden, E. L., Ebesutani, C., Young, J., Becker, K. D., Nakamura, B. J., . . . Starace, N. (2011), Evidence-based treatments for children and adolescents: An updated review of indicators of efficacy and effectiveness. Clinical Psychology: Science and Practice, 18, 154 172.

Clement, S., Schauman, O., Graham, T., Maggioni, F., Evans-Lacko, S., Bezborodovs, N., . . . Thornicroft, G. (2014, February 25). What is the impact of mental health-related stigma on help-seeking? A systematic review of quantitative and qualitative studies. Psychological Medicine, l 17.

Cuijpers, P., Sijbrandij, M. Koole, S. L., Andersson, G., Beekman, A. T., & Reynolds, C. F. (2014). Adding psychotherapy to antidepressant medication in depression and anxiety disorders: a meta-analysis. World Psychiatry, 13(1), 56 67.

Daniel, D. (n.d.). Rational emotive in behavior therapy the context of modern psychlogical research. albertellis.org/rebt-in-the-context-of-modern-psychological-research

Davidson, W. S. (1974). Studies of aversive conditioning for alcoholics: A critical review of theory and research methodology. Psychological Bulletin, 81(9), 571 581.

DeRubeis, R. J., Hollon, S. D., Amsterdam, J. D., Shelton, R. C., Young, P. R., Salomon, R. M., . . . Gallop, R. (2005). Cognitive Therapy vs medications in the treatment of moderate to severe depression. Archives of General Psychiatry, 62(4), 409 416.

DeYoung, S. H. (2013, November 14). The woman who raised that monster. http://www.huffingtonpost.com/suzy-hayman-deyoung/the-woman-who-raised-t hat_b_4266621.html

Dickerson, F. B., Tenhula, W. N., & Green-Paden, L. D. (2005). The token economy for schizophrenia: Review of the literature and recommendations for future research. Schizophrenia Research, 75(2), 405 416.

Donahue, A. B. (2000). Electroconvulsive therapy and memory loss: A personal journey. The Journal of ECT, 162, 133 143.

Elkins, R. L. (1991). An appraisal of chemical aversion (emetic therapy) approaches to alcoholism treatment. Behavior Research and Therapy, 29(5), 387 413.

Gary, F. A. (2005). Stigma: Barrier to mental health care among ethnic minorities. Issues in Mental Health Nursing, 26(10), 979 999.

Gerardi, M., Cukor, J., Difede, J., Rizzo, A., & Rothbaum, B. O. (2010). Virtual reality exposure therapy for post-traumatic stress disorder and other anxiety disorders. Current Psychiatry Reports, 12(298), 299 305.

Harter, S. (1977). A cognitive-developmental approach to children's expression of conflicting feelings and a technique to facilitate such expression in play therapy. Journal of Consulting and Clinical Psychology, 45(3), 417 432.

Hemphill, R. E. (1966). Historical witchcraft and psychiatric illness in Western Europe. Proceedings of the Royal Society of Medicine, 59(9), 891 902.

Ivey, S. L., Scheffler, R., & Zazzali, J. L. (1998). Supply dynamics of the mental health workforce: Implications for health policy. Milbank Quarterly, 76(1), 25 58.

Jang, Y., Chiriboga, D. A., & Okazaki, S. (2009). Attitudes toward mental health services: Age group differences in Korean American adults. Aging & Mental Health, 13(1), 127 134.

Jones, M. C. (1924). A laboratory study of fear: The case of Peter. Pedagogical Seminary, 31, 308 315.

Kalff, D. M. (1991). Introduction to sandplay therapy. Journal of Sandplay Therapy, 1(1), 9.

Leblanc, M., & Ritchie, M. (2001). A meta-analysis of play therapy outcomes. Counselling Psychology Quarterly, 14(2), 149 163.

Lovaas, O. I. (1987). Behavioral treatment and normal educational and intellectual functioning in young autistic children. Journal of Consulting & Clinical Psychology, 55, 3 9.

Lovaas, O. I. (2003). Teaching individuals with developmental delays: Basic intervention techniques. Pro-Ed.

Lowinger, R. J., & Rombom, H. (2012). The effectiveness of cognitive behavioral therapy for PTSD in New York City Transit Workers. North American Journal of Psychology, 14(3), 471 484.

Madanes, C. (1991). Strategic family therapy. In A. S. Gurman and D. P. Kniskern (Eds.), Handbook of Family Therapy, Vol. 2. (pp. 396 416). Brunner/Mazel.

Marques, L., Alegr a, M., Becker, A. E., Chen, C. N., Fang, A., Chosak, A., & Diniz, J. B. (2011). Comparative prevalence, correlates of impairment, and service utilization for eating disorders across US ethnic groups: Implications for reducing ethnic disparities in health care access for eating disorders. International Journal of Eating Disorders, 44(5), 412 420.

Martin, B. (2007). In-Depth: Cognitive behavioral therapy. http://psychcentral.com/lib/in-depth-cognitive-behavioral-therapy/000907

Mayo Clinic. (2012). Tests and procedures: Transcranial magnetic stimulation. http://www.mayoclinic.org/tests-procedures/transcranial-magnetic-stimula tion/basics/definition/PRC-20020555

McGovern, M. P., & Carroll, K. M. (2003). Evidence-based practices for substance use disorders. Psychiatric Clinics of North America, 26, 991 1010.

McGrath, R. J., Cumming, G. F., Burchard, B. L., Zeoli, S., & Ellerby, L. (2009). Current practices and emerging trends in sexual abuser management: The safer society North American survey. The SaferSociety Press.

McLellan, A. T., Lewis, D. C., O Brien, C. P., & Kleber, H. D. (2000). Drug dependence, a chronic medical illness: Implications for treatment, insurance, and outcomes evaluation. JAMA, 284(13), 1689 1695.

Minuchin, P. (1985). Families and individual development: Provocations from the field of family therapy. Child Development, 56(2), 289 302.

Mullen, E. J., & Streiner, D. L. (2004). The evidence for and against evidence-based practice. Brief Treatment and Crisis Intervention, 4(2), 111 121.

Mu oz-Cuevas, F. J., Athilingam, J., Piscopo, D., & Wilbrecht, L. (2013). Cocaine-induced structural plasticity in frontal cortex correlates with conditioned place preference. Nature Neuroscience, 16, 1367 1369.

National Association of Cognitive-Behavioral Therapists. (2009). History of cognitive behavioral therapy. http://nacbt.org/historyofcbt.htm.

National Institute of Mental Health. (2017). Any disorder among children. http://www.nimh.nih.gov/statistics/1ANYDIS_CHILD.shtml

National Institute of Mental Health. (n.d.). Use of mental health services and treatment among children. http://www.nimh.nih.gov/statistics/1NHANES.shtml

National Institutes of Health. (2013, August 6). Important events in NIMH history. http://www.nih.gov/about/almanac/organization/NIMH.htm

National Institute on Drug Abuse. (2008). Addiction science: From Molecules to managed care. http://www.drugabuse.gov/publications/addiction-science/relapse

National Institute on Drug Abuse. (2011). Drug facts: Comorbidity: Addiction and other mental disorders. http://www.drugabuse.gov/publications/drugfacts/comorbidity-addiction-ot her-mental-disorders

National Institute on Drug Abuse. (2012). Principles of drug addiction treatment: A research-based guide (3rd ed.). http://www.drugabuse.gov/publications/principles-drug-addiction-treatmen t-research-based-guide-third-edition/principles-effective-treatment

Nelson, P. (1993). Autobiography in Five Short Chapters. In There s a Hole in my Sidewalk: The Romance of Self-Discovery. Beyond Words Publishing.

O Connor, K. J. (2000). The play therapy primer (2nd ed.). Wiley.

Page, R. C., & Berkow, D. N. (1994). Unstructured group therapy: Creating contact, choosing relationship. Jossey Bass.

Pagnin, D., de Queiroz, V., Pini, S., & Cassano, G. B. (2004). Efficacy of ECT in depression: A meta-analytic review. Journal of ECT, 20, 13 20.

Prins, S. J., & Draper, L. (2009). Improving outcomes for people with mental illnesses under community corrections supervision: A guide to research-informed policy and practice. Council of State Governments Justice Center.

Prochaska, J. O., & Norcross, J. C. (2010). Systems of psychotherapy (7th ed.). Wadsworth.

Prudic, J., Peyser, S., & Sackeim, H. A. (2000). Subjective memory complaints: A review of patient self-assessment of memory after electroconvulsive therapy. The Journal of ECT, 16(2), 121 132.

Rathus, J. H., & Sanderson, W. C. (1999). Marital distress: Cognitive behavioral treatments for couples. Jason Aronson.

Reti, I. R. (n.d.). Electroconvulsive therapy today. Johns Hopkins Medicine. http://www.hopkinsmedicine.org/psychiatry/specialty_areas/brain_stimulat ion/docs/DepBulletin407_ECT_extract.pdf

Richman, L. S., Kohn-Wood, L. P., & Williams, D. R. (2007). The role of discrimination and racial identity for mental health service utilization. Journal of Social and Clinical Psychology, 26(8), 960 981.

Rizzo, A., Newman, B., Parsons, T., Difede, J., Reger, G., Holloway, K., . . . Bordnick, P. (2010). Development and clinical results from the Virtual Iraq exposure therapy application for PTSD. Annals of the New York Academy of Sciences, 1208, 114 125.

Rogers, C. (1951). Client-centered psychotherapy. Houghton-Mifflin.

Sackett, D. L., & Rosenberg, W. M. (1995). On the need for evidence-based medicine. Journal of Public Health, 17, 330 334.

Sallows, G. O., & Graupner, T. D. (2005). Intensive behavioral treatment for children with autism: Four-year outcome and predictors. American Journal of Mental Retardation, 110(6), 417 438.

Scott, L. D., McCoy, H., Munson, M. R., Snowden, L. R., & McMillen, J. C. (2011). Cultural mistrust of mental health professionals among Black males transitioning from foster care. Journal of Child and Family Studies, 20, 605 613.

Shechtman, Z. (2002). Child group psychotherapy in the school at the threshold of a new millennium. Journal of Counseling and Development, 80(3), 293 299.

Shedler, J. (2010). The efficacy of psychodynamic psychotherapy. American Psychologist, 65, 98 109.

Simpson D. D. (1981). Treatment for drug abuse. Archives of General Psychiatry, 38, 875 880.

Simpson D. D, Joe, G. W, & Bracy, S. A. (1982). Six-year follow-up of opioid addicts after admission to treatment. Archives General Psychiatry, 39, 1318 1323.

Snowden, L. R. (2001). Barriers to effective mental health services for African Americans. Mental Health Services Research, 3, 181 187.

Stensland, M., Watson, P. R., & Grazier, K. L. (2012). An examination of costs, charges, and payments for inpatient psychiatric treatment in community hospitals. Psychiatric Services, 63(7), 66 71.

Stewart, S. M., Simmons, A., & Habibpour, E. (2012). Treatment of culturally diverse children and adolescents with depression. Journal of Child and Adolescent Psychopharmacology, 22(1), 72 79.

Streeton, C., & Whelan, G. (2001). Naltrexone, a relapse prevention maintenance treatment of alcohol dependence: A meta-analysis of randomized controlled trials. Alcohol and Alcoholism, 36(6), 544 552.

Sue, D. W. (2001). Multidimensional facets of cultural competence. Counseling Psychologist, 29(6), 790 821.

Sue, D. W. (2004). Multicultural counseling and therapy (MCT). In J. A. Banks and C. Banks (Eds.), Handbook of research on multicultural education (2nd ed., pp. 813 827). Jossey-Bass.

Sue, D. W., & Sue, D. (2007). Counseling the culturally different: Theory and practice (5th ed.). Wiley.

Sussman, L. K., Robins, L. N., & Earls, F. (1987). Treatment seeking for depression by Black and White Americans. Social Science & Medicine, 24, 187 196.

Thomas, K. C., & Snowden, L. R. (2002). Minority response to health insurance coverage for mental health services. Journal of Mental Health Policy and Economics, 4, 35 41.

Tiffany, F. (2012/1891). Life of Dorothea Lynde Dix (7th ed.). Houghton, Mifflin.

Torrey, E. F. (1997). Out of the shadows: Confronting America's mental illness crisis. Wiley.

Torrey, E. F., Zdanowicz, M. T., Kennard, A. D., Lamb, H. R., Eslinger, D. F., Biasotti, M. C., & Fuller, D. A. (2014, April 8). The treatment of persons with mental illness in prisons and jails: A state survey. Treatment Advocacy Center. http://tacreports.org/storage/documents/treatment-behind-bars/treatment-behind-bars.pdf

Townes D. L., Cunningham N. J., & Chavez-Korell, S. (2009). Reexaming the relationships between racial identity, cultural mistrust, help-seeking attitudes, and preference for a Black counselor. Journal of Counseling Psychology, 56(2), 330 336.

U.S. Department of Agriculture. (2013, December 10). USDA announces support for mental health facilities in rural areas [Press release No. 0234.13]. http://www.usda.gov/wps/portal/usda/usdahome?contentid=2013/12/0234.xml

U.S. Department of Health and Human Services. (1999). Mental health: A report of the Surgeon General. Rockville, MD: U.S. Department of Health and Human Services, Substance Abuse and Mental Health Services Administration, Center for Mental Health Services, National Institutes of Health, National Institute of Mental Health.

U.S. Department of Health and Human Services, Health Resources and Services Administration, Office of Rural Health Policy. (2005). Mental health and rural America: 1984-2005. ftp://ftp.hrsa.gov/ruralhealth/RuralMentalHealth.pdf

U.S. Department of Health and Human Services, Substance Abuse and Mental Health Services Administration, Center for Behavioral Health Statistics and Quality. (2017). Results from the 2016 National Survey on Drug Use and Health: Mental Health Findings. https://www.samhsa.gov/data/sites/default/files/NSDUH-DetTabs-2016/NSDUH-DetTabs-2016.pdf

U.S. Department of Health and Human Services, Substance Abuse and Mental Health Services Administration, Center for Behavioral Health Statistics and Quality. (2011, September). Results from the 2010 National Survey on Drug Use and Health: Summary of National Findings (NSDUH Series H-41, HHS Publication No. [SMA] 11-4658). http://www.samhsa.gov/data/NSDUH/2k10ResultsRev/NSDUHresultsRev2010.htm

U.S. Department of Health and Human Services, Substance Abuse and Mental Health Services Administration, Center for Behavioral Health Statistics and Quality. (2013, September). Results from the 2012 National Survey on Drug Use and Health: Summary of National Findings (NSDUH Series H-46, HHS Publication No. [SMA] 13-4795). http://www.samhsa.gov/data/NSDUH/2012SummNatFindDetTables/NationalFindings/NSDUHresults2012.htm#ch2.2

U.S. Department of Housing and Urban Development, Office of Community Planning and Development. (2011). The 2010 Annual Homeless Assessment Report to Congress. http://www.hudhre.info/documents/2010HomelessAssessmentReport.pdf

U.S. Department of Labor. (n.d.). Mental health parity. http://www.dol.gov/ebsa/mentalhealthparity/

U.S. Public Health Service. (2000). Report of the Surgeon General s conference on children s mental health: A national action agenda. Department of Health and Human Services.

Wagenfeld, M. O., Murray, J. D., Mohatt, D. F., & DeBruiynb, J. C. (Eds.). (1994). Mental health and rural America: 1980 1993 (NIH Publication No. 94-3500). U.S. Government Printing Office.

Wampold, B. E. (2007). Psychotherapy: The humanistic (and effective) treatment. American Psychologist, 62, 857 873. doi:10.1037/0003-066X.62.8.857

Weil, E. (2012, March 2). Does couples therapy work? The New York Times. http://www.nytimes.com/2012/03/04/fashion/couples-therapists-confront-th e-stresses-of-their-field.html?pagewanted=all&_r=0

Weiss, R. D., Jaffee, W. B., de Menil, V. P., & Cogley, C. B. (2004). Group therapy for substance abuse disorders: What do we know? Harvard Review of Psychiatry, 12(6), 339 350.

Willard Psychiatric Center. (2009). Echoes of Willard. http://www.echoesofwillard.com/willard-psychiatric-centre/

Wolf, M., & Risley, T. (1967). Application of operant conditioning procedures to the behavior problems of an autistic child: A follow-up and extension. Behavior Research and Therapy, 5(2), 103 111.

Wolpe, J. (1958). Psychotherapy by reciprocal inhibition. Stanford University Press.

Attribution and Production Record

OpenStax attribution

Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett. OpenStax, Rice University, 2020. Used under Creative Commons Attribution 4.0 International. Source repository: https://github.com/cnx-user-books/cnxbook-psychology-2e. Pinned snapshot: f5c9cf50686749d28cc5a594e0fdc487398f8307.

License: https://creativecommons.org/licenses/by/4.0/. The full license text is included in the source archive accompanying the production files. OpenStax and Rice University do not endorse this adaptation.

Changes made for this course

Selected and reorganized modules for CAC.901.

Converted CNXML to a new PDF layout.

Added new course lectures, application tasks, and twenty original vector diagrams.

Source photographs and illustrations are not reproduced; their alt text and captions are retained as descriptions.

Removed obsolete video-promotion boxes and source learning-objective boxes from the reading stream.

Reference lists are collected at the back and excluded from the substantive word count.

Module register

Chapter 1: m82218 What Is Lifespan Development?.

Chapter 2: m82167 Why Is Research Important?, m82170 Ethics.

Chapter 3: m82168 Approaches to Research, m82169 Analyzing Findings.

Chapter 4: m82219 Lifespan Theories.

Chapter 5: m82220 Stages of Development.

Chapter 6: m82172 Human Genetics, m82173 Cells of the Nervous System.

Chapter 7: m82175 The Brain and Spinal Cord, m82176 The Endocrine System.

Chapter 8: m82199 What Is Learning?, m82200 Classical Conditioning, m82201 Operant Conditioning, m82202 Observational Learning (Modeling).

Chapter 9: m82205 What Is Cognition?, m82206 Language, m82207 Problem Solving.

Chapter 10: m82208 What Are Intelligence and Creativity?, m82209 Measures of Intelligence, m82211 The Source of Intelligence.

Chapter 11: m82213 How Memory Functions, m82215 Problems with Memory, m82216 Ways to Enhance Memory.

Chapter 12: m82225 Motivation, m82229 Emotion.

Chapter 13: m82233 What Is Personality?, m82235 Neo-Freudians: Adler, Erikson, Jung, and Horney, m82239 Trait Theorists, m82240 Cultural Understandings of Personality.

Chapter 14: m82273 What Is Social Psychology?, m82247 Self-presentation, m82249 Conformity, Compliance, and Obedience.

Chapter 15: m82250 Prejudice and Discrimination, m82251 Aggression, m82252 Prosocial Behavior.

Chapter 16: m82259 What Is Stress?, m82260 Stressors, m82262 Regulation of Stress.

Chapter 17: m82265 What Are Psychological Disorders?, m82266 Diagnosing and Classifying Psychological Disorders, m82267 Perspectives on Psychological Disorders, m82268 Anxiety Disorders, m82274 Posttraumatic Stress Disorder.

Chapter 18: m82278 Disorders in Childhood.

Chapter 19: m82282 Types of Treatment, m82283 Treatment Modalities, m82285 The Sociocultural Model and Therapy Utilization.

Rights and authorship distinction

The newly written counseling lectures, fictional applications, seminar tasks, glossary, and original vector diagrams are course-specific additions. The foundational reading text remains attributed to the OpenStax authors. Third-party material is not relicensed by this project. No source photographs are reproduced. No endorsement by OpenStax, Rice University, or a university degree program is implied.

Publication verification

This PDF is generated from the saved manuscript and source blocks. The production check verifies the individual PDF s instructional word count, page count, chapter navigation, diagrams, and absence of empty pages. The reference section is excluded from the minimum-length check. See the machine-readable verification report in the production workspace for the exact count and file hash.